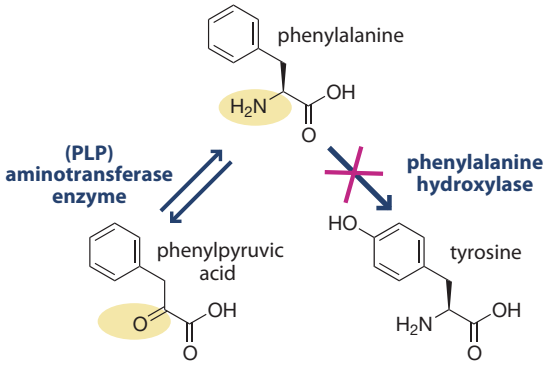



**Figure 6.81 A deficiency in phenylalanine hydroxylase is diagnosable by the buildup of phenylpyruvic acid.** PLP, pyridoxal phosphate.

phenylketonuria. Children diagnosed with phenylketonuria can develop normally if they adhere to a diet low in phenylalanine. Packages of the synthetic sweetener aspartame contain a warning to phenylketonurics because aspartame is a dipeptide composed of phenylalanine and aspartic acid.

The equilibrium between phenylpyruvic acid and phenylalanine is coupled to the equilibrium between pyridoxal phosphate and pyridoxylamine. This transamination reaction, catalyzed by a group of amino acid aminotransferases, is key to the biosynthesis of all amino acids. The mechanism involves the formation of an imine between the amino acid and pyridoxal phosphate, followed by a stereoselective tautomerization (**Figure 6.82**).

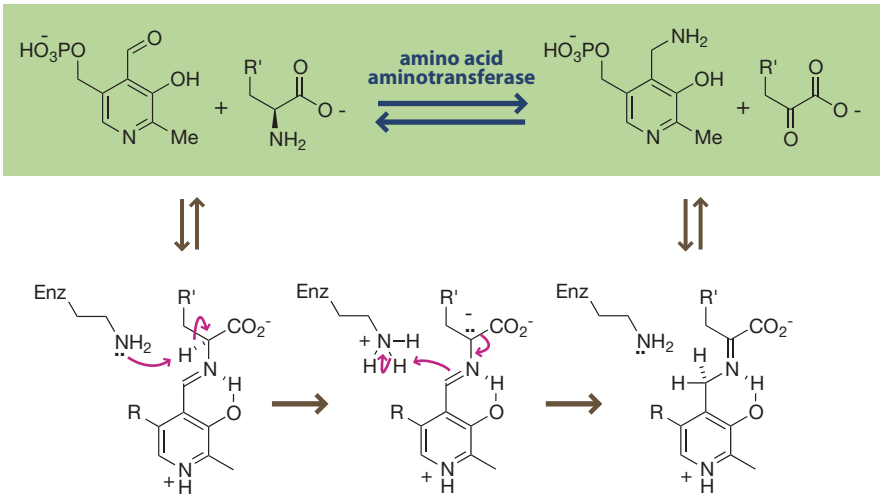
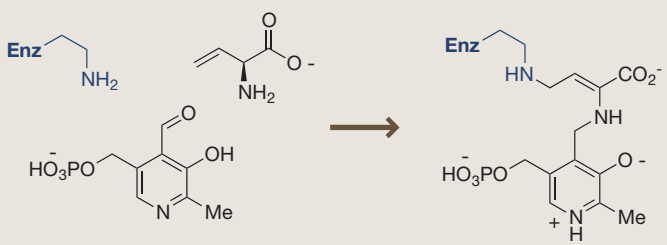



**Figure 6.82 Aminotransferase mechanism.** Aminotransferases catalyze tautomerizations that convert alkylamines to ketones and pyridoxal to pyridoxylamine.

## Problem 6.14

Vinylglycine is an irreversible inhibitor of alanine aminotransferase that forms a covalent crosslink between the enzyme and pyridoxal. Suggest a plausible arrow-pushing mechanism for formation of the crosslink.



# 6.5 ENGINEERING IMPROVED PROTEIN FUNCTION

## Protein engineering provides power tools for the dissection of protein function and the development of hyperfunctional molecules

Given the remarkable capabilities of the average protein, one might question the need to equip them with new functions. After all, each enzyme, the result of billions of years of evolution, offers a finely tuned, powerful catalyst for a particular chemical reaction. Furthermore, antibodies and hybridoma technology can provide a potentially limitless source of high-affinity receptors for binding to essentially any target. However, not all naturally occurring proteins are up to the tasks required by chemical biologists. Antibodies from mice, for example, cause immunogenic reactions in humans. Furthermore, no one wants to wait for the ideal enzyme to be identified from the proteome of, for example, the red squirrel. The quest to tinker with protein sequences to alter protein functions has also provided important insights into the mechanisms underlying protein function. A number of different computational approaches can contribute to the tailoring of protein function. Such experiments remain at the frontiers of protein engineering, and are not discussed here because of their evolving capabilities, which can quickly make even the most rudimentary discussion obsolete.

Unfortunately, the label "protein engineering" implies a much higher degree of precision than is available with current technologies. No civil engineer would plan to build a bridge or skyscraper with materials as unpredictable as proteins, which are held together and functioning primarily through the capricious properties of molecular recognition. Every change made to a protein sequence could have unpredictable consequences for the protein folding pathways, and to the ultimate conformation and function of the altered protein. In addition, proteins are large, complex machines. It is typically nonobvious where to introduce changes to obtain a desired new activity.

Many an elegant protein engineering project has foundered on misplaced assumptions and too little knowledge of the complete properties and consequences of changes to amino acid side chains. Despite such strong caveats, considerable progress has been made in the field by literally leveraging the uncertainty. In other words, if you cannot know in advance the exact properties of a particular alteration to the structure, make a large number of changes. Then, select from this collection or library for some powerful new function, and prepare (or hope?) for surprises. As we will see in this section, despite its misnomer protein engineering offers power tools for the transmogrification of protein function.

## Alanine scanning assigns function to side chains and motifs

In reverse engineering, a complicated machine is taken apart piece by piece to dissect how the pieces fit together and work. This technique is often used in the computer chip and automobile industries to uncover the secrets of a competitor's product. The typical experiment begins with the thought, "Hmm… I wonder what this part does." Then, after removing it, you test how well the machine functions without it. Insight from reverse engineering can guide development of both outright copies and improved versions of a product or process. Building an improved version is an especially important part of the innovation loop, because it allows the testing of hypotheses suggested by close examination of each component of the machine.

Many experiments in protein engineering apply this reverse engineering approach. For example, both reverse engineering and protein engineering begin with a mysterious structure having a particular ability. The primary structure or protein sequence, in fact, is sometimes all that is available at the beginning of a protein engineering project. Then, changes are made to specific side chains, allowing one to deduce the contributions made to protein function by the altered side-chain functionalities. In such experiments, the naturally occurring protein found in the wild is typically referred to as the wild-type protein. In contrast, a protein that is longer, shorter, or has different amino acids as a result of genetic mutation should be referred to as a variant or a mutant protein.

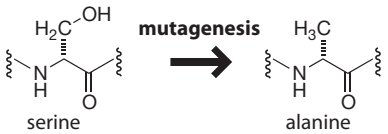



**Figure 6.83 Reverse engineering.** The importance of a serine hydroxyl group can be revealed by mutating the serine to an alanine.

One common alteration to protein structure, replacement of a particular amino acid with alanine, effectively truncates the side chain at the β-carbon (**Figure 6.83**). Thus, the alanine-substituted protein can be tested to investigate the contributions to protein function made by the atoms past the β-carbon. For example, substitution of alanine for a serine residue allows one to test the contributions, if any, made by the hydroxyl functionality of the serine. Alanine provides a particularly useful substitution; the change clips off the side-chain functionalities without introducing excessive additional flexibility into the protein backbone, which could result from replacement with glycine.

Some of the earliest peptide synthesis experiments applied this reverse engineering approach to understanding protein function. After inventing solid-phase peptide synthesis, Merrifield and coworkers chemically synthesized a wide variety of peptide hormones and proteins, including the 124-residue ribonuclease A (RNase A). In 1969, the latter *tour de force* galvanized the community around solid-phase synthesis. Overcoming objections of skeptics who held fast to the paradigm of solution-phase synthesis, Merrifield synthesized the enzyme using solid phase synthesis as described in Chapter 5. The resulting product mixture was allowed to generate the four disulfide crosslinks through air oxidation. Incorrect disulfide formation led to oligomers and misfolded protein. After isolation of monomers by gel electrophoresis and digestion of misfolded monomers with trypsin, the resultant enzyme convincingly possessed 78% of its expected enzymatic activity. In those early days of peptide synthesis, Merrifield also synthesized modified variants of RNase A with specific alterations designed to probe the functionalities required for enzymatic catalysis.

In these early mutagenesis studies, Merrifield did not have to synthesize the entire RNase A chemically to introduce each mutation. Unusually, RNase A can be split into two pieces, providing a more tractable synthetic target. The larger piece consists of RNase A digested with the highly specific protease subtilisin. The subtilisin clips off a 20-residue stretch of the N terminus (termed the S-peptide), rendering the enzyme inactive. Mixing this truncated enzyme with chemically synthesized S-peptide restores enzymatic activity to reasonable levels (more than 50%) (**Figure 6.84**). Thus, Merrifield could synthesize short peptides containing mutations and could probe the effect of each mutation on the activity of a good model for the full-length enzyme. Protein variants synthesized by this approach removed hydrogen bond-donating hydroxyl functionalities. For example, substituting an *O*-methyl ether for the hydroxyl group of a serine converts the side chain from a hydrogen-bond donor into a hydrogen-bond acceptor (**Figure 6.85**). Similar experiments form the basis for the field of protein engineering, which seeks to understand protein structure and function.

**Figure 6.84 Ribonuclease A active site.** The 20 N-terminal residues of the S-peptide are colored purple. In addition to His12 (purple van der Waals spheres), the active site includes Lys41 and His119 (yellow). (PDB 1FS3)

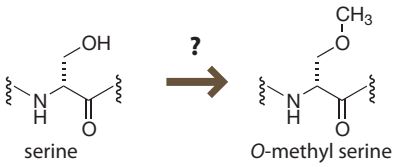



**Figure 6.85 Replacing the serine hydroxyl group with an ether.** This conversion of a hydrogen-bond donor to a hydrogen-bond acceptor requires chemical synthesis or another technique for incorporation of the non-natural amino acid *O*-methyl serine into the protein.

## Alanine scanning allows reverse engineering of protein function

The alanine mutagenesis described above provides a powerful test for contributions to protein function by amino acid atoms past the β-carbon. Systematic alanine mutagenesis across long stretches of the protein sequence, termed **alanine scanning**, has provided seminal insights into the structure and function of numerous proteins. The binding energy ($\Delta G$) for receptor–ligand interactions is typically evaluated by measuring the equilibrium constant ($K_d$) for the interaction and applying the following equation:

$$\Delta G = -RT \ln K_d \qquad (6.1)$$

When alanine scanning a protein interface, the binding energy for the alanine-substituted variant ($\Delta G_{Ala}$) can be measured and compared with that of the wild-type

receptor ($\Delta G_{wt}$) by the following equation, which calculates the difference in binding energy between the wild-type protein and the alanine-substituted variant:

$$\Delta\Delta G_{Ala-wt} = \Delta G_{Ala} - \Delta G_{wt} \quad (6.2)$$

This $\Delta\Delta G_{Ala-wt}$ value quantifies in kcal mol$^{-1}$ exactly how much binding energy the side-chain atoms past the β-carbon contribute to the interaction. Large $\Delta\Delta G_{Ala-wt}$ values (more than 1.4 kcal mol$^{-1}$) indicate a strong contribution to the binding energy (more than 10-fold effect on the $K_d$). Low values (less than 0.5 kcal mol$^{-1}$) are often within the error of the measurement and are associated with negligible contributions to protein binding.

Human growth hormone (hGH), for example, has been extensively studied through alanine scanning to uncover how hGH binds to two hGH receptors to activate cell signaling through a JAK–STAT pathway (**Figure 6.86**). The first hGH receptor covers a large area of the hGH surface, and directly contacts 19 amino acid side chains. With so many hGH residues touching the surface of the receptor, one might expect each side chain to provide a tiny and equal fraction of binding energy to the overall interaction (like the dozens of Velcro hooks sticking into the fuzzy material, in which each weak interaction adds to the overall strength holding the Velcro together). Alanine scanning, however, definitively shows this *not* to be the case. Several contact residues provide more than 1.5 kcal mol$^{-1}$ per residue of binding energy to the overall interaction, and many contact residues contribute essentially no energy to stabilize binding to hGH receptor.

**Figure 6.86 Ligand-mediated dimerization of human growth hormone receptor.** Human growth hormone receptor transduces growth signals from outside the cell to the inside of the cell. Two identical receptor molecules (blue) bind to nonidentical sides of the hormone (green). Each receptor has an additional transmembrane domain on the bottom (not shown) that extends through the membrane into the cytoplasm. (PDB 3HHR)

The bulk of the binding energy gluing together hGH and hGH receptor is provided by a small, tightly clustered set of 6 out of 19 residues. This **hotspot of binding energy** looks like a core sample of a protein; the outside diameter of the hotspot has hydrophilic functionalities, which surround a hydrophobic core (**Figure 6.87**). Alanine scanning of other proteins has revealed that many, but not all, proteins bind to their targets by using a similar hotspot strategy.

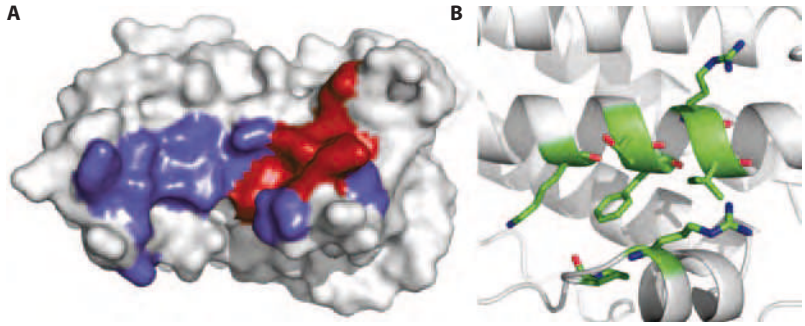



**Figure 6.87 A hotspot of binding energy on the hGH surface.** (A) Binding to the first human growth hormone (hGH) receptor buries 19 hGH side chains (red and blue surfaces). However, only six tightly clustered side chains contribute significantly to the binding energy (red). The other buried side chains (blue) fail to contribute much binding energy, but might provide shape requirements for further specificity in differentiating between several related protein hormones. (B) A close-up view of the hGH hotspot reveals that the side chains form a bullseye pattern with the hydrophobic aryl and alkyl carbons (green) in the center surrounded by hydrophilic functionalities with oxygen (red) and nitrogen (blue) atoms.

**Figure 6.88 Thermostable proteases.** Many detergents contain engineered thermostable proteases that degrade protein-based stains. (Courtesy of Proctor & Gamble.)

## Problem 6.15

Shotgun alanine scanning is a combinatorial technique for alanine scanning large numbers of residues across a protein interface. After mutagenesis to introduce a 1:1 ratio of wild-type protein to Ala variants, selections (usually with phage display) are used to identify the binding members of the library, which are sequenced to identify residues that are intolerant of alanine substitutions. The site-directed mutagenesis to create the alanine shotgun scanning library requires that degenerate oligonucleotides encode the wild-type/alanine substitutions. For each of the following sequences, design degenerate oligonucleotides for alanine shotgun scanning (denote 1:1 ratios of two DNA bases as {base1/base2}). For some amino acids, up to two other residues are allowed, as a result of degeneracy in the genetic code (for example A below).

**A** Hexahistidine affinity tag: $H_6$

**B** Initial five residues of mannan-binding tectin serine protease 2: MRLLT

**C** From human leukocyte antigen: HPVSD

## Protein engineering enables improvement of protein function

The different approaches to site-directed mutagenesis described in earlier chapters (Section 3.8) apply the power of chemical synthesis of DNA to alter a specific residue in a target protein. The use of site-directed mutagenesis allows one to test specific hypotheses. Each oligonucleotide for site-directed mutagenesis can encode either a single point mutation or a battery of different mutations. Site-directed mutagenesis requires synthetic oligonucleotides. The current state of the art in oligonucleotide synthesis reaches its limits around 115 base pairs. Thus, one oligonucleotide can encode up to 25 amino acid changes. Ideally, mutagenesis with a single oligonucleotide, even one encoding multiple changes, should test a specific hypothesis: the design of the mutant protein requires careful consideration, and can be very challenging.

As an early target of protein engineering efforts, proteases offer an enzyme with both interesting theoretical challenges and practical applications. For example, the enzyme subtilisin has been extensively modified to explore altered specificity and stability. Variants of subtilisin with enhanced thermal stability have long been deployed in common laundry detergents and stain removers (**Figure 6.88**). In addition, subtilisin can be expressed in large quantities by fermentation—often more than 1 g of subtilisin is obtained per liter of bacteria grown (a whopping 30% or more of the protein mass secreted by the organism). This beguiling combination of properties has prompted protein engineers to modify every aspect of subtilisin activity, including its catalytic rate, thermal stability, optimal temperature and pH for activity, and substrate specificity. The new abilities were typically introduced by rational design, and uncovered important insights into how proteases and other enzymes and proteins work.

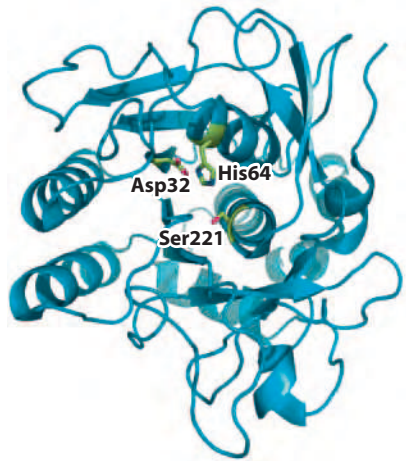



**Figure 6.89 Subtilisin.** The catalytic triad (sticks and labeled) accelerates peptide bond hydrolysis by using the serine protease mechanism described earlier in this chapter. (PDB 2ST1)

Given the field's uncertainties, protein engineering projects should always start with mutations aimed at testing the mechanistic contributions made by key residues. The bacterial serine protease subtilisin catalyzes the hydrolysis of protein amide bonds, and accelerates the reaction about $10^9$-fold versus the uncatalyzed spontaneous reaction (**Figure 6.89**). Replacing any of the catalytic triad residues (Ser221, His64, or Asp32) with alanine nearly disables the enzyme, decreasing $k_{cat}$ $10^4$–$10^6$-fold. Remarkably, the triple mutant protein with alanine substituted for all three residues of the catalytic triad retains some residual catalytic activity, catalyzing peptide bond hydrolysis 1000-fold over the spontaneous reaction. Although the triple mutant protein is about $10^6$-fold slower than subtilisin, the residual catalytic activity demonstrates that subtilisin relies on additional functionalities outside the catalytic triad for catalysis. Indeed, further mutagenesis has demonstrated contributions to subtilisin function by hydrogen bonds from other residues.

All enzymes have their preferred substrates, which result in the fastest enzymatic turnover rates ($k_{cat}$) and most effective binding ($K_m$). For proteases, the preferred

substrates are peptide sequences that can optimally fit into the active site. Determining this specificity, the enzyme cradles substrate side chains in pockets with shapes and functionalities evolved to fit specific amino acids. The amino acids N-terminal to the cleaved amide bonds are labeled P1, P2, P3, ... P*n*, and P1′, P2′, P3′, ..., P*n*′ denote residues on the C-terminal side of the scissile peptide bond (**Figure 6.90**).

Subtilisin evolved to chew apart essentially any protein, exhibiting broad substrate specificity. However, by judicious substitution of the residues lining the substrate-specificity pockets, the specificity of subtilisin can be limited to specific peptide sequences. For example, engineering subtilisin to hydrolyze only the peptide bond after the sequence RAKR required the incorporation of negatively charged residues in the P4, P2, and P1 binding pockets to complement the positively charged arginine and lysine residues of the substrate. The resultant subtilisin variant then rejects sequences that the wild-type enzyme would eagerly digest. Such large changes to enzyme specificity often require additional mutations to either regulatory sequences or other residues responsible for shifting the positioning of the substrate in the active site.

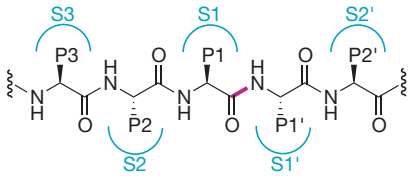



**Figure 6.90 Protease specificity.** The hydrolyzed bond (red) is anchored in the protease active site by amino acid side chains to either side (P1, P1′, etc.), which nestle into complementary pockets provided by the protease (S1, S1′, etc.).

## Protein engineering enables a change of protein function

In addition to tinkering with the specificity and activity of proteases, similar techniques can transform a nonhydrolytic enzyme into a functional protease. Cyclophilin catalyzes the *cis–trans* isomerization of proline, a key step in protein folding, and completely lacks proteolytic activity (**Figure 6.91**). To accomplish the proline isomerization, the cyclophilin active site has evolved exquisite specificity for proline-containing peptides. Reasoning that such binding abilities would be perfect for a high-specificity protease, Quéméneur and coworkers attempted to transform cyclophilin from a *cis–trans* proline isomerase into a proline-specific protease. First, cyclophilin residues near the proline amide were mutated to serine to provide a nucleophile for serine protease activity. Mutation of Ala91 to Ser (A91S) provided some protease activity, which, although modest, represented a more than $10^5$-fold gain in activity over the lowest limit of detection for the wild-type cyclophilin (**Table 6.7**). Additional mutagenesis identified positions amenable to hosting the Asp and His of the serine protease catalytic triad (F104H–N106D). The resultant triple mutant protein is a surprisingly efficient Xaa-Pro peptidase (where Xaa represents any amino acid). Despite the elegance of this effort, the catalytic efficiency and rate enhancement of a naturally occurring proline-specific peptidase remain far more impressive (prolyl oligopeptidase in Table 6.7).

**Figure 6.91 Cyclophilin.** The enzyme catalyzes rotation around the amide bond of proline. Like all enzymes, cyclophilin catalyzes both the forward and reverse reactions.

**Table 6.7 Engineering protease activity into cyclophilin.**

| Enzyme | Efficiency ($k_{cat}/K_m$) ($M^{-1}\,s^{-1}$) | $K_m$ ($10^{-3}$ M) | $k_{cat}$ ($s^{-1}$) | Rate enhancement ($k_{cat}/k_{uncat}$) |
|---|---|---|---|---|
| Cyclophilin | $<10^{-3}$ | ND | | |
| Cyclophilin (A91S) | 73.1 | 0.6 | 0.044 | $9 \times 10^6$ |
| Cyclophilin (A91S-F104H-N106D) | 1675 | 2.4 | 4.0 | $8.33 \times 10^8$ |
| Prolyl oligopeptidase | 1,000,000 | 0.06 | 60.5 | $1.26 \times 10^{10}$ (est.) |

ND, not determined; est., estimated.

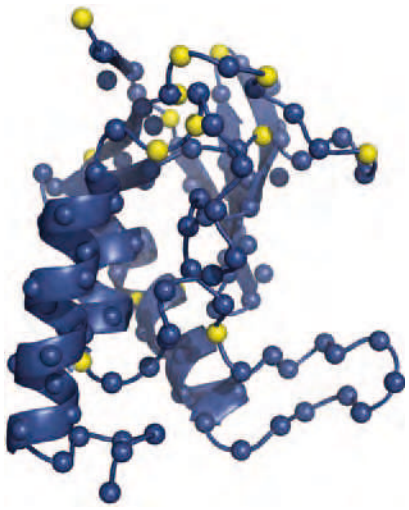


**Figure 6.92 Tolerant positions.** Relatively few sites in staphylococcal nuclease tolerate random mutations (yellow) that arise during error-prone PCR. Most residues (blue) are critical. Such residues stabilize secondary structure, are packed within the interior of a protein, or involve critical turn residues. Such sites do not tolerate random substitutions. (PDB 1STN)

## Most random mutations debilitate rather than enhance protein function

Introducing random changes into DNA sequences is quite straightforward using **error-prone PCR**. Many error-prone PCR methods are based on the ability of additives to interfere with the proof-reading mechanisms of DNA polymerase. For example, the addition of dimethylsulfoxide to the PCR buffer disrupts base-pair hydrogen bonding, which leads to a higher rate of incorrect base incorporation. When $Mn^{2+}$ is added to the PCR mixture, the $Mn^{2+}$ displaces $Mg^{2+}$ in the active site of DNA polymerase and increases the error rate through an incompletely understood mechanism. Because the G•C base pairs are stronger than A•T base pairs, altered ratios of dNTPs are often used to limit bias and favor incorporation of the less common mutations. Nucleotide mutation rates of a few percent are typical.

A few small, highly stable proteins such as egg lysozyme tolerate a wide range of random mutations. More typically, mutations lead to inactive enzymes that either do not fold or do not catalyze. For example, when random mutagenesis was applied to plasmids encoding staphylococcal nuclease, only 3% of the proteins retained enzymatic activity (**Figure 6.92**). Tightly packed side chains in the center of proteins are particularly sensitive to mutations. If you mutate an alanine to a leucine, you need to make space for the new isopropyl substituent by mutating another residue—close in space, but farther along the peptide chain—to something smaller. That is not always possible. Even if it is physically possible to make such a compensatory mutation, the chances of compensatory mutations occurring simultaneously are very low when one relies on random changes.

## Recombination generates new combinations of existing mutations

Living organisms use redundancy to mitigate the effects of deleterious point mutations. A colony of staphylococci can thrive even if a single bacterium experiences a lethal point mutation. Because human cells are diploid, with two homologous copies of each gene, the cell can often tolerate a point mutation that codes for a nonfunctional protein, as long as the other copy of the gene codes for a functional protein. Most organisms, from bacteria to humans, use genetic recombination to enhance genetic diversity without the perils of random mutagenesis. Pairwise recombination of homologous genes generates new combinations of the mutations that were already present in the original strands.

Exceptionally diverse combinatorial libraries of homologous variants can be accessed by an artificial form of recombination called **DNA shuffling** to form chimeric sequences (**Figure 6.93**). Imagine that you would like to develop a thermostable version of subtilisin that is stable in hot water or active at high pH. Because subtilisin is produced by *Bacillus subtilis*, you might start by collecting and amplifying genes for homologous proteases from other species of *Bacillus*. Then you would expose this mixture of gene copies to limited, random digestion with DNase I. After allowing the gene fragments to rehybridize randomly, you could use PCR to fill in the gaps and generate a staggering number of different genes with completely new combinations of the natural mutations. With natural recombination, two strands of DNA recombine to give just two new rearranged strands. Because DNA shuffling uses many copies of the original strands, it can generate a vast number of new combinations; some of the resulting proteins are more fit and some are less fit. The diverse population of genes can be transformed into bacteria to facilitate selection and amplification.

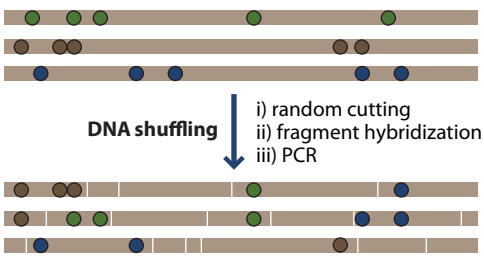



**Figure 6.93 DNA shuffling.** DNA shuffling breaks apart and recombines homologous DNA sequences, each with a different set of mutations. Each of the resultant chimeras contains a unique combination of the mutations that were present in the original sequences.

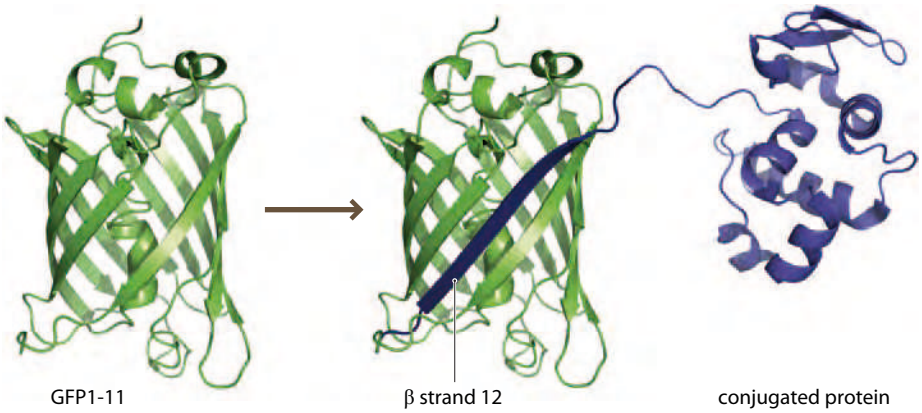



**Figure 6.94 Split GFP for complementation screens.** GFP1-11 (residues 1–214, in green) becomes fluorescent only when the structure is complementated by residues from the twelfth β strand (GFP12, residues 216–230, in dark blue). β Strand 12 complements GFP1-11 even when it is conjugated to another protein (light blue). (PDB 1GFL)

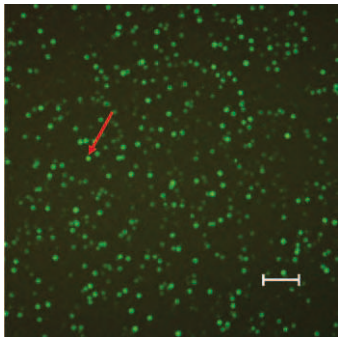


**Figure 6.95 Bacterial colonies in a folded protein screen.** Colonies of bacteria expressing proteins conjugated to GFP β strand 12 show varying levels of fluorescence, depending on the solubility of the protein (scale bar = 1 cm). Correctly folded, soluble proteins lead to highly fluorescent colonies. The colony with the red arrow was picked for further study. (From S. Cabantous and G.S. Waldo, *Nat. Meth.* 3:845–854, 2006. With permission from Macmillan Publishers Ltd.)

### Screens work well for modest numbers of protein variants, but exceptionally diverse libraries require selections

DNA shuffling was used to improve a truncated version of green fluorescent protein (GFP) for use in **complementation assays**. The truncated form of GFP becomes fluorescent only when the C-terminal β strand 12 is present in a soluble form (**Figure 6.94**). The ability of β strand 12 to complement GFP1-11 makes the split GFP system useful for assays. For example an assay was developed using the split GFP system to distinguish soluble proteins from insoluble proteins. When a library of proteins conjugated to GFP β strand 12 was expressed in *E. coli*, the colonies expressing soluble proteins could be easily identified through fluorescence simply by holding the plate above an ultraviolet light (**Figure 6.95**). The approach allows quick screening of a large number of protein variants to identify the well-folded variants.

Screens can work well for up to a few million different protein variants. However, the theoretical diversity of combinatorial protein libraries can exceed the number of particles in the Universe; although chemical biologists do not access numbers even close to such diversity, exceptionally diverse libraries of billions to trillions are not unusual. To examine all members of such massive libraries, screens must usually be coupled to some kind of selection that amplifies only the hits. Many auxotrophic selections couple successful traits such as catalysis to bacterial survival and growth. Hits or **selectants** isolated from phage-displayed protein libraries can be transfected into *E. coli*, leading to marked amplification. Selectants from *in vitro* libraries of DNA can be easily amplified with PCR.

## 6.6 SUMMARY

Every chemical conversation in the human body is initiated by touch, by some kind of noncovalent interaction between molecules. Proteins bind with high affinity and specificity to a much broader range of molecules than any other type of biooligomer. Thus, proteins are capable of performing a broad range of roles in living cells. Dose–response curves offer critical insight into the effects of ligand, substrate, and drug concentration on binding, and ultimately on the effects of binding. At sufficiently low concentrations, acute toxins are not toxic and natural cures will not cure you; however, at sufficiently high concentrations, anything can be toxic, including such otherwise harmless substances as sugar and salt.

It is usually difficult to measure an association constant directly. Chemical biologists rely on $IC_{50}$ values, $EC_{50}$ values, and $K_m$ values—indirect measures of lability—to describe protein binding interactions. As a consequence we are stuck with the awkward semantic situation in which lower values correspond to tighter binding.

We looked at enzymes in two distinct ways: using abstract kinetic parameters and with mechanistic arrow-pushing. The kinetic parameters $K_m$ and $k_{cat}$ describe how tightly the enzyme binds a substrate and how fast it is converted to product. If the substrate concentration (either in your cells or in your microtiter plate) is not at least equal to $K_m$, most of the enzyme is idling. The $k_{cat}$ rate constant relates to the individual steps that make up the arrow-pushing mechanism. We looked at detailed arrow-pushing mechanisms for two of the most common classes of human enzymes: kinases and proteases. Enzymes bind to and stabilize transition states. Even when an enzyme mechanism involves more than one transition state, the high-energy transition states along the reaction pathway tend to be very similar, and we should draw arrow-pushing mechanisms to reflect those similarities. Many enzymes use a combination of metal cations, cationic side chains, and helical dipoles to stabilize the buildup of negative charge in transition states. Enzyme cofactors extend the capabilities of enzymes, usually involving covalent catalysis. Hydrogen bonding has a critical role in catalysis. Hydrogen bonds often immobilize the substrate, allowing catalytic residues to act with surgical precision. The fact that hydrogen bonding precedes proton transfers blurs the line between concerted and stepwise arrow-pushing mechanisms. If you are unsure, draw all proton transfers as distinct mechanistic steps.

Even as we endeavor to understand how proteins fold, how they assemble, how they recognize, how they cleave, graft and create, we are empowered to improve them. Site-directed mutagenesis allows us to study the roles of individual residues in enzymes or to redesign enzymes for enhanced performance. DNA-based techniques such as error-prone PCR and DNA shuffling can be used to generate vast libraries of protein mutants from which one can select proteins with improved properties, such as an enzyme with high thermal stability, or a protein drug with longer serum half-life. Now that we have a clearer view of binding and catalysis, we can explore the three remaining classes of biooligomers that are assembled by enzymes: oligosaccharides, polyketides, and terpenes.

## LEARNING OUTCOMES

- Master the equations describing binding equilibrium.
- Use equations for on-rate and off-rate to relate kinetics to the equilibrium dissociation constant.
- Determine $IC_{50}$ and $EC_{50}$ from dose response curves.
- Based on $EC_{50}$ or $LD_{50}$, determine the concentrations of a chemical at which no significant biological effect is expected.
- Use the kinetic variables $K_m$ and $k_{cat}$ to describe binding, catalytic turnover, and catalytic efficiency.
- Draw arrow-pushing mechanisms for serine/cysteinyl proteases, metalloproteases, and aspartyl proteases.
- Provide mechanisms for acid and base catalysis by various functionalities in enzyme active sites.
- Recognize the most common enzyme cofactors, and draw arrow-pushing mechanisms for PLP, NADH, and thiazolium.
- Diagram the dual roles of $Zn^{2+}$ ions in enzyme structure and activity.
- Explain the mechanistic basis for the regulation of protease and kinase activities.
- Relate equilibrium constants to the thermodynamic energies governing receptor–ligand interactions.
- Sketch the steps required for the design and introduction of amino substitutions into proteins.
- Plan selections of proteins for fluorescence or other readily recognized phenotypes.

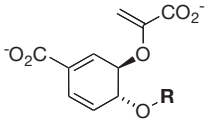
# PROBLEMS

***6.16** Chorismate mutase from *E. coli* catalyzes the Claisen rearrangement of both chorismate and the *O*-methyl derivative **1** with the kinetic parameters below.

**A** Which substrate binds more tightly?

**B** Which bound substrate rearranges more rapidly?

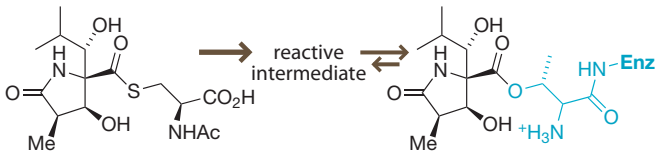
**C** Overall, which substrate is rearranged more quickly by the enzyme?



| | | $K_m$ (mM) | $k_{cat}$ ($s^{-1}$) |
|---|---|---|---|
| Chorismate | R = H | 0.14 | 29 |
| Derivative **1** | R = $CH_3$ | 1.9 | 0.56 |

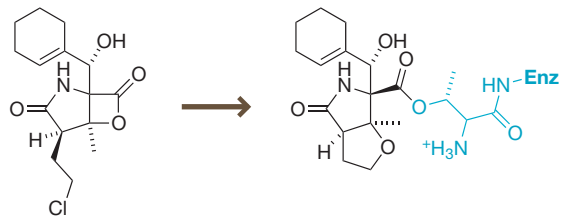
***6.17** In eukaryotic cells, the turnover of intracellular proteins is mediated mainly by the ubiquitin–proteasome pathway, a nonlysosomal proteolytic pathway. The 26S proteasome is a 2.5 MDa multiprotein complex found in both the nucleus and cytosol of all eukaryotic cells. The 20S particle is a threonine protease.

**A** The natural product lactacystin forms a strained reactive intermediate that acylates the N-terminal threonine in the active site of the 20S proteasome. Suggest a plausible structure for the reactive intermediate that modifies the enzyme and a mechanism for the acylation of the enzyme.



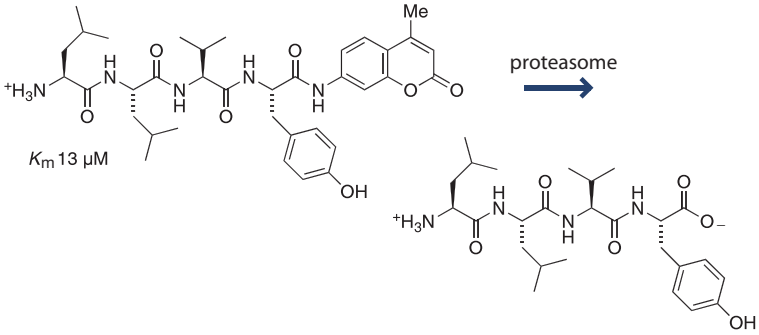

**B** The acylation of the proteasome by lactacystin is reversible. Suggest a plausible arrow-pushing mechanism for the reverse reaction that regenerates the reactive intermediate and the free enzyme.

**C** The natural product salinosporamide forms an irreversible adduct with the 20S proteasome. Suggest a plausible arrow-pushing mechanism for the reaction.




salinosporamide A

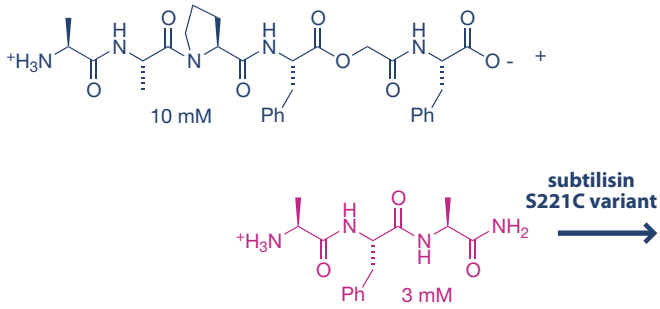
**D** The fluorogenic substrate below is used to monitor the activity of the proteasome. Using resonance structures, speculate on why the nonpeptide product is fluorescent but the substrate is not.



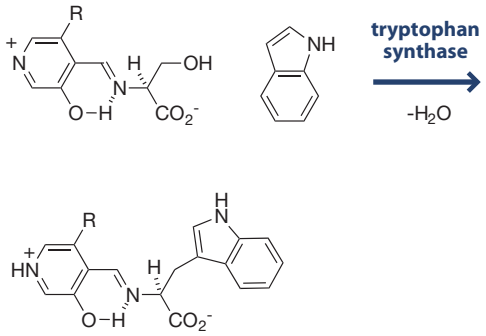

**6.18**

**A** Suggest a plausible arrow-pushing mechanism for the cleavage of the peptide AAPFGF by the serine protease subtilisin to give AAPF and GF.

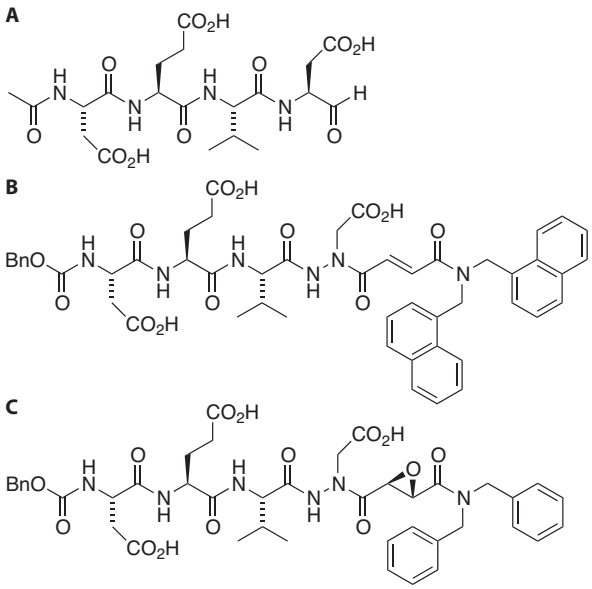
**B** Mutation of the key serine in the catalytic triad of subtilisin to cysteine generates a variant enzyme that will catalyze aminolysis of esters. Draw the expected products of the following amide-bond-forming reaction catalyzed by the S221C variant of subtilisin.



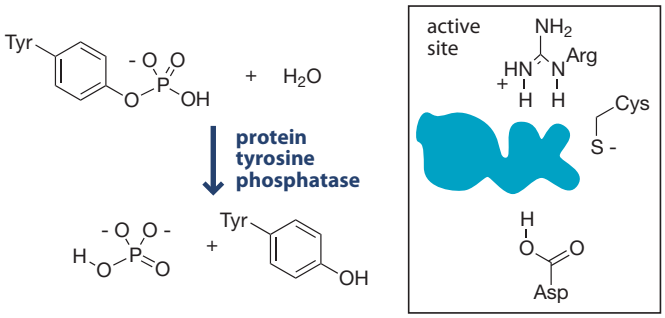

**6.19** Tryptophan synthase has a catalytic amine base in the active site to facilitate the following transformation. Suggest a plausible arrow-pushing mechanism for the reaction.



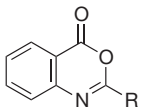

***6.20** Caspase-3 is a cysteine protease that cleaves at the sequence DEVD. Suggest a plausible mechanism for covalent inhibition of caspase-3 by the following inhibitors.



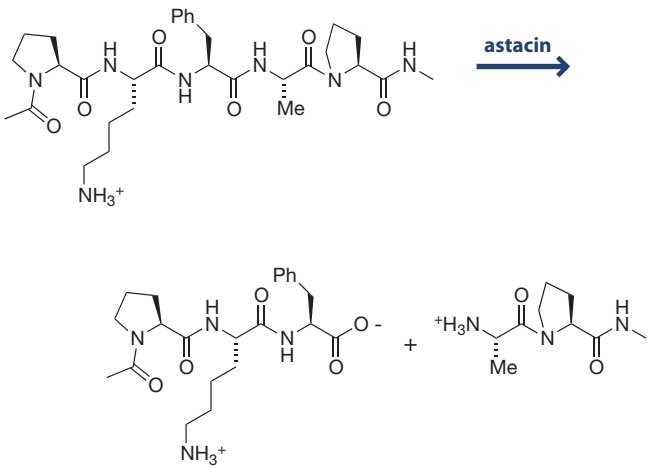
**6.21** The active site of bovine protein tyrosine phosphatase has three key residues in the active site: an arginine, a cysteine, and an aspartic acid. Suggest a plausible arrow-pushing mechanism for the enzyme-catalyzed hydrolysis of tyrosine monophosphate esters that involves these three active-site residues. Expect a mechanism with at least seven steps.



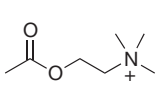

***6.22** Benzoxazinones form covalent adducts with serine proteases. Suggest a plausible structure for the adduct and a plausible arrow-pushing reaction mechanism for formation of the covalent adduct.



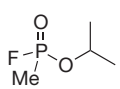

**6.23** Design a stable transition-state analog that might inhibit the enzyme astacin, which catalyzes the following transformation.



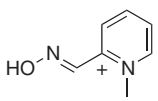

**6.24** In 1995, the Aum Shinrikyo cult released sarin nerve gas in a crowded Tokyo subway. Recall that the fluorophosphonate sarin is a potent inhibitor of acetycholinesterase. Pralidoxime can be used to treat persons affected with sarin. Suggest a plausible arrow-pushing reaction mechanism for the dephosphorylation of the acetycholinesterase–sarin adduct.



**acetylcholine**
neurotransmitter

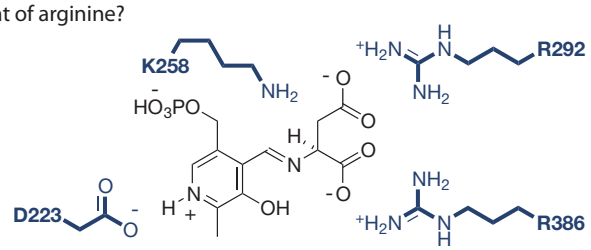



**sarin**
"nerve gas"




**pralidoxime**
antidote

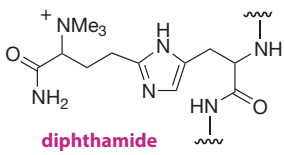
***6.25** Suggest a plausible amino acid substitution in aspartate aminotransferase that would change the substrate specificity from aspartate to arginine. How many times faster does aspartate aminotransferase catalyze the transamination of aspartate relative to that of arginine?




**Wild-type enzyme**

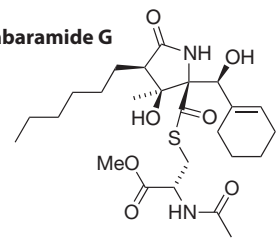
| Substrate | $\frac{k_{cat}}{K_m}$ ($M^{-1}\,s^{-1}$) |
|---|---|
| Aspartate | 34,500 |
| Arginine | 0.0695 |

**6.26** The protein elongation factor 2 is essential for ribosomal translation in all eukaryotes. It possesses a unique residue named diphthamide. From what two amino acids is diphthamide probably derived?




***6.27** Cinnabaramides A and G, isolated from *Streptomyces* strain JS360, are potent inhibitors of the human proteasome. Cinnabaramide G is believed to generate cinnabaramide A, with an unusual carbonyl stretch at 1812 $cm^{-1}$. Suggest a structure for cinnabaramide A and a plausible mechanism for covalent inhibition of the proteasome (a threonine protease).

**cinnabaramide G**




**cinnabaramide G**
IR $\bar{\nu}$ 1749, 1682,1656 $cm^{-1}$

**cinnabaramide A**
IR $\bar{\nu}$ 1812, 1702 $cm^{-1}$

***6.28** 8-Oxo-7,8-dihydroGTP is generated through oxidative damage to guanosine. Unfortunately, 8-oxoG base-pairs with cytosine and adenine with nearly equal efficiency. The *mutT* gene encodes a phosphatase that hydrolyzes 8-oxo-7,8-dihydroGTP before it can be incorporated into DNA. However, in bacteria that lack a functional *mutT* gene, misincorporation of 8-oxoG into growing DNA strands leads to C to A transversion mutations. Such cell lines are useful for (semi)random mutagenesis of proteins.

**A** Draw two base-pairing diagrams to show how 8-oxo-7,8-dihydroguanine might base-pair with either cytosine or adenine.

**B** Which amino acids are not susceptible to mutation in a strain that lacks a functional *mutT* gene. Ignore cases in which two or more A to C transversions occur within a single codon.

**Table (for Problem 6.29) Biological activity of IL-8 variants.**

| Variant | $EC_{50}$ (nM) | Variant | $EC_{50}$ (nM) |
|---|---|---|---|
| (Wild type) | 5 | P32A | 11 |
| E4A | >500 | N36A | 14 |
| L5A | 165 | T37A | 18 |
| R6A | >1000 | I39A | 10 |
| I10A | 83 | I40A | 50 |
| P16A | 7 | V41A | 3 |
| F17A | 198 | L43A | 14 |
| F21A | 33 | S44A | 2 |
| I22A | 73 | L49A | 8 |
| I28A | 6 | L51A | 9 |
| S30A | 3 | P53A | 6 |

**6.29** The cytokine interleukin-8 (IL-8) binds as a dimer to a pair of interleukin-8 receptor molecules. Alanine scanning mutagenesis was performed to generate a series of 21 variants that were assessed for biological activity; the results are given in the table above.

**A** On the basis of the solution NMR structure of IL-8 (PDB 2IL8), identify which regions of the IL-8 dimer are the most important for receptor activation.

**B** Is the greater activity of Phe17 relative to Phe21 due to contacts with the receptor?

***6.30** Identify the three residues that make up the catalytic triad of the serine protease neuropsin (PDB 1NPM).

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# 7 Glycobiology

## LEARNING OBJECTIVES

- Master the basic vocabulary of human glycobiology, including abbreviations and anomeric stereochemistry.
- Understand the structure and reactivity of acetals as they relate to glycans.
- Understand the basic mechanisms of enzymes that break and make glycosidic bonds.
- Differentiate the biosynthesis of O-linked glycoproteins from that of N-linked glycoproteins.
- Contrast the heterogeneity of human glycans with the homogeneity of the template-based biooligomers—DNA, RNA, and proteins.
- Distinguish the range of human glycans from those found on pathogenic organisms.
- Appreciate the role of cell-surface glycans in cell–cell communication.
- Understand the relationship between hyperglycemia and nonenzymatic glycation.

The field of glycobiology is connected to the very origins of organic chemistry through the monumental work of Hermann Emil Fischer (**Figure 7.1**). At a time when horses and trains were the common mode of transportation, Fischer worked out the relative stereochemical structures of all the common sugars, and completed the total syntheses of three stereochemically rich monosaccharides: glucose, fructose, and mannose. These achievements, made over the course of a single decade (1884–1894), are all the more remarkable when one considers that the tetrahedral nature of carbon had only been proposed 10 years earlier by van 't Hoff and Le Bel.

Despite this fast start, the field of glycobiology has been slow to mature, lagging behind the chemical biology of nucleic acids and proteins. At the time of writing of this book, there is no simple method for sequencing oligosaccharides, there is no general way to express homogeneous oligosaccharides, there is no simple way to amplify small amounts of oligosaccharides, and there is no general method for the synthesis of oligosaccharides. These inadequacies are arrayed against formidable objects of study—naturally occurring oligosaccharides are precious in quantity, dazzling in complexity, and maddening in heterogeneity. In spite of these challenges, heroic efforts of the past 50 years have elucidated the role of many of the carbohydrates in the human body, particularly in cell–cell communication and cooperation. These efforts empower chemical biologists to have an expansive view of human glycobiology, in which the oligosaccharides on the outside of cells are perhaps even more interesting than the glucose that fuels them from the inside.

**Figure 7.1 Hermann Emil Fischer in his laboratory around 1900.** (Courtesy of Archiv der Max-Planck-Gesellschaft, Berlin-Dahlem.)

## 7.1 STRUCTURE

### There are 10 common monosaccharide building blocks for human glycans

The monomeric sugar building blocks, called **monosaccharides**, that make up carbohydrates are more complex than building blocks of DNA, RNA, or proteins, so we will start this chapter with an introduction to the 10 monosaccharide building blocks that are common in human **oligosaccharides**. The common term **carbohydrate** is derived from the empirical formula of monosaccharides such as ribose ($C_5H_{10}O_5$) and

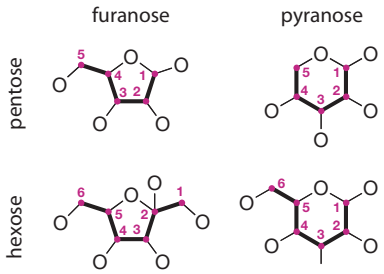



**Figure 7.2 Numbering and descriptors for monosaccharides.** Monosaccharide descriptors convey the number of carbons in the monosaccharide and the size of the ring.

glucose ($C_6H_{12}O_6$). A more general formula $[C(H_2O)]_n$ suggests a "hydrate of carbon." The term carbohydrate is usually used to refer to molecules or moieties composed of one or more monosaccharides. The collection of all molecules that contain carbohydrate functional groups, such as glycolipids, glycoproteins, and **polysaccharides**, are referred to as **glycans**.

The acetal carbon of monosaccharides, called the **anomeric center**, is the focal point for chemical reactivity. Carbohydrate rings are numbered such that the end of the chain closest to the anomeric center is numbered as carbon 1 (**Figure 7.2**). Only nine monosaccharides are incorporated into human glycans by **glycosyltransferases** (**Figure 7.3**), and their structures and names should be committed to memory. Without this common vocabulary, we cannot carry on a meaningful discussion of glycobiology. The actual diversity of the human **glycome** is greatly amplified by post-glycational modifications. For example, glucuronic acid is sometimes epimerized at the C5 position to generate iduronic acid. Similarly, **deacetylases** and **sulfotransferases** can drastically alter the ionic state of various sugars. Later, we will use a combination of monosaccharide names and positional numbering to abbreviate the structures of stunningly complex oligosaccharides.

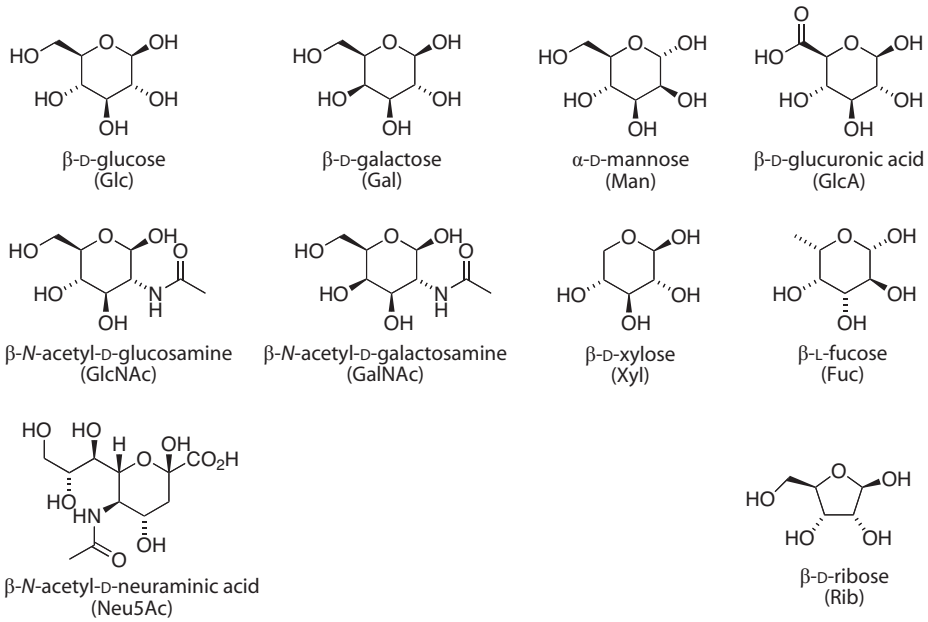
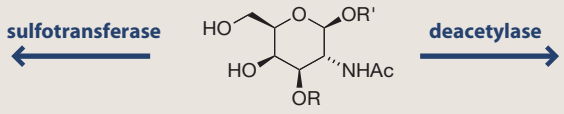



**Figure 7.3 Key carbohydrate building blocks that are important in human oligosaccharides.** Ribose, which is found in nucleic acids, is absent from human oligosaccharides.

## Problem 7.1

Draw Lewis structures of the charged functional groups resulting from each of the enzymes below. Assume physiological pH (7.4).




The chemistry of ribose and its nucleoside derivatives was discussed in detail in Chapters 3 and 4. However, five-membered ring monosaccharides (furanoses) have not yet been identified in human oligosaccharides. If one considers all the organisms in Nature, the set of known oligosaccharide substituents becomes considerably more diverse. For example, in microorganisms one finds glycans containing subunits with five-membered rings, six-membered rings, and five, six, seven, eight, or nine carbon

atoms. Exploiting the differences between the oligosaccharides of human cells and the oligosaccharides of pathogenic organisms is a promising avenue for the development of new vaccines and new medicines.

## Problem 7.2

Draw the following sugars in the preferred chair conformation.

**A** β-D-Xylose
**B** β-D-Galactose
**C** α-D-Mannose
**D** α-L-Fucose

### Glycobiology uses a compact form of nomenclature

Consider the IUPAC names that distinguish the cyclic forms of glucose and galactose: (3*R*,4*S*,5*S*,6*R*)-tetrahydro-6-(hydroxymethyl)-2*H*-pyran-2,3,4,5-tetraol and (3*R*,4*S*,5*R*,6*R*)-tetrahydro-6-(hydroxymethyl)-2*H*-pyran-2,3,4,5-tetraol. It is immediately obvious that the IUPAC system of chemical nomenclature, based on Cahn–Ingold–Prelog (*R*/*S*) assignments of stereogenic centers, is too clumsy for glycobiology. Thus, glycobiologists have retained the venerable naming system that was introduced in the 1800s. In the traditional naming system, the *relative* configurations of the stereogenic centers of monosaccharides are embodied by prefixes, such as gluco-, galacto-, manno-, and xylo-. The *absolute* configuration of the monosaccharide molecule is then designated using the (D/L) system of nomenclature (**Figure 7.4**). Usually, only a single enantiomeric form of each sugar is present in an organism (typically the D-enantiomer), and it is common practice to leave out the D or L descriptor. Configurational assignments such as D or L can be made by examining the single stereogenic carbon atom farthest from the acetal or ketal moiety. If this stereogenic center has an *R* configuration, the entire sugar is a D-sugar; if the stereogenic center has an *S* configuration, the entire sugar is an L-sugar. To maintain the analogy with the parent compounds, the uronic acids glucuronic acid and iduronic acid are not assigned stereochemistry as carboxylic acid derivatives. Instead the uronic acids are assigned D/L stereochemistry like the parent compounds, glucose and idose respectively (**Figure 7.5**).

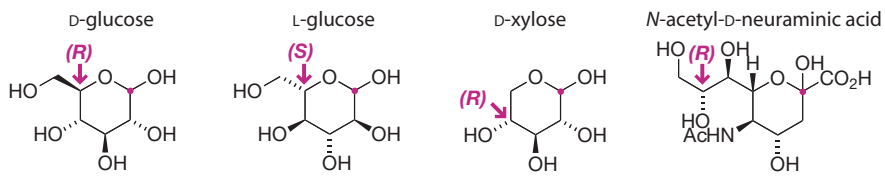
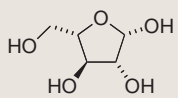


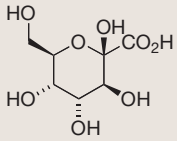

**Figure 7.4 D and L configurations.** For the common glycan building blocks, assignment of D and L enantiomers can be made on the basis of the stereogenic center (*R* or *S*) farthest from the acetal or ketal (indicated with a pink dot).

## Problem 7.3

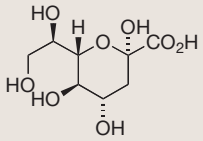
Assign the absolute configuration (D or L) of the following naturally occurring non-human monosaccharides.

arabinose





KDO



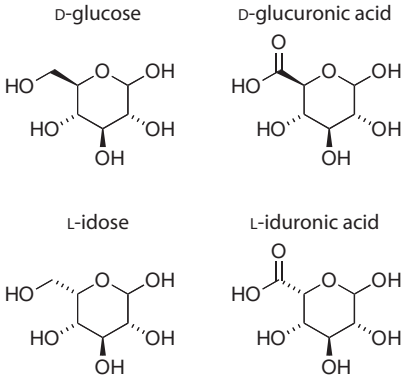



**Figure 7.5 Uronic acids.** The names and D/L assignments of uronic acids are derived from the parent compounds.

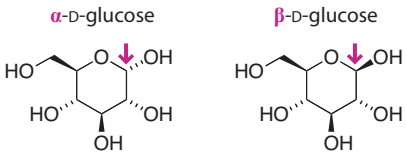



**Figure 7.6 Anomers of glucose.** The anomeric center is indicated with an arrow.

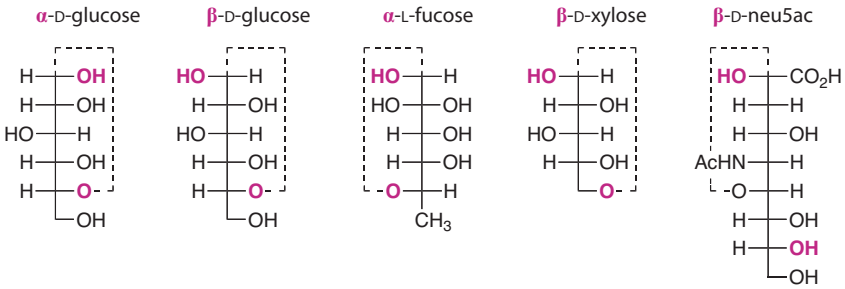



**Figure 7.7 α and β configurations are based on Fischer projections.** Anomeric α and β configurations are designated on the basis of the relative orientation of the anomeric hydroxyl and the substituent on the stereocenter farthest from the anomeric center. Since Fischer projections are no longer used in modern organic chemistry, it is easier simply to memorize α and β configurations for each of the monosaccharides in Figure 7.3.

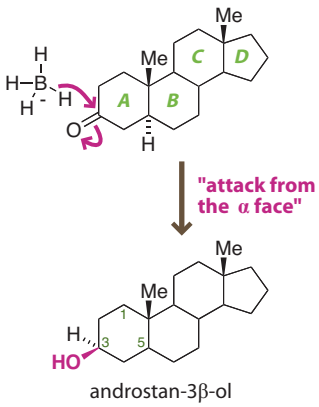



**Figure 7.8 α and β faces.** α and β are used in organic chemistry to distinguish between faces of a molecule as drawn on a flat surface.

The acetal or ketal carbon is a stereogenic center in all naturally occurring glycoconjugates. That means that it can exist in two different configurations, referred to as **anomers** (**Figure 7.6**). According to IUPAC rules, the two anomers are labeled as α or β (see Figure 7.6) depending on the relative orientation between the D/L reference carbon (farthest from the acetal or ketal) and the anomeric carbon in the Fischer projection representation (see Figure 7.3 and **Figure 7.7**). When the substituents at the anomeric center and the D/L reference center are on the same side of the carbon backbone, the stereoisomer is referred to as the α anomer. When the substituents at the anomeric center and the D/L reference center are on the opposite sides of the carbon backbone, the stereoisomer is referred to as the β anomer. This book does not use Fischer projections, because they do not convey meaningful spatial or conformational information. Therefore it is best to memorize the structures of the α and β anomers of each of the common human sugars in Figure 7.3. Fortunately, most of these sugars are D-hexopyranoses, like D-glucose.

Chemists also use an α/β system of nomenclature to differentiate the faces of molecules drawn in the plane of a two-dimensional surface—like a sheet of paper, a writing board, or a computer screen. The α face of a molecule is the face below the sheet of paper; the β face of a molecule is the face above the sheet of paper. A useful mnemonic device is that fish swim down (α resembles a fish) and bees fly up (β resembles the letter B). The α/β facial descriptors are highly effective whenever the molecule is drawn on a chalkboard for an audience, but are ambiguous if the molecule is not drawn. The α and β facial designations have been adopted as unambiguous descriptors for steroids because, by convention, chemists draw steroids with the A, B, C, and D rings in a left-to-right orientation (**Figure 7.8**).

## Polar effects and stereoelectronic effects determine the relative stability of α and β anomers

Axial substituents are never preferred in monosubstituted cyclohexane rings. A few substituents with long bonds, such as acetoxymercurio groups (AcOHg), show no preference, but most substituents on cyclohexane rings favor equatorial conformations. In some cases the preferences are small; in other cases the preferences are large. Alkoxy groups show modest preferences for equatorial positions over axial positions; in methoxycyclohexane the equatorial isomer is favored over the axial isomer by only about 0.6 kcal $mol^{-1}$. However, for methoxytetrahydropyran the axial anomer is preferred over the equatorial anomer by 1.0 kcal $mol^{-1}$ (**Figure 7.9**). Similarly, glucopyranosides such as methyl D-glucopyranose show a slight preference for the axial anomer. The axial anomer is stabilized over the equatorial anomer by two effects: a lower dipole moment and more favorable donation of filled orbitals into unfilled orbitals. By developing a fuller understanding of these effects, we are empowered to understand the enzymes that process glycosidic bonds.

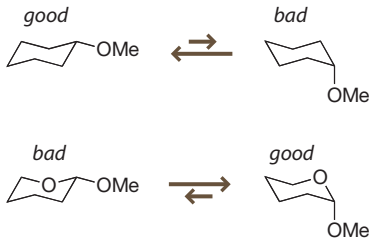



**Figure 7.9 A stereoelectronic effect.** The equatorial and axial preferences of alkoxy substituents are different for cyclohexane rings and pyranose rings.

Axial and equatorial conformers of methoxycyclohexane have similar dipole moments, between 1.2 and 1.3 debyes. However, the axial and equatorial anomers of

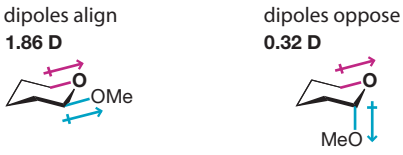



**Figure 7.10 A polar effect.** Axial anomers have lower dipole moments.

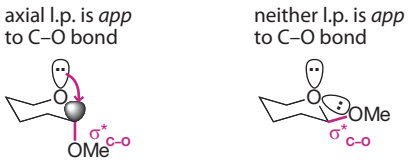



**Figure 7.11 Origin of the stereoelectronic effect.** Axial anomers are better aligned for donation into the σ* orbital of the anomeric substituent. *app*, antiperiplanar; l.p., lone pair.

2-methoxypyranose (IUPAC numbering) have very different dipole moments. In the equatorial anomer the dipoles align to create an overall dipole moment of 1.9 debyes (**Figure 7.10**). In the axial anomer the dipoles are almost in opposition, leading to a much lower dipole moment of 0.32 debyes. A larger dipole moment implies more charge separation, and Coulomb's equation predicts that charge separation is more costly in environments with a low dielectric constant. In nonpolar solvents, therefore, the more polar equatorial anomer is destabilized relative to the less polar axial anomer. The overall preference for axial anomers is diminished in polar solvents because the dipole penalty is decreased. For example, the axial form of 2-methoxytetrahydropyran is favored over the equatorial form by a 5:1 ratio in benzene, but in polar solvents like methanol and acetonitrile the ratio of the axial conformer to the equatorial conformer is only 2:1.

The **anomeric effect** also has a stereoelectronic component that we discussed in Chapter 5. In the axial anomer, the axial pyranose lone pair is antiperiplanar to the polar anomeric bond. Thus, the lone pair is *perfectly* aligned with the antibonding orbital of the anomeric C–O bond, and because of this alignment there is significant donation from the filled nonbonding orbital of the pyranose oxygen into the empty σ* orbital of the anomeric substituent (**Figure 7.11**). In the equatorial anomer, neither lone pair of the pyranose oxygen is aligned with the C–O bond. Note that only the hydroxyl groups (or alkoxy groups) at the 1 position of the pyranose ring can enjoy this effect; hydroxyl groups at other positions of a pyranose ring prefer conventional equatorial conformations.

The anomeric effect is evident in crystal structures in which donation into antibonding orbitals lengthens the anomeric C–O bond of axial anomers relative to equatorial anomers. The same donation that lengthens the bonds helps to push out the anomeric leaving group in $S_N1$ solvolysis reactions. The methyl-α-D-glucopyranoside hydrolyzes twice as fast as the β anomer. Small differences in bond length correlate with large differences in reactivity (**Figure 7.12**). A 0.01 Å increase in bond length corresponds to a 3 kcal $mol^{-1}$ decrease in the activation energy for ionization of the anomeric leaving group.

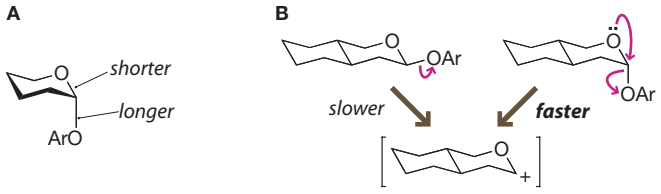



**Figure 7.12 The anomeric effect in structure and reactivity.** (A) Structural changes due to the anomeric effect have been observed in crystal structures. (B) The anomeric effect alters the reactivity of the anomeric carbon.

Lone-pair donation into lower-energy orbitals is more effective than donation into higher-energy orbitals. Recall that the energy of $\sigma^*_{C-X}$ orbitals is affected by electronegativity. For example, $S_N2$ attack on $CH_3$–$NH_2$ is much slower than $S_N2$ attack on $CH_3$–F. For the same reason, the anomeric effect is most pronounced for pyranosyl fluorides, because fluorine is the most electronegative atom (**Figure 7.13**). Similarly, the anomeric effect for an alkoxy substituent should be greater when the alkoxy substituent is protonated.

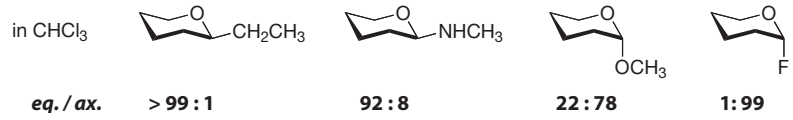
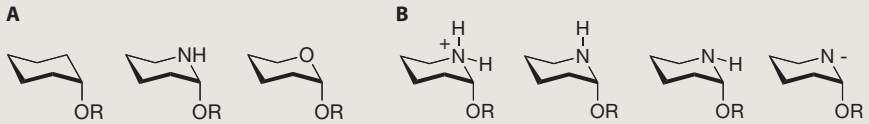



**Figure 7.13 The anomeric effect depends on electronegativity.** The equatorial/axial preference of anomeric substituents increases with electronegativity.

**Problem 7.4**

For each series of compounds, rank each compound from strongest axial preference for the OR group to lowest axial preference.




## 7.2 THE CHEMISTRY AND ENZYMOLOGY OF THE GLYCOSIDIC BOND

### Monosaccharide carbonyl groups form hemiacetals

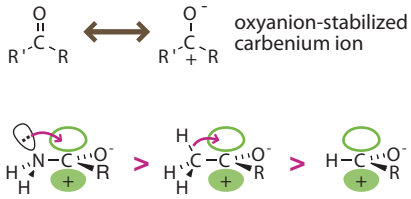



**Figure 7.14 Donation.** Carbenium ions are stabilized by donation of filled orbitals into the empty *p* orbital.

All carbohydrates possess latent carbonyl groups. So before we begin our discussion of carbohydrate structure, it is worth considering the reactivity of carbonyls with hydroxylic nucleophiles. Most aldehydes and ketones have a slight tendency to form tetrahedral hydrates by addition of water. To understand the factors that favor hydrate formation, it is best to treat the carbonyl group as an oxyanion-stabilized carbenium ion and recall that carbenium ions are stabilized by overlap of filled orbitals with the empty *p* orbital (**Figure 7.14**). Recall that hydrogen atom substituents on a carbenium ion have no lone pairs or bonds that can overlap with the empty *p* orbital. That is why $H_3C^+$ is less stable and more reactive than $(CH_3)_3C^+$ . Similarly, $H_2C{=}O$ is less stable and more reactive than $(CH_3)_2C{=}O$. Therefore, the stability of hydrates is inversely correlated with the stability of the carbonyl starting material: ketone < aldehyde < formaldehyde (**Figure 7.15**). In addition, acyl groups and inductively electron-withdrawing groups such as trifluoromethyl enhance the rates and extent of hydration. In some cases, special features of the hydrate product might contribute to stability. For example, the greater tendency of pyruvic acid to undergo hydration might be attributable to the greater capacity of carboxylic groups to form hydrogen bonds than ketones.

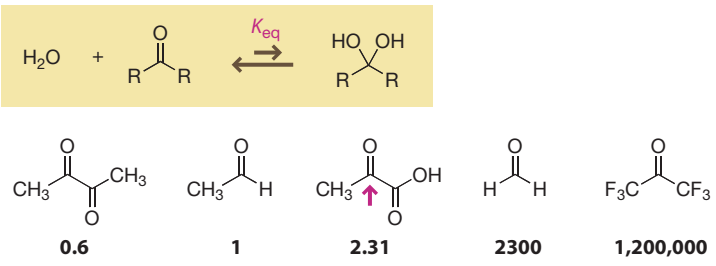



**Figure 7.15 Dramatic differences in ketone reactivity.** Tetrahedral adducts are more stable when ketones (or aldehydes) are electron deficient. The ketone carbonyl of pyruvic acid, indicated with an arrow, is more reactive than an ester of pyruvic acid.

### Six- and five-membered ring hemiacetals are common

When ketones and aldehydes form adducts with alcohols they are referred to as hemiketals and hemiacetals, respectively. To simplify our discussion we will use the term acetals to refer collectively to acetals and ketals. Similarly, we will use the term hemiacetals to refer to either hemiacetals or hemiketals. The trends in stability for acetal and hemiacetal formation with alcohols follow the trends for hydrate formation with water shown in Figure 7.15.

The mechanism for ring opening of hemiacetals depends on the pH of the solvent (**Figure 7.16**). Under acidic conditions, protonation of the ring oxygen facilitates ring opening; under basic conditions, deprotonation of the anomeric hydroxyl anion facilitates ring opening.

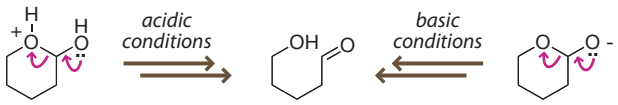
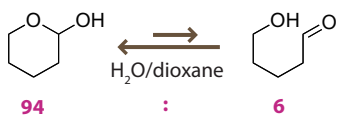



**Figure 7.16 Opening cyclic hemiacetals.** There are two different pH-dependent mechanisms for opening hemiacetal rings, acid-catalyzed and base-catalyzed.

Acyclic acetals and hemiacetals are thermodynamically unstable in water, but cyclic acetals and hemiacetals formed through intramolecular reactions can enjoy substantial stability (**Figure 7.17**). The equatorial hydroxyl groups of glucose give the cyclic form added stability relative to the open-chain aldehyde because of steric considerations. In aqueous solution, glucose exists almost exclusively (99.9%) in the cyclic form. However, although the concentration of the open-chain aldehyde is usually small, the reactive carbonyl of the aldehyde is often a key functional group in many reactions of carbohydrates.




| ring size | percentage closed |
|---|---|
| **4** | <1 |
| **5** | 89 |
| **6** | 94 |
| **> 6** | ≤20 |

**Figure 7.17 Furanose and pyranose are special.** Cyclic hemiacetals exhibit an intrinsic preference over acyclic hydroxyaldehydes for five- and six-membered rings.

Most naturally occurring carbohydrates have many hydroxyl groups and can, in theory, form more than one size of ring; however, among the possible cyclic forms, one is always preferred. For example, D-glucose can, in theory, exist in a five-membered ring furanose form, a six-membered ring pyranose form, or even a seven-membered ring form (**Figure 7.18**). In practice, only the glucopyranose form is observed under aqueous conditions. Only two types of cyclic acetals are found in oligosaccharides: six-membered ring pyranoses and five-membered ring furanoses. All human oligosaccharides are made up of six-membered ring monosaccharides. Oligosaccharides from other plants and microorganisms also include five-membered ring monosaccharides. No naturally occurring monosaccharides are known to prefer a seven-membered ring form.

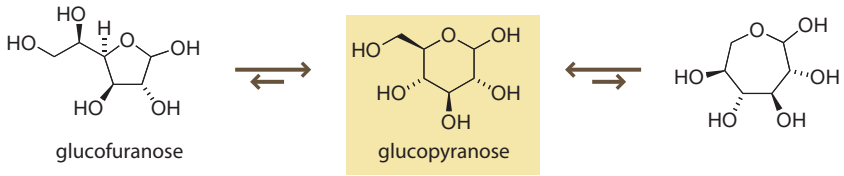
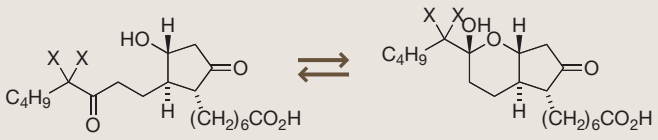



**Figure 7.18 The pyranose form of D-glucose is preferred over other ring sizes.** Seven-membered rings are highly disfavored because of the entropic costs required for their formation.

## Problem 7.5

The drug lubiprostone has a carefully positioned difluoromethylene group (X = F). What effect would these substituents have on the equilibrium between the open-chain ketone and the cyclic hemiketal, relative to the parent compound (X = H)?



Human macrophages swallow up pathogenic organisms into phagocytic vesicles called phagosomes. When these phagosomes fuse with the protease-rich lysosomes, a lethal environment is created that attacks the pathogen. However, mycobacteria can inhibit the fusion of their phagosomal prison with lysosomes, allowing the mycobacteria to survive and replicate inside their would-be assassin. Much of the stealth of *Mycobacterium tuberculosis* is associated with its waxy, protective coat. Lipoarabinomannans are one component of the *M. tuberculosis* membrane that helps it avoid

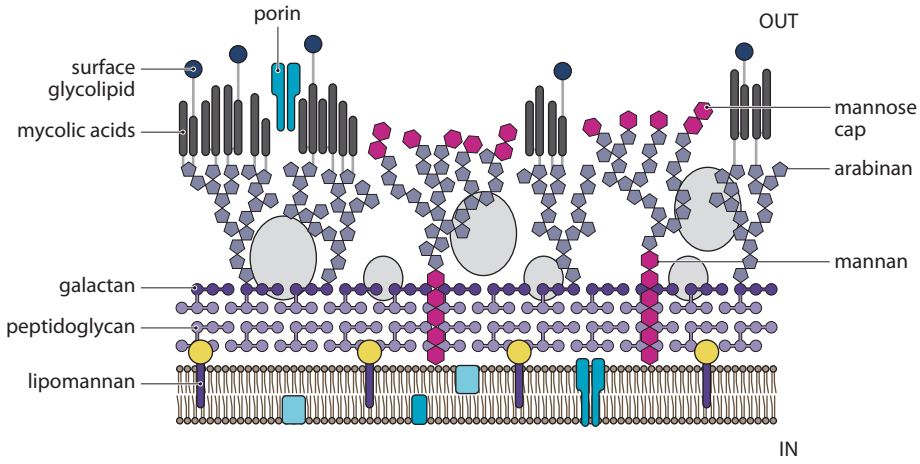



**Figure 7.19 Furanose rings in mycobacterial oligosaccharides.** Five-membered ring arabinofuranose sugars (pentagons) are present in two types of highly branched glycans that make up the cell wall of *Mycobacterium tuberculosis*, lipoarabinomannans and arabinogalactans. Hydrophobic long-chain mycolic acids attached to the outer branches of arabinogalactans create a waxy external barrier. (Adapted from D. Chatterjee and K.H. Khoo, *Glycobiology* 8:113–120, 1998. With permission from Oxford University Press.)

unwanted attention from the human immune system (**Figure 7.19**). The oligosaccharide portion of mycobacterial lipoarabinomannans is composed of a mannose oligomer called a mannan and a highly branched arabinofuranose (Araf) oligomer called an arabinan. Polysaccharides composed of five-membered ring furanose sugars are unique to mycobacteria (**Figure 7.20**). Furanoses are sometimes found in oligosaccharides in lower organisms, but not in human oligosaccharides. Of course, furanoses do have a role in other aspects of cell function. Recall that the backbones of DNA and RNA are based on the five-membered ring sugar ribose.

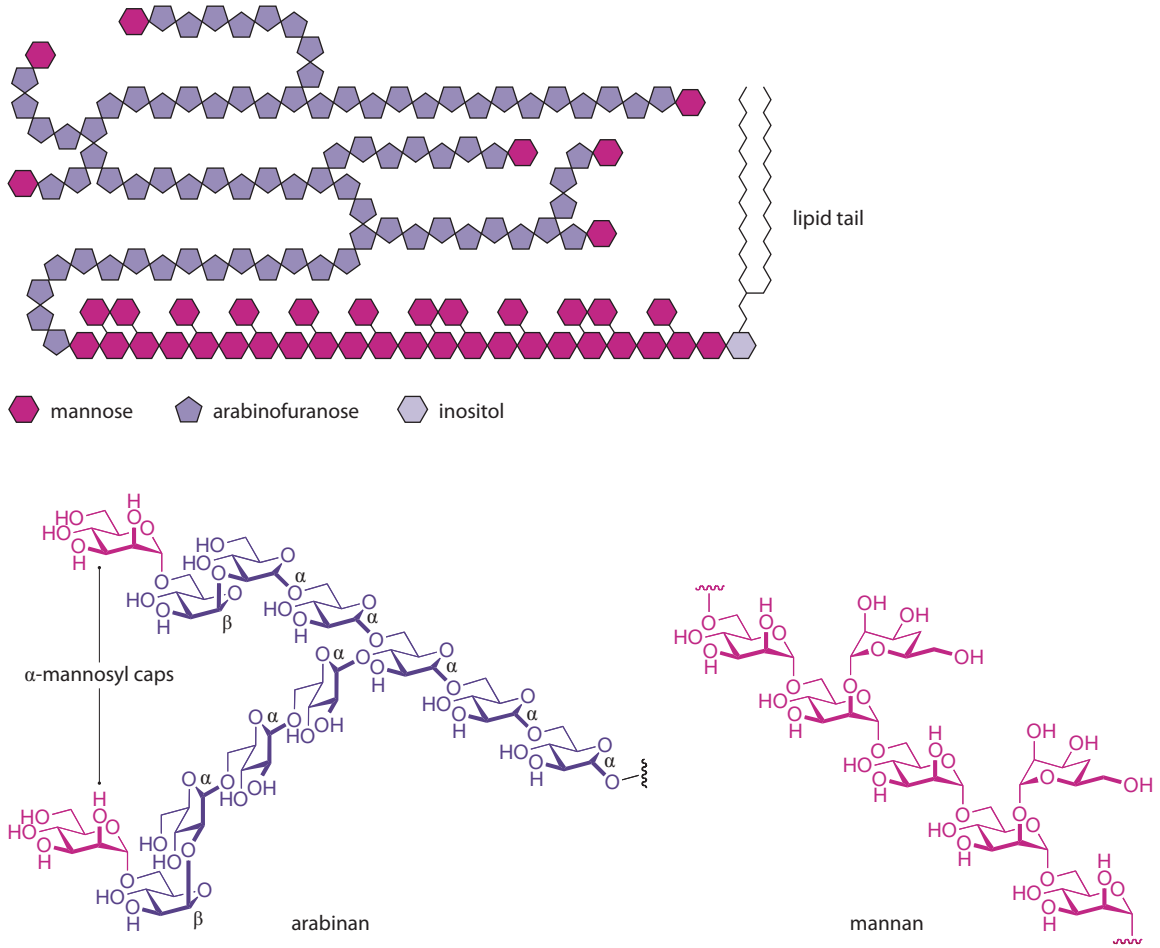



**Figure 7.20 Lipoarabinomannans.** Mycobacterial lipoarabinomannans are tree-like structures with a lipid for roots, a mannan trunk, arabinan branches, and α-mannosyl caps for leaves. Human cells don't make oligosaccharides out of five-membered ring sugars.

## Chemical hydrolysis of glycosidic bonds involves $S_N1$ reactions

In the laboratory, chemical cleavage of glycosidic bonds is conducted under acid-catalyzed conditions and usually involves the ionization of a leaving group to produce a carbocation intermediate. The mechanism can be drawn as an unassisted solvolysis leading to a carbocation or as an assisted solvolysis leading to an **oxonium ion** (**Figure 7.21**). Because the carbocation and the oxonium ion are resonance structures, the two mechanistic depictions are equally acceptable for use in arrow-pushing. There are two types of carbocations: trivalent carbenium ions such as $R_3C^+$ and pentavalent carbonium ions such as $R_5C^+$. Carbonium ions are protonated alkanes and are so unusual that chemical biologists usually use the terms carbocation and carbenium ion interchangeably. The term **oxocarbenium ion** is very precise; it describes a carbenium ion with an oxygen substituent. Although the oxocarbenium ion resonance structure and the oxonium ion resonance structure are equally acceptable for arrow-pushing, they are not equal. Students who are learning about the stability, conformation, and bonding of carbocations for the first time may find that the oxonium ion depiction is preferable to the oxocarbenium ion depiction. The formal charges in the oxonium ion resonance structure are wildly misleading (**Figure 7.22**). Electronic structure calculations reveal that the positive charge is distributed among the hydrogen atoms and carbenium carbon atom, whereas the oxygen atom and all of the $sp^3$ carbon atoms have partial negative charges. It is most often the case that students are interested in chemical reactivity and arrow-pushing, and for these applications the oxocarbenium ion is more useful than the oxonium ion resonance structure; the oxocarbenium representation emphasizes the carbenium center as the site of nucleophilic attack. Sadly, all organic chemistry textbooks cling to the oxonium ion representation even though it misleads many students to attack the positively charged oxygen atom.

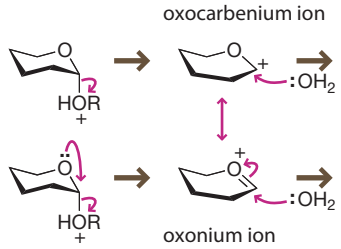



**Figure 7.21 Equivalent depictions.** The two arrow-pushing mechanisms above are equivalent, showing alternative resonance structures for the starting materials and intermediates.

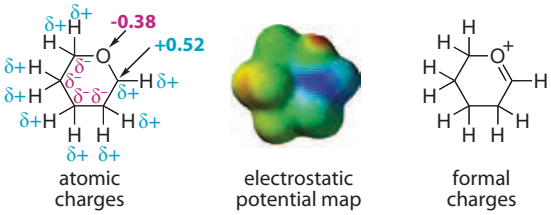
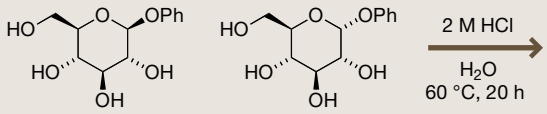



**Figure 7.22 Where is the charge?** The formal charges in the oxonium ion representation are at odds with the calculated atomic charges. The electrostatic potential map reveals the positive charge (colored blue) to be centered on the carbenium atom.

### Problem 7.6

Identify the α and β anomers shown below. On the basis of stereoelectronic effects, predict which anomer would ionize faster in an $S_N1$ solvolysis reaction.




The hydroxyl substituents of sugars drastically slow the rate of acid-catalyzed hydrolysis. Acid-catalyzed hydrolysis of glucosides is 10 million times slower than the hydrolysis of a tetrahydropyranoside with no hydroxyl substituents. Tetrahydropyranyl groups are commonly used as protecting groups for alcohols in organic synthesis because they are removed under such mild acidic conditions. The inductive effect of the hydroxyl substituents is most significant for hydroxyl groups at the 2 position, next to the anomeric center. Given the robustness of glycosidic bonds, it is surprising that **glycosylhydrolases** can hydrolyze them with such apparent ease. For example, enzyme-catalyzed hydrolysis rates can be up to 100,000,000,000,000,000 ($10^{17}$) times faster than background hydrolysis rates under neutral conditions. In all known glycosylhydrolases and glycosyltransferases, a nucleophile is poised on the side opposite the leaving group. In such an arrangement the reactants seem poised for a concerted $S_N2$ displacement; however, the system is also poised to form a very stable cation in an $S_N1$ process. The question of whether glycosylhydrolases and glycosyltransferases

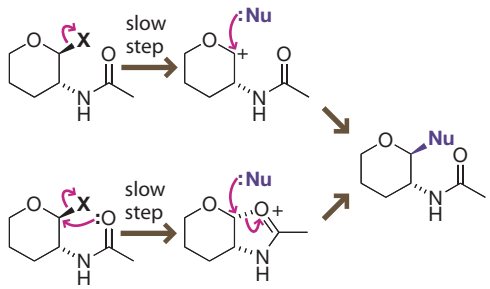



**Figure 7.23 $S_N1$ or $S_N2$?** Both of these reaction pathways will proceed with reaction kinetics that follow a first-order rate law: rate ∝ [R–X]. Therefore, both mechanisms correspond to $S_N1$ reactions.

act via concerted displacements, via discrete oxocarbenium ions, or via some hybrid mechanism is not easy to answer.

The widely used descriptions $S_N1$ and $S_N2$ can be misleading when used to label elementary reaction steps. For example, the solvolyses of 2-acetamido-2-deoxyglycosides involve $S_N1$ reactions (**Figure 7.23**). However, the term $S_N1$ does not necessarily mean that the mechanism involves a discrete carbocation. If a neighboring group pushes out the leaving group in an intramolecular $S_N2$ reaction to generate a highly reactive substrate for a faster intermolecular $S_N2$ reaction, the reaction will also obey first-order kinetics which makes it an $S_N1$ reaction. In fact, hydrolysis of β-GlcNAc and β-GalNAc derivatives proceeds through this neighboring-group participation mechanism. Enzymes that hydrolyze and form β-GlcNAc and β-GalNAc derivatives favor this type of neighboring-group displacement mechanism.

## Enzymatic hydrolysis of glycosidic bonds involves $S_N1$-like $S_N2$ reactions

To understand how glycosylhydrolases achieve such phenomenal rate accelerations, it is essential to understand how the pyranose ring oxygen affects both $S_N1$ ionizations and concerted $S_N2$ displacements. The effect of the ring oxygen on $S_N1$ ionizations has already been discussed in the context of the anomeric effect, so we will now turn to the more challenging issue of concerted $S_N2$ displacements. Alkoxy substituents have two opposite effects on the trigonal bipyramidal transition state for an $S_N2$ reaction. Recall from Chapter 2 that electronegative equatorial substituents destabilize trigonal bipyramidal $S_N2$ transition states. However, the oxygen lone pair can donate into the empty *p* orbital of the $S_N2$ transition state, leading to stabilization. Thus the pyranose oxygen can significantly accelerate the $S_N2$ displacement of anomeric substituents by $S_N2$ mechanisms for reasons that are analogous to the way in which it can accelerate $S_N1$ ionization. Electronic structure calculations suggest that glycosylhydrolases act via concerted $S_N2$ mechanisms (**Figure 7.24**). However, the nucleophile is not really pushing out the leaving group. The leaving group is almost completely dissociated by the time the nucleophile (usually water, a hydroxyl group, or a carboxylic acid side chain) has formed a bond. It might be more appropriate to picture the nucleophile getting sucked into the empty orbital of the oxocarbenium ion as it forms.

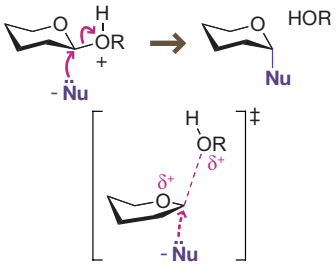



**Figure 7.24 Glycosylhydrolase mechanism.** The arrangement of groups in glycosylhydrolases favors an $S_N2$ displacement mechanism with the hallmarks of an $S_N1$ reaction.

## Members of all classes of glycosylhydrolases have two carboxylic acids in the active site

Mechanistically, glycosylhydrolases fall into two distinct classes, inverting and retaining, which reflect the stereochemistry of the product compared with the starting material (**Figure 7.25**). Both types of glycosylhydrolases involve the carefully orchestrated

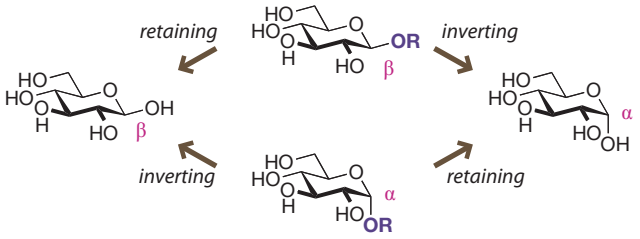
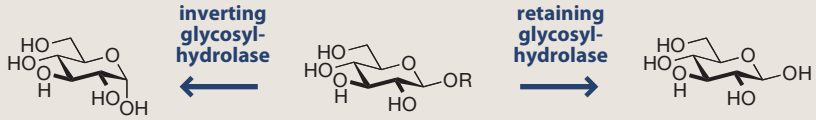



**Figure 7.25 Two types of glycosylhydrolases.** Glycosylhydrolase enzymes are described as "retaining" or "inverting," depending on the stereochemical outcome of the reaction.

actions of two carboxylic acid functional groups. One carboxylic acid protonates the leaving group. The role of the other carboxylate is dependent on the type of glycosylhydrolase. For inverting glycosylhydrolases, the second carboxylate deprotonates a water molecule that displaces the leaving group through an $S_N2$ reaction (see Figure 7.24). In retaining glycosylhydrolases, the second carboxylate acts as a nucleophile, displacing the protonated leaving group. The covalently bound carboxylate is then displaced by an attacking water molecule. Retaining glycosylhydrolases lead to net retention through double inversion.

## Problem 7.7

Provide arrow-pushing mechanisms for both retaining and inverting β-D-glucosylhydrolases, each with two acidic residues in the active site. Do not worry about changes in pyranose conformation.




The mechanism for glycosylhydrolases involves several steps of indeterminate sequential order. As we have discussed, X-ray crystal structures usually do not reveal the positions of the protons. It is therefore difficult to know whether the nucleophilic water is deprotonated before it attacks, as it attacks, or after it attacks. Students who are looking for a single correct arrow-pushing mechanism for enzyme-catalyzed reactions may find this disheartening, but the order of proton transfers is moot if the protons are already engaged in a hydrogen bond (which is often the case in enzyme–substrate complexes). As we have discussed, proton transfers are really just ultra-fast vibrations of hydrogen bonds; the precise order of sequential ultra-fast steps is less important than the order of the slow, rate-determining steps.

### Substrate distortion is important in glycosylhydrolase enzymes

What is the origin of the phenomenal rate accelerations of glycosylhydrolases? In mechanistic terms, the inverting α-glycosylhydrolases are the simplest to understand because the anomeric bond starts off in the stereoelectronically activated α configuration. For example, the yeast enzyme processing α-glucosidase I is an inverting α-glycosylhydrolase that removes glucosyl residues from the $Glc_3Man_9GlcNAc_2$ oligosaccharide that is added to many newly synthesized proteins. Removal of these exterior glucosyl residues is critical because these carbohydrates serve as signals for glycoprotein folding and degradation in the endoplasmic reticulum. The enzyme class I α1–2 mannosidase (PDB 1FO2) is also an inverting enzyme and has a role in the degradation of misfolded glycoproteins in the endoplasmic reticulum. The enzyme binds to its substrate in a skew-boat conformation as the $Ca^{2+}$ complex (**Figure 7.26**). The $Ca^{2+}$ ion also coordinates to the nucleophilic water molecule (**Figure 7.27**). Two glutamate residues drive the key proton transfers.

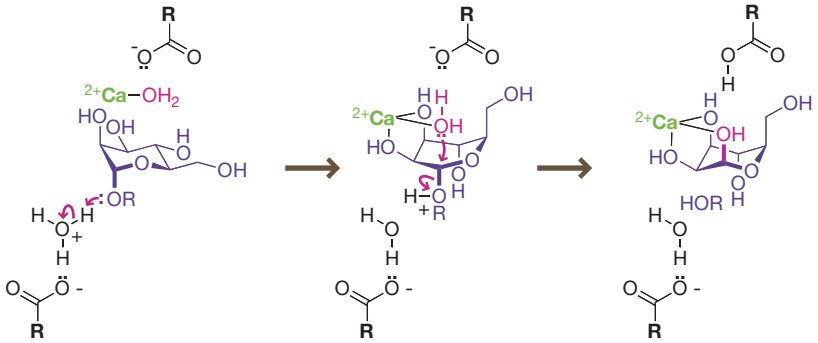


**Figure 7.26 Special delivery.** A $Ca^{2+}$ ion in class I α1–2 mannosidase forms a distorted substrate complex and coordinates a nucleophilic water molecule (red) that is delivered to the anomeric carbon.

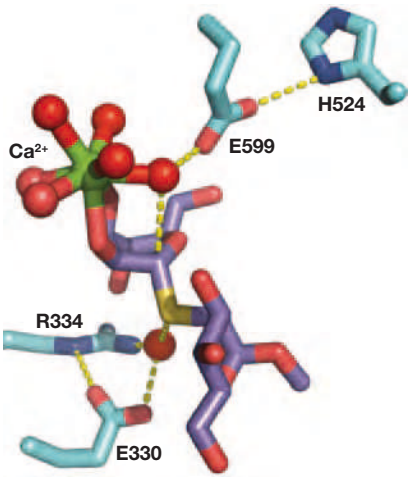



**Figure 7.27 Freeze frame.** Co-crystallization of class I α1–2 mannosidase (PDB 1X9D) with a nonhydrolyzable thiodisaccharide substrate reveals the precise placement of functional groups in the active site. The sulfur atom is yellow; water molecules are rendered as red spheres; a $Ca^{2+}$ ion is rendered as a green sphere.

The mechanism of β-glycosylhydrolases is less obvious because equatorial anomeric substituents are not aligned to benefit from the stereoelectronic effect of the pyranose oxygen. However, if the ring flips into a boat conformation, an advantageous alignment of the pyranose oxygen lone pair with the antibonding orbital of the anomeric substituent can be achieved (**Figure 7.28**). Unfortunately this imposes an energetic cost on the enzyme, because boat conformations are generally less stable than chair conformations. Protonation of the anomeric substituent increases the anomeric effect markedly. In a simple pyranose ring, protonation of an anomeric hydroxyl group will help to drive the anomeric substituent into an axial conformation that is stereoelectronically favorable toward hydrolysis. Importantly, the stereoelectronic benefits of an axial leaving group are realized regardless of whether the reaction proceeds through an $S_N1$ or an $S_N2$ pathway.

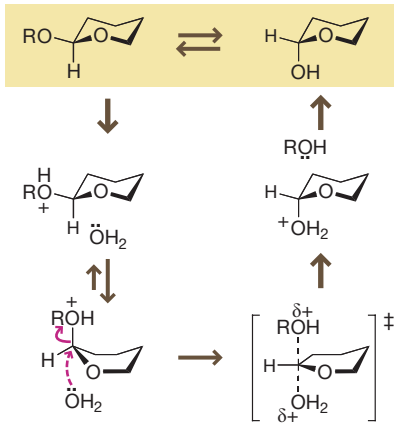


**Figure 7.28 Acid catalysis.** Protonation of the anomeric substituent can drive a conformational change that facilitates substitution.

Studies of glycosylhydrolases have shown that they distort β-glycosides into conformations that force the anomeric leaving group into a pseudo-axial orientation. It is rarely possible to obtain crystal structures of glycosylhydrolases bound to true reactive intermediates, but one can infer a plausible mechanism from the structures of enzymes bound to substrate analogs. A crystal structure of a fungal β-glucanase with a nonhydrolyzable thioglycoside substrate revealed that the inhibitor is distorted such that the pyranose oxygen lone pair is aligned with the leaving group, weakening the bond for cleavage in an $S_N2$ reaction (**Figure 7.29**). Similar results have been observed for other β-glycosylhydrolases.

Lysozyme, a retaining β-glycosylhydrolase, selectively cleaves a polysaccharide that is found in bacterial cell walls but not in humans. An abundant supply of lysozyme in chicken eggs and human tears helps to make these nutrient-rich environments inhospitable to bacteria. Lysozyme was the first enzyme for which a crystal structure was obtained, dating back to 1965. Yet it took 36 years to provide convincing evidence that the mechanism involves a double inversion and a covalent intermediate. The enzyme positions an acidic glutamate residue in a perfect position to protonate the

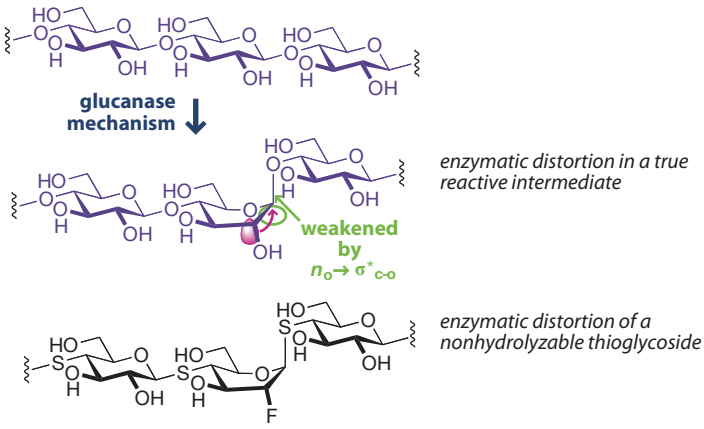



**Figure 7.29 Transition state analog inhibitor.** Structural studies of a synthetic analog bound to β-glucanase I from *Fusarium oxysporum* shows that substrate distortion orients and weakens the anomeric bond, facilitating $S_N2$ displacement of the leaving group.

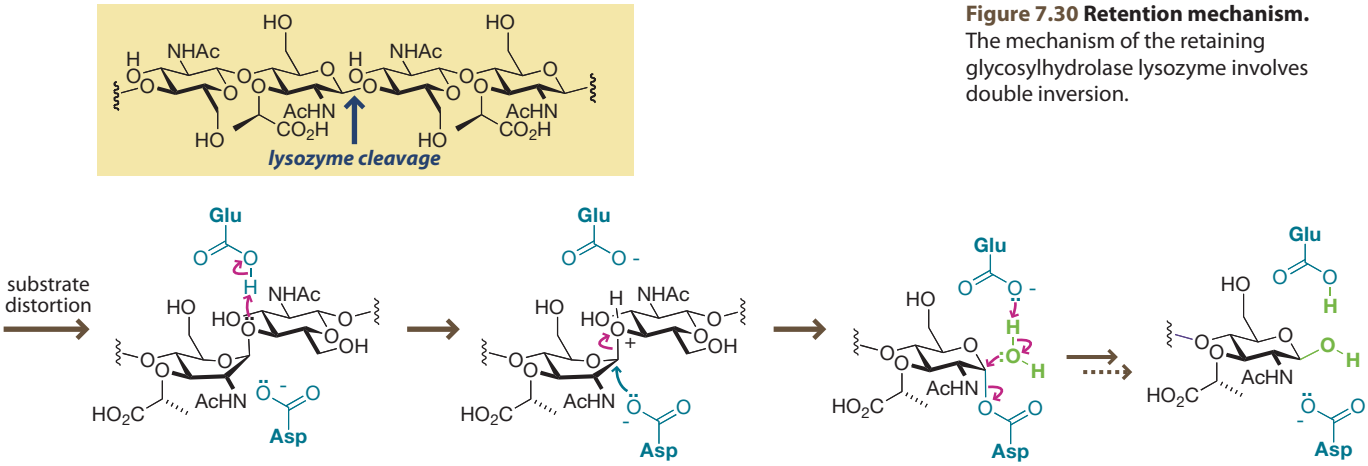



**Figure 7.30 Retention mechanism.** The mechanism of the retaining glycosylhydrolase lysozyme involves double inversion.

oxygen leaving group (**Figure 7.30**). Once protonated, an aspartate pushes the leaving group out from an axial orientation. A water molecule then binds to the site vacated by the GlcNAc leaving group, where the basic glutamate residue enhances the nucleophilicity of a water molecule through deprotonation. The glutamate might deprotonate the water as it attacks as part of a concerted mechanism; however, if we doubt that the breaking and forming bonds vibrate in perfect synchrony, then we would be better off depicting attack of water and deprotonation as individual mechanistic steps.

## Inhibiting glycosylhydrolase enzymes fight influenza

Inhibition of pathogenic glycosidases represents a rich area for treatment of infectious diseases, including viral infections. For example, the newly packaged virions of human influenza virus display two unique proteins on the viral coat: a hemagglutinin that affects fusion with target cells, and a neuraminidase that affects release from the host cell. *N*-Acetylneuraminic acid (Neu5Ac) is the most important member of a broader class of nine-carbon monosaccharides called sialic acids but *N*-Acetylneuraminic acid is the only sialic acid we will discuss in this book. New influenza virions are bound to host cells through the interaction of *N*-acetylneuraminic acid-binding proteins on the virion coat with *N*-acetylneuraminic acid residues on mammalian cells (**Figure 7.31**). The viral enzyme neuraminidase cleaves these *N*-acetylneuraminic acid residues, releasing the infectious virion.

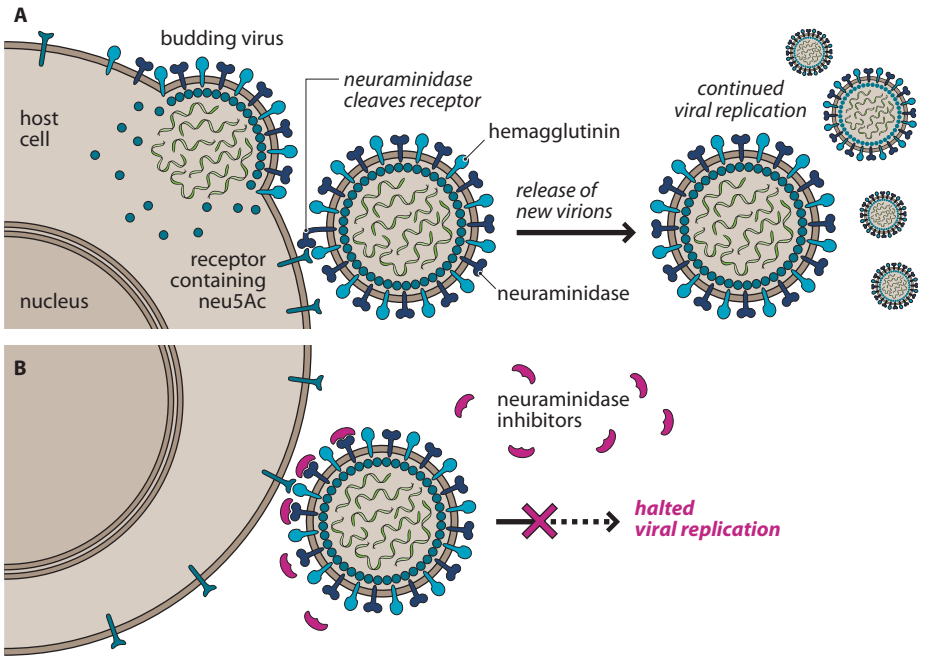
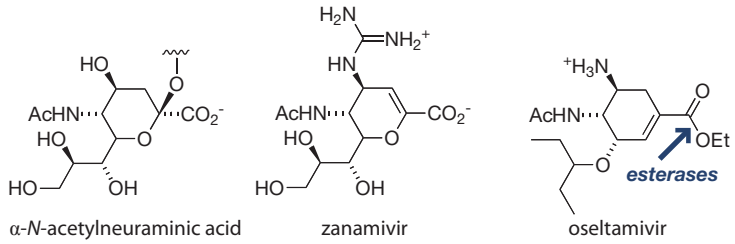



**Figure 7.31 Interfering with release of virions.** (A) The enzyme neuraminidase affects release of influenza virions by cleaving Neu5Ac linkages. (B) Inhibition of viral neuraminidase prevents release. (Adapted from A. Moscona, *N. Eng. J. Med.* 353:1363–1373, 2005.)

**Figure 7.32 Flu fighters.** Structures of substrates and inhibitors of influenza neuraminidase.



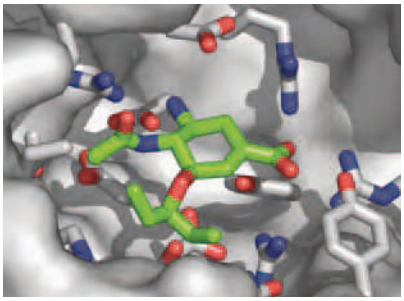



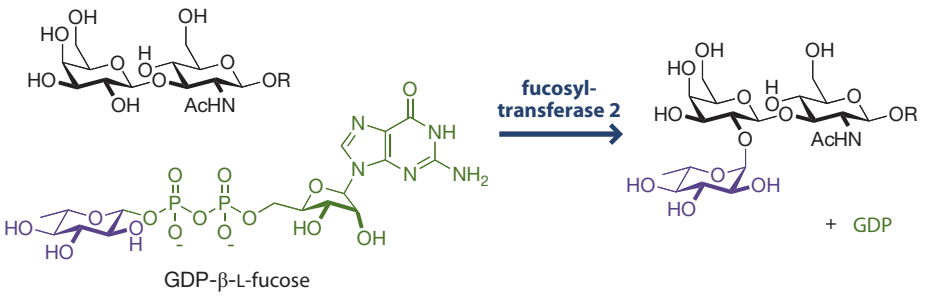
**Figure 7.33 The active site of influenza neuraminidase N1.** In this crystal structure oseltamivir carboxylic acid is bound to the same site as the neuraminic acid substrate (PDB 2HU0). See PDB 1NNC or 2HTQ for zanamivir.

Influenza α-neuraminidase is a retaining glycosidase in which a tyrosyl hydroxyl group is believed to form a β glycoside with Neu5Ac. The mechanism has been proposed to involve a carbocation, in contrast with other glycosidases. Because neuraminic acids are the only components of the human glycome that are connected through ketal rather than acetal linkages, the mechanism may indeed involve discrete carbocations rather than concerted $S_N2$ displacements (described in the previous section). Two neuraminidase inhibitors are available for the prevention of influenza and in early-stage treatment: zanamivir and oseltamivir. Both of these neuraminidase inhibitors, developed independently, include an $sp^2$-hybridized atom at the anomeric center (**Figure 7.32**). Zanamivir is a relatively close structural analog of *N*-acetylneuraminic acid, and inhibits the enzyme as a substrate mimic. However, this zwitterionic compound is not orally available and must be delivered by an inhaler. Oseltamivir is an orally available ester prodrug that is cleaved by human esterases to generate a carboxylic acid that ultimately binds to neuraminidase (**Figure 7.33**).

## Glycosyltransferases transfer monosaccharides from glycosyl phosphate donors

In humans, glycosidic bonds are generated by glycosyltransferases that use glycosyl donors with phosphate leaving groups at the anomeric position. Most glycosyltransferases, whether transferring glycosyl groups to amino acid side chains, lipids, or other sugars, use glycosyl donors with nucleoside phosphate leaving groups. Uridyl diphosphate (UDP) is the most common leaving group and is present on the donors UDP-α-D-glucose, UDP-α-D-GlcNAc, UDP-α-D-GalNAc, UDP-α-D-xylose, and UDP-α-D-glucuronic acid. Mannosyltransferases and fucosyltransferases use GDP-α-D-mannose and GDP-β-L-fucose, respectively, as substrates. For example, fucosyltransferase 2 transfers α-fucosyl residues specifically to the 2-hydroxyl group of certain Galβ1−3GalNAcβ groups (**Figure 7.34**). This modification correlates with the presence of blood group antigens on proteins secreted into saliva, gastrointestinal mucus, and pulmonary fluid. Sialyltransferases use the cytidyl monophosphate derivative of *N*-acetylneuraminic acid (CMP-Neu5Ac) as the glycosyl donor. In a few cases, as in the case of protein mannosyltransferases, the monosaccharide is transferred from a membrane-bound dolichol phosphate derivative.

**Figure 7.34 Enzymatic glycosylation.** Fucosyltransferase 2 transfers an α-fucosyl group from GDP-β-L-fucose.




## Glycosyltransferases transfer glycosyl groups from phosphates

The mechanisms of glycosyltransferases involve $S_N2$ displacement mechanisms, reminiscent of glycosylhydrolases. Glycosyltransferases are also classified as either inverting or retaining based on the stereochemical outcome. The inverting enzyme

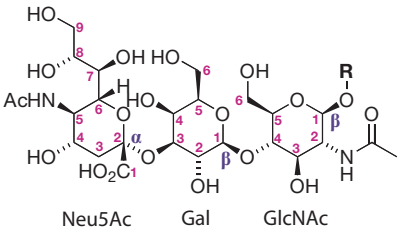



**Figure 7.35 Shorthand notation for oligosaccharides.** Unbranched oligosaccharides can be described with a compact nomenclature that uses the abbreviation for each monosaccharide unit, followed by the anomeric stereochemistry, followed by the connectivity (based on numbering of the monosaccharide backbone). The trisaccharide moiety shown would be Neu5Acα2–3Galβ1–4GlcNAcβ; alternatively, arrows or other symbols can be used to designate the connectivity Neu5Acα2→3Galβ1→4GlcNAcβ.

β1−4 galactosyltransferase adds β-galactosyl residues to the 4-hydroxyl group of GlcNAc to generate Galβ1−4GlcNAc. The retaining enzyme α1−3 galactosyltransferase then adds α-galactosyl groups to the 3-hydroxyl group of the galactose to generate Neu5Acα2−3Galβ1−4GlcNAc (**Figure 7.35**). Both the inverting and retaining galactosyltransferases use the same UDP-Gal substrate.

The structures of glycosylhydrolase enzymes are diverse, but the structures of glycosyltransferases that use nucleotidyl diphosphate donors are limited to two basic folds, GT-A and GT-B (**Figure 7.36**). Unfortunately there is no correlation between the protein fold and the stereochemical outcome of the reaction. For example, the retaining glycosyltransferase α1−3 *N*-acetylgalactosaminyltransferase and the inverting glycosyltransferase β1−4 galactosyltransferase both have GT-A folds.

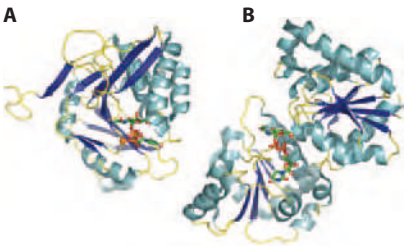



**Figure 7.36 Two canonical structures for glycosyltransferases.** (A) Mouse α1–4 *N*-acetylhexosaminyl-transferase complexed with UDP-GalNAc has a GT-A fold (PDB 1OMZ). (B) *E. coli* MurG complexed with UDP-GlcNAc has a GT-B fold (PDB 1NLM).

Glycosyltransferase enzymes have a catalytic base, usually a carboxylate side chain, that can remove the proton from the nucleophilic hydroxyl at some point during the reaction. In the active site, the phosphate leaving group is usually coordinated to divalent ions such as $Mg^{2+}$ or $Mn^{2+}$ (**Figure 7.37**). Divalent $Mg^{2+}$ ions have no ligand field and can accept ligands with any coordination geometry; in contrast, a divalent $Mn^{2+}$ ion prefers to adopt octahedral geometries. In a few cases, amino acid side chains act as Brønsted acids to facilitate loss of the anomeric phosphate.

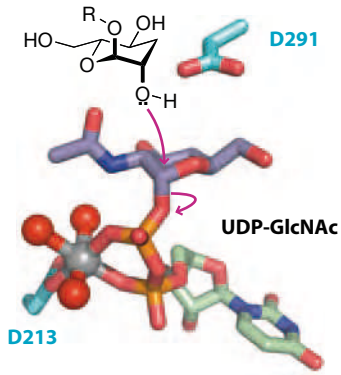



**Figure 7.37 Functional groups in a glycosyltransferase active site.** An aspartate side chain (D291) in the inverting glycosyltransferase *N*-acetylglucosaminyltransferase I is poised to deprotonate the 2-hydroxyl group of the mannose acceptor. A manganese ion (gray) coordinates to the diphosphate, making it a better leaving group. (PDB 1FOA)

Retaining glycosyltransferases, such as UDP-GalNAc:polypeptide transferase 1, catalyze **glycosylation** with retention of stereochemistry. Retaining enzymes are often presumed to proceed via a covalent glycosyl-enzyme intermediate, but even with several good crystal structures it is still unclear whether retaining glycosyltransferases proceed through a double inversion mechanism or through an attack on a discrete oxocarbenium ion with retention of anomeric configuration. In support of a glycosyl-enzyme intermediate, when the glutamate carboxylate of E137 of the retaining glycosyltransferase, α1−3 galactosyltransferase, was mutated to alanine, the E317A enzyme was essentially inactive; however, when azide anion was present at concentrations higher than 100 mM in the reaction medium, it was able to rescue the activity of the enzyme by forming a temporary β-azido intermediate (**Figure 7.38**).

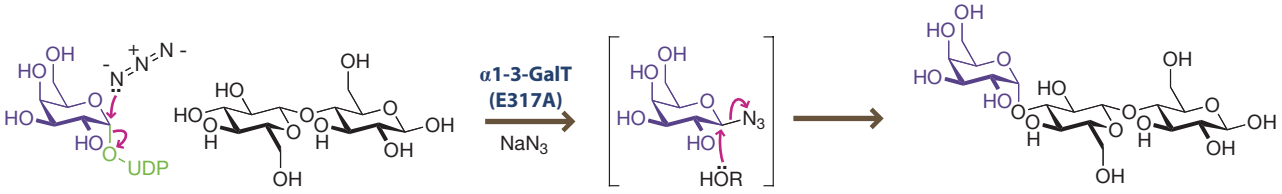



**Figure 7.38 Chemical assistance.** Azide anion rescues the catalytic activity of a retaining glycosyltransferase from which the nucleophilic glutamate has been removed.

**Problem 7.8**

Suggest a plausible $S_N2$ arrow-pushing mechanism for the transfer of α-UDP-GlcNAc to the 3 position of galactose. What is the preferred conformation of the GlcNAc pyranose ring in the transition state?

## 7.3 POLYSACCHARIDES

### Diastereomers of glucose polymers have very different properties

From the standpoint of human glycobiology, oligomers of monosaccharides tend to be more interesting than repeating polymers, known as polysaccharides. However, polysaccharides such as starch, cellulose, and chitin are highly familiar carbohydrates, even to scientific lay persons (**Figure 7.39**). Starch is commonly found in potatoes, bread, and other foodstuffs that are nutritionally labeled as carbohydrates. The cell walls of plants are composed of cellulose, a fiber found in paper, wood, and cotton. Chitin forms the exoskeletons of insects and shellfish. In humans the important polysaccharides are glycogen, an oligomer of glucose, and polysialic acid $[\text{-Neu5Ac}\alpha2\text{–}8]_n$, which is found in developing embryonic tissues. Glycogen is analogous in structure to the starch found in foods.

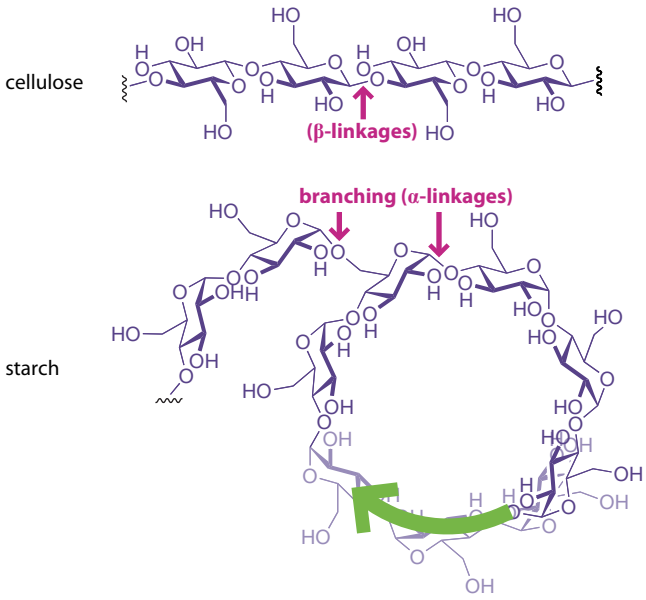



**Figure 7.39 Polymers.** Structures of the common polysaccharides cellulose and starch.

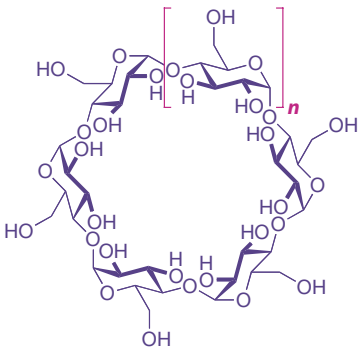


**Figure 7.40 Cyclodextrins.** Cyclic oligomers of glucose, $[\text{Glc}\alpha1\text{–}4]_n$, are called cyclodextrins.

Cellulose and starch are polymers of D-glucose linked through the 4-hydroxyl group. The principal structural difference between starch and cellulose is that cellulose is linked through β-linkages, whereas starch, also known as α-amylose, is linked through α-linkages. The structure of cellulose can be succinctly abbreviated as $[\text{-Glc}\beta1\text{–}4]_n$; the structure of starch can be abbreviated as $[\text{-Glc}\alpha1\text{–}4]_n$ (**Figure 7.40**). Starch is often branched, in which case it is referred to as amylopectin. The α1–4 linkages of starch force the polymer to adopt a helical structure. Since 1814, starch has been known to form dark blue inclusion complexes with iodine. These complexes involve the binding of the linear triiodide $I_3^-$ anion in the center of the cycloamylose helix. Starch is usually added to redox titrations with iodine to ensure a sharp endpoint.

If you wet a piece of bread with your saliva and then taste it later, it will taste sweet. Human saliva contains the enzyme **α-amylase**, which catalyzes the hydrolysis

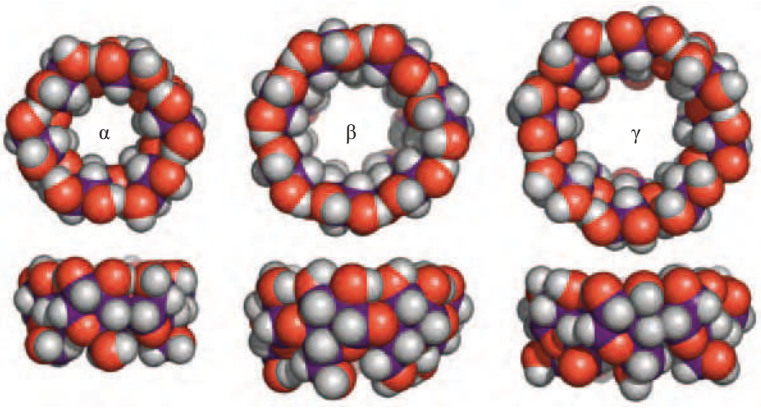



**Figure 7.41 Crystal structures of α-, β-, and γ-cyclodextrins.**

of amylose into sweet-tasting glucose. α-Amylose is also a substrate for the enzyme cyclodextrin glycosyltransferase. Cyclodextrin glycosyltransferase converts partly hydrolyzed starch helices into rings with six, seven, or eight glucose subunits. These cyclic molecules are referred to as α-, β-, and γ-**cyclodextrins**, respectively (**Figure 7.41**). The individual ringlets in medieval chain mail armor were made through an analogous process of cutting wire spirals into small wire rings. The hydrophobic cavity of cyclodextrins can bind to hydrophobic molecules. Cyclodextrins are adept at binding benzene rings, which are about 7 Å in diameter. For example, 4-bromopropiophenone binds to α-cyclodextrin with a $K_d$ of 1.3 mM. β-Cyclodextrin binds aromatics with even higher affinity. The ability to encapsulate aromatic molecules has been commercialized by Procter & Gamble in two products. Bounce® dryer sheets have perfumes encapsulated that are slowly released in the heat of a dryer. Febreze® deodorizer traps foul-smelling molecules in cyclodextrins.

## Problem 7.9

Draw each of the following structures as dash–wedge drawings. Do not worry about the conformations, but clearly indicate the configuration of each stereogenic center.

**A** Fucα1–2Galβ1–3GlcNAcβ1–3Galβ-OR

**B** Galβ1–4GlcNAcβ1–2Manα-Ser

## Chitin is a resilient polymer in insect cuticles

**Chitin** is a β1−4 linked polymer of *N*-acetyl-D-glucosamine (**Figure 7.42**). It is the highly resilient material upon which arthropod (insects, crabs, shrimp) exoskeletons are based. When the acetyl groups of chitin are chemically hydrolyzed, the resulting polymer, known as chitosan, has a number of materials applications. For example, the ability of chitosan to strongly induce blood clotting has led to its application in battlefield dressings.

From the perspective of human nutrition, mammals seem to have missed out on a rich potential food source. All mammals have enzymes that can hydrolyze the α1−4 linkages of starch. However, no mammals have the enzymes necessary to hydrolyze the β1−4 linkages of cellulose or chitin. Fortunately, grazing animals have developed a symbiotic relationship with microorganisms that can hydrolyze cellulose into digestible fragments.

chitin
$[GlcNAc\beta1\text{–}4]_n$

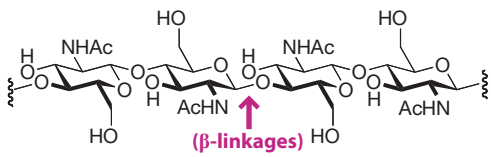



**Figure 7.42 Chitin.** A polymer of *N*-acetylglucosamine gives the shells of insects and crustaceans great resiliency.

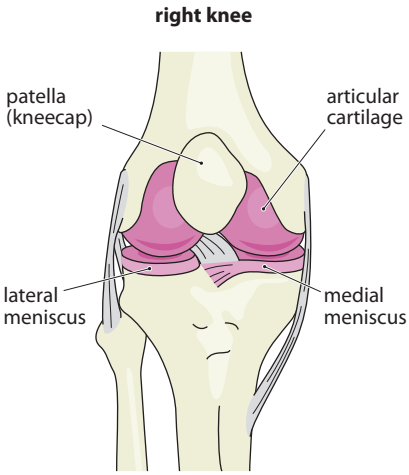



**Figure 7.43 Glycosaminoglycans.** The meniscal and articular cartilages of the knee joint are made up of glycosaminoglycans.

## Some tissues are cushioned by the polysaccharide hyaluronan

The chondrocytes that make up human cartilage express two important types of glycosaminoglycans: hyaluronan and chondroitin sulfate. Together, these glycosaminoglycans form a gel-like layer that reduces friction in articulating joints and cushions joints from impact (**Figure 7.43**). Hyaluronan is a glycosaminoglycan composed of repeating units of [GlcAβ1−3GlcNAcβ1−4] that occurs as a free oligosaccharide (**Figure 7.44**). A typical hyaluronan chain has thousands of disaccharide repeats. Hyaluronan is distinct in two ways from the structurally related glycosaminoglycans. First, the oligosaccharide is not attached to a protein. Instead, it is extruded directly into the extracellular milieu. Second, hyaluronan synthase is a polymerizing enzyme that has two active sites: one for UDP-GlcA and one for UDP-GlcNAc. The growing chain has a UDP leaving group on the reducing end that is displaced with each addition of a subunit. As the chain grows it bounces from one active site to the other.

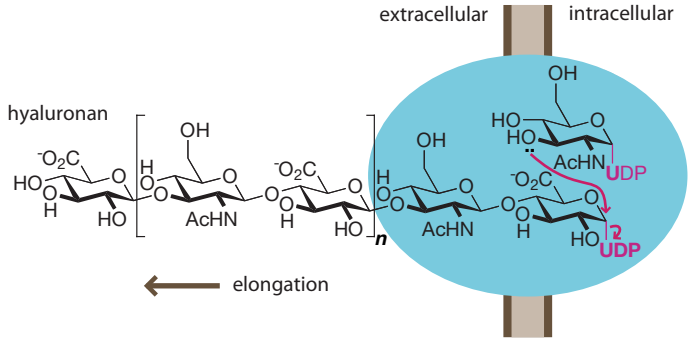



**Figure 7.44 Extrusion polymerization.** Hyaluronan synthase is a polymerase that extrudes the alternating copolymer hyaluronan into the extracellular medium.

Osteoarthritis involves a loss of cartilage and commonly affects the knee joint. Even though cartilage is composed of chondroitin sulfate and hyaluronan (both of which contain glucosamine) it would be naive to think that the disease is caused by a simple dietary deficiency. However, by 2007, sales of chondroitin and glucosamine in the United States exceeded half a billion dollars. Unfortunately, recent clinical trials have shown that oral chondroitin sulfate and glucosamine supplements are no better than placebos at reducing pain or damage to osteoarthritic knees.

## Meningococci are coated with polysialic acids like those found on neurons

Meningococci are one of many types of infectious agents that can lead to meningitis, an inflammation of the membranes that surround the central nervous system, called the meninges. Young toddlers are at greatest risk for the disease, which can present in various ways, often as quickly spreading necrotic lesions. The cellular capsules of *Neisseria meningitidis* are coated with polymers that contain *N*-acetylneuraminic acids either as repeating polymers or as repeating copolymers; the type of polymer varies with the strain of *N. meningitidis*. For example, serogroup B is coated with $[Neu5Ac\alpha2{-}8]_n$; serogroup C is coated with $[Neu5Ac\alpha2{-}9]_n$; serogroup Y is coated with $[Glc\alpha1{-}4Neu5Ac\alpha2{-}6]_n$; serogroup W135 is coated with $[Gal\alpha1{-}4Neu5Ac\alpha2{-}6]_n$ (**Figure 7.45**). The branches of many human glycans terminate in sialic acid residues, and the meningococcal polysialic acids are believed to help the bacterium evade the host immune response (**Figure 7.46**). In fact, poly-α2−8Neu5Ac, is found on human neuronal cell adhesion molecule (NCAM) but on no other human protein. The serogroup C polysialic acid $[Neu5Ac\alpha2{-}9]_n$ is a component of several vaccines against *N. meningitidis* (Meningitec®, Menjugate®, and NeisVac-C®).

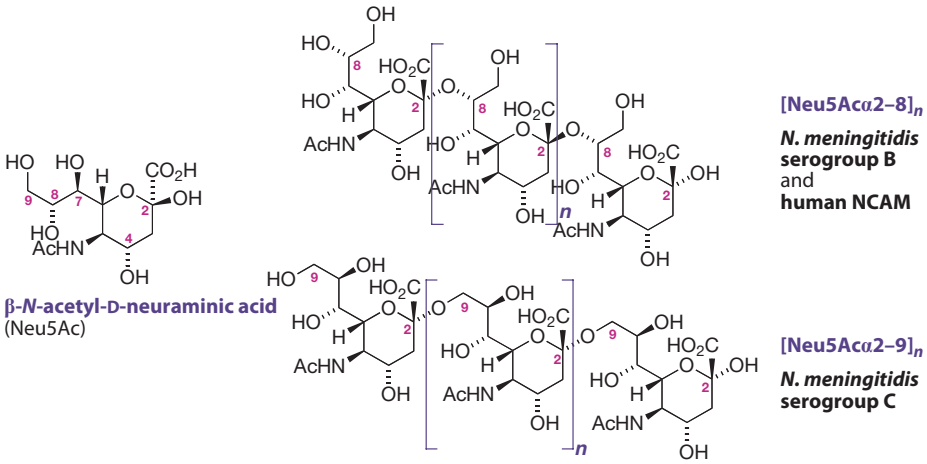



**Figure 7.45 Polysialic acids.** Two different types of Neu5Ac polymers are found in the capsular coats of meningococci.

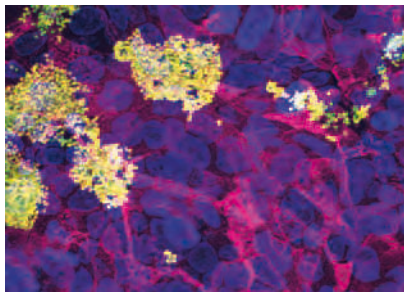


**Figure 7.46 Sleeper cells**. *Neisseria meningitidis* live in the nasopharynx of 5–10% of the population. Infections are problematic when *N. meningitidis* crosses the blood-brain barrier or blood-cerebrospinal fluid barrier. The fluorescence image above shows adherence of *N. meningitidis* (stained yellow) to human choroid plexus papilloma cells which serve as a model of the blood-cerebrospinal fluid barrier. (From C. Schwerk et al., *PLoS One* 7(1): e30069, 2012.)

## 7.4 GLYCOPROTEINS

### Glycosylation of human proteins occurs in the vesicles of the secretory pathway

**Glycoproteins** are rare in prokaryotes but common in eukaryotic organisms. Virtually all glycosylation of human proteins occurs in the vesicles of the secretory pathway, the endoplasmic reticulum (ER), and the Golgi apparatus, which are not found in prokaryotic cells (**Figure 7.47**). The ER is a dynamic network of organelles. The contents of the ER are isolated from the cytosol by a membrane, and the surfaces of the rough ER are distinguished by the presence of membrane-bound ribosomes that "squirt" proteins directly into the ER during translation. The lamellae of the ER are dynamic structures that bud off as ribosome-free organelles, referred to as the Golgi apparatus. Interestingly, this process can be inhibited by the natural product brefeldin A. The Golgi vesicles migrate toward the plasma membrane and ultimately fuse, releasing soluble proteins from the cell. This process has two additional topological consequences. First, soluble proteins in the secretory vesicles are exported from the cell. Second, the cytosolic domain of proteins bound to the ER membrane remains on the cytosolic side of the plasma membrane; conversely, the domain of membrane-bound proteins that is directed into the ER ends up on the extracellular side of the plasma membrane. Because glycosylation of proteins occurs inside the ER and Golgi apparatus, the attached oligosaccharides are destined to be presented on the extracellular surface and never within the cytosol. Proteins and lipids decorated with complex glycans are meant for communication between cells, not for communication within cells.

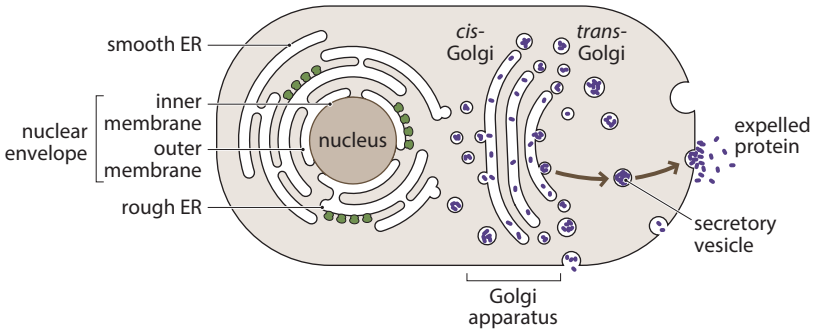



**Figure 7.47 N-glycosylation of proteins occurs in the endoplasmic reticulum (ER).** The glycans are further modified by enzymes in the Golgi vesicles. (Adapted from B. Alberts et al., Molecular Biology of the Cell, 5th ed. New York: Garland Science, 2008.)

There are two basic categories of glycoproteins: N-linked and O-linked. N-linked oligosaccharides are attached through asparagine side chains, and are usually complex, highly branched structures. N-linked glycans are added while proteins are being translated into the ER. O-linked oligosaccharides are attached through serine (or threonine) side chains and tend to be linear, repeating oligomers. They are added to proteins at a later stage of vesicular trafficking within the Golgi network. Conceptually, the biosynthesis of O-linked glycoproteins is simpler than the biosynthesis of N-linked glycoproteins, so we will discuss O-linked glycoproteins first.

## Synthesis of O-linked glycoproteins begins with the addition of xylose or *N*-acetylgalactose

O-linked oligosaccharides are added to proteins in the Golgi apparatus starting with the transfer of either xylose or *N*-acetylgalactose to a serine or threonine hydroxyl group. These are further glycosylated to generate polymeric or branched glycans, which we will discuss shortly. The biosynthesis of proteoglycans is initiated by the enzyme UDP-D-xylose:protein β-D-xylosyltransferase. In this common form of enzyme nomenclature the name of the enzyme is preceded by the two substrates, which are separated by a colon. Humans have genes for two different isoforms of UDP-D-xylose:protein β-D-xylosyltransferase; the expression levels depend on the cell type. The biosynthesis of mucins, and that of many other O-linked glycoproteins, is initiated by UDP-*N*-acetyl-D-galactosamine:polypeptide *N*-acetylgalactosaminyltransferase. Fourteen of these polypeptide GalNAc transferases have been enumerated in humans (ppGalNAc T1–T14) and there are a handful of other genes that could code for isoforms. The expression of these various polypeptide GalNAc transferases is dependent on cell type and other factors. For example, the mucin protein MUC1 is initially glycosylated by the ubiquitous polypeptide ppGalNAc T1 and T2. In fibroblasts, the extracellular matrix protein fibronectin is glycosylated by ppGalNAc T3. In neurons, the post-translational modification of syndecan is initiated by the glycosyltransferase ppGalNAc T13. The GalNAcβ1–Ser group is further modified in healthy cells. Unmodified GalNAcα1–Ser groups are often present on the surface of cancer cells and leukemias and are referred to as the Tn antigen.

There are no simple peptide consensus sequences that allow one to identify O-glycosylation sites in proteins, but such sites can be predicted with complex computer algorithms. Localization of the substrate and the glycosyltransferase can be as important as any overt structural feature on the substrate (**Figure 7.48**). For example, human β1–4 galactosyltransferase 3 attaches galactosyl residues to any terminal GlcNAc residue in a membrane-bound acceptor, regardless of whether it is an O-linked glycoprotein, an N-linked glycoprotein, or a glycolipid (**Figure 7.49**).

Less commonly, extracellular human proteins are O-glycosylated by other monosaccharides. The α-dystroglycan component of the dystrophin glycoprotein complex is O-mannosylated by *O*-mannosyltransferase, which transfers an α-mannosyl residue from dolichol phosphate β-D-mannose to serine and threonine residues on the protein. In all mammals, this is further elaborated to oligosaccharides with a common trisaccharide core: Galβ1–4GlcNAcβ1–2Manα–Ser/Thr. The enzyme GDP-L-fucose:polypeptide fucosyltransferase transfers α-fucosyl residues to serine and threonine residues in the EGF domains of extracellular proteins like Notch.

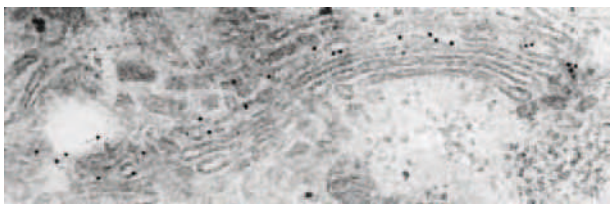


**Figure 7.48 Evidence that glycosyltransferases are localized in the Golgi network.** A transmission electron microscopy image of human HeLa cells stained with rabbit antibodies against galactosyltransferase. The rabbit antibodies were revealed with mouse antibodies conjugated to 10 nm gold nanoparticles (the dark dots in the image). (From T. Nilsson et al., *J. Cell Biol.* 120:5–13, 1993. With permission from Rockefeller University Press.)

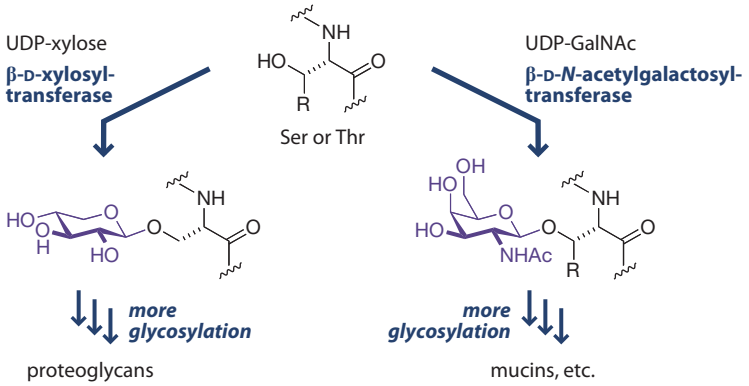



**Figure 7.49 Creating the O-linkage.** The biosynthesis of O-linked glycoproteins with long oligosaccharides begins with the addition of β-D-xylosyl or β-D-*N*-acetylgalactosyl residues.

## O-linked proteoglycans are polyanions

Proteoglycans are proteins with long O-linked glycosaminoglycan chains. Some examples include chondroitin sulfate, heparin/heparan sulfate, and keratan sulfate. Proteoglycans are generated through the initial addition of β-xylose to serine, followed by elaboration to a trisaccharide core: Galβ1–3Galβ1–4Xylβ-OSer (**Figure 7.50**). This trisaccharide core is then elaborated to various longer oligosaccharides.

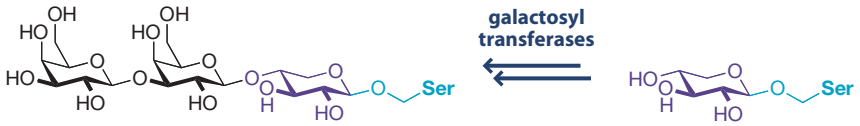



**Figure 7.50 Making the trisaccharide core.** The trisaccharide core of O-linked proteoglycans is assembled one monosaccharide at a time, prior to an uncontrolled polymerization.

### Problem 7.10

Using chair conformations, draw the structure of the following pentasaccharide segment of a proteoglycan: GlcNAcβ1–4GlcAβ1–3Galβ1–3Galβ1–4Xylβ-OSer.

Glycosaminoglycans are hygroscopic proteoglycans that are formed from repeating disaccharide units: a uronic acid such as D-glucuronic or L-iduronic acid, and an acetamido sugar such as *N*-acetylglucosamine or *N*-acetylgalactosamine. Found on the surface of nearly all cell types, glycosaminoglycans determine the physicochemical and mechanical properties of the tissue (**Figure 7.51**). Glycosaminoglycans always occur as heterogeneous mixtures as a result of varying levels of sulfation. Heterogeneity is a general property of many resilient materials, because it avoids the brittleness associated with crystallinity.

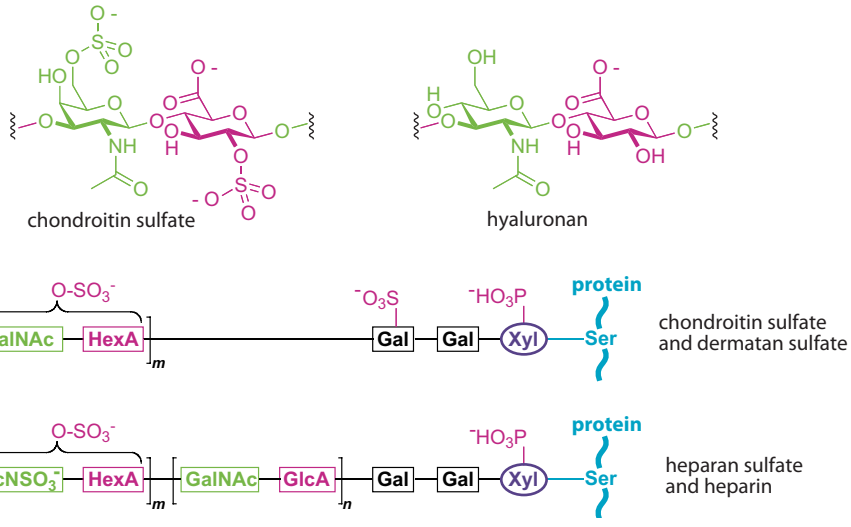



**Figure 7.51 Glycosaminoglycans.** The glycosaminoglycans in human cartilage (HexA = GlcA or IdoA) are made up of repeating disaccharide subunits.

Recall that chondrocytes produce two types of glycosaminoglycans: the free polysaccharide hyaluronan and the proteoglycan chondroitin sulfate. Chondroitin sulfate is a glycoprotein made up of a repeating disaccharide [GalNAcβ1–4GlcAβ1–3] similar to that in hyaluronan, but chondroitin has varying levels of sulfation on the 2 position of glucuronic acid residues and the 4 and 6 positions of the *N*-acetylglucosamine residues. Dermatan sulfate, found primarily in skin, is related to chondroitin sulfate except that it is made of repeating subunits of [GalNAcβ1–4IdoAβ1–3]. Recall that iduronic acid is the 5-epimer of glucuronic acid and is therefore an L-sugar. Iduronic acid is generated from glucuronic acid through post-glycational epimerization.

Heparin sulfate is a glycosaminoglycan composed of repeating units of iduronic acid or glucuronic acid and glucosamine *N*-sulfate, with high levels of sulfation at the 2 position of the uronic acid and the 3 position and 6 position of glucosamine. Heparin binds to and activates the thrombin inhibitor antithrombin-III. Antithrombin-III then forms a covalent complex with various serine proteases in the clotting cascade (Section 6.3). Antithrombin-III is best described as a very poor substrate that is turned over slowly by the protease. Antithrombin-III is most strongly activated by a specific pentasaccharide sequence of heparin (**Figure 7.52**). Low-molecular-weight heparin, isolated from pig intestinal mucosa, is used clinically as an injectable anticoagulant during surgery when there is a high risk of unwanted blood clotting. In 2008, a major pharmaceutical company recalled batches of heparin that were contaminated with a related glycosaminoglycan. Allegedly, the original supplier had augmented the batches of heparin with oversulfated chondroitin sulfate, which was 10 times cheaper to obtain. More recently, a synthetic pentasaccharide, fondaparinux, has been introduced as an alternative to heparin isolated from pigs.

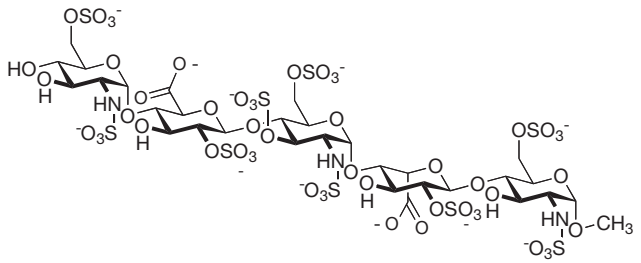


**Figure 7.52 A heparin-like drug**. Fondaparinux is based on the antithrombin-III binding sequence of heparin.

The slippery hydrophilic properties of mucous secretions such as snot are characteristic of polyanionic mucin glycoproteins. Mucins are heavily glycosylated oligomeric proteins often held together by disulfide bonds. For example, the group of human salivary mucins, MG1, has 292 O-linked oligosaccharides attached to the large MUC5B protein (more than 3500 amino acids). The structure is often depicted like a bottle brush with oligosaccharides for bristles. The polysaccharide moieties of MG1 are typically composed of type 1 repeats $[Gal\beta1–3GalNAc\beta1–3]_n$ or type 2 polylactosamine repeats $[Gal\beta1–4GalNAc\beta1–3]_n$ with varying levels of sulfation on the Gal residues. Almost half of the O-linked oligosaccharides of MG1 have Neu5Ac on the nonreducing end.

The slippery mucus produced by land snails and slugs is rich in mucins (**Figure 7.53**). In secretory vesicles, the polyanionic mucins are held in a compact form by $Ca^{2+}$ ions. When $Ca^{2+}$ ions are stripped away, the mucins rapidly expand through hydration and are released from the cells. This system illustrates a general property of materials. The hydrophilicity of polymers generally correlates with anionic charge density. For

**Figure 7.53 Mucin trail**. The slime trail of snails is rich in polyanionic mucin glycoproteins. (Courtesy of Photo Researchers.)

example, the superabsorbant polymers found in disposable diapers are simply polyacrylates, $[CH_2CH(CO_2Na)]_n$. They absorb up to 500 times their weight in water.

Hagfish are one of the most notorious sources of mucin-based slime (**Figure 7.54**). When stressed, hagfish produce voluminous quantities of a thick slime composed of approximately 99.996% seawater, 0.0015% mucin, and 0.002% protein threads. The protein imparts elasticity, and the mucin imparts viscosity. The slime has been hypothesized to clog the gills of potential predators.

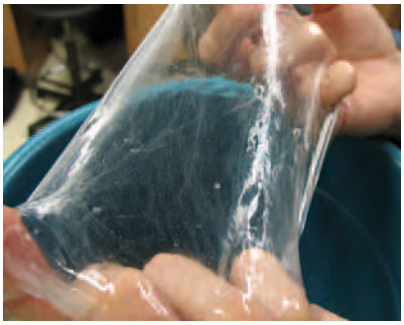


**Figure 7.54 Instant slime.** Dried hagfish slime, composed of polyanionic mucins, can be reconstituted from seawater. (Courtesy of Jamie Miller, University of Guelph.)

## The carbohydrate moiety of N-linked glycoproteins is initially added as an oligosaccharide

The N-glycosylation of proteins takes place on the asparagine residue of Asn-Xxx-Ser and Asn-Xxx-Thr sequences as the proteins are translated into the endoplasmic reticulum (**Figure 7.55**). Initially, the branched tetradecasaccharide $Glc_3Man_9GlcNAc_2$ is transferred from a membrane-bound precursor by the enzyme oligosaccharyltransferase (**Figure 7.56**). The structure of the initial mannose-rich oligosaccharide varies among eukaryotes, but the $Man_3GlcNAc_2$ core is common to all of them.

As the N-linked glycoprotein is trafficked through the vesicular network of the endoplasmic reticulum and Golgi apparatus, various glycosylhydrolases trim the

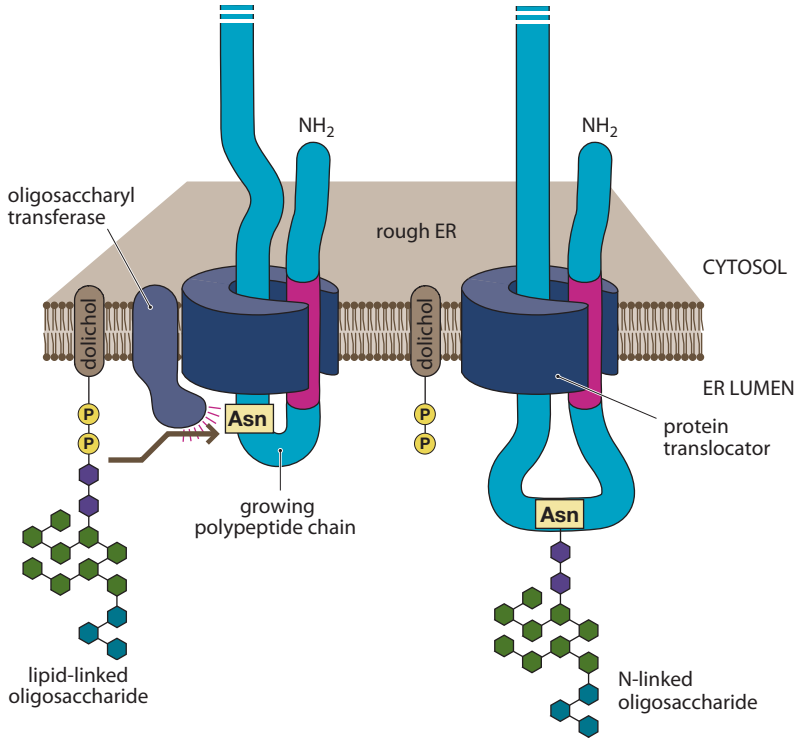



**Figure 7.55 Grafting the scion onto the stock.** Transfer of a complex oligosaccharide to the tripeptide sequence AsnXxxSer occurs as the ribosome extrudes the protein chain into the endoplasmic reticulum. (Adapted from B. Alberts et al., Molecular Biology of the Cell, 5th ed. New York: Garland Science, 2008.)

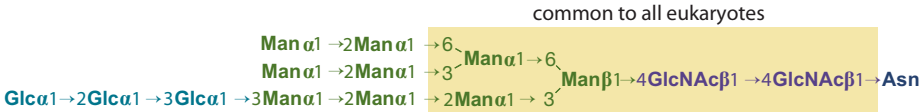



**Figure 7.56 The initial N-glycan core.** The core oligosaccharide that is added to asparagine side chains of human proteins is composed of 14 monosaccharide subunits.

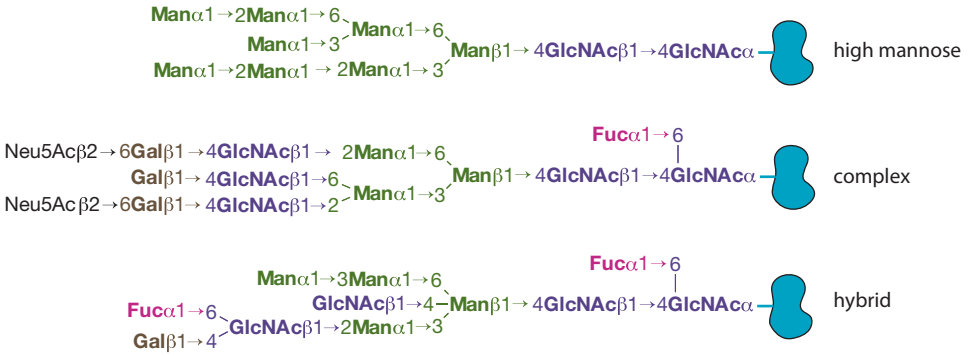



**Figure 7.57 Man kind.** The three types of N-glycans that ultimately emerge after vesicular trafficking are classified on the basis of the mannose core: high mannose ($Man_8$), complex ($Man_3$), and hybrid ($Man_{3–8}$).

structure, and various glycosyltransferases graft on additional monosaccharide units. The resulting N-linked glycoproteins can be subdivided into three types: oligomannose (high mannose), hybrid, and complex (**Figure 7.57**). Oligomannose structures are found in all eukaryotes, including yeast, whereas hybrid and complex N-linked glycoproteins are characteristic of multicellular organisms.

The oligosaccharide substrate for the oligosaccharyltransferase is synthesized starting from UDP-GlcNAc and dolichol phosphate. Dolichol is a simple terpene alcohol up to 85 carbons in length—significantly longer than the width of a typical lipid bilayer. The membrane-bound substrate, GlcNAc-PP-dolichol, is then expanded to a complex oligosaccharide through the action of a series of glycosyltransferases. The mannose-rich oligosaccharide with a dolichol diphosphate leaving group is added to the asparagine side chain (**Figure 7.58**). The initial coupling of UDP-GlcNAc and dolichol phosphate is inhibited by the tunicamycins, which have become a widely used tool in glycobiology to shut down N-glycosylation. The tunicamycins A, B, C, and D are a set of homologous natural products with lipophilic acyl chains of various lengths.

## An Asn-Xxx-Ser motif adopts a reactive conformation in the N-glycosylation of proteins

The asparagine side chain amide does not have a nucleophilic lone pair because of resonance donation of the $H_2N$ lone pair into the C=O π* orbital. Imperiali and coworkers noted that, in reactive conformations, Asn-Xxx-Ser substrates place the side-chain hydroxyl and backbone amide in proximity to the asparagine side-chain carbonyl group. The amide and hydroxyl groups of Ser/Thr are believed to facilitate the formation of an imidate tautomer of the Asn residue, making a basic nitrogen lone pair

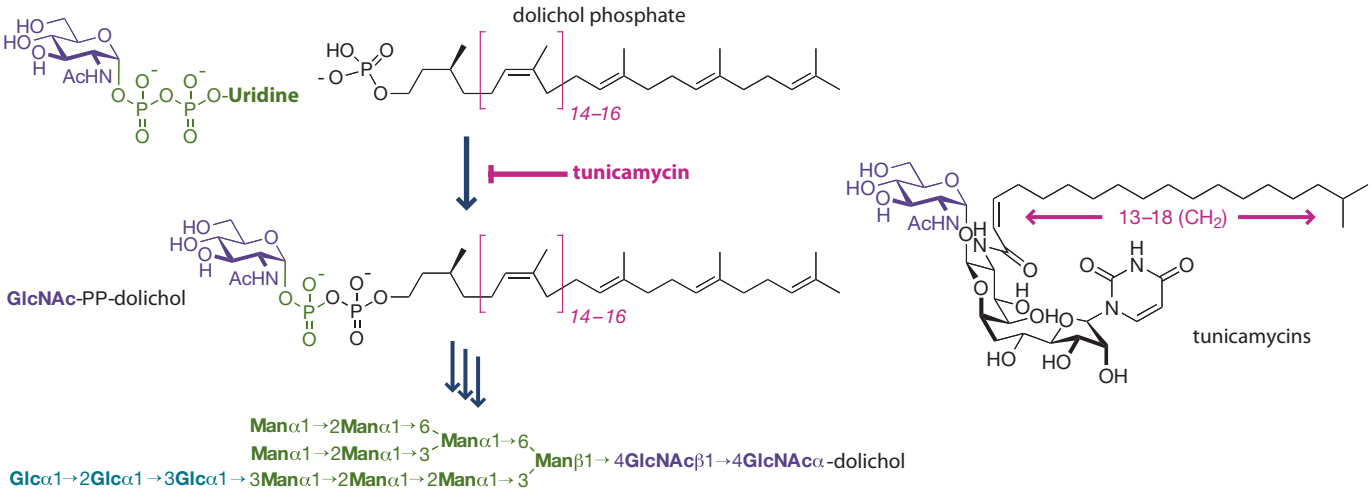



**Figure 7.58 Blocking N-glycan synthesis.** Tunicamycin blocks the synthesis of N-linked glycoproteins by inhibiting the initial attachment of a lipophilic dolichol chain to the first GlcNAc residue in the glycan core.

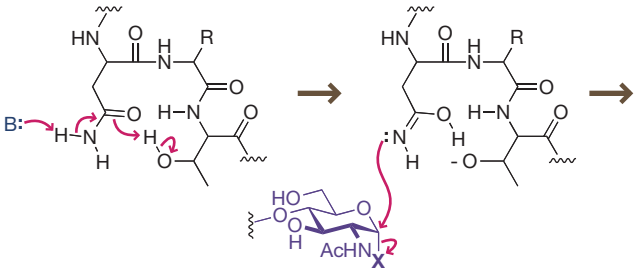


**Figure 7.59 The mechanism for N-glycan attachment.** The mechanism of asparagine glycosylation by oligosaccharyl transferase is believed to proceed through an imidate tautomer.

available for glycosylation (**Figure 7.59**). This example of **substrate-assisted catalysis** illustrates how sequence specificity for post-translational processes results from both the preferences of modifying enzymes and the functional groups of the enzyme's substrate placed in close proximity to the targeted side chain.

## The processing of glycans occurs during vesicular trafficking

The processing of N-linked oligosaccharides begins with trimming of the branches within the endoplasmic reticulum. Further removal and addition of individual monosaccharides occurs as the glycans are trafficked outward toward the cell surface (see Figure 7.47), first through the endoplasmic reticulum and then through the Golgi network toward the cell surface. At no time are the glycans exposed to the cytoplasm, and most of the trimming reactions are performed by membrane-bound glycosylhydrolases with their active sites oriented toward the interior of the vesicular compartments. To appreciate why N-linked glycans are so heterogeneous and complex, it is helpful to imagine an automobile assembly line staffed by workers who perform their operation only when certain parts are missing. Now imagine speeding up the assembly line so that workers do not have sufficient time to work on every vehicle.

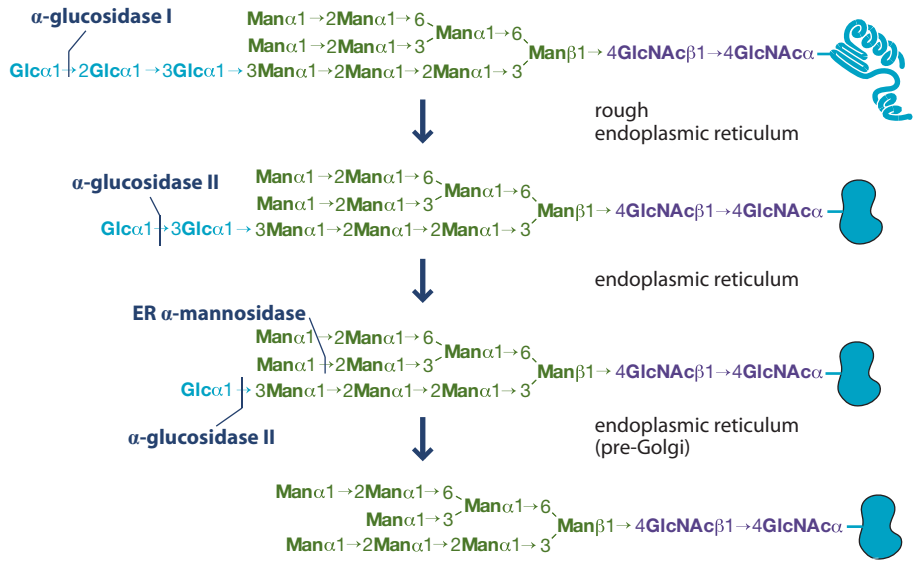



**Figure 7.60 Early stages of trimming.** Trimming reactions of N-linked oligosaccharides that take place within the endoplasmic reticulum.

Trimming starts within the rough endoplasmic reticulum (covered with ribosomes), as proteins are being translated by ribosomes, where α-glucosidase I removes the outermost α1–2-linked glucosyl residue (**Figure 7.60**). The next two α-1–3-linked glucosyl residues are removed by α-glucosidase II, which is weakly bound to the membranes of the smooth endoplasmic reticulum and the pre-Golgi vesicles. α-Glucosidases I and II are inhibited by the natural products 1-deoxynojirimycin and castanospermine (**Figure 7.61**).

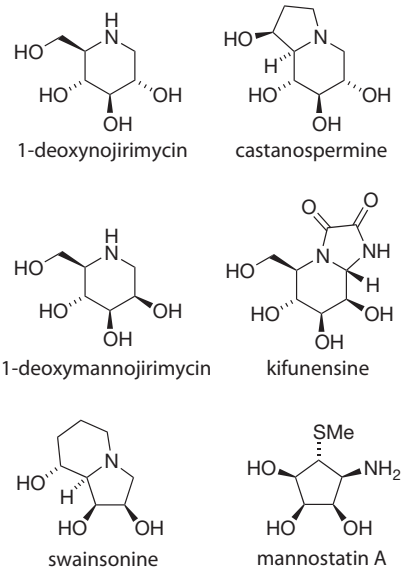



**Figure 7.61 Tools from nature.** A number of natural product inhibitors of glycosidases exhibit sufficient selectivity against various glycosidases to be useful as tools for studying N-glycan processing in human cells.

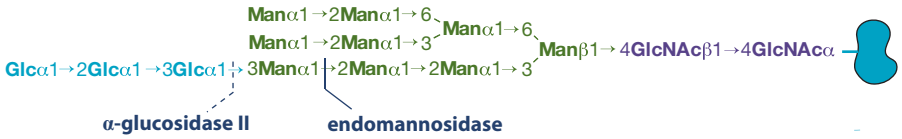



**Figure 7.62 Alternative early stage trimming.** If any of the outer glucosidase residues remain then endomannosidase takes care of any unfinished business by cleaving an internal glycosidic bond.

The second α1–3-linked glucosyl residue is hydrolyzed by α-glucosidase II more slowly than the first one. If the glucosyl residue is removed too slowly, an endomannosidase can cleave the Glcα1–3Man disaccharide as the glycans enter the Golgi network (**Figure 7.62**). Even if both of the glucosidases are inhibited by 1-deoxynojirimycin or castanospermine, endomannosidase can still cleave off the entire $Glc_3Man$ tetrasaccharide. Endomannosidase is the only mammalian trimming enzyme that cleaves an internal glycosidic bond.

As the pre-Golgi vesicles fuse with the Golgi vesicular network, they are set upon by α1–2 mannosidases IA, IB, or IC (**Figure 7.63**). The face of the Golgi network that is closest to the nucleus is referred to as *cis*; the face of the Golgi network that is closest to the cell surface is referred to as *trans*. Since Golgi vesicles slowly bubble outward toward the cell surface, glycosidases in the *cis*-Golgi act before glycosidases in the *trans*-Golgi. The three α1–2 mannosidases IA, IB, and IC seem to have the same function, cleaving the three α1–2-linked mannosyl residues, but their expression is tissue-specific. α-Mannosidases are inhibited by the natural products 1-deoxymannojirimycin and kifunensine. Glycans that are trimmed back to the $Man_5GlcNAc_2$ core are elaborated by glycosyltransferases. Those "high-mannose" N-glycans that are not completely trimmed by α1–2 mannosidases within the *cis*-Golgi may eventually make their way to the cell surface without further modification.

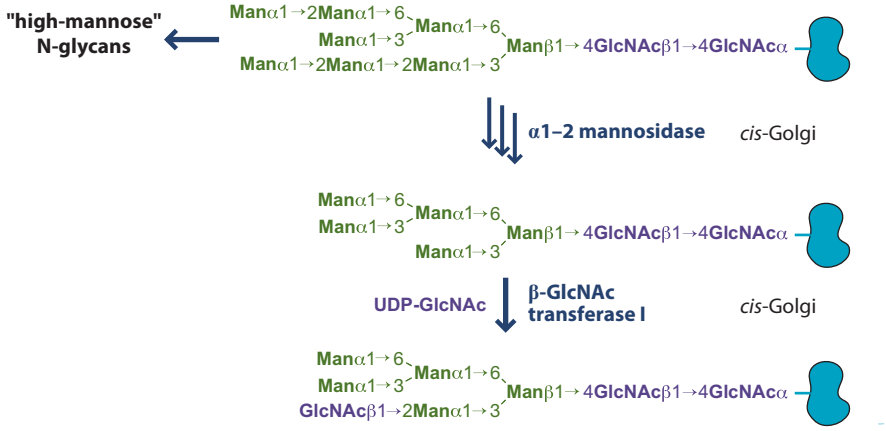



**Figure 7.63 Addition or subtraction.** As they enter the *cis*-Golgi N-glycans are either elaborated to high-mannose glycans or trimmed further. Small molecules (shown in Figure 7.61) can inhibit N-glycan modifying enzymes.

In the *cis*-Golgi network, the branch of the $Man_5GlcNAc_2$ core that formerly possessed the glucosyl residues is then glycosylated by the inverting glycosyltransferase β-*N*-acetylglucosaminyltransferase I, which tags it for further trimming by α-mannosidase II; this enzyme removes the outer Manα1–3 and Manα1–6 groups (see Figure 7.63). N-glycans that are trimmed by α-mannosidase II to a branched $GlcNAcMan_3GlcNAc_2$ core are further elaborated to "complex" N-glycans through the addition of β-*N*-acetylglucosaminyl, α-fucosyl, β-galactosyl, and α-Neu5Ac groups (**Figure 7.64**). N-glycans that are not trimmed by α-mannosidase II are also further elaborated, but because they have more than three mannosyl residues they are referred to as "hybrid" N-glycans because they have the extra mannose residues of a high-mannose N-glycan and the glycosylated branches of a complex N-glycan.

Golgi α-mannosidase II can be selectively inhibited by the natural products swainsonine and mannostatin A, diverting N-glycans toward hybrid cores rather than complex cores. The $Man_3$ and $Man_5$ cores are glycosylated by a number of β-*N*-acetylglucosaminyltransferases, which add up to six GlcNAc branches. For example, β-GlcNAc transferase II adds GlcNAc residues to the 2-hydroxyl groups of mannosyl residues (those that were not glycosylated by β-*N*-acetylglucosaminyltransferase II); β-GlcNAc transferases IV and V generate branches by adding a second GlcNAc residue to the 6-hydroxyl groups of mannosyl residues; β-GlcNAc transferase I adds a

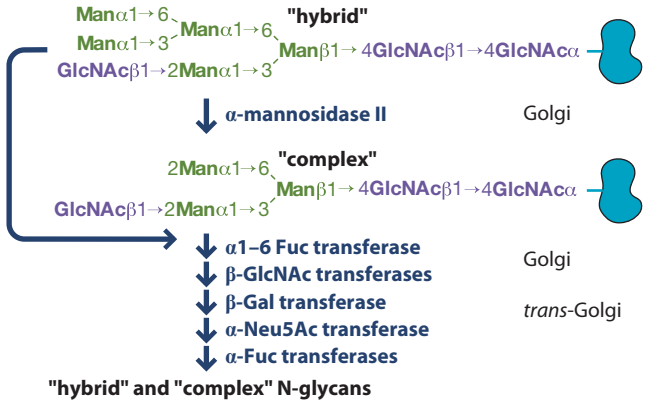



**Figure 7.64 Late-stage glycosylation in the Golgi network.** Further glycosylation reactions add additional sugars to the mannosyl branches and to the outer GlcNAc residue.

third GlcNAc residue to the 4-hydroxyl groups of mannosyl residues. As the N-glycans reach the *trans* face of the Golgi network, α1–6 fucosyltransferase VIII adds a fucosyl residue to the *N*-acetylglucosamine residue that is directly connected to the protein.

The outer GlcNAc branches of the N-glycans are then further adorned with β-galactosyl residues by two of the seven human β1–4 galactosyltransferases, II or V (**Figure 7.65**). The 3-hydroxyl or 6-hydroxyl group of the galactosyl residues can be modified by any of 20 different human α-sialyltransferases. There are 11 different fucosyltransferases in the human genome; they selectively add α-fucosyl groups to the 2-hydroxyl group of β-galactosyl residues, whereas other fucosyltransferases selectively add α-fucosyl groups to the 3-hydroxyl or 4-hydroxyl group of β-GlcNAc residues. More than 100 different glycosyltransferases are known to be expressed in Golgi compartments of different human cells. Most of the late-stage glycosylation reactions are performed by glycosyltransferases that recognize just a few monosaccharide subunits in the substrate rather than the full-length oligosaccharide. Thus a glycosyltransferase might recognize a GlcNAc residue in the acceptor, but the enzyme is unable to tell whether the acceptor residue is attached to a glycolipid or a glycoprotein. For example, β1–4 galactosyltransferase V targets GlcNAc residues in both glycolipids and proteins. Similarly, α1–2 fucosyltransferase II adds fucosyl groups to galactosyl residues on both glycolipids and glycoproteins.

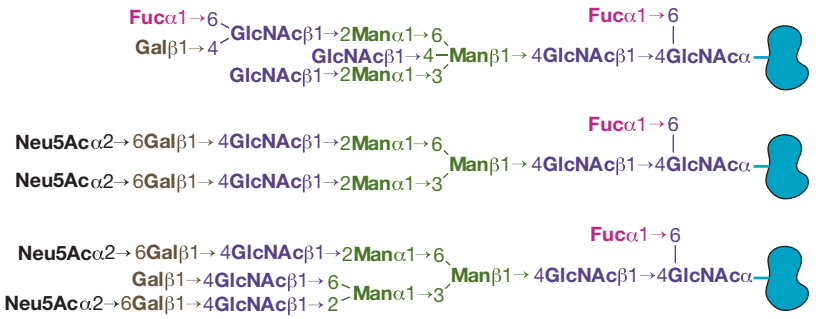



**Figure 7.65 Examples of complex N-glycans.** Protein glycans are further elaborated one monosaccharide at a time.

The time spent in transit through the Golgi network is limited, and there is not sufficient time for all of the potential glycosylation reactions to reach completion. As a result, N-linked glycans are always produced as heterogeneous combinatorial mixtures, making it difficult to purify them, characterize them, or even accurately represent the structures on paper. To understand the degree of heterogeneity in glycosylated proteins it is useful to look at an example of the glycans that are present on just one type of protein. When the soluble form of intracellular adhesion molecule-1 (sICAM-1) is expressed in Chinese hamster ovary (CHO) cells it is produced as a mixture of N-glycans, most with a complex $Man_3GlcNAc_2$ core (**Figure 7.66**). Increasingly, glycobiologists are turning to a symbolic representation system, using colored shapes to represent the individual monosaccharide subunits. The symbolic representation

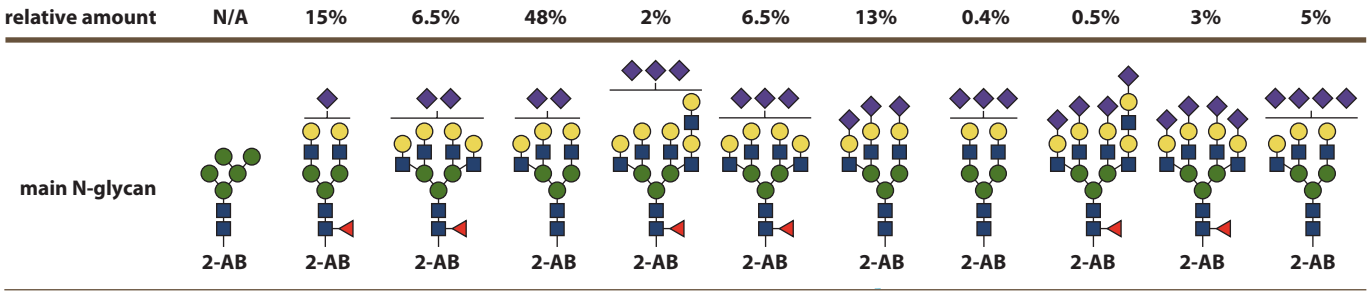



**Figure 7.66 Heterogeneity**. In order to reveal the wide range of glycan structures on a single type of protein, glycans were enzymatically cleaved from the expressed glycoprotein sICAM-1 and then fluorescently labeled with 2-aminobenzamide (2-AB) to facilitate chromatographic analysis. The connectivity of the monosaccharide subunits in each glycan was determined by mass spectrometry ( blue square, GlcNAc; green circle, mannose; red triangle, fucose; yellow circle, galactose; purple diamond, *N*-acetylneuraminic acid).

system makes it much easier to compare the structures of oligosaccharides. However, the absence of explicit atoms and bonds deprives chemists of potentially useful information.

## A few human proteins are C-mannosylated on tryptophan residues

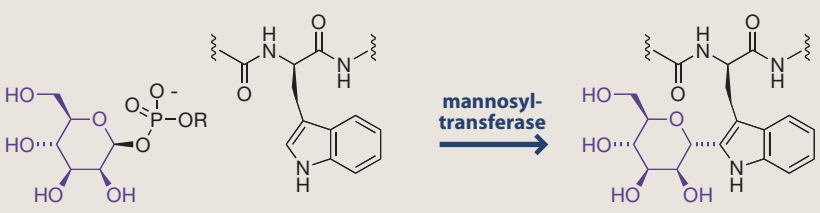
Surprisingly, a few secreted human proteins have been found to be post-translationally glycosylated on tryptophan residues within the endoplasmic reticulum. The modification is an α-mannosyl group attached to the 2-position of the indole ring. Ribonuclease 2 and interleukin-12 are C-mannosylated on the first tryptophan in a WXXW tetrapeptide motif. Several proteins in the complement system have been found to be C-mannosylated enzymatically. The generic name of the complement system belies its ruthless lethality toward foreign cells that have been targeted for destruction. The complement system is a set of proteolytically activated proteins that self-assemble into gaping channels in the membranes of pathogenic organisms. These channels allow the contents of the cell to leak out. The complement proteins C6, C7, C8α, C8β, and C9 are multiply C-mannosylated: six in C6, four in C7, C8α, and C8β, and two in C9. All of these proteins are glycosylated on WXXWXXW sequences in a common thrombospondin type I repeat domain; however, other tryptophan residues that are not part of this domain are also C-mannosylated.

## Glycosylation of proteins sometimes, but not always, affects the intrinsic function of the protein

Most membrane-bound proteins that traffic through the endoplasmic reticulum and Golgi are N-glycosylated. In Chapter 9 we will discuss a variety of receptor proteins that are important for signal transduction. For these receptors glycosylation is sometimes, but not always, important for ligand binding. For example, the granulocyte/macrophage colony-stimulating factor (GM-CSF) receptor requires intact N-glycosylation for high-affinity binding to the ligand. However, the glycosyl groups of the vascular endothelial growth factor (VEGF) receptor are not essential for high-affinity binding to VEGF. Epidermal growth factor receptor, a dimerizing receptor tyrosine kinase, has multiple N-glycosylation sites. The Asn240 glycosylation site is particularly important for transducing binding events into intracellular signals, but not for ligand

### Problem 7.11

Suggest a plausible arrow-pushing mechanism for the enzymatic transfer of a mannosyl residue from β-mannosyl dolichol phosphate to tryptophan.



binding. The N240Q mutant of the EGF receptor, which lacks glycosylation at position 240, undergoes spontaneous autophosphorylation even in the absence of the ligand; a mutation with this phenotype would be termed constitutively active.

More than 95% of 7TM receptors are glycosylated. About two-thirds of the 7TM receptors that have been analyzed have glycosylation sites in the second extracellular loop; the remaining one-third have glycosylation sites in the first or third extracellular loop. The role of these oligosaccharides is still a subject of investigation. In many cases, the oligosaccharides on cell-surface receptors appear to contribute to membrane localization and ligand binding. For example, when two of the three glycosylation sites (Asn4 and Asn188) of the angiotensin II receptor were mutated to non-glycosylatable threonine residues, most of the membrane expression and ligand affinity of the receptors were retained. However, mutation of the third glycosylation site Asn176 led to a significant reduction in both membrane expression of the receptor and affinity of the receptor for the peptide ligand.

During a viral infection, human cells fragment viral proteins and present the peptide fragments bound to major histocompatibility complex (MHC) receptors on the cell surface for recognition by T cells. A Braille-like communication takes place between the two cells, as proteins touch each other. Productive binding results from protein–protein molecular recognition at the interface between the two cells, termed the immunological synapse. Through this mechanism, killer T cells can distinguish between friendly cells and enemy cells, and between healthy cells and infected cells. A key interaction between the antigen-presenting cell and the cytotoxic T lymphocyte is mediated by the B7-1 protein of the antigen-presenting cell and the cytotoxic T-lymphocyte-associated antigen 4 (CTLA-4) (**Figure 7.67** and **Figure 7.68**). Both proteins are heavily glycosylated (CTLA-4 has two glycosylation sites, and B7-1 has ten). However, the carbohydrates do not make key interactions responsible for specificity. Instead, protein–protein recognition triggers a T-cell response. Although the glycan moieties on CTLA-4 do not make contacts with B7-1, they are still important. The drug abatacept, which is a dimeric form of an antibody Fc domain fused to CTLA-4, has a longer half-life when produced in CHO cells than in other cell lines, primarily because of the presence of terminal Neu5Ac residues.

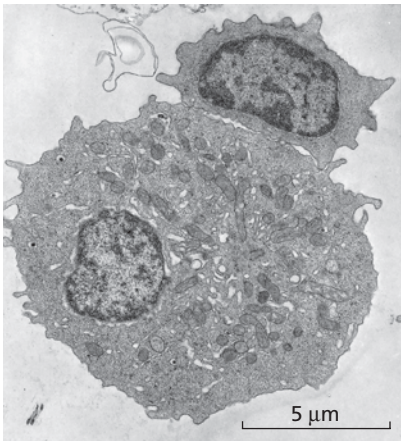



**Figure 7.67 An intimate chemical conversation.** Cytotoxic T lymphocyte (top) interacting with an antigen-presenting cell (bottom). (From B. Alberts et al., Molecular Biology of the Cell, 5th ed. New York: Garland Science, 2008.)

## Most extracellular signaling proteins are glycosylated with oligosaccharides

The majority of soluble signaling proteins are glycosylated. Most of the cytokine signaling proteins that we discuss in Chapter 9 (such as interleukins, interferons,

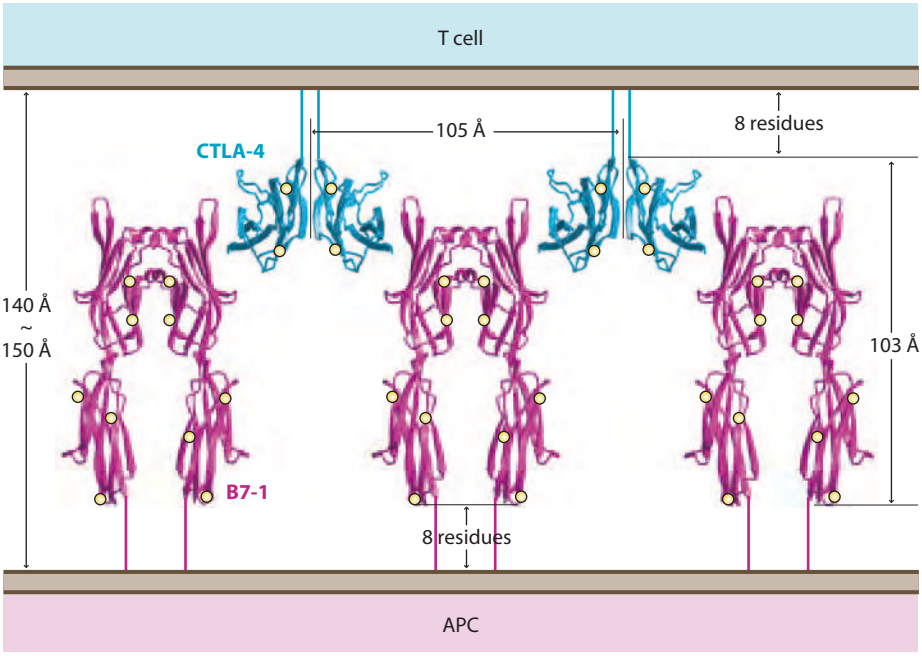



**Figure 7.68 Sweet greeting.** The recognition interface between a T cell and an antigen-presenting cell (APC) has many glycosylation sites (yellow), but none are important for binding. (Courtesy of S. Ikemizu and S.J. Davis.)

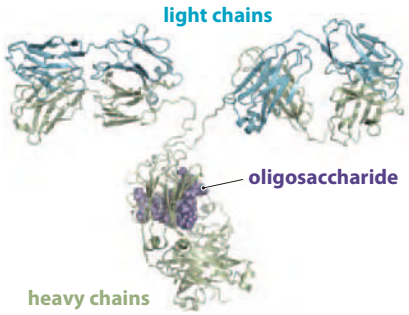



**Figure 7.69 An inside out glycoprotein.** The oligosaccharides of most glycoproteins are attached to the outside, but the opposite is true for antibodies. In this crystal structure of a full-length type G immunoglobulin the oligosaccharide is clearly embedded deeply within the constant region of the heavy chains: two heavy chains (green), two light chains (blue), and two N-linked oligosaccharides (purple). The sialic acid residues are missing from the oligosaccharide. (PDB 1IGT)

and colony-stimulating factors) are glycosylated, which extends the half-life of the cytokine in the blood. Such glycosylation usually has no effect on protein binding. With respect to serum half-life, the structure of the oligosaccharide fusion matters. For example, glycoproteins stripped of Neu5Ac residues to reveal terminal galactosyl residues (asialoglycoproteins) are taken up by the liver within minutes.

In some cases the oligosaccharide is also important for function. For example, β-human chorionic gonadotropin (Pregnyl®) induces ovulation. When deglycosylated, the protein binds to its cognate receptor with similar affinity, but the binding fails to stimulate adenylate cyclase activity inside the target cell. The most common antibodies, gamma immunoglobulins (IgG), are glycosylated on Asn297 of the heavy chain fragment (**Figure 7.69**). The oligosaccharides are present in various related forms (**Figure 7.70**). Surprisingly, although the added oligosaccharides are heterogeneous, the addition of the carbohydrates is still important for the function of the protein. Glycosylation is essential for binding to immunoglobulin Fc receptors and to the C1q complement protein that leads to lysis of foreign cells.

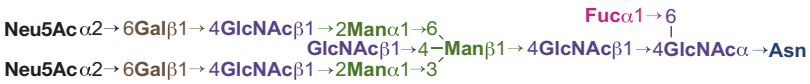



**Figure 7.70 Sequence of an IgG oligosaccharide.** Truncated forms of this oligosaccharide are also observed.

Aldehydes condense with acyl hydrazides to form stable hydrazone linkages, even in water. A common method for creating aldehyde functional groups is to cleave oligosaccharides oxidatively with sodium periodate (pronounced per-iodate) (**Figure 7.71**). Because the cleavage reaction involves a cyclic five-membered ring periodate ester, the vicinal hydroxyl groups must be able to adopt a *syn* orientation. Acyclic 1,2-diols can adopt an appropriate conformation through rotation about a carbon-carbon bond. However, cyclic 1,2-diols will only react if the hydroxyl groups are on the same face of the ring (*syn*); cyclic *anti* 1,2-diols are much less susceptible to cleavage by periodate.

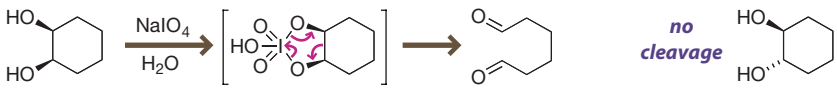



**Figure 7.71 Periodate cleavage.** Sodium periodate ($NaIO_4$) generates potentially reactive aldehyde functional groups from *syn* 1,2-diols.

## Problem 7.12

Which common monosaccharide subunits in a typical IgG oligosaccharide (Figure 7.70) would you expect to readily generate reactive aldehydes after treatment with sodium periodate ($NaIO_4$), a reaction shown in Figure 7.71?

## Many protein pharmaceuticals are glycosylated

Most glycoprotein pharmaceuticals, such as Aranesp®, are produced in CHO cells, a stable mammalian cell line that can be grown on a large scale. Mammalian cells are more costly to grow than bacteria, but mammalian cells provide the post-translational enzymes and compartmentalized environments that are essential for the efficient production of many human proteins. However, the glycosylation patterns that result from expression in CHO cells, or any heterologous (nonhuman) cell line, are not necessarily the same as those produced in the native, human cell. For example, when the T-cell antigen CTLA-4 is expressed in CHO cells, an extra Asn glycosylation occurs. Avonex® (β-interferon 1a) is produced as a glycoprotein in CHO cells, whereas Betaseron® (β-interferon 1b) is produced in *Escherichia coli* without glycosylation. Avonex® has been reported to be more than 10 times as potent as Betaseron®, mainly

as a result of the presence of an N-linked oligosaccharide rather than the small differences in sequence between the two pharmaceuticals. The carbohydrate group of Avonex® confers greater resistance to protein aggregation. The glycosylation patterns of proteins expressed in cultured cells such as CHO cells are not necessarily the same as those found in proteins isolated from humans. For example, the interferon β1, isolated from humans, is a mixture of four sialylated triantennary and tetraantennary glycoproteins (**Figure 7.72**). CHO cells produce the same mixture of glycans, but in slightly different ratios. Fortunately, these differences do not seem to have adverse effects on the therapeutic benefit. Significant advances have been made in the engineering of yeast (*Saccharomyces cerevisiae* and *Pichia pastoris*) cell lines that can produce glycoproteins. Through screening it is possible to identify cell lines that produce essentially homogeneous glycoforms of biologically active erythropoietin, although not the major glycoform produced in humans. There is every reason to expect that yeast will become a major source of glycoprotein pharmaceuticals in the future.

| | human | CHO |
|---|---|---|
| (Neu5Acα2→3)Galβ1→4GlcNAcβ1→2Manα1→6<br>(Neu5Acα2→3)Galβ1→4GlcNAcβ1→2Manα1→3 Manβ1→ | 74% | 68% |
| (Neu5Acα2→3)Galβ1→4GlcNAcβ1→2Manα1→6<br>(Neu5Acα2→3)Galβ1→4GlcNAcβ1→4<br>(Neu5Acα2→3)Galβ1→4GlcNAcβ1→2 Manβ1→3 Manβ1→ | 10% | 0% |
| (Neu5Acα2→3)Galβ1→4GlcNAcβ1→4<br>(Neu5Acα2→3)Galβ1→4GlcNAcβ1→2 Manβ1→6<br>(Neu5Acα2→3)Galβ1→4GlcNAcβ1→2Manα1→3 Manβ1→ | 8% | 27% |
| (Neu5Acα2→3)Galβ1→4GlcNAcβ1→2Manα1→6<br>(Neu5Acα2→3)Galβ1→4GlcNAcβ1→3 Galβ1→4GlcNAcβ1→2Manα1→3 Manβ1→ | 8% | 0% |
| (Neu5Acα2→3Galβ1→4GlcNAcβ1→3)Galβ1→4GlcNAcβ1→4<br>(Neu5Acα2→3Galβ1→4GlcNAcβ1→3)Galβ1→4GlcNAcβ1→2 Manβ1→6<br>(Neu5Acα2→3Galβ1→4GlcNAcβ1→3)Galβ1→4GlcNAcβ1→2Manα1→3 Manβ1→ | 0% | 4% |

**Figure 7.72 Different glycoforms from different cells.** The glycoforms of human interferon β1 expressed in CHO cells are slightly different from the glycoforms isolated from humans. Each of the N-glycans shown is attached to interferon β1 through 4GlcNAcβ1–4GlcNAcα-Asn (not shown) with varying amounts of α-fucosylation on the first GlcNAc. The outer branches exhibit varying degrees of sialylation with Neu5Ac on the 3- and 6-positions of the terminal Gal residues.

## Cell–cell recognition is often mediated by glycoproteins

The spikes that project from an infectious HIV virion bind selectively to helper T cells (**Figure 7.73**). These spikes are composed of trimers of a heavily glycosylated protein gp120, which undergoes a drastic rearrangement when the spikes bind to the CD4 receptor of T cells (**Figure 7.74**). HIV has evolved several different tricks to disguise this unique spike glycoprotein from the human immune system. Rapid mutation is the chief mechanism. After infection with HIV, the viral population rapidly diversifies, and

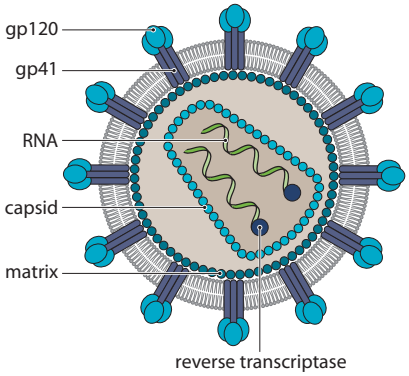



**Figure 7.73 Structure of the HIV virion.**

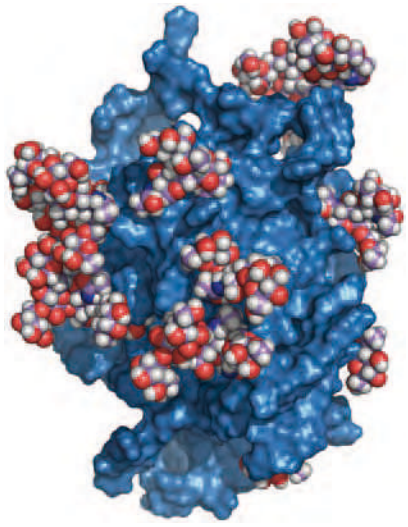


**Figure 7.74 Many glycosylation sites.** A crystal structure of gp120 from simian immunodeficiency virus, a monkey homolog of HIV, shows the extent of glycosylation. (PDB 2BF1)

viruses that are no longer recognized by neutralizing antibodies emerge. When these mutations were analyzed, the sequences often corresponded to changes in N-glycosylation patterns.

## Introduction of N-glycosylation sites can improve protein pharmaceuticals

The protein pharmaceutical erythropoietin is a cytokine that induces the proliferation of red blood cells. More red blood cells translate into more oxygen intake, a key parameter for wound healing and aerobic sports. In the late 1990s, the pharmaceutical erythropoietin was used by athletes to gain a competitive advantage over rivals. Many organizations now test athletes for recombinant erythropoietin. Eythropoietin is a simple four-helix bundle protein with three N-glycosylation sites and one O-glycosylation site; 40% of the total mass is carbohydrate. Ironically, glycosylation of erythropoietin decreases the affinity of the protein for its receptor by a factor of 10. However, the carbohydrate increases serum half-life. The mixture of N-linked oligosaccharides on recombinant human erythropoietin expressed in CHO cells is similar to the pattern found on human interferon β1 (**Figure 7.75**); note the presence of Gal-GlcNAc disaccharide repeats, sometimes abbreviated as LacNAc because the disaccharide is Galβ1–4Glc.

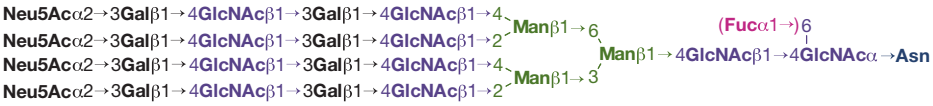



**Figure 7.75 Glycosylation of Epogen™.** The most abundant oligosaccharide on erythropoietin expressed in CHO cells.

The O-linked oligosaccharide on Ser126 of erythropoietin is a mixture of O-glycosides: Neu5Acα2–3Galβ1–4GalNAc and Neu5Acα2–6[Neu5Acα2–3]Galβ1–4GalNAc. When the terminal Neu5Ac residues (up to 14 of them) of erythropoietin are removed to reveal terminal galactosyl residues, the modified erythropoietin is rapidly cleared. An improved version of erythropoietin, sold as Aranesp®, has been engineered to include two additional AsnXxxSer N-glycosylation sites. The additional glycosylation results in slightly weaker binding to the erythropoietin receptor, but the weaker binding is more than made up for by a threefold increase in serum half-life.

N-linked oligosaccharides can be selectively released from proteins by peptide: N-glycosidase F (PNGase F) as long as the asparagine-linked GlcNAc lacks a Fucα1–3 group. The enzyme cleaves the side-chain amide of asparagine and does not affect O-linked oligosaccharides. O-linked oligosaccharides can be selectively cleaved with an alkaline solution of sodium borohydride.

## Modified sugars can carry reactive groups through the glycoprotein biosynthesis pathway

In human cells, monosaccharides are synthesized through aldol reactions catalyzed by class I aldolases. The mechanism of class I aldolases involves the formation of a nucleophilic enamine intermediate that adds to an aldehyde carbonyl group (**Figure 7.76**). Enamines are more nucleophilic than enols, but less nucleophilic than

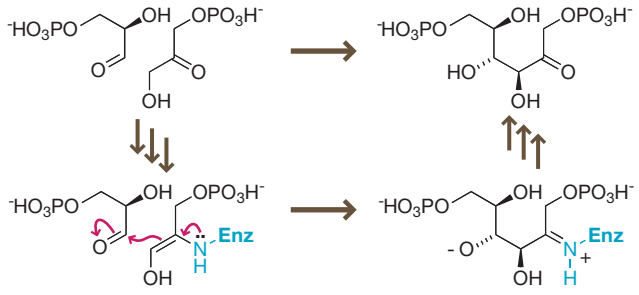



**Figure 7.76 Aldolase mechanism.** The biosynthesis of monosaccharides by class I aldolases involves enamines.

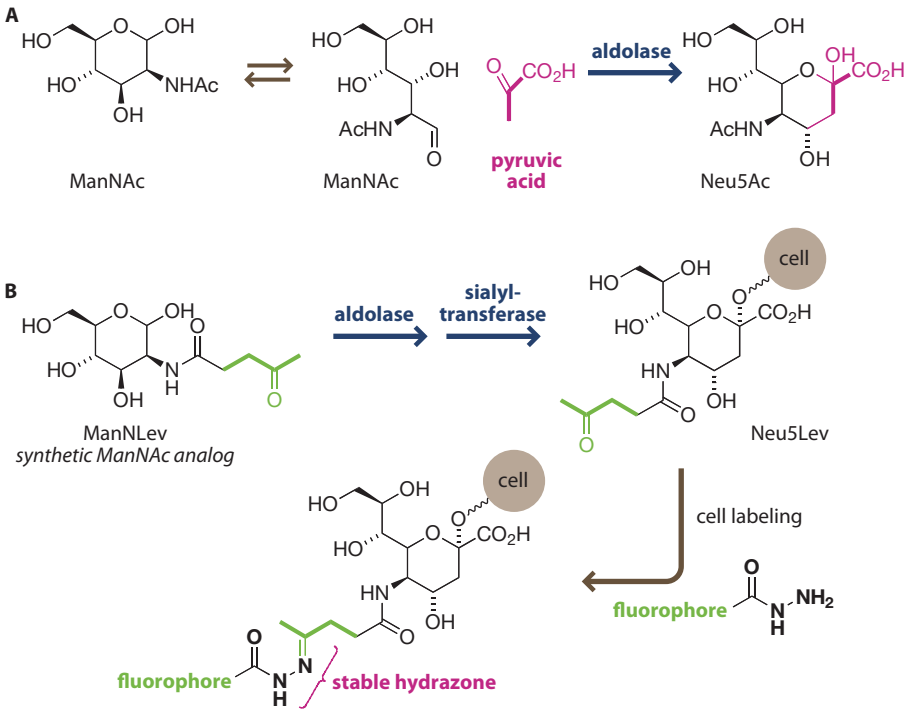


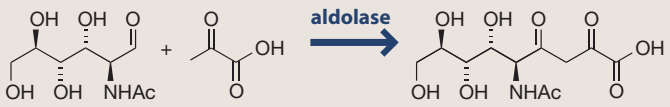

**Figure 7.77 Metabolic incorporation of reactive ketone groups.** (A) Biosynthesis of *N*-acetylneuraminic acid (Neu5Ac) from ManNAc. (B) Incorporation of levulinamide groups into cell-surface glycoproteins.

enolates. Fungi and bacteria also have class II aldolases that catalyze the addition of a zinc enolate to an aldehyde carbonyl group.

In human cells, *N*-acetylneuraminic acid is biosynthesized through an aldol reaction of a two-carbon fragment to *N*-acetylmannosamine (**Figure 7.77**). The enzymes that catalyze *N*-acetylneuraminic acid biosynthesis and glycosyl transfer are tolerant enough to allow additional substituents on the *N*-acetyl group. For example, when a synthetic ManNAc analog, ManNLev, is taken up by cells, it is converted first into a neuraminic acid derivative Neu5Lev by an aldolase and then into an activated CMP derivative. The CMP-Neu5Lev serves as a substrate for sialyltransferases, which incorporate the Neu5Ac analog into N-glycans, including N-glycans on membrane-bound cell-surface proteins. Free ketones are more reactive as electrophiles than esters or amides are. Like aldehydes generated through periodate cleavage of glycans, ketones can be selectively condensed with hydrazine derivatives in water to generate stable hydrazone linkages. Thus, after cells are allowed to display the levulinic derivatives of Neu5Ac on their surface, the free ketone groups can be selectively conjugated to biotin, fluorophores, or other synthetic molecules with acylhydrazide functional groups. This powerful approach developed by Bertozzi and coworkers allows the specific chemical decoration of cell-surface glycans.

## Problem 7.13

Lys165 has a key role in the mechanism of sialic acid aldolase. Suggest a plausible arrow-pushing mechanism for the transformation of ManNAc to Neu5Ac by sialic acid aldolase.




Cells displaying fluorescently labeled oligosaccharides can be identified by using **flow cytometry**. Flow cytometers use ultra-fast electronics to measure the fluorescence of cells in a hydrodynamically focused stream of medium containing cells. A typical flow cytometer can easily measure the fluorescence of tens of thousands of cells per second. When fluorescence intensity is plotted against the number of cells, it

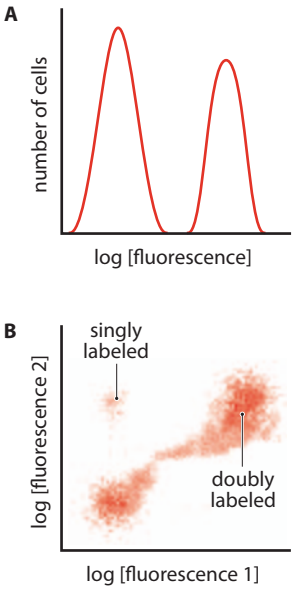



**Figure 7.78 Flow cytometry.** (A) Flow cytometry can distinguish between populations of cells with low fluorescence and high fluorescence. (B) Two-dimensional flow cytometry allows more than one type of fluorescent label to be read on each cell.

is easy to distinguish labeled cells from unlabeled cells (**Figure 7.78**). The technique is applicable to cells stained with specific antibodies (as long as the antibodies carry a fluorescent tag), autofluorescent proteins such as green fluorescent protein, or cells labeled with fluorogenic agents. Many flow cytometers can detect more than one fluorescence emission wavelength, and cell sorters use similar principles to separate cells into distinct populations based on the fluorescence of the cell (for example, fluorophore-labeled cells can be isolated from a mixture of cells).

Fluorescence-activated cell sorting (FACS) is based on the principle of flow cytometry (**Figure 7.79**). If tiny droplets are charged according to the presence or absence of a fluorescence signal, they can be diverted electrostatically for collection. For example, a solution of white blood cells can be labeled with a fluorescently labeled antibody that is specific for one type of B cell; the fluorescently labeled B cells can then be separated from the unlabeled cells by using a FACS sorter.

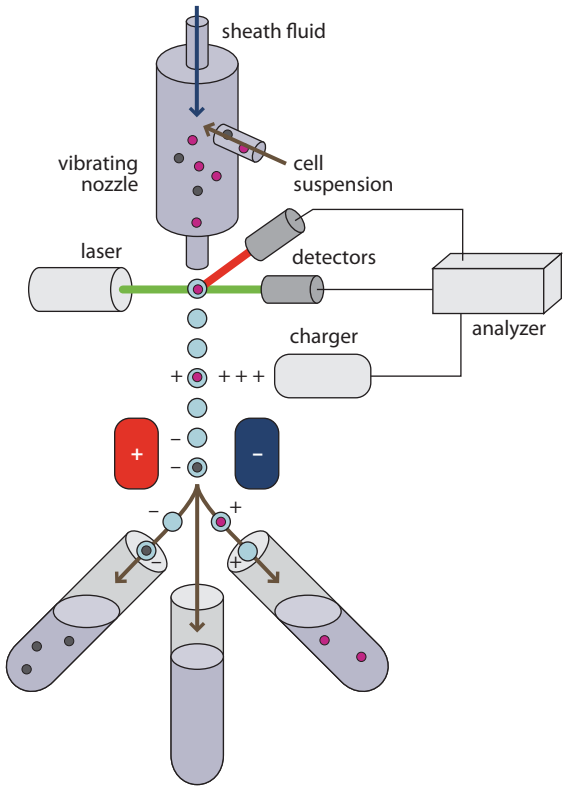



**Figure 7.79 FACS.** Diagram of a fluorescence-activated cell sorter. (Adapted from B. Alberts et al., Molecular Biology of the Cell, 5th ed. New York: Garland Science, 2008.)

## 7.5 GLYCOLIPIDS

### Glycosphingolipids are lipid-like glycoconjugates

Some cell-surface glycans are attached to lipids rather than membrane-bound proteins. **Glycosphingolipids** are oligosaccharides attached to the lipid ceramide. There are two classes of glycosphingolipids. Galactocerebrosides are found primarily in neural tissue, whereas glucocerebrosides are more widely distributed (**Figure 7.80**). A deficiency in a specific glycosidase, glucocerebrosidase, leads to the accumulation of fatty glucocerebrosides in various tissues, a condition known as Gaucher's disease.

The hemagglutinin protein of the influenza virus recognizes a particular glucocerebroside called a ganglioside. Gangliosides are glucocerebrosides with the tetrasaccharide Galβ1–3GalNAcβ1–4Galβ1–4Glcβ1 bound to ceramide. Gangliosides usually have one or more Neu5Ac residues on the oligosaccharide.

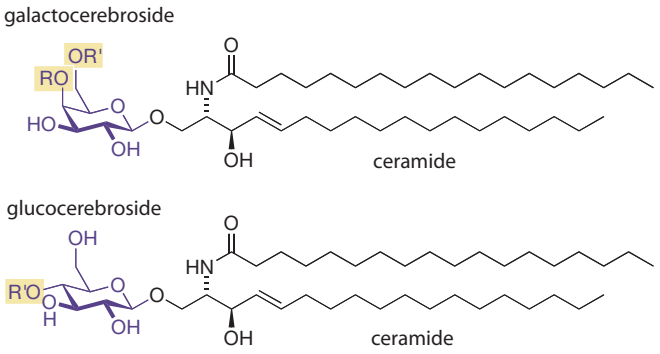



**Figure 7.80 Cerebrosides.** Cerebrosides are an important class of glycosphingolipids in brain tissue. The ceramide moieties anchor the glycans to cell membranes. In galactocerebrosides, glycan chains are attached to either the 4- or 6-hydroxyl of the galactosyl residue. In glucocerebrosides, glycan chains are attached to the 4-hydroxyl of the glucosyl residue.

Globosides are glucocerebrosides with a GalNAcβ1–3Galα1–4Galβ1–4Glcβ1 tetrasaccharide bound to ceramide. The globo H antigen is a hexasaccharide that is heavily expressed on globosides and glycoproteins on the surface of prostate cancer cells. When covalently linked to nonhuman proteins (such as keyhole limpet hemocyanin), the carbohydrate can be used to raise an immune response in humans; there is significant interest in using globo H in cancer vaccines (**Figure 7.81**).

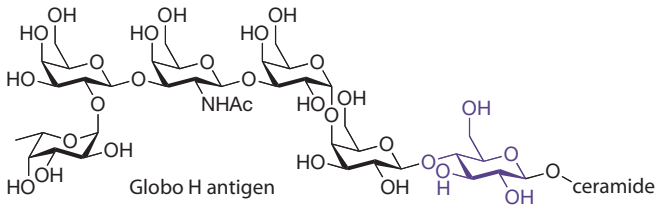



**Figure 7.81 Anticancer carbohydrate vaccines.** A hexasaccharide known as the globo H antigen is characteristic of human cancer cells. Vaccines based on globo H may help the human body fight cancers.

## Glycosylphosphatidylinositols from pathogens are potential vaccines

The human immune system is optimized to respond to pathogens through two unique molecular signatures: their oligosaccharide coats and their polypeptide debris. Sometimes the immune system needs help in learning to recognize the molecular signatures of pathogens. Glycosylphosphatidylinositols (GPIs) serve as anchors that link proteins and carbohydrates to the cell membrane of eukaryotes, but the glycosylated lipids are more common in single-celled eukaryotes than in humans. The core of GPI is the cyclohexane derivative inositol. The inositol of GPIs is linked to a lipid moiety. The GPI core resembles the lipids that make up the cell membrane (**Figure 7.82**).

Less common on human cells, GPI anchors are often distinctive features of infectious microorganisms such as *Toxoplasma gondii*, *Leishmania*, *Trypanosoma brucei*, and *Plasmodium falciparum* (the causative agent of malaria). The pathogen *T. gondii* displays a set of oligosaccharides with a unique conserved GPI core. *T. gondii* is

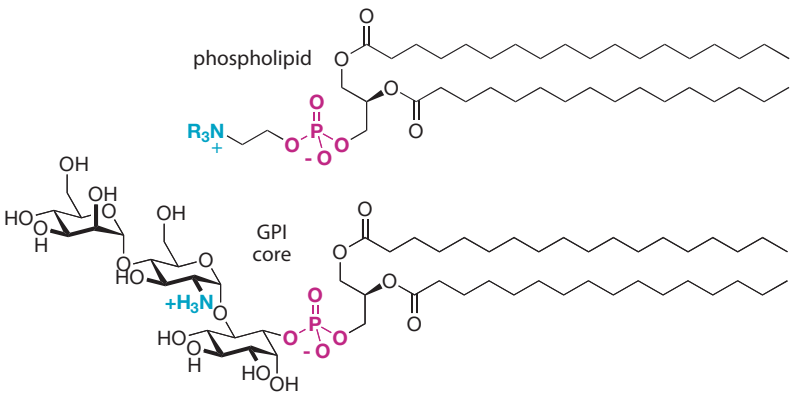



**Figure 7.82 Glycosylphosphatidylinositols.** The zwitterionic glycan moiety of a glycosylphosphatidylinositol (GPI) resembles the polar headgroup of a phospholipid.

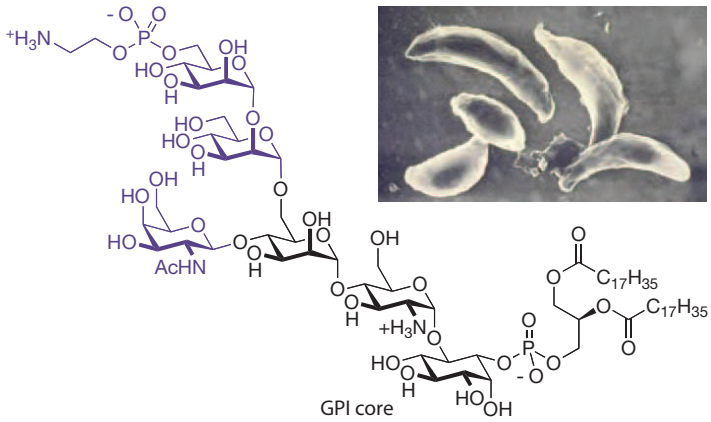



**Figure 7.83 Targeting the glycans in infectious diseases.** Some of the glycosylphosphatidylinositols on the surface of *Toxoplasma gondii* (inset), which causes toxoplasmosis, are unique to the pathogen. These nonhuman glycans can be used as vaccines against the disease. (Inset, courtesy of Michael J. Cuomo. With permission from Public Health Source.)

carried by cats; pregnant women and individuals with a compromised or suppressed immune system are at risk of infection (toxoplasmosis) from cat feces (**Figure 7.83**). In advanced stages of infection, *T. gondii* forms cysts inside human skeletal muscle and brain cells, which are not easily accessible to the immune system. However, the human immune system can mount an initial response toward a GPI oligosaccharide of *T. gondii*, and that oligosaccharide has been synthesized for use as a potential vaccine against toxoplasmosis.

Quite a few human cell-surface proteins are linked to the plasma membrane by GPI anchors. For example, Thy-1 is a GPI-linked protein that is abundant on the surface of human T cells and brain cells (**Figure 7.84**). Thy-1 is translated as a 161 amino acid protein. The GPI anchor is then attached to Thy-1 by transamidation of the ethanolamine group of the GPI at a cysteine residue. The C terminus is cleaved, leaving the GPI covalently linked to Thy-1. The bacterial toxin aerolysin binds to the carbohydrate portion of human GPI anchors, such as those found on Thy-1, with high affinity. As the concentration of aerolysin increases on the surface of the human cell, the aerolysin molecules form a heptameric pore in the cell membrane, killing the cell.

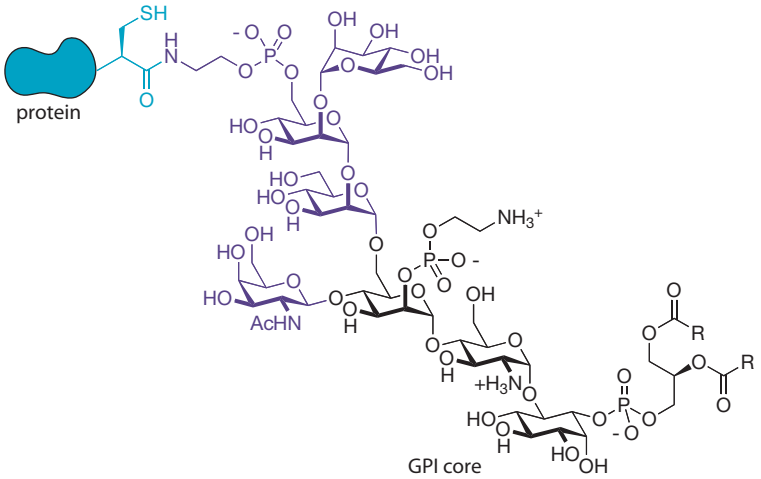



**Figure 7.84 A GPI anchor.** The glycoprotein Thy-1 is anchored to the membrane of T cells by a glycosylphosphatidylinositol.

## 7.6 GLYCOSYLATION IN THE CYTOSOL

### O-glycosylation of proteins in the cytosol with β-GlcNAc is analogous to phosphorylation

In human cells, virtually all enzymatic glycation occurs in the endoplasmic reticulum and Golgi network on molecules destined for the extracellular environment.

One type of protein glycosylation occurs in the cytosol of eukaryotic cells: the addition of GlcNAc to specific serine and threonine residues. The glycosylation of serine and threonine residues by GlcNAc bears a strong functional analogy to protein phosphorylation. For example, the modification is highly specific. Indeed, many of the glycosylated residues are also sites of protein phosphorylation. Phosphorylation and glycosylation are mutually exclusive modification reactions. Thus, the actual structure of the *O*-GlcNAc residue may be less important for protein function than the fact that it prevents the addition of a phosphate group. Phosphorylation of proteins is controlled by many enzymes, but glycosylation of intracellular proteins is mediated by just one enzyme, UDP-*O*-GlcNAc transferase, using the substrate UDP-GlcNAc (**Figure 7.85**). Dephosphorylation of proteins is controlled by several phosphatases, but the hydrolytic removal of *O*-GlcNAc groups from proteins is mediated by just one enzyme: *O*-GlcNAcase.

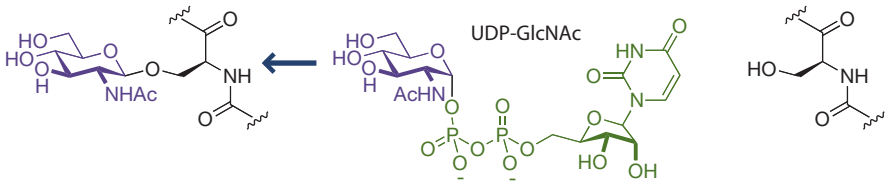



**Figure 7.85 Glycation within the human cytosol.** *O*-GlcNAc groups are transferred from UDP-GlcNAc to serine sidechains.

Nuclear pore proteins are extensively modified with *O*-GlcNAc residues. Other proteins known to be modified include transcription factors, polymerases, RNA-binding proteins, kinases, and cytoskeletal proteins. Glycosylation with *O*-GlcNAc tends to correlate with extracellular glucose concentrations. Thus, *O*-GlcNAc modification may be a cellwide signal to ramp up the metabolism and storage of glucose.

A fluorescence polarization assay has been developed to identify chemical inhibitors of *O*-GlcNAc transferase (OGT) that could compete with UDP-GlcNAc (**Figure 7.86**). When fluorophores are excited with light, it takes a while ($10^{-7}$–$10^{-9}$ second) for them to relax to the ground state and emit a photon. When fluorescent molecules are irradiated with polarized light, they emit light with the same polarization, as long as they do not change orientation. Small molecules rapidly change orientation in solution by tumbling, but small molecules bound to macromolecules such as DNA or proteins tumble slowly. With the use of fluorescence polarization it is easy to tell when fluorophores are bound to macromolecules or free in solution. In the absence of a suitable glycosyl acceptor, UDP-GlcNAc forms a stable complex with OGT. A fluorescein derivative was synthesized and shown to bind to the transferase enzyme in the same way as unmodified UDP-GlcNAc. A fluorescence polarization assay was used to identify inhibitors that could displace the protein-bound fluoresceinylated UDP-GlcNAc derivative.

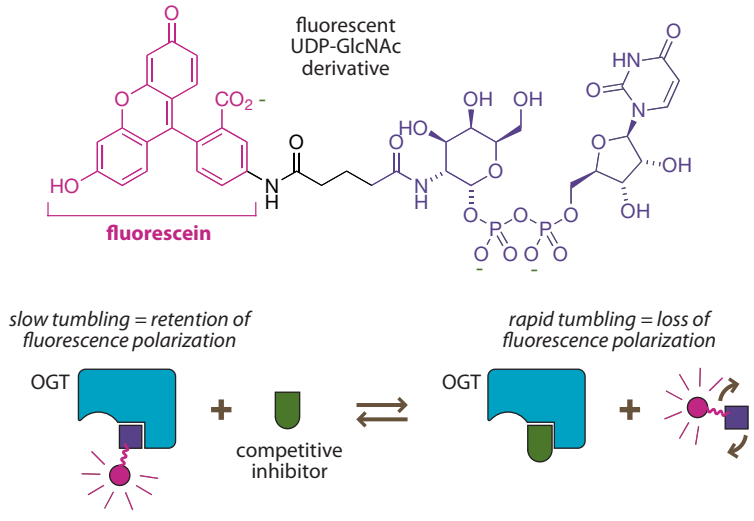



**Figure 7.86 Fluorescent *O*-GlcNAc probe.** Emission of polarized fluorescent light is greater when molecules tumble slowly. A fluorescence polarization assay was developed to screen for compounds that could prevent the binding of UDP-GlcNAc.

### Drugs are targeted for export by glucuronidation

The human liver is a major processing center for the elimination of foreign molecules. Recall that the liver, for example, is rich in cytochrome P450 (CYP) enzymes that catalyze the oxidation of benzo[*a*]pyrene, aflatoxin, cyclophosphamide, and other xenobiotics (that is, foreign molecules). The liver is also rich in UDP-glucuronosyltransferases (UGTs) that transfer β-D-glucuronic acid groups to various compounds, a process that targets them for excretion. These enzymes transfer glucuronic acid to various nucleophilic functional groups, but hydroxyl groups and carboxylic acids are the most common substrates. Some endogenous alcohols, such as estradiol, testosterone, and bile acids, are good substrates. Lipophilic carboxylic acids such as retinoic acid are also glucuronidated. Glucuronidation is a major clearance mechanism for many drugs. For example, the phenolic hydroxyls of morphine and acetominophen (paracetamol) are glucuronidated (**Figure 7.87**). Similarly, the carboxylates of aspirin, ibuprofen, and naproxen are glucuronidated. Polar, anionic compounds are more rapidly processed through the kidneys than lipophilic compounds, so glucuronidated compounds tend to end up in the urine.

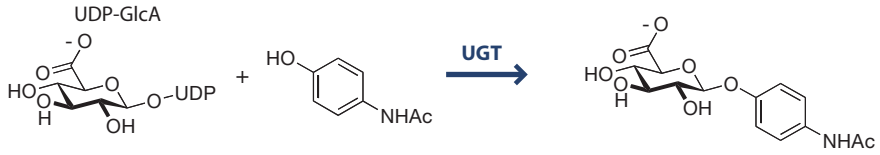



**Figure 7.87 Glucuronidation.** Drugs with phenolic groups, like acetominophen (also known as paracetamol) are targeted for cellular export by glucuronidation.

**Problem 7.14**

Find the structures of ibuprofen, naproxen, and morphine and draw the structure of the glucuronide derivatives that would be generated by UDP-glucuronosyltransferase.

There are at least 13 UDP-glucuronosyltransferases in the human genome, and the selectivity of each isoform is different. For example, morphine is rapidly glucuronidated by UGT 2B7, slowly glucuronidated by UGTs 1A3 and 1A8, and not glucuronidated at all by other UGTs. Some UGTs are expressed in other tissues instead of the liver. For example, UGT 1A7 is expressed in the stomach, UGT 1A8 is expressed in the small intestine and in the colon, but neither of these UGTs are expressed in the liver.

There are related mechanisms for the clearance of unwanted compounds. For example, *O*-sulfation of steroids facilitates excretion. A related mechanism for the clearance of electrophilic drugs is the conjugation with glutathione by glutathione S-transferase.

## 7.7 CHEMICAL SYNTHESIS OF OLIGOSACCHARIDES

### Anomeric stereochemistry is controlled by the anomeric leaving group and the 2 substituent

The chemical synthesis of oligosaccharides is not yet fully developed, in contrast with the synthesis of oligonucleotides or peptides, in which it is common for the average yield per step to exceed 98%. Oligosaccharides are typically synthesized through $S_N1$ reactions of selectively protected monosaccharides. It is usually important to match the stereochemistry of the anomeric leaving group with the identity of the 2′ substituent so as to exert good control over the resulting stereochemistry of the anomeric carbon. For typical pyranose sugars with clear equatorial preferences, axial trajectories are preferred in the transition states for both leaving-group ionization and nucleophilic attack. To favor α-linkages, it is best to use a glycosyl donor with a protected

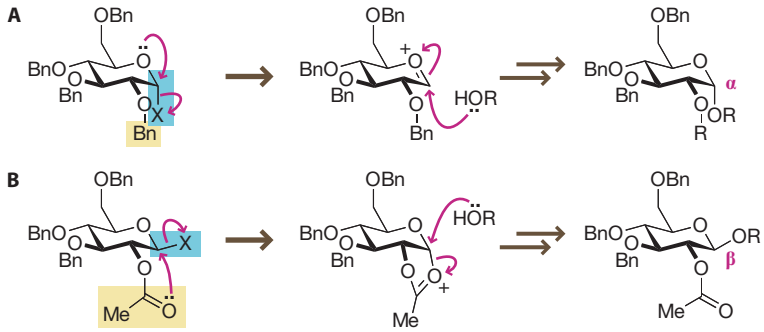



**Figure 7.88 Careful choice of glycosyl donor can lead to good stereocontrol.** (A) A non-participatory protecting group at the 2′ position directs the displacement to proceed through an $S_N1$ reaction, and favors α-linkages. (B) Neighboring-group displacement at the 2′ position and β leaving groups favors β-linkages.

2′ substituent such as benzyloxy that cannot displace the anomeric leaving group (**Figure 7.88**A). Under polar conditions that favor $S_N1$ over $S_N2$ with inversion of configuration, good selectivity of 10–20:1 α/β can be expected. To favor the β anomer, it is best to use a glycosyl donor with a 2′ substituent such as an ester or an amide that can generate a five-membered ring oxonium intermediate (see Figure 7.88B). The nucleophile then attacks, resulting in equatorial selectivity. Such glycosylations usually proceed with high selectivity (more than 20:1 β/α). Interestingly, mechanistic studies of some enzymes that hydrolyze *N*-acetylglucosamine residues have revealed the same type of neighboring-group participation.

Neighboring-group participation can also be used to synthesize α-mannosyl linkages because an axial 2′ nucleophile can push out the axial anomeric leaving group in an analogous manner to the mechanism in **Figure 7.89**. Efficient incorporation of β-mannosyl sugars into complex oligosaccharides on a solid phase represents an unmet synthetic challenge. Because the axial 2′ substituent of a mannose sugar is on top, it cannot push out an equatorial leaving group from the bottom. If the nucleophile pushes out the leaving group directly, it generates the α anomer. If the leaving group ionizes without assistance to generate an oxacarbenium ion, the nucleophile will still prefer an axial trajectory. The β-mannosyl problem seems close to resolution. Mannosyl triflates have been shown to favor β-mannosyl adducts in solid-phase glycosylations as long as the glycosyl donor is protected by a 4,6-*O*-benzylidene protecting group.

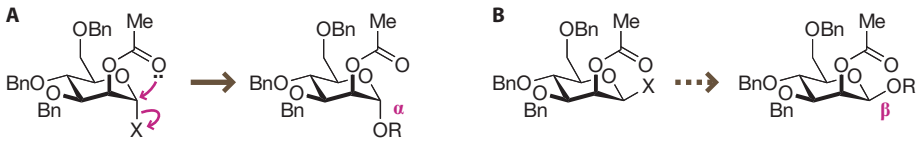
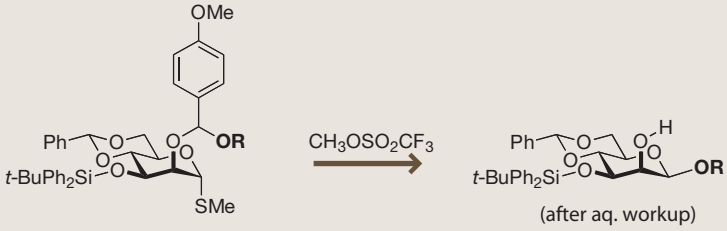



**Figure 7.89 The β-mannosyl problem.** (A) α-Mannosyl linkages are not a problem because neighboring-group participation leads to substitution with retention of anomeric configuration. (B) β-Mannosyl linkages, however, present a big synthetic problem.

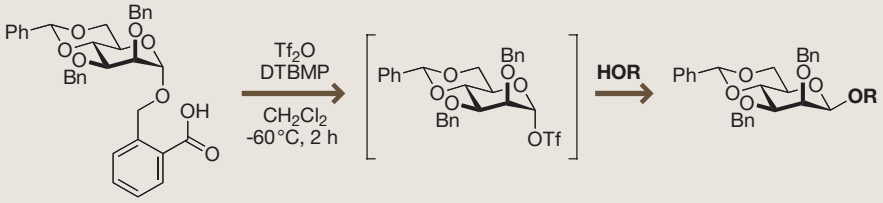
## Problem 7.15

Suggest a plausible arrow-pushing mechanism to explain the formation of the β-mannosyl linkage in the following reaction.



## Problem 7.16

Suggest a plausible arrow-pushing mechanism to explain the formation of the β-mannosyl linkage in the following reaction. DTBMP is a hindered base, 2,6-di-*t*-butyl-4-methylpyridine.




## Modern oligosaccharide synthesis takes advantage of activatable leaving groups

The earliest chemical glycosylation method was developed, shortly on the heels of Emil Fischer's landmark studies of monosaccharides, by his contemporary Wilhelm Koenigs. The Koenigs–Knorr glycosylation method is based on a combination of halide leaving groups and halophilic silver ions. Pyranosyl halides are extremely reactive, even in the absence of silver ions, and are prone to degradation. A wide range of stable glycosyl donors have been developed that are stable until they are activated *in situ* (**Figure 7.90**). The most common examples of activatable anomeric functional groups include fluorides, thioethers, sulfoxides, imidates, phosphates, and *n*-pentenyloxy groups.

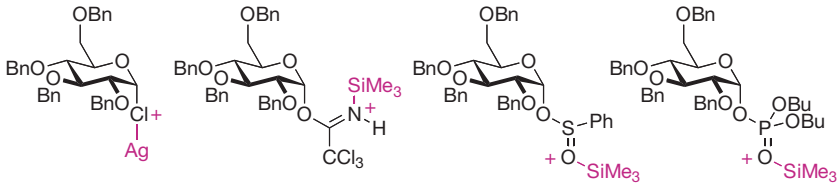
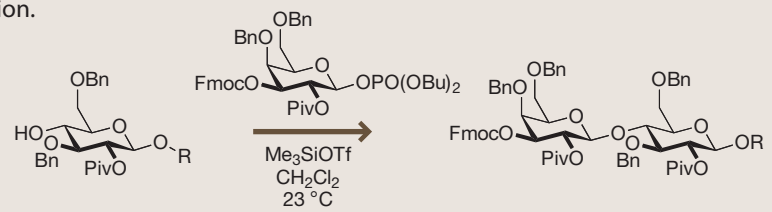



**Figure 7.90 Chemical activation of glycosyl donors.** In chemical glycosylation reactions, leaving groups are activated by reagents that confer a positive charge.

Until recently, glycosyl donor building blocks were only useful to skilled artisans of organic synthesis, and the synthesis of any pure oligosaccharide was a harrowing endeavor. An insatiable demand for precisely defined oligosaccharides has spurred efforts to develop automatable synthetic methods based on solid-phase oligosaccharide synthesis. Coupling reactions of sulfoxides, trichloroacetimidates, and phosphate esters have been optimized for solid-phase synthesis. Koenigs–Knorr glycosylation, and related glycosylation reactions, are unsuitable for solid-phase synthesis because they involve insoluble precipitates. Trichloroacetimidates, sulfoxides, and phosphates are usually activated by silyl trifluoromethanesulfonates (triflates), which ultimately form very stable silicon–oxygen bonds in the by-products. Protic substrates or adventitious water react aggressively with trimethylsilyl triflate to generate triflic acid. In the absence of proton-specific bases (such as 2,6-di-*t*-butylpyridine), the silyl group of silyl triflates drives the reaction thermodynamically, while the proton of triflic acid drives the reaction kinetically.

## Problem 7.17

Suggest a plausible arrow-pushing mechanism for the following glycosylation reaction.



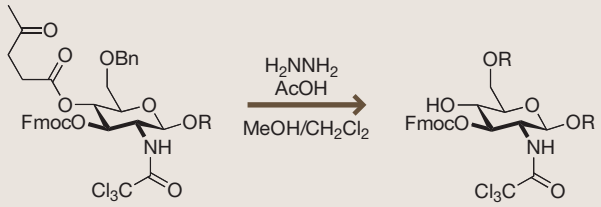
## Synthesis of oligosaccharides still requires a skilled synthetic organic chemist

Monosaccharide building blocks for oligosaccharide synthesis are highly specific. For example, two different classes of galactosyl donors are needed for the addition of either Galα or Galβ residues. For each of these classes of galactosyl donors, four different protecting-group schemes are required to allow the selective deprotection of one of the four hydroxyl groups at the 2, 3, 4, or 6 position. Even more protecting-group combinations are needed for the addition of a branching galactosyl residue. The synthesis of a single selectively protected glycosyl donor requires about 10 steps from the unprotected monosaccharide. Compare this with the synthesis of a typical amino acid building block for peptide synthesis, which takes on average two steps and usually does not require chromatography. As of 2009, few selectively protected monosaccharide building blocks for oligosaccharide synthesis were commercially available. However, as the demand for custom-designed oligosaccharides increases, commercial starting materials may find their way to the market.

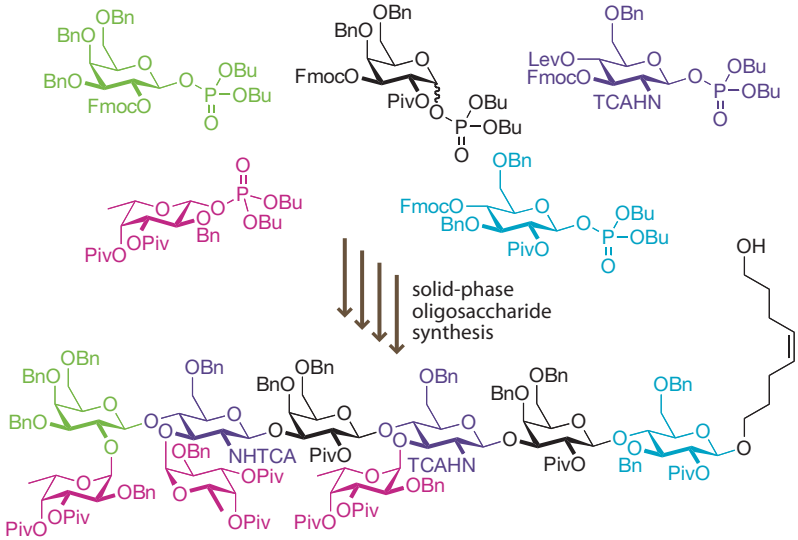
Benzyl protecting groups are usually removed at the end of the synthesis by hydrogenolysis with hydrogen and catalytic Pd-C. The deprotected oligosaccharide is easily removed from the catalyst by filtration, and the hydrogenolysis by-product, toluene, by rotary evaporation. Fmoc groups can be selectively removed from alkoxy substituents in the presence of hindered esters such as pivalates (*t*-BuCOO) by using piperidine.

### Problem 7.18

Levulinoyl groups can be selectively removed by using hydrazine buffered with acetic acid. Suggest a plausible arrow-pushing mechanism for the deprotection, and show the structure of the by-product.




The first automated oligosaccharide synthesizer was designed to use glycosyl phosphate donors. Starting from five monosaccharide building blocks, the Lewis$^y$–Lewis$^x$ nonasaccharide tumor antigen (**Figure 7.91**) was prepared in 23 hours. Each




Piv = *t*-BuCO; Bn = $PhCH_2$; TCA = $Cl_3CCO$; Lev = $CH_3COCH_2CH_2CO$

**Figure 7.91 Solid-phase oligosaccharide synthesis**. Automated synthesis of the branched Lewis$^y$–Lewis$^x$ tumor antigen involves carefully orchestrated reactions with protected glycosyl donors. (Adapted from P.H. Seeberger and D.B. Werz, *Nature Rev. Drug Disc.* 4:751–763, 2005. With permission from Macmillan Publishers Ltd.)

coupling was performed with five equivalents of glycosyl phosphate donor and five equivalents of trimethylsilyl triflate. After washing, the coupling was repeated to effect the glycosylation of any unreacted hydroxyl acceptors. After purification by high-performance liquid chromatography, the protected oligosaccharide was isolated in 6.5% overall yield. The hydroxyolefin tail is usually not removed from the oligosaccharide product. It is a useful linker for conjugation to biomolecules, and it protects the terminal anomeric hydroxyl from side reactions of the open-chain form under basic deprotection conditions. The use of excess reagents and double couplings requires a large amount of the glycosyl donor. However, it is easier to synthesize glycosyl donors than to purify a full-length oligosaccharide from a mixture of incomplete products. For typical applications such as vaccine testing, only a small amount of the final product is necessary.

## 7.8 PROTEINS THAT BIND TO CARBOHYDRATE LIGANDS

### Glycans differentiate the surfaces of human cells

At the molecular level, the language of glycobiology is spoken through carbohydrate–protein interactions. A wide variety of oligosaccharide structures are now known to differentiate different human cells not only within an individual but also at different stages of development of an individual and between different individuals. In some cases, the determinants are distinguished by multivalent lectin proteins and in other cases the determinants are recognized by antibodies, which are bivalent. Richard D. Cummings recently tabulated some of the prominent determinants of identity and compatibility in the human glycome. This impressive list is reproduced in **Table 7.1**, not for memorization but to impress on the reader the great diversity of oligosaccharides that adorn human cells.

### Most carbohydrate-binding proteins are multivalent

To a first-order approximation, monosaccharides bear a strong resemblance to aqueous solvent, so individual interactions between monosaccharides and proteins are usually weak. Most high-affinity carbohydrate–protein interactions are mediated by multivalent interactions. Protein receptors for carbohydrates are referred to as lectins; all known lectins bind carbohydrates through multivalent interactions (**Table 7.2**). Plants are a rich source of lectins. Lectins are often toxic because they crosslink cells, leading to aggregation. Many uncooked beans contain lectins that can be used to hemagglutinate red blood cells; examples are fava beans, lima beans, lentils, string beans, and kidney beans.

The snowdrop plant (*Galanthus nivalis*) produces a tetrameric lectin that binds to 12 mannosyl residues (**Figure 7.92**). Snowdrop lectin has been investigated for use in transgenic corn because it binds to cells in the intestine of aphid nymphs and inhibits development. The human body produces a defensive lectin, mannose-binding protein, as a hexamer of trimeric subunits. The trimeric form is enforced by a three-helix coiled-coil domain (**Figure 7.93**).

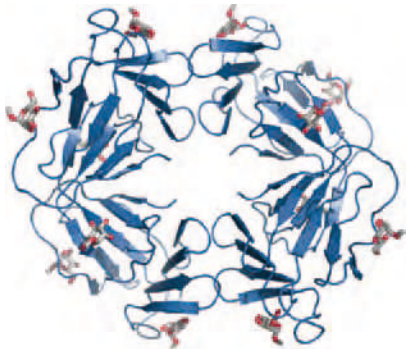


**Figure 7.92 Lectins bind glycans.** All lectins are polyvalent. This lectin from the snowdrop plant is bound to 12 molecules of methyl α-D-mannose. (PDB 1MSA)

**Figure 7.93 Mannose-binding protein.** This trimeric lectin is held together by a three-helix bundle.

**Table 7.1 Human glycan determinants in N-linked glycoproteins, O-linked glycoproteins and glycolipids.**

| Glycan structure | Trivial name* |
|---|---|
| Galβ1–4GlcNAcβ1–R | Type 2 LN (*N*-acetyllactosamine) |
| Galβ1–4GlcNAcβ1–3Galβ1–4GlcNAcβ1–R | Type 2 $LN_2$ ("I" antigen) |
| Galβ1–4GlcNAcβ1–3(Galβ1–4GlcNAcβ1–6)Galβ1–4GlcNAcβ1–R | "I" antigen |
| Galβ1–4GlcNAcβ1–3(Galβ1–4GlcNAcβ1–3)Galβ1–4GlcNAcβ1–R | "I" antigen |
| Galβ1–3GlcNAcβ1–R | Type 1 LN |
| Galβ1–3GlcNAcβ1–3Galβ1–3GlcNAcβ1–R | Type 1 $LN_2$ |
| Galβ1–3GlcNAcβ1–3Galβ1–3GlcNAcβ1–3Galβ1–3GlcNAcβ1–R | Type 1 $LN_3$ |
| GalNAcα1–Ser/Thr | Tn antigen |
| Galβ1–3GalNAcα1–Ser/Thr | T (also TF) antigen |
| Fucα1–2Galβ1–3GalNAcα1–Ser/Thr | H antigen (type 3) |
| GalNAcα1–3(Fucα1–2)Galβ1–3GalNAcα1–Ser/Thr | A antigen (type 3) |
| Neu5Acα2–6GalNAcα1–R | Sialyl Tn antigen |
| Neu5Acα2–3Galβ1–3GalNAcα1–Ser/Thr | Sialyl T antigen |
| Neu5Acα2–3Galβ1–3(Neu5Acα2–6)GalNAcα1–Ser/Thr | Di-sialyl T antigen |
| Galα1–3Galβ1–R | α-Gal antigen |
| GalNAcβ1–4GlcNAc–R | LacdiNAc (LDN) |
| GalNAcβ1–4(Fucα1–3)GlcNAc–R | Fucosylated LDN (LDNF) |
| Galβ1–4(Fucα1–3)GlcNAc–R | Lewis x ($Le^x$) (SSEA-1) |
| Fucα1–2Galβ1–4(Fucα1–3)GlcNAc–R | Lewis y ($Le^y$) |
| Galβ1–3(Fucα1–4)GlcNAc–R | Lewis a ($Le^a$) |
| Fucα1–2Galβ1–3(Fucα1–4)GlcNAc–R | Lewis b ($Le^b$) |
| Neu5Acα2–3Galβ1–4(Fucα1–3)GlcNAcβ1–R | Sialyl Lewis x ($SLe^x$) |
| Cyclic Neu5Acα2–3Galβ1–4(Fucα1–3)(Su–6)GlcNAcβ1–R† | Cyclic sialyl 6-sulfo Lewis X |
| Neu5Acα2–3Galβ1–3(Fucα1–4)GlcNAcβ1–R | Sialyl Lewis a ($SLe^a$) (CA19-9 antigen) |
| Neu5Acα2–3(Su–6)Galβ1–4(Fucα1–3)GlcNAcβ1–R | 6′-Sulfo-sialyl Lewis x (6′-sulfo $SLe^x$) |
| Neu5Acα2–6Galβ1–4(Su–6)GlcNAcβ1–R | 2,6-sialyl-Sulfo-LN (6-sialyl-6-sulfo LN) |
| Neu5Acα2–3Galβ1–4(Fucα1–3)(Su–6)GlcNAcβ1–R | 6-Sulfo-sialyl Lewis x (6-sulfo $SLe^x$) |
| Neu5Acα2–3(Su–6)Galβ1–4(Fucα1–3)(Su–6)GlcNAcβ1–R | 6,6′-bisSulfo-Lewis x (6,6′-bissulfo $Le^x$) |
| Su–3Galβ1–4(Fucα1–3)GlcNAcβ1–R | 3′-Sulfo-Lewis x (3′-sulfo $Le^x$) |
| Su–3Galβ1–3(Fucα1–4)GlcNAcβ1–R | 3′-Sulfo-Lewis a (3′-Sulfo $Le^a$) |
| Galβ1–4(Fucα1–3)GlcNAcβ1–3Galβ1–4(Fucα1–3)GlcNAcβ1–R | $Le^x$–$Le^x$ |
| Neu5Acα2–3Galβ1–4(Fucα1–3)GlcNAcβ1–3Galβ1–4(Fucα1–3)GlcNAcβ1–R | $SDLe^x$ |
| Neu5Acα2–3Galβ1–4GlcNAcβ1–3Galβ1–4(Fucα1–3)GlcNAcβ1–R | VIM-2 |
| Neu5Acα2–3Galβ1–4GlcNAc–R | 3-Sialyl-LN (type 2) |
| Neu5Acα2–6Gal(NAc)β1–R | 6-Sialyl-GalNAc or 6-Sialyl LN (type 1 or 2) |
| Fucα1–2Galβ1–4GlcNAcβ1–3Galβ1–R | Blood group H (type 2) |
| Galα1–3(Fucα1–2)Galβ1–4GlcNAcβ1–3Galβ1–R | Blood group B (type 2) |
| GalNAcα1–3(Fucα1–2)Galβ1–4GlcNAcβ1–R | Blood group A (type 2) |
| GalNAcα1–3(Fucα1–2)Galβ1–4(Fucα1–3)GlcNAcβ1–R | Blood group A (type 2) (A-$Le^y$) |
| GalNAcα1–3(Fucα1–2)Galβ1–3GalNAcα1–3(Fucα1–2)Galβ1–4GlcNAcβ1–R | Blood group A1 (type 3) |
| Fucα1–2Galβ1–3GalNAcα1–3(Fucα1–2)Galβ1–4GlcNAcβ1–R | Blood group A2 (A-associated H type 3) |
| Fucα1–2Galβ1–3GlcNAcβ1–3Galβ1–R | Blood group H (type 1) |
| Galα1–3(Fucα1–2)Galβ1–3GlcNAcβ1–R | Blood group B (type 1) |
| GalNAcα1–3(Fucα1–2)Galβ1–3GlcNAcβ1–R | Blood group A (type 1) |

| Glycan structure | Trivial name* |
|---|---|
| Fucα1–2Galβ1–4(Fucα1–3)GlcNAcβ1–R | Blood group H (type 2) ($Le^y$) |
| Galα1–3(Fucα1–2)Galβ1–3(Fucα1–4)GlcNAcβ1–R | Blood group B (type 1) (B-$Le^y$) |
| GalNAcα1–3(Fucα1–2)Galβ1–3(Fucα1–4)GlcNAcβ1–R | Blood group A (type 1) (A-$Le^b$) |
| Fucα1–2Galβ1–4(Fucα1–3)GlcNAcβ1–R | Lewis b ($Le^b$) |
| Su–4GalNAcβ1–4GlcNAc–R | 4′-sulfated LDN |
| Neu5Acα2–6GalNAcβ1–4GlcNAc–R | Sialylated LDN |
| Neu5Acα2–3(GalNAcβ1–4)Galβ1–4GlcNAcβ1–R | $Sd^a$/CT antigen |
| Neu5Acα2–8(Neu5Acα2–8)*n*Neu5Acα2–3Galβ1–4GlcNAcβ1–R | Polysialic acid |
| (Neu5Acα2–3Galβ1–4(Fucα1–3)GlcNAcβ1–6)(Neu5Acα2–3Galβ1–3)GalNAcα1–Ser/Thr | $SLe^x$ Core 2 *O*-glycan glycan |
| P–6–Manα1–2Manα1–3(Manα1–3)(P–6–Manα1–6)Manα1–6Manβ1–4GlcNAcβ–R | Diphosphorylated $Man_6$ |
| Neu5Acα2–3Galβ1–4GlcNAcβ1–2Manα1–Ser/Thr | O-linked mannose |
| Neu5Acα2–3Galβ1–4GlcNAcβ1–2(Neu5Acα2–3Galβ1–4GlcNAcβ1–6)Manα1–Ser/Thr | 2,6-Branched *O*–mannose |
| Galβ1–4(Fucα1–3)GlcNAcβ1–2Manα1–Ser/Thr | *O*-Mannose $Le^x$ |
| Su–3GlcAβ1–3Galβ1–4GlcNAcβ1–R | HNK-1 antigen |
| Su–3GlcAβ1–3Galβ1–4GlcNAcβ1–2Manα1–Ser/Thr | HNK-1 on *O*-mannose |
| Glcβ1–Cer | Glucosylceramide |
| Galβ1–Cer | Galactosylceramide |
| Su–3Galβ1–alkyl–2–acyl–s–glycerol | Seminolipid |
| Su–3Galβ1–Cer | Sulfatide |
| Galβ1–4Glcβ1–Cer | Lactosylceramide |
| Su–3Galβ1–4Galβ1–Cer | Ceramide dihexosyl sulfate |
| Su–3Galβ1–3GalNAcβ1–4Galβ1–4Glcβ1–Cer | Monosulfated gangliotetraosylceramide |
| GlcNAcβ1–3Galβ1–4Glcβ1–Cer | Lactotriaosylceramide ($Lc_3$) |
| Galα1–4Galβ1–4Glcβ1–Cer | $P^k$ antigen ($Gb_3$, globotriaosylceramide) |
| Galα1–4Galβ1–4GlcNAcβ1–3Galβ1–4Glcβ1–Cer | P1 antigen |
| GalNAcβ1–3Galα1–4Galβ1–4Glcβ1–Cer | Globoside (P antigen) ($Gb_4$) |
| GalNAcβ1–3Galα1–3Galβ1–4Glcβ–Cer | Isoglobotetraosylceramide |
| Su–GalNAcβ1–3Galα1–3Galβ1–4Glcβ1–Cer | Monosulfated globopentaosylceramide |
| Su–3GalNAcβ1–3Galα1–4Galβ1–4Glcβ–Cer | Monosulfated globotetraosylceramide |
| Su–3GalNAcβ1–3Galα1–3Galβ1–4Glcβ–Cer | Sulfo-isogloboside |
| GalNAcβ1–4(GlcNAcβ1–3)Galβ1–4Glcβ1–Cer | LcGg4 |
| Galβ1–3GalNAcβ1–3Galα1–4Galβ1–4Glcβ1–Cer | $Gb_5$ |
| Galβ1–4GlcNAcβ1–3Galβ1–4Glcβ1–Cer | Paragloboside |
| Fucα1–2Galβ1–3GalNAcβ1–3Galα1–4Galβ1–4Glcβ1–Cer | GL-6 fucosylated (globoH) |
| Galβ1–3GalNAcβ1–3Galα1–4Galβ1–4Glcβ1–Cer | SSEA-3 |
| Neu5Acα2–3Galβ1–3GalNAcβ1–3Galα1–4Galβ1–4Glcβ1–Cer | GL-7 globoseries ganglioside (SSEA-4) |
| Neu5Acα2–3GalNAcβ1–3Galβ1–4GlcNAcβ1–3Galβ1–4Glcβ1–Cer | Sialosyl paragloboside |
| Neu5Acα2–3Galβ1–3(Neu5Acα2–6)GalNAcβ1–3Galα1–4Galβ1–4Glcβ1–Cer | Disialosyl globopentaosylceramide |
| GalNAcβ1–3Galα1–3Galβ1–4Glcβ1–Cer | Cytolipin R |
| GalNAcα1–3GalNAcβ1–3Galα1–4Galβ1–4Glcβ1–Cer | Forssman glycolipid |
| GalNAcβ1–3GalNAcβ1–3Galα1–4Galβ1–4Glcβ1–Cer | *para*-Forssman glycolipid |
| Fucα1–2Galβ1–3GalNAcβ1–3Galα1–4Galβ1–4Glcβ1–Cer | Blood group H (type 4) |
| Galα1–3(Fucα1–2)Galβ1–3GalNAcβ1–3Galα1–4Galβ1–4Glcβ1–Cer | Blood group B (type 4) |
| GalNAcα1–3(Fucα1–2)Galβ1–3GalNAcβ1–3Galα1–4Galβ1–4Glcβ1–Cer | Blood group A (type 4) |
| Neu5Acα2–3Galβ1–Cer | GM4 |

| Glycan structure | Trivial name* |
|---|---|
| Neu5Acα2–3Galβ1–4Glcβ1–Cer | GM3 |
| Neu5Acα2–8Neu5Acα2–3Galβ1–4Glcβ1–Cer | GD3 |
| Neu5Acα2–8Neu5Acα2–8Neu5Acα2–3Galβ1–4Glcβ1–Cer | GT3 |
| 9–O–Neu5Acα2–8Neu5Acα2–3Galβ1–4Glcβ1–Cer | 9-*O*-Acetyl GD3 |
| GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GD2 |
| GalNAcβ1–4(Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GM2 |
| GalNAcβ1–4(Neu5Acα2–3)Galβ1–4Glcβ1–Cer | *N*-Glycolyl-GM2 |
| GalNAcβ1–4(Neu5Acα2–3)Galβ1–4GlcNAcβ1–3Galβ1–4Glcβ1–Cer | Sialopentaosylceramide |
| Galβ1–3GalNAcβ1–4(Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GM1 |
| Galβ1–3GalNAcβ1–4Galβ1–4Glcβ1–Cer | Asialo-GM1 |
| Neu5Acα2–3Galβ1–3GalNAcβ1–4Galβ1–4Glcβ1–Cer | *cis*GM1 (GM1b) |
| Fucα1–2Galβ1–3GalNAcβ1–4(Neu5Acα2–3)Galβ1–4Glcβ1–Cer | 2-Fucosyl-GM1 |
| Galα1–3(Fucα1–2)Galβ1–3GalNAcβ1–4(Neu5Acα2–3)Galβ1–4Glcβ1–Cer | B-GM1 |
| Fucα1–2Galβ1–3GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | 2-Fucosyl-GD1b |
| Galα1–3(Fucα1–2)Galβ1–3GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | B-GD1b |
| Neu5Acα2–8Neu5Acα2–3Galβ1–3GalNAcβ1–4Galβ1–4Glcβ1–Cer | GD1 (GD1c) |
| Neu5Acα2–3Galβ1–3(Neu5Acα2–6)GalNAcβ1–4Galβ1–4Glcβ1–Cer | GD1α |
| Neu5Acα2–3Galβ1–3GalNAcβ1–4(Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GD1a |
| Galβ1–3GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GD1b |
| GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GT2 |
| Neu5Acα2–8Neu5Acα2–3Galβ1–3GalNAcβ1–4(Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GT1a |
| Neu5Acα2–3Galβ1–3(Neu5Acα2–6)GalNAcβ1–4(Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GT1aα |
| Neu5Acα2–3Galβ1–3GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GT1b |
| Galβ1–3GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GT1c |
| Neu5Acα2–8Neu5Acα2–3Galβ1–3(Neu5Acα2–6)GalNAcβ1–4(Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GQ1aα |
| Neu5Acα2–8Neu5Acα2–3Galβ1–3GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GQ1b |
| Neu5Acα2–3Galβ1–3(Neu5Acα2–6)GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GQ1bα |
| Neu5Acα2–3Galβ1–3GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GQ1c |
| Neu5Acα2–8Neu5Acα2–3Galβ1–3GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GP1c |
| Neu5Acα2–3Galβ1–3(Neu5Acα2–6)GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GP1cα |
| Neu5Acα2–8Neu5Acα2–3Galβ1–3(Neu5Acα2–6)GalNAcβ1–4(Neu5Acα2–8Neu5Acα2–8Neu5Acα2–3)Galβ1–4Glcβ1–Cer | GH1cα |
| Galβ1–4GlcNAcβ1–3Galβ1–4(Fucα1–3)GlcNAcβ1–3Galβ1–4GlcNAcβ1–3Galβ1–4GlcNAcβ1–3Galβ1–4Glcβ1–Cer | Mono-fucosyl $LN_5$ |
| Galβ1–4GlcNAcβ1–3Galβ1–4(Fucα1–3)GlcNAcβ1–3Galβ1–4Glcβ1–Cer | Mono-fucosyl $LN_3$ |
| (Neu5Acα2–3Galβ1–4GlcNAcβ1–6)(Neu5Acα2–3Galβ1–4GlcNAcβ1–3)Galβ1–4GlcNAcβ1–R | Disialyl-branched Type 2 |
| Fucα1–2Galβ1–3(Fucα1–4)GlcNAcβ1–3(Fucα1–2)Galβ1–3(Fucα1–4)GlcNAcβ1–3Galβ1–4Glcβ1–Cer | Extended tetrafucosyl-$Le^b$ |
| Fucα1–2Galβ1–3(Fucα1–4)GlcNAcβ1–3Galβ1–4(Fucα1–4)GlcNAcβ1–3Galβ1–4Glcβ1–Cer | Extended trifucosyl-$Le^b$ |
| Galβ1–3(Fucα1–4)GlcNAcβ1–3Galβ1–3(Fucα1–4)GlcNAcβ1–3Galβ1–4Glcβ1–Cer | Dimeric $Le^a$ |
| Neu5Acα2–3Galβ1–3(Fucα1–4)GlcNAcβ1–3Galβ1–4Glcβ1–Cer | Sialyl $Le^a$ glycolipid |
| Neu5Acα2–6Galβ1–4GlcNAcβ1–3Galβ1–4Glcβ1–Cer | Monosialylganglioside LSTb |
| Su–3GlcAβ1–3Galβ1–4GlcNAcβ1–3Galβ1–4GlcNAcβ1–3Galβ1–4Glcβ1–Cer | HNK-expressing SGLPG |

*Common names and symbols for different glycans are used.
†Su = sulfate.
(Data from R. Cummings, *Molecular BioSystems* 5:1087–1104, 2009.)

**Table 7.2 Lectin examples.**

| Lectin group | Typical ligands | Examples of functions |
|---|---|---|
| Calnexin | $Glc_1Man_9$ | Protein sorting in the endoplasmic reticulum |
| M-type lectins | $Man_8$ | Endoplasmic reticulum-associated degradation of glycoproteins |
| L-type lectins | Various | Protein sorting in the endoplasmic reticulum |
| P-type lectins | Man-6-phosphate | Protein sorting post Golgi |
| C-type lectins | Various | Cell adhesion (selectins); glycoprotein clearance; innate immunity (collectins) |
| Galectins | β-Galactosyl | Glycan crosslinking in the extracellular matrix |
| I-type lectins | Neu5Ac | Cell adhesion (siglecs) |
| R-type lectins | Various | Enzyme targeting; glycoprotein hormone turnover |

(Data from Kurt Drickamer, Imperial College London, 2012. A Genomics Resource for Animal Lectins. http://www.imperial.ac.uk/research/animallectins)

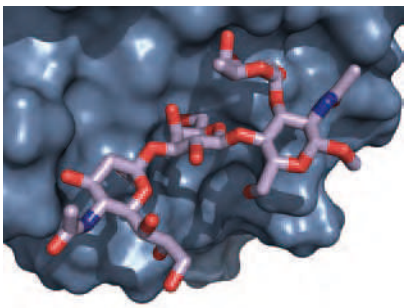


**Figure 7.94 Protein–glycan recognition.** Sialyl-Lewis$^x$ binds selectively to P-selectin without being fully enveloped by the protein. (PDB 1G1R)

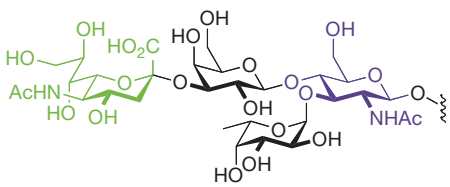



**Figure 7.95 The sialyl-Lewis$^x$ antigen.** This tetrasaccharide mediates the adhesion of leukocytes to the P-selectins on vascular endothelium.

## Human lectins mediate selective adhesion of leukocytes

One well-known human glycan determinant is important for the spread of cancer cells. On a cellular scale, cells in the bloodstream move at dizzying speeds, so the ability of blood cells to engage in meaningful molecular recognition while in transit is truly remarkable. The interaction of blood cells with the walls of blood vessels is mediated by a group of lectins referred to as selectins: L-selectins are found on leukocytes (white blood cells), P-selectins are found on platelets and endothelial cells, and E-selectins are found on endothelial cells. As part of the inflammatory response, cells in the vascular endothelium display P-selectin and E-selectin on their surface. Leukocytes display P-selectin glycoprotein ligand, which has a tetrasaccharide carbohydrate antigen, sialyl-Lewis$^x$, on its O-linked glycans (**Figure 7.94** and **Figure 7.95**). The interaction between P-selectin glycoprotein ligand and P-selectin allows leukocytes to slow down and roll along the vessel wall before finally squeezing between endothelial cells to invade the surrounding tissue, where they can provide assistance. Unfortunately, many metastatic tumors have been found to express glycoproteins with the sialyl-Lewis$^x$ antigen. The sialyl-Lewis$^x$ antigen allows them to invade tissues other than those in which they are normally localized.

The lymph nodes express the proteoglycan GlyCAM-1, which is a ligand for L-selectin on the surface of leukocytes. The interaction between GlyCAM-1 and L-selectin allows leukocytes to localize to the lymph nodes. The key epitope of GlyCAM-1 is believed to be a sulfated form of a sialyl-Lewis$^x$ with a sulfate group on the 6′ position of galactose.

## Human blood group antigens are found on glycolipids and glycoproteins

The most familiar antigens are those that distinguish the ABO blood groups. When persons with type A blood receive a transfusion from persons with type B blood, antibodies of the person with type A blood attack the type B blood cells. The converse is also true. When persons with type B blood receive a transfusion from persons with type A blood, antibodies of the person with type B blood attack the type A blood cells. Neither type A nor type B blood is compatible with persons with type O blood. However, persons with type A or B blood can receive transfusions from persons with type O blood. For this reason, persons with type O blood are referred to as universal donors.

The basis for these compatibilities rests in a particular class of oligosaccharide structures, known as the ABH antigens, present on the surface of erythrocytes. The ABH antigens are found on the ends of polylactosamines (e.g., $[Gal\beta1–3GlcNAc\beta1–3]_n$ or $[Gal\beta1–4GlcNAc\beta1–3]_n$) present on three types of membrane-bound molecules: N-linked glycoproteins, O-linked glycoproteins, and GPIs. The two most abundant carriers of the ABH antigens on erythrocytes are the oligosaccharides attached to

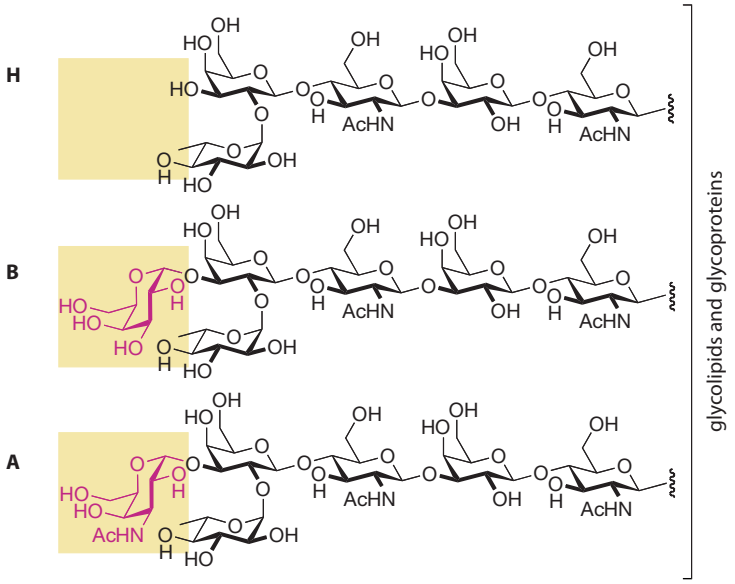



**Figure 7.96 Glycans as determinants of ABO blood type.** Subtle differences in the ABH antigens distinguish various blood types.

the anion transporter and the glucose transporter. The ABH antigenic determinants are only subtly different (**Figure 7.96**). The immunogenic response to these antigens rests on the presence or absence of a single sugar or acetamido group. All of the ABH antigens share a common pentasaccharide core. Persons with type O blood have the type H pentasaccharide antigen. Persons with type B blood have the type B antigen characterized by one additional Galα1–3 residue. Persons with type A blood have the type A antigen characterized by one additional GalNAcα1–3 residue.

## Problem 7.19

Draw a diagram of each of the ABH antigens, depicted in Figure 7.96, using abbreviated notation.

### Some toxins enter cells through multivalent carbohydrate recognition

Cholera is an intestinal disease mediated by the microorganism *Vibrio cholerae*. This organism produces a potent toxin composed of a catalytic A subunit and a pentameric B subunit that acts as a delivery vehicle (**Figure 7.97**). The pentameric B subunit acts as a lectin that binds selectively to ganglioside GM1 on the surface of intestinal cells (**Figure 7.98**). The catalytic subunit enters the intestinal cell, where the protein then begins to conjugate the ADP moiety of $NAD^+$ molecules to the α subunit of heterotrimeric $G\alpha_s$ proteins, which will be discussed in Chapter 9. The persistently activated $G\alpha_s$ proteins then generate cAMP without control. The massive diarrhea that results has two consequences. First, the affected individual suffers from lethal dehydration. Second, the cholera-rich effluent is highly infectious and can easily contaminate water supplies. Several other catalytic toxins attach to cellular targets through a carbohydrate-binding domain: pertussis toxin, Shiga toxin, ricin, and *E. coli* heat-labile enterotoxin A.

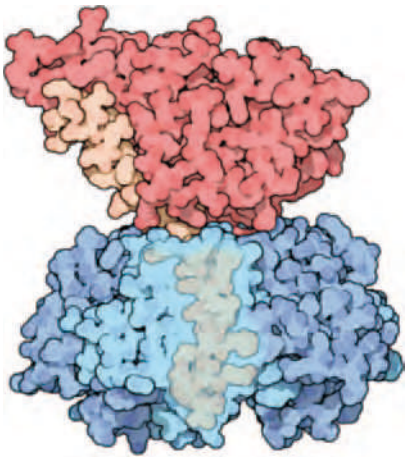


**Figure 7.97 Cholera toxin.** The catalytic A subunit is rendered in red and the pentameric B, which binds to the ganglioside GM1, is rendered in blue. (Courtesy of David Goodsell and RCSB Protein Data Bank.)

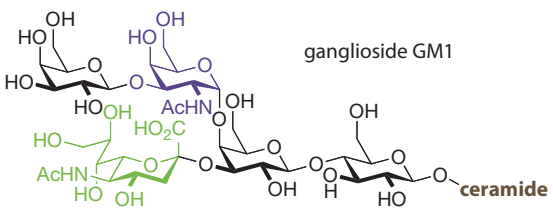



**Figure 7.98 Ganglioside GM1.** Each of the five B subunits of cholera toxin specifically recognizes the pentasaccharide moiety of GM1.

## Microarray technology facilitates the analysis of protein–glycan interactions

The great success of microarray technology in the analysis of DNA, RNA, and proteins inspired the development of microarrays for the analysis of protein–glycan interactions. Glycan arrays have been used to profile a wide range of carbohydrate-binding proteins, either purified or in complex samples. Two types of arrays are common: arrays of glycans and arrays of proteins that bind glycans (**Figure 7.99**).

Glycoproteins readily adsorb (noncovalently) on plastic surfaces such that their oligosaccharide moieties are exposed to solution. A variety of methods have been developed for immobilizing protein-free oligosaccharides on surfaces. The idea has its origins in the immobilization of semisynthetic oligosaccharide lipid conjugates in the wells of 96-well polyvinyl chloride microtiter plates. In this early work, reactive aldehyde groups were generated on the reducing end of oligosaccharides—both synthetic and isolated—through cleavage with periodate (see Figure 7.71), and the aldehydes were then reductively coupled to an aminophospholipid that adsorbed strongly on plastic surfaces. Using periodate to generate aldehyde functional groups is intrinsically destructive, so it is much more advantageous to make use of the aldehyde that is present on the end of reducing sugars. Recall that the closed form of sugars

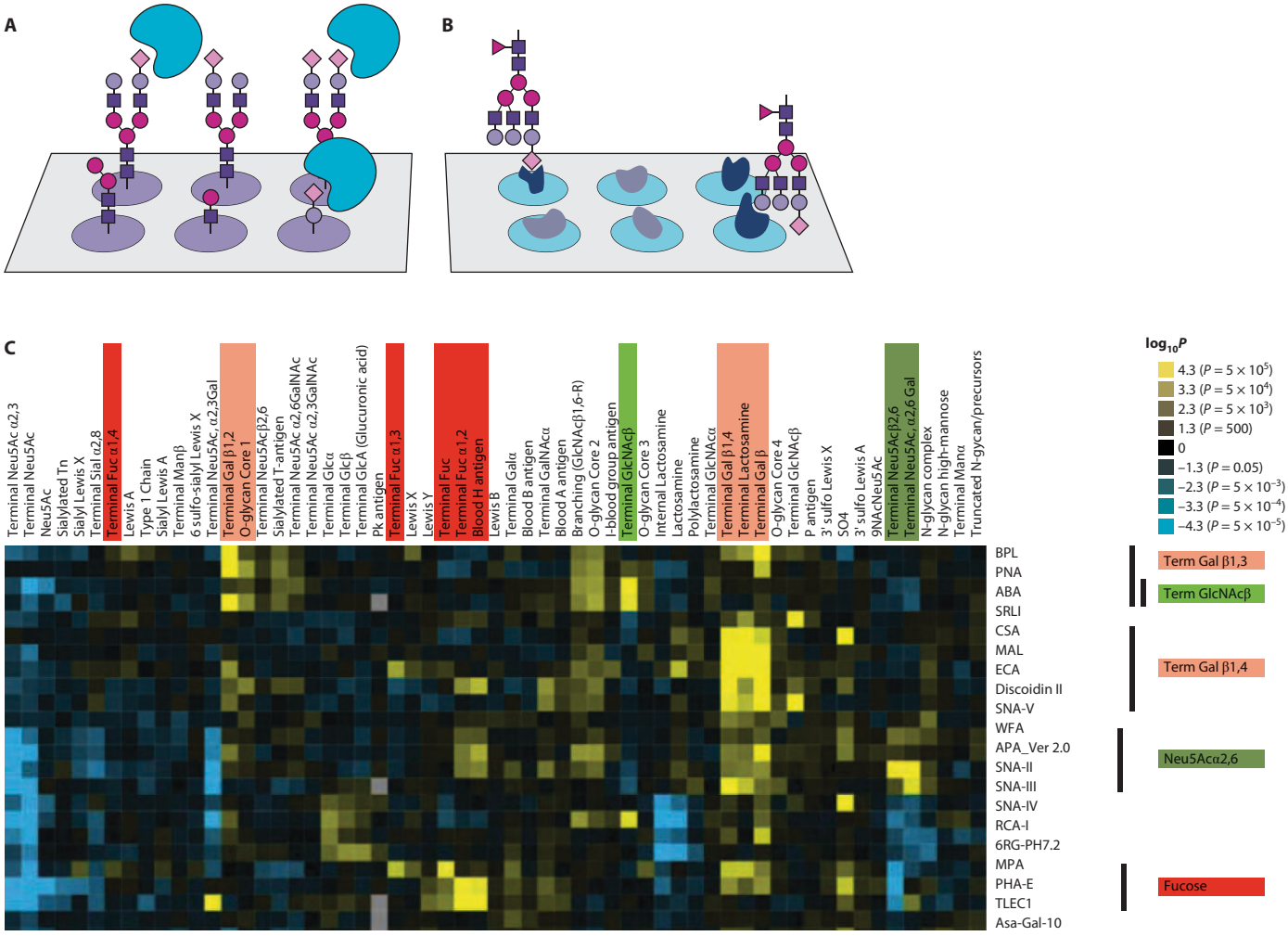
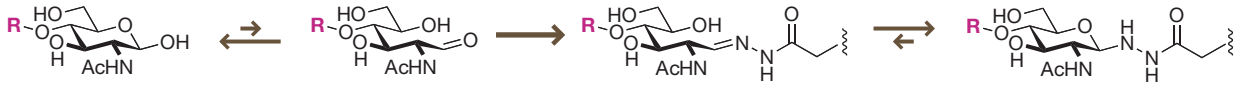



**Figure 7.99 Glycan microarray.** (A) Microarrays of oligosaccharides can be used to detect proteins that selectively bind glycans. (B) Microarrays of lectins can be used to identify structural features of glycans. (C) Actual data from a microarray of glycans treated with various fluorescently labeled lectins. Yellow correlates with high-affinity binding; blue correlates with low-affinity binding. (C, from A. Porter et al., *Glycobiology* 20:369–380, 2010. With permission from Oxford University Press.)



predominates over the open-chain aldehyde form; nevertheless, alkoxyamines and hydrazides react so aggressively with aldehydes that they can be used to form N-glycosides of reducing sugars, even in water. Such reactions have been used to attach complex oligosaccharides covalently to spot surfaces on microarrays (**Figure 7.100**). Glycans can also be immobilized by using surfaces derivatized with reactive epoxides or *N*-hydroxysuccinimido esters, but the linking reaction is not selective among the many hydroxyl groups that exist on typical glycans. The ability to synthesize complex oligosaccharides, with or without linkers for immobilization, has been a major advance.

**Figure 7.100 Glycan capture.** The reducing end of sugars reacts readily with acylhydrazides.

The attachment of polyfluorinated alkyl chains to glycans facilitates interactions that are neither hydrophilic nor hydrophobic but involve interactions between fluorocarbon functional groups. In fact, when fluorocarbons are mixed with oil and water, the fluorocarbons form a third phase that is distinct from the aqueous and organic layers (**Figure 7.101**). The nonstick polymer Teflon® is the most common example of a perfluoroalkane that repels aqueous and organic compounds. Compounds with polyfluorinated alkyl chains (for example -$(CF_2)_7CF_3$), often referred to as fluorous tags, can be extracted from organic solvents into fluorocarbon solvents or adsorbed on fluorous surfaces when making spot arrays. Various types of fluorinated media are commercially available for fluorous phase applications: solvents, supports for chromatography, and fluorocarbon-coated glass slides for microarray applications. Intermediates in the synthesis of complex oligosaccharides can be separated from reaction by-products through solid-phase fluorous extraction when one of the monosaccharide units (usually the first) bears a perfluoroalkane tag. Fluorous-tagged synthesis of biooligomers offers the advantages of solution phase and solid phase. Reactions can be conducted in solution phase, where yields are high, and separated from by-products on a solid phase, facilitating rinse-based purification.

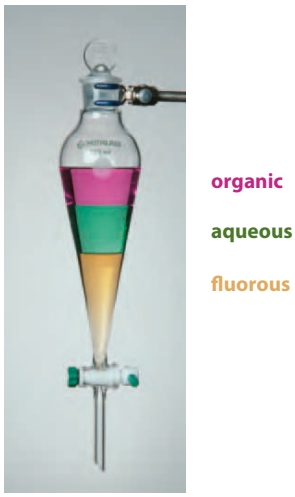



**Figure 7.101 Separation of liquid phases.** Each phase contains a different dye that is soluble only in that phase. (Courtesy of Fluorous Technologies.)

To gain a better appreciation for the power of microarrays it is useful to consider an example in which a glycan microarray was used to study how mammalian immune systems target the glycans of the SARS (severe acute respiratory syndrome) coronavirus. A glycan array was made by spotting, in triplicate, 51 structurally characterized glycoproteins and glycolipids onto nitrocellulose-coated glass slides; proteins and lipids adhere strongly to nitrocellulose. Horse antibodies against SARS coronavirus were covalently modified with fluorescein isothiocyanate (FITC), which reacts randomly with lysine amino groups (**Figure 7.102**); only a small number of the lysine side chains are modified on the antibodies. As a control experiment, antibodies were also isolated from a horse immunized with a bacterial glycan and fluorescently labeled with FITC. When the array was treated with the anti-SARS antibodies, the antibodies bound to an asialo-orosomucoid glycan with a weaker binding to lacto-*N*-tetraose, Galβ(1,3) GlcNAcβ1–3Galβ1–4Glc. The antibodies against the bacterial glycan did not bind to

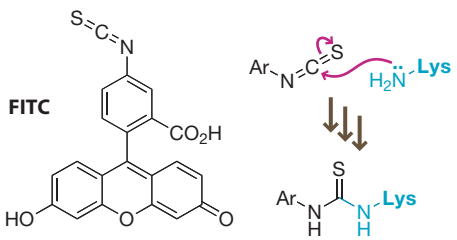



**Figure 7.102 Lysine side chains react with the isothiocyanate of FITC.** Isocyanates, R–N=C=O, hydrolyze too quickly for use in protein labeling.

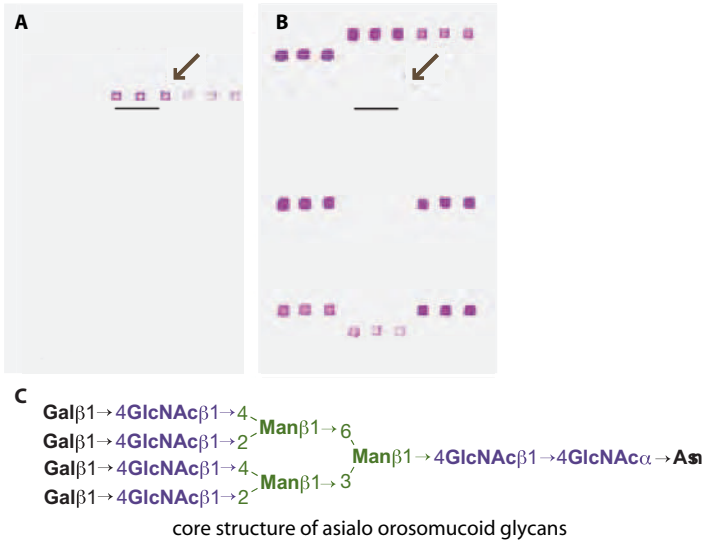



**Figure 7.103 A glycan array reveals antibody selectivities.** (A) Glycan selectivities for horse antibodies against the SARS coronavirus. (B) Glycan selectivities for horse antibodies raised against a bacterial glycan from *Streptococcus pneumoniae* serotype 18. (C) The anti-SARS antibodies seem to target the Galβ1–4GlcNAc branches of the orosomucoid glycans. (A, from D. Wang and J. Lu, *Physiol. Genomics* 18:245–248, 2004. With permission from the American Physiological Society.)

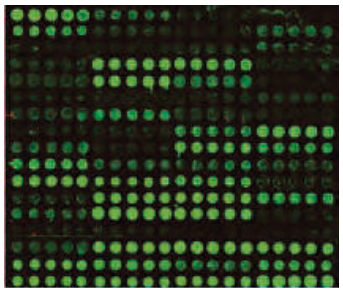


**Figure 7.104 A lectin microarray reveals glycan epitopes.** This array contains about 70 discrete carbohydrate-binding proteins with known selectivity. After treatment with fluorescently labeled HIV, one can determine which glycan epitopes are present on the virus. (From L. Krishnamoorthy et al., *Nat. Chem. Biol.* 5:244–250, 2009. With permission from Macmillan Publishers Ltd.)

the orosomucoid glycan (**Figure 7.103**). Orosomucoid is a serum glycoprotein, about 1–3% of total plasma protein, that is produced mainly by hepatocytes; it binds nonselectively to cationic and/or lipophilic drugs. Further characterization of the anti-SARS antibodies with glycan arrays (not shown) suggested that the antibodies recognize the Galβ1–4GlcNAc branches of the asialo-orosomucoid glycans, implying that related glycans are present on the coat of the SARS virus.

Lectin microarrays are used to reveal the carbohydrate composition of a biological sample—either pure oligosaccharides or complex mixtures of oligosaccharides isolated from, or displayed on, a cell membrane (**Figure 7.104**). A wide range of lectins (and antibodies) with known carbohydrate binding selectivity are commercially available, readily expressed, or readily isolated from plant sources. These lectins can be readily spotted onto glass slides coated with reactive aldehyde or epoxide groups. Lectins usually recognize only a small part, often the outer branches, of large oligosaccharides; however, these are the same parts that any other biological entity, like a cell or virus, would see.

## 7.9 GLUCOSE HOMEOSTASIS AND DIABETES

### Human metabolism and paper-burning are related transformations

Aerobic organisms combine oxygen with glucose to produce carbon dioxide and water. The transformation seems nothing short of miraculous until we realize that a burning piece of paper achieves the same effect with even greater brilliance. The human biochemical process of glucose-burning, referred to as respiration, is miraculous because it harnesses enthalpy and subdues entropy. One result of this process, the production of ATP, is the subject of most undergraduate biochemistry texts.

Glucose and oxygen exist harmoniously in the human bloodstream. Oxygen is sequestered and carried by hemoglobin molecules in erythrocytes. Glucose is kept at a relatively low concentration (3–7 mM) in human serum. After meals, pancreatic beta cells produce the hormone insulin, which then acts as a signal for cells to take in glucose. Two of the most important glucose-utilizing cell types are muscle cells, which burn glucose to generate mechanical energy, and adipocytes, in which glucose is metabolized to generate fat via pyruvate. Such cells have empty vesicles that fuse with the cell membrane when insulin activates the insulin receptor. These empty vesicles have GLUT4 glucose transporters directed into the cytosol so that, when fused with the cell membrane, the transporters are directed outside where glucose is plentiful. In this way, insulin signaling can rapidly notify cells to take in glucose (**Figure 7.105**).

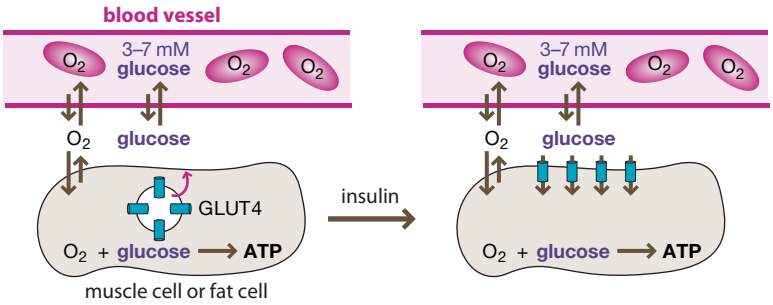



**Figure 7.105 Glucose transporters.** An important aspect of glucose homeostasis involves the import of glucose into cells. Glucose import is increased when vesicles lined with glucose transporters fuse with the cell membrane.

Patients with type I diabetes do not produce functional insulin, and must take insulin supplements. Patients with type II diabetes produce functional insulin, but have other problems in the glucose homeostasis pathway. The protein hormone incretin induces the glucose-dependent release of insulin from the pancreatic cells. Interestingly, exenatide, a 39 amino acid peptide toxin from the venom of the Gila monster, was shown to mimic the effect of incretin (**Figure 7.106**). It has been approved in the United States for the treatment of type II diabetes.

Any mutation that interferes with glucose uptake is likely to lead to diabetes. For example, insufficient insulin, insulin mutations, and insulin receptor mutations can all cause diabetes with dangerous consequences. Under hypoglycemic conditions (not enough glucose) cells cannot get the glucose required to produce energy. Glucose concentrations rise drastically after meals, and high levels of glucose are passed out in the urine. The most prevalent condition is referred to as diabetes mellitus, a Greek term referring to the flow of honey.

**Figure 7.106 Venom of the Gila monster is a source for the drug exenatide.** (Courtesy of Digital West Media, Inc., 2012. Desert USA, http://www.desertusa.com)

## Glucose reacts with proteins over time

The chronic effects of diabetes are attributable to hyperglycemia (too much glucose in the bloodstream). Glucose is dangerously reactive. The open-chain form of glucose is an aldehyde, which can condense with amines to form highly reactive iminium ions. This nonenzymatic reaction of proteins with glucose is referred to as glycation and should be distinguished from the enzymatic process of glycosylation. Every serum protein that has been studied has been shown to be glycated by glucose, usually on lysine side chains. For example, more than 11% of human serum albumin, which makes up 67% of the serum protein, is glycated (**Figure 7.107**). About half of the glycation of human serum albumin occurs on one particular lysine side chain, Lys525 (**Figure 7.108**). The high concentration of serum albumin in the blood (about 40 g $l^{-1}$) makes it important in the pharmacokinetics of many drugs. The protein readily binds neutral or anionic compounds, and most drugs that remain in human plasma for extended periods do so by binding to albumin. The corresponding albumin from cow's blood, bovine serum albumin (BSA), is one of the least expensive soluble proteins available to chemical biologists. It is often used as a generic protein platform for the conjugation of biomolecules and for blocking nonspecific hydrophobic interactions through the addition of high concentrations of BSA to experimental conditions.

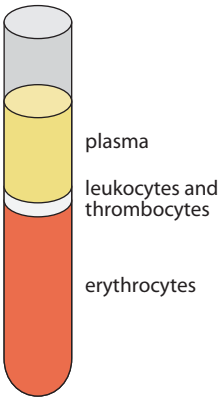



**Figure 7.107 Blood components.** When whole blood is spun down in a centrifuge the plasma (yellow) is separated from the white and red blood cells. Serum is plasma that has been depleted of fibrinogen and clotting factors.

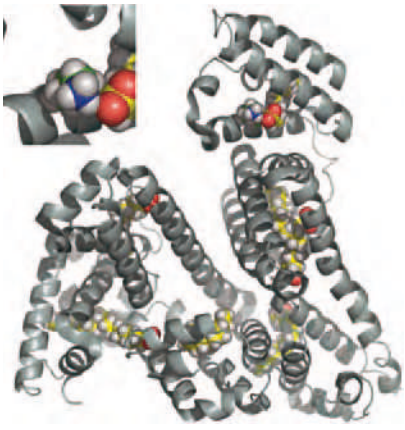
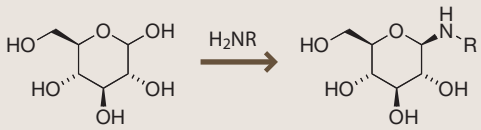


**Figure 7.108 A protein sponge.** This crystal structure of human serum albumin is bound to five molecules of the fatty acid myristate (yellow). Lys525 of human serum albumin (inset) is readily modified by glucose.

Albumin reacts with other agents as well. Within 90 minutes, 2% of human serum albumin is acetylated at various sites by 100 μM aspirin (acetylsalicylic acid), which is less than the peak concentration resulting from two regular aspirin tablets. Even when albumin turnover is taken into account, it is estimated that chronic dosing of aspirin would lead to complete acetylation of the circulating albumin within 2 months.

## Problem 7.20

Suggest a plausible arrow-pushing mechanism for the acid-catalyzed glycosylation of lysine side chains at pH 7.




**Nonenzymatic glycation** has two detrimental effects: first, the modification creates immunogenic structures on proteins capable of raising a deleterious immune response; and second, glycation can generate undesirable protein crosslinks. Covalent crosslinking reduces the rate of protein degradation, which is a necessary part of the ongoing process of renewal that prevents the buildup of damaged proteins. The most common protein–protein crosslinks involve lysine (**Figure 7.109**). Retro-aldol reactions lead to smaller fragments of glucose (glyoxal and methylglyoxal) that can also crosslink proteins. Nonenzymatic glycation ultimately produces many chemical structures, referred to as advanced glycation end-products. A variety of cells in the human body express a receptor (RAGE) that binds to advanced glycation end-products. A variety of ligands have been identified for RAGE. Carboxymethyllysine ($LysNHCH_2CO_2$-) is a prominent end-product of glycation that activates RAGE. RAGE activation initiates an inflammatory response and has been implicated in the pathogenesis of diabetes, Alzheimer's disease, and aging.

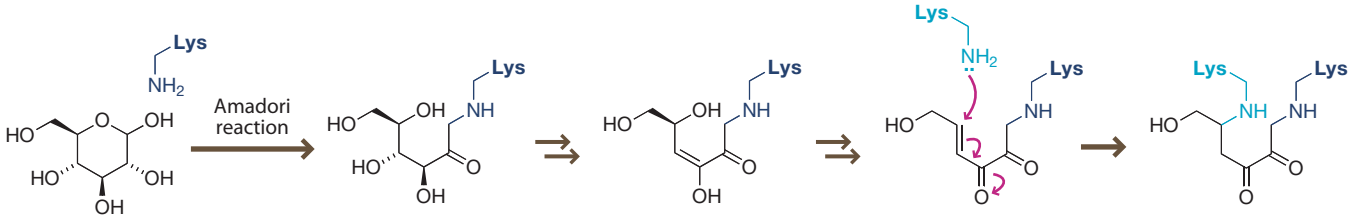
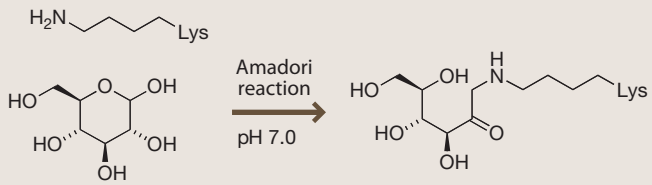



**Figure 7.109 Glycation.** A mechanism for protein crosslinking by glucose.

## Problem 7.21

Glucose can form a fructosyl adduct with lysine side chains through the Amadori reaction. Suggest a plausible arrow-pushing mechanism for the reaction at pH 7.




### Glucose-derived protein crosslinks are not necessarily permanent

The investigational drug alagebrium is a thiazolium derivative that has been shown to catalytically cleave 1,2-dicarbonyls such as those that are sometimes found between proteins. Unlike thiamine, alagebrium has an extra acidic functional group on the nitrogen that can form a second C–C bond. One of many plausible mechanisms for the cleavage of protein crosslinks is shown in **Figure 7.110**.

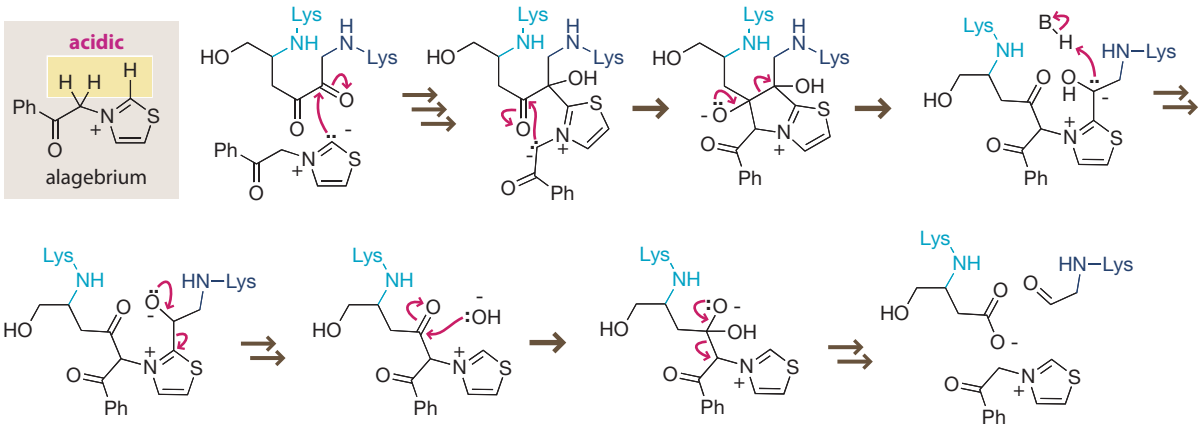



**Figure 7.110 One of many possible mechanisms for 1,2-dicarbonyl cleavage by alagebrium.**

Mature erythrocytes have no nucleus and do not actively produce proteins, but they last a long time in circulation, with a lifespan of between 3 and 5 months. Because the proteins inside erythrocytes are not turned over, those proteins are subject to glycation, just as proteins in the serum are. One of the most common tests for diabetes analyzes the fraction of a glycated form of hemoglobin, referred to as hemoglobin A1c (HbA1c).

## There is a big market for artificial ligands for human taste receptors

In part as a result of the reactivity of glucose described here, the demand for sweet foods in the American diet is a contributory factor in obesity and diabetes. Much of the dietary sugar is consumed as the sweetener sucrose, or table sugar. Most sucrose comes from sugar cane, but sugarbeet now accounts for almost one-third of the world's table sugar production. Sucrose is a 1,1′-disaccharide derived from glucose and the isomeric ketosugar fructose (**Figure 7.111**). Mixtures of glucose and fructose are available in grocery stores under the name corn syrup (**Figure 7.112**). The 1,1′-α,α′-disaccharide of glucose called trehalose is the blood sugar found in insects, much like glucose in humans.

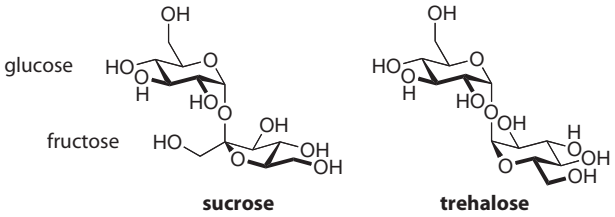



**Figure 7.111 Two common 1,1′-disaccharides.**

**Figure 7.112 Cooking with glucose.** This 1909 advertisement for Karo corn syrup, a glucose-rich sweetener made from corn starch, describes some of the culinary applications.

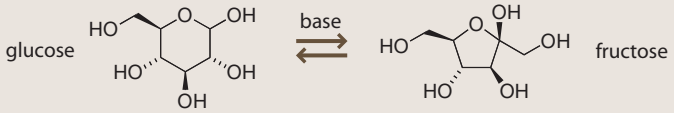
Fructose is slightly sweeter than sucrose. Thanks to farm subsidies and the efficiency of the US agricultural–industrial complex, high-fructose corn syrup has become a much cheaper alternative to sucrose. High-fructose corn syrup has become ubiquitous in packaged foods, particularly soda. The human GLUT5 transporter is selective for fructose, which is also taken in by the nonselective GLUT2 transporter. However, the effects of fructose and glucose on human metabolism are essentially the same.

In humans, sweet-tasting molecules are sensed by three different 7TM G protein-coupled receptors, T1R1, T1R2, and T1R3, which act in pairs. Most of the sweet-tasting compounds are sensed by cells that co-express T1R2 and T1R3 receptors. These G protein-coupled receptors are believed to act as a heterodimeric complex; its structure has not yet been fully defined.

The earliest artificial sweeteners to be discovered were the sulfamates, the oldest of which is saccharin. The anionic salts have good solubility and dissolve readily in cold beverages. The sweetness of saccharin, 300 times that of sucrose, was discovered

## Problem 7.22

Suggest a plausible arrow-pushing mechanism for the nonenzymatic isomerization of glucose to fructose under basic conditions.




by accident by a chemist at Johns Hopkins University in 1878. The sweetness of sodium cyclamate, which is 30–50 times sweeter than sucrose, was discovered in 1937 by a graduate student at the University of Illinois at Urbana-Champaign. The sweetness of acesulfame was discovered in 1967 by a chemist at the German chemical company Hoechst AG. Acesulfame potassium is up to 200 times sweeter than sugar. There is widespread public concern over the safety of these sweeteners as a result of some early animal studies. There is no worldwide consensus on the safety of these sulfamates in humans. For example, saccharin is banned in Canada, whereas cyclamate is banned in the United States. The pharmaceutical company Abbott Laboratories has petitioned the US Food and Drug Administration to approve cyclamate (**Figure 7.113**).

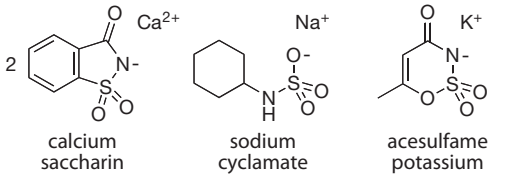



**Figure 7.113 Sulfamate sweeteners.**

The dipeptide sweetener aspartame was discovered at G.D. Searle & Company in 1965. With a sweetness 180 times that of sucrose, aspartame looks almost like the natural fragment of a protein except that it has a C-terminal methyl ester (**Figure 7.114**). Under basic conditions, the N terminus rapidly cyclizes onto the ester to form a diketopiperazine. Diketopiperazine formation can complicate the solid-phase synthesis of peptides linked through C-terminal ester linkages. At pH 7, the half-life for this cyclization has been reported to be 12 days at room temperature. However, in other studies a half-life of 49 hours at pH 7.0 and 25 °C has been reported. In acidic medium, such as cold carbonated soft drinks, the compound is considerably more stable. Although the diketopiperazine derivative of aspartame has been shown to be safe in numerous studies, the compound does not have the sweetness of the acyclic dipeptide. The aspartame derivative neotame is more than 10,000 times sweeter than sucrose, and only reluctantly forms a diketopiperazine. Several naturally occurring proteins are known to have an intensely sweet taste: thaumatin, monellin, mabinlin, pentadin, brazzein, and curculin (**Figure 7.115**). Monellin was discovered in 1972 in

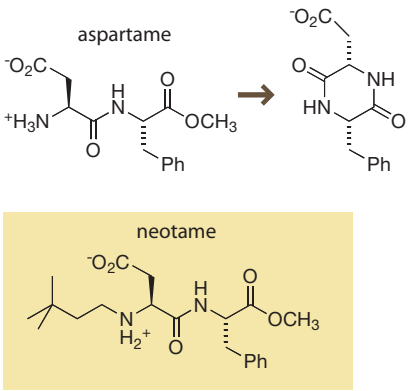



**Figure 7.114 Aspartame cyclization.** The dipeptide aspartame spontaneously cyclizes to form a diketopiperazine. The more recently developed analog neotame resists cyclization.

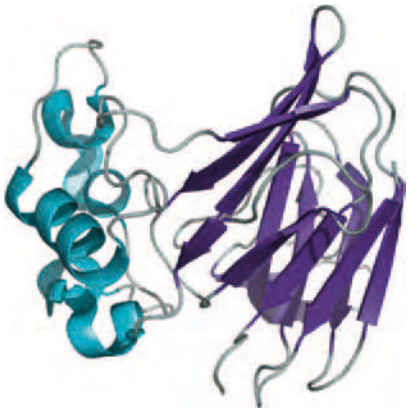


**Figure 7.115 Thaumatin.** This protein from the fruit of the African katemfe plant is sold as the sweetener Talin®. (PDB 1RQW)

the fruit of the west African shrub called the serendipity berry, also known as miracle fruit (**Figure 7.116**). The glycoprotein miraculin from the miracle fruit does not itself taste sweet. Instead, the name reflects the compound's remarkable ability to cause sour foods to taste sweet long after exposure to miraculin, a result of strong binding to taste receptors. Thaumatin is marketed under the trade name Talin®, but proteins have limited application as low-calorie sweeteners because of their tendency to denature at elevated temperatures.

**Figure 7.116 Miracle fruit.** A glycoprotein in the miracle fruit changes taste perception for hours after consumption. (Courtesy of Hamale Lyman.)

Sucralose, a chlorodeoxysucrose derivative, is 600 times sweeter than sucrose and is more stable than aspartame. Many chlorinated compounds are toxic to the liver because the compounds are metabolized into DNA-alkylating agents. However, extensive studies of sucralose have revealed no long-term exposure risks. The compound is poorly absorbed in the gastrointestinal tract, and because of its hydrophilicity it is rapidly cleared.

Sucralose is more economical to synthesize than aspartame because the compound can be made directly from sucrose (**Figure 7.117**). An unusual chlorodehydration of sucrose is performed with sulfuryl chloride ($SO_2Cl_2$), which is not to be confused with thionyl chloride ($SOCl_2$). Sulfuryl chloride resembles $Cl_2$ in its reactivity and is rarely used for substitution of hydroxyl groups. Sulfonate esters of sucrose have differential reactivity: 6-$OSO_2R$ ≈ 6′-$OSO_2R$ >> 4-$OSO_2R$ > 1′-$OSO_2R$. Thus, practical routes for the synthesis of sucralose always involve a selective protection of the 6-hydroxyl group of sucrose so that the 6′-OH is not sulfonated and converted to a chloride.

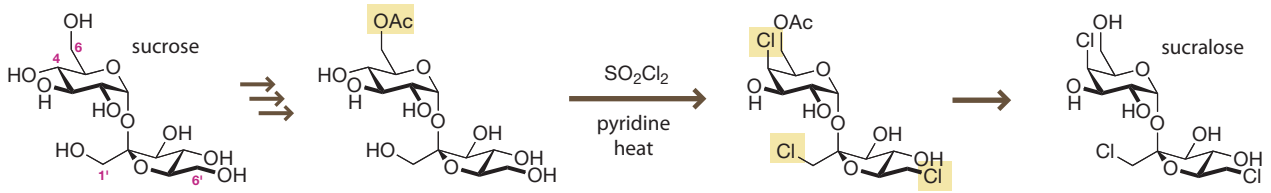



**Figure 7.117 Sucralose.** Sucralose is a chlorinated compound that is chemically synthesized from sucrose.

The South American plant known as sweetleaf (*Stevia rebaudiana* Bertoni) produces a group of terpene glycosides with an intensely sweet taste. Extracts of this plant are sold as a low-calorie sweetener. The most abundant of these sweet-tasting terpenes, stevioside, makes up about 5–10% of the steviol glycosides; the sweetest derivative, rebaudioside A, is up to 300 times sweeter than the equivalent weight of sucrose (**Figure 7.118**).

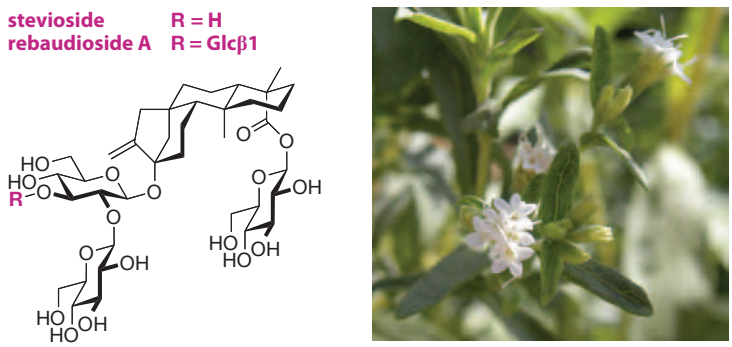



**Figure 7.118 Sweet terpenes.** Steviol glycosides (left) account for the intensely sweet properties of the sweetleaf plant (right). (Right, courtesy of Ethel Aardvark.)

## 7.10 SUMMARY

Since the 1930s, most students of biochemistry have been indoctrinated with a myopic view of glycobiology, focusing on the role of glucose as an energy source and as a molecular building block. However, by the end of the past millennium, the confluence of biological problems and emerging techniques has galvanized the field of glycobiology toward large, complex oligosaccharides.

Human oligosaccharides are constructed from about half as many building blocks as proteins, yet oligosaccharides are structurally more complex than proteins. The structural complexity of each oligosaccharide is easily traced to the potential for branching, and is further complicated by the fact that the formation of glycosidic bonds creates a new stereogenic center. These challenges are further exacerbated by the fact that mammalian cells are designed to produce mixtures of oligosaccharides. The oligosaccharides are assembled by enzymes with varying levels of selectivity, and there is no proofreading system to ensure the fidelity of glycosylation reactions.

Human cells have evolved to use glucose as a carbon source and as the molecular currency of energy in the bloodstream. They wear a coat of glycoproteins and glycolipids that mediate cell–cell interactions. Cells speak with glycans and listen with lectins. Many pathogenic organisms recognize their human cellular targets through protein–oligosaccharide interactions. Chemists can exploit the differences between human oligosaccharides and those of pathogenic organisms to design vaccines and therapeutic drugs. Many of the proteins that circulate in the bloodstream are glycosylated, and the oligosaccharide structure is important; for example, the half-life of proteins in the serum can be altered by glycosylation.

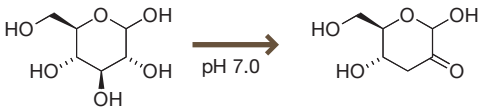
The confectionery of evolution has produced some sweet treats indeed, and our view of carbohydrate chemistry and biology has also evolved. In a little more than 100 years, the glycobiologist's perspective has progressed from a malt-ball view of human cells, with sugar on the inside and fat on the outside, to a gumdrop model, with protein on the inside and sugar on the outside. Unfortunately, the importance of glucose metabolism is on the rise, driven by an alarming increase in the incidence of diabetes. The continuing efforts of glycobiologists to elucidate the structures and roles of complex carbohydrates and to understand and control glucose/lipid metabolism will propel this field forward into new frontiers.

## LEARNING OUTCOMES

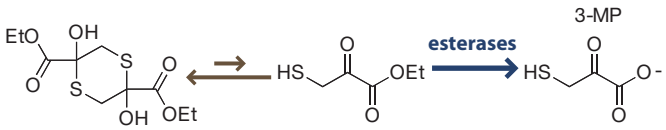
- Draw the structures of the 10 monosaccharide subunits found in human oligosaccharides, including anomeric stereochemistry and conformation.
- Draw mechanisms for inverting and retaining glycosylhydrolases.
- Distinguish polysaccharides from oligosaccharides.
- Explain how vesicular compartmentalization controls the cellular topology of protein glycosylation processes.
- Distinguish the synthesis and structure of N-linked glycoproteins from those of O-linked glycoproteins.
- Contrast the presence of oligosaccharides on extracellular/membrane-bound proteins with the absence of oligosaccharides on intracellular proteins.
- Describe the functional/physiological significance of protein glycosylation.
- Draw the lipid anchors of glycolipids.
- Contrast the structure of glycolipids with the structure of glycoproteins.
- Explain why vaccines are an important therapeutic application of oligosaccharides.
- Draw the mechanisms for the chemical formation of glycosidic bonds.
- Understand the importance of multivalency in lectin–carbohydrate interactions.
- Describe some of the physiological functions of human lectins.
- Explain how insulin affects glucose homeostasis.
- Draw mechanisms for nonenzymatic protein glycation.
- Draw the structures of common artificial sweeteners.

# PROBLEMS

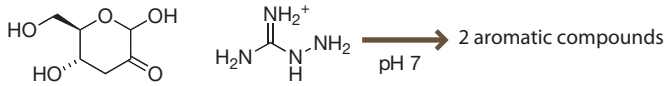
***7.23** Suggest a plausible base-catalyzed arrow-pushing mechanism for the formation of 3-deoxyglucosone from glucose.



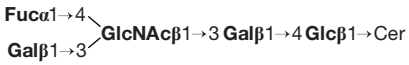

***7.24** Methyl α-L-fucoside selectively forms a single bicyclic bis-ketal because it minimizes 1,3-diaxial interactions and maximizes anomeric effects. Draw the structure of the preferred ketal, using chair conformations.



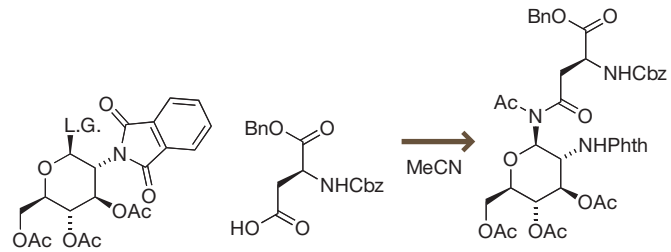

**7.25** 3-Deoxyglucosones are highly reactive and can form crosslinks between lysine side chains. A developmental anti-diabetic drug called pimagedine (aminoguanidine) reacts with 3-deoxyglucosones to form two different stable aromatic products. Suggest a plausible structure for either aromatic product and a mechanism for its formation.



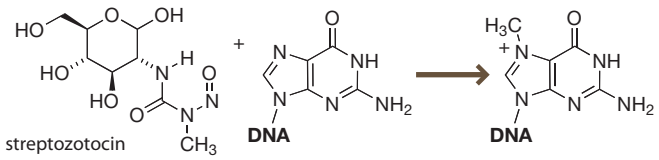

***7.26** Draw the structure of the following glucosphingolipid with the Lewis[a] antigen, using chair conformations.



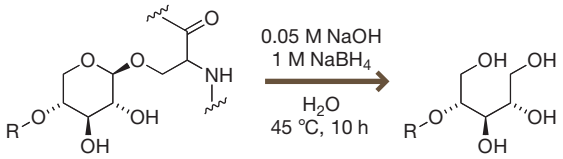

**7.27** One strategy for the synthesis of asparagine-linked oligosaccharides involves the use of the Ritter reaction. Suggest a plausible arrow-pushing mechanism for the reaction. Do not worry about the identity of the leaving group.




**7.28** The human pancreas produces insulin—a hormone that controls serum glucose levels. Streptozotocin is a glucose derivative isolated from the soil microorganism *Streptomyces achromogenes* that exhibits selective cytotoxicity toward pancreatic beta cells. It has been approved for the treatment of pancreatic cancer (Zanosar®) and is even used to induce diabetes in rats. Streptozotocin exerts selective cytotoxicity against pancreatic beta cells through a mechanism that includes DNA alkylation. Suggest a plausible arrow-pushing mechanism (base-catalyzed) for the methylation of guanine bases in DNA by streptozotocin.

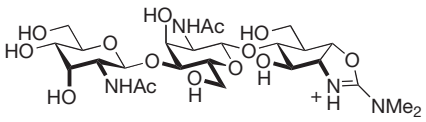



***7.29** O-linked oligosaccharides can be selectively released from glycopeptides (derived from digestion with the peptidase pronase) under mild basic conditions in the presence of excess sodium borohydride without affecting N-linked oligosaccharides. Suggest a plausible arrow-pushing mechanism.

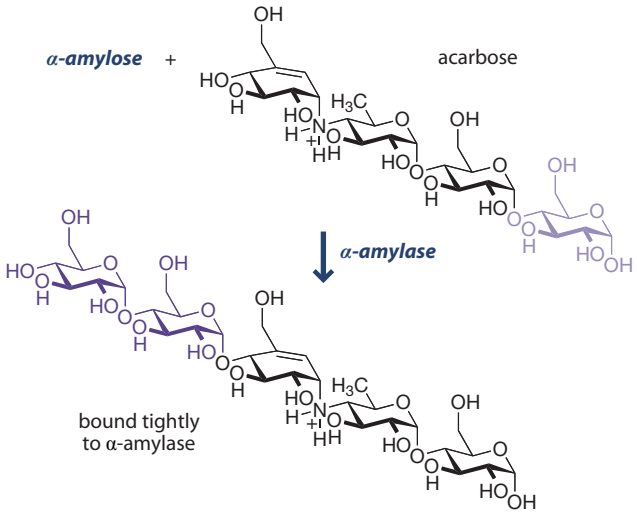



**7.30** The enzyme chitinase is selectively inhibited by the natural product inhibitor allosamidin. On the basis of the analogy between allosamidin and a key mechanistic intermediate, suggest a plausible arrow-pushing mechanism for chitinase.

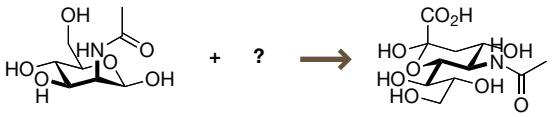
allosamidin



**7.31** The natural product acarbose is an inhibitor of human pancreatic α-amylase, which is a retaining glucosidase. The mechanism of inhibition involves the hydrolysis and transglycosylation of acarbose to produce a pseudopentasaccharide that binds tightly in the active site. Suggest a plausible arrow-pushing mechanism for the formation of the tightly bound inhibitor.

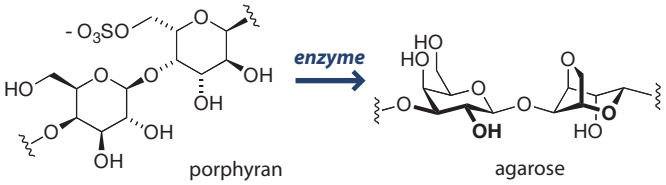


***7.32** Predict the structure of the missing starting material for the aldolase-catalyzed biosynthesis of *N*-acetylneuraminic acid from *N*-acetylmannosamine.



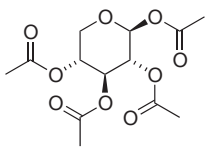

ManNAc Neu5Ac

**7.33** The agarose gel used for the electrophoresis of DNA is seaweed polysaccharide containing an unusual bicyclic subunit. An enzyme isolated from the red seaweed that produces the related polysaccharide porphyran has been shown to convert galactose 6-sulfate residues into the anhydrogalactosyl residues of agarose. Suggest a plausible arrow-pushing mechanism for the enzymatic reaction.




porphyran agarose

**7.34** Draw the following xylose derivative in a chair conformation and then consider the conformation of the ester substituents. On the basis of $A_{1,3}$ strain, which carbonyl groups do you expect to point above the plane of the pyran ring, and which do you expect to point below the plane of the pyran ring?



# 8 Polyketides and Terpenes

## LEARNING OBJECTIVES

- Distinguish between the chemical mechanisms for iterative carbon–carbon bond formation in polyketide biosynthesis and terpene biosynthesis.
- Recognize the limited structural range of human polyketides and human terpenes.
- Appreciate the unlimited structural range of polyketides and terpenes found in lower organisms and plants.
- Describe the origin and effects of eicosanoid and sphingoid signaling molecules.
- Understand the origin of diversity in polyketide metabolites from lower organisms.
- Explain the structure and reactivity of nonclassical carbocations.
- Describe how cationic cyclizations and rearrangements amplify the diversity of terpene natural products.

Carbon–carbon bonds are as ubiquitous in organic chemistry as the notes in music, the words in English, and the mathematics in physics. Yet, despite their foundational centrality, carbon–carbon bonds seem slighted by the central dogma, which emphasizes the roles of bonds to and between heteroatoms, such as the phosphorus–oxygen bonds in DNA and RNA, the carbon–oxygen bonds in glycans, and the carbon–nitrogen bonds in proteins. Why is carbon–carbon bond formation relegated to a relatively minor role in the central dogma?

Naive answers to this question, right or wrong, might point to an incompatibility between carbon nucleophiles or carbon electrophiles, and the aqueous environment, but that kind of thinking loses traction when the effortless nature of enzymatic carbon–carbon bond formation is revealed. There is also no reason to impugn the stability of carbon–carbon bonds. Aliphatic carbon–carbon bonds are exceptionally robust—perhaps too robust. They are not readily cleaved by any single reaction, either chemical or enzymatic, whereas DNA, RNA, proteins, and glycans are easily hydrolyzed by enzymes into their molecular components for eventual reuse by the cell. The durability of aliphatic biooligomers makes them ideal for the lipid chainmail armor that protects all living cells, the long-distance signals that coordinate the actions of human cells, and the advanced weaponry used by the fittest microorganisms.

**Polyketides** and **terpenes** range in complexity and form, from simple linear hydrocarbons (**Figure 8.1**) to ornate polycyclic architectures; however, the iterative reactions used to construct these two classes of natural products are quite different. Polyketides are built up from two-carbon and three-carbon subunits through the reactions of enolates with carboxylic esters and thioesters. Terpenes, in contrast, are built up from five-carbon subunits through the addition of carbocations to olefins. We have

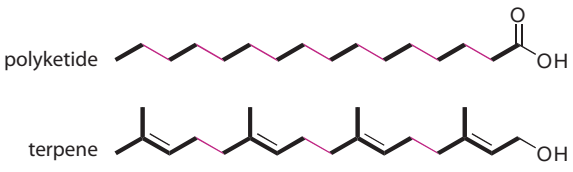



**Figure 8.1 Rudimentary examples of a polyketide and a terpene.** The repeating subunits are shown in bold, and the bonds formed through oligomerization are highlighted in red.

already discussed a wide range of polyketide and terpene natural products, such as daunomycin, aflatoxin, ptaquiloside, erythromycin, rapamycin, parthenolide, and steviol. These natural products are so complex that it can sometimes be difficult to discern the monomeric units from which they are assembled.

Cellular biosynthesis produces both primary and secondary metabolites. **Primary metabolites** are the molecules considered essential for normal cellular function, such as ATP, glyceraldehyde 3-phosphate, or cholesterol. Signaling molecules such as dopamine and estrogen are also considered primary metabolites. In contrast, **secondary metabolites** are nonessential for the immediate survival of a cell; instead, cells produce secondary metabolites for their chemical defense. Bacteria favor the polyketide pathway for the production of bioactive secondary metabolites. Plants, in contrast, tend to favor the terpene biosynthesis pathway for the production of secondary metabolites.

The evolutionary arms race has led to some truly ingenious polyketide and terpene weaponry. All organisms use polyketide and terpene biosynthesis to make primary metabolites, but humans seem to have missed out on the defensive potential of polyketide and terpene secondary metabolites. Instead, we developed agile physiques to fight with or flee from predators, and sophisticated protein and cellular defenses to fend off pathogens. This powerful combination of defenses has ensured the survival of our species, but not the survival of individuals who must battle infectious diseases, metabolic diseases, or cancer. After millions of years of evolutionary neglect, humans have finally started to harness the power of polyketide and terpene biosynthesis pathways, borrowing natural products from lower organisms and weaponizing them against human afflictions through a combination of molecular biology and chemical synthesis. Thus, over the past hundred years, we have moved beyond the molecular hunter–gatherer phase to the domestication phase. Perhaps over the next hundred years, through harnessing such remarkably complex biosynthetic technology, we might reach a level of molecular sophistication on a par with bacteria, fungi, and plants.

## 8.1 THE CLAISEN REACTION IN POLYKETIDE BIOSYNTHESIS

### The diverse structures of polyketide natural products belie their iterative construction

Polyketides were first defined by the mountaineer–chemist J. Norman Collie in 1907 (**Figure 8.2**) as formal polymers of ketene ($H_2C{=}C{=}O$), which he described as a ketide group. Collie did not investigate any specific biochemical pathways, but he recognized that a wide range of chemical structures were accessible through further functionalization of the basic polyketide backbone $(CH_2CO)_n$.

**Figure 8.2 Mountaineer and chemist.** J. Norman Collie coined the term "ketide." (Courtesy of the Whyte Museum of the Canadian Rockies.)

### Polyketides are derived from two-carbon and three-carbon building blocks

Polyketides are built up through repeated Claisen reactions. Each enzyme-catalyzed Claisen reaction of either two-carbon acetate esters or three-carbon propionate esters generates a β-ketoester (**Figure 8.3**). Such an intermolecular dimerization is usually referred to as a Claisen condensation, even though the reaction generates an alcohol, rather than water, as its by-product. In a base-catalyzed Claisen condensation, the

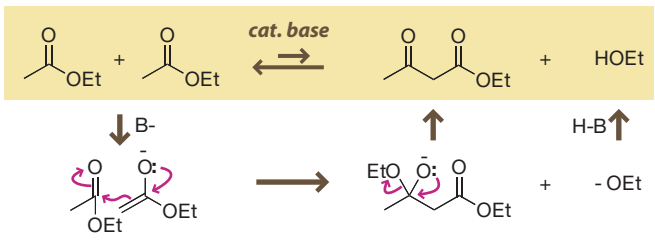


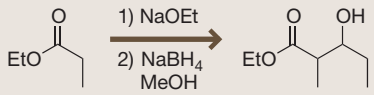

**Figure 8.3 Claisen condensation mechanism.** The base-catalyzed Claisen condensation involves the addition of an enolate to a carboxylic acid ester.

base—usually an alkoxide—deprotonates the ester, generating low concentrations of a highly reactive ester enolate. The enolate then attacks one of the unreacted esters, and the alkoxide anion that is ejected from the tetrahedral intermediate recaptures a proton from the protonated base.

Unfortunately, the Claisen condensation of ethyl acetate is energetically uphill by over 20 kcal $mol^{-1}$. In this reaction, one of the ester functional groups in the starting material, stabilized by resonance, is converted to a ketone, which does not enjoy such resonance stabilization. The retro-Claisen condensation is facile in the presence of alkoxides, so the yield of β-ketoester at equilibrium is low in an alkoxide-catalyzed Claisen condensation. The successful laboratory version of this reaction, developed by Rainer Ludwig Claisen in 1887, involves the use of *stoichiometric* ethoxide anion ($pK_a \approx 15.9$) to help drive the reaction toward a highly stabilized enolate of ethyl acetoacetate ($pK_a \approx 10.7$). Even when a stoichiometric base is employed, the reaction is not high-yielding with respect to the ester, but the yield is decent with respect to the sodium alkoxide base. Fortunately, ethyl acetate is an inexpensive solvent. This and/or other special tricks are needed to make the Claisen condensation efficient in the laboratory. One notable trick introduced by Walter Dieckmann was to apply the reaction to tethered esters that can cyclize to form highly favorable five- and six-membered rings.

## Problem 8.1

Draw all of the stereoisomeric products that are expected from the following synthetic sequence.




Nature favors the Claisen condensation for the formation of carbon–carbon bonds, but uses thioesters instead of carboxylic acid esters. The thioesters used by Nature are derivatives of **coenzyme A** (**Figure 8.4**). Coenzyme A is composed of two modular subunits: a phosphopantetheinyl unit and adenosine 3,5-diphosphate. To emphasize the sulfur atom, coenzyme A is often abbreviated as HS-CoA or CoA-SH. CoA is readily recognized by enzymatic machinery, and it may eventually turn out to be a ligand for microRNA regulation like many of the other vitamin cofactors. As far as we know, none of the rococo elements of the phosphopantetheinyl linker are important in the Claisen condensations of acyl-CoA esters. Nevertheless, enzymes that assemble polyketides use phosphopantetheinyl linkers just like the one in coenzyme A.

Thioester-based Claisen condensations are different in a couple of ways from Claisen condensations of carboxylic acid esters. First, the condensation of thioesters is energetically unfavorable, but less unfavorable than the corresponding condensation of carboxylic acid esters, by a very large amount, 11.7 kcal $mol^{-1}$. Second, the α-protons of thioesters are more acidic than the α-protons of traditional esters. For example, ethyl thioacetate has a $pK_a$ of 21, which is closer to the $pK_a$ of a ketone

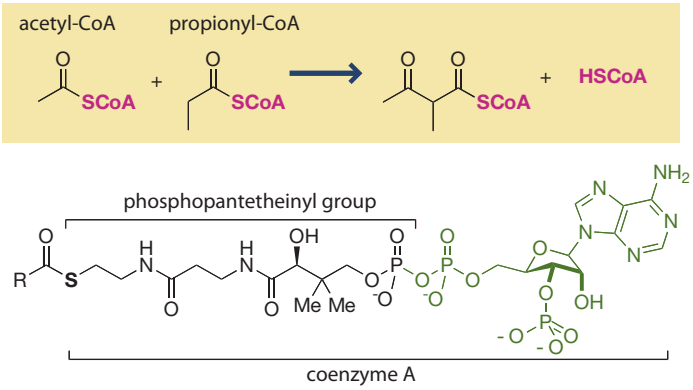



**Figure 8.4 Carbon-carbon bond formation with thioesters.** Nature performs Claisen condensations with thioesters derived from the thiol coenzyme A.

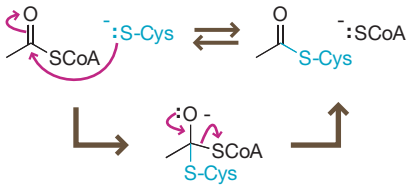



**Figure 8.5 Thioester exchange mechanism.** Unlike carboxylic esters derived from alcohols, acyl group exchange rapidly between thiols under physiological conditions.

(for example, p$K_a$ = 20 for acetone) than a carboxylic ester (for example, p$K_a$ = 24 for ethyl acetate). Enolates of thioesters are commonly used in organic synthesis, but the Claisen condensation of thioesters, usually requiring Grignard reagents as bases, has no synthetic advantages over the Claisen condensation of the much more readily available carboxylic acid esters. As we will see, the method of enolate formation in biology is much less direct than deprotonation.

Most enzymatic Claisen condensations involving acyl-CoA esters involve the formation of an acyl-enzyme intermediate through acyl transfer to a cysteine thiolate (**Figure 8.5**). The transesterification of thioesters is more facile than the transesterification of carboxylic esters. Thiol ester exchange between acyl-CoA and glutathione (GSH) occurs with a second-order rate constant of 0.02 $M^{-1}$ $s^{-1}$ (**Figure 8.6**), which is about 100 times faster than exchange between regular carboxylic acid esters and GSH. This rate of transesterification is significantly faster than thioester hydrolysis, which is slow at physiological pH. The latter attribute also explains why the native chemical ligation of peptide thioesters works. Thiols are more reactive than water in large part because a significant fraction (about 1%) of thiols (p$K_a \approx 9$) are deprotonated at physiological pH.

**Figure 8.6 Water resistant.** Chemical studies reveal that thioester exchange (right) is much faster than hydrolysis (left).

## 8.2 THE BIOSYNTHESIS OF FATTY ACIDS IS A PARADIGM FOR POLYKETIDE BIOSYNTHESIS

### Fatty acids have varying levels of unsaturation

Fatty acids are the prototypical polyketides. They have the simplest chemical structures, yet their biosyntheses involve most of the steps common to the synthesis of more complex polyketide natural products, both human and nonhuman. Human fatty acids have even numbers of carbons. The saturated fatty acids myristic acid, palmitic acid, and stearic acid, for example, have 14, 16, and 18 carbons, respectively (**Figure 8.7**). Fatty acids are important in chemical biology because they are essential components of the **lipid** molecules that make up cellular membranes and are precursors to signaling molecules. Human fatty acids have *cis*, but never *trans*, double bonds. Sometimes fatty acids have many *cis* double bonds, but they are never in conjugation with each other. Notice in Figure 8.7, for example, that there is a methylene group between the double bonds of both linoleic and arachidonic acids.

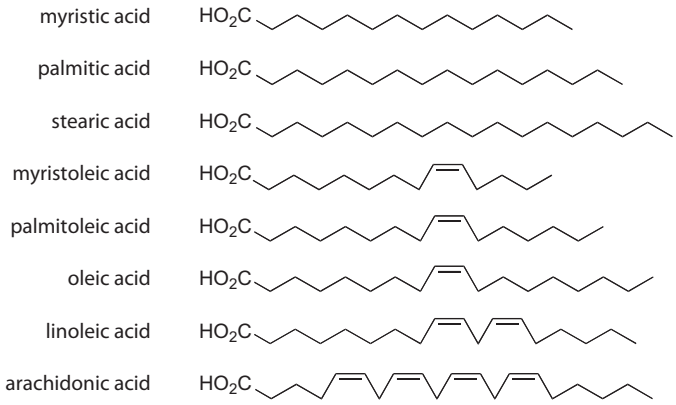



**Figure 8.7 Human fatty acids.** Fatty acids in humans are saturated or have nonconjugated *cis* double bonds.

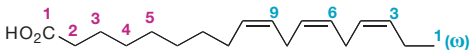



**Figure 8.8 ω-n nomenclature.** There are a number of alternatives to IUPAC chemical nomenclature. Nutritional labels tend to favor the ω-n system in which positions of unsaturation are numbered from the last (ω) carbon. α-linolenic acid above is an ω-3 fatty acid, specifically, 3ω-3 octadecatrienoic acid.

Traditional rules of IUPAC nomenclature enumerate the carbons starting from the carboxylic acid end. Lipid chemists, in contrast, have devised a specialized naming system, which numbers the carbon atoms of lipids starting from the end farthest from the carboxylic acid group (**Figure 8.8**). The last position is named ω-1 (pronounced "omega one"). This naming system is used to categorize ω-3 and ω-6 **fatty acids**, the two types of fatty acids that cannot be biosynthesized by humans. Because these fatty acids must be obtained from the diet, they are called essential fatty acids. In ω-3 fatty acids, there is a *cis* double bond three carbons away from the ω-terminus, regardless of the overall length of the fatty acid. Fish oil is a common source of ω-3 fatty acids. The ratio of ω-6 to ω-3 fatty acids in the human diet is considered critical, because the ω-6 fatty acids can displace ω-3 fatty acids in cell membranes. An ideal human diet would have a 4:1 ratio, but very high levels of ω-6 fatty acids in safflower (50:1 ratio), corn (250:1), and other common plant seed oils make the average Western diet closer to an overall ratio of 10:1 to 20:1. Cooking with canola oil (2:1) can help maintain the balance of ω-6 to ω-3 fatty acids.

## Fatty acid/polyketide synthases are categorized on the basis of their supramolecular structure

The enzymatic machinery that constructs polyketides through iterative carbon–carbon bond-forming reactions works in a highly cooperative fashion, passing synthetic intermediates from one enzymatic subunit to another. Most commonly, the growing polyketide chain is shuttled between clustered enzymatic subunits by a carrier protein. In some cases, all of the enzymatic subunits are ribosomally synthesized as a single polypeptide chain, and these assemblies are referred to as **type I polyketide synthases**. In other cases, the enzymatic subunits are ribosomally synthesized as individual peptide chains that fold independently and then associate through noncovalent interactions; such complexes are referred to as **type II polyketide synthases**. **Type III polyketide synthases** are smaller, more compact enzymes, which lack the domain organization of type I and type II polyketide synthases.

Human **fatty acid synthase** is a type I polyketide synthase, so the catalytic domains are tethered together, as shown schematically in **Figure 8.9**. These multidomain catalytic factories pair up as symmetrical dimers. Mammalian fatty acid synthase is optimized to produce palmitic acid, a saturated aliphatic acid with 16 carbon atoms. This synthesis requires repetitive cycles of aliphatic chain extension by a series of catalytic domains surveyed briefly here and in more detail below.

The **transacylase domain** (MAT) catalyzes the transfer of malonyl (HO2CCH2C(=O)) and acetyl (CH3C(=O)) groups from coenzyme A to the **acyl carrier protein** (ACP), which carries the synthetic intermediates between the enzymatic subunits. The **ketosynthase domain** (KS) catalyzes the Claisen condensation of malonyl and acetyl thioesters, which results in the formation of a β-ketoester (RC(=O)CH2C(=O)SR′) (hence the name of the enzyme). The **ketoreductase domain** (KR) catalyzes the reduction of the β-keto group in RC(=O)CH2C(=O)SR′, following the Claisen condensation, to a β-hydroxyl group. The **dehydratase domain** (DH) catalyzes a dehydration to yield an enone (RCH=CHC(=O)SR′). The **enoyl reductase** (ER) domain catalyzes conjugate reduction of the enone to give a saturated alkanoyl group (RCH2CH2C(=O)SR′. Finally, when the fatty acid reaches an appropriate length after iterative reactions catalyzed by the enzymatic domains, the **thioesterase** (TE) **domain** catalyzes the hydrolysis of the enzyme-bound acyl chain to generate a free fatty acid. In the X-ray crystal structure of the porcine fatty acid synthase, the ACP and TE domains were not well resolved, suggesting a high degree of mobility, which would be expected for the ACP in its role as a transporter. Most bacteria and plants synthesize fatty acids through type II synthases. The domains of fungal fatty acid synthases are less conveniently categorized because the various catalytic domains are present on two polypeptide chains.

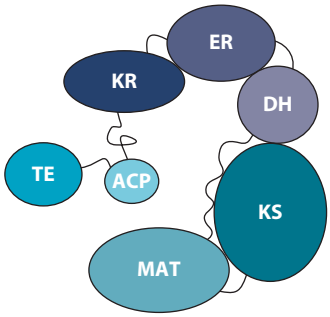



**Figure 8.9 Type I domain organization of mammalian fatty acid synthase.** For clarity, only half of the homodimeric complex is shown. The fully assembled dimeric complex resembles a large "X." TE, thioesterase; ACP, acyl carrier protein; KR, ketoreductase; ER, enoyl reductase; DH, dehydratase; KS, ketosynthase; MAT, malonyl-CoA/acetyl-CoA transacylase domain.

## The acyl carrier protein shuttles the growing polyketide chain from one catalytic domain to another

The chemical transformations catalyzed by polyketide synthases take place exclusively while acyl chains are covalently bound as thioesters to protein subunits. Type I and type II polyketide synthases use an acyl carrier protein (ACP) to shuttle the thioester intermediate between various catalytic subunits. The acyl group of the nascent polyketide is not attached directly to a cysteine thiol of the ACP; instead, the acyl intermediate forms a thioester with a phosphopantetheinyl group that is added post-translationally to a serine residue of the ACP by the enzyme phosphopantetheinyl transferase. In type I polyketide synthases, the acyl carrier protein is part of the same long polypeptide chain composing all of the catalytic domains. It is a wonder that such large multidomain proteins can fold correctly! In such polyketide synthases, the ACP domain is attached by a long flexible peptide linker, which allows the domain to carry the growing polyketide chain to each of the catalytic domains (**Figure 8.10**).

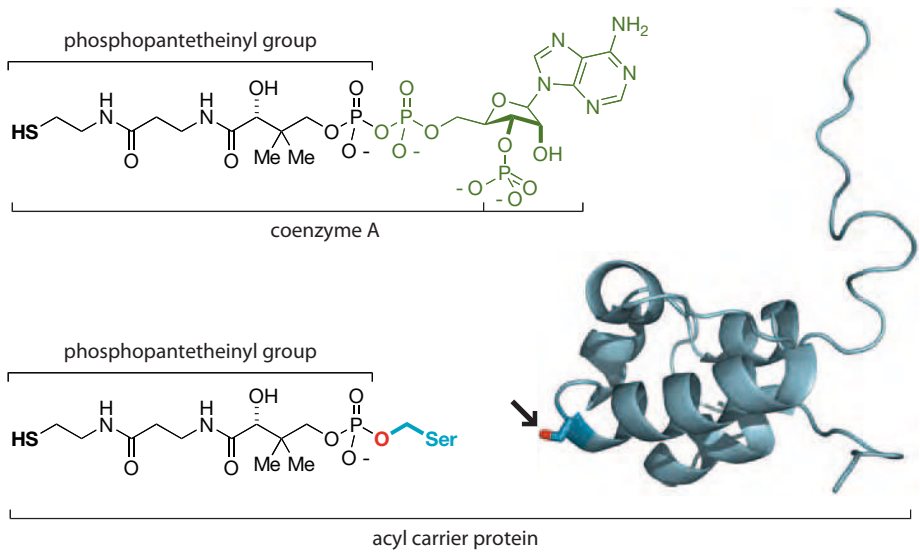



**Figure 8.10 The thiol leash.** The polyketide chain is bound to the acyl carrier protein via a phospho-pantetheinyl linker attached to a serine hydroxyl group (red). The free thiol at the end of the phosphopantetheinyl group provides a reactive site for conjugation to the acyl group of the growing polyketide. (PDB 2PNG)

## A transacylase loads monomeric subunits onto the carrier protein

The biosynthesis of polyketides begins with the loading of the first thioester subunit onto the acyl carrier protein by a transacylase domain. For fatty acids, the first subunit is always acetyl-CoA (**Figure 8.11**). Chemically, this is an unremarkable reaction because thioester exchange is usually facile. However, during each subsequent round of elongation, malonyl-CoA/acetyl-CoA transacylase (MAT) then loads a malonyl subunit, not an acetyl subunit, onto the carrier protein.

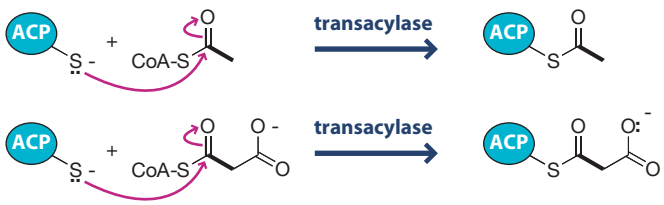



**Figure 8.11 Batter up.** The transacylase loads one of two possible acyl groups onto the carrier protein ACP. Acetyl-CoA (top) is used for the first subunit in fatty acid synthases, and malonyl-CoA (bottom) is used for subsequent rounds of elongation.

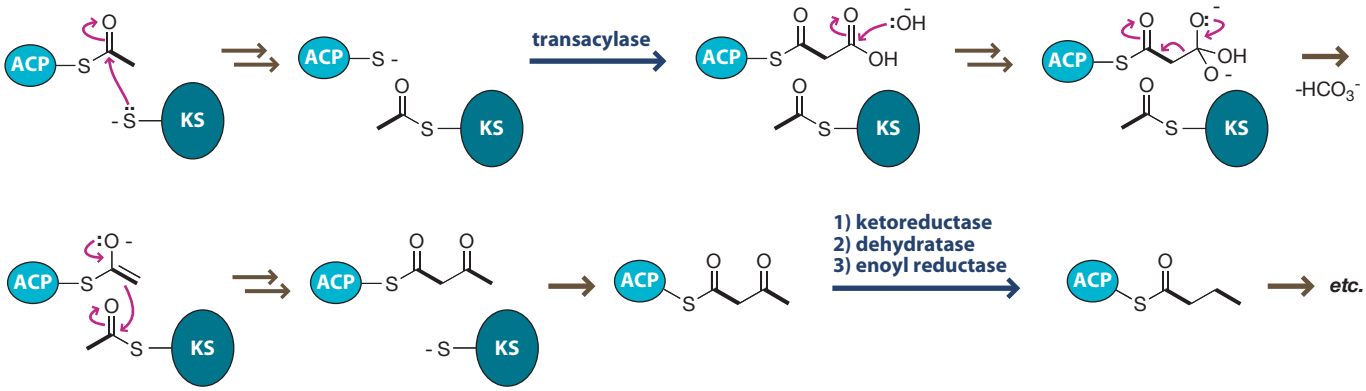



**Figure 8.12 Forming a carbon-carbon bond.** The ketosynthase domain (KS) catalyzes the Claisen condensation during each cycle of polyketide biosynthesis.

## Ketosynthases catalyze a decarboxylative Claisen condensation

Our understanding of the key mechanisms for the Claisen condensation in polyketide biosynthesis comes primarily from studies of FabH, a bacterial fatty acid ketosynthase enzyme. Both the electrophilic thiol ester and the nucleophilic thiol ester enolate are bound to proteins via cysteine side chains (**Figure 8.12**). The electrophilic thiol ester is transferred to the ketosynthase enzyme through thiol ester exchange. The thiol enolate precursor is transferred to an acyl carrier protein as a malonate derivative. Decarboxylation of the malonate in the active site of the ketosynthase enzyme leaves a nucleophilic enolate. Studies of a bacterial ketosynthase strongly support the idea that decarboxylation is initiated by the addition of hydroxide to the carboxylic acid moiety of the malonate, followed by a loss of bicarbonate, in contrast to the more conventional decarboxylation resulting in a loss of carbon dioxide. The resulting enolate is stabilized by two imidazolium ions from histidine residues that can form hydrogen bonds with the enolate oxygen. This stabilization strongly suggests that the ketosynthase mechanism can best be drawn as the reaction of an enol with a thioester. The tetrahedral intermediate generated in the key carbon–carbon bond-forming event is stabilized by two amide N–H bonds—one from the backbone of the ketosynthase and the other from the nearby phosphopantetheine.

Cerulenin, platensimycin, and thiolactomycin (**Figure 8.13**) are natural-product inhibitors of ketosynthases. Cerulenin is a general inhibitor of type I and type II ketosynthases. It covalently reacts with one of the thiolates in the active site. Cerulenin leads to profound weight loss and feeding inhibition in mice, and it garnered a lot of interest as a potential weight-control medication. Although the mechanisms of ketosynthase domains in type I and type II fatty acid synthases are believed to operate through the same mechanisms, the type II ketosynthase used to synthesize fatty acids in bacteria can be selectively inhibited by platensimycin and thiolactomycin. Both platensimycin and thiolactomycin possess chemically reactive functional groups, but they have been shown to be reversible inhibitors.

### Problem 8.2

Cerulenin has four potentially electrophilic sites for reaction with a thiol in the ketosynthase active site (numbered in the structure shown in Figure 8.13). Assuming that the thiol of the ketosynthase active site reacts irreversibly, provide an arrow-pushing mechanism for the reaction that indicates the correct stereochemistry of the product. How did you choose the more chemically reactive carbon of the epoxide?

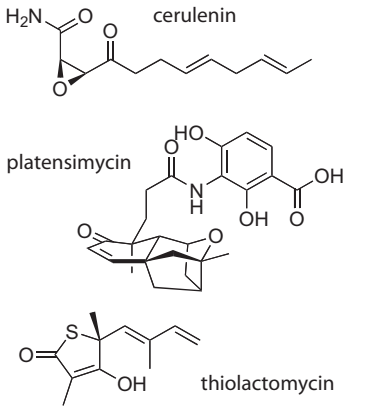



**Figure 8.13 Fat fighters.** A variety of natural products inhibit the ketosynthase activity of fatty acid synthases.

## Ketoreductases catalyze hydride transfer from NADPH

After the Claisen condensation, the second step in fatty acid synthesis is the reduction of the β-ketone to an alcohol. This reaction is catalyzed by the ketoreductase (KR) domain, using NADPH as a hydride donor. The hydride transfer from the

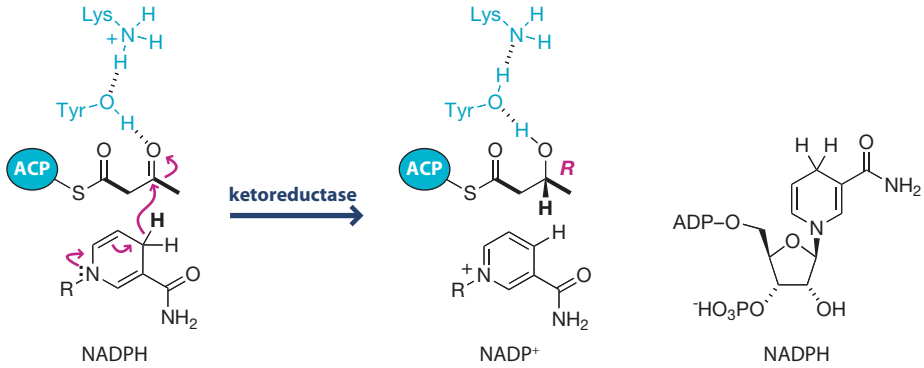



**Figure 8.14 Ketoreductase mechanism.** Formation of an aromatic pyridinium ion drives hydride transfer from NADPH to the β-keto group.

dihydropyridine to the carbonyl group is essentially the reverse mechanism for alcohol dehydrogenase discussed in Chapter 6, except that the carbonyl group undergoing hydride addition is activated not by a $Zn^{2+}$ ion but by a tyrosyl group that is acidified by hydrogen bonding to a lysine ammonium group (**Figure 8.14**). As expected for an enzymatically catalyzed reaction, the hydride addition reaction stereoselectively adds to only one side of the carbonyl carbon; the two faces of planar, $sp^2$-hybridized carbons are designated either *re* or *si* for a clockwise or a counter-clockwise arrangement, respectively, of functional groups, which are prioritized by the Cahn–Ingold–Prelog rules. In mammalian fatty acid synthase, the hydride adds from the *si* face of the carbonyl group to generate an *R* stereocenter.

NADPH differs from NADH by the presence of a single phosphate on the 3′-hydroxyl group. It is important for cells to have two independent pools of reversible hydride donor/acceptors. In humans, NADPH is generally used as a hydride donor/acceptor in biosynthetic transformations, whereas NADH is generally involved in redox equilibria. In a healthy mammalian cell, NADPH is usually used as a hydride donor because the ratio of [NADPH] to [$NADP^+$] is about 200:1. When a hydride acceptor is needed, as it is for alcohol dehydrogenase, it makes more sense for an enzyme to use $NAD^+$ because the ratio of [NADH] to [$NAD^+$] is closer to 1.

## Dehydratases catalyze β-elimination

The third step in fatty acid synthesis involves elimination of the β-hydroxyl group. Two residues have a key role in this reaction, an aspartic acid and a histidine residue. The aspartic acid protonates the hydroxyl group. Then the histidine catalyzes an elimination reaction. In a solution-phase reaction, such an elimination would certainly proceed via a two-step E1cB mechanism involving an enolate as the conjugate base. However, there are no trivial experiments that allow one to rule out a fully concerted E2 mechanism. A useful pattern is revealed by highlighting the individual two-carbon subunits—mainly that the double bonds generated by dehydratases are generally found between the acetyl and propionyl subunits rather than within them (**Figure 8.15**).

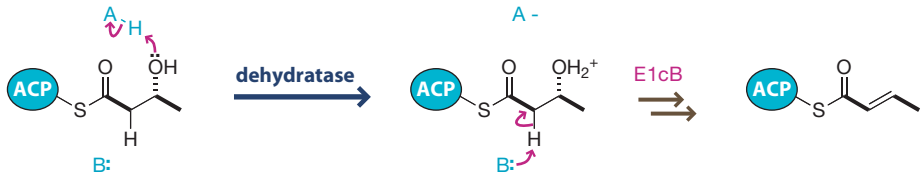



**Figure 8.15 Going and gone.** The β-elimination involves acid and base catalysis. The darkened bonds emphasize the two-carbon units derived from acetyl groups.

## Enoyl reductases catalyze a conjugate reduction

The final step in a single elongation cycle by the fatty acid synthase involves the conjugate reduction of the enone by the enoyl reductase (ER) domain. NADPH again serves as the hydride donor, and the resulting enolate is protonated by a neighboring residue

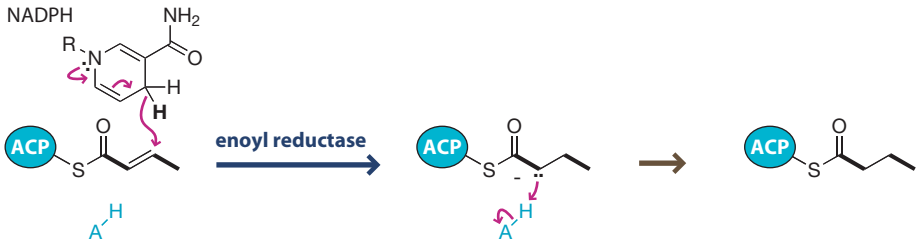



**Figure 8.16 Hydrogenation by enoyl reductase.** In the final step of the elongation of fatty acids, the enoyl reductase domain uses a hydride from NADPH to reduce the conjugated double bond.

(**Figure 8.16**). Both an aspartic acid and a lysine ammonium group are in close proximity in the active site, and it is difficult to know which of these residues actually protonates the enolate intermediate.

Several different enzymes help to maintain the equilibrium between NADPH and $NADP^+$. In *Mycobacterium tuberculosis*, the enzyme KatG oxidizes NADH and NADPH to their oxidized forms. The reaction is slow with oxygen and much faster with peroxide or superoxide. The powerful antibiotic isoniazid, used against tuberculosis, is oxidatively activated by KatG (**Figure 8.17**). The resulting reactive intermediate is believed to be an acyl radical, which then undergoes a Minisci radical addition reaction with the pyridinium ion of $NADP^+$. The resulting adduct is a potent inhibitor of the enoyl reductase in *M. tuberculosis*. Thus, patients taking isoniazid are tricking the bacterium into synthesizing the instruments of its own destruction.

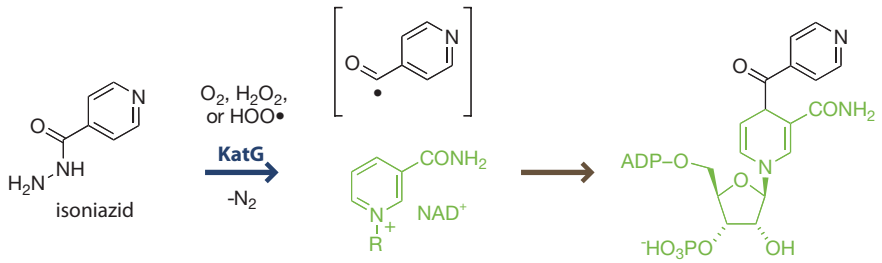
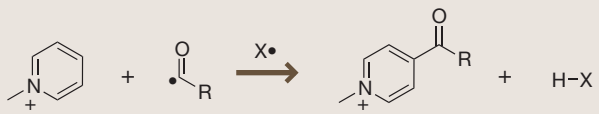



**Figure 8.17 Booby trap.** The bacterial enzyme KatG unwittingly synthesizes a deadly inhibitor from isoniazid.

## Problem 8.3

Suggest a plausible arrow-pushing mechanism (using fishhook half-arrows) for the Minisci reaction of acyl radicals with pyridinium ions.




## A thioesterase uses a catalytic triad to cleave the acyl group from the acyl carrier protein

Synthesis of the $C_{16}$ palmitic acid requires eight cycles through the fatty acid synthase machinery described above. Once the chain has reached the intended length (as determined by the size of the active site), the acyl chain is directed to the active site of the thioesterase domain. The thioesterase uses a catalytic triad—composed of aspartate, histidine, and serine—like those found in serine proteases (**Figure 8.18**). The nucleophilic serine residue attacks the thioester linkage that attaches the acyl chain to the carrier protein. Water then replaces the thiol in the active site and the mechanistic sequence is reversed.

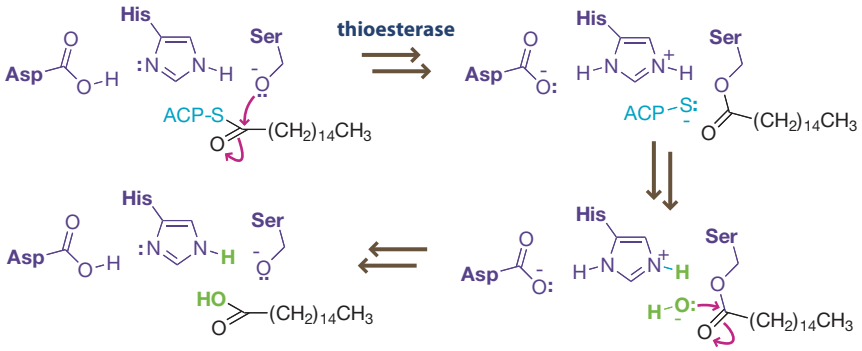



**Figure 8.18 Release.** Thioesterase cleaves the finished fatty acid from acyl carrier protein by using a catalytic triad analogous to that found in a serine protease.

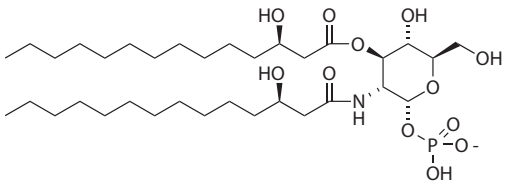
### Enzymes associated with the endoplasmic reticulum put the finishing touches on fatty acids

Molecules of palmitic acid that are released from the acyl carrier protein of human fatty acid synthase can be incorporated directly into lipids through an enzymatic esterification reaction. However, such intermediates are also subject to further transformations by enzymes associated with the endoplasmic reticulum. They can be extended by two carbons to stearic acid through a series of reactions analogous to those just described, or *cis* double bonds can be introduced through oxidative enzymes called dehydratases (as with palmitoleic acid).

## 8.3 THE BIOLOGICAL ROLE OF HUMAN POLYKETIDES

### Eight categories of lipids are found in biology

By definition, a lipid is any cellular molecule that dissolves in nonpolar organic solvents and not in water. Based on that definition, eight categories of lipids are found in living organisms—namely, fatty acids, glycerolipids, glycerophospholipids, sphingolipids, saccharolipids, sterols, prenols, and polyketides. Most molecules that are present in the human **lipidome** (the totality of lipids in cells) derive their hydrophobic character from polyketide chains, particularly fatty acids, and we will focus on such lipids in this section. Saccharolipids are comparatively rare, with only about a dozen bacterial examples known and characterized (**Figure 8.19**). The broadest class of lipids, polyketides, includes around 7000 examples. The field of lipidomics relies largely on mass spectrometry and other tools to characterize the complete set of lipids from an organism, tissue, or cell type. As yet, the complete lipidome for humans has not been completely defined, and might be complicated if it includes lipids from the tremendous numbers of different microorganisms that colonize humans.




saccharolipids: lipid X
(2,3-bis-(3*R*-hydroxy-tetradecanoyl)-α-D-glucosamine 1-phosphate)

**Figure 8.19 Molecular anchor bolt.** Lipid X is one of the basic building blocks of larger bacterial saccharolipids, which anchor long polysaccharide chains to the outer bacterial membrane.

### Lipid membranes are composed of lipids with a polar head group and a nonpolar tail

From the simplest category of lipids, fatty acids are present in two of the three general classes of molecules that make up mammalian lipid bilayers (**Figure 8.20** and **Figure 8.21**): phospholipids and glycosphingolipids, but not cholesterol. Lipids are present

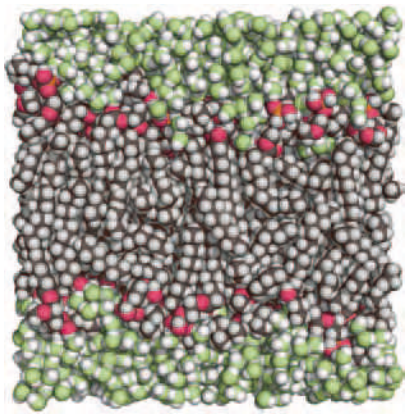


**Figure 8.20 Lipid barrier.** A model of a lipid bilayer generated through molecular dynamics reveals substantial disorder within the lipid chains, consistent with the fluid-like nature of the lipid bilayer. Water molecules are shown in green.

**Table 8.1 Lipid composition of an average mammalian cell membrane.**

| Lipid | Percentage |
|---|---|
| Phosphatidylcholine | 45–55 |
| Phosphatidylethanolamine | 15–25 |
| Phosphatidylinositol | 10–15 |
| Cholesterol | 10–20 |
| Sphingomyelin | 5–10 |
| Phosphatidylserine | 5–10 |
| Cardiolipin | 2–5 |
| Phosphatidic acid | 1–2 |

in membranes as combinatorial mixtures, with different combinations of head groups and fatty acids. Phospholipids have a phosphatidic acid tail with two fatty acid ester groups. Sphingolipids have a ceramide tail, with one fatty acid attached to the amino group of the aminoalcohol sphingosine. There is significant variation in the polar head groups of phospholipids. Choline is the most common head group of mammalian phospholipids, but ethanolamine, inositol, and serine are also common (**Table 8.1**).

The two leaflets of the plasma membrane have distinctly different phospholipid compositions. The leaflet facing the cytoplasm, for example, has an excess of anionic phospholipid head groups. The appearance of such anionic head groups on the outside of the cell is one indicator of apoptosis. Maintaining this asymmetry requires energy, because the hydrophilic and charged head group cannot readily diffuse through the aliphatic-packed center of the membrane. Enzymes called flippases catalyze the flip-flop of phospholipids from one side of the membrane to the other. These P-type ATP-binding transporters both maintain the membrane and have a role in the synthesis of the glycoconjugated phospholipids.

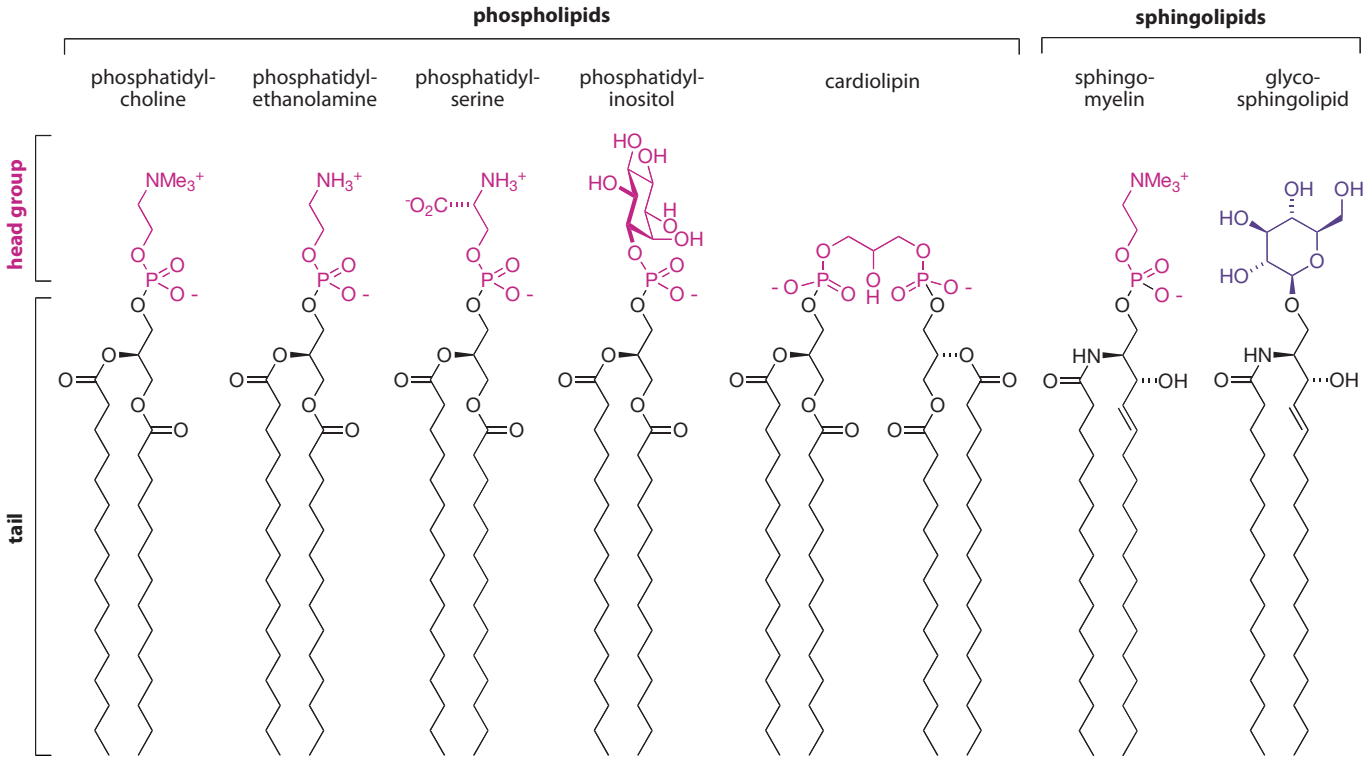


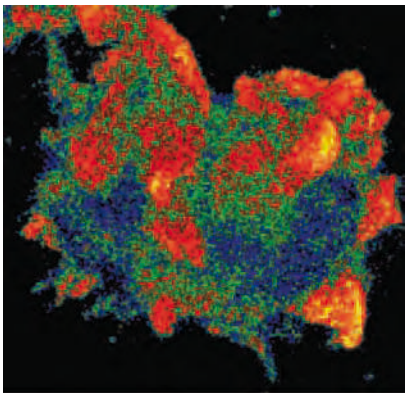

**Figure 8.21 Common lipid components of human cell membranes.** These lipids exist as mixtures with acyl groups of various lengths and various degrees of unsaturation. Most of the head groups are anions or zwitterions, with the exception of the glycosyl groups of glycosphingolipids.

## The lipid bilayer entropically favors interactions between embedded molecules

Proteins embedded within lipid bilayers have fewer degrees of freedom than molecules in solution. Molecules bound to lipid bilayers have essentially two degrees of translational freedom (side-to-side movement) and at most one degree of rotational freedom (spinning perpendicular to the membrane). Membrane-bound proteins are translated directly into the membrane with a precise orientation, which remains fixed. Glycans always point outward from the cell, SH2 protein domains always point inward into the cytoplasm, and so on. With fewer degrees of freedom, interactions between membrane-bound molecules should have a natural entropic advantage over solution-phase reactions. However, the complexity of cell membranes makes this entropic advantage difficult to quantify. Lipids diffuse up to two orders of magnitude more slowly in real cell membranes than in artificial membranes, which reflects the considerable heterogeneity of the plasma membrane and contacts to the actin skeleton.

A wide range of signaling proteins, such as Hedgehog, are anchored to the plasma membrane through ester, thioester, and amide linkages to long-chain fatty acids such as $C_{14}$ myristic acid and $C_{16}$ palmitic acid. Other signaling proteins, such as Ras and Raf, are anchored to membranes by terpenes. These signaling proteins usually lose their function when the lipid anchor is removed and the protein is allowed to float aimlessly within the cytoplasm.

Until the late 1980s, the fluid mosaic model of membranes held that phospholipids, glycolipids, cholesterol, and proteins were randomly and evenly distributed throughout the lipid bilayer. However, most membrane-bound proteins associate with proteins or microfilaments on the cytosolic side of the membrane. Those structures impose order on the components of the membrane. Furthermore, it is now known that the outer membranes of human cells possess microdomains called **lipid rafts** that are particularly rich in glycosphingolipids and cholesterol (**Figure 8.22** and **Figure 8.23**). Many transmembrane receptors, such as immunoglobulin E receptors, T-cell antigen receptors, and B-cell receptors, are clustered in lipid rafts. Thus, lipid rafts are often the focal points for cell signaling.



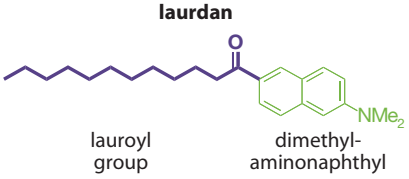



**Figure 8.23 Paint the cell. T**he fluorescence of the lauric acid derivative laurdan is dependent on the polarity of the environment. A human macrophage stained with laurdan reveals areas of high generalized polarization in yellow. Areas of high polarization correlate with lipid rafts. (Top, from K. Gaus et al., *Proc. Natl. Acad. Sci. USA* 100:15554–15559, 2003. With permission from the National Academy of Sciences.)

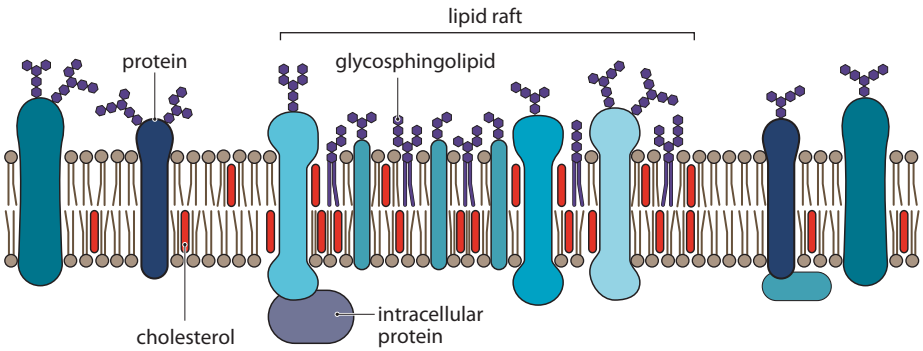



**Figure 8.22 Lipid rafts are strongly associated complexes of lipids, proteins, and cholesterol.** (Adapted from Inside the Cell, NIH Publication No. 05-1051, 2005.)

## Phospholipases generate distinct chemical signals by hydrolyzing various bonds of phospholipids

Many of the phospholipids in mammalian membranes serve as substrates for highly specific hydrolases (**Figure 8.24**). Any enzyme that selectively hydrolyzes the acyl group from the 1 position of the glycerol core is called a **phospholipase** A1 (PLA1). One version of human phospholipase A1 is specific for phosphatidylserine. Any enzyme that selectively hydrolyzes the acyl group from the 2 position of the glycerol core is called a phospholipase A2 (PLA2). Some versions of phospholipase A2 are secreted from human cells, whereas others are localized in the cytosol. Some types of phospholipase A2 cleave arachidonoyl esters from the C2 position of phospholipids. The arachidonic acids are converted to various signaling molecules that are important in inflammatory immune responses. Wasp, hornet, and snake venoms contain various

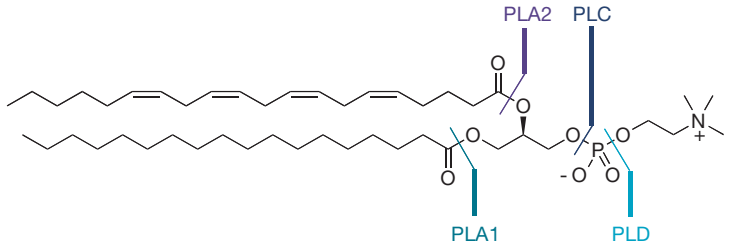



**Figure 8.24 Regioselectivity of various classes of phospholipases acting on a generic lipid substrate.** PLA1 is phospholipase A1, PLA2 is phospholipase A2, PLC is phospholipase C, and PLD is phospholipase D.

phospholipases (A1 and A2) that act on all kinds of phospholipids. Phospholipases C and D (PLC and PLD, respectively) target phosphodiester bonds. Phospholipase C cleaves the polar lipid head group from the hydrophobic lipid tail, whereas phospholipase D cleaves the other phosphodiester bond from phosphatidylcholine to generate phosphatidic acid and choline. The science of lipid signaling molecules has been slow to evolve. Simple ideas like concentration, measured in mol $L^{-1}$, aren't readily applied to molecules that reside in two-dimensional membranes.

## Phospholipase Cβ generates two signaling molecules

Recall from our discussion of protein kinase regulation in Chapter 6 that the activity of protein **kinase** C (PKC) is controlled by two regulatory domains. The C2 domain acts as a calcium sensor, whereas the C1 domain is activated by 1,2-diacyl-*sn*-glycerol (often referred to as diacylglycerol). Don't worry about the *sn* (*stereospecific numbering*) nomenclature, which is specific for lipids based on the prochiral molecule glycerol (**Figure 8.25**). Although most lipids are made through the attachment of phosphoryl head groups to diacylglycerols, the concentration of diacylglycerols is low in a resting cell. Activation of some 7TM G protein-coupled receptors coupled to the heterotrimeric G protein $G_q$ leads to the activation of phospholipase Cβ. Phospholipase Cβ cleaves phosphatidylinositol 4,5-bisphosphate ($PIP_2$) to give 1,2-diacyl-*sn*-glycerol and inositol 1,4,5-triphosphate (**Figure 8.26**). Inositol triphosphate leads to the opening of calcium storage vesicles, whereas diacylglycerol directly binds to PKC. PKC is generally active in cells that are actively performing their physiological function, such as the contraction of smooth muscle cells; secretion by stomach, salivary, and lacrimal cells; and the generation of glucose by fat and liver cells.

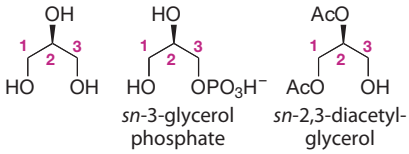



**Figure 8.25 Numbering glycerol.** With stereospecific numbering (*sn*), the positions of glycerol are assigned fixed numbers, regardless of the Cahn–Ingold–Prelog priorities of the substituents.

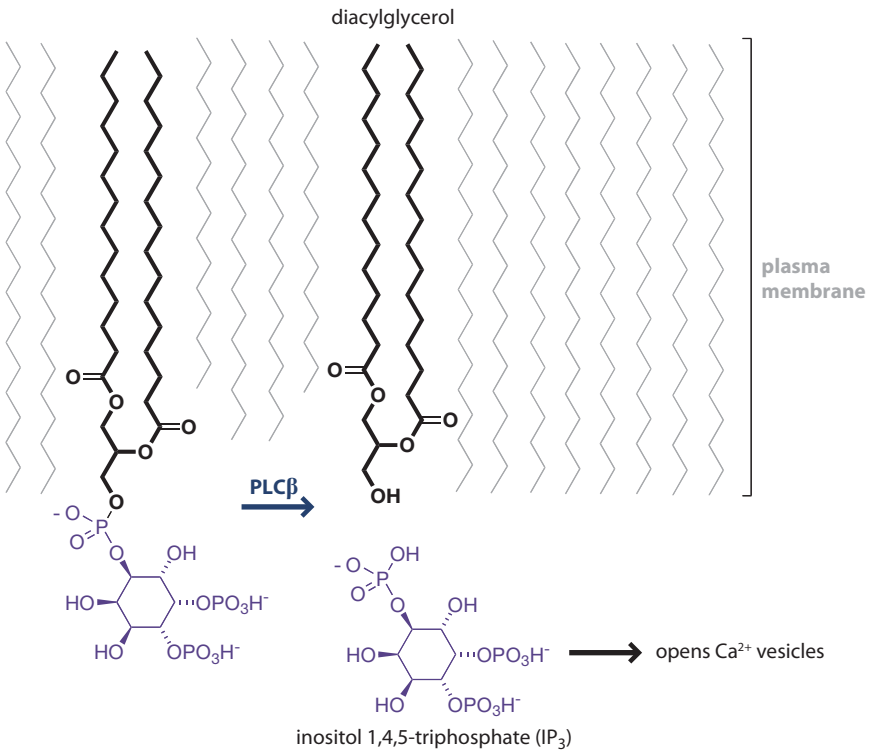



**Figure 8.26 PLCβ initiates a cytosolic calcium burst.** The activation of some seven-transmembrane receptors leads to the dissociation of the heterotrimeric G protein subunits. Depending on the cell, either the Gα or Gβγ subunit activates phospholipase Cβ (PLCβ), which generates diacylglycerol (DAG), an activator of protein kinase C.

The binding of diacylglycerol to the C1 domain of PKC increases the affinity for the plasma membrane. This activation subsides as the diacylglycerol is converted back into lipids or hydrolyzed. Some bioactive natural products, such as phorbol esters, debromoaplysiatoxins, ingenols, teleocidins, and bryostatins, lead to persistent activation by binding to the diacylglycerol-binding site of PKC. Persistent activation is associated with uncontrolled cell growth and consequent tumor formation.

## Arachidonic acids are converted into diverse signaling molecules during inflammation

A wide range of human cells convert the fatty acid arachidonic acid into eicosanoid signaling molecules, usually in response to inflammatory signals (**Figure 8.27**). There are four general classes of eicosanoid signaling molecules: thromboxanes, prostacyclins, prostaglandins, and leukotrienes. These signaling molecules usually act locally, on nearby tissues. Most eicosanoids affect two types of cells—namely, smooth muscle cells and cells derived from hematopoiesis, such as platelets, white blood cells, and bone-forming cells. The first step in the biosynthesis of thromboxanes, prostacyclins, and prostaglandins is the conversion of arachidonic acid (released by phospholipase A2) into prostaglandin $H_2$ ($PGH_2$) by the enzyme cyclooxygenase. Humans have two variants of cyclooxygenase, COX-1 and COX-2, both of which are the target of aspirin and other nonsteroidal anti-inflammatory drugs such as ibuprofen and naproxen. COX-1 is expressed constitutively in all cells, whereas the expression of COX-2 is upregulated in response to proinflammatory signals. Aspirin is an irreversible inhibitor of both cyclooxygenases, because aspirin acylates a serine hydroxyl group in the active site (**Figure 8.28**). In some patients, the high doses of aspirin needed for chronic pain relief lead to bleeding in the stomach as a result of COX-1 inhibition, so new selective inhibitors of COX-2 were developed (for example rofecoxib, valdecoxib, and celecoxib). Unfortunately, these selective COX-2 inhibitors were unexpectedly found to cause cardiovascular injuries in some patients.

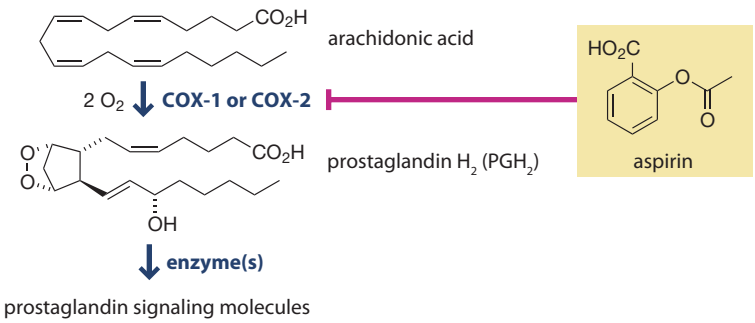



**Figure 8.27 The start of prostaglandin signaling.** The first step in prostaglandin biosynthesis is the conversion of arachidonic acid to prostaglandin $H_2$ ($PGH_2$). The red line with a bar at the end implies inhibition.

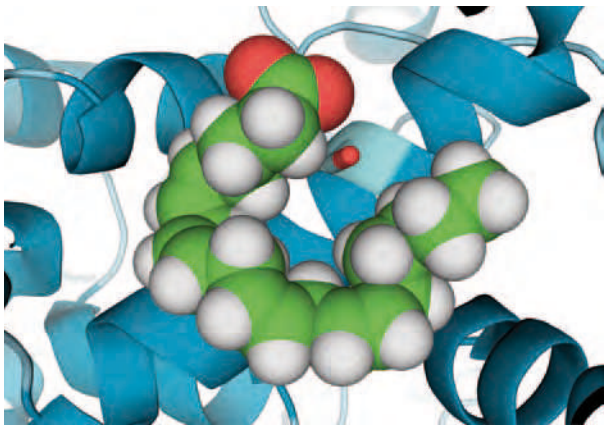


**Figure 8.28 Active site for inflammation.** Arachidonic acid (space-filling) nestles within the active site of cyclooxygenase-2 (COX-2). Ser530 (sticks) in the active site of COX-2 is susceptible to acylation by aspirin. (PDB 1CVU)

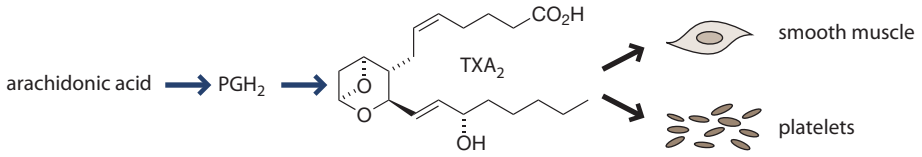



**Figure 8.29 Signaling by a strained bicyclic acetal.** Thromboxane $A_2$ ($TXA_2$) is produced by activated platelets.

The $PGH_2$ is further converted into other prostaglandins, dependent on cell type, that can signal neighboring cells. Most of these prostaglandin signaling molecules, or chemical analogs, are sold as prescription pharmaceuticals. When platelets are activated as part of a blood clotting response, they convert prostaglandin $H_2$ into thromboxane $A_2$ ($TXA_2$; **Figure 8.29**), which has a half-life of 32 seconds, as a result of the hydrolysis of the strained bicyclic acetal. The hydrolysis product, thromboxane $B_2$, has no known biological activity. Because of the brief half-life, $TXA_2$ acts only on nearby smooth muscle, causing vasoconstriction, and on other nearby platelets, resulting in a localized autocatalytic effect that amplifies clot formation.

## Problem 8.4

Thromboxane $A_2$ is unstable at physiological pH, with a half-life of less than 1 minute. Predict the product of thromboxane $A_2$ hydrolysis.

Cells of the vascular endothelium that line blood vessels convert prostaglandin $H_2$ into prostaglandin $I_2$, more commonly called prostacyclin (**Figure 8.30**), which counters the effects of $TXA_2$. Like $TXA_2$, prostacyclin is unstable at physiological pH because of the hydrolyzable enol ether, and has a half-life of under 1 minute at physiological pH. It acts on platelets to inhibit aggregation and also acts on nearby smooth muscle cells, causing vasodilation.

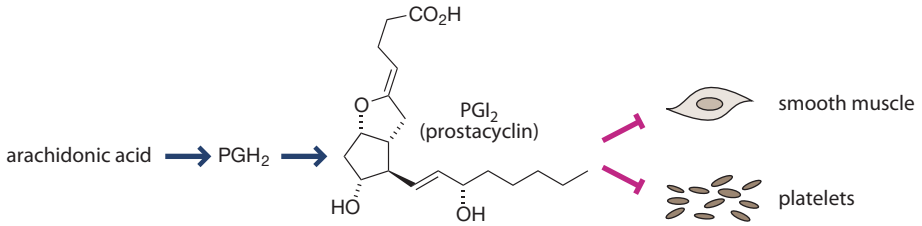



**Figure 8.30 Signaling by an enol ether.** Prostacyclin ($PGI_2$) is produced by vascular endothelial cells.

## Problem 8.5

Prostacyclin is unstable at physiological pH, with a half-life of less than 1 minute. Predict the product of prostacyclin hydrolysis.

Mast cells are a type of white blood cell that have an important role in allergic responses. In response to proinflammatory signals, the cells release the allergic signaling molecule histamine and convert prostaglandin $H_2$ into prostaglandin $D_2$ ($PGD_2$; **Figure 8.31**). To mimic this effect in the laboratory, lipopolysaccharides derived from bacterial cell walls are commonly used as a proinflammatory signal, which can

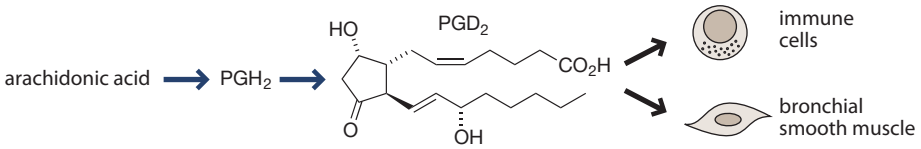



**Figure 8.31 Prostaglandin $D_2$ ($PGD_2$) produced in mast cells leads to several allergic responses.**

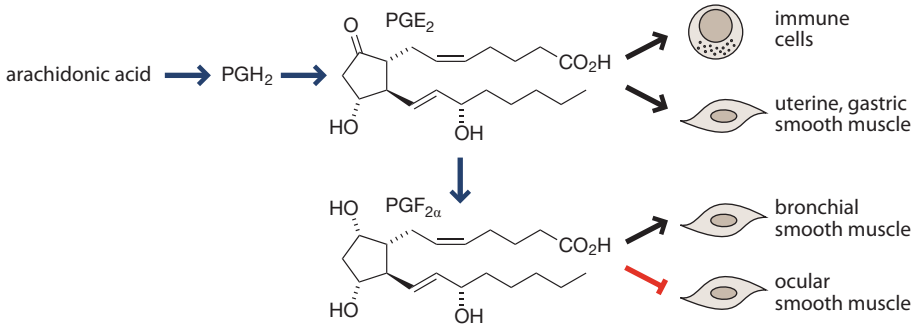



**Figure 8.32 Prostaglandin signaling.** Prostaglandin $E_2$ ($PGE_2$) induces rapid inflammatory responses and slower responses affecting smooth muscle.

be added to the cell culture. $PGD_2$ activates several types of immune cells, such as T helper cells, eosinophils, and basophils. It also induces the contraction of bronchial smooth muscle, which causes the shortness of breath associated with asthma. $PGD_2$ acts on cells within the nervous system and on mucus-releasing goblet cells.

Prostaglandin $E_2$ ($PGE_2$) is the most abundant prostanoid in the human body and is generated from prostaglandin $H_2$ by prostaglandin $E_2$ synthase (**Figure 8.32**). Depending on the cell type and location, $PGE_2$ can affect inflammation, the integrity of the gastric mucosa, fertility, and parturition (labor and delivery). A cytosolic version of prostaglandin $E_2$ synthase is primarily associated with rapid inflammatory responses. In stomach, uterine, and other cells, a microsomal (that is, derived from vesicles of the endoplasmic reticulum) form of prostaglandin $E_2$ synthase, usually associated with inducible COX-2 activity, is associated with slower smooth muscle responses such as stomach distress, the menstrual cycle, and childbirth. However, the expression of microsomal $PGE_2$ synthase is also upregulated in response to proinflammatory signals. $PGE_2$ has several effects related to childbirth. It relaxes smooth muscle in the cervix, for example, while inducing uterine contractions. It also acts on osteoblasts, which then recruit osteoclasts to assist with pelvic bone remodeling.

In some cells, $PGE_2$ is further converted into prostaglandin $F_{2\alpha}$, and the two tend to act in concert. $PGF_{2\alpha}$ leads to the contraction of uterine smooth muscle cells and to the degradation of the luteal cells that support the developing egg before ovulation in the menstrual cycle. Like $PGD_2$, $PGF_{2\alpha}$ is also important in asthma, because the compound leads to the contraction of bronchial smooth muscle cells. In the human eye, $PGE_2$ and $PGF_{2\alpha}$ inhibit the contraction of smooth muscle cells, lowering the intraocular pressure.

Leukotrienes represent the fourth class of eicosanoid signaling molecules. They generally act to sustain the rapid initial inflammatory response and are important in diseases such as asthma. Leukotrienes are produced by inflammatory immune cells—namely mast cells, macrophages, eosinophils, and neutrophils. Leukotrienes do not arise through the action of cyclooxygenase or the intermediation of prostaglandin $H_2$. Instead, the leukotrienes are generated by the action of 5-lipoxygenase on arachidonic acid to generate 5-hydroperoxyeicosatetraenoic acid (5-HPETE) (**Figure 8.33**). 5-HPETE is converted into leukotriene $A_4$, which has two potential fates. In some mast cells and monocytes, leukotriene $A_4$ is hydrolyzed to leukotriene $B_4$; in other mast cells, eosinophils, and basophils, leukotriene $A_4$ is converted into cysteinyl leukotrienes. Leukotriene $B_4$ exerts a powerful chemotactic response, attracting leukocytes and some cytotoxic T lymphocytes to sites of inflammation.

The synthesis of the cysteinyl leukotrienes begins with an enzyme-catalyzed $S_N2$ reaction of leukotriene $A_4$ with glutathione. The amide bonds of the glutathione moiety are cleaved to generate leukotriene $D_4$ and leukotriene $E_4$ (see Figure 8.33). The cysteinyl leukotrienes act on cysteinyl leukotriene 1 receptors, causing contraction of bronchial smooth muscle. The blockbuster asthma drug montelukast blocks this key interaction. The cysteinyl leukotrienes also act on cysteinyl leukotriene 2 receptors expressed on, for example, vascular endothelial cells, neuronal cells, adrenal gland cells, and cardiac Purkinje cells.

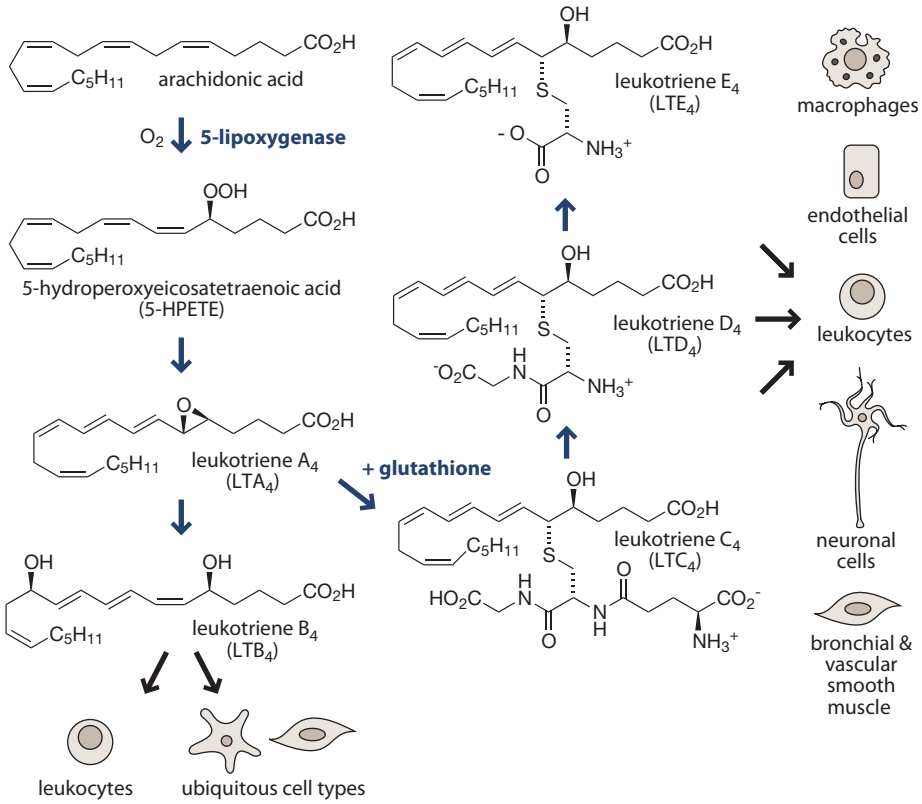
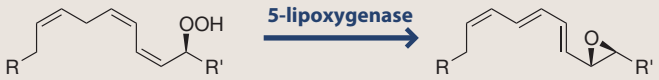



**Figure 8.33 Signaling by leukotrienes.** Leukotrienes derived from arachidonic acid mediate sustained inflammatory responses.

## Problem 8.6

Suggest a plausible arrow-pushing mechanism for the enzymatic conversion of 5-HPETE into leukotriene $A_4$, assuming that the enzyme 5-lipoxygenase has acidic and basic residues in its active site. The following is an abbreviated version of the reaction shown in Figure 8.33.

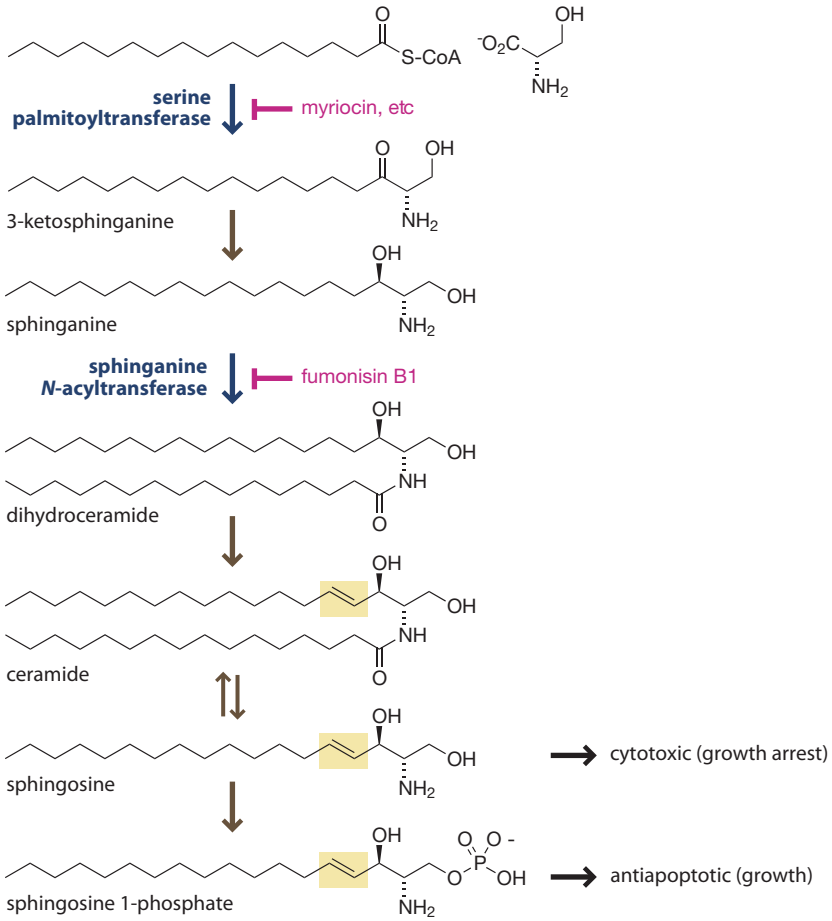



## Sphingosine derivatives are important in intracellular signaling

Sphingolipids are essential components of human cell membranes, yet the intermediates in sphingolipid biosynthesis also have important effects on cell function (**Figure 8.34**). Many of these sphingoid molecules exhibit profound biological activities when added to cells. Ceramide halts cell growth, for example, and is cytotoxic as a result of interference with cell cycle regulation via binding to protein phosphatases and the protease cathepsin D. Sphingosine has similar effects on cells. Interestingly, sphingosine 1-phosphate has the opposite effect, inducing the inhibition of apoptosis and causing cell growth. The enzyme sphingosine kinase 1 required to phosphorylate sphingosine is recruited to the plasma membrane by phosphatidic acid, a product of phospholipase D.

A variety of natural mycotoxins synthesized by microfungi target early steps in sphingoid biosynthesis (**Figure 8.35**). Myriocin, sphingofungins, and lipoxamycin inhibit the first step in sphingoid biosynthesis, which is a decarboxylative C-acylation reaction of serine with palmitoyl-CoA. These compounds exhibit immunosuppressant activity and have been shown to inhibit the proliferation of mouse T cells.

**Figure 8.34 Stop and go.** The biosynthesis of sphingolipid signaling molecules involves several enzyme-catalyzed steps. Mycotoxin inhibitors of this pathway (red) are immunosuppressants.




Fumonisin B1 inhibits the further acylation of sphinganine to produce dihydroceramide. The toxin is produced by a fungus (*Fusarium moniliforme*) that infects corn. The fungus has been shown to cause pulmonary edema in pigs, leukoencephalomalacia in horses, and tumors in rats.

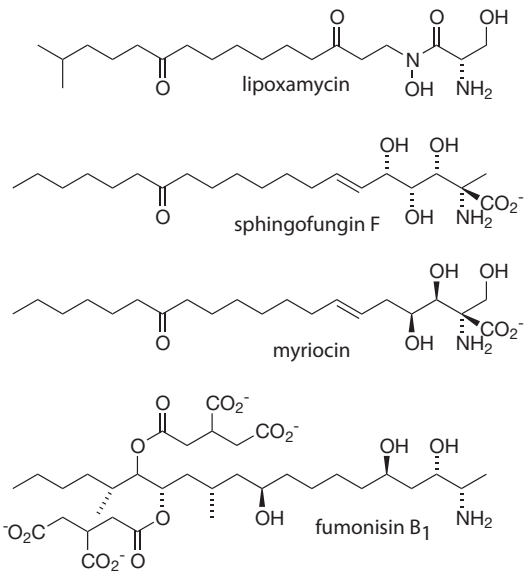


**Figure 8.35 Mycotoxins produced by the mold *Fusarium moniliforme*.** This mold infects corn, leading to the ear rot shown in the photograph. (Photograph courtesy of Alison Robertson, Iowa State University.)

## Metal-catalyzed hydrogenation of unsaturated fats changed the human diet

Obesity and diabetes are two of the most challenging health problems facing industrialized nations. Whereas most diseases of developing countries are diseases of deficiency (that is, too little food, water, and vitamins), most diseases of developed countries are diseases of excess (that is, too much food, cholesterol, sugar, and fat). The major problem with the modern human diet is not merely the quantity of fat, but the type of fat. Polyunsaturated fatty acids and *cis* fatty acids are an essential part of the human diet, and they are healthier than *trans* fatty acids and saturated fatty acids, which contribute to blockage of the coronary arteries (**Figure 8.36**).

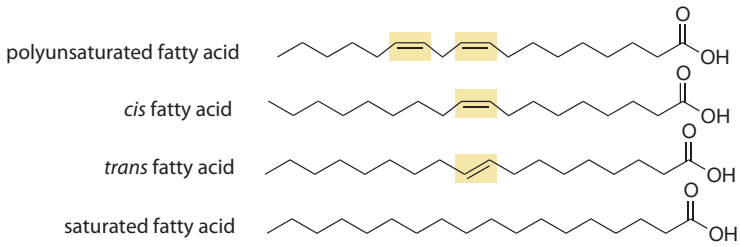



**Figure 8.36 Classifying fatty acids.** The number of double bonds and stereochemistry (*cis* or *trans*) distinguish the types of fatty acids used in nutritional labeling.

Saturated fatty acids and *trans* fatty acids can pack tightly together and tend to be solids at room temperature. In contrast, *cis* fatty acids cannot readily pack together and tend to be liquids. Unfortunately, highly liquid fats based on polyunsaturated fatty acids and *cis* fatty acids readily undergo autoxidation. In 1820, the Swiss chemist Nicolas-Théodore de Saussure noted that a thin layer of walnut oil was able to absorb 150 times its volume of oxygen over a one-year period. Back in the days of horse-drawn wagons, oils and fats often went rancid owing to a combination of short shelf life and slow transport. Solid fats derived from saturated fats tend to have longer shelf lives, however, and they give foods a smooth velvety texture that coats the tongue.

In 1909 Procter & Gamble began to market Crisco®, a form of partly hydrogenated cottonseed oil with a long shelf life (**Figure 8.37**). Partial hydrogenation converts

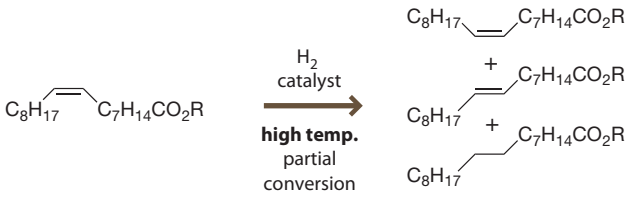
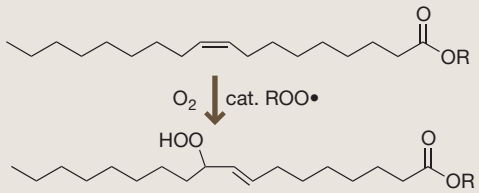


**Figure 8.37 Cooking cottonseed oil.** Partly hydrogenating vegetable oil turns *cis* fats into *trans* fats and saturated fats.

good, healthy, *cis* fats to unhealthy *trans* fats and unhealthy saturated fats. Before the development of partly hydrogenated vegetable oils, the only sources of *trans* fats in a natural diet were meat and dairy products from cows and sheep (less than 5%). Partly hydrogenated vegetable oils have become a cheap staple of western diets. Before 2008, a large order of french fries from the world's largest hamburger chain contained 8 g of *trans* fat (out of 30 g), but since then the company has started making its french fries with oil that is free of *trans* fats. In modern times, it is unnecessary to rely on partly hydrogenated oils, because transportation is fast and shelf life is not an issue. Healthy diets tend to be made up of oils such as olive oil, in which 55–85% of the fatty acids are oleic acid, a *cis* fatty acid.

## Problem 8.7

Using fishhook-style arrows, suggest a plausible arrow-pushing mechanism for the radical chain oxidation of lipids.




## Some lipids from lower organisms contain cyclopropane rings

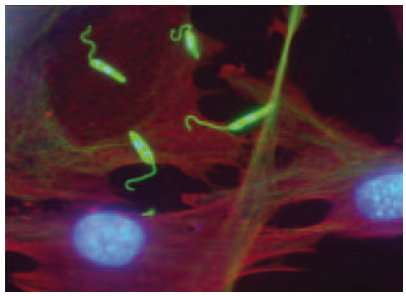


**Figure 8.38 The free-swimming form of *Leishmania* (green), called a promastigote, infecting a macrophage.** (Courtesy of the Beverley Laboratory, Washington University School of Medicine, St. Louis.)

In the 1950s, the bacteria in the genus *Lactobacillus* were found to contain lipids that incorporated cyclopropane rings. More than 16% of the fatty acid content of lipids from the common strain *Lactobacillus casei* is composed of the cyclopropane-containing fatty acid lactobacillic acid. *Leishmania* cells produce another cyclopropane fatty acid, dihydrosterculic acid. Sandflies transmit the protozoal parasite *Leishmania*, which causes leishmaniasis, a disease characterized by disfiguring open sores. The *Leishmania* cells enter and proliferate inside macrophages, which normally kill microorganisms (**Figure 8.38**).

Cyclopropane fatty acids are biosynthesized in two steps from saturated fatty acids (**Figure 8.39**). The first step involves the oxidative introduction of a *cis* double bond by a desaturase enzyme, and the second step involves cyclopropanation by the sulfonium ion *S*-adenosylmethionine. Interestingly, the biosynthesis of dihydrosterculic acid was shown to be most strongly inhibited by 9-thiastearic acid, although actual growth inhibition of *Leishmania* was strongest for the 10-isomer. In similar studies against the parasite *Trypanosoma cruzi*, it was speculated that the thia fatty acids were inhibiting the desaturase enzyme.

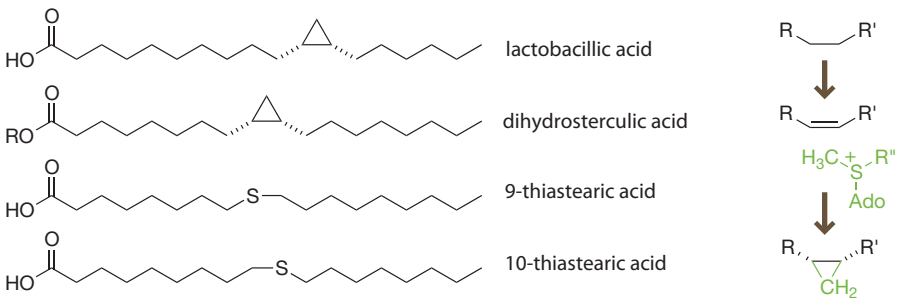
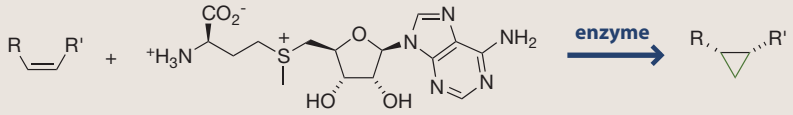



**Figure 8.39 Cyclopropane- and thia-based lipids.** Some lipids from microorganisms, such as lactobacillic acid and dihydrosterculic acid, contain cyclopropane rings generated through the two-step process shown on the right. 9-Thiastearic acid and 10-thiastearic acid, in contrast, inhibit the biosynthesis of dihydrosterculic acid.

## Problem 8.8

Suggest a plausible arrow-pushing mechanism for the following base-promoted reaction.



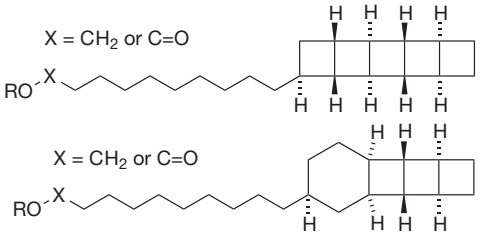
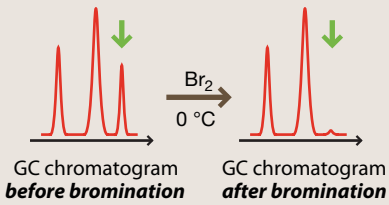



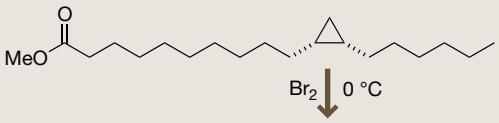

**Figure 8.40 Lipid ladders.** Ladderane fatty acids from *Candidatus Brocadia anammoxidans* contain cyclobutane rings.

An amazing set of lipid structures comes from the ammonotrophic bacterium *Candidatus Brocadia anammoxidans*. About half of the total lipid content in these bacteria is composed of ethers and fatty acids containing concatenated four-membered rings (**Figure 8.40**). Such compounds are named ladderanes for their resemblance to ladders in a two-dimensional depiction.

## Problem 8.9

Cyclopropane fatty acids can be distinguished from saturated fatty acids by hydrogenating mixtures of fatty acid esters and then subjecting them to gas chromatography (GC) before and after treatment with bromine; after the treatment, the peaks for the cyclopropane-containing molecules (green arrows) disappear from the chromatogram. Suggest a plausible arrow-pushing mechanism for the reaction of lactobacillic acid with bromine to yield two diastereomeric dibromides.




### Acylation of human proteins induces membrane localization

Proteins participating in signal transduction pathways interact with each other through specific high-affinity interactions. Association with membranes is often a key requirement for these kinds of interactions; such binding confines the protein to a particular subcellular location and can cause a change in conformation. For many cell-surface receptors, hydrophobic residues confer a permanent affinity for the plasma membrane. For other proteins, membrane affinity is a conditional property that initiates, propagates, or terminates a signaling pathway. Reversible association with lipophilic molecules can control membrane affinity. Recall that the binding of diacylglycerol to the C1 domain of PKC confers membrane affinity. Later we shall see how vitamin K brings about the binding of phosphatidylserine to Gla protein domains, thus conferring membrane affinity.

Other proteins are covalently modified with fatty acids, which provides a greasy tail for binding to the membrane. For example, some human proteins possess an N-terminal signal that causes the N terminus to be acylated with the 14-carbon fatty

**Figure 8.41 Unmasked with calcium.** When recoverin binds $Ca^{2+}$ ions (green spheres), the N-terminal myristoyl group (van der Waals spheres) is available to interact with membranes.

acid myristic acid. All such proteins start with $H_2N$-Met-Gly-, but the myristoylation signal is not simply a particular short peptide sequence. Most myristoylated proteins are modified cotranslationally, during protein synthesis. The N-terminal methionine is cleaved as the proteins are translated by the ribosome, and the exposed N terminus of the glycine residue is then myristoylated by an N-terminal myristoyltransferase enzyme. The protein recoverin, which is important for human vision, normally sequesters its N-terminal myristoyl group; however, when it binds calcium, the myristoyl group is extruded, thus allowing it to interact with membranes (**Figure 8.41**).

Less commonly, myristoylation is a post-translational event. For example, recall that the protein Bid associates with mitochondria as part of caspase-mediated apoptosis (see Figure 6.37). Myristoylation is induced when the cleavage of Bid by caspase-8 reveals an N-terminal glycine residue. The acylation of serine and threonine residues is rare in human proteins. One of the few examples is the 28-amino-acid gastric peptide hormone ghrelin. It is enzymatically acylated on Ser3 with dietary octanoyl groups and then released into the circulation, where it stimulates a hunger response.

The bio-orthogonal Staudinger ligation has been used to identify myristoylated proteins in cells (**Figure 8.42**). The azido derivative, 12-azidododecanoic acid, can replace myristic acid during protein myristoylation. After treatment with this compound, the cells are lysed, and their proteins are run on a polyacrylamide gel. The myristoylated proteins can be selectively detected by treating the gel with a 2-phosphinobenzoate ester carrying a biotin handle. 2-Phosphinobenzoates react selectively with azido functional groups, leading to robust amide linkages and tagging the myristoylated proteins with biotin. The presence of biotin can then be readily detected on the gel with a Western blot using the high affinity between streptavidin and biotin to deliver a fluorescent or other label to the bands belonging to myristoylated proteins. The experiment demonstrates that adding a reactive chemical label specific for the target population can be used to isolate an entire class of protein from the cell.

A number of proteins undergo enzyme-catalyzed palmitoylation of cysteine residues. The cysteines are often located at an N terminus or C terminus that already has a lipophilic modification. For example, the protein tyrosine kinase $p56^{lck}$ is

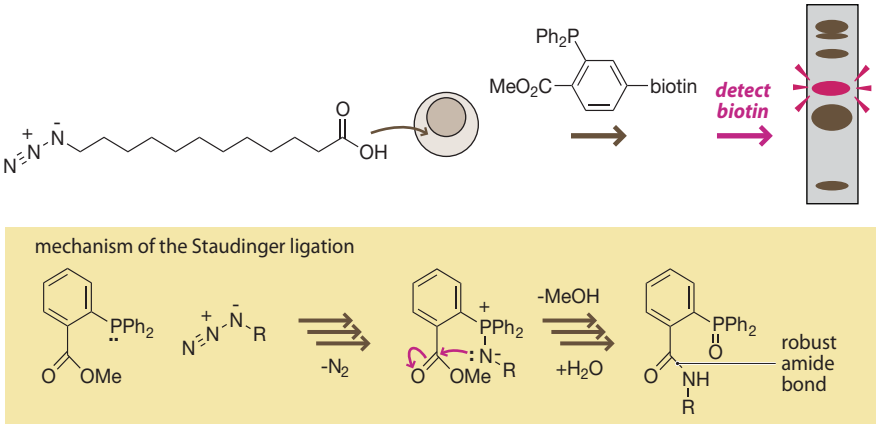



**Figure 8.42 Pinned by a ligation reaction.** The Staudinger ligation can be used to detect proteins that are acylated with an azido analog of myristate.

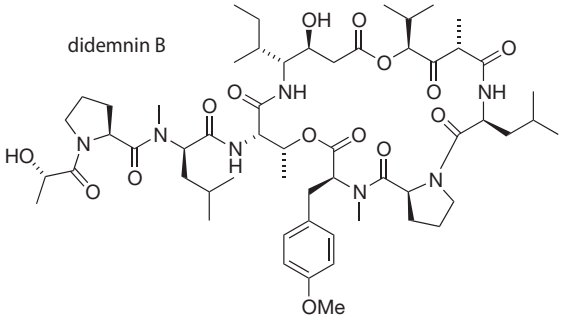



**Figure 8.43 Didemnin B, an inhibitor of palmitoyl protein thioesterase.**

palmitoylated at two of the cysteines near the N-terminal myristoyl group. In a protein with the sequence myristoyl-GCGCSS, for example, the underlined cysteine residues are modified. The *S*-palmitoylation reaction is catalyzed by the enzyme palmitoyl protein transferase, which catalyzes a transacylation with palmitoyl-CoA. In some instances (for example mannose-6-phosphate receptor) depalmitoylation occurs on a reasonably fast timescale ($t_{1/2}$ = 2 hours). Depalmitoylation can be catalyzed by the enzyme palmitoyl protein thioesterase. The cyclic peptide natural product didemnin B (**Figure 8.43**), produced by a primitive ocean organism called a tunicate, is a potent inhibitor ($K_i$ = 92 nM) of palmitoyl protein thioesterase. Inhibition of this enzyme confers strong cytotoxic activity against cells infected with RNA viruses and against tumor cells. Unfortunately, the compound failed to progress beyond phase II clinical trials because of anaphylactic toxicity.

### Problem 8.10

Most *S*-palmitoyl groups are readily cleaved by thiolates, but the palmitoyl group at the N-terminal cysteine of Sonic hedgehog protein is not readily cleaved. Suggest a plausible reason for this stability of the Sonic hedgehog palmitoyl group.

## Chemical transformation of fats generates useful compounds

Humans store fatty acids as triesters of glycerol, more commonly referred to as triacylglycerols, or triglycerides. Such compounds are stored in adipocytes for subsequent retrieval as energy sources. Triacylglycerols are the component of cooked foods that we typically associate with the word fat. Soybean oil (marketed as vegetable oil) and other cooking oils, chicken fat, and bacon grease are mainly composed of triacylglycerols. Because of their high boiling points, such fats are convenient solvents for frying foods.

In the search for a more economical use of natural resources, there has been a growing trend toward the replacement of fossil fuels with fuels based directly on renewable resources. Early in this millennium, in response to rapidly escalating fuel prices, there was a push toward the reclamation of used cooking oils for use as automotive fuels. A century earlier, Rudolf Diesel demonstrated a diesel engine that operated on peanut oil, but high-viscosity triacylglycerols are not well suited for modern diesel engines; however, it is simple to convert triacylglycerols to esters of simple alcohols through base-catalyzed transesterification (**Figure 8.44** and **Figure 8.45**). Methyl and ethyl esters of fatty acids have lower viscosities and lower flash points and are conveniently burned in diesel automobiles with little or no modification to the engine.

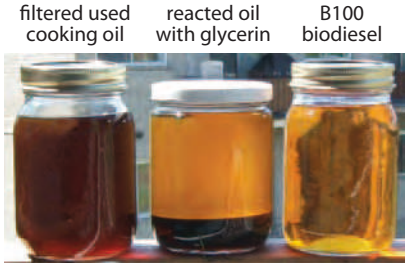



**Figure 8.44 Biodiesel in various stages of preparation.** (Courtesy of Luc S. Blais.)

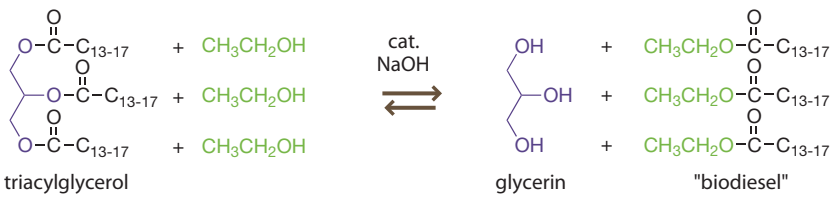



**Figure 8.45 The synthesis of biodiesel through transesterification.**

**Figure 8.46 A woman from the Mississippi valley, ca. 1911, making lye soap.** (From C. Johnson, Highways and Byways of the Mississippi Valley. New York: Macmillan, 1913.)

Humans have manufactured soap for at least 2000 years. Traditionally, soap is made through the base-promoted hydrolysis of fat with aqueous base (**Figure 8.46**). Alkaline solutions of sodium carbonate were generated by extracting wood ash with water. The term alkali is derived from the Arabic phrase *al-qaly*, referring to ash. The shortened term lye can refer to sodium hydroxide (caustic soda) or potassium hydroxide (potash). Hydrolysis of fat with lye generates glycerin and soap (sodium salts of long-chain fatty acids). The residual glycerin from the saponification of fat is valued as a moisturizer (**Figure 8.47**). In Chuck Palahniuk's novel *Fight Club*, human fat from liposuction clinics was transformed into glycerin soap, which was then resold in high-end boutiques. The base-promoted hydrolysis of esters, known as saponification, is a rapid and general reaction. Solutions of sodium hydroxide feel "slippery" when you rub them between your fingers because the hydroxide is saponifying the lipids in your outer layer of dead skin cells.

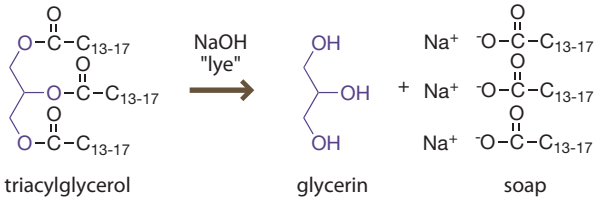



**Figure 8.47 The synthesis of glycerin soap by the hydrolysis of fat.**

## 8.4 NONHUMAN POLYKETIDE NATURAL PRODUCTS

### Several tricks amplify the potential diversity of polyketide natural products

Unlike simpler organisms that produce a structurally diverse range of polyketides, humans never really tapped the full biosynthetic potential of polyketide biosynthesis in their metabolism. A variety of tricks are used by some organisms to expand the potential diversity of polyketides. One trick that expands the diversity of polyketides is to skip some of the steps that lead to an aliphatic backbone. Recall that fatty acid synthase follows each Claisen condensation with a carbonyl reduction, a β-elimination, and a conjugate enone reduction. Omitting any of these steps generates an alternative functional group. For example, leaving out the conjugate reduction leaves a double bond between the ketide subunits (**Figure 8.48**). Many mammalian fatty acids have double bonds at specific positions, but such double bonds are introduced by a dehydratase domain, long after the fatty acid synthase has completed its work. Recall that the ketoreductase in human fatty acid synthase creates a stereogenic carbinol center during each cycle, and then destroys it through a β-elimination. What a waste! Leaving out both the conjugate reduction and the β-elimination leaves a hydroxyl group with a defined configuration. Repeatedly skipping these steps generates 1,3-diols, which are a distinguishing feature of many polyketide natural products. Leaving out all of the steps after the Claisen condensations would generate a polymer of ketide subunits. Such a polymer, containing skipped carbonyl groups, would be unstable because of

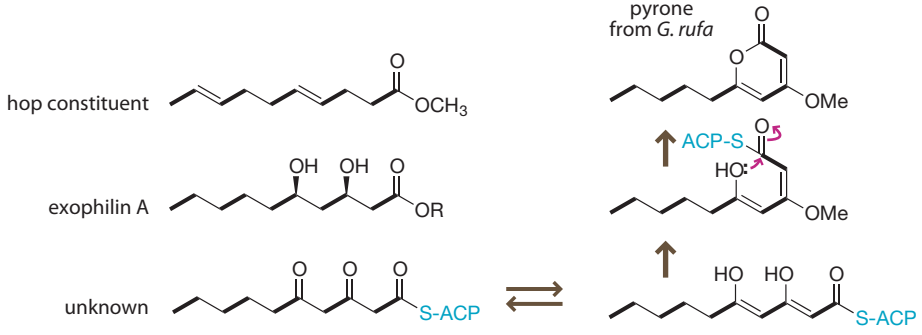



**Figure 8.48 Diversifying polyketides.** Leaving out conjugate reductions (hop constituent), β-eliminations (exophilin A), and carbonyl reductions (unknown) increases the potential diversity of polyketides. Reactive compounds resulting from such omissions can undergo rearrangements (for example, to give the pyrone from *Galiella rufa*).

the potential for cyclization reactions. The enolic forms of 3,5-diketoesters are susceptible to cyclization to generate pyrones, and some polyketide synthases harness these cyclization reactions. For example, the fungus *Galiella rufa* produces a 3-methoxypyrone through a route involving a 3,5-diketoester intermediate.

Polymers containing skipped carbonyl groups are also highly susceptible to intramolecular aldol condensations. For example, the naturally occurring benzoic acid derivative orsellinic acid is a tetraketide derived from an aldol condensation followed by double tautomerization to an aromatic compound (**Figure 8.49**). Similarly, curvulinic acid is derived from a pentaketide through an aldol condensation and aromatization.

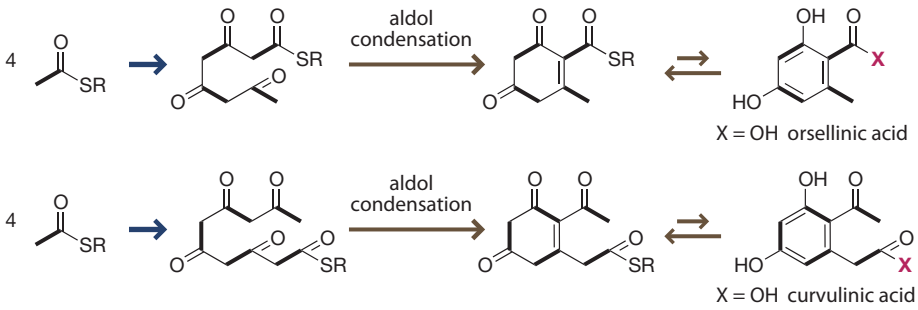
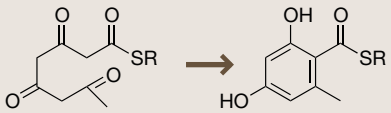



**Figure 8.49 Synthesis of aromatic polyketides.** Polyketide synthases use intramolecular aldol condensations to generate aromatic compounds.

## Problem 8.11

Suggest a plausible arrow-pushing mechanism for the following transformation using a catalytic base.




A wide range of aromatic natural products can arise through this type of aldol condensation/aromatization cascade. Usually, the synthases that catalyze the formation of aromatic polyketides perform additional steps both before and after the aromatization steps. The reduction of ketone groups before aromatization leads to the β-elimination of water during the aromatization step, leaving the aromatic carbon without a hydroxyl substituent (**Figure 8.50**). Many of the anthraquinone antibiotics that bind to DNA are generated by the oxidation of electron-rich anthracyclines (**Figure 8.51**). During the biosynthesis of nogalamycin, for example, two of the aromatic positions are further oxidized by enzymes coded for by the nogalamycin biosynthesis gene cluster.

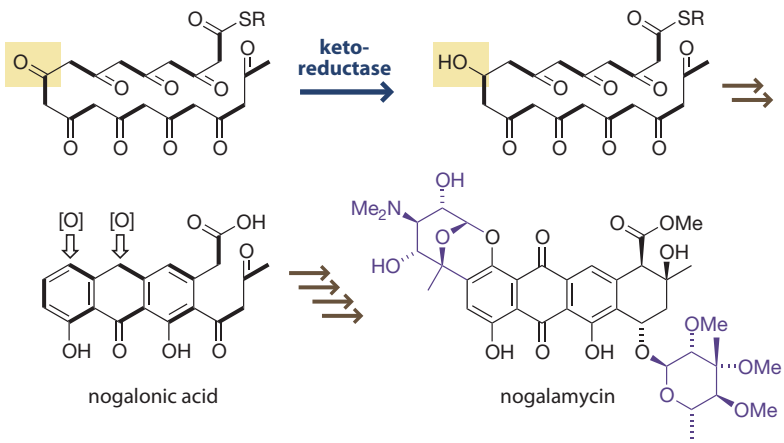



**Figure 8.50 Making nogalamycin.** Nogalamycin biosynthesis involves additional steps before and after aromatization of the decaketide.

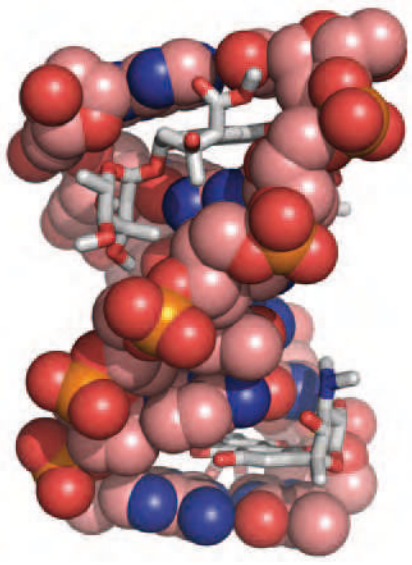


**Figure 8.51 Nogalamycin in action.** Two molecules of nogalamycin (depicted as stick models) are intercalated between the base pairs of double-stranded DNA with the sequence CGTACG (rendered as spheres). (PDB 182D)

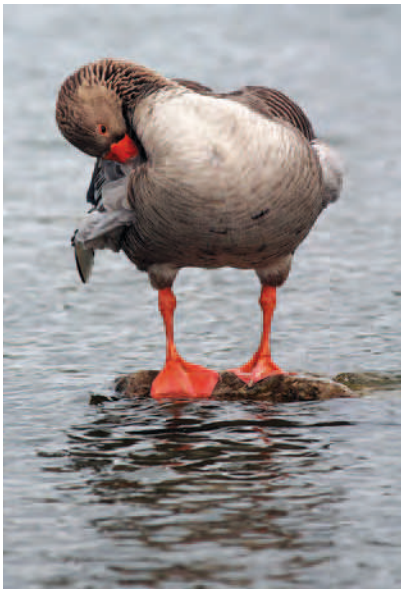
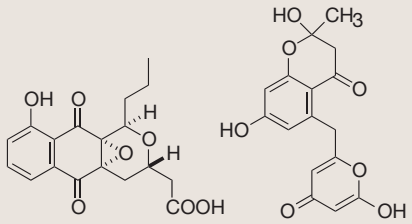


**Figure 8.52 Water proofing.** Waterfowl such as greylag geese wax their feathers by transferring water-repellent waxes from their preening gland. (Courtesy of Alan Weaver.)

## Problem 8.12

For the following two aromatic polyketides, identify the backbone resulting from aldol condensation reactions, highlighting the two-carbon building blocks.




frenolicin B SEK4

Another trick that expands the potential diversity of polyketide natural products is to incorporate both acetyl and propionyl subunits. Animal-based polyketides rarely contain propionyl subunits, but one exception is the wax from the preening glands of waterfowl. The fastidious preening of waterfowl such as geese transfers waxes from the preening gland to the feathers, making them water repellent (**Figure 8.52**). Some waxes are composed of mixtures of acetyl and propionyl subunits, whereas others contain fatty acids composed solely of propionyl subunits (**Figure 8.53**). The incorporation of propionyl subunits can lead to stereocenters with *R* or *S* configurations. Each of the methyl groups in goose preening wax is attached to a stereogenic center with the *R* configuration.

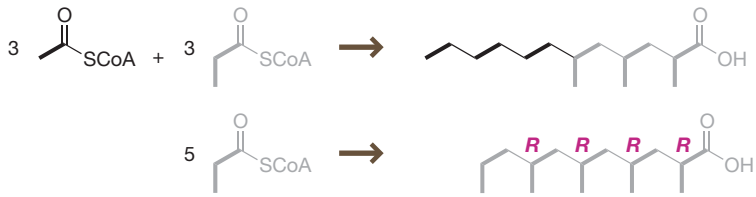



**Figure 8.53 Waxy fatty acids from geese are made up from both acetyl and propionyl subunits.**

Polyketide synthases rarely construct polyketides from subunits with more than three carbon atoms. However, there is considerable variability in the starter unit that is loaded by the loading module (**Figure 8.54**). For example, some polyketides start with the four-carbon isobutyryl-CoA and others start with *p*-hydroxycinnamoyl-CoA.

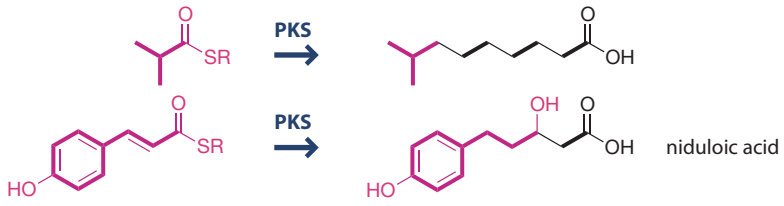
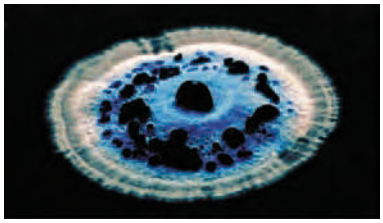



**Figure 8.54 Variation in the starter units of polyketides: isobutyryl-CoA (top) and *p*-hydroxycinnamoyl-CoA (bottom).** PKS, polyketide synthase.

## *Streptomyces* has mastered polyketide biosynthesis

By omitting steps, varying the subunits, and varying the starter units, microorganisms access a vastly greater array of polyketide structures than humans. Recall that human fatty acid synthase uses just one ketosynthase, one ketoreductase, one dehydratase, and one enoyl reductase for each of the two-carbon ketide subunits in the resulting oligomer. The resulting short polymers are monotonous. The most significant trick exploited by microorganisms to produce diverse polyketides is to devote a unique set



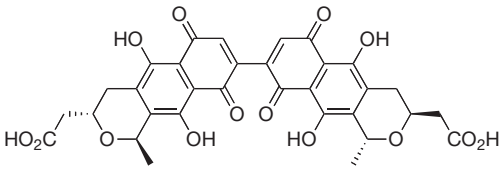


**Figure 8.55 Colonies of *Streptomyces coelicolor* A3 excreting droplets containing the blue antibiotic actinorhodin, depicted on the right.** (Photograph courtesy of Sir David Hopwood, John Innes Centre.)

of enzymatic subunits to the synthesis of each and every ketide unit. This makes it possible to expand polyketide biosynthesis beyond repeating polymers to oligomers with a defined sequence of functionalities and structures.

Soil bacteria in the phylum Actinobacteria are some of the premier architects of polyketide natural products. Microbiologists have had considerable success in culturing Actinobacteria in the laboratory and isolating bioactive natural products, often polyketides. Among the various species and strains of Actinobacteria that have been collected around the world, those of the genus *Streptomyces* are particularly common sources of bioactive polyketide natural products (**Figure 8.55**). Using the same basic reactions as those employed to make fatty acids, *Streptomyces* produces a dazzling array of bioactive polyketide natural products, such as amphotericin B, tetracycline, tylosin, and avermectin $B_{1a}$ (**Figure 8.56**). Because the backbone is generated through iterative Claisen condensations, vestigial repeating patterns of 1,3-oxygenation are apparent, particularly in natural products such as amphotericin B.

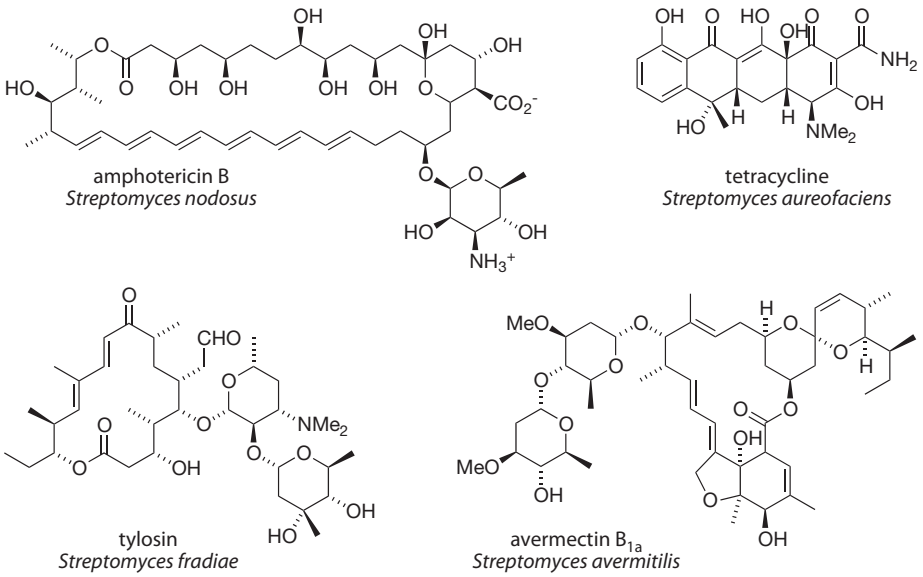
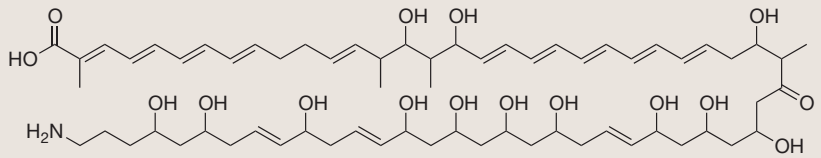



**Figure 8.56 *Streptomyces* and their natural products.** Clinically useful natural products from *Streptomyces* illustrate the power of polyketide biosynthesis to access a wide range of structures.

## Problem 8.13

Highlight each of the individual two-carbon acetate and three-carbon propionate subunits in linearmycin A.



## The modular genetic organization of type I polyketide synthases facilitates genetic reprogramming

In RNA transcription there is largely a one-to-one sequence correspondence between the DNA template and the RNA transcript (not counting splice variants or other alterations to the RNA sequence). In protein translation there is also a one-to-one correspondence between the nascent protein and the mRNA sequence. Similarly, in the biosynthesis of many polyketides there is a one-to-one correspondence between the polyketide structure and the sequence of enzymatic domains involved in biosynthesis. In theory, the modular genetic organization of many polyketide synthases offers the potential to (re)program the biosynthesis of any polyketide by deleting or replacing genetic elements. It is easier and more efficient to cut and splice DNA than to re-engineer the total synthesis of complex polyketide natural products.

The best-studied example of a modular type I polyketide synthase is that involved in the synthesis of the macrolide antibiotic erythromycin. Several thousand tons of erythromycin are produced annually through microbial fermentation. The key macrocycle, 6-deoxyerythronolide B, is generated by three very large, multisubunit proteins—namely, 6-deoxyerythronolide B synthases 1, 2, and 3 (DEBS1, DEBS2, and DEBS3). Each of the DEBS proteins is composed of enzymatic domains that perform the same reactions used to produce fatty acids in the order required to program the erythronolide structure. The single exception is the thiolesterase domain, which catalyzes a macrolactonization rather than a hydrolysis of the acyl-enzyme linkage that terminates fatty acid synthesis (**Figure 8.57**). The 6-hydroxyl group is added by a cytochrome P450 hydroxylase, eryF, to produce the aglycone of erythromycin B. Three more enzymes are required to convert erythronolide B to erythromycin A: two glycosyltransferases and one cytochrome P450 hydroxylase. The genes that code for these additional enzymes, including enzymes for the biosynthesis of the specialized sugars, are part of the erythromycin A gene cluster.

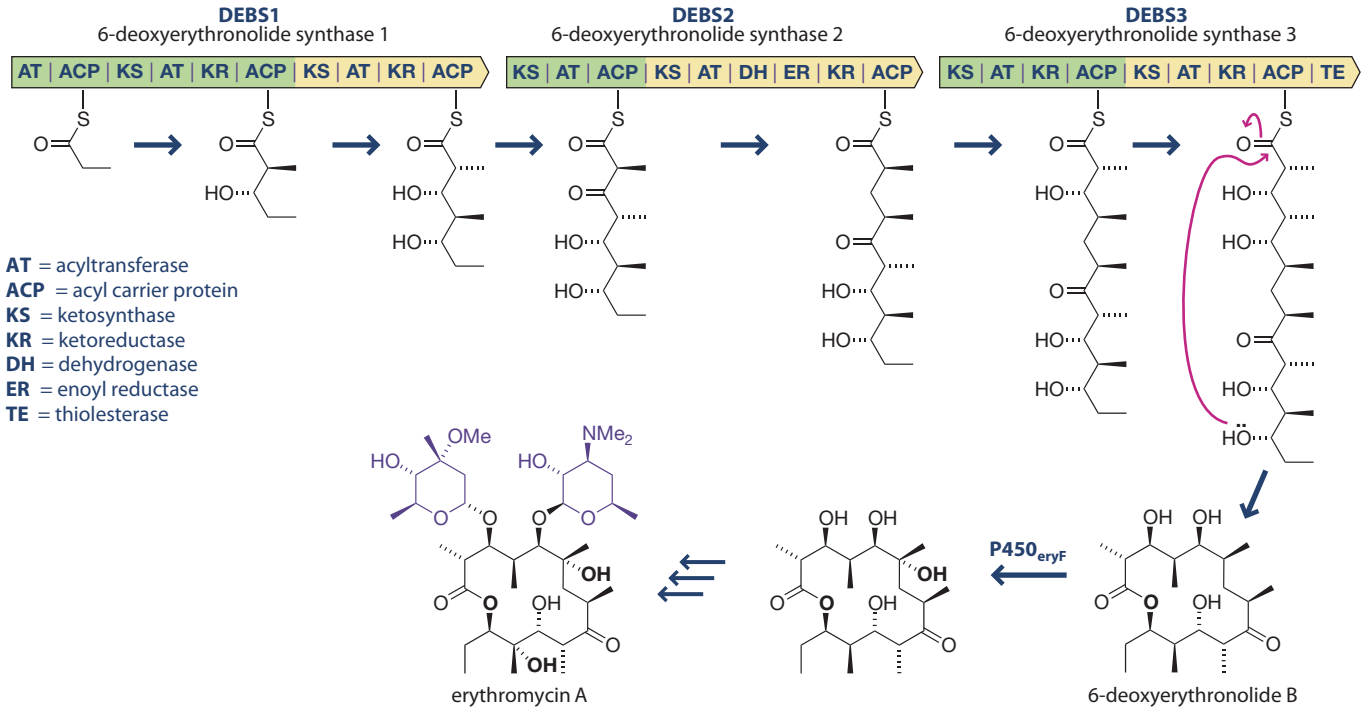
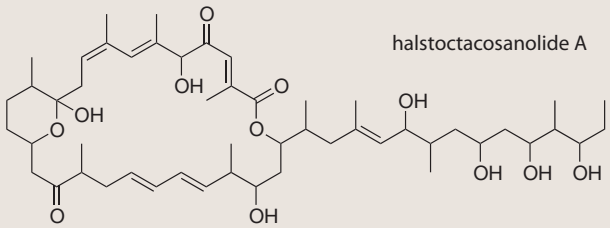



**Figure 8.57 A polyketide assembly line.** The biosynthesis of 6-deoxyerythronolide B requires three multidomain proteins and several other enzymes such as hydroxylases and glycosylases. Each enzymatic domain (colored green or beige) constructs one ketide subunit. Each domain is composed of multiple enzymatic subunits, each with an independent catalytic function (see the key).

## Problem 8.14

Highlight each of the acetate and propionate subunits in halstoctacosanolide A. Which of the carbon atoms in halstoctacosanolide A requires additional oxidation reactions by an oxidase?




When the modules are deleted or exchanged through genetic engineering, they can often still produce functional multienzyme complexes. Thus, the size, functionality, and stereochemistry of polyketides can be controlled through reliable genetic methods. For example, the thiolesterase subunit that cyclizes the final 6-deoxyerythronolide chain can be moved to an alternative location, resulting in a smaller ring (**Figure 8.58**).

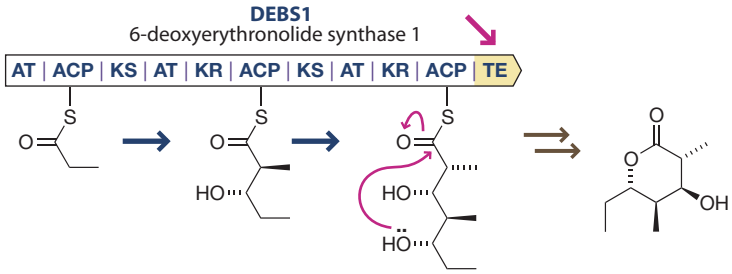



**Figure 8.58 Changed outcome.** Moving the thiolesterase domain (TE) from DEBS3 to DEBS1 leads to a new lactone natural product through premature cyclization.

When additional modules are genetically introduced without optimization, the biosynthetic machinery happily introduces the new ketide units but the results are unpredictable. For example, when an acetate genetic module was transferred from the operon for rapamycin synthase to the operon for erythromycin synthase, new macrolide products were produced with an additional ketide unit; surprisingly, the original heptaketide **1** was still the major product of the reaction (**Figure 8.59**). Two

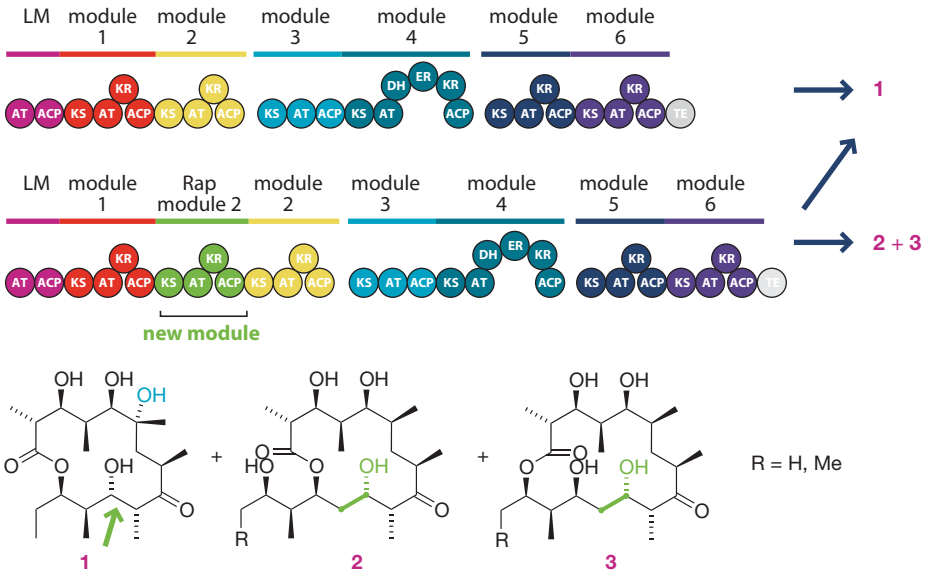



**Figure 8.59 New natural products through genetic engineering.** A new module was introduced into the operon for erythromycin synthase through genetic engineering, resulting in the biosynthesis of new "unnatural products." (Adapted from C.J. Rowe et al., *Chem. Biol.* 8:475–485, 2001. With permission from Elsevier.)

new types of octaketides (structures **2** and **3** in Figure 8.59) were detected, differing in the site of macrolactonization. It seems as though the synthase struggled to adapt to the newly introduced module. On the basis of the structures of the products, the loading module was frequently misloading acetyl-CoA instead of propionyl-CoA. Furthermore, the two new cyclic products were not being hydroxylated at the C6 position. Thus, the size and shape of the intermediates are probably important, both in the thiolesterase cyclization and in the subsequent hydroxylation reactions conducted by cytochrome P450 enzymes.

To access further polyketide diversity in the laboratory, polyketide synthases have been driven to load non-natural thiol ester starter units for incorporation into polyketides. *Streptomyces coelicolor* produced 6-deoxyerythronolide B when transformed with a plasmid coding for the synthase DEBS. When the first ketosynthase domain of DEBS1 was deactivated through a C729A mutation it was unable to initiate the assembly of 6-deoxyerythronolide B. However, when the growth medium was supplemented with chemically synthesized *N*-acetylcysteamine thiol esters (which mimic CoA thiol esters), those thiol esters were picked up by the first acyl carrier protein domain and transformed into macrolactone products. Through this combination of selective genetic deactivation and chemical rescue it was possible to create semisynthetic analogs of erythronolide B that would have been challenging to create with biological or chemical techniques alone (**Figure 8.60**).

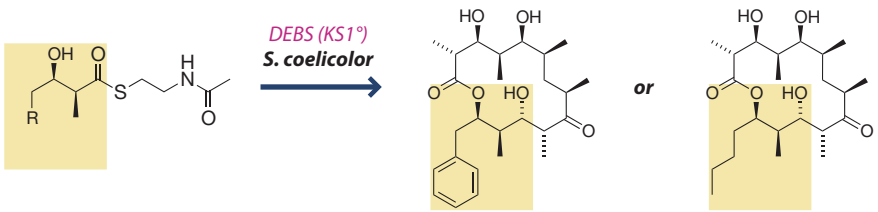


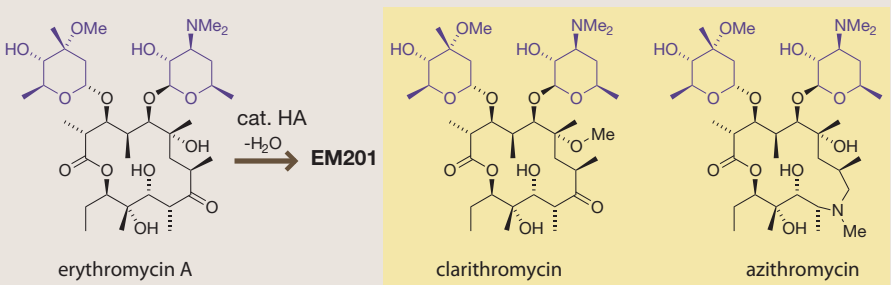

**Figure 8.60 Nonnatural starter units.** When *S. coelicolor* is transformed with a plasmid encoding the C279A variant of the DEBS ketosynthase domain [*DEBS (KS1°)*], the ketosynthase will accept non-natural thioesters, including synthetic analogs.

## Problem 8.15

The antibiotic erythromycin A undergoes a dehydrative cyclization in the stomach to generate a by-product, EM201, that is a potent agonist of the gastrointestinal motilin receptor. Two synthetic analogs of erythromycin A—clarithromycin and azithromycin—were developed that cannot undergo this side reaction.

**A** Given the structures of the two analogs, suggest a plausible structure for EM201.

**B** What simple changes could you make to the erythromycin A gene cluster to produce an analog that cannot form a motilin agonist such as EM201?



Microbial expression has obvious advantages over chemical synthesis, particularly in the ease of scale-up and the costs of starting materials. However, mixing and matching polyketide synthase modules in novel combinations often results in low yields of the polyketide natural products. It is expected that the expression of worthwhile polyketides can be optimized through combinations of random mutation and selection, much like the expression of erythromycin A. In reality, most newer versions of erythromycin (such as clarithromycin and azithromycin) are produced through a combination of optimized microbial expression and chemical modification.

### Sometimes additional methyl groups are added to the polyketide backbone

Sometimes polyketides possess additional methyl groups whose presence cannot be rationalized on the basis of acetate or propionate subunits. Often, these methyl groups are added through alkylation of the enzyme-bound polyketide with the sulfonium ion *S*-adenosylmethionine. For example, in the biosynthesis of the microtubule-stabilizing agent epothilone C, alkylation of a β-ketothioester intermediate by a methyltransferase domain generates the quaternary center that appears in the natural product (**Figure 8.61**). *S*-Adenosylmethionine is not used merely to create quaternary centers; sometimes methyl groups that seem to derive from propionate subunits are actually derived from alkylation with *S*-adenosylmethionine. For example, all three carbon atoms that dangle from the chain of alternaric acid are derived from *S*-adenosylmethionine (**Figure 8.62**). Although it is true that methyl substituents on polyketide backbones sometimes arise from *S*-adenosylmethionine, this is not always so. In the absence of specific data, such as feeding organisms isotope-labeled propionate and observing the incorporation of the isotope, you should assume that methyl groups are introduced as a part of a propionate subunit.

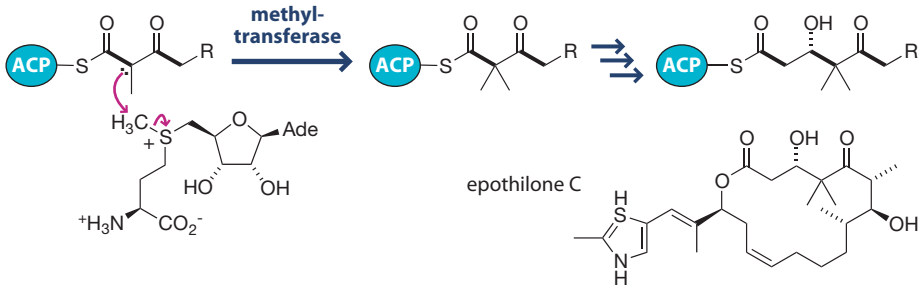



**Figure 8.61 Give me a $CH_3$.** The quaternary center in epothilone C arises through alkylation of a β-ketothioester with S-adenosylmethionine.

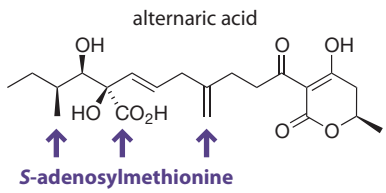



**Figure 8.62 One methyl at a time.** Methyl groups derived from *S*-adenosylmethionine masquerade as parts of propionate subunits.

## 8.5 NONRIBOSOMAL PEPTIDE SYNTHASES

### Ribosomal translation is suited to the production of large proteins, not short peptides

The human body contains a wide variety of bioactive peptides that act as signaling molecules. All of these short peptides are produced by ribosomal translation of a full-length protein followed by some type of proteolytic cleavage reaction. For example, the hormone ghrelin (28 amino acids) is derived from a much larger protein called proghrelin, which has 99 amino acids. The short opioid signaling peptides enkephalin, melanocortin, and dynorphin are generated from the differential cleavage of the large protein precursors proenkephalin, proopiomelanocortin, and prodynorphin, respectively. The protein precursors to the enkephalins, preproenkephalins A and B, each have an N-terminal signal peptide sequence that directs localization to places such as the Golgi vesicles. Depending on the tissue of origin, the full-length protein (211

amino acids) can be cleaved to yield Met-enkephalin (YGGFM), Met-enkephalin variants, and/or Leu-enkephalin (YGGFL) (**Figure 8.63**). Enkephalins bind to δ-opioid receptors in the pons, the amygdala, the olfactory bulb, and the cortex areas of the brain.

**Figure 8.63 Short human peptide hormones, snipped from long peptide chains.** Proteolytic processing of preproenkephalin A involves proteolytic cleavage of an N-terminal signal peptide followed by tissue-specific cleavage to generate various signaling peptides, such as Leu-enkephalin, Met-enkephalin, and variants of Met-enkephalin. The preproenkephalin A, depicted schematically here as the blue sequence from N to C terminus, is translated and proteolyzed into the indicated short peptide sequences.

## Most bioactive peptide secondary metabolites are generated by peptide synthases, not by ribosomes

Bioactive peptides in humans are always derived from ribosomal translation. In contrast, many of the bioactive peptide metabolites found in fungi and microorganisms arise from nonribosomal pathways. The immunosuppressant peptide drug cyclosporin A was isolated from the soil fungus *Tolypocladium inflatum*. More so than any surgical technique, cyclosporin A made it possible to successfully transplant human organs between nonidentical patients by inhibiting the rejection of the foreign tissue mediated by human T cells (**Figure 8.64**). The 1:1 complex of cyclosporin A and cyclophilin (**Figure 8.65**) binds to, and inhibits, the phosphatase calcineurin in T cells. Various features of cyclosporin A suggest that the compound was not synthesized by a ribosome. It is small and cyclic, and has unique amino acids that differ from the 20 common ribosomal amino acids—in particular, *N*-methylated amino acids, a D-amino acid (D-Ala), an amino acid with an ethyl side chain, and an eight-carbon amino acid abbreviated MeBmt that is quite clearly generated by polyketide biosynthesis domains. The *N*-methyl groups help to reduce solvation, making cyclosporin A membrane permeable. In addition, *N*-methylation and the cyclic structure of cyclosporin A make the compound resistant to proteolysis, and the compound can therefore be taken orally. Other nonribosomal peptides have ester linkages, β-amino acids, and amino acids modified with glycan, *N*-formyl, halogen, and hydroxyl functionalities.

Most nonribosomal peptides are synthesized by synthetases, which are structurally and functionally akin to modular polyketide synthases. As in the case of cyclosporin A, nonribosomal peptide synthetases sometimes include modules for polyketide and peptide assembly. In the biosynthesis of cyclosporin A, the first amino acid that is loaded onto the peptidyl carrier protein (PCP) domain of the synthetase is the D-Ala residue. The peptidyl carrier protein serves the same role as the acyl carrier protein in polyketide synthases. Both the amino acid subunits and the growing peptide chain are bound to the PCP through a phosphopantetheinyl linker. The amino

**Figure 8.64 Immunosuppression for organ transplantation.** The first successful heart transplant recipient survived for 18 days before succumbing to pneumonia. The state-of-the-art immunosuppressant azathioprine was administered to prevent rejection. Azathioprine is a DNA synthesis inhibitor that kills all rapidly dividing cells, which includes the entire immune system. Cyclosporin A is a much more effective immunosuppressant than azathioprine for heart transplant patients. (Courtesy of the Christiaan Barnard Division of Cardiothoracic Surgery at the University of Cape Town.)

**Figure 8.65 Cyclophilin D (stick model) bound to cyclosporin A (ribbon model).** (PDB 2Z6W)

acids of cyclosporin are added one at a time through aminolysis reactions, including the unusual polyketide amino acid MeBmt. The final cyclic structure is generated by a thiolesterase domain, which catalyzes a macrolactamization (**Figure 8.66**).

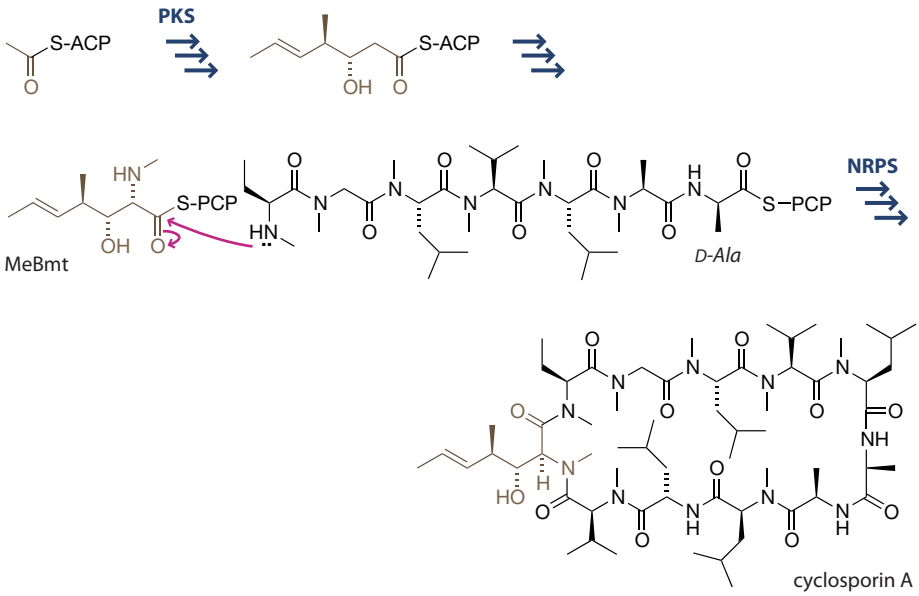
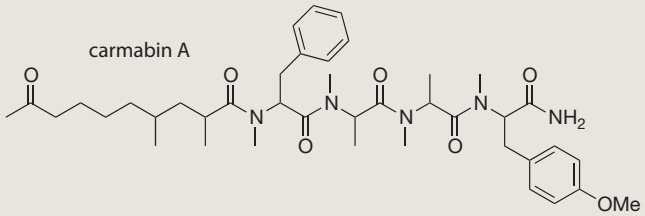



**Figure 8.66 Forging the ring.** The biosynthesis of cyclosporin A involves the aminolysis of thioesters attached to carrier proteins. PKS, polyketide synthase; NRPS, nonribosomal peptide synthetase.

**Problem 8.16**

Highlight each of the acetate, propionate, or amino acid subunits in carmabin A.




## 8.6 HUMAN TERPENES

### Early chemists recognized terpenes as oligomers of isoprene

In the 1880s, the chemist Otto Wallach recognized that a wide range of natural products seemed, conceptually, to be oligomers of the five-carbon building block isoprene, $H_2C{=}C(CH_3)CH{=}CH_2$. Leopold Ružička expanded on this hypothesis, pointing out that the arrangement of isoprene units was not random; those natural products were best formulated as arising through the cyclization of linear oligomers of isoprene connected in a head-to-tail fashion (**Figure 8.67**). Ružička was awarded the 1939 Nobel

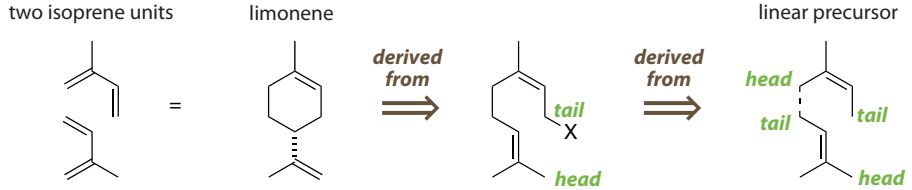



**Figure 8.67 Head-to-tail.** According to the biogenetic isoprene rule, terpene natural products are most likely to derive from a *linear* precursor composed of prenyl units joined in a head-to-tail fashion.

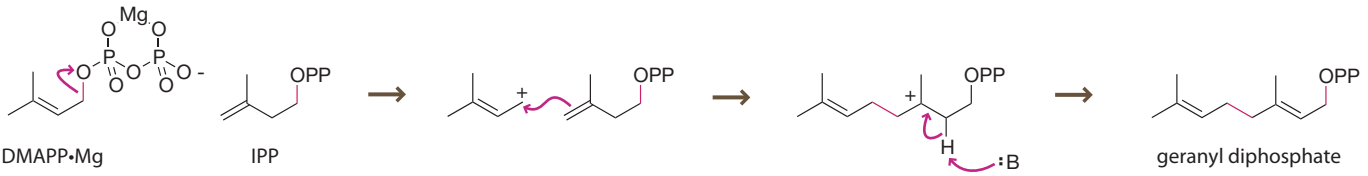



**Figure 8.68 Connecting the prenyl units.** The mechanism for enzyme-catalyzed carbon–carbon bond formation in terpene biosynthesis resembles the Friedel-Crafts alkylation reaction. DMAPP, dimethylallyl pyrophosphate; IPP, isopentenyl pyrophosphate.

Prize in Chemistry for this work. Ružička's *biogenetic isoprene rule* was formulated empirically, but we now know that all terpene natural products are in fact assembled through a head-to-tail oligomerization of five-carbon subunits.

## Cationic additions lead to linear chains

All terpenes are built up enzymatically from dimethylallyl pyrophosphate (DMAPP) and isopentenyl pyrophosphate (IPP). Monoterpenes contain two prenyl units (10 carbons), sesquiterpenes contain three prenyl units (15 carbons; the Latin prefix sesqui, which means one and a half), diterpenes contain four prenyl units (20 carbons), and triterpenes contain six prenyl units (30 carbons).

As we have already seen in the case of polymerases and kinases, pyrophosphates make excellent leaving groups, particularly when coordinated to Lewis acidic $Mg^{2+}$ ions. In terpene biosynthesis, dimethylallyl pyrophosphate forms the initial primary carbocation in close proximity to the π bond of the other reactant IPP. The π bond of IPP then adds to the nascent dimethylallyl cation to generate a tertiary carbocation (**Figure 8.68**). β-Elimination or deprotonation of the proton adjacent to the tertiary carbocation generates geranyl pyrophosphate. Each of these allylic diphosphates can serve as a substrate in further, iterative enzyme-catalyzed reactions with IPP. In all of these reactions, the growing oligomer generates a cation, and IPP serves as a cation trap. The formation of geranyl diphosphate illustrates the key steps in terpene assembly—namely, Lewis-acid-catalyzed loss of the diphosphate, formation of a carbocation, a series of carbocation-based reactions or rearrangements, and then a β-elimination step to quench the carbocation.

The suffix pyrophosphate is slowly giving way to the simpler suffix diphosphate. In this book, the term "diphosphate" will be favored, except in the case of the inorganic dimer pyrophosphate ($H_2P_2O_7^{2-}$) or the terpene precursors DMAPP and IPP. To simplify our structures, we will also use OPP as an abbreviation for pyrophosphate or, in cation-forming reactions, the pyrophosphate•$Mg^{2+}$ complex.

Bisphosphonate drugs, used to treat osteoporosis, potently inhibit farnesyl diphosphate synthase in humans through substrate mimicry (**Figure 8.69**). Biphosphonates work especially well as substrate mimics, because the structure provides a nonhydrolyzable analog of a diphosphate. Inhibition of farnesyl synthesis prevents protein prenylation, a process by which farnesyl or geranylgeranyl groups are transferred to key proteins, such as small GTPase signaling proteins (described further below). The loss of protein prenylation interferes with the normal cell signaling processes in osteoclasts, eventually causing cell death. Bisphosphonates bind strongly to hydroxyapatite, the calcium phosphate mineral that makes up bones. Thus, the compounds are selectively taken up by osteoclasts, which are the only cells actively involved in bone resorption.

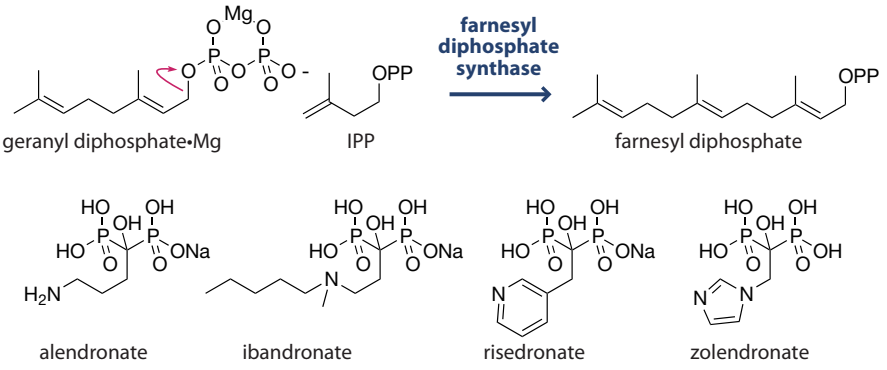



**Figure 8.69 Bisphosphonates.** The enzyme farnesyl diphosphate synthase (top) promotes ionization of the magnesium pyrophosphate leaving group to generate an allylic carbocation. Bisphosphonates (bottom) used to treat osteoporosis mimic essential features of the reactive intermediates in the enzymatic reaction.

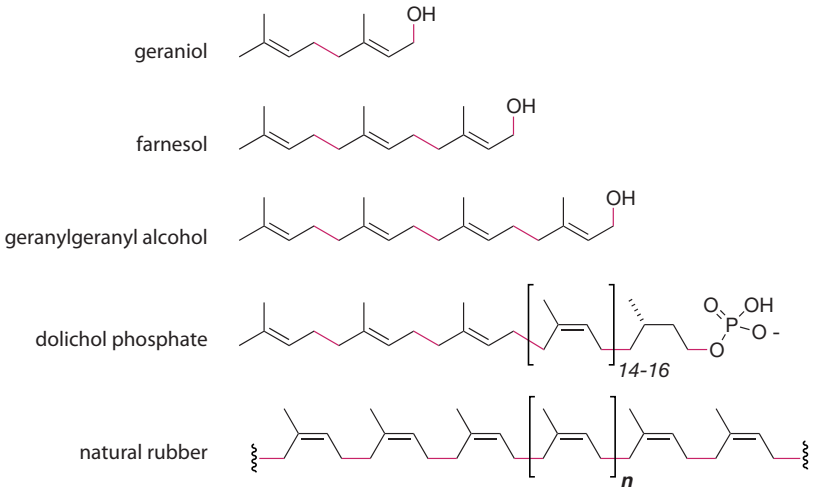



**Figure 8.70 Structures and names of common linear terpenes.** Note the saturation of the terminal isoprene group in dolichol. The five-carbon prenyl subunits shown in black and the bonds linking the subunits are shown in red.

The alcohols derived from the linear terpene diphosphates are found in a variety of fragrant essential plant oils (**Figure 8.70**). Geraniol is found in rose oil. Farnesol is present in citronella, lemon grass, rose, and neroli oil. Farnesol is active against some pests, and has been shown to inhibit transcription in the bacterial pathogen *Pseudomonas aeruginosa*. In humans, the most important terpenes are the prenyl chains dolichol, prenylated quinones, retinoids, and steroids. Farnesyl and geranylgeranyl groups are added post-translationally to protein cysteinyl groups. Recall that dolichol phosphate is the leaving group in the initial *N*-glycosylation of proteins. The dolichol keeps the glycosyl donor firmly anchored in the membrane of the endoplasmic reticulum, where it can be best utilized by the membrane-bound enzyme oligosaccharyltransferase. Dolichol and natural rubber are much longer oligomers of prenyl subunits. The first three prenyl units of dolichol have an *E* configuration, but the remaining prenyl units have the *Z* configuration like rubber. This suggests that dolichol is biosynthesized from an initial farnesyl diphosphate precursor. The terminal prenyl unit in dolichol is saturated, making it resistant to the formation of carbocations and the spontaneous loss of the phosphate or other leaving group.

Natural latex rubber is a polymer of prenyl units with the *Z* configuration. Synthetic rubber can be made by the polymerization of the diene isoprene, with butyllithium as an initiator. When anionic polymerization is performed in aliphatic solvents, 97% of the double bonds have the *Z* configuration, because the *Z*-allyllithium intermediate reacts faster than the *E*-allyllithium intermediate (**Figure 8.71**).

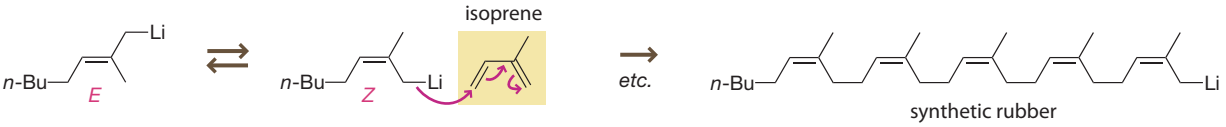



**Figure 8.71 Synthesis of rubber.** The anionic polymerization of isoprene is stereoselective because the *Z* isomer of the allyllithium intermediate reacts faster than the *E* isomer.

## Prenyl subunits arise through enolate chemistry

In humans, the biosynthesis of terpenes initially begins with enolate chemistry, which looks a lot like polyketide biosynthesis. The biosynthesis of isopentenyl pyrophosphate from three molecules of acetyl-CoA (**Figure 8.72**) involves a six-step pathway. Each step is catalyzed by an enzyme.

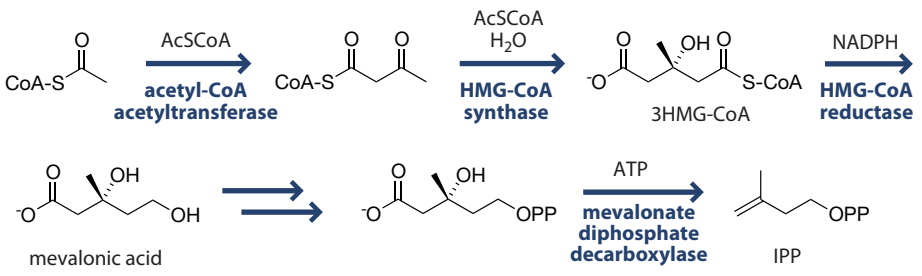



**Figure 8.72 Building the building blocks.** The biosynthesis of prenyl monomers involves an aldol addition reaction. HMG, hydroxymethylglutaryl.

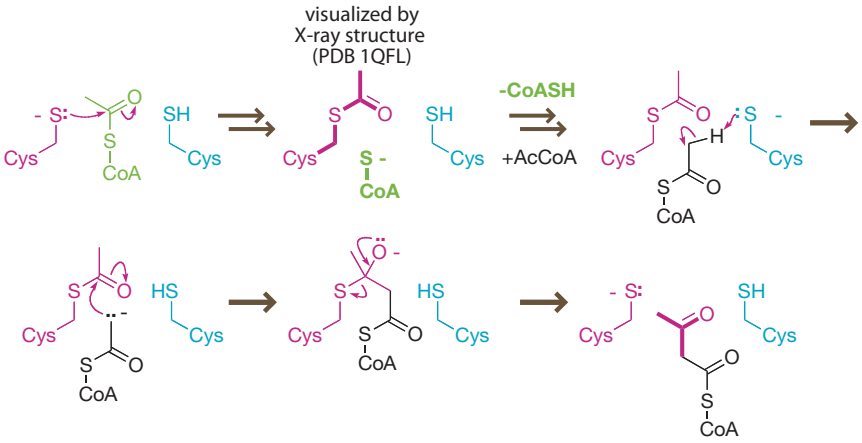



**Figure 8.73 The mechanism of the Claisen condensation catalyzed by acetyl-CoA acetyltransferase.** The initial acetyl thioester intermediate was captured in a crystal structure as shown in Figure 8.74.

**Figure 8.74 A bacterial thiolase that catalyzes the Claisen condensation of acetyl-CoA.** To solve this crystal structure of a tetrameric thiolase from *Zoogloea ramigera*, the crystals were soaked briefly in a solution of acetyl-CoA before being flash-frozen. The acetylated residue Cys89 is rendered in red and the CoA leaving group is rendered in green. (PDB 1QFL)

The first two reactions in the synthesis of IPP involve enzymes called thiolases. The first thiolase, acetyl-CoA acetyltransferase, catalyzes the Claisen condensation of two acetyl-CoA esters via a covalently bound thiol ester intermediate (**Figure 8.73**). Initially, the acetyl group of acetyl-CoA is transferred to Cys89 in the active site, followed by Claisen condensation with an enolate derived from acetyl-CoA. When crystals of the thiolase from the bacterium *Zoogloea ramigera* (**Figure 8.74**) were soaked with substrate, X-ray diffraction revealed the presence of the acetylated cysteine residue. Unlike the ketosynthases involved in polyketide biosynthesis, thiolases generate enolates directly by deprotonation of the thiol ester. A second key difference between thiolases and ketosynthases is that the substrates for the thiolases involved in terpene biosynthesis (namely acetyl-CoA and acetoacetyl-CoA) are free in solution, whereas the substrates for modular polyketide synthases remain covalently attached to the ACP domain of the multienzyme complex at all times.

Acetyl-CoA, and hence acetoacetyl-CoA, arises from the well-known breakdown of glucose to produce energetic molecules such as ATP and NADH. In a key step, the enzyme 3-hydroxy-3-methylglutaryl-CoA synthase catalyzes an aldol reaction and hydrolysis to generate 3-hydroxymethylglutaryl-CoA (3HMG-CoA; **Figure 8.75**). However, 3HMG-CoA can also be produced by HMG-CoA lyase, which catalyzes the aldol reaction of acetyl-CoA to acetoacetate, a product of fat metabolism. Thus, the fundamental building block 3HMG-CoA can be generated from two different molecular feedstocks.

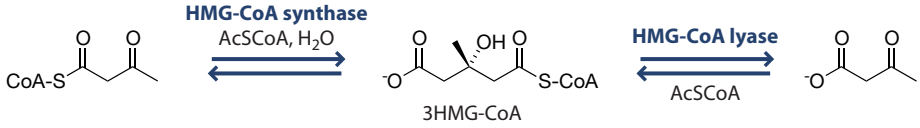



**Figure 8.75 Backup plan.** Inhibition of HMG-CoA synthase does not shut down terpene production in humans. The liver can generate 3-hydroxymethylglutaryl-CoA using an alternative enzyme.

The thioester group of 3HMG-CoA is reduced to a primary alcohol by the enzyme HMG-CoA reductase (**Figure 8.76**). As described in the next section, this humble transformation was enormously important in the ascendancy of the pharmaceutical industry at the end of the previous millennium and well into the current one. The mechanism of HMG-CoA reductase involves two reductions by the hydride donor NADPH. As with the reactions of NADH catalyzed by alcohol dehydrogenase, hydride donation is driven by the restoration of aromaticity in $NADP^+$. The thioester moiety is activated by a glutamic acid residue in the active site.

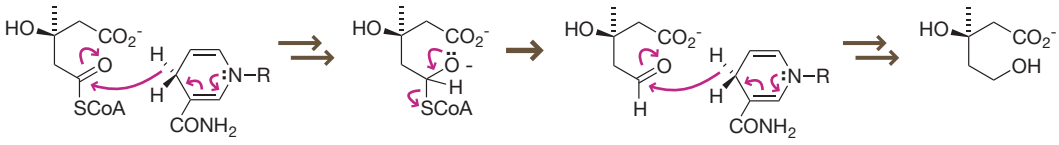



**Figure 8.76 The reduction of a thioester to a primary alcohol by HMG-CoA reductase.**

## Inhibition of terpene biosynthesis is the number one treatment for heart disease

HMG-CoA reductase catalyzes the committed, rate-determining step in terpene biosynthesis and ultimately in the biosynthesis of cholesterol. High levels of cholesterol and lipids in the serum are associated with the occlusion of arteries, referred to as atherosclerosis, which eventually leads to heart attacks and/or strokes (**Figure 8.77**). Unhealthy diets contribute significantly to the prevalence of high cholesterol levels in western populations. However, a normal diet (300 mg cholesterol per day) contributes far less cholesterol than biosynthesis by the liver (800 mg per day). Thus, the inhibition of cholesterol biosynthesis is as essential as a decrease in dietary cholesterol in the treatment of hypercholesterolemia.

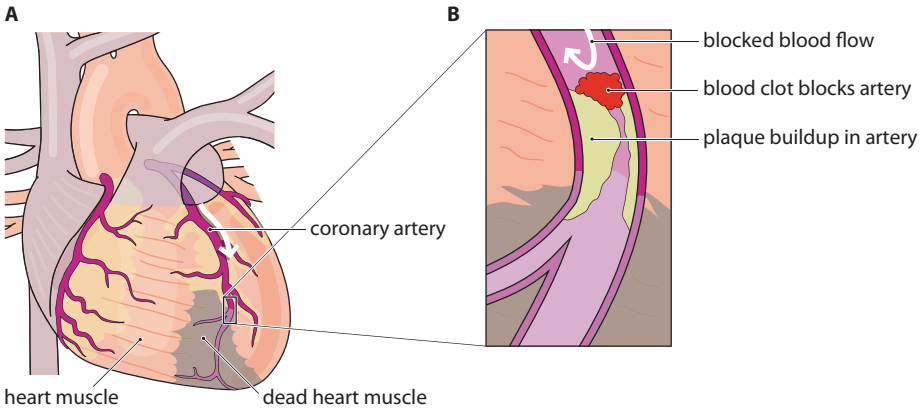



**Figure 8.77 Heart attack!** Hypercholesterolemia can result in blockage of coronary arteries and ultimately to heart attacks. (Adapted with permission from the National Heart Lung and Blood Institute.)

In humans, both cholesterol and fats are transported through the bloodstream as lipoprotein complexes. The key lipoproteins in cardiovascular disease are formed through the association of one molecule of apoprotein B-100 with many triacylglycerols and many cholesteryl esters. Lipoproteins can have varying ratios of protein to fats and cholesteryl esters, and such ratios determine their overall density. Low-density lipoproteins (LDLs), with low ratios of protein to cholesterol and fat, are particularly prone to forming plaques in arteries. Those plaques can grow until they occlude blood flow through the arteries. Both cholesterol biosynthesis and lipoprotein processing take place in the liver. When cholesterol biosynthesis is inhibited, the low levels of cholesterol lead to proteolytic release of the transcription factor sterol regulatory element binding protein from the endoplasmic reticulum membrane, allowing it to translocate to the nucleus, where it initiates the production of LDL receptors. The LDL receptors cause liver cells to take in low-density lipoproteins. Thus, the inhibition of cholesterol biosynthesis directly decreases serum levels of cholesterol and, more importantly, low-density lipoproteins.

In the 1970s, a screen for natural products that inhibit HMG-CoA reductase led to the discovery of the polyketide natural product mevastatin (**Figure 8.78**). The lactone

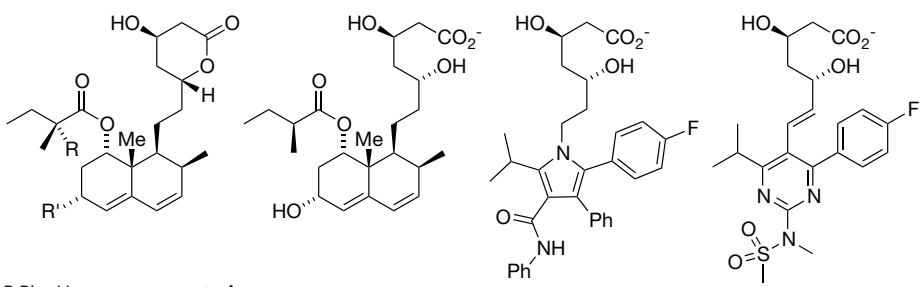



**Figure 8.78 The evolution of statin drugs.** Over several decades, the statin natural products mevastatin and lovastatin were carefully crafted into potent medicines that inhibit the biosynthesis of terpenes, and ultimately cholesterol.

**Figure 8.79 Food as a pharmaceutical.** Red yeast rice contains lovastatin. It is prepared by fermenting rice with the fungus *Monascus purpureus* and has been used in Chinese traditional medicine for more than 1000 years. In 2007 the US Food and Drug Administration declared that red yeast rice preparations containing significant quantities of lovastatin must be regulated as drugs. (Courtesy of Photo Researchers.)

moiety in mevastatin and similar structures found in all other members of the "statin" class of drugs resemble the cyclic form of mevalonic acid. Following on from this work, two pharmaceutical companies isolated related natural products for the treatment of cardiovascular disease—namely lovastatin (Mevacor™) at Merck and pravastatin (Pravachol™) at Sankyo (**Figure 8.79**). Merck improved upon lovastatin by replacing the ester with one possessing an additional methyl group, resulting in simvastatin (Zocor™). Chemists at Warner-Lambert showed that the decalin ring system could be replaced with a substituted heterocycle, resulting in the wildly successful pharmaceutical Lipitor™. These efforts culminated in the development of rosuvastatin (Crestor™) by AstraZeneca (see Figure 8.78). The introduction of cheaper generic versions of these effective pharmaceuticals may dissuade further incremental advances in statin development.

Catalyzing the last step in the synthesis of IPP, the enzyme mevalonate diphosphate decarboxylase seems to stabilize the considerable buildup of cationic character in the transition state (**Figure 8.80**). Buildup of cationic character does not necessarily prove the existence of a discrete carbocation intermediate; the decarboxylation could occur in concert with the loss of the phosphate group.

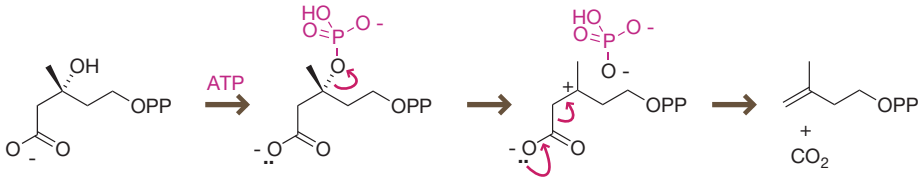
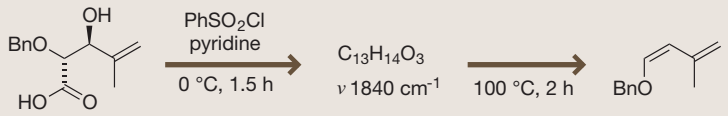



**Figure 8.80 The mechanism of mevalonate diphosphate decarboxylase.** The formation of isopentenyl pyrophosphate involves a decarboxylative elimination.

## Problem 8.17

Suggest a plausible structure for the intermediate in the following two-step decarboxylation of β-hydroxy acids.




Isopentenyl pyrophosphate isomerase isomerizes the double bond from a less stable terminal position to a more stable internal position (**Figure 8.81**). A $Mg^{2+}$ ion is coordinated to two glutamate residues in the active site. One of these glutamic acids protonates the double bond of IPP. On the opposite face, a cysteine thiolate abstracts a proton from the resulting tertiary carbocation. When the methylene carbon of IPP is labeled with $^{13}C$, most of the isotopic label ends up in the *trans* methyl group of DMAPP; however, a small amount ends up in the *cis* methyl group of DMAPP. The origin of this stereochemical infidelity is unclear. Either the substrate binds, and is protonated, in a minor conformation, or else the carbocation undergoes sigma bond rotation on the timescale of enzymatic turnover.

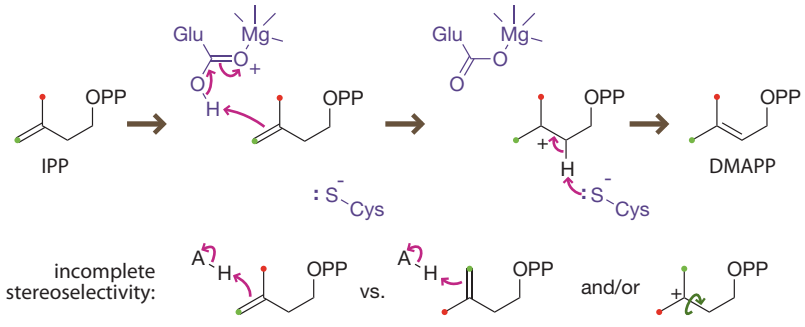
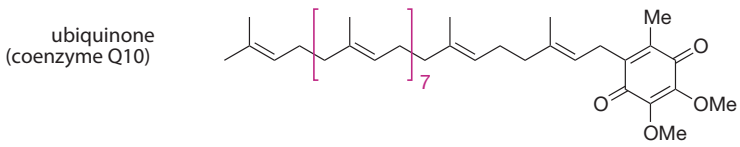
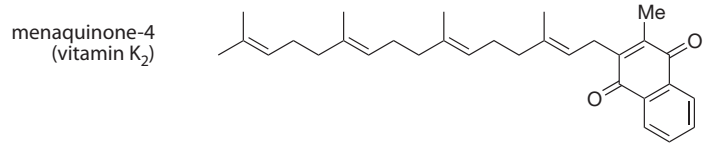


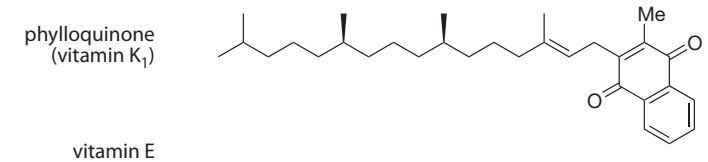

**Figure 8.81 A critical double bond isomerization.** Mammalian isopentenyl pyrophosphate isomerase is mostly stereoselective. The carbon atom labeled with a green dot is $^{13}C$; the carbon atom labeled with a red dot is the normal isotope $^{12}C$.

## Problem 8.18

If you assume that isopentenyl pyrophosphate isomerase is completely stereoselective, which of the carbon–carbon bonds in farnesyl diphosphate arises through cationic additions, and which arises through enolate reactions of acetyl thioesters?




vitamin E

| | $R_1$ | $R_2$ | $R_3$ |
|---|---|---|---|
| α-tocopherol | Me | Me | Me |
| β-tocopherol | Me | H | Me |
| γ-tocopherol | Me | Me | H |
| δ-tocopherol | H | H | H |

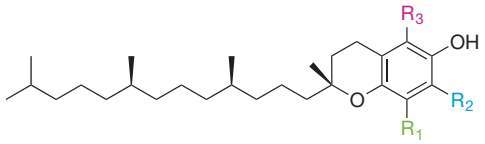



**Figure 8.82 Prenylated quinones in humans.**

## Prenylated quinones serve important roles in redox chemistry

The human body uses two types of prenylated quinones (**Figure 8.82**): ubiquinones and vitamin K. The ubiquinones are a group of prenylated benzoquinones, which are important as redox carriers in mitochondrial membranes. The length of the prenyl tail varies from six to ten prenyl units. The variant with ten prenyl units, called coenzyme Q10 or ubiquinone, is most important in humans. The redox equilibrium between quinones and dihydroquinones is balanced and facile (**Figure 8.83**). The quinone form is favored by the exceptional stability of C=O π bonds relative to C=C π bonds. The benzoquinone form of coenzyme Q10 is further stabilized by resonance donation from the methoxy groups. The dihydroquinone form is favored by aromaticity.

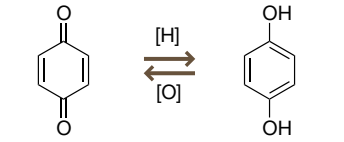



**Figure 8.83 Redox messenger.** Benzoquinone is easily reduced to dihydroquinone and reoxidized back to benzoquinone.

The term "vitamin K" describes a group of naphthoquinones with terpenoid tails that includes both menaquinones and phylloquinones. Humans do not possess genes for the biosynthesis of these quinones. Menaquinones are produced by obligate gastrointestinal bacteria. Phylloquinones are produced by plants. Thus, humans have two possible sources of vitamin K, which is essential for coagulation. The vitamin K group of naphthoquinones serve as cofactors in the enzyme-catalyzed carboxylation of glutamate residues in the Gla domains of some proteins. Once the glutamate residues of Gla domains are converted to γ-carboxyglutamate residues, the side chains can bind $Ca^{2+}$, leading to a change in protein conformation. For example, the carboxylated Gla domain of prothrombin binds to $Ca^{2+}$ and creates a binding pocket for

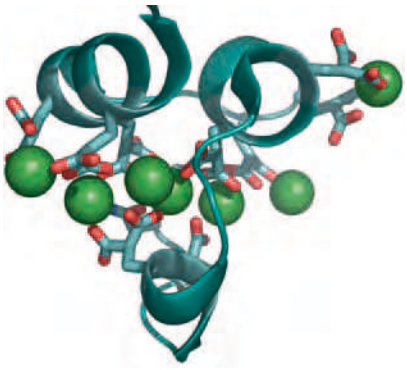


**Figure 8.84 Ionic glue.** The Gla domain of prothrombin binds to $Ca^{2+}$ ions (green spheres). The $Ca^{2+}$ ions serve as the glue makes prothrombin stick to cell membranes where it can be proteolytically activated by factor Xa. (PDB 1NL1)

phosphatidylserine, which confers membrane affinity (**Figure 8.84**). Thus, the post-translational generation of γ-carboxyglutamate residues is essential for normal blood clotting.

A series of reactions transfer carbon dioxide to the glutamic acid side chain and then recycle the then-modified vitamin K (**Figure 8.85**). The carboxylation of glutamate residues by γ-glutamyl carboxylase requires the initial reduction of the cofactor to the dihydroquinone form by a reductase enzyme. Each enzyme-catalyzed carboxylation event consumes a molecule of oxygen and converts the dihydroquinone to an epoxyquinone called vitamin K oxide. A consensus motif (EXXXEXC) seems to be essential for the initial recognition of the substrate protein by the carboxylase enzyme. The vitamin K cofactor is regenerated from vitamin K oxide by another reductase. Vitamin K oxide reductase is essential for the regeneration of vitamin K. The rodenticide warfarin (sold as the anticoagulant Coumadin®) acts by inhibiting vitamin K oxide reductase. Given the potential for uncontrolled bleeding due to the inhibition of blood clotting, the therapeutic index for warfarin is frighteningly narrow.

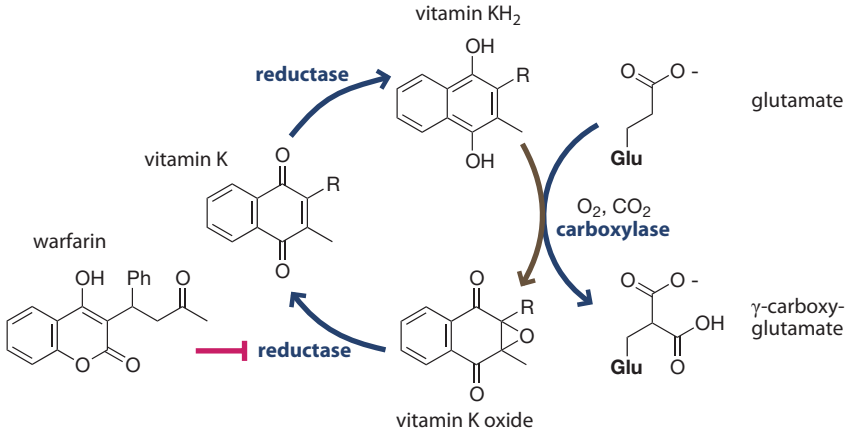



**Figure 8.85 Cycling of vitamin K.** The vitamin K-catalyzed carboxylation of glutamate residues in Gla domains leads to the binding of calcium (not shown).

The mechanism of action for γ-glutamyl carboxylase has been proposed to involve intramolecular epoxidation of the quinone by a hydroperoxy hemiketal (**Figure 8.86**). This transformation strongly resembles the base-catalyzed epoxidation of enones by hydrogen peroxide. The hemiketal alkoxide is proposed to deprotonate the glutamate side chain before it can eject hydroxide. The highly reactive enolate of the glutamate side chain then reacts with carbon dioxide to generate the γ-carboxyglutamate.

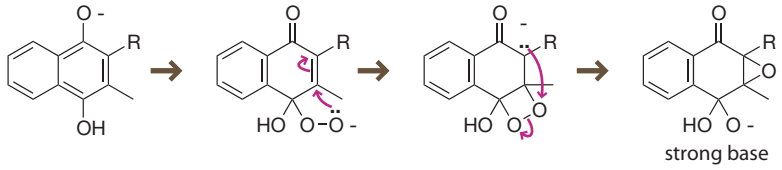



**Figure 8.86 The mechanism for the formation of a powerful alkoxide base by γ-glutamyl carboxylase.** The resultant strong base can then deprotonate α to the carboxylic group on the side chain of glutamic acid to generate a highly reactive enolate for reaction with carbon dioxide.

Vitamin E is not a quinone, but it is structurally related to ubiquinone and vitamin K (see Figure 8.82). Vitamin E includes any or all of the tocopherols, which are prenylated benzopyrans with various methyl substituents. Vitamin E is generally regarded as an antioxidant. Like all phenol derivatives, the aromatic hydroxyl group of the tocopherols serves as a hydrogen atom donor that can quench peroxy radical intermediates in lipid peroxidation. Among the tocopherols, only the trimethyl isomer, α-tocopherol, is nutritionally important. Not surprisingly, only α-tocopherol has substituents at both *ortho* positions, preventing radical dimerization (see Problem 5.41).

## Prenylation of proteins confers membrane affinity

Recall that the acylation of proteins on their N termini or cysteine residues confers membrane affinity, which can be essential for function. An alternative, related post-translational modification involves the enzymatic prenylation of cysteine residues with 15-carbon farnesyl groups by farnesyltransferase or with 20-carbon geranylgeranyl groups by one of two geranylgeranyltransferases. The most prominent class of proteins that are prenylated are the small GTPases such as Ras and Rho, which control the duration of individual biochemical signaling events. These proteins are discussed more fully in Chapter 9.

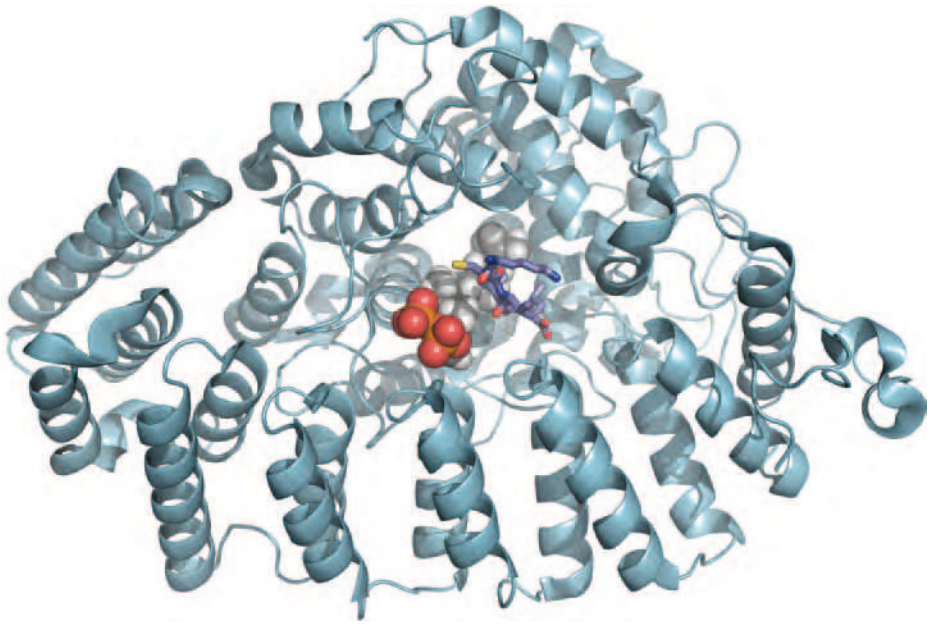


**Figure 8.87 Bioconjugal visit.** Two substrates bind in the active site of the enzyme geranylgeranyltransferase I (cyan): the electrophile geranylgeranyl diphosphate (spheres) and the nucleophilic tail of a protein substrate. In this crystal structure, the protein fragment KCVIL (sticks) is bound in the active site. The cysteine thiolate (yellow) is poised to displace the diphosphate leaving group in an SN2 reaction. (PDB 1N4Q)

Farnesyltransferase and geranylgeranyltransferase I (**Figure 8.87**) catalyze the prenylation of proteins with a C*aa*X tetrapeptide motif at the C terminus, where *a* tends to be a hydrophobic amino acid such as valine, leucine, isoleucine, or alanine, and where X is any amino acid except proline. After formation of the robust thioether linkage, several enzymatic modification reactions take place to rearrange the linkage to the lipid (**Figure 8.88**). First, an endopeptidase removes the three terminal amino acids. Then, the C terminus of the prenylated cysteine is enzymatically alkylated with

**C*aa*X recognition site**
***a*** = Leu, Val, Ile, Ala
**X** ≠ Pro

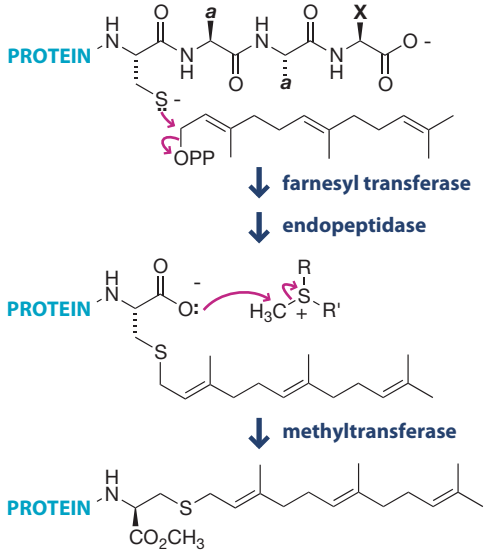



**Figure 8.88 Adding a membrane anchor.** Two modifications increase the lipophilicity of the protein C terminus. The cysteine sidechain is prenylated and the tripeptide-carboxylate terminus is replaced with a carboxymethyl ester.

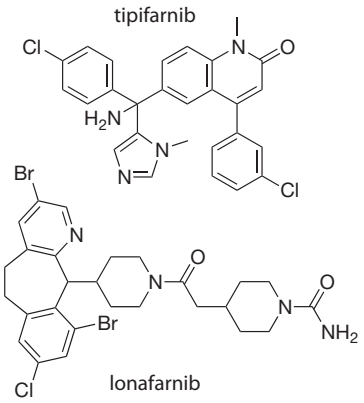



**Figure 8.89 Farnesyltransferase inhibitors.** Synthetic inhibitors of protein farnesyltransferases prevent prenylation of signal transduction proteins like ras. Without a membrane anchor, Ras is unable to associate with the cell membrane where it can transduce extracellular signals into intracellular instructions for cell proliferation.

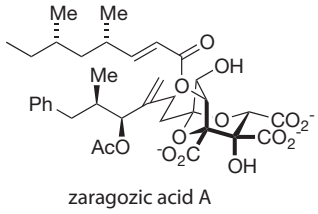



**Figure 8.90 Shutting down cholesterol synthesis.** Zaragozic acid A, a polyketide natural product, is a potent inhibitor of squalene synthase.

*S*-adenosylmethionine. The resultant highly modified C-terminal cysteine is quite hydrophobic. It is common to find that S-palmitoylated cysteines are close to the farnesylated cysteine at the C terminus. Geranylgeranyltransferase II is highly specific and is only known to target the small GTPases in the Rab family, which have CC or CXC at the C terminus.

The prenylation of Ras is particularly important, because the protein amplifies signals for growth and differentiation. About one-quarter of all human tumors include mutations to the Ras protein, which leave the protein permanently stuck in an "on" position. Prenylation is essential for the function of Ras, which must interact with proteins in the plasma membrane. Inhibitors of farnesyltransferase have shown promise as anticancer drugs. Two of the leading clinical candidates, lonafarnib and tipifarnib, do not bear an obvious resemblance to the C*aa*X motif or to farnesyl diphosphate (**Figure 8.89**).

## Tail-to-tail coupling of terpenyl diphosphates generates precursors of higher-order terpenes

Monoterpenes, sesquiterpenes, and diterpenes are generated from linear head-to-tail oligomers of prenyl phosphates. However, the higher-order terpenes, such as steroids and carotenoids, are generated through an initial tail-to-tail dimerization of terpene diphosphates. For example, the enzyme-catalyzed reductive dimerization of farnesyl diphosphate generates squalene, which is the precursor to triterpenes. The dimerization of geranylgeranyl diphosphate generates phytoene, which is the precursor to carotenoids. In comparison with these late-stage steps in the biosynthesis of cholesterol, the efficacy of statins such as Lipitor is somewhat surprising, because the drug inhibits the biosynthesis of all terpenes in the human body, not just cholesterol. The pharmaceutical industry has invested significant efforts to develop inhibitors of cholesterol biosynthesis that target enzymes further downstream in the cholesterol biosynthesis pathway. In a screen for inhibitors of squalene synthase, zaragozic acid A, also called squalestatin 1, was identified as a potent ($IC_{50}$ = 13 nM), selective inhibitor of mammalian squalene synthase (**Figure 8.90**). Several potent synthetic inhibitors of squalene synthase have been developed, but none has yet made it to the clinic.

The dimerization of farnesyl diphosphate to generate squalene is deceptively simple, but the mechanism for this reductive coupling is complex. The first stage of the reaction involves a cyclopropanation reaction to generate presqualene diphosphate (**Figure 8.91**). In the synthetic chemistry laboratory, cyclopropanes can be formed by carbene attack on olefins. Such carbenes are generated through an α-elimination reaction, such as base-promoted cyclopropanations with chloroform. However, α-elimination mechanisms involve an initial deprotonation followed by the loss of a leaving group, as opposed to loss of the leaving group followed by deprotonation (**Figure 8.92**). In the water-soaked environment of the cell, imagining conditions that favor deprotonation of a carbocation over the addition of water or other quencher to a

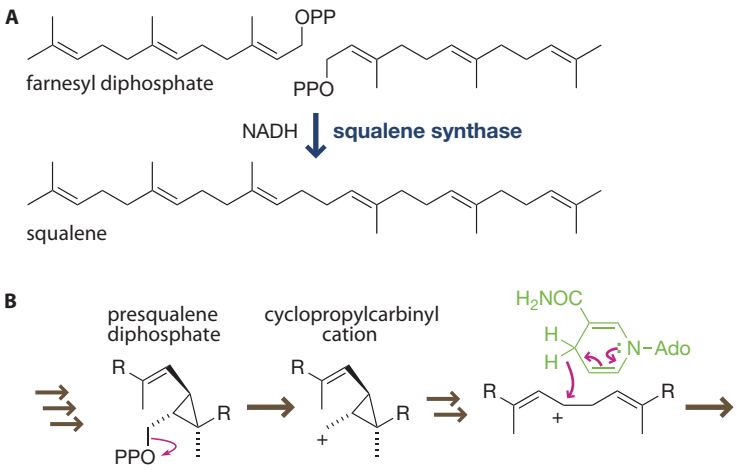



**Figure 8.91 Mechanism of squalene formation.** (A) Squalene synthase catalyzes the tail-to-tail reductive dimerization of farnesyl diphosphate. (B) The mechanism involves a cyclopropane intermediate.

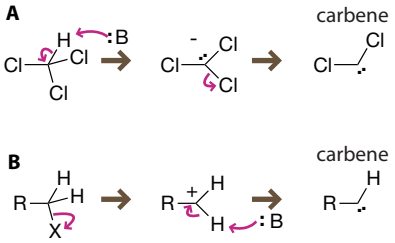



**Figure 8.92 Generating free carbenes.** α-Elimination mechanisms generally involve a carbanion (A), not a carbocation (B).

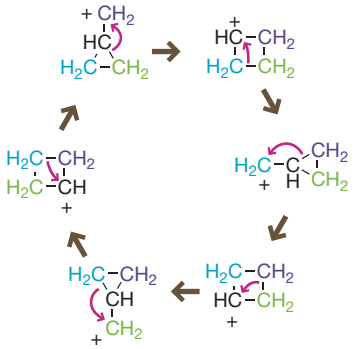


**Figure 8.93 Rapid scrambling.** Facile rearrangements between cyclopropylcarbinyl and cyclobutyl cations.

carbocation requires some creativity. However, the active site of an enzyme provides exactly the necessary environment where such processes can be controlled. Thus, the formation of presqualene diphosphate involves the loss of the diphosphate group, followed by deprotonation to generate a carbene. This carbene can then add to the olefin in the nearby farnesyl diphosphate to form the cyclopropane of presqualene diphosphate.

Presqualene diphosphate is poised to ionize, and in so doing to generate a cyclopropylcarbinyl cation. Recall that cyclopropylcarbinyl cations are exquisitely stable, as a result of the nucleophilicity of the strained carbon–carbon bonds in the three-membered ring. An unsubstituted cyclopropylcarbinyl cation rearranges effortlessly to a cyclobutyl cation, which is only slightly higher in energy (**Figure 8.93**). The energy barrier for these interconversions has been calculated to be less than 2 kcal $mol^{-1}$, and at equilibrium the ratio of cyclopropylcarbinyl cation to cyclobutyl cation should be 1.6:1.0. These facile rearrangements lead to immediate scrambling of the $CH_2$ positions. Undoubtedly, such rearrangements are occurring in the active site of squalene synthase. The formation of acyclic products is understood by looking at chlorination reactions of cyclopropylcarbinyl alcohol or the reaction of cyclobutanol with $SOCl_2$. Both reactions give very similar mixtures of products—namely cyclopropylcarbinyl chloride and cyclobutyl chloride in about a 2:1 ratio, with a much smaller amount of the acyclic homoallyl chloride. The acyclic homoallyl chloride arises not from an acyclic primary carbocation but from attack on one of the two cyclic cations (**Figure 8.94**). The situation is different when the cyclopropylcarbinyl cation can open to generate a stable carbocation, however, as occurs with presqualene, which can open to generate a stable, secondary allylic carbocation.

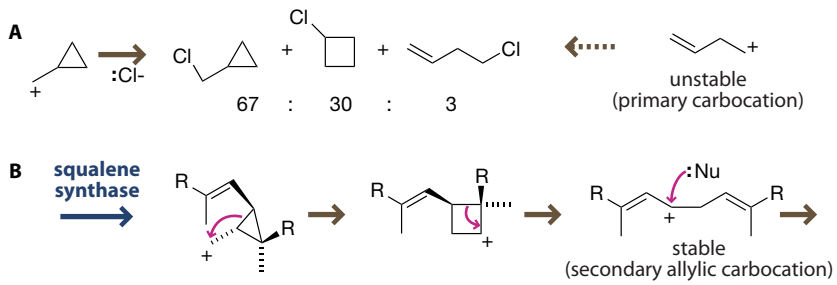
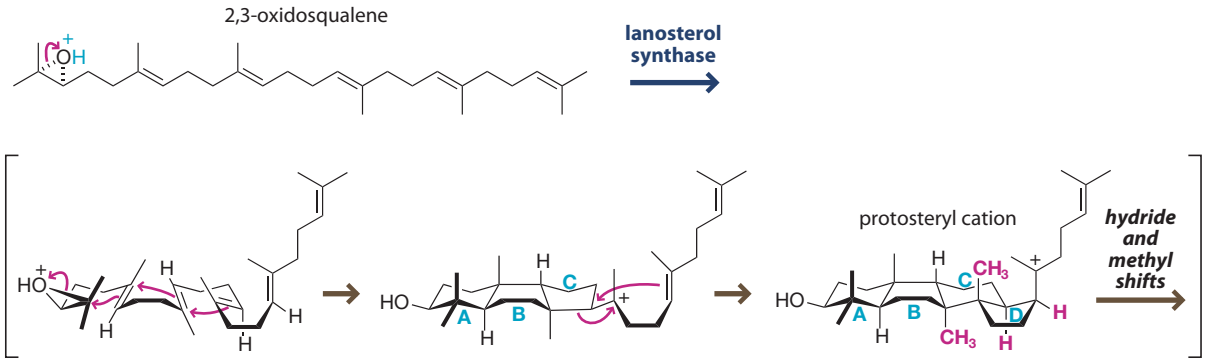



**Figure 8.94 The different fates of cyclopropylcarbinyl cations.** (A) Unsubstituted cyclopropylcarbinyl cations primarily give strained ring products. (B) The presqualene cation generates a ring-opened product.

## Polyene cyclizations generate many rings in a single reaction

All steroids in humans are generated from squalene. The enzyme squalene monooxygenase catalyzes the epoxidation of squalene to generate 2,3-oxidosqualene, using oxygen and NADPH. Then, lanosterol synthase catalyzes a dramatic cascade cyclization of 2,3-oxidosqualene to produce four rings and six stereogenic centers



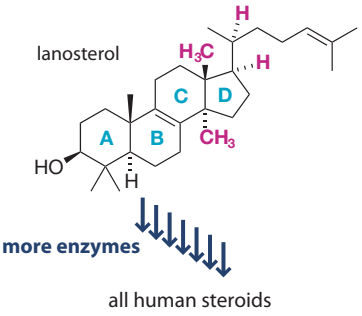




**Figure 8.95 The biosynthesis of lanosterol.** Lanosterol synthase protonates the epoxide of 2,3-oxidosqualene, which initiates the carbocation cascade. The active site bends 2,3-oxidosqualene into the depicted conformation to guide the stereochemistry of the resultant product.

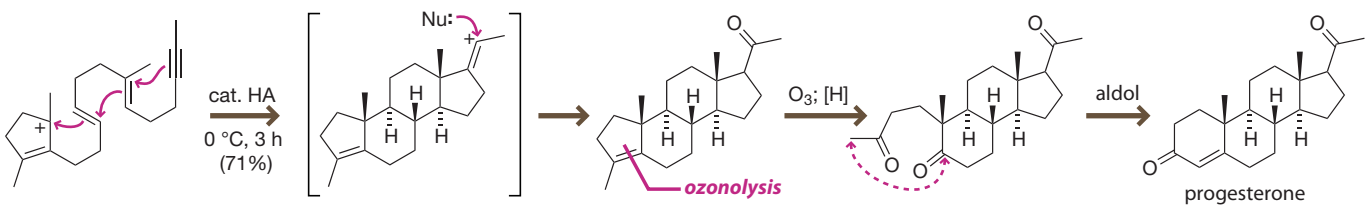
(**Figure 8.95**). The reaction can be drawn as a concerted reaction, but as with any enzymatic reaction it is difficult to tell whether all bonds are breaking and cleaving in the timespan of a single bond vibration, as opposed to a stepwise mechanism involving short-lived cationic intermediates. Protonation of the epoxide is clearly the challenging step for the enzyme. Model studies suggest that the A ring does not open to give a discrete carbocation; calculations suggest that it is *pushed* open by the C6–C7 π bond, generating ring A, followed effortlessly by the B ring and a five-membered C ring. Formation of the D ring is accompanied by a migratory ring expansion of the C ring to generate the full A-B-C-D ring system of the protolanosteryl cation. In these types of ultra-fast reactions, one cannot predict whether the individual steps are concerted or stepwise, and for practitioners of arrow-pushing the distinction is not very useful. However, experiments conducted on oxidosqualene analogs, with oxygen in place of methylene groups, support the idea of discrete carbocations (see Problem 8.34). The most fascinating aspect of the enzyme mechanism is that the protosteryl cation is generated such that ring B favors a boat conformation, and in this conformation the bonds to the bridgehead substituents are aligned to facilitate a precise series of hydride and methyl shifts that culminate in the formation of lanosterol with the correct stereochemistry.

## Problem 8.19

Suggest a plausible arrow-pushing mechanism for the conversion of the protosteryl cation to lanosterol as a series of discrete [1,2] migration steps.

The polyene cyclization that generates lanosterol in the active site of lanosterol synthase has inspired synthetic organic chemists. The ability to generate the A-B-C-D ring system of steroids through polyene cyclizations can be recapitulated *in vitro* without the aid of enzymes, as long as one uses the right substrate. 2,3-Oxidosqualene does not cyclize efficiently without the aid of an enzyme. The synthesis of progesterone is highly efficient when a five-membered ring is used to initiate the polyene cyclization and an alkyne is used as the nucleophile (**Figure 8.96**). The product in this reaction is of course racemic.

**Figure 8.96 Borrowing from nature.** Many of the most impressive synthetic achievements have been inspired by reactions in nature. A highly efficient chemical synthesis of progesterone takes advantage of a biomimetic polyene cyclization.



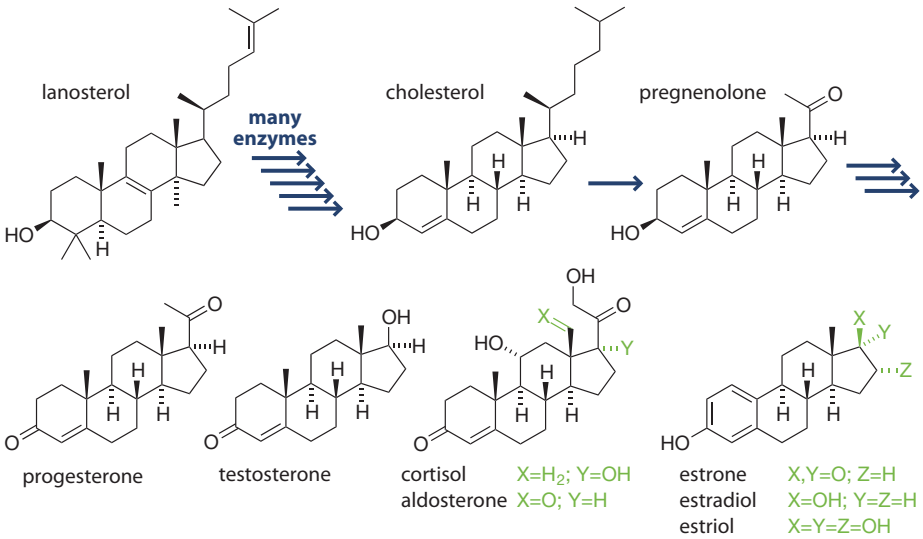



**Figure 8.97 The wrong steroid.** Humans don't need lanosterol; they need cholesterol. A host of enzymes is required to convert lanosterol into cholesterol and pregnenolone. All human steroid hormones are produced biosynthetically from pregnenolone.

The human conversion of oxidosqualene to lanosterol seems a wondrous trick—until one realizes that it is the wrong steroid. Nineteen more enzymes are required to convert lanosterol into the right steroid—cholesterol—as it is trafficked within the endoplasmic reticulum (**Figure 8.97**). Those enzymes perform hydroxylations, redox hydride transfers catalyzed by nicotinamide-dependent enzymes, decarboxylations, and hydrogenations. These same kinds of transformations are used by microorganisms and plants in the biosynthesis of terpene natural products. The removal of three methyl groups from lanosterol makes it difficult to trace the squalene backbone within the structure of cholesterol. Similarly, enzymatic modifications to other terpene natural products can also mask the location of the prenyl units within the structure. Most cholesterol winds up in cell membranes, but a small amount is converted into signaling molecules. The side chain of cholesterol is oxidatively cleaved to generate pregnenolone, the common precursor to all of the steroids that target human nuclear receptors (see Figure 8.97).

## Humans lack genes for retinoid biosynthesis

Retinoids are an important class of terpenes used by the human body (**Figure 8.98**). *Cis* and *trans* retinoic acids serve as signaling molecules for nuclear receptors. Retinal serves as the covalently bound ligand for the vision receptor rhodopsin. These receptors are discussed in Chapter 9. Humans lack enzymes for the *de novo* synthesis of retinoids. Instead, human retinoids are derived from two dietary sources—β-carotene and retinyl esters, which are hydrolyzed to generate retinol, also known as vitamin A. Plants produce β-carotene from the tail-to-tail dimer phytoene. Phytoene synthase catalyzes the generation of phytoene from two molecules of geranylgeranyl diphosphate. The mechanism of phytoene synthase is believed to involve a cyclopropane intermediate, like the mechanism of squalene synthase. The penultimate carbocation is deprotonated to generate an alkene linkage, however, and does not involve hydride addition from NADH (**Figure 8.99**). To solve nutritional deficiency caused by diets low in β-carotene, scientists have genetically engineered a strain of rice, called Golden

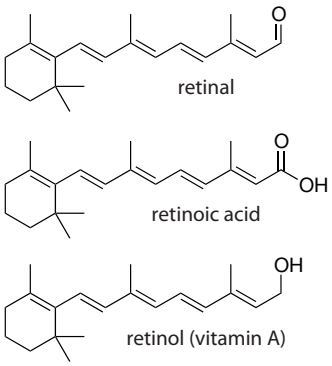



**Figure 8.98 Important human retinoids.** Retinal, retinoic acid, and retinol (vitamin A) are essential retinoids in humans, but we don't have enzymes that can construct the six-membered ring.

### Problem 8.20

Suggest a plausible arrow-pushing mechanism for the enzyme-catalyzed cyclization of lycopene to β-carotene (for the necessary structures, see Figure 9.99). Assume that the active site of the enzyme, lycopene β-cyclase, has a catalytic acid residue.

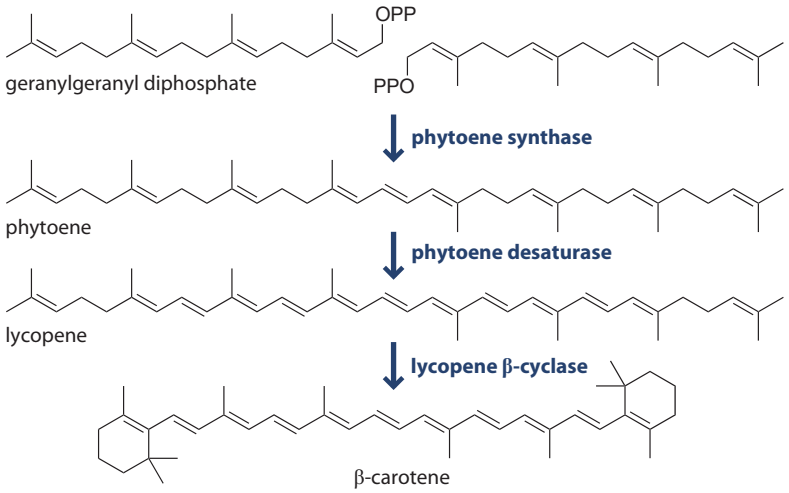



**Figure 8.99 Key compounds from vegetables.** Carotenoids are derived from the tail-to-tail dimer phytoene.

**Figure 8.100 Golden rice.** Golden rice (right) has been genetically modified to include the genes required for the biosynthesis and accumulation of β-carotene in the edible grains. (Courtesy of Golden Rice Humanitarian Board.)

Rice, that produces β-carotene (**Figure 8.100**). To do so, the relevant genes from maize and a bacterium found in soil were transfected into rice, which was then bred into different varieties of rice important to vitamin A-deficient communities. In humans, β-carotene is oxidatively cleaved to retinal with $O_2$ by a monooxygenase enzyme. Retinoic acids are generated through the oxidation of retinal.

The red hues of many animals arise from dietary carotenoids. Wild salmon has a deep red hue due to the presence of the carotenoid pigment astaxanthin, which is accumulated from the oceanic diet of krill and shrimp (**Figure 8.101**).

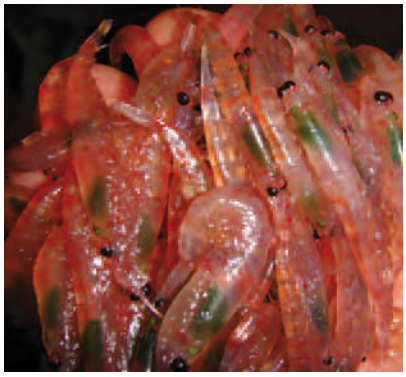
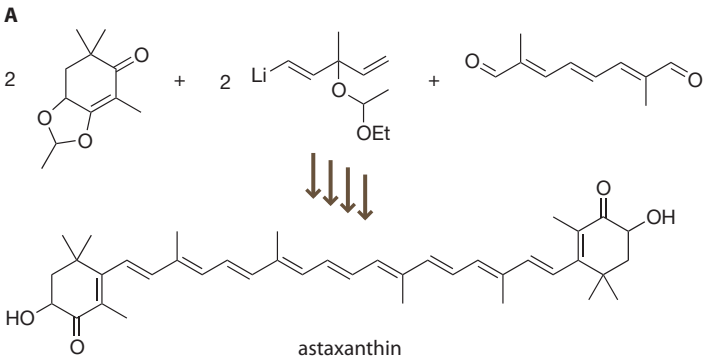


**Figure 8.101 Red shrimp.** The red color of krill results from the accumulation of the dye astaxanthin. (Courtesy of Joseph Warren, Stony Brook University.)

**Figure 8.102 Painting salmon with synthetic astaxanthin.** (A) A set of starting materials used to make synthetic astaxanthin for coloring farm-raised salmon. (B) The right shade of red is judged by the SalmoFan™ set of color standards. (B, courtesy of DSM Nutritional Products, Ltd.)

Similarly, the brilliant pink hue of flamingos is attributable to their natural diet of brine shrimp. Farm-raised salmon do not naturally acquire the deep red color of wild oceanic salmon, because they are raised on a diet of other fish. To increase the value of farm-raised salmon, synthetic astaxanthin (DSM's Carophyll® Pink 10% CWS or BASF's Lucantin® Pink) is added to the fish food shortly before harvest; a color gauge called the SalmoFan™ is used to judge the coloration of the fish (**Figure 8.102**).

A




B

# 8.7 NONHUMAN TERPENE NATURAL PRODUCTS

## Plants and microorganisms produce a much wider range of terpene natural products than humans

Terpene natural products have a much wider range of polycyclic structures than polyketides. Most terpene natural products are produced by, and isolated from, plants. In contrast, polyketide natural products are produced by a variety of organisms, but are most commonly isolated from readily cultured microorganisms. Terpenes were among the first natural products to be structurally studied. The name is associated with turpentine, which is the distillate from the resin of certain trees. Turpentine was originally distilled from the resin of the terebinth tree (*Pistacia terebinthus*) but is now more commonly obtained from pine trees. Of the 101 components in turpentine from *P. terebinthus*, the major components are α-pinene (39.6%), β-pinene (19.5%), sabinene (6.5%), terpinen-4-ol (3.8%), and δ-3-carene (3.3%). Modern turpentine produced from pine trees is more than 90% α- and β-pinenes (**Figure 8.103**).

**Figure 8.103 Collecting terpenes.** Turpentine is distilled from tree sap. (Left, courtesy of Wikimedia Commons; right, courtesy of Speedball Art Products.)

Terpene natural products inspire an irresistible allure in chemists, because a limited range of linear precursors generates an astounding number of complex cyclic structures. Higher diterpenes, such as triterpenes and tetraterpenes, tend to arise from tail-to-tail dimers of squalene or phytoene. Recall that the formation of lanosterol began with the opening of an epoxide to give a tertiary carbocation. Monoterpenes, sesquiterpenes, and diterpenes arise from the ionization of linear diphosphates to yield allyl cations (**Figure 8.104**). Polycyclic terpene natural products challenge the ingenuity of organic chemists who seek to demonstrate the efficiency of chemical synthesis.

The biosynthesis of terpenes by terpene synthase enzymes fascinates organic chemists because the synthetic steps involve the rich chemistry of carbocations, ionization reactions, $S_N1$ substitutions, rearrangements, and migrations. All known terpene synthases share a common structural fold (**Figure 8.105**) and must perform

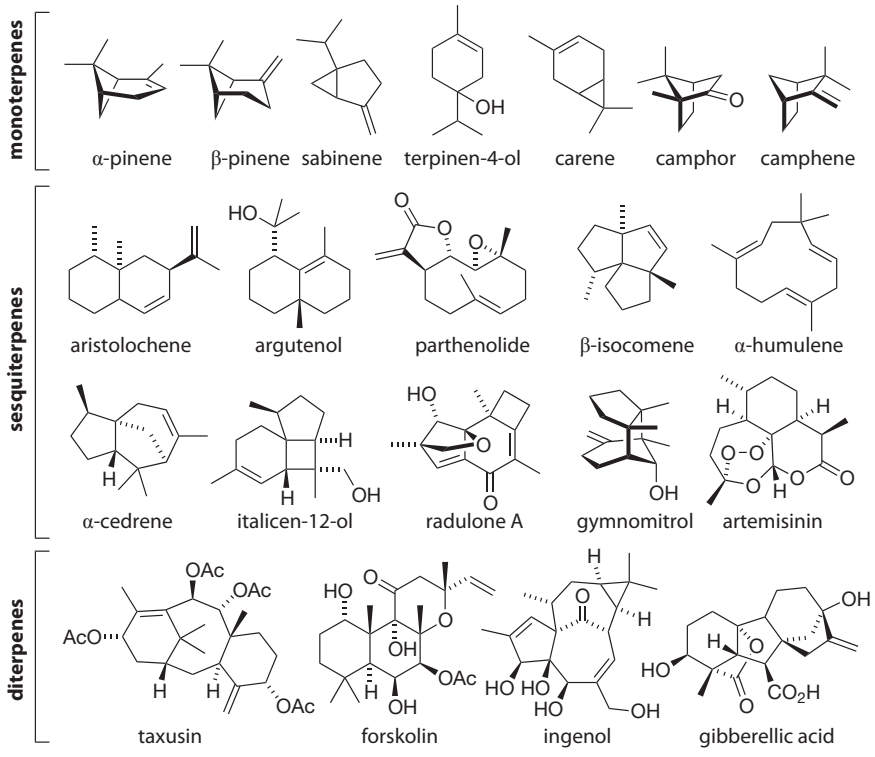



**Figure 8.104 Examples of the many cyclic terpene natural products synthesized by plants and microorganisms.**

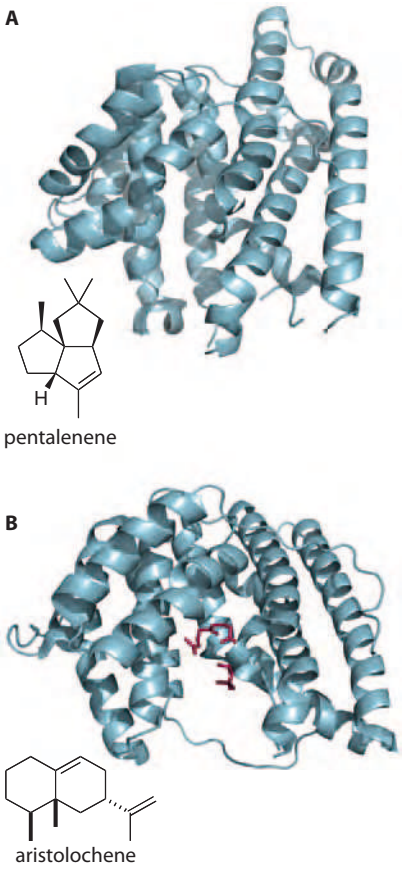



**Figure 8.105 The structure of terpene synthases.** Pentalenene synthase (A) (PDB 1HM7) and aristolochene synthase (B) (PDB 2BNY) have a common structural fold, despite their very low sequence homology and different products. The substrate mimic 2-fluorofarnesyl diphosphate (red) highlights the location of the enzyme's active site.

the following steps. First, the enzymes bind to the diphosphate substrate. During this binding event, water must be squeezed out of the binding site and the active site must be sealed shut; any water present during the cyclization would otherwise react quickly with any carbocation intermediates, shutting down the reaction. Thus, the active sites of terpene synthases resemble a hydrophobic deep cave. Binding to the back of this cave, the isoprenoid tail of the diphosphate allows the walls of the cave to collapse around it. The active site pushes the substrate into an approximate shape of the product, which helps to guide the product specificity. The mouth of the cave includes $Mg^{2+}$ ions chelated by the carboxylate side chains of aspartic and glutamic acid residues. As Lewis acids, the $Mg^{2+}$ ions bind to the diphosphate of the substrate, which seals the mouth of the cave shut.

Next, the Lewis acidic $Mg^{2+}$ ions promote loss of diphosphate, which results in the generation of the carbocation intermediate. This intermediate requires careful shepherding to avoid premature quenching and the enzyme guides the formation of the correct product. Through cation–π and other interactions, the terpene synthase active site can stabilize specific intermediates, favoring particular reaction pathways and thus products. Illustrative of the power of this class of enzymes, examples of terpene synthases favoring a secondary over a tertiary carbocation are known. In the penultimate step, the reactive carbocation is quenched, either by enzyme-directed reaction with a nucleophile such as water, or through β-elimination to yield an olefin. Finally, the hydrophobic active site must unseal, allow the entry of water, and resolvate the hydrophobic product. Given the resistance of such hydrophobic compounds to dissolve readily in water, this last step, although perhaps the least complex from the view of chemical reactivity, can be the rate-determining step of the reaction.

The simplest terpene natural products are the linear monoterpenes. Geranyl diphosphate is the universal substrate among plant terpene synthases. Linalyl diphosphate synthase is among the simplest terpene synthase. The enzyme catalyzes the $S_N1$ isomerization of geranyl diphosphate to linalyl diphosphate (**Figure 8.106**). Although the mechanistic details are not yet available, it is tempting to suggest that the oxygen that attacks the allyl cation is different from the one that was initially bound, making it possible for the reaction to proceed through a highly facile transposition.

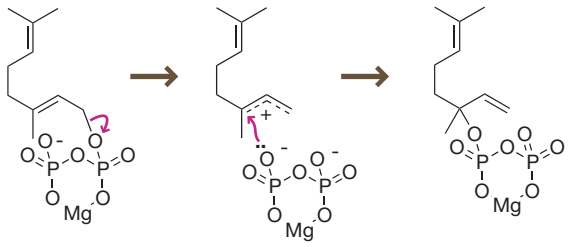


**Figure 8.106 Allylic transposition.** The mechanism of linalyl diphosphate synthase involves formation of an allyl cation.

## Isomerization of geranyl diphosphate to linalyl diphosphate facilitates cyclization

The mechanism of linalyl diphosphate synthase is not profound, but the two steps—ionization of geranyl diphosphate and addition to give linalyl diphosphate—are used by all enzymes that generate cyclic monoterpenes from geranyl diphosphate. The initial isomerization to form linalyl diphosphate is essential, because geranyl diphosphate ionizes to yield an *E*-geranyl cation, which cannot directly cyclize. To understand this limitation, recall that allylic cations have partial double-bond character, which prevents bond rotation (**Figure 8.107**); thus, the cyclization of a geranyl cation

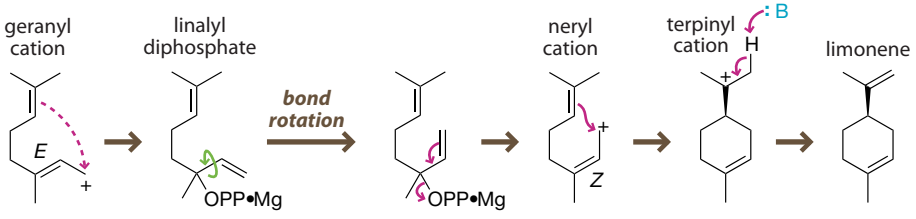



**Figure 8.107 The mechanism of limonene synthase.** The first step shown on the left is disallowed, because it would result in the formation of a *trans* (*E*) olefin in a six-membered ring.

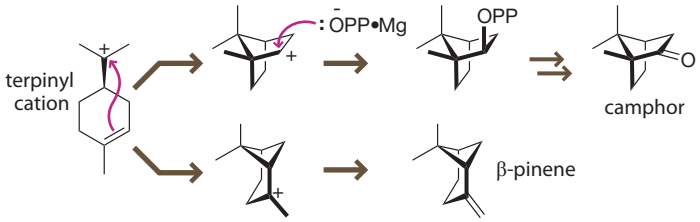



**Figure 8.108 Two ways to cyclize.** The transannular cyclization of a terpinyl cation can give two different ring systems.

would generate an impossibly strained six-membered ring containing a *trans* double bond. Trapping the geranyl cation to generate linalyl diphosphate provides a freely rotatable vinyl group. Linalyl diphosphate can ionize to form a neryl cation with a *Z* configuration. The allylic terminus of the neryl cation is poised to react with the trisubstituted olefin. After the cyclization of the neryl cation, the resulting terpinyl cation, common to many monoterpene biosynthetic pathways, can undergo many possible fates. In the enzyme limonene synthase, a base deprotonates one of the methyl groups to generate limonene, a common component in the oil of citrus fruits.

All terpenes are generated through enzymatic reactions, but the mechanistic pathways that lead to the diverse terpene skeletons can generally be explained by simple arrow-pushing mechanisms composed of cationic additions and [1,2] migrations. A large number of different terpene natural products can arise from a single precursor. For example, the terpinyl cation that generates limonene in limonene synthase can further cyclize to bicyclic structures in other enzymes (**Figure 8.108**), thus generating a [2.2.1] bornyl cation that is trapped by pyrophosphate or the [3.3.1] ring system of α-pinene and β-pinene.

The mechanism for the formation of (+)-sabinene hydrate by the cyclase enzyme from sweet marjoram involves an initial isomerization to (*R*)-linalyl diphosphate, followed by ionization to generate a terpinyl cation in a boat conformation (**Figure 8.109**). In theory this cation could undergo a **transannular** cyclization to generate a bornyl or pinanyl ring system. Instead, the cation undergoes a hydride migration to generate a homoallyl cation that then cyclizes to generate a cyclopropylcarbinyl cation. Ultimately, the cyclopropylcarbinyl cation is trapped by water to generate sabinene hydrate as a mixture of two epimers.

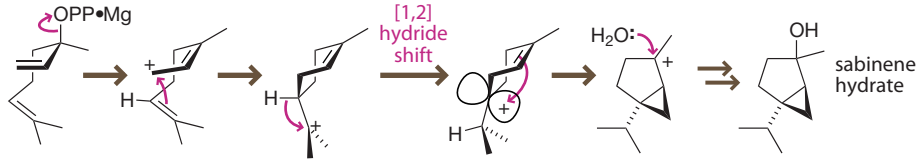



**Figure 8.109 Cyclopropane formation.** The formation of sabinene hydrate involves a hydride shift, facilitating formation of a highly stable tertiary cyclopropylcarbinyl cation.

While on vacation in Florence, Italy, the biochemist Konrad Bloch noticed the honey blonde hair of women depicted in the paintings of fifteenth-century Europe (**Figure 8.110**). Bloch found their hair color puzzling, especially since the women lacked the blue eyes expected for Nordic blondes. Italian women today, as then, typically have brunette hair, and the most common method for artificially lightening hair

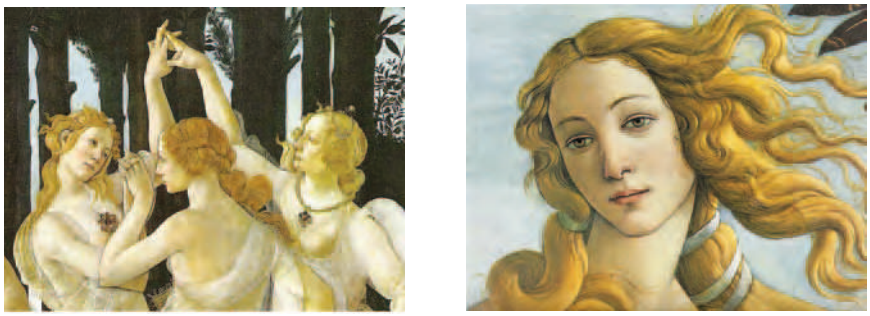


**Figure 8.110 Blondes in Venetian paintings.** These selections from *Primavera* (1482, left) and *The Birth of Venus* (1486, right), both by Sandro Botticelli, illustrate that Italian women had access to blonde hair colors well before the invention of hydrogen peroxide.

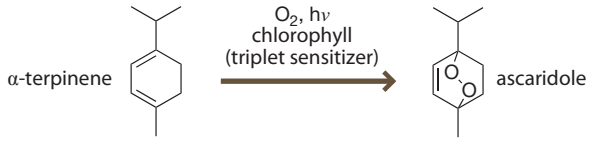



**Figure 8.111 Diels-Alder reactions with air.** Sensitized photooxidation has been proposed as a source of alkyl peroxides used in early hair bleaches. For example, the natural product ascaridole can be formulated as the product of a [4+2] cycloaddition with singlet oxygen.

color requires peroxides; hydrogen peroxide would not be available until three centuries later. Awarded the 1964 Nobel Prize in Physiology or Medicine for his investigations of cholesterol biosynthesis, Bloch was well aware of the chemical properties of terpenes. On further investigation, he learned that the women depicted in Botticelli's paintings would comb plant extracts containing terpenes into their hair and sit in sunlight. Through sensitized photooxygenation reactions, alkyl peroxides could be generated from the terpenes in pine oil (**Figure 8.111**). However, the inefficient process probably required multiple treatments and hours spent in sunlight.

## The 2-norbornyl cation exhibits exceptional behavior

If the methyl groups are removed from the [2.2.1] ring system of borneol, the compound is said to possess a norbornyl ring system. 2-Norbornyl cations derived from terpenes and related synthetic substrates have exceptional properties that helped chemists understand the properties of carbocations and what makes them stable (**Figure 8.112**). There are two possible diastereomers of 2-norbornyl derivatives. When the substituent is on the same side as the *methylene* bridge, the compound is called the exo isomer. Conversely, when the substituent is on the same side as the *ethylene* bridge, it is called the endo isomer. Surprisingly, when an optically pure endo isomer of a 2-norbornyl sulfonate was subjected to a substitution reaction with the acetate ion, the exo isomer was isolated *in racemic form*. The formation of a racemic mixture could not be explained by a simple $S_N2$ reaction with inversion of configuration. When the optically pure exo isomer was subjected to the same conditions, the reaction was 350 times faster, and the exo acetate was formed with retention of configuration.

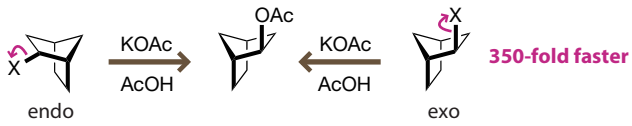



**Figure 8.112 Reactivity of endo and exo isomers.** Exo isomers of 2-norbornyl derivatives (X = sulfonate or other leaving group) undergo $S_N1$ reactions much faster than the corresponding endo isomers.

We now know that the exo isomer ionizes faster as a result of hyperconjugative donation by a perfectly aligned axial C–C bond, which is antiperiplanar to the leaving group. In fact, this alignment is so perfect that the 2-norbornyl cation actually exists as a symmetrical system with a three-center two-electron bond, as opposed to two carbocations that equilibrate through a higher-energy transition state (**Figure 8.113**). The exo isomer facilitates the formation of this three-center two-electron bond owing to the alignment of the axial C–C bond, which is not aligned correctly in the endo isomer. Bond alignment is also a critical factor in determining the ease by which [1,2] migration reactions can take place. If we recall the mechanism of lanosterol synthase, the impressive aspect of this reaction is not just the four rings and many stereocenters generated: an equally impressive achievement by lanosterol synthase is the grip that the enzyme exerts on the protosteryl cation to hold intermediates in a conformation that facilitates four precisely controlled [1,2] shifts.

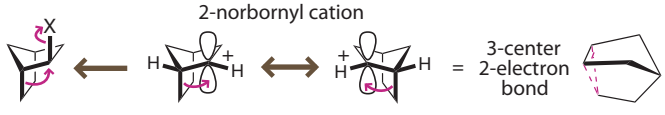
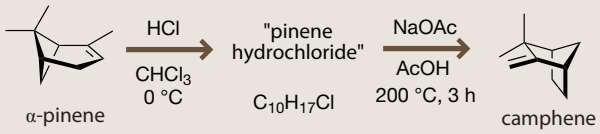



**Figure 8.113 The importance of bond alignment.** Precise alignment of a C–C bond facilitates ionization of the leaving group in exo-2-norbornyl derivatives and in the two resonance depictions of the 2-norbornyl cation.

## Problem 8.21

When α-pinene is treated with anhydrous HCl, it generates an adduct called pinene hydrochloride that is structurally analogous to camphor. When pinene hydrochloride is treated with sodium acetate at high temperature, camphene is produced. Suggest a plausible arrow-pushing mechanism for the formation of pinene hydrochloride from α-pinene and the subsequent conversion to camphene.




## Minor products offer clues to the enzymatic mechanisms of terpene cyclases

Terpene cyclases are not always perfect. Recombinant enzymes are often shown to generate mixtures of terpenes when supplied with an appropriate substrate. For example, *epi*-cedrol synthase was shown to convert farnesyl diphosphate into *epi*-cedrol along with cedrol in a 96:4 ratio. About 3% of the product mixture consists of other olefinic products, such as α-cedrene, β-cedrene, acoradiene, (*E*)-α-bisabolene, and (*E*)-β-farnesene (**Figure 8.114**). This molecular debris trail offers clues to the enzyme's mechanism.

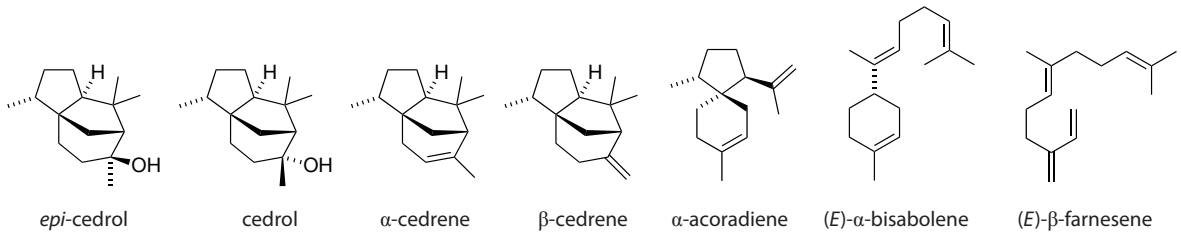



**Figure 8.114 Debris trail.** Major and minor products generated by *epi*-cedrol synthase.

For example, the mechanism of *epi*-cedrol synthase is believed to involve an initial isomerization to nerolidyl diphosphate that ionizes to yield a *Z*-allylic cation (**Figure 8.115**). Deprotonation of nerolidyl cation gives (*E*)-β-farnesene. Cyclization of the nerolidyl cation generates a six-membered ring and deprotonation of this cation generates (*E*)-α-bisabolene, but the preferred enzymatic pathway involves a [1,2] hydride shift to generate a cyclohexyl cation. Cyclization of the pendant olefin generates a spirocyclic [5.4.0] acorane ring system. The isopropyl cation is poised to cyclize with the cyclohene olefin, but a small amount of the isopropyl cation is deprotonated to generate acoradiene. The appropriate cyclization generates the cedrane ring system. Stereoselective attack by water favors the formation of *epi*-cedrol, but a small amount undergoes elimination to give α- and β-cedrene.

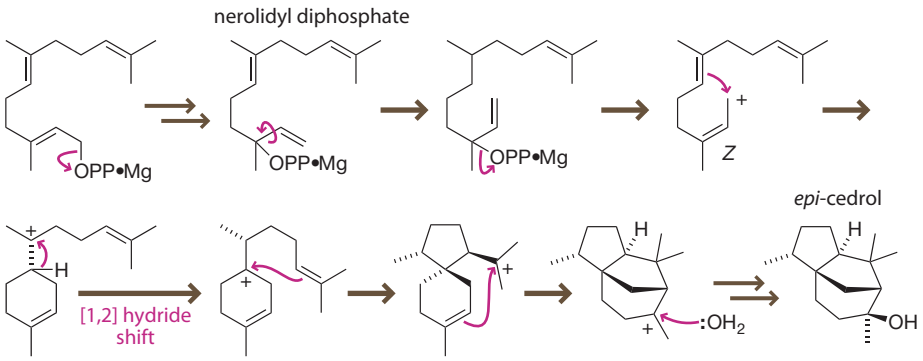



**Figure 8.115 *epi*-Cedrol synthase.** The mechanism of *epi*-cedrol synthase accounts for the other terpene natural products produced by the enzyme from nerolidyl diphosphate.

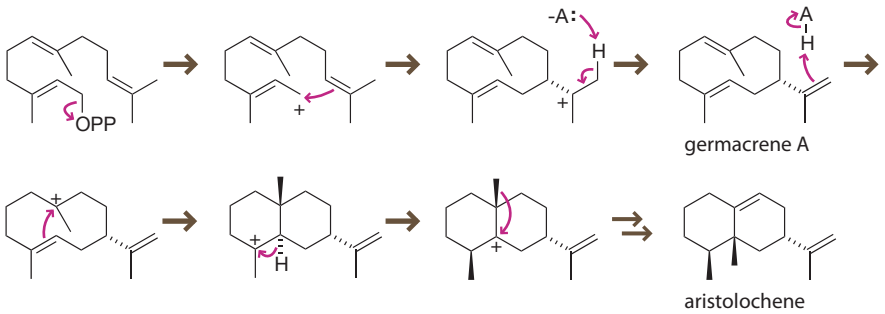


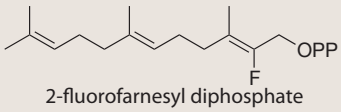

**Figure 8.116 Aristolochene synthase.** Medium-sized rings with 8–11 carbon atoms are among the most difficult to synthesize due to transannular steric interactions. Aristolochene synthase initially catalyzes the cyclization of farnesyl diphosphate to produce the 10-membered ring intermediate garmacrene A.

## Some terpene cyclases generate medium-sized rings

Eight-, nine-, and ten-membered rings are among the most difficult to generate through chemical cyclization, because of the strain produced by transannular steric interactions. Recall that the direct cyclization of geranyl diphosphate, without isomerization to linalyl diphosphate, is virtually impossible because it would place a *trans* double bond in a small ring. Some enzymes generate cyclic sesquiterpenes without first isomerizing the *E*-allylic diphosphate to a tertiary allylic diphosphate. For example, germacrene A synthase catalyzes the cyclization of farnesyl diphosphate leading to a strained 10-membered ring (**Figure 8.116**). Aristolochene synthase performs this same cyclization to yield germacrene, but takes the transformation even further. An acidic functional group protonates one of the trisubstituted double bonds in the ring, leading to a tertiary carbocation that undergoes a transannular cyclization to generate a decalin ring system. The acidic functionality in the active site responsible for the protonation of the olefin is most probably a phenolic hydroxyl group of a tyrosine side chain, a reaction unlikely to be considered outside an enzyme active site. A [1,2] hydride shift and a [1,2] methyl shift result in a bridgehead carbocation that is deprotonated to generate aristolochene.

### Problem 8.22

Why is 2-fluorofarnesyl diphosphate an inhibitor of aristolochene synthase?




2-fluorofarnesyl diphosphate

The enzyme pentalenene synthase generates pentalenene from farnesyl diphosphate. The reaction proceeds via initial cyclization to an 11-membered ring humulyl cation (**Figure 8.117**). A [1,2] hydride shift would generate a secondary carbocation poised for transannular cyclization. For many years it was speculated that the humulyl cation was deprotonated to generate the well-known natural product α-humulene, and then reprotonated to generate an isomeric cation. However, there is no evidence for a catalytic base in the enzyme. Transannular cyclization generates a fused 5–8 ring system. A [1,2] hydride shift would generate the bridgehead carbocation needed for the final cyclization to generate the tricyclic skeleton of pentalenene. When isotopically labeled [8-$^3$H]-farnesyl diphosphate was transformed by pentalenene synthase, the tritium label appeared at the expected position.

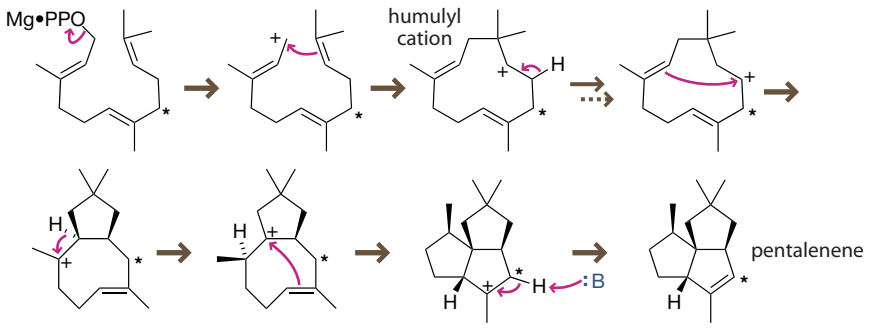



**Figure 8.117 Isotoping labeling.** The mechanism for the formation of pentalenene from farnesyl pyrophosphate is supported by tritium labeling. The tritium-substituted carbon is indicated with an asterisk.

## The biosynthesis of some terpenes involves nontraditional [1,3] hydride shifts

Hydride and alkyl shifts are highly responsive to bond alignment. Thus, terpene synthases can facilitate certain [1,2] migrations by holding cationic intermediates in specific conformations. Similarly, terpene synthases can also disfavor [1,2] shifts by holding carbocations in specific conformations. In such cases, less obvious cation reactions can occur. [1,2] Migration reactions dominate in solution-phase carbocation chemistry, but in a few systems there is clear evidence for hydride transfer over longer distances. In solvolytic reactions of strained cycloalkyl tosylates, for example, the initial secondary carbocation undergoes a transannular [1,5] hydride transfer to generate a more stable tertiary carbocation (**Figure 8.118**).

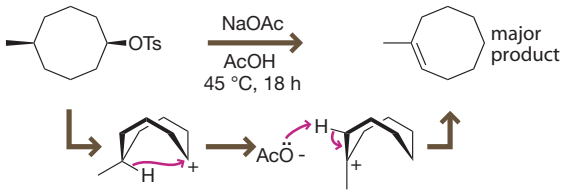



**Figure 8.118 Proximity effect.** Transannular [1,5] hydride migrations are observed in eight-membered rings.

The mechanisms of some terpene cyclases are believed to involve [1,3] hydride migration. In the case of longifolene synthase, for example, the mechanism is believed to involve an initial ionization to generate the tertiary allylic diphosphate nerolidyl diphosphate, which can ionize to form a *Z*-allylic cation (**Figure 8.119**). Contrary to chemical intuition, the cation cyclizes to generate a 13-membered ring with a secondary carbocation. A [1,3] hydride shift next generates a more favorable allylic cation. The allylic cation could undergo transannular cyclization to generate the [5.4.0] ring system. A second transannular cyclization would generate a norbornyl cation, which could be deprotonated to generate longifolene. Thus, terpene synthases have numerous powerful points of control over the reaction pathway: such enzymes can control the stereochemistry of the initial allylic cation, which atoms form the initial ring, which atoms are aligned to migrate, and which protons are set to eliminate.

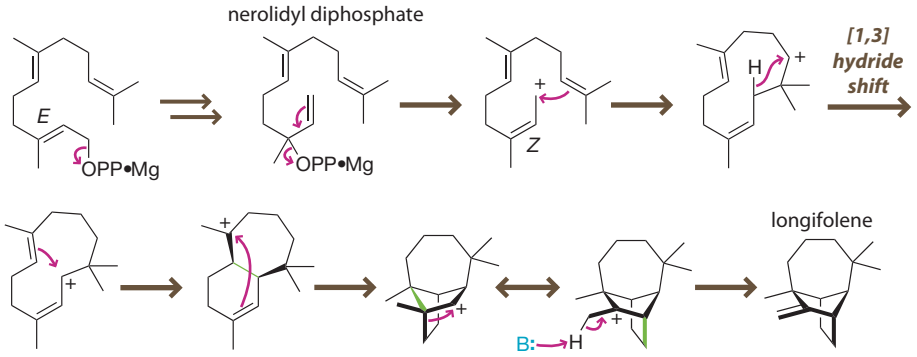



**Figure 8.119 Mechanism of longifolene biosynthesis.** Longifolene biosynthesis is believed to involve a [1,3] hydride shift. To follow the final cyclization to form the three-ring system, note the formation of a new bond (shown in green) between two carbons attached to methyl groups.

## Plants can also make complex triterpenes from squalene

The cyclization of oxidosqualene to lanosterol is a truly amazing reaction. However, the range of possible structures is far greater than the repertoire of human steroids would suggest. Plants also make terpenes from tail-to-tail dimers such as oxidosqualene, and triterpenes have been isolated with more than 90 different skeletons

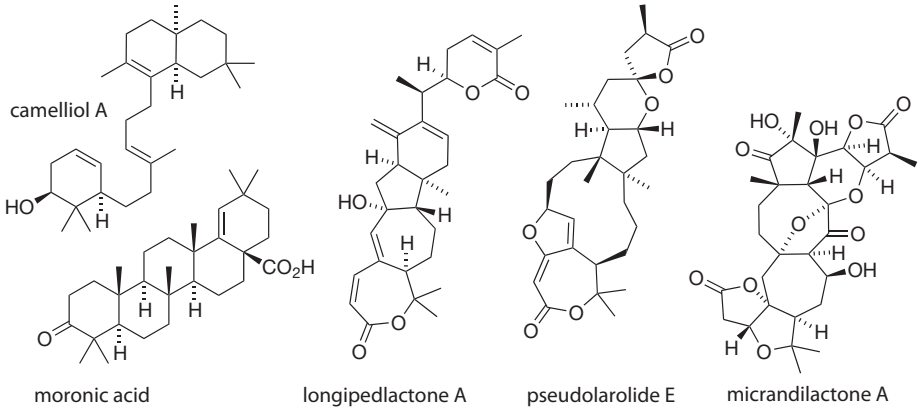



**Figure 8.120 Dazzling complexity.** Plant triterpenes, such as the five shown here, have a much wider range of structures than human steroids.

(**Figure 8.120**). Post-cyclization oxidation reactions greatly enhance the degree of functionality and the diversity of these natural products.

## Hyperthermophilic archaebacteria produce cyclic lipids from terpenes

**Figure 8.121 The heat is on.** This boiling volcanic spring in Campi Flegrei, Italy, is home to the hyperthermophile *Sulfolobus solfataricus*. (Courtesy of Science@NASA.)

Heat disrupts lipid bilayers, yet many microorganisms in the primitive domain Archaea—which is distinct from Eukaryotes and prokaryotic Eubacteria—thrive at high temperatures (**Figure 8.121**). How do they maintain the integrity of their membranes? Hyperthermophilic bacteria that thrive at temperatures above 60 °C have been found to produce a structurally complex set of lipids, relative to the diacylglycerol motif of human lipids. Some hyperthermophiles prosper within the withering heat of an autoclave. The membranes of hyperthermophiles are predominantly cyclic glycerol dialkyl glycerol tetraether lipids, some containing additional five-membered rings (**Figure 8.122**). These lipids arise from terpene biosynthetic pathways rather than fatty acid synthases. Interestingly, the tetraterpene chains are linked head-to-head, whereas squalene and phytoene are linked tail-to-tail. Only a few strains of traditional Eubacteria have been found to contain glycerol dialkyl glycerol tetraether lipids, which are abundant in hyperthermophilic archaebacteria.

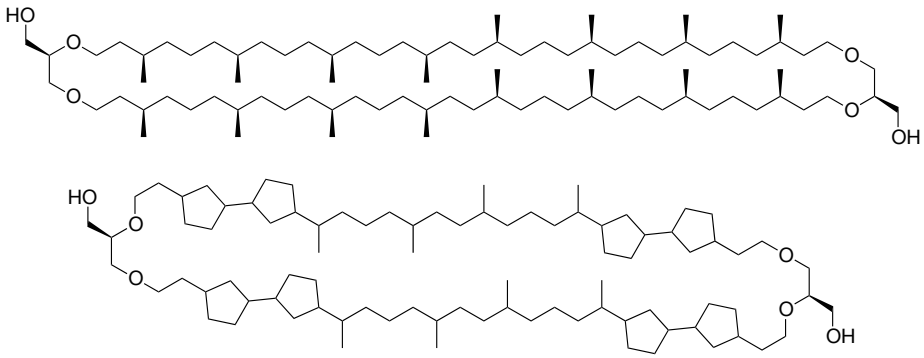
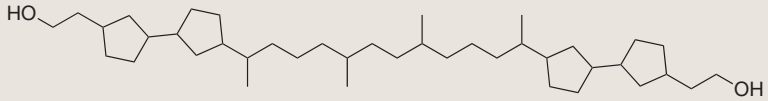


**Figure 8.122 Super-sized rings.** Some hyperthermophilic archaebacteria contain macrocyclic tetraether lipids that are structurally and biosynthetically distinct from traditional lipids based on fatty acids.

### Problem 8.23

Identify each of the terpene subunits in this diol portion of an archaeal cyclic ether lipid.



## 8.8 SUMMARY

We humans pride ourselves on the ingenuity of our species. We take simple resources and craft them into ingenious devices such as axles, slingshots, bicycles, telescopes, guns, engines, generators, pharmaceuticals, airplanes, computers, and spacecraft. Given our ability to invent so prolifically, it is disappointing that the catalog of human polyketides should be such a meager affair—namely, straight-chain fatty acids, linear sphingoid molecules, and some mildly interesting eicosanoids. The best that might be said of human polyketides is that they fulfill their intended function as primary metabolites. How can one not be awestruck by the spectacular range of polyketides produced by "primitive" organisms—namely, polycyclic aromatics, polyenes, polyynes, enediynes, heterocycles, spirocycles, strained rings, and macrocycles, which bristle with stereogenic centers and glisten with novel sugars? Our artless implementation of polyketide biosynthesis is even more embarrassing when we realize that microorganisms generate their resplendent polyketides by using the same reactions that we use to generate boring saturated fatty acids.

Humans have done an equally minimal job when it comes to the design and production of terpenes. Our cells generally tinker with simple prenyl chains, using them to alkylate proteins and quinones. We seem to have chanced upon one exciting cyclization—the conversion of squalene epoxide into lanosterol—but rather than develop new cyclization reactions our bodies simply modified lanosterol, investing 19 laborious steps to convert it to cholesterol, and even more effort to make the various steroid hormones with a monotonous A-B-C-D ring system. Vitamin D is a notable departure from this theme because it lacks a B ring. A quick glance at the range of monoterpene secondary metabolites found in plants inspires envy. Plants have managed to make every possible ring system that can be accessed from the 10 carbons in geranyl diphosphate. Starting with longer precursors—that is, with 15, 20, and 30 carbons—Nature has accessed a boundless array of structures.

Evolution has clearly pushed otherwise defenseless organisms to maximize the potential of polyketide and terpene biosynthesis. Somehow, the human combination of physical agility, a complex immune system, and mental computing power has obviated the need for chemical weaponry to ensure the survival of our species. However, we need these kinds of sleek, membrane-permeable chemical weapons to wage a war of survival against infectious diseases, metabolic diseases, and even cancer. Humans aspire to reach a level of synthetic sophistication on a par with plants and microorganisms. We are slowly catching up, not through human genetic evolution but through human intellect, by harnessing the combined power of organic synthesis, biochemistry, microbiology, and molecular biology.

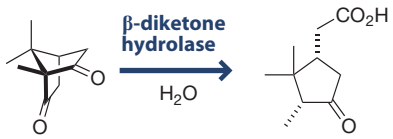
### LEARNING OUTCOMES

- Understand the chemical mechanisms of each step in fatty acid biosynthesis—namely, the Claisen condensation, carbonyl reduction, β-elimination, and conjugate reduction.
- Appreciate the limited range of human fatty acids.
- Recognize the presence of polyketides in human phospholipids and sphingolipids.
- Understand how lipases can generate signaling molecules from lipids.
- Recognize the four classes of eicosanoid signaling molecules derived from arachidonic acid—namely, thromboxanes, prostacyclin, prostaglandins, and leukotrienes.
- Draw the structure of ceramide and understand how it is formed from palmitoyl-CoA.
- Understand the effects of N-myristoylation and S-palmitoylation on protein localization.
- Understand the chemical transformation of fats into partly hydrogenated vegetable oil, soap, and biofuels.
- Understand the origin of 1,3-diol motifs in polyketide natural products.
- Draw the mechanism for aldol condensations that lead to aromatic polyketides.
- Understand how the incorporation of propionate subunits generates methyl substituents on polyketide chains.
- Identify the acetate and propionate subunits in polyketide natural products.
- Understand how the modular genetic organization of type I polyketide synthases facilitates genetic reprogramming.
- Recognize the similarities between polyketide biosynthesis and nonribosomal peptide synthesis.

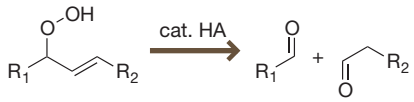
- Draw the structures of the linear terpenes geraniol, farnesol, geranylgeranyl alcohol, and squalene.
- Identify which step of cholesterol biosynthesis is inhibited by statins.
- Draw the sequence of post-translational modifications that accompany prenylation.
- Recognize the prenylated quinones used by humans.
- Understand the structure and reactivity of cyclopropylcarbinyl cations.
- Draw the arrow-pushing mechanism of the transformation catalyzed by lanosterol synthase.
- Recognize human steroid signaling molecules.
- Appreciate the dietary origin of human retinoids.
- Understand how cationic cyclizations and rearrangements amplify the diversity of terpene natural products.
- Describe the key steps catalyzed by a terpene synthase in the biosynthesis of terpenes.
- Understand the structure and reactivity of 2-norbornyl cations.
- Recognize the complexity of cyclic ether lipids from hyperthermophilic archaebacteria.

# PROBLEMS

**8.24** Suggest a plausible arrow-pushing mechanism for the retro-Claisen reaction of 6-oxo-camphor to (2*R*,4*S*)-α-campholinic acid by the enzyme β-diketone hydrolase. Assume that the active site contains basic and acidic residues.

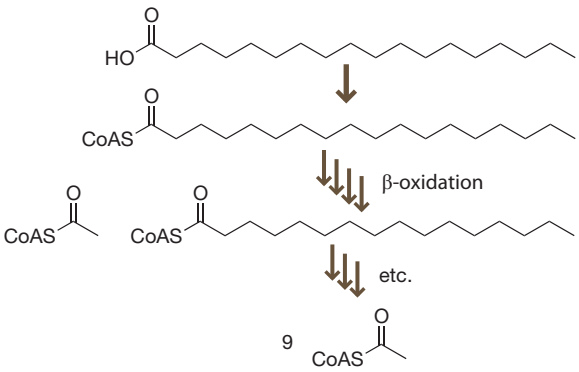



***8.25** The autoxidation of lipids leads to unstable allylic peroxides that can oxidatively cleave via the Criegee mechanism. Suggest a plausible arrow-pushing mechanism for this reaction.

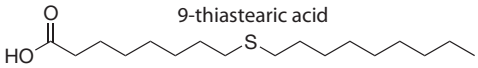



**8.26** Humans and other organisms metabolize fatty acids back into acetyl-CoA through a process called β-oxidation, which is the reverse of the steps used to make fatty acids.

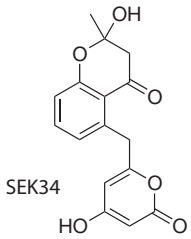
**A** Draw the three intermediates in the first cycle of β-oxidation of stearic acid.



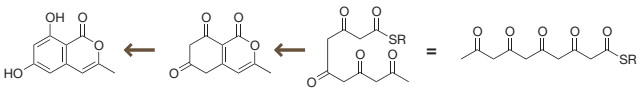

**B** Thia-fatty acids, which have sulfur atoms at odd-numbered positions, cannot be broken down all the way by β-oxidation. Draw the intermediate derived from 9-thiastearic acid that is resistant to further reaction.



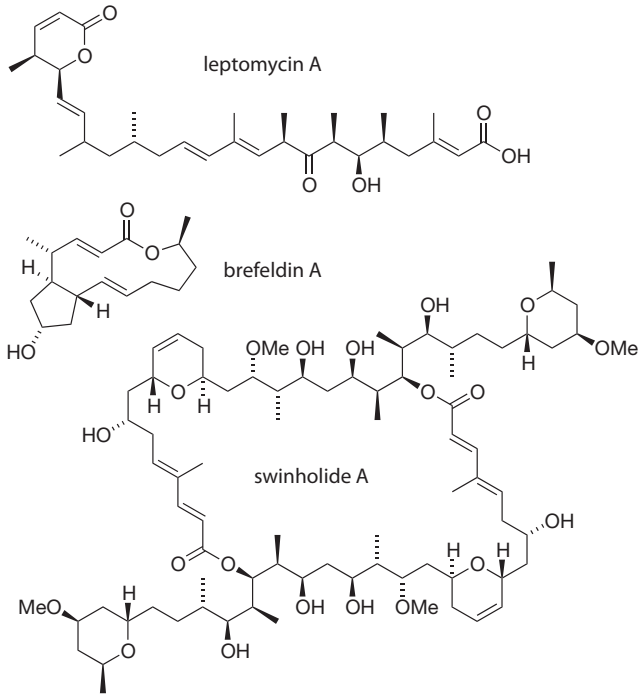

***8.27** Suggest a plausible acyclic polyketide precursor to SEK34, assuming that it arises through aldol reactions, dehydrations, and enolizations.



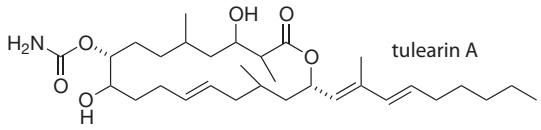

**example:**



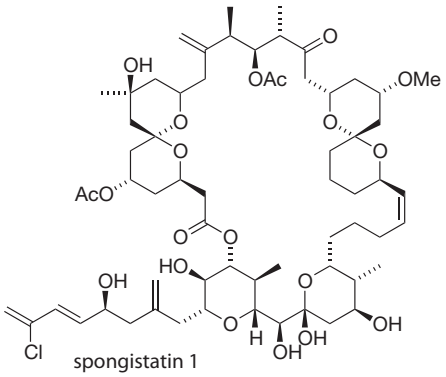

***8.28** Highlight each of the acetate and propionate subunits in the natural products brefeldin A, leptomycin A, and swinholide A.



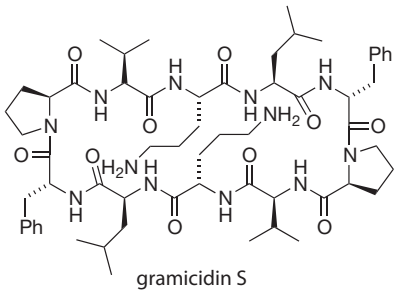
**8.29** Which part of tulearin A is the most difficult to rationalize on the basis of an iterative Claisen condensation of acetyl and propionyl subunits?



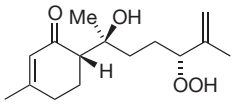

**8.30** The polyketide spongistatin 1 is active at femtomolar concentrations. Highlight each of the acetate and propionate subunits in spongistatin 1 and circle the carbon atoms derived from the carbonyl carbon of each subunit. Note which methyl groups do not seem to arise from a propionate subunit.

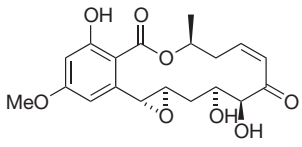


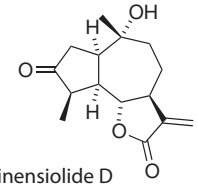

**8.31** Which of the amino acids in the antibiotic gramicidin S is not normally incorporated into proteins by a ribosome?

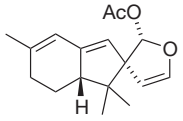


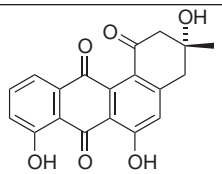

***8.32** For each of the following compounds, predict whether it most plausibly arises through a polyketide pathway or a terpene pathway.

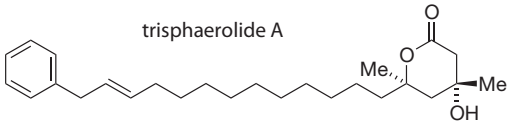

peroxylippidulcine B

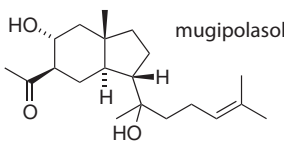

hypothemycin

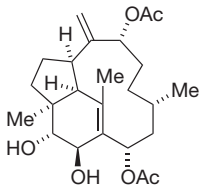

chinensiolide D


haterumadysin A


rabelomycin

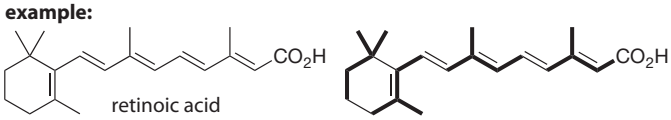

trisphaerolide A

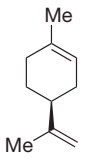

mugipolasol

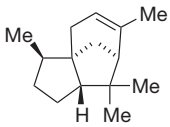

9α,14α-diacetoxy-2β,3α-dihydroxy-1(15),8(19)-trinervitadiene

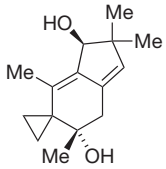
***8.33** Identify the five-carbon terpene subunits in the natural products (–)-limonene, α-cedrene, illudin M, and camphor. Illudin M is extra difficult; for experts only.

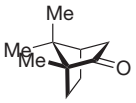
**example:**



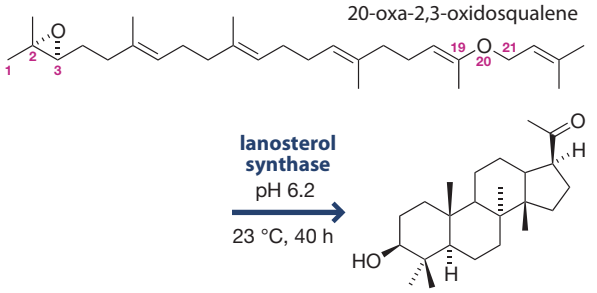


(–)-limonene


α-cedrene

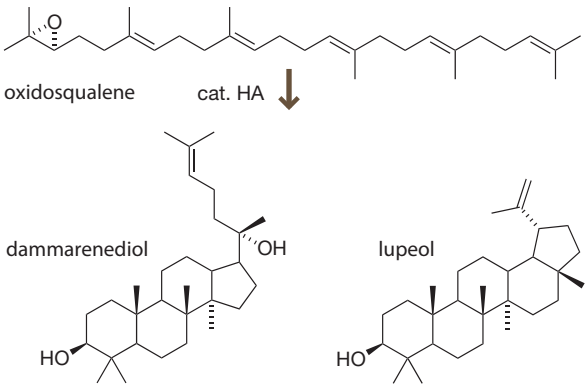

illudin M


camphor

***8.34** Propose an arrow-pushing mechanism for the following transformation of the 20-oxa analog of 2,3-oxidosqualene.

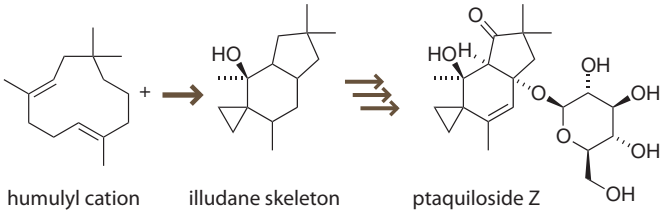



**8.35** The natural products dammarenediol-I and lupeol are both generated from oxidosqualene via the same cationic intermediates.

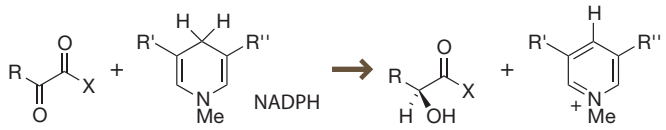



Suggest a plausible arrow-pushing mechanism for the biosynthesis of each triterpene.

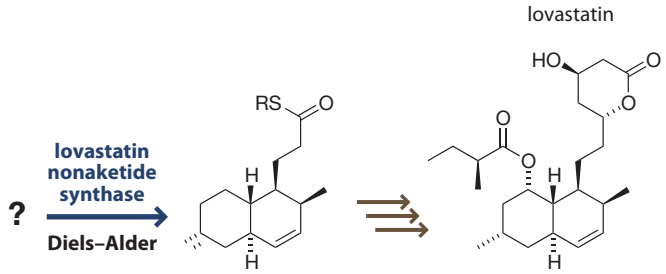
***8.36** Suggest a plausible arrow-pushing mechanism for the enzymatic synthesis of illudane terpenes via the humulyl cation. Assume that either catalytic acids or bases are present.




**8.37** Design a mimic of NADPH that would stereoselectively reduce an α-keto ester or α-ketoamide. Make use of noncovalent interactions such as π stacking, hydrogen bonding, and base pairing.



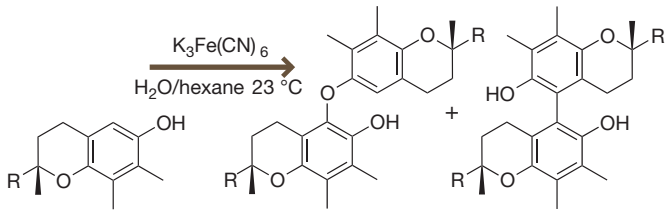

**8.38** Lovastatin nonaketide synthase is believed to catalyze an intramolecular Diels–Alder cyclization reaction. Draw the structure of the acyclic chain precursor that would give rise to the following bicyclic intermediate.



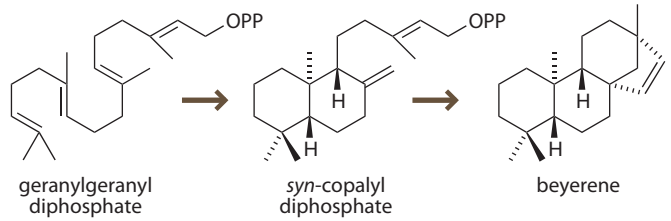

***8.39**

**A** γ-Tocopherol dimerizes in the presence of a one-electron ferric oxidant under basic conditions. Suggest a plausible arrow-pushing mechanism, representing the oxidant as X•.

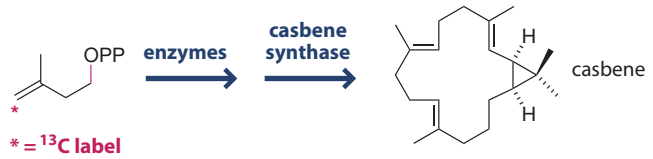
**B** Two diastereomers of the biaryl are generated in the reaction. Clarify the nature of these two diastereomers.



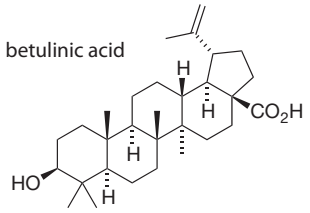

***8.40** The mechanisms of several terpene cyclases are believed to involve the initial formation of *syn*-copalyl diphosphate. Suggest a plausible arrow-pushing mechanism for the formation of beyerene from geranylgeranyl diphosphate through an acid-catalyzed mechanism that involves *syn*-copalyl diphosphate.



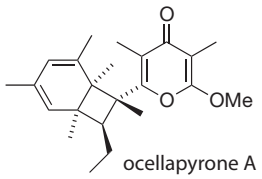

**8.41** When $^{13}$C-labeled isopentenyl pyrophosphate was used in a cell-free extract containing casbene synthase and other enzymes, eight carbons in casbene synthase were labeled. Suggest a plausible arrow-pushing mechanism for the synthesis of casbene by casbene synthase and show which carbons would end up labeled.



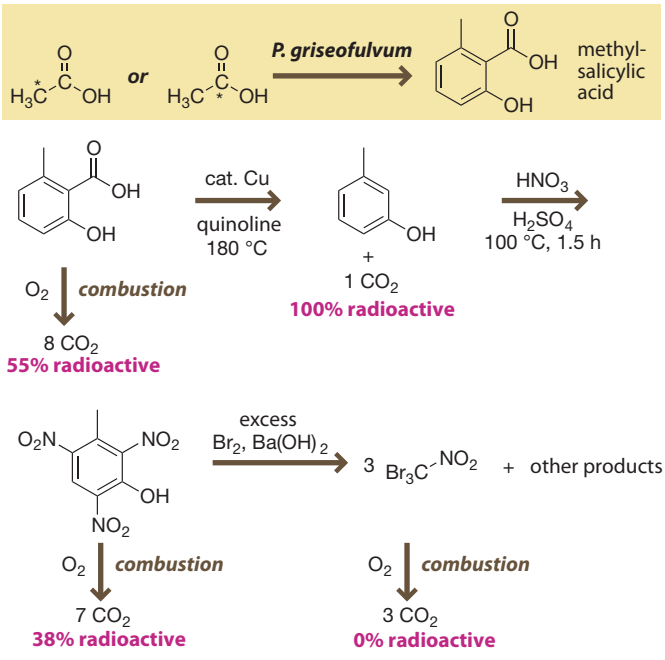

**8.42** Suggest a plausible arrow-pushing mechanism for the formation of the triterpene betulinic acid skeleton from oxidosqualene. The mechanism must include initial formation of the protosteryl cation (from Figure 8.95).




**8.43** Suggest a plausible acyclic polyketide precursor that could give rise to ocellapyrone A through two consecutive electrocyclic ring-closure reactions in the absence of light. If you are familiar with pericyclic selection rules, you may wish to think about the stereochemistry of the double bonds.




***8.44** In the earliest isotopic labeling study of polyketide biosynthesis, $^{14}$C-labeled acetic acid was fed to *Penicillium griseofulvum*. The resulting methylsalicylic acid was isolated and subjected to various chemical degradation and combustion reactions. On the basis of the radioactivity in the liberated $CO_2$, one can deduce the distribution of radioactive label in the methylsalicylic acid and each of the degradation products.




**A** Which carbon atoms in the methylsalicylic acid were isotopically labeled?

**B** Which carbon atom in the acetic acid was isotopically labeled?

# Chemical Control of Signal Transduction

# 9

## LEARNING OBJECTIVES

- Recognize primary signal transduction pathways in the scientific literature.
- Predict the effects of activators and inhibitors on signaling outcomes.
- Describe two rapid nontranscriptional processes that involve $Ca^{2+}$ ions.
- Identify and describe the seven major signal transduction pathways that control transcription in human cells.
- Distinguish the three major subpathways affected by receptor tyrosine kinases.
- Outline the two types of pathways affected by G protein-coupled receptors.
- Recognize the ionic imbalance maintained by human cells.

At the outset of this book, we proposed to use the central dogma of molecular biology as a framework for understanding how human cells work. Genes are transcribed into RNA; RNA is translated into enzymes; enzymes catalyze the formation of glycans, lipids, and terpenes. Our relentless focus on the central dogma led us to skirt *the* fundamental question: *Which genes*? Gene expression determines how cells differentiate, allowing each cell to have a precise role in the human organism. Gene expression also determines the way in which fully differentiated cells respond to their environment for the maximum benefit of the organism. Ultimately, extracellular cues determine which genes are expressed both during and after differentiation. In this concluding chapter, we focus on the biochemical pathways that transduce signals from outside the cell to changes in gene expression (**Figure 9.1**). Chemists who understand and control these pathways are empowered to remediate errant cells that cause disease and to harness cells for new applications.

The human quest to control cellular signaling with small molecules is an ancient one. The earliest molecular tools came from natural sources such as plants, and their use pre-dates written records. Archaeological evidence suggests that residents of the Nanchoc valley in Peru were orally freebasing cocaine in the eighth millennium BC by chewing on coca leaves along with lime (calcium carbonate). Furthermore, poppies, a source of bioactive compounds, were cultivated on the Eurasian continent as early as the fourth millennium BC, and were prominent in Egyptian art by the Eighteenth Dynasty (1549–1298 BC).

The earliest comprehensive written pharmacopeia is contained within the Ebers papyrus, an ancient Egyptian papyrus written around 1550 BC. Many formulations in the Ebers papyrus involve plants with plausible medicinal value—such as pomegranate root bark, poppy plants, and juniper berries; and many more of the formulations

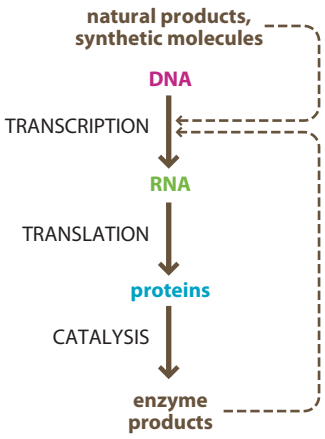



**Figure 9.1 Small molecules at work.** RNA, proteins, and enzyme products exert control over transcription from within the cell. Natural products and small synthetic molecules that originate outside the cell can also affect transcription.

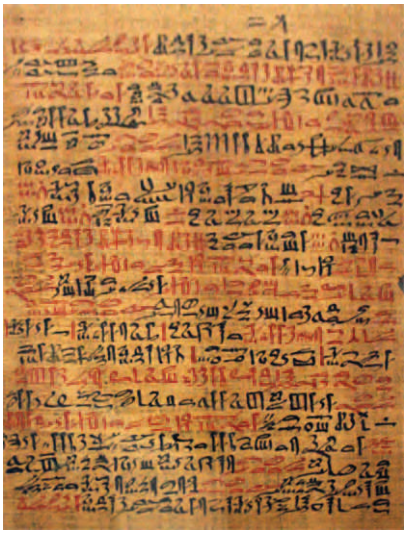


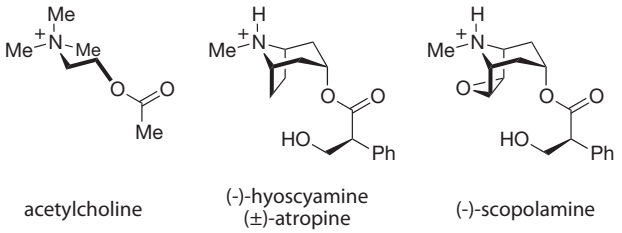
**Figure 9.2 The ancient Egyptian pharmacopeia.** The Ebers medical papyrus is written in hieratic script. Page 33, column 89, line 13 describes a treatment for tooth pain involving the netherworld plant.

**Figure 9.3 A Babylonian painkiller.** A cuneiform tablet from the library of King Ashurbanipal describes an ancient treatment for toothache involving a plant believed to be henbane (right). The exact date of origin of the treatment cannot be assigned because the scribes (*ca* 650 BC) were copying and recording much earlier writings dating back many thousands of years. (Left, courtesy of Wikimedia; right, courtesy of Martin Vidner.)

involve plants that cannot be identified. For example, the Ebers papyrus describes a recipe for dental maladies involving celery, beer, and the netherworld plant (**Figure 9.2**). A similar prescription was found on a cuneiform tablet, penned a thousand years later, from the library of Ashurbanipal in Nineveh: a mixture of oil, beer, and the *sa-kil-bir* plant (**Figure 9.3**). The Babylonian *sa-kil-bir* plant has been interpreted as henbane, which contains large amounts of the tropane alkaloid hyoscyamine (L-atropine). Wisely, these treatments were not intended for ingestion, because hyoscyamine is toxic at high concentrations. Unfortunately, plausible formulations in the Ebers papyrus are listed side by side with questionable treatments involving, for example, incantations and animal excrement. Even today, desperate patients are faced with an equally wide variation in choices, ranging from clinically proven drugs to fake medicines with deceptive labels such as "natural" and "herbal."

The ancient formulations of Chinese traditional medicine are many thousands of years old and, unlike ancient Egyptian medicines, have endured to the present day. The precise age of the formulations is difficult to ascertain. The original copies of *Shénnóng Běn Cǎo Jīng*—the ancient materia medica of Chinese traditional medicine—have never been found, and the oldest copies are only about 2000 years old.

The earliest examples of natural medicines and poisons involve compounds that interfere with neuronal signaling, either between pairs of neurons or between neurons and muscle cells. Neuronal signaling is rapid, making it easy for primitive cultures to establish cause-and-effect relationships. For example, plants in the family Solanaceae (henbane, mandrake, belladonna, and nightshade, among others) produce tropane alkaloids such as hyoscyamine and scopolamine (**Figure 9.4**). The racemic form of (–)-hyoscyamine, produced by the plant belladonna, is referred to as (±)-atropine, but only the (–) enantiomer is active. Hyoscyamine binds to a specific class of receptors in parasympathetic neurons by mimicking the structural features of the neurotransmitter acetylcholine. Parasympathetic neuronal pathways cause the body to "rest and




acetylcholine (-)-hyoscyamine (±)-atropine (-)-scopolamine

**Figure 9.4 Neurotransmitters and poisons.** The Solonaceae tropane alkaloids mimic acetylcholine.

digest." Hyoscyamine has many systemic effects. It is a local anesthetic, and causes dilation of the pupils, increased heart rate, decreased perspiration, decreased salivation, and cessation of bowel movement.

The most ancient of medicines—for example cocaine, morphine, atropine, and scopolamine—clearly have profound effects on cell function, but the immediate effects of these drugs do not involve the transcription of DNA into RNA. Medicines that exert their primary effects on transcription are much more recent. For example, *Croton tiglium*, mentioned in the *Shénnóng Běn Cǎo Jīng*, is a source of phorbol esters that activate protein kinase C, an enzyme that controls many transcriptional processes.

Our ability to "see" the inner workings of human cells is only a few decades old and is still evolving. The most common methods in use today still involve either tools that report on only one gene at a time or on post-mortem analysis of cellular debris such as DNA, RNA, and proteins. Chemical biologists do not yet have the ability to design potent selective inhibitors by simply looking at the structure of a protein target; however, by screening diverse collections of natural and synthetic compounds, we are discovering molecular tools that control the pathways that direct the transcription of human genes. Most of those tools that come from screening campaigns are rather blunt, but some can be sharpened for pinpoint accuracy. As we progress in the new millennium, one can expect to see new tools for inhibiting disease pathways, enhancing beneficial pathways, and directing the differentiation of stem cells to generate a tissue of choice.

## 9.1 SIGNAL TRANSDUCTION

### Chemical signaling is universal

Communication is fundamental to all animals on the planet Earth and involves many different types of signals: visual, auditory, chemical, and tactile. Male Indian peafowl boast with colorful displays. Siberian huskies growl out warnings. Avocado seed moths beckon with alluring pheromones (**Figure 9.5**). African elephant mothers reassure their calves with gentle caresses from their massive trunks.

All examples of signaling between organisms involve the relay of information from the outside of a cell to the inside. Although the types of signals used by various organisms may differ, all cells in all organisms—even single-celled organisms such as bacteria—communicate using chemical signals. The ability to receive signals is essential for cells because they must be able to respond to changes in their environmental conditions: conditions such as crowding, famine, and safety. The ability to send signals is just as important as receiving signals because even the simplest unicellular organism needs a way to allow its clonal brethren to distinguish it from predators, competitors, and prey. Intercellular communication is particularly important between human cells, because each human cell has a distinct function. Muscle cells contract. Neurons forge connections. Goblet cells spew mucus. Epithelial cells proliferate. B cells differentiate. Such cellular actions are conditional and occur only when the cells receive the correct chemical signals. Understanding how molecular signals control cell behavior, or misbehavior, is a central goal of molecular biology. Using small molecules as precise tools to control these processes is a major goal of chemical biology. Honing these small molecules into safe, effective medicines is the goal of the pharmaceutical industry.

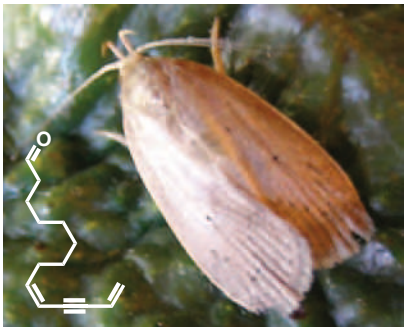


**Figure 9.5 Chemical sex appeal.** The avocado seed moth *Stenoma catenifer* responds to the pheromone 9(*Z*),13-tetradecadien-11-ynal. The larvae devastate crops by feeding on avocado pits and pulp. Adult moths can be lured into traps laced with the pheromone. (Courtesy of Mark S. Hoddle, University of California, Riverside.)

### The field of biology is full of cryptic acronyms and ambiguous symbols

The conversion of one type of chemical signal into another type of chemical signal is called **signal transduction**. The student interested in understanding the signal transduction networks that control cells faces a seemingly impenetrable morass of cryptic acronyms. For example, SOS1 stands for "son of sevenless homolog 1" and PAK1 is an abbreviation for "p21/Cdc42/Rac1-activated kinase 1." Often, the names have nothing to do with our current understanding of the role of the protein, like the ryanodine receptor—named after a molecule not found in the human body. Human signal transduction is further complicated by the redundancy in naming that proliferated in

the late 1990s. Researchers often christened their newly identified proteins with exotic new names, even when closely related proteins were already known in the literature. To understand how multiple names arise, consider the enzyme Akt. In 1991 the protein kinase Akt was identified in a virus (designated Akt-8) that causes cancer in the Ak strain of mouse. Around the same time, a nearly identical kinase was discovered independently in two different human cell lines and given two different names: protein kinase B and RAC protein kinase. Scientists published papers for many years on this kinase without agreeing on which name to use! Fortunately, one can now search vast online databases for any protein sequence to prevent redundant naming. However, the tendency to infer the function of a protein from a name is perilous. If you find yourself becoming disheartened by the large number of new names, then you might take comfort from the fact that you have learned the names of your friends and family members and developed a predictive understanding of their relationships. The names of human beings are just as arbitrary as the names of human proteins. Clearly time and patience will breed an easy familiarity with complex names and complex relationships.

## Problem 9.1

Look at the most recent table of contents for any of the top scientific journals or any specialized journal of biological chemistry, chemical biology, or biochemistry. List the biological molecules that are found in the titles. When molecules are designated by acronyms, use the Internet to find the full name on which the acronym is based.

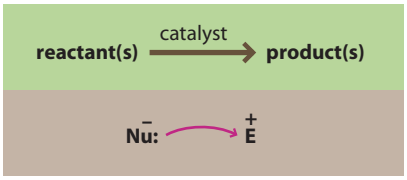



**Figure 9.6 Semantics of arrows.** The two types of arrows used by chemists have precise meanings. Straight, horizontal arrows indicate a chemical reaction. Curved arrows indicate the interaction of filled orbitals with unfilled orbitals.

New students in chemical biology tend to confuse the arrows in a signal transduction diagram with the arrows used in reaction diagrams. Organic chemists have agreed on a precise system of representation to depict chemical reactions. They use straight, horizontal arrows to show the transformation of reactants to products, and curved arrows depict the interaction of filled orbitals with unfilled orbitals (**Figure 9.6**). A signal transduction diagram makes use of arrows to depict the known and hypothesized molecular interactions that are responsible for translating extracellular signals into cellular responses. Arrows with a single arrowhead start from some molecule and end on either a function that is facilitated or a second molecule with a function resulting from the first molecule. Arrows that end with a small perpendicular bar generally indicate that the function of the second molecule is inhibited by the presence of the first. These relationships can be established through genetic deletion or chemical inhibition studies.

In signal transduction diagrams, an arrow can have many chemical meanings (**Figure 9.7**). An arrow from Abc to Xyz could mean that proenzyme Abc is converted

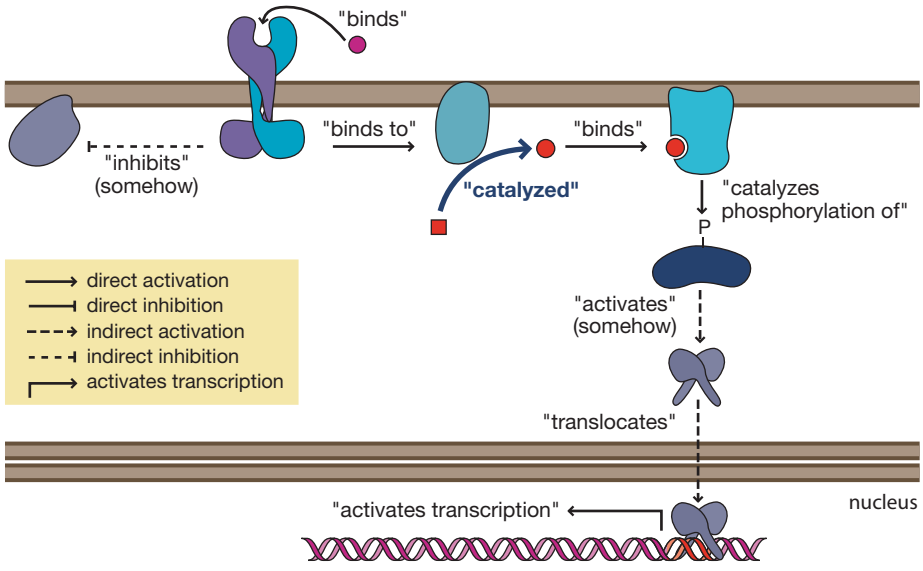



**Figure 9.7 Deciphering signal transduction diagrams.** Arrows in signal transduction diagrams have various chemical meanings, rarely corresponding to the transformation of starting materials to products. It is up to the reader to intuit the type of interaction that is being depicted. In this book we will use solid arrows to indicate processes that involve direct interactions between molecules (for example binding, chemical reactions, catalytic transformation, and covalent modification).

to the active form Xyz, or it could mean that enzyme Abc binds to a co-catalyst Xyz; alternatively, it might mean that enzyme Abc acts on substrate Xyz; of course, one cannot exclude the possibility that it means that enzyme Abc generates an active product Xyz. Arrows in signal transduction diagrams do not always imply a direct interaction; they are sometimes used to indicate spatial translocation, for example across a membrane. In other cases signal transduction arrows represent a series of complex biochemical phenomena in which the key components have been omitted. For example, an arrow from *Abc* to Xyz could mean that gene *Abc* is expressed as protein Xyz and necessarily involves a large number of intermediate steps. Note that the names of genes are usually italicized. Recently, there have been attempts to develop unambiguous notation for signal transduction diagrams, but these notations have not yet been widely adopted.

## Fast cellular responses do not involve the production of proteins

Before we learn about human signaling pathways that affect the transcription of genes (and ultimately the translation of proteins), we need to distinguish those pathways that affect transcription from those that do not. Some signals lead to responses by the biochemical machinery already in place in the cell; such signaling does not involve transcriptional regulation (**Figure 9.8**A). For example, after acetylcholine receptors in intestinal goblet cells bind to acetylcholine, the cells rapidly release the oligosaccharide mucin from storage vesicles (**Figure 9.9**). The mucin helps to ensure a slippery pathway for debris traveling down the gastrointestinal tract. Like it or not, you resemble a slug turned inside out.

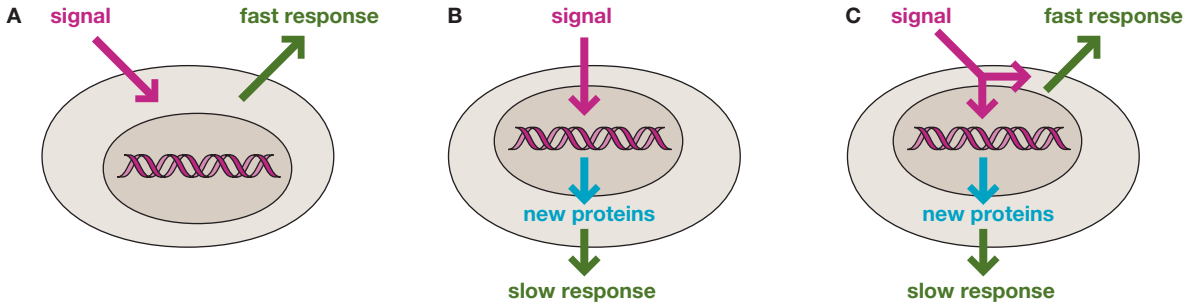



**Figure 9.8 Scenarios for signaling.** (A) Some signaling pathways lead to a response without affecting transcription. (B) Other signaling pathways involve the transcription of new proteins, and are necessarily slow on a cellular timescale. (C) Many signaling pathways affect both immediate nontranscriptional responses and slower responses that involve the transcription of new proteins.

In other cases, extracellular signals turn on signaling networks that penetrate the nucleus, and either activate or repress transcription of specific genes (Figure 9.8B). Processes that require the production of new proteins are generally slower than processes that do not. For example, when you cut your skin the binding of fibroblast growth factor to the fibroblast growth factor receptor leads to mitotic proliferation of skin cells; cell division is heavily reliant on the translation of new proteins. The transcriptional processes leading to mitosis take many hours.

In other cases, signaling pathways can bifurcate, leading to both a fast response and a slow response involving transcription (Figure 9.8C). For example, insulin exerts both fast and slow responses in liver cells. When insulin binds to the insulin receptor, it leads to the fusion of vesicles adorned with glucose transporters, allowing liver cells to rapidly vacuum up glucose from the medium. In addition, insulin induces slower responses through the regulation of genes. These slower effects include the upregulation of genes responsible for lipid biosynthesis: genes encoding fatty acid synthase, acetyl-CoA carboxylase, and others.

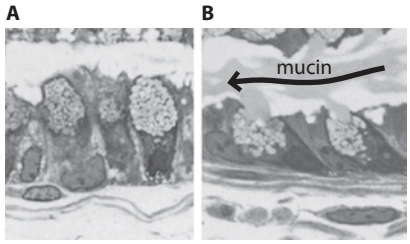



**Figure 9.9 Spilling their guts.** (A) Normally, mucin is released from goblet cells at a very slow rate. (B) After treatment with acetylcholine and physostigmine (an acetylcholinesterase inhibitor), the goblet cells rapidly release their mucus, which streams into the colon. (Adapted from R.D. Specian and M.R. Neutra, *J. Cell Biol.* 85:626–640, 1980. With permission from Rockefeller University Press.)

### Problem 9.2

Which of the following compounds would greatly reduce the acetylcholine-induced release of mucin by intestinal goblet cells? Cycloheximide, a ribosome inhibitor; physostigmine, an inhibitor of acetylcholinesterases; atropine, an acetylcholine receptor antagonist.

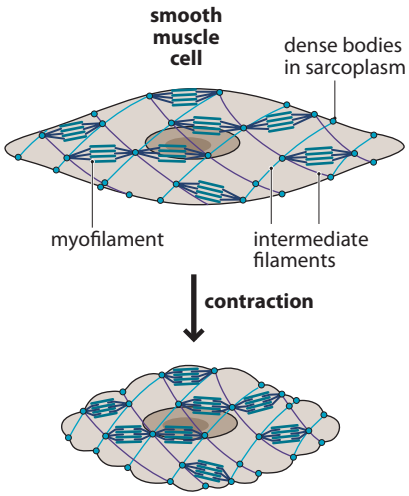



**Figure 9.10 Smooth operator.** The contractile state of smooth muscle cells arises from the contraction of myofilaments containing myosin, actin, and other proteins. (Adapted from E. Marieb and K. Hoehn, Human Anatomy and Physiology, 7th ed. Benjamin Cummings, 2007.)

## Cell contraction and vesicle fusion: fast calcium-dependent responses that do not involve changes in transcription

Two of the most common examples of fast cellular responses (that do not involve transcription) are muscle cell contraction and vesicular export. Both responses are triggered by $Ca^{2+}$ ions. These calcium-triggered cellular responses account for most of the physiological effects that can be monitored by a physician, including heart rate, blood pressure, breathing, pupil dilation, knee reflexes, lifting your arm, seeing, and speaking. Most fast-acting natural toxins act on nontranscriptional pathways. If you want to kill a potential predator before it eats you, or eat prey before it escapes, you would not want to target transcription. It is too slow. Both muscle contraction and the fusion of secretory vesicles are fast and involve $Ca^{2+}$ ions; we will focus on these mechanisms now, and the origins of those $Ca^{2+}$ ions later in this chapter.

The contraction of all muscle cells is due to the crawling of myosin filaments along actin filaments (**Figure 9.10**)—which is mediated by an increase in cellular $Ca^{2+}$ ion concentration. The human body has two classes of muscle cells. Smooth muscle cells, such as those present in bronchial airways and vascular endothelia, are important in asthma and hypertension, respectively. The other class of muscle cells includes cardiac and skeletal muscle cells, which are important in heart disease and muscular dystrophy, respectively. Myosin proteins resemble dimeric bean sprouts; myosin dimers exist as filaments that are grouped into much larger bundles. The "heads" of the myosin proteins protrude from the myosin bundles. The myosin head is an ATP-driven motor that crawls along actin filaments within the cell through a rapid, hinge-like motion. Each stroke is powered by the hydrolysis of one ATP molecule (**Figure 9.11**).

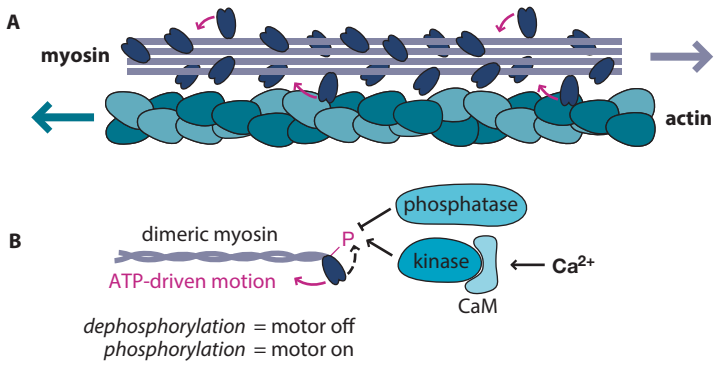



**Figure 9.11 Powering muscles.** (A) The head domains that protrude from the myosin bundle serve as ATP-driven motors that drag the myosin filament along an actin fiber. (B) Each myosin motor is controlled through the competing actions of two species: a phosphatase and a $Ca^{2+}$-dependent kinase complex. CaM, calmodulin. The magenta letter P represents a phosphate group.

Each individual myosin motor is controlled through phosphorylation; the motor is "on" when it is phosphorylated and "off" when it is dephosphorylated. The ratio between the phosphorylated and dephosphorylated states is determined by a kinase and a phosphatase, respectively. The phosphatase is always active. The activity of the kinase depends indirectly on the concentration of $Ca^{2+}$ ions. When cytosolic $Ca^{2+}$ levels increase, the $Ca^{2+}$ binds to the calcium sensor protein calmodulin (CaM), leading to a marked change in conformation. The distorted $Ca^{2+}$•CaM complex binds to and activates the kinase, forming an active complex that phosphorylates myosin proteins, turning on the motors. The ratio of on-motors to off-motors is controlled with high gain, because each kinase and phosphatase acts on many myosin substrates. Controlling states through competing enzyme levels is common among all organisms from bacteria to humans.

The molecular mechanisms that control vesicle fusion are not as well understood as the mechanisms that lead to contraction. Vesicle fusion is most closely associated with the release of neurotransmitters such as acetylcholine, dopamine, and

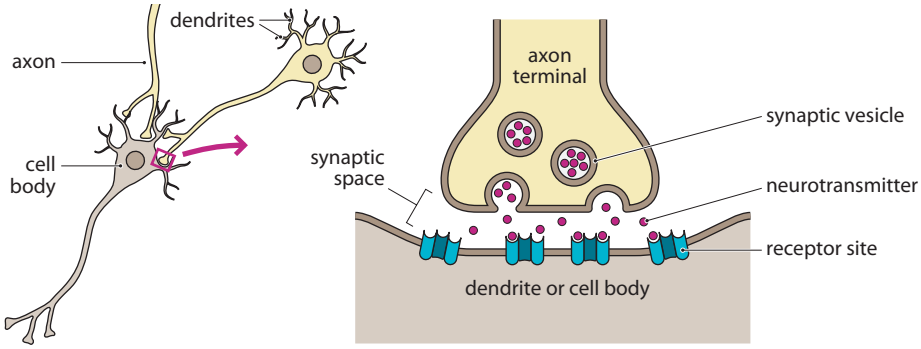



**Figure 9.12 Signaling across the synapse.** Neurotransmitters are released on one side of the synaptic gap and detected by 7TM GPCRs on the other side. (Adapted from US National Institutes of Health, National Institute on Aging.)

5-hydroxytryptamine from neurons (**Figure 9.12**), but it is also important for the secretion of signaling molecules from non-neuronal cells such as the release of histamine from mast cells. In addition, recall from Chapter 7 that vesicle fusion leads to rapid insertion of fully assembled glucose transporters (GLUT4) into the plasma membrane after stimulation with insulin. In neurons, neurotransmitters are prepackaged within vesicles that dock with the synaptic membrane (**Figure 9.13**). A complex of several proteins holds the vesicle close to the cell membrane, but not close enough to fuse accidentally. When the neuronal signal reaches the synaptic cleft, a flood of $Ca^{2+}$ ions binds to synaptotagmin. $Ca^{2+}$ ions have a high affinity for phosphate groups, such as those found on lipids; up to 50% of bone is a mineralized form of calcium phosphate. Once the vesicle has been drawn close to the plasma membrane, fusion occurs, releasing neurotransmitters into the synaptic space.

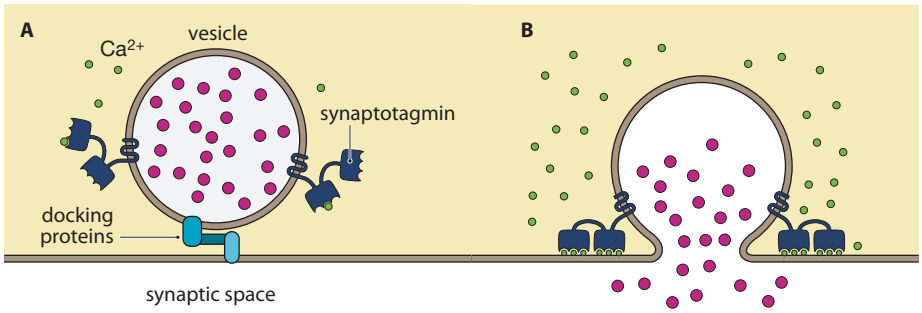



**Figure 9.13 Release the hounds.** (A) Neurons release neurotransmitters from vesicles that are docked with synaptic membranes. (B) When $Ca^{2+}$ binds to synaptotagmin, the $Ca^{2+}$ helps pull the vesicle into the cell membrane.

## Cell signaling can involve pathways within cells and/or between cells

When you add a selective enzyme inhibitor to a cell or delete the gene for the enzyme, it will usually affect many other proteins within the cell. The great challenge is to distinguish *direct* interactions—mediated by binding, catalysis, or other chemical phenomena—from *indirect* interactions. A typical cell contains thousands of different proteins; how can one tell which proteins are responsible for the effects of an inhibitor on a cell? *In vitro* assays allow one to assess the effects of one protein on another directly, without the complications of complex cellular mixtures. However, there are many challenges to setting up unambiguous *in vitro* assays that can truly mimic the situation inside a cell. First, it is difficult—and sometimes impossible—to express and purify stably folded proteins for *in vitro* studies, particularly in the case of membrane-bound proteins. Second, highly sensitive functional assays are required to detect changes at physiological concentrations, usually in the low micromolar or nanomolar range. Third, many proteins exist as subunits in multiprotein complexes; once removed from the complex, the actions of an isolated protein might be physiologically irrelevant. Given the challenges inherent in assays *in vitro*, it is not surprising that so many studies are conducted in living cells.

Cell biologists are mainly interested in understanding how cells work and how they manage to work together. In many cases, the distinction between these two goals is blurry. Most studies of signal transduction are performed on thousands of cells in culture, often in plastic multiwell plates. In such circumstances, it is difficult to know

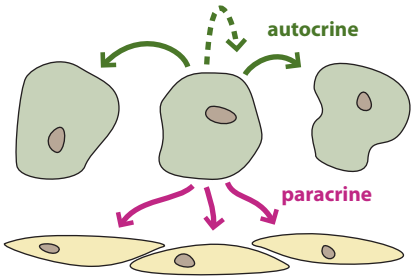



**Figure 9.14 *Entre nous*.** Autocrine signaling occurs between cells of the same type. Paracrine signaling occurs between cells of different types.

whether signaling pathways are strictly localized within the cytosol or whether they involve signaling between cells. Those signals can be diffusible molecules or molecules displayed on the cell membrane that require intimate cell-to-cell contact. When one cell sends a signal to another cell, the recipient cells can be of the same type (for example, signals from muscle cells to other muscle cells) or of a different cell type (for example, signals from dendritic cells to T cells). Such signaling pathways are referred to as **autocrine** or **paracrine**, respectively (**Figure 9.14**). Molecular signals released into the bloodstream are broadly referred to as endocrine signals.

## 9.2 AN OVERVIEW OF SIGNAL TRANSDUCTION PATHWAYS IN HUMAN CELLS

### There are seven major signal transduction pathways in humans

If you have taken a traditional course in biochemistry, you probably experienced the harrowing process of memorizing biosynthetic pathways and metabolic pathways such as the citric acid cycle, amino acid biosynthesis, gluconeogenesis, and glycolysis. These were cutting-edge pathways 50–100 years ago—but no longer. The chemical mechanisms involved in metabolism are fascinating, but they are not the focus of this book. Instead, we choose to focus on how cells receive signals, particularly those signals that affect transcription—not just any cells, but *human* cells. Transcription is the start of the central dogma: DNA is transcribed to RNA; RNA is translated into enzymes; enzymes catalyze reactions of small (and big) molecules.

A quick glance at the literature would suggest that there are thousands of unique signaling pathways in human cells; the complexity is misleading. In humans, seven major signal transduction systems transduce extracellular signals into changes in transcription. The seven can be recognized and arranged by the type of ligands/receptors involved (**Table 9.1**). Each acronym will be fully discussed later in the chapter.

**Table 9.1 Human signal transduction pathways that modulate transcription.**

| Receptor type | Ligands |
|---|---|
| Nuclear receptors | Steroids, retinoic acid, thyroxine |
| Two-component pathways | TGF-β, interleukins, interferons |
| Receptor tyrosine kinases | Growth factors (EGF, FGF, VEGF, NGF) |
| Trimeric death receptors | TNFα, FasL |
| G-protein-coupled receptors | Neurotransmitters, hormones, odorants, tastes, photons, enzymes |
| Ion channel receptors | Glutamate, $Na^+$ ions |
| Diffusible gas receptors | $O_2$, NO |

In **Figure 9.15** we summarize the seven principal signal transduction pathways that transduce signals from outside human cells into changes in transcription inside human cells. The reader should commit these to memory. The dashed lines imply the involvement of multiple steps, and we will explain the details in the following sections. As you read those sections, return to this figure so that you are able to keep the big picture in perspective. Figure 9.15 is the forest; keep it in sight as we introduce you to the major trees.

Our understanding of human signal transduction is historically biased toward pathways associated with cancer as opposed to, for example, metabolic diseases. Cancer-derived cell lines are easily grown in the laboratory, whereas primary or normal human tissue is difficult to obtain and culture. Furthermore, the lethality of compounds toward cells is easily assayed, whereas more subtle effects on cell function and human physiology can be challenging to assess. Empowered by new tools from

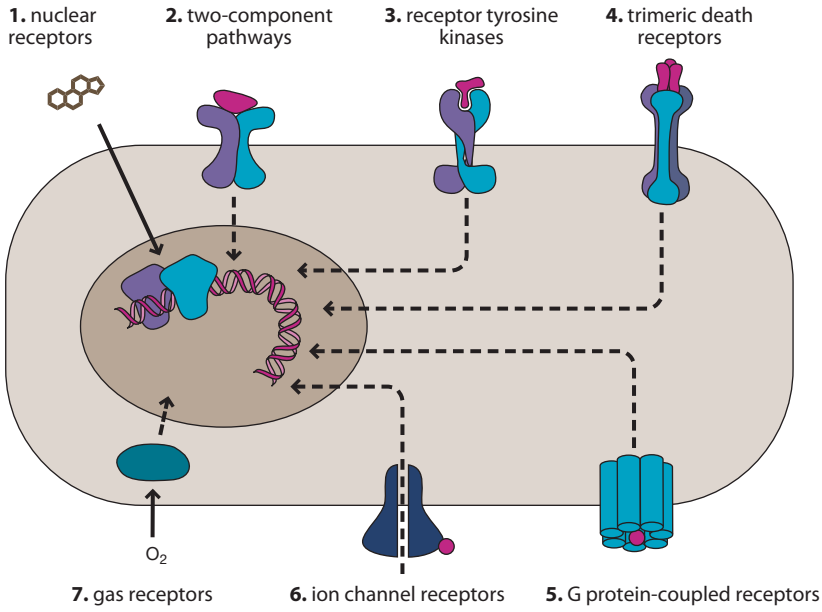



**Figure 9.15 Seven pathways.** Seven major signal transduction pathways control transcription of human genes. The dashed arrows imply multiple chemical steps.

**Figure 9.16 The color of their eyes.** The *white* (eye) gene of *Drosophila* codes for a protein that transports molecules such as guanine and tryptophan—precursors of red eye pigments called drosopterins—across membranes. (Courtesy of Wikimedia.)

genomics, cell biology, and chemical biology, the field of cell biology is slowly overcoming the bias toward cell growth pathways. Thus, the view of which pathways are important continues to evolve.

**Problem 9.3**

Which of the seven major signal transduction pathways involve ligands that pass through the cell membrane and enter the cell?

## Chemical genetics involves the use of small molecules to understand gene function

*Drosophila* has been a fruitful system (pun intended) for the field of genetics, which is primarily concerned with the hunt for genes linked to phenotypic mutations. Phenotypic mutations can enhance the function of a protein, but in most cases the phenotype results from a debilitating effect of the mutation. For example, the *Drosophila white* mutation impairs the transporter that brings pigment precursors into cells of the eye (**Figure 9.16**). It is easy to generate mutations in *Drosophila* through irradiation, and then breed the mutant flies to sort out the effects of diploidy (two copies of each gene per fly). In **forward genetics**, such stochastic mutations are introduced into the organism or cells, and then DNA sequencing of the mutant phenotype identifies the genotype. The corollary, **reverse genetics**, involves examining a specific protein by making mutations to its encoding gene, and then examining the resultant function.

In **chemical genetics**, one seeks to trace the phenotypic effects of small molecules back to a protein target and, ultimately, the gene that controls expression of the protein target. A well-known example is the antifungal natural product rapamycin, discovered in a soil microorganism on Rapa Nui, also known as Easter Island. Interest in rapamycin was raised when it was shown to inhibit the activation of T cells. Rapamycin binds FKBP (FK506-binding protein), and the resultant composite surface then binds the prosaically named protein mTOR (mammalian target of rapamycin) (**Figure 9.17**). Interfering with mTOR suppresses the immune system, and rapamycin has been used both as a pharmaceutical immunosuppressant (for preventing the rejection of transplanted organs) and as an important tool in the laboratory for taking apart the related immune system pathways, a technique called reverse chemical genetics..

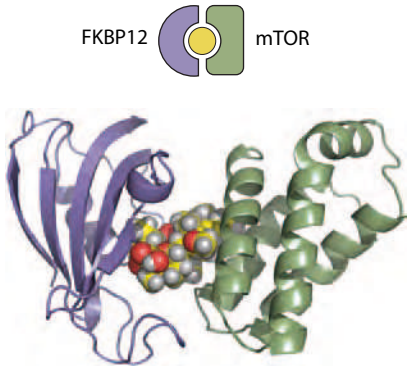



**Figure 9.17 A small molecule dimerizer.** The polyketide natural product rapamycin (yellow) induces the dimerization of FKBP12 (blue) and mTOR (green). Rapamycin is used clinically as an immunosuppressant. (PDB 1FAP)

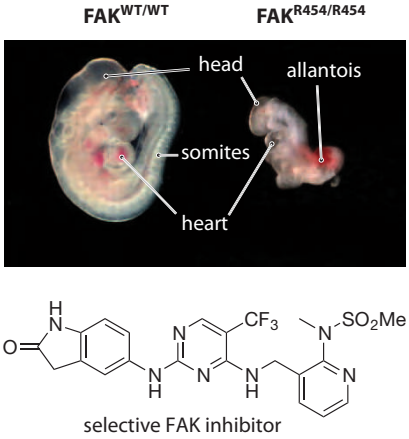



**Figure 9.18 Small-molecule control over embryo development.** You cannot study mice with a K454R mutation in focal adhesion kinase (FAK) because they are never born. On the left is a wild-type embryo (WT/WT); on the right is a nonviable deformed embryo resulting from the mutation of K454 to R454. Bottom: a selective FAK inhibitor could be applied *after* the wild-type mouse has developed into a healthy adult. (Adapted from S.T. Lim et al., *J. Biol. Chem.* 285:21526–21536, 2010. With permission from The American Society for Biochemistry and Molecular Biology.)

Rapid DNA-sequencing methods have revolutionized the field of genetics. Such sequences, for example, facilitate both forward and reverse genetics. In reverse genetics, the gene (sequence) is known, but the effects of mutations to the gene are not. After oligonucleotide-directed mutagenesis, the resultant mutant cells or organism are studied to determine how the modified piece contributes to the protein function. This process is analogous to reverse engineering—change one part at a time and then examine what results from the change.

Small-molecule agonists and antagonists offer several important advantages over the addition or deletion of genes. First, the small molecules can be added to a particular place—say an organ or cluster of cells—at a set time point, and can be expected to act shortly after the addition of the compounds. In addition to spatial and temporal control, small molecules can be added at specific concentrations for the characterization of dose-dependent effects. Furthermore, small diffusible molecules, unlike genetic constructs, often have reversible effects and can be removed from the cell under study simply by washing with excess buffer.

Small molecules also solve another problem inherent in reverse chemical genetics. It is rarely possible to study the effect of a mutation that is lethal to the developing organism. If you wish to understand the role of focal adhesion kinase (FAK) you cannot simply delete the gene or inactivate the protein through a genetic point mutation. Genetic mutations that deactivate FAK (for example K454R) are lethal at the embryonic stage (**Figure 9.18**). However, one can assess the importance of FAK in a mouse at any stage of development by treating it with a selective small-molecule inhibitor of FAK such as the synthetic small molecule PF-562,271 (see Figure 9.18). Thus, chemical genetics provides a powerful toolkit of reagents for dissecting the tremendous complexity inherent in the overlapping, cross-communicating, and feedback-regulated cellular networks.

## Screening identifies small molecules for use in chemical genetics

A major challenge in chemical biology and pharmaceutical development is finding small molecules that are selective for a particular protein target. We have seen throughout this book that Nature generates an abundance of small molecules that target cellular machinery, but natural inhibitors are rarely potent and selective against the full array of proteins in a human. Finding selective small-molecule modulators usually involves setting up a high-throughput assay and testing many thousands of compounds in a single screening campaign.

To illustrate a typical workflow, let us take an example of a screen to identify compounds that inhibit mitosis without affecting microtubules. With regard to the latter criterion, we now have many potent anticancer drugs that affect both microtubule assembly and disassembly; we do not need more examples in this class of compounds. The antimitotic assay was based on the use of an enzyme-linked immunosorbent assay (ELISA) to detect mitosis in whole cells, as opposed to solutions of proteins. Epithelial cells growing in 384-well plates were screened with a diverse library of 16,320 chemical compounds. Cells were fixed with 70% ethanol to wash away lipids and displace water, leaving most proteins insoluble; the fixed cells were then subjected to an ELISA with an antibody selective for phosphonucleolin, which is present only in mitotic cells. Fewer than 1% of the compounds were active (**Figure 9.19**A). When these 139 hits were individually assayed for the ability to disrupt microtubule formation, 86 were found to have no effect on microtubules (Figure 9.19B). The effects of each of these 86 compounds on various phases of the cell cycle were then assessed individually by fluorescence microscopy. Kidney cells were treated with each of the compounds, and then the cells were fixed and stained with fluorescent antibodies against microtubules (stained green) and chromatin (stained blue). Various effects were observed by microscopy. Among the five hits that affected only mitosis, monastrol led to drastic aberrations in the mitotic spindle (Figure 9.19C). Whereas normal cells align chromosomes along a scrimmage line during mitosis, in cells treated with monastrol chromosomes emanate from a central point, like an exploding star. Further studies revealed the true target of monastrol to be kinesin, a motor protein that drags DNA along microtubules (**Figure 9.20**). It is not uncommon for pharmaceutical companies or large academic

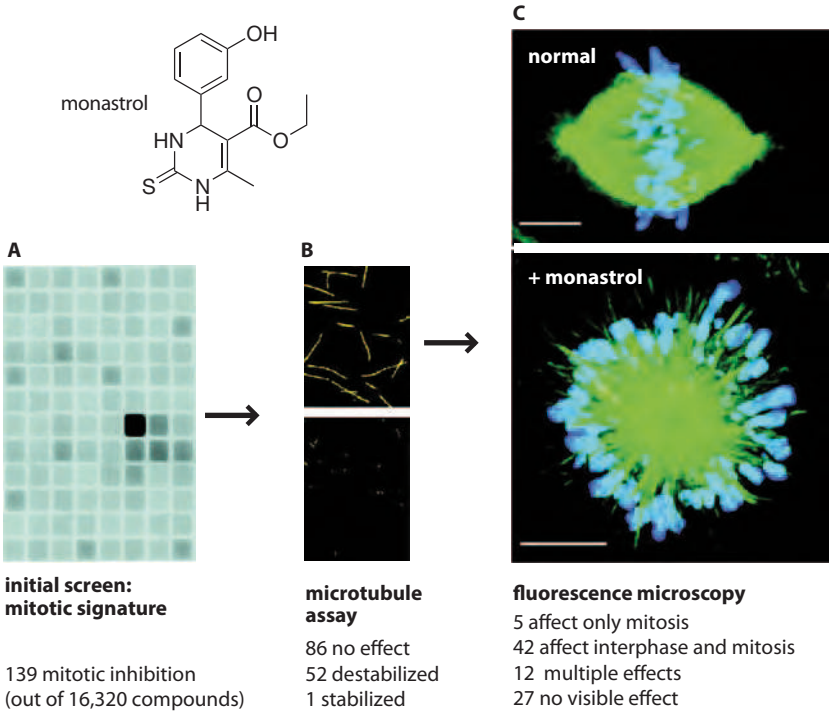



**Figure 9.19 High-throughput screening for a cellular phenotype.** (A) Screening of 16,320 chemical compounds for antimitotic effects led to 139 hits. (B) The antimitotic hits were further assayed for activity against microtubules. (C) Of the 86 compounds that had no effect on microtubules, five were selective for mitosis. Fluorescent staining of microtubules (green) and chromatin (blue) revealed an aberrant mitotic spindle induced by one of the compounds, which was dubbed monastrol. (From T.U. Mayer et al., *Science* 286:971–974, 1999. With permission from AAAS.)

centers to screen hundreds of thousands of compounds in a single screening campaign. These screens are a rich source of leads for drug development and for new tools for biologists.

How important are small-molecule screens? Recall from Chapter 4 that RNA interference makes it possible to stop the expression of proteins solely on the basis of mRNA sequence without the need to carry out high-throughput screens. Small interfering RNAs against any protein can be easily designed and synthesized, but RNA interference has two drawbacks. First, RNA is not membrane permeable. At present, RNA molecules lack the oral availability and pharmacokinetics of typical pharmaceuticals. Second, stopping the expression of a protein is not the same as inhibiting the function of protein that has already been expressed. Small molecules can solve these problems, and thus provide an invaluable tool in chemical biology studies.

## 9.3 NUCLEAR RECEPTORS

### Binding of small-molecule ligands activates nuclear receptor transcription factors

We will begin our discussion of signal transduction pathways by starting with the simplest: transcription factors that bind directly to signaling molecules. Bacteria use transcription factor receptors for quorum sensing, engaging in activities such as luminescence when they reach a high population density. In humans, transcription factor receptors are commonly referred to as **nuclear receptors**. The nuclear receptor pathways are simple but the ligands are structurally complex and interesting, so we will spend more time talking about the ligands than the pathway itself.

Analysis of the human genome has revealed 28 types of nuclear receptors (48 if you include subtypes such as α, β, γ, that bind to the same ligand). Naturally

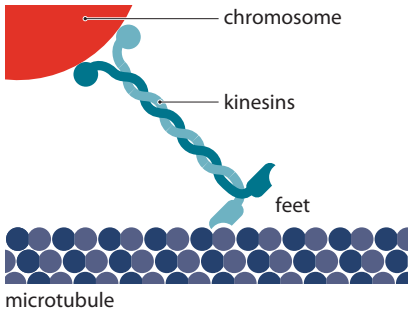
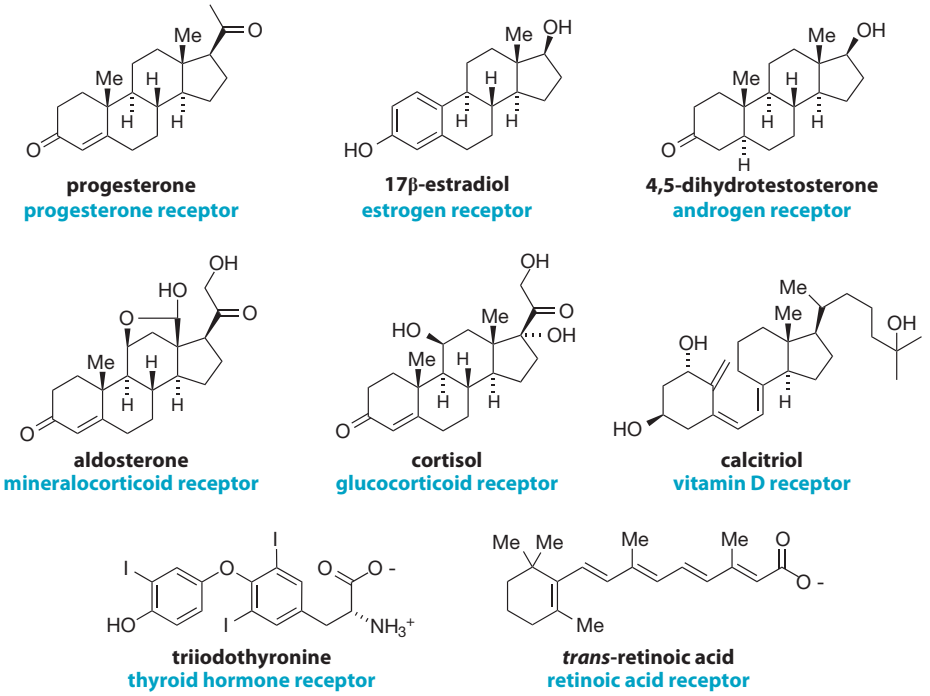



**Figure 9.20 The longest journey begins with a single step.** ATP hydrolysis drives a conformational change that prevents the kinesin from stepping backward. As the kinesin dimers waddle along the microtubule, they drag the massive chromosomes along with them. (Courtesy of C. Asbury and Steven M. Block.)

**Figure 9.21 Ligands and their receptors.** These examples of nuclear receptor ligands (black labels) bind to human nuclear receptors (blue labels) with high affinity and high specificity.




occurring high-affinity ligands have been identified for only 8 of these 28 types (**Figure 9.21**). Steroid hormones are the best-known ligands for nuclear receptors: progesterone, estrogen, testosterone, aldosterone, and cortisol. In addition, vitamin D, triiodothyronine, and *trans*-retinoic acid are also high-affinity ligands. Nuclear receptors bind hormones with high affinity (nanomolar $K_d$ values) and high specificity. Crystal structures of nuclear receptor–hormone complexes show that the ligand is completely sequestered within the ligand-binding domain (**Figure 9.22**).

Receptors for which the endogenous ligand has not yet been identified are referred to as orphan nuclear receptors. Extensive efforts have not revealed high-affinity, high-specificity ligands for 20 of the 28 human nuclear receptors. It has been speculated that the orphan nuclear receptors bind with relatively low affinity to abundant intracellular metabolites. Examples of nuclear receptors that are believed to bind to abundant intracellular metabolites include the liver X receptor (LXR), the farnesoid X receptor (FXR), the retinoid X receptor (RXR), the peroxisome proliferator activator receptor (PPAR), the aromatic hydrocarbon receptor, the prostanoid X receptor, and the constitutive androstane receptor (CAR). The liver X receptor can be activated by cholesterol metabolites, referred to as *oxysterols*, at concentrations below their physiological levels (**Figure 9.23**). There is no nuclear receptor for cholesterol, which is a highly abundant component of lipid bilayers, and the liver X receptor is believed to

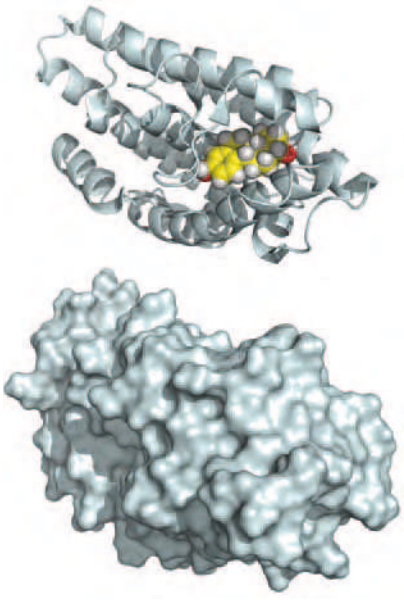


**Figure 9.22 Trapped!** The ligand-binding domain of human estrogen receptor-α (PDB 1QKU) rendered in ribbon mode (top) and as a solvent-accessible surface (bottom) reveals that the ligand, 17β-estradiol, is completely encapsulated by the receptor.

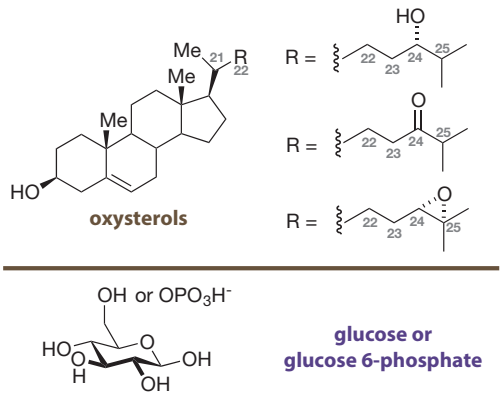



**Figure 9.23 Ligands for the liver X receptor.** Two very different types of ligands, sterols and monosaccharides, activate the liver X receptor at physiological concentrations.

serve as an indirect sensor for cholesterol levels. Surprisingly, the liver X receptor has also been shown to be activated by glucose and glucose 6-phosphate at physiological levels. Because the liver X receptor activates genes for lipid metabolism and binds to oxysterols, the receptor seems to connect lipid metabolism with glucose levels which could have relevance to diabetes.

Nuclear receptors are intracellular, and the ligands for nuclear receptors must be either membrane permeable or intracellular. The known high-affinity ligands for nuclear receptors are lipophilic, and therefore have low solubility in water. These ligands are shuttled through the bloodstream by proteins with lipophilic pockets, such as retinol-binding protein and thyroxine-binding globulin. Some cells contain retinoic-acid-binding proteins that assist in transport to the nucleus.

## Some nuclear receptors translocate from cytoplasm to the nucleus, and bind DNA as homodimers

The signal transduction pathway for 17β-estradiol illustrates how nuclear receptors work. The ligand for the estrogen receptor, estradiol, is most abundant in women. During reproductive years it is produced primarily within the ovaries, but smaller amounts are produced by the brain in both sexes and in the testes in men. Estradiol has a key role in the menstrual cycle, targeting cells in both the uterus and the brain. Estradiol diffuses across the cell membrane of hypothalamic cells, binding tightly to the estrogen receptor-α, which is sequestered within the cytoplasm as a dimeric complex with heat shock protein 90 (Hsp90). Subsequent dissociation of Hsp90 allows the estrogen receptor dimer to be actively transported into the nucleus. The estrogen receptor possesses a short peptide sequence, called a **nuclear localization signal** (NLS), that can bind to karyopherin β. This binding event allows the protein to be ferried through a nuclear pore complex. Once in the nucleus, the ligand-bound form of the estrogen receptor-α binds tightly to a specific DNA sequence, referred to as the estrogen response element, and enhances transcription of the gene for the protein hormone gonadotropin-releasing hormone (**Figure 9.24**). The signal transduction pathways for other ligands to nuclear receptors, including progesterone, 4,5-dihydrotestosterone, cortisol, and aldosterone, follow similar pathways, but the tissue distribution and target genes are different.

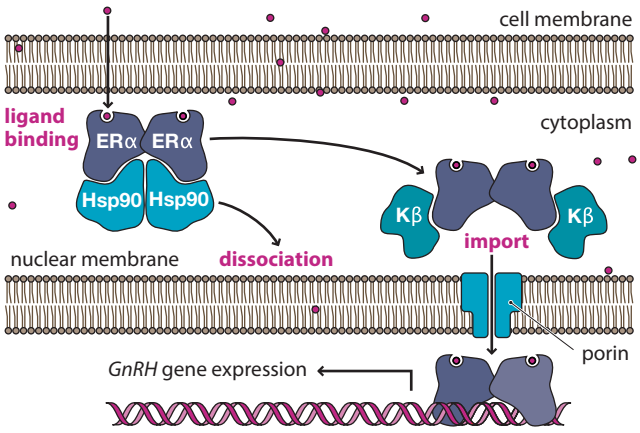



**Figure 9.24 Road map for estrogen.** The estrogen receptor pathway in hypothalamic neurons involves dissociation from Hsp90, association with karyopherin β (Kβ), and import through a nuclear porin. The estrogen receptor dimer (ERα) binds to DNA and promotes transcription of the *GnRH* gene.

## Some nuclear receptors are localized in the nucleus and bind to DNA as heterodimers

The second major variant of the nuclear receptor signal transduction pathway involves the heterodimerization of nuclear receptors that remain within the nucleus. This pathway is best appreciated by considering the signal transduction pathway for the endogenous ligand retinoic acid. *Trans*-retinoic acid (isotretinoin) is prescribed for the treatment of severe cystic acne, but like other nuclear receptor ligands it has profound

**Figure 9.25 Profound birth defects by retinoids.** Treatment of pregnant mice with massive doses of retinol, a precursor to retinoic acid, causes embryonic malformation. (Courtesy of the Nobel Foundation.)

effects on cell growth, particularly in the developing fetus (**Figure 9.25**). Retinoic acid is a potent teratogen, known to cause human birth defects including hydrocephalus (an accumulation of cerebrospinal fluid in the brain) and malformed or missing ears, so prescriptions of retinoic acid are carefully controlled. In the USA, women must submit two negative pregnancy tests before the initial prescription, and all patients are advised not to donate blood while taking retinoic acid.

Retinoic acid diffuses across both the cell membrane and nuclear membrane. In the nucleus of developing embryonic neuronal cells, retinoic acid binds to the retinoic acid receptor α (RARα), which binds as a heterodimer with the retinoid X receptor (RXR). Together, the RARα–RXR heterodimer binds to specific DNA sequences enhancing the transcription of some genes and repressing the transcription of others. For example, in developing neuronal cells, the RARα–RXR heterodimer enhances the expression of the *homeobox b-1* gene (*Hoxb-1*). Hox genes code for transcription factors that further control cell growth and differentiation (**Figure 9.26**). This type of heterodimerization pathway is also found with the thyroid receptor, vitamin D receptor, the peroxisome proliferator activator receptor, liver X receptor, and the constitutive androstane receptor.

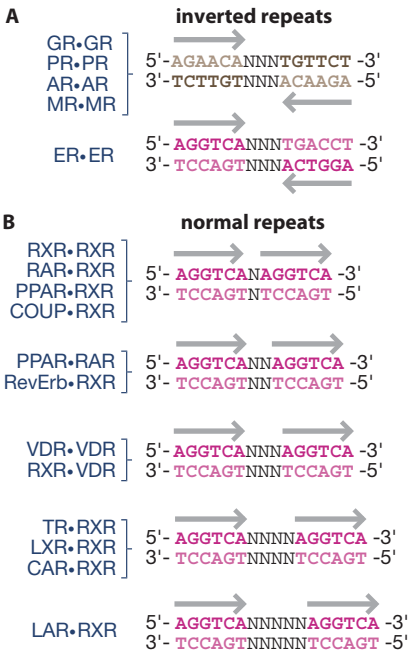



**Figure 9.27 DNA binding by nuclear receptors.** Unique DNA sequences are recognized by homodimers or heterodimers of nuclear receptors. Some DNA response elements are inverted repeats (A); others are simply repeats (B) separated by one to five base pairs. See Figure 9.21 for the full names of the high affinity nuclear receptors. Some orphan nuclear receptors have cryptic acronyms: COUP, chicken ovalbumin upstream promoter; RevErb, reverse orientation of the c-erbA gene.

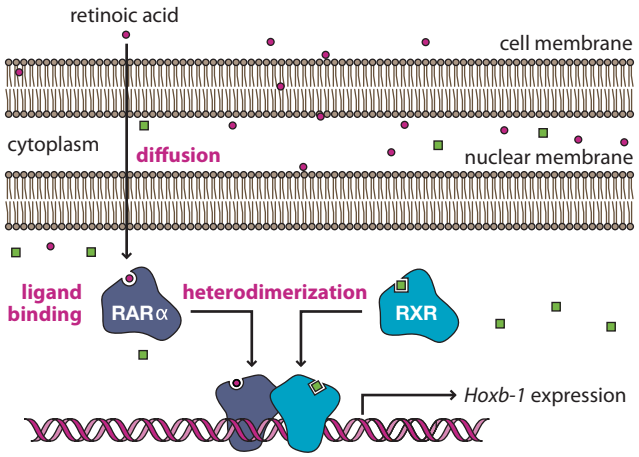



**Figure 9.26 Trajectory for retinoic acid.** The retinoic acid receptor pathway in hypothalamic neurons is a canonical nuclear receptor pathway.

## The mode of nuclear receptor dimerization determines DNA sequence selectivity

The DNA-binding domain of each nuclear receptor is a zinc-finger domain that recognizes one of only two different six-base-pair sequences (**Figure 9.27**): 5′-AGAACA-3′ or 5′-AGGTCA-3′. A significant level of selectivity in gene regulation arises from the fact that nuclear receptors bind as dimers recognizing a total of 12 unique base pairs. These DNA recognition sequences are referred to as response elements because they determine the genetic response to transcription factors. Examples of nuclear receptor response elements include the estrogen response element, the androgen response element, and the vitamin D response element.

Some nuclear receptors form head-to-head homodimers that bind to inverted repeats of the six-base-pair recognition sequences. For example, the glucocorticoid receptor form homodimers that bind to the inverted repeat 5′-AGAACA*NNN*TGTTCT-3′. Note that the complementary strand of this sequence is the same when read in the 5′ to 3′ direction. Estrogen receptors bind as homodimers to the inverted repeat 5′-AGGTCA*NNN*TGACCT-3′. Other nuclear receptors bind six-base-pair repeats such as 5′-AGGTCA$(N)_{1-5}$AGGTCA-3′, for example retinoid X receptor, retinoic acid receptor, peroxisome proliferator activator receptor (PPAR), vitamin D receptor, thyroxine

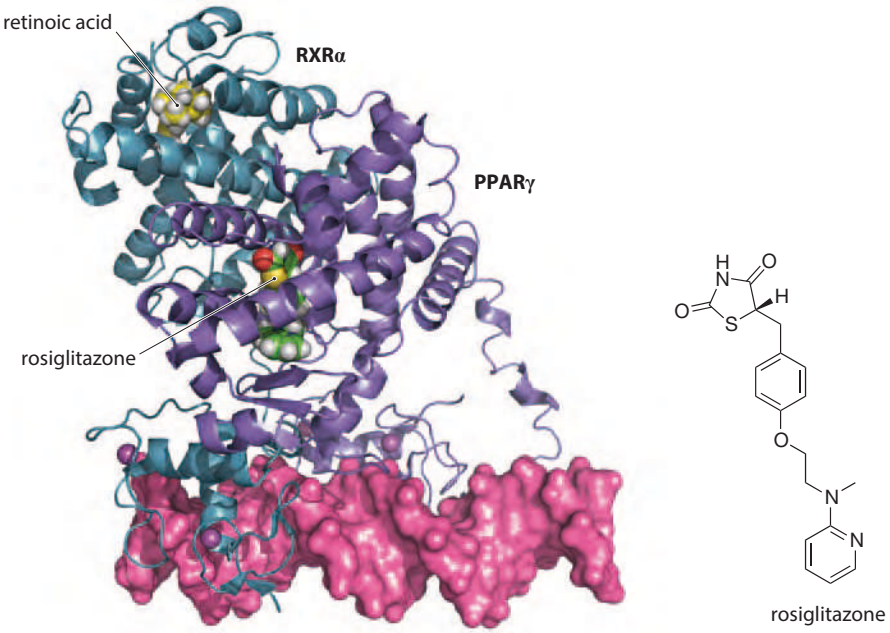



**Figure 9.28 Structure of a full-length nuclear receptor bound to DNA.** The structure of the PPARγ–RXRα complex bound to the ligand rosiglitazone and retinoic acid and DNA. The DNA includes the following consensus binding sequence (underlined) AGGTCAAAGGTCA. (PDB 3DZY)

receptor, and liver X receptor. The number of intervening base pairs matters; a span of five base pairs represents a 180° twist of the helical axis of B-form DNA.

A wide range of partial structures of nuclear receptors have been solved through X-ray crystallography of either the ligand-binding domain or the DNA-binding domain, but not both. So far, there is only one crystal structure of full-length nuclear receptors bound to DNA—the RXRα–PPARγ heterodimer. In the crystal structure, retinoic acid is bound to the RXRα and the pharmaceutical (*S*)-rosiglitazone is bound to PPARγ. As expected, the zinc fingers of each nuclear receptor bind to the sequence 5′-AGGTCA-3′ (**Figure 9.28**).

## Human cells can be rewired for control by *Drosophila* nuclear receptors

Nuclear receptors have a common structure and mechanism across a range of different organisms. However, they do not bind to precisely the same ligands in different organisms. For example, in humans, steroid hormones such as testosterone and estrogen have a key role in development during puberty. However, in *Drosophila*, progression from nonreproductive juveniles to reproductive adults is controlled by the binding of the steroid 20-hydroxyecdysone to the ecdysone receptor and the binding of juvenile hormone III to the ultraspiracle receptor. These insect nuclear receptors form a heterodimer that controls differentiation (**Figure 9.29**). Nearly all plants produce

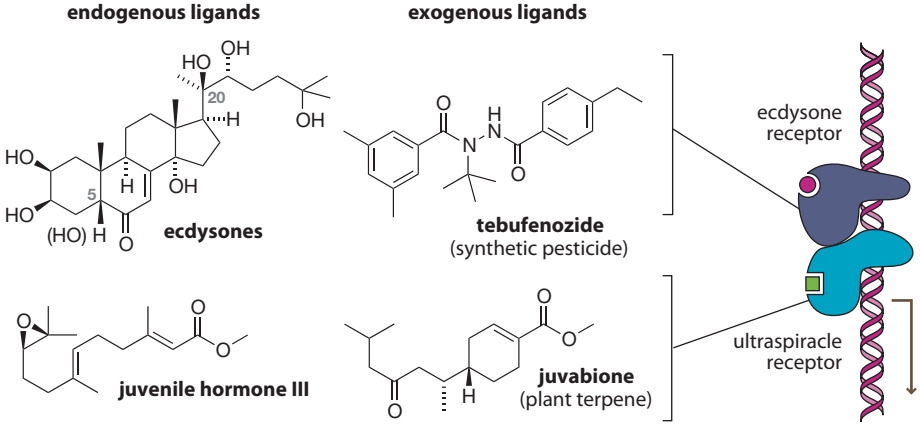



**Figure 9.29 Controlling insects with ligands to nuclear receptors.** Endogenous ligands like ecdysones and juvenile hormones activate nuclear receptors in insects. Exogenous ligands like synthetic and natural pesticides also target these insect nuclear receptors, inducing a premature molt.

**Figure 9.30 Nuclear receptor targeting by an insecticide.** Sixth-instar (molting phase) larvae of the spruce budworm 48 hours after molt. Top, control; bottom, treated with tebufenozide. (From A. Retnakaran et al., *Pest Manag. Sci.* 57:951–957, 2001. With permission from John Wiley & Sons.)

ecdysteroids that target the ecdysone receptor and induce abnormal molting in larvae. Each gram of spinach contains 0.1 mg of 20-hydroxyecdysone and 5,20-dihydroxyecdysone (also known as polypodine B). Plants also contain nuclear receptor agonists that look nothing like the insect ligand. For example, the terpene natural product juvabione, found in fir trees, mimics juvenile hormone III and adversely affects molting. When firebugs are treated with juvabione before the final larval molt, they turn into giant supernumerary larvae instead of developing into adults. The insecticide tebufenozide (Mimic®) targets the ecdysone receptor of moth (Lepidoptera) larvae, and induces a lethal premature molt (**Figure 9.30**).

Neither 20-hydroxyecdysone, juvenile hormone III, nor their cognate receptors are present in humans, but each *Drosophila* receptor is a homolog of a human nuclear receptor. On the basis of amino acid sequence and protein structure, the ecdysone receptor is most homologous to the farnesoid X receptor in humans, whereas the ultraspiracle receptor is homologous to the retinoid X receptor in humans. Surprisingly, the ecdysone receptor can form functional heterodimers with human retinoid X receptor. When human cells are transfected with plasmids expressing ecdysone receptor and then treated with ecdysone, the ecdysone receptors associate with human retinoid X receptors, and bind to DNA (**Figure 9.31**).

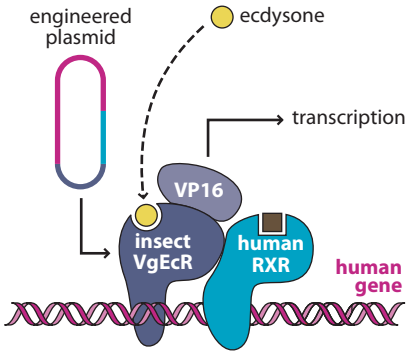



**Figure 9.31 Re-engineering human cells to respond to an insect hormone.** Human cell lines can be transformed with a plasmid encoding a special form of the ecdysone receptor (bearing the human transcription flag VP16).

The ecdysone receptor has been re-engineered to include a mammalian VP16 domain that recruits human transcriptional machinery (as opposed to *Drosophila* transcriptional machinery), and three point mutations that confer the same DNA binding selectivity as the glucocorticoid receptor: 5′-AGAACA-3′. When expressed in human cells, this re-engineered nuclear receptor VgEcR forms a heterodimer with the retinoid X receptor and can be used to turn on the expression of genes driven by a novel response element, 5′-AGGTCANAGAACA-3′. This combination of *Drosophila* nuclear receptor and novel response element can be used to turn on transcription in human cells by using an insect hormone, without affecting other human genes.

## Problem 9.4

Draw a cartoon depiction showing how you could use the DNA-binding domains of both the androgen receptor and the estrogen receptor to engineer a new pair of proteins that would bind to the DNA sequence 5′-AGGTCATAGTGTTCT-3′ in the presence of rapamycin.

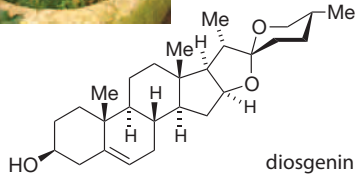



**Figure 9.32 The massive tubers of the barbasco yam.** Two percent of the tuber is a single steroid, diosgenin. (Courtesy of Nhu Nguyen, University of California, Berkeley.)

## Steroids make highly potent pharmaceuticals

Chemical interest in steroid hormones drove the field of nuclear receptors, leading to the first contraceptive drug, a progesterone agonist, in the 1960s. Today, a wide range of steroid drugs are synthesized and sold. Initially, the main starting material for pharmaceutical steroids was a steroid called diosgenin isolated from massive Mexican barbasco yams (**Figure 9.32**). Today, steroid starting materials are isolated from soybeans. The phytosterol content is low, but soybeans are grown on a massive scale. In 2009, for example, US soybean production was more than 80 billion kilograms. Soybeans contain 0.3% phytosterols by weight (3:1:1 β-sitosterol, stigmasterol, campesterol). After processing, the phytosterol content of soybean oil (sold in supermarkets as "vegetable oil") is still between 0.2% and 0.3% (**Figure 9.33**). The sterols are present as a mixture of alcohols and fatty acid esters. Because of the low abundance and diverse composition of sterols in soybeans, the phytosterol route to steroids did not compete with the diosgenin route until 1976, when Mexican diosgenin prices rocketed by 250%. Soy phytosterols are converted into useful starting materials through microbial cleavage of the steroid side chain.

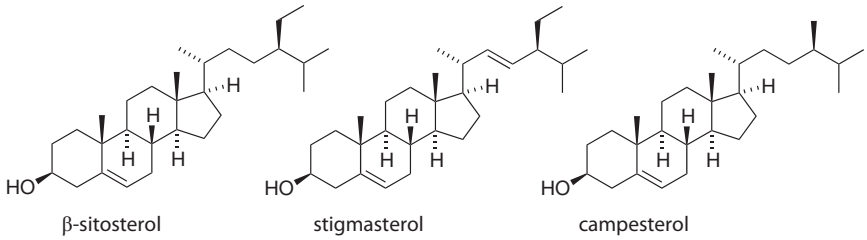



**Figure 9.33 Steroids from soybeans.** A mixture of three phytosterols derived from soybean oil is now the most common starting material for pharmaceutical steroid synthesis.

Steroids are still widely prescribed for a range of indications beyond birth control (**Figure 9.34**). For many years, a mixture of estrone conjugates, mainly *O*-sulfate esters, isolated from the urine of pregnant mares (sold as Premarin™) was widely used for hormone replacement therapy to ameliorate post-menopausal symptoms such as hot flashes. At its peak in 2002, US sales of Premarin were more than $2 billion a year. After 2002, sales declined due to mounting evidence that long-term use of estrogen and progestin increases the risk of cardiovascular disease and cancer.

Glucocorticoids are widely used as anti-inflammatory drugs. Hydrocortisone creams (1% hydrocortisone) are widely available over the counter as anti-itch medications. The powerful synthetic glucocorticoid fluticasone is a component of Flonase™ nasal spray and the two-component asthma medication Advair™. US sales of Advair™ were almost $4 billion in 2008.

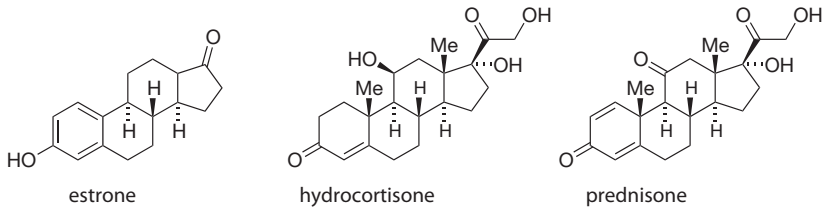
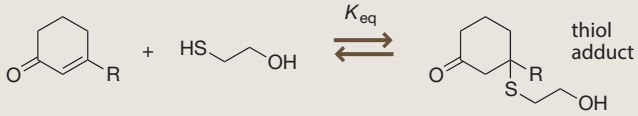



**Figure 9.34 Steroids as therapeutics.** Estrone derivatives and glucocorticoids (such as hydrocortisone) have been widely prescribed as medicines.

## Problem 9.5

Many steroid hormones and steroid pharmaceuticals possess enone functional groups that can react with thiols such as glutathione or cysteine side chains. The equilibrium is exquisitely sensitive to sterics. Given the equilibrium constants below and a thiol concentration of 5 mM, what percentage of progesterone would be present as the thiol adduct at equilibrium? What percentage of prednisone would be present as the thiol adduct at equilibrium?




| R | $K_{eq}$ |
|---|---|
| H | 7000 |
| $CH_3$ | 2 |

## Nonsteroidal ligands for nuclear receptors are also widely used as drugs

Nonsteroidal ligands for nuclear receptors are both biosynthesized and prescribed widely as drugs. The thyroid gland in the neck produces triiodothyronine and tetraiodothyronine, which is converted into the more potent triiodothyronine by enzymes in the target cells (**Figure 9.35**). Triiodothyronine controls metabolism throughout the

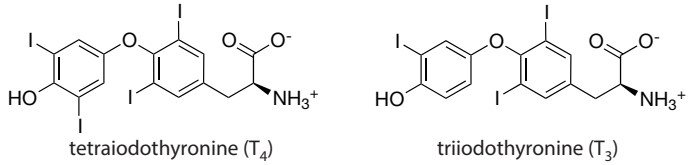



**Figure 9.35 Regulators of basal metabolic rate.** Synthesized by the thyroid, the iodo-substituted, nonsteroidal thyronine derivatives are derivatives of tyrosine, and exert control over diverse physiological responses.

**Figure 9.36 Food for your nuclear receptors.** Iodized salt was introduced in 1924 to prevent goiter, which is characterized by enlargement of the thyroid gland. (Left, courtesy of Morton Salt.)

body by binding to nuclear receptors. Such binding can stimulate the body's basal metabolic rate, spurring increased usage of glucose and lipids. Triiodothyronine also stimulates increases in cellular protein synthesis, heart rate, the force of heart contractions, the systolic component of blood pressure, and body temperature. Taken together, the two compounds mediate a broad range of physiological responses through direct binding to nuclear receptors.

When the human diet is deficient in iodine, the cells in the thyroid gland proliferate, leading to a condition known as goiter (**Figure 9.36**). Since 1924, grocery salt has been "iodized" with the addition of 0.006–0.010% KI. Tetraiodothyronine is an important prescription pharmaceutical, with almost $1 billion a year in US sales.

Nuclear accidents generate radioactive $^{131}I$, which ends up being concentrated in the thyroid gland, leading to thyroid cancer. Taking potassium iodide pills overwhelms the body with the nonradioactive natural isotope $^{127}I$, preventing the accumulation of $^{131}I$ in the thyroid gland.

## Drugs can be designed to target specific mutations of nuclear receptors

The extraordinary affinity of nuclear receptor ligands allows them to act at vanishingly low concentrations, which makes such compounds good candidates for therapeutic treatments. For example, the small molecule calcitriol, sold as Rocaltrol®, is prescribed in minute doses of 0.25 and 0.50 μg. The human body uses two pericyclic photochemical reactions to generate the vitamin D ring system of cholecalciferol (**Figure 9.37**):

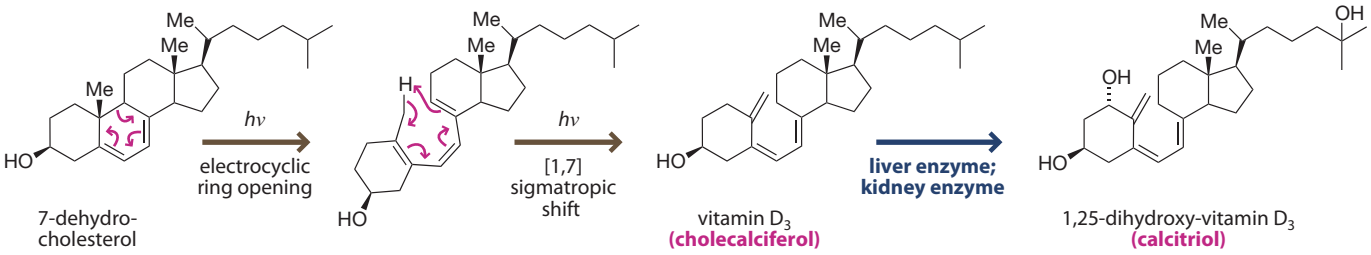



**Figure 9.37 Hormones for bone growth.** The biosynthesis of calcitriol involves pericyclic reactions.

a 6-π-electron electrocyclic ring-opening reaction and a suprafacial [1,7]-sigmatropic reaction. The enzyme for this transformation is conveniently located in the skin, where photons are plentiful. The 25-hydroxyl group is then added by an enzyme in the liver, followed by the addition of the 1-hydroxyl group by an enzyme in the kidney. The final form of vitamin $D_3$ (calcitriol) binds tightly to the vitamin D receptor. Calcitriol, and the genes it controls, are essential for proper growth of bones. A diet deficient in vitamin D leads to a disease known as rickets. In growing children with rickets, the weight of the body distorts the leg bones, causing the legs to bow outward (**Figure 9.38**).

Mutations that affect the ligand-binding site of the vitamin D receptor are associated with vitamin D-resistant rickets. For example, the R274L mutation makes the vitamin D receptor more than 1000 times less responsive to vitamin D because it removes a hydrogen bond and creates a void that cannot be filled by the natural ligand (**Figure 9.39**). However, the addition of *O*-benzyl groups to the 1-hydroxyl group of vitamin D generates derivatives that are capable of restoring up to 80% of the inducible activity.

**Figure 9.38 Rickets.** Vitamin D deficiency in children leads to bones that are too soft to support the weight of the body. The resulting disease, rickets, is characterized by bow-shaped legs. (Courtesy of Photo Researchers.)

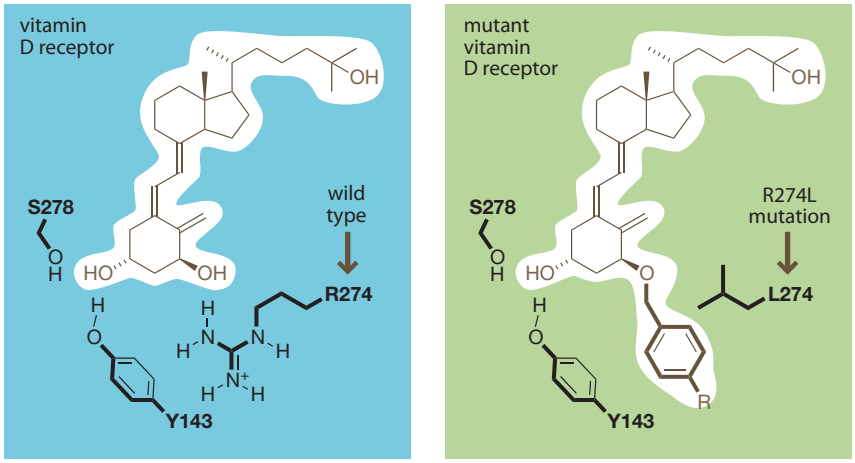
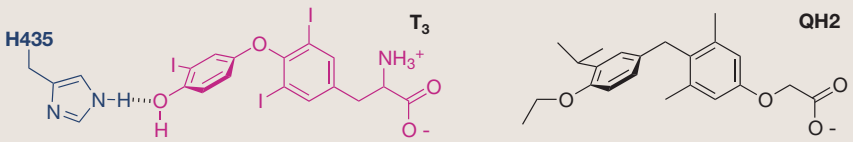



**Figure 9.39 A synthetic ligand to restore biological function.** Calcitriol binds tightly to the wild-type vitamin D receptor (left) by making numerous contacts, including a key hydrogen bond with Arg274; however, calcitriol cannot fill the void left by an R274L mutation. A synthetic *O*-benzyl derivative of calcitriol fills the void in the R274L variant (right), rescuing patients from vitamin D-resistant rickets.

**Problem 9.6**

Triiodothyronine ($T_3$) binds tightly to the wild-type thyroid hormone receptor, forming a stabilizing hydrogen bond with His435 (H435) in the ligand-binding site. An orthogonal, non-metabolizable analog (QH2) was designed to be inactive against the natural receptor, but highly active against a compensatory mutant. Which thyroid receptor mutant would be most likely to respond to QH2: H435A, H435F, H435G, H435K, H435W, or H435R?




## 9.4 CELL-SURFACE RECEPTORS THAT INTERACT DIRECTLY WITH TRANSCRIPTION FACTORS

### Hematopoietic proliferation and differentiation are controlled by molecular signals

The ability to respond to membrane-impermeable ligands requires cell-surface receptors. Bacteria such as *Staphylococcus aureus* use peptides, called auto-inducing peptides, for quorum sensing. These impermeable peptides activate a two-component signal transduction system that involves a membrane-bound receptor and a phosphorylating protein in addition to the transcription factor. The bacterial phosphorylating protein acts on histidine residues as opposed to serine, threonine, or tyrosine. Humans use a two-component signal transduction system to control the proliferation and differentiation of blood cells—a process referred to as hematopoiesis (**Figure 9.40**). The bone marrow is the central factory for hematopoiesis. It contains a family of cells that originate from a self-reproducing cell line known as multipotent hematopoietic stem cells. Multipotent stem cells can proliferate to produce more cells of the same type or, as they divide, differentiate to become a variety of essential cell types: red blood cells, platelets, antibody-producing B lymphocytes, killer T lymphocytes, bacteria-killing macrophages, bone-forming osteoclasts, and bone-remodeling osteoblasts. If you suffer a traumatic injury you probably need to clot the wound with platelets, fight off infection with white blood cells, and replace the lost erythrocytes. However, you do not always need equal amounts of the various blood types. For example, in response to a viral infection you need B cells, not platelets; to assist with clotting you need platelets, not T cells. Pluripotent stem cells can differentiate to produce any type of cell in the organism: for example, blood, brain, muscle.

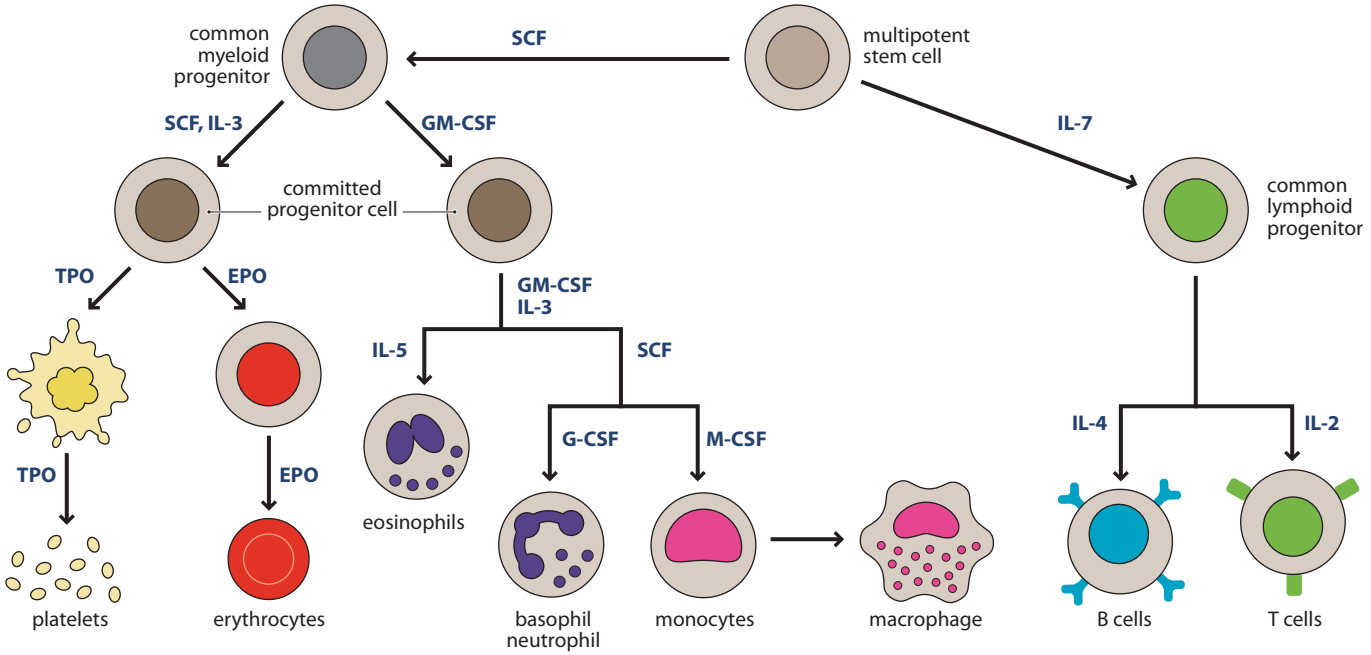



**Figure 9.40 Cytokines that control the differentiation of blood cells.** Proteins called cytokines (for example erythropoietin (EPO), interleukin (IL)-3, and IL-6) control which types of blood cells are being made. (SCF, stem cell factor; TPO, thrombopoietin; GM-CSF, granulocyte/macrophage colony-stimulating factor; G-CSF, granulocyte colony-stimulating factor; M-CSF, macrophage colony-stimulating factor.)

The proliferation and differentiation of hematopoietic cell types are induced by endogenous protein signals called **cytokines** or by extracellular cues from pathogenic organisms. Many of the cytokines that control hematopoiesis are named **interleukins** because most of them were first identified in studies of white blood cells, which are also referred to as leukocytes. When multipotent stem cells divide, they can produce identical copies, but when the right mixture of cytokines are present they can differentiate to produce various specialized blood cells. The presence of interleukin-7 induces differentiation to produce lymphoid progenitor cells. The presence of stem cell factor induces differentiation to produce myeloid progenitor cells. Depending on the mixture of cytokines in the environment, myeloid progenitors can further differentiate as they divide. For example, the presence of interleukin-5 induces differentiation to produce eosinophils, which help to fight off infections that include multicellular parasites.

Most cytokines mediate their intracellular effects through a closely related set of pathways that involve the activation of a kinase and the phosphorylation of a transcription factor. As we will see, the organization of the receptor assemblies varies widely from cytokine to cytokine, and this variation can permit activation of and cross-talk with more than one signaling pathway.

**Problem 9.7**

Which cytokines would be necessary to ensure the production of (a) eosinophils, (b) B cells?

## Human cytokines can be used as pharmaceuticals

Naturally occurring ligands for cytokine receptors have proven to be a rich source of protein pharmaceuticals. Whereas most pharmaceuticals inhibit biochemical or cellular processes, cytokines can be used to induce cellular processes, such as differentiation. Erythropoietin, sold as Epogen® and Aranesp®, induces growth of red blood cells, and is used to treat anemia. Athletes have taken injections of erythropoietin to increase their supply of red blood cells, and hence the oxygen-carrying capacity of their blood. Thus, the use of erythropoietin is banned by most international sporting organizations. Colony-stimulating factor induces the proliferation of a wide range of white blood cells. Granulocyte/macrophage colony-stimulating factor, sold as

Neupogen®, is particularly important for patients undergoing chemotherapy, whose immune systems have been decimated. Interferon-α (IFN-α), sold as Roferon® and Intron A®, is used to boost the immune system of patients with various forms of hepatitis and various forms of cancer. Interferon-β, sold as Avonex® and Betaseron®, is used to modulate the overactive immune system in multiple sclerosis. Interferon-γ, sold as Actimmune®, is used to treat chronic granulomatous disease, in which neutrophils take pathogens into granulocytic vesicles but are unable to destroy them—rather like a garbage disposal unit without a blade.

Other hormones, not related to hematopoietic cell lines, also act through receptor pathways that resemble those of cytokines. Recombinant human growth hormone 1, sold as Nutropin®, is used to promote growth in children who grow slowly as the result of a deficiency of the natural human growth hormone. Excess growth hormone, taken inappropriately by athletes seeking faster recovery from injury, can cause acromegaly (**Figure 9.41**).

**Figure 9.41 Excess growth hormone.** This photo depicts acromegaly caused by genetic overproduction of human growth hormone. (Courtesy of Biophoto Associates. With permission from Photo Researchers.)

## The JAK–STAT pathway involves a receptor, a kinase, and a transcription factor

The signal transduction pathway involved in cytokine signaling is the closest human analogy to the primitive bacterial signal transduction pathway that mediates quorum proliferation. The human kidneys monitor oxygen and extracellular volume; when more erythrocytes are needed, they release the cytokine erythropoietin. The signal transduction pathway by which erythropoietin induces the differentiation and proliferation of erythrocytes is representative of all the other intracellular cytokine signal transduction pathways. Erythropoietin binds to a pair of erythropoietin receptors in the membrane of erythrocyte progenitor cells found in the bone marrow. Cytokines such as erythropoietin bind to two receptor molecules in a 1:2 complex. Other cytokines, such as interleukin-6, bind to their receptors in a 2:2 stoichiometry. Because erythropoietin lacks two-fold symmetry, each half of the dimeric erythropoietin receptor is binding to a different part of the erythropoietin molecule (**Figure 9.42**). Each receptor protein in the dimer is constitutively complexed to a Janus kinase, abbreviated JAK (Figure 9.42 and **Figure 9.43**). There are four different Janus kinases—JAK1, JAK2, JAK3, and TYK2—in the human genome. A key conformational change occurs upon binding of the cytokine ligand that causes each Janus kinase to catalyze the transfer

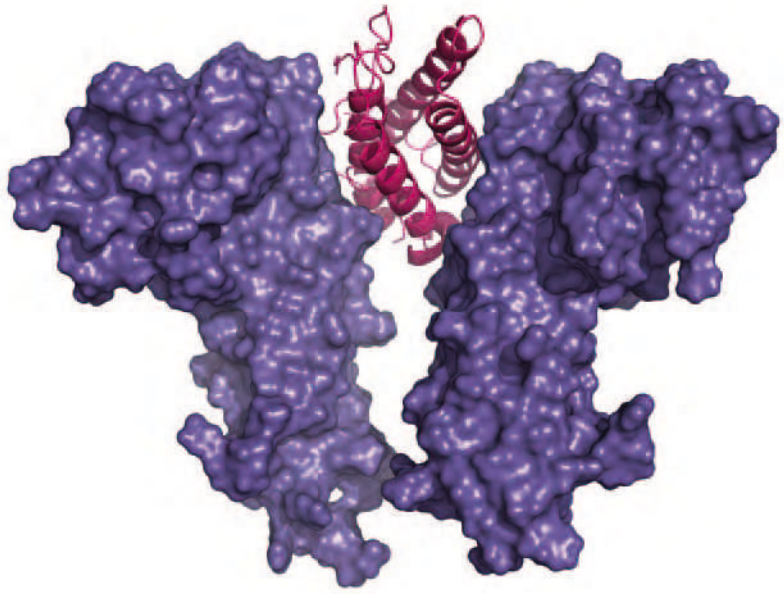


**Figure 9.42 A receptor–ligand sandwich.** A 2:1 complex forms between the extracellular domain of the erythropoietin receptor (blue) and the aglycone of erythropoietin (violet); the extracellular domain represents only about 40% of the full-length receptor. (PDB 1CN4)

**Figure 9.43 Two-faced.** Janus, the god of doorways, has two faces. Janus kinases are part of the doorway for cytokine signaling.

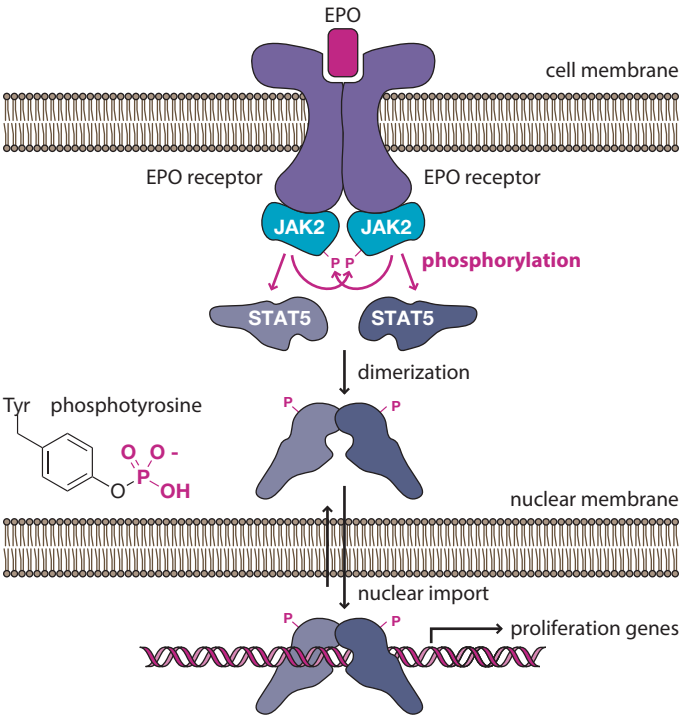



**Figure 9.44 JAK–STAT signaling.** The erythropoietin receptor–Janus kinase complex and the STAT transcription factor constitute a two-component signal transduction system. The red symbol P represents a phosphorylated tyrosine, serine, or threonine side chain. The red arrows indicate phosphorylation reactions.

of phosphate groups from ATP to a specific tyrosine residue of the opposing Janus kinase. This type of reciprocal phosphorylation is referred to as **autophosphorylation** because the Janus kinases seem to have phosphorylated themselves when they are analyzed by electrophoresis; however, the phosphorylation is actually catalyzed by the other half of the receptor pair.

Once the Janus kinases have become phosphorylated, the erythropoietin receptors catalyze the transfer of phosphoryl groups from ATP to a tyrosine on the transcription factor STAT5 (signal transducer and activator of transcription). STAT5 has an SH2 (Src homology 2) domain that allows it to bind to phosphotyrosines, either on JAK2 or on another STAT5 protein (**Figure 9.44**). Humans have genes for six different STAT transcription factors. The precise combination of JAK and STAT proteins in any signal transduction pathway depends on the cell type. STAT5 is actively transported in and out of the nucleus both before and after phosphorylation; however, once phosphorylated, STAT5 accumulates in the nucleus by binding tightly to DNA (**Figure 9.45**). Three other cytokines act through a pathway much like that of erythropoietin: thrombopoietin, human growth hormone, and prolactin.

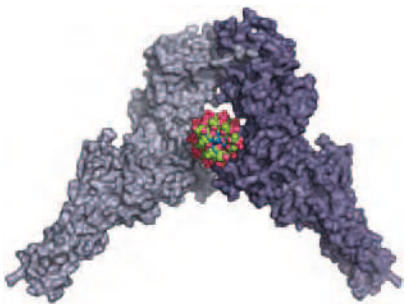


**Figure 9.45 STAT-DNA binding.** The STAT3B dimer straddles double-stranded DNA. (PDB 1BG1)

## Small-molecule dimerizers can be used to demonstrate functional relationships between proteins

Coumermycin A1 has been used to demonstrate that the activation of JAK2 does not require highly precise positioning by the cytokine receptor (**Figure 9.46**). Rather, activation merely requires proximity between two JAK2 kinase domains. Each end of the pseudosymmetrical antibiotic coumermycin A1 can bind tightly to a B subunit of bacterial DNA gyrase (GyrB), giving coumermycin A1 the ability to dimerize any proteins that are attached to GyrB. A non-natural conjugate between JAK2 and GyrB did not exhibit autophosphorylation. However, in the presence of coumermycin A1, autophosphorylation of the GyrB–JAK2 conjugate was observed, as was phosphorylation of STAT5.

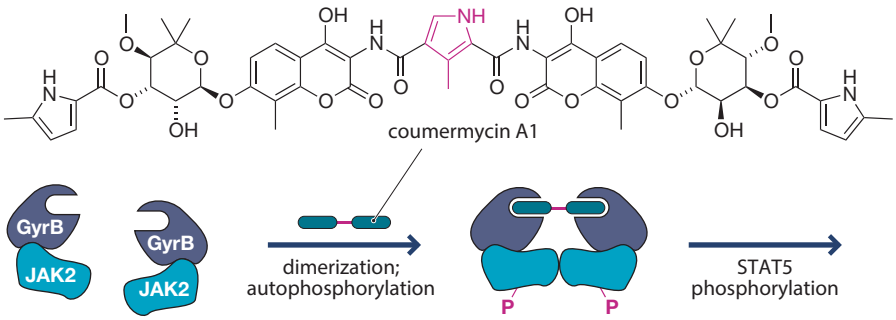



**Figure 9.46 Dimerization by a pseudosymmetrical antibiotic.** Chemically induced dimerization of GyrB–Jak2 fusion proteins with coumermycin A1 leads to autophosphorylation and downstream signaling.

## Other interferons bind to heterodimeric and higher-order receptor assemblies

Interferons are cytokines that were first noted for their ability to interfere with viral and bacterial infections by activating cells of the immune system. However, these cytokines are now known to have more diverse functions. Interferons like interferon-α and inteferon-β are five-helix bundles that bind to heterodimeric receptors on human cells. Interferon-α binds tightly to the interferon-α receptor 2 subunit (IFN-αR2) and ultimately as a ternary complex including the interferon-α receptor 1 subunit (**Figure 9.47**). Each of these two receptor subunits binds to a different Janus kinase: IFN-αR2 binds to the Janus kinase JAK1, whereas IFN-αR1 binds to the Janus kinase TYK2. The two Janus kinases activate each other through transphosphorylation. TYK2 then phosphorylates Tyr466 on the IFN-αR1 and those phosphotyrosines serve as a recognition site for the SH2 domain of STAT2 (see Figure 9.47). Once STAT2 has bound to the receptor, it is phosphorylated by TYK2 and recruits STAT1 to the receptor complex. JAK1 then phosphorylates STAT1, leading to a heterodimeric STAT1–STAT2 complex. Other cytokines act through higher-order ligand–receptor assemblies. The interleukin-6 ligand–receptor complex is rather like a dimer of the interferon-α receptor complex, composed of interleukin-6, the receptor gp130, and the interleukin-6 co-receptor in a 2:2:2 ratio. Other interferons, such as interferon-γ, bind to heterotrimeric receptor assemblies involving one cytokine and three different receptor proteins.

## Synthetic *N*-hydroxysuccinimidyl esters can acylate proteins in aqueous solution

*N*-Hydroxysuccinimidyl esters (NHS esters) are activated esters that react rapidly with amines and much more slowly with hydroxyl groups. The difference in reactivity is sufficient that the functional group can be used to acylate protein amines even in buffered aqueous solution. With two reactive esters connected by a short linker, bis-NHS esters can provide a tool for spot-welding together complexes found in cells; addition of the bis-NHS linker can form covalent crosslinks between closely associated proteins. The approach, however, can also turn up nonspecific interactions (for example proteins at high concentration that bind only weakly to each other), and the results need to be validated by additional techniques.

A bis-NHS ester was used to demonstrate the formation of a ternary complex between IFN-β, IFN-β receptor 2, and IFN-β receptor 1. When treated with a large excess of the crosslinking agent, random crosslinks were formed within and between proteins in the ternary receptor complex even when the proteins were present at relatively low concentrations (**Figure 9.48**). The covalently crosslinked complex between IFN-β, IFN-β receptor 2, and IFN-β receptor 1 was stable enough to be observed by denaturing SDS-polyacrylamide gel electrophoresis.

The chemical yield of such protein acylation reactions is usually low with respect to the NHS ester, as a result of hydroxide-dependent hydrolysis by water and other OH groups. The half-life for hydrolysis of an NHS ester is about 1 hour at pH 8.0 and 25 °C. pH is a critical variable in the acylation of lysine side chains. If the pH is too low, lysine side chains are protonated and unreactive. If the pH is too high, the concentration of hydroxide anions is so high that the NHS esters will be rapidly hydrolyzed.

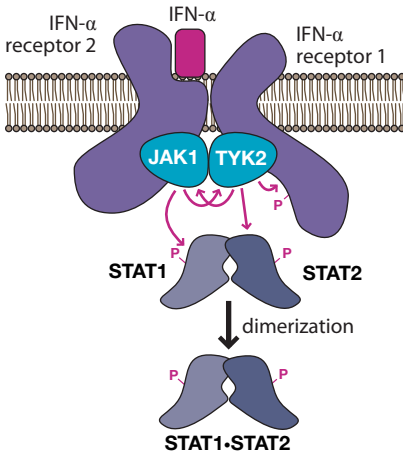



**Figure 9.47 Heterodimerization during signal transduction.** Type I interferon (IFN) pathways involve two different receptor subunits, two different Janus kinases, and two different STAT transcription factors.

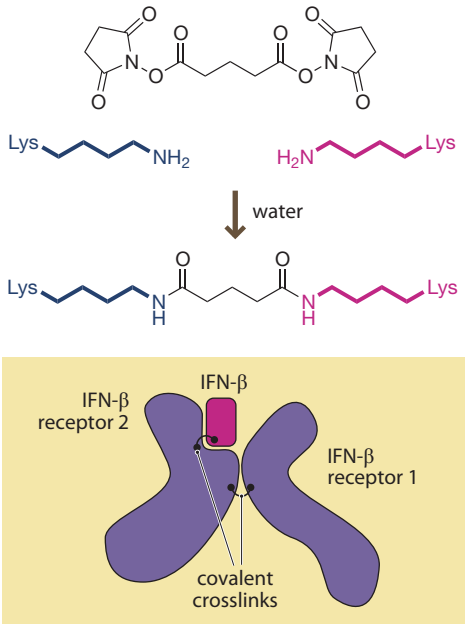



**Figure 9.48 Covalent, but nonspecific, crosslinking.** A bis-*N*-hydroxysuccinimidyl ester generates crosslinks between proteins in the interferon-β receptor complex.

NHS esters are used extensively in chemical biology to form covalent bonds to proteins in aqueous solutions. NHS esters can provide a straightforward route to modify protein surfaces through covalent bond formation. For example, recombinant human interferon-$\alpha_{2b}$ boosts the activity of the immune system and is prescribed to decrease the possibility of relapse from malignant melanoma. The half-life in serum is significantly increased by acylating interferon-$\alpha_{2b}$ with polyethylene glycol (PEG) chains. In general, covalent modification with PEG increases the molecular mass of the protein, thus slowing excretion from the body; in addition, modification with PEG often increases the solubility of the modified protein.

Polyethylene glycols are not readily available in monodisperse form, because the polymers are synthesized through the polymerization of oxirane. Instead, polyethylene glycols are used as mixtures with a known average molecular mass. When human interferon-$\alpha_{2b}$ is acylated in pH 6.5 phosphate buffer with a mixture of polyethylene glycols (average molecular mass 12 kDa; 273 oxyethylene units) bearing an NHS carbonate ester, the reaction generates a mixture of mono-pegylated proteins, of which 95% possess a single polyethylene glycol chain. This "pegylated," long-half-life form of interferon-$\alpha_{2b}$ is sold commercially as PegIntron® (**Figure 9.49**). The acylation reaction targets lysine, histidine, serine, threonine, and tyrosine side chains, but more than 45% of the pegylated species are acylated on a single residue, His34.

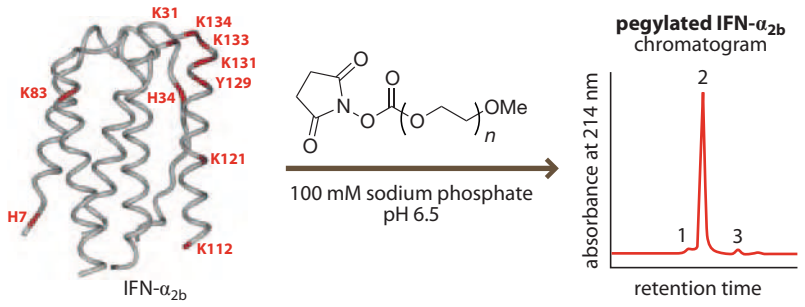



**Figure 9.49 A mixture of different products from a nonspecific PEGylation reaction.** Acylation of interferon-$\alpha_{2b}$ with PEG–*N*-hydroxysuccinimidyl carbonate esters generates a mixture of many different species, acylated on different nucleophilic side chains. Following chromatography, the main product consists of a mixture of many different monoacylated species (peak 2), along with a small amount of diacylated species (peak 1) and unacylated IFN-$\alpha_{2b}$.

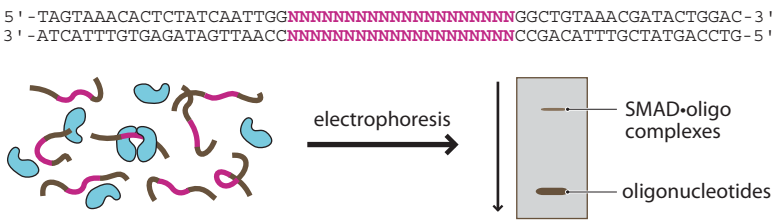
## Problem 9.8

The small molecule "stattic" selectively blocks the binding of the 5-carboxyfluorescein-G(pY)LPQTV-$NH_2$ to the SH2 domain of STAT3. Surprisingly, the saturated analog had no activity and the degree of inhibition was found to increase at longer incubation times of STAT3 with stattic. What do the data suggest about the interaction between stattic and STAT3?

### Transforming growth factor-β receptors possess built-in serine/threonine kinase domains

Transforming growth factor-β (TGF-β) is a key signaling protein that exerts opposing effects on the cell's balance between proliferation and apoptosis, depending on the cell type. Mechanistically, the pathway resembles that of interleukin-6 (with two ligands, two receptors, and two co-receptors) except that the kinase domain is built into the receptor protein (**Figure 9.50**). The transcription factor in the TGF-β pathway is named SMAD (or RSMAD) instead of STAT. Whereas phosphotyrosines mediate dimerization of STAT transcription factors, it is a phosphoserine that mediates interactions between RSMAD and its partner co-SMAD. As we will see shortly, built-in kinase domains are usually associated with **receptor tyrosine kinases** (discussed in the next section) and not cytokine receptors.

The DNA consensus sequence for the binding of SMAD was determined through affinity selection. A library of oligonucleotides, with 20 random nucleotides, was prepared using a DNA synthesizer. The random region in each nucleotide was flanked by constant regions with a known sequence, permitting amplification by PCR. The library of oligonucleotides was mixed with the DNA-binding domain of SMAD4 and the mixture was subjected to polyacrylamide gel electrophoresis. The larger protein–oligonucleotide complexes moved more slowly on the gel than uncomplexed oligonucleotides. The area of the gel containing DNA complexes was then removed, and the oligonucleotides were extracted and amplified by PCR (**Figure 9.51**). The sequences were determined and analyzed, revealing an idealized consensus sequence for SMAD binding: 5′-GTCTAGAC-3′.




## 9.5 RECEPTOR TYROSINE KINASES

### Receptor tyrosine kinases control tissue growth

Growth of human tissues is carefully controlled. During fetal development, tissue growth is exquisitely organized to allow development of the human body plan. As we grow from infancy to adulthood, tissue growth is directed (mostly) upward and outward. In adulthood, tissue growth is mostly restricted to and triggered by response to injury. If you cut your finger, you need immune cells to fight off bacterial infection. You need new skin, made up of epidermal cells and fibroblasts, to protect the body. You need new blood vessels to supply the growing tissue and new nerves to innervate the tissue. To support rapid proliferation, these cells must also metabolize nutrients at enhanced rates. These needs are generally communicated by growth factor proteins

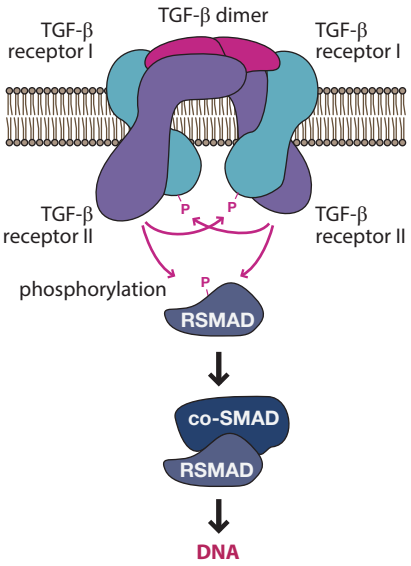



**Figure 9.50 Ser/Thr kinase.** The transforming growth factor-β receptor has a serine/threonine kinase built into the receptor. The red arrows in this pathway indicate phosphorylation reactions.

**Figure 9.51 Selecting DNA sequences that bind to a protein.** When a library of oligonucleotides is mixed with a tightly binding transcription factor, the DNA•protein complexes can be selected by electrophoresis, amplified by PCR, and sequenced to reveal the DNA target sequence. This selection method was used to identify the DNA sequence of genetic SMAD response elements.

**Figure 9.52 Fetal bovine serum (or fetal calf serum).** The liquid serum from the blood of cow fetuses is a rich source of protein growth factors and cytokines. Once purified, this relatively inexpensive mixture of protein hormones is usually used in culturing mammalian cells. (Courtesy of ZenBio, Inc.)

**Table 9.2 Subclasses of receptor tyrosine kinases with known ligands.**

| RTK class name | Ligand(s) |
|---|---|
| EGF receptor | Epidermal growth factor |
| Insulin receptor | Insulin |
| PDGF receptor | Platelet-derived growth factor, stem cell factor |
| FGF receptor | Fibroblast growth factor |
| VEGF receptor | Vascular endothelial growth factor |
| HGF receptor | Hepatocyte growth factor |
| Trk receptor | Nerve growth factor |
| Ephrin receptor | Ephrin |
| AXL receptor | Vitamin K-dependent growth-arrest specific gene 6 product |
| TIE receptor | Angiopoietin |
| DDR receptor | Collagen? |
| *RET* receptor | Glial cell line-derived neurotrophic factor |
| MuSK receptor | Agrin |

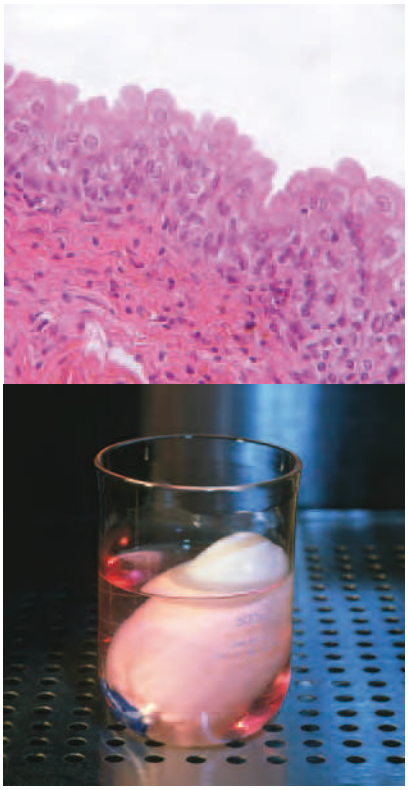


**Figure 9.53 Growing a bladder.** Top: transitional epithelial cells of the bladder. Bottom: forming an artificial bladder through the application of growth factors to bladder cells growing on a bladder-shaped form. (Top, courtesy of Wikimedia; bottom, courtesy of Photo Researchers.)

(**Figure 9.52**). Growth factors act on a class of receptors that function in ways similar to JAK–STAT receptors, but activate very different downstream pathways. Analysis of the human genome has revealed 20 subclasses of these growth factor receptors, referred to as receptor tyrosine kinases (RTKs), but ligands have not yet been identified for all of these receptors (**Table 9.2**). The names can be misleading or uninformative. For example, tropomyosin receptor kinase (Trk) receptor is activated by nerve growth factor, and the muscle-specific kinase (MuSK) receptor is activated by the proteoglycan agrin.

Given the central role of RTKs in tissue growth, it should not be surprising to find that the receptors and their associated signaling pathways contribute critically to all steps in the progression of cancer. Many different cancer types are associated with mutations in the RTK pathway, including breast, lung, and skin cancers. RTK pathways can go awry as a result of any mutations that decouple growth factor signals from downstream events, including the overexpression of proteins that positively regulate growth factor signaling and the overproduction of growth factors capable of autocrine signaling—fanning the flames of the fire. Such changes can disrupt the control and feedback mechanisms keeping check on the RTK pathways, and trigger undesirable cell division, growth, and other processes associated with the formation of tumors.

## Growth factors have a role in proliferation of urothelial cells

The human bladder has an amazing ability to expand and contract as urine accumulates and is eliminated. A specialized type of endothelial cell, termed the transitional epithelium (also known as the urothelium), is specific to the bladder, and can change from flat for an empty bladder to cubical when the bladder fills. Biomedical engineers have recently succeeded in growing artificial bladders in the laboratory by culturing a mixture of urothelial cells and smooth muscle cells around a bladder-shaped mold. Growth factors, which stimulate growth through RTKs, are an essential component of the growth media used to culture these artificial bladders (**Figure 9.53**).

The propensity for the urothelium to grow has a more sinister side—90% of bladder cancers are carcinomas of the urothelial cells, not fibroblasts or other cells found in bladder tissue. Many of the mutagens that enter our body ultimately end up in the bladder, where they can contribute to the formation of bladder cancers. Smoking is

one of the biggest risk factors for bladder cancer. Several easily cultured urothelial carcinoma cell lines have been isolated from humans and each of those cell lines has a unique set of mutations that permit uncontrolled growth. Most urothelial cell lines exhibit mutations in the fibroblast growth factor receptor (FGFR). There are four genes for FGFR in the human genome, namely *FGFR1–4*; urothelial cells express *FGFR3*. Postnatal mutations in FGFR3 contribute to cancer. Three FGFR3 mutations appear with relatively high frequency in urothelial carcinomas: S249C (67%), Y375C (15%), and R248C (10%). Congenital mutations of FGFR3 lead to problems in embryological development. Mutations to FGFR3 that contribute to the cancer phenotype usually arise spontaneously after birth, usually through lifelong exposure to carcinogens that collect in the urine.

**Problem 9.9**

Suggest a plausible rationale for why an S249C mutation in the extracellular domain might lead to constitutive activation of the FGFR3 receptor.

## Comparing receptor tyrosine kinases and cytokine receptors reveals useful commonalities

Receptor tyrosine kinases share many features with the cytokine receptors discussed in the previous section. First, all receptor tyrosine kinases form dimers in response to ligand binding. Second, the cytoplasmic domain of each receptor molecule is coupled to a kinase that transphosphorylates tyrosine residues on the partner. The phosphotyrosines then serve as beacons for the binding of proteins with SH2 domains, alerting the cell to gear up for big changes in protein expression. Third, some receptor tyrosine kinases act as homodimers, whereas others act as heterodimers. All growth factor ligands bind with RTKs in a 2:2 ratio.

There are also significant differences between RTKs and cytokine receptors. The first difference is that the tyrosine kinase is a *domain* of the receptor as opposed to an independent protein; we have already seen an example of a kinase domain embedded in the TGF-β receptor, but it was a serine/threonine kinase. A second difference between cytokine receptors and receptor tyrosine kinases is that the phosphotyrosine residues are coupled to four subpathways: mitogen-activated protein (MAP) kinase, phospholipase Cγ (PLCγ), phosphoinositide 3-kinase (PI3K), and STAT (**Figure 9.54**). From our discussion of TGF-β, you are already familiar with the recruitment of STAT transcription factors by phosphotyrosines on membrane-bound receptors. So we will focus on the other three subpathways, all of which involve interactions with the cell membrane.

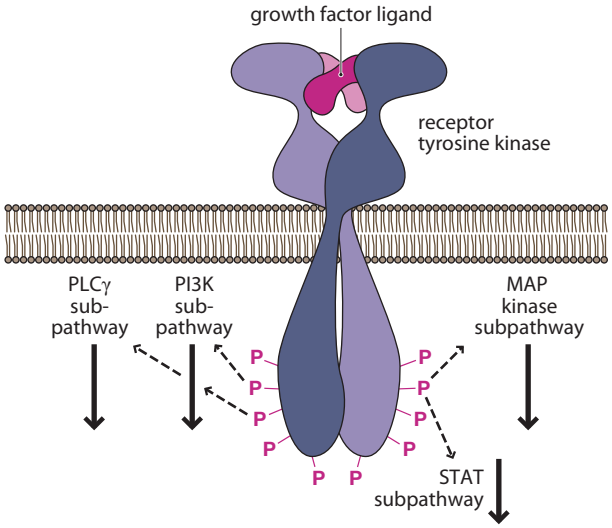
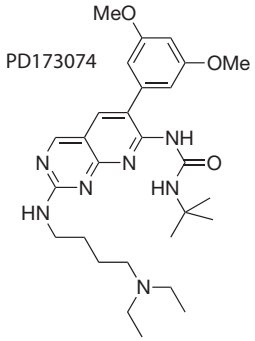



**Figure 9.54 Four subpathways.** Receptor tyrosine kinases activate four different subpathways. Dimerization of receptor tyrosine kinases leads to the sequential autophosphorylation of tyrosine residues, each marked with a P. Some of these phosphotyrosines serve as beacons that turn on subpathways.




A

| RTK | $IC_{50}$ (nM) |
|---|---|
| FGFR1 | 4 |
| FGFR2 | 3 |
| FGFR3 | 5 |
| VEGFR2 | 100 |
| PDGFR | 17,600 |
| c-SRC | 19,800 |
| EGFR | >50,000 |
| InsR | >50,000 |

B

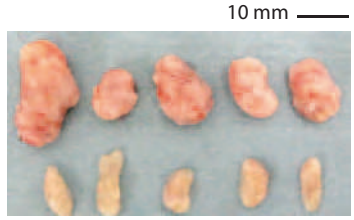



**Figure 9.55 Selective inhibition of RTKs.** (A) The synthetic inhibitor PD173074 (left) shows significant selectivity among receptor tyrosine kinases. (B) The inhibitor slows the growth of urothelial carcinomas. (B, from M. Miyake et al., *J. Pharmacol. Exp. Ther.* 332:795, 2010. With permission from American Society for Pharmacology and Experimental Therapeutics.)

## The ATP-binding sites of receptor tyrosine kinases are sufficiently different that they can be selectively inhibited by small molecules

All protein kinase catalytic domains use Mg•ATP as a phosphoryl donor and have similar folded structures. Yet the ATP-binding pockets are not identical. The subtle differences between the various tyrosine kinases are sufficient to allow the development of selective kinase inhibitors through a combination of empirical screening and targeted synthesis. Chemists at Pfizer developed a selective inhibitor of the tyrosine kinase domain of the fibroblast growth factor receptor (**Figure 9.55**). When screened against 244 different protein kinases, PD173074 was shown to be active against only FGFR1, FGFR2, FGFR3 (but not FGFR4), and the γ isoform of CaM kinase II (see Figure 9.55). When administered orally, PD173074 significantly decreases the growth of the human transitional cell carcinoma cell line MGHU3 transplanted into nude mice, which lack an immune system to reject human cells. Other cell lines (T24, KU7, UM-UC-2, UM-UC-3, UM-UC-6, and J82) were completely unaffected by PD173074. In these cell lines, the aberrant signaling is not due to mutations in an FGF receptor. For example, a GGC (Gly) to GTC (Val) mutation in the Ras protein downstream of Ras in T24 cells causes persistent activation of the FGFR3 pathway.

### Problem 9.10

Which tyrosine kinases in Figure 9.55 would be inhibited by more than 90% at 1 μM PD173074?

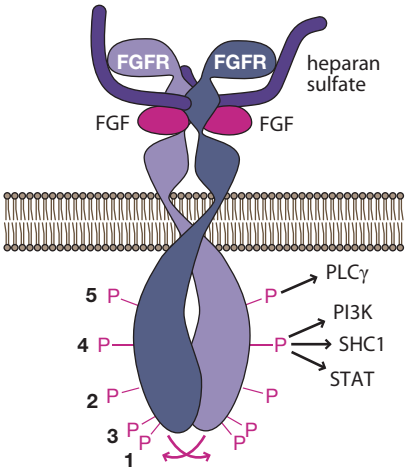



**Figure 9.56 Sequential phosphorylation.** Tyrosine residues on FGFR3 are phosphorylated in a precise order (1–5). The fourth and fifth phosphotyrosines activate specific subpathways. The actual locations of the phosphotyrosines are different from that shown in this schematic diagram.

## Transphosphorylation of tyrosine residues is sequential

The various growth factor receptors show strong mechanistic similarities and subtle structural differences in the way in which they bind the growth factor ligands. All growth factor receptors bind growth factors as 2:2 ligand–receptor complexes. All growth factor receptors transphosphorylate: a kinase domain on one receptor molecule phosphorylates tyrosines on the other receptor.

To understand phosphorylation better, we will focus on the FGF receptor in urothelial cells, namely FGFR3. Activation of fibroblast growth factor receptor requires two types of ligands (**Figure 9.56**): fibroblast growth factor and the anionic glycosaminoglycan heparan sulfate. The surfaces of bladder and other cells are coated with the proteoglycan heparan sulfate. This anionic glycan limits adherence by bacteria. Heparan sulfate sifts the extracellular milieu, binding weakly to cationic growth factors such as FGF and VEGF; these growth factors have p*I* values of 8.2 and 7.7, respectively, and are thus positively charged at pH 7. The coulombic interactions between heparan sulfate and the growth factors ensure that the ligands are in close proximity when they encounter the FGF receptor. The two molecules of FGF make no contact with each other in the 2:2:2 receptor–growth factor–heparan sulfate complex.

After our discussion of cytokine receptors in Section 9.4, transphosphorylation of tyrosine residues on receptors should be a familiar story (**Figure 9.57**). Within urothelial cells, phosphotyrosines that are generated on FGFR3 beckon proteins equipped with SH2 domains to the cell membrane; the cytoplasmic side of the cell membrane

is where the signal transduction pathway begins. Much of our understanding of FGFR3 comes from studies of the closely related receptor FGFR1. The kinase domain of FGFR3 is kept at bay by an inhibitory loop with the sequence $Y_{647}Y_{648}KK$. The net positive charge confers a strong affinity for the active site, but this effect is abrogated by phosphorylation of Y647. Phosphorylation of Y647 allows the kinase to phosphorylate Y577 on the opposing receptor, followed by the second tyrosine (Y648) in the inhibitory loop. The effect of phosphorylating tyrosines in the inhibitory loop has been quantified for FGFR1: phosphorylation of the first tyrosine increases activity 50-fold, and phosphorylation of the second tyrosine further increases activity 500-fold. Once both tyrosines in the inhibitory loop are phosphorylated, the affinity of the loop for the active site is fully reversed and two additional tyrosines (Y724 and Y760) are then phosphorylated. These two residues have a critical role in recruiting signaling proteins with SH2 domains. Tyr724 recruits STAT3, PI3K, and SH2-containing transforming protein 1 (SHC1). Tyr760 recruits the enzyme phospholipase Cγ.

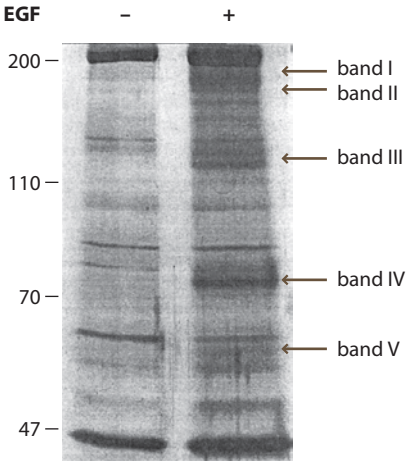



**Figure 9.57 Immunoblot.** Antibodies that specifically bind to phosphotyrosine residues can be used to label proteins in an SDS–PAGE gel. Cells treated (+) with epidermal growth factor show new patterns of tyrosine phosphorylation compared with untreated cells (–), as indicated by the presence of new bands in the (+) lane. (From H. Steen et al., *J. Biol. Chem.* 277:1031–1039, 2002. With permission from The American Society for Biochemistry and Molecular Biology.)

## Receptor tyrosine kinases signal growth via a MAP kinase cascade

Early studies of mitogens pointed to a connection between protein growth factors (such as FGF) and protein kinases. Whereas some protein kinases seemed to phosphorylate many proteins (protein kinase C, protein kinase A), the protein kinases that were activated by mitogenic growth factors seemed to be highly specific. Studies of mitogen-activated protein kinases (MAP kinases) gave birth to the idea that mitogenic signals are transduced through a cascade of successive protein phosphorylations. In these MAP kinase cascades, which we discussed in Chapter 6, one protein kinase activates, through phosphorylation, a second protein kinase, and the second protein kinase activates a third kinase, and so on.

In urothelial cells, FGFR3 transduces growth signals to a MAP kinase cascade through a complex of three adaptor proteins: SHC1, Grb2, and SOS. The ubiquitous adaptor protein SHC1 (SH2 domain-containing protein 1) binds the phosphorylated form of Tyr724 on the FGFR3 receptor, and undergoes tyrosine phosphorylation. The adaptor protein Grb2 has an SH2 domain that binds to the phosphotyrosine on SHC1 (**Figure 9.58**). Grb2 also has two SH3 domains (Chapter 5) that bind specifically to

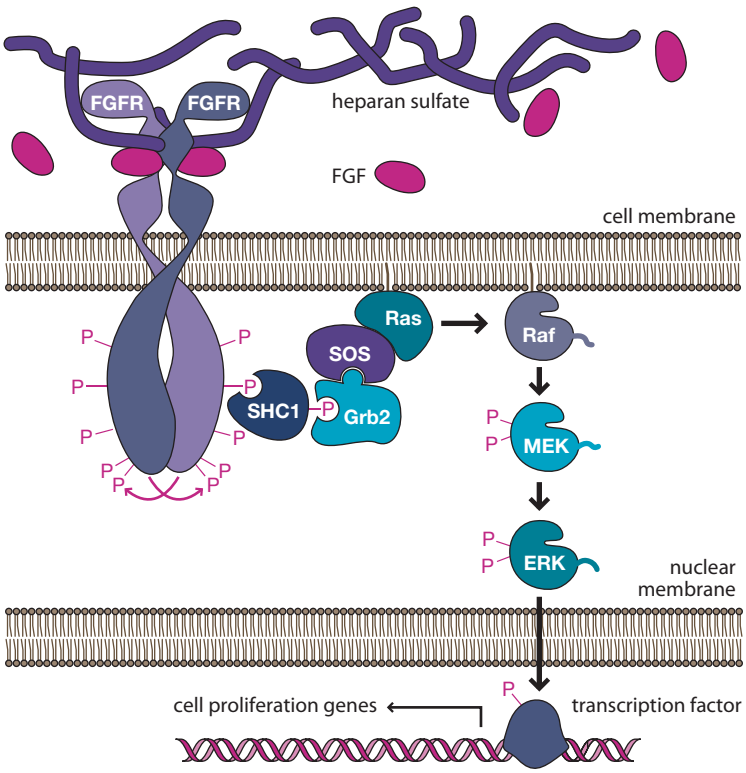



**Figure 9.58 The MAP kinase subpathway.** Receptor tyrosine kinases such as FGFR promote genes for proliferation via a cascade of protein phosphorylations by the kinases Raf, MEK, and ERK.

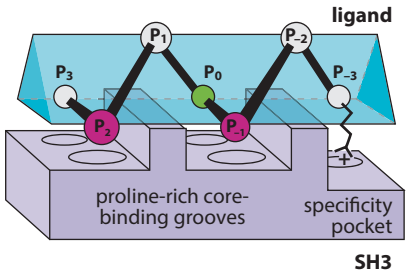



**Figure 9.59 Binding to a polyproline helix.** The tight turns of a polyproline helix allow extensive interactions between an SH3 domain and its ligand. (Adapted from J.T. Nguyen et al., *Science* 282:2088–2092, 1998.)

helical polyproline segments (PPXPPR) on SOS (**Figure 9.59**). SOS catalytically activates a G protein called Ras, which is held in the plasma membrane by a farnesyl group.

The kinases in the MAP kinase pathways phosphorylate serine and threonine residues. Ras activates the serine/threonine protein kinase called Raf, which specifically phosphorylates the kinase MEK. MEK phosphorylates the protein ERK (which stands for extracellular signal regulated kinase). ERK translocates to the nucleus, where it phosphorylates transcription factors, which turns on genes for cell proliferation and for inhibition of the FGFR signaling pathway (*Sprouty*, *Sef*, and *MKP3* phosphatase). In this pathway, ERK is considered the mitogen-activated protein kinase. The kinase that phosphorylates a MAP kinase (such as ERK) is called a MAP kinase kinase. The kinase that phosphorylates a MAP kinase kinase (such as MEK) is called a MAP kinase kinase kinase. Thus, Raf is a MAP kinase kinase kinase. There are two other homologous MAP kinases, JNK and p38 MAP kinase, and each of these kinases is activated by a specific series of MAP kinase kinases and MAP kinase kinase kinases.

The exquisite specificity of kinases in MAP kinase pathways is due to the presence of docking grooves in the kinase, and corresponding docking peptide segments in either the upstream or downstream kinase. The interaction of the docking peptide on the MAP kinase kinase MEK with the docking groove on the MAP kinase ERK is essential for specificity and binding. The devious enzyme anthrax lethal factor, produced by *Bacillus anthracis*, specifically castrates the docking peptide from MEK and other MAP kinase kinases, leaving them impotent (**Figure 9.60**).

**Figure 9.60 Chemical castration.** Anthrax lethal factor is a protease that cleaves the docking peptide from MAP kinase kinases such as MEK.

## Problem 9.11

**A** Which proteins in the MAP kinase pathway (Figure 9.58) are phosphorylated when the FGF is added to urothelial cells?

**B** Which proteins in the MAP kinase pathway would be phosphorylated if both FGF and vemurafenib, a selective inhibitor of Raf, were added to urothelial cells?

**C** Which proteins in the MAP kinase subpathway would you expect to interact with an antibody that binds selectively to phosphotyrosine residues, but not phosphoserine or phosphothreonine residues? You may have to refer back to Chapter 6.

## Many signal transduction pathways involve abundant small molecules and scarce proteins

Most of the signal transduction networks in cells are mediated by protein–protein interactions, but a variety of small molecules transmit signals between proteins. In general, small molecules are much more diffusible than proteins and, unlike proteins, their concentrations can be increased to relatively high levels without involving the translational machinery or requiring such high protein concentrations that they turn the cytoplasm into a gelatinous paste. Small molecules operate in two environments: membranes and cytosol. To appreciate the effect of membrane localization in the PI3K and PLCγ subpathways, we need to make a distinction between the signaling molecules that reside in the plasma membrane and the signaling molecules that reside in the cytosol (**Figure 9.61**).

In Chapter 8 we discussed a range of fatty acids and sphingosine derivatives that mediate signaling within membranes: for example diacylglycerols, phosphatidylinositols, and sphingosines. Terms such as diacylglycerol, phosphatidylinositol, and phosphatidylinositol 4,5-diphosphate ($PIP_2$) refer to classes of molecules composed of various combinations of fatty acids (see Figure 9.61). $PIP_2$ has an essential role in growth factor signaling. The fatty acids present on phosphatidylinositols vary in length

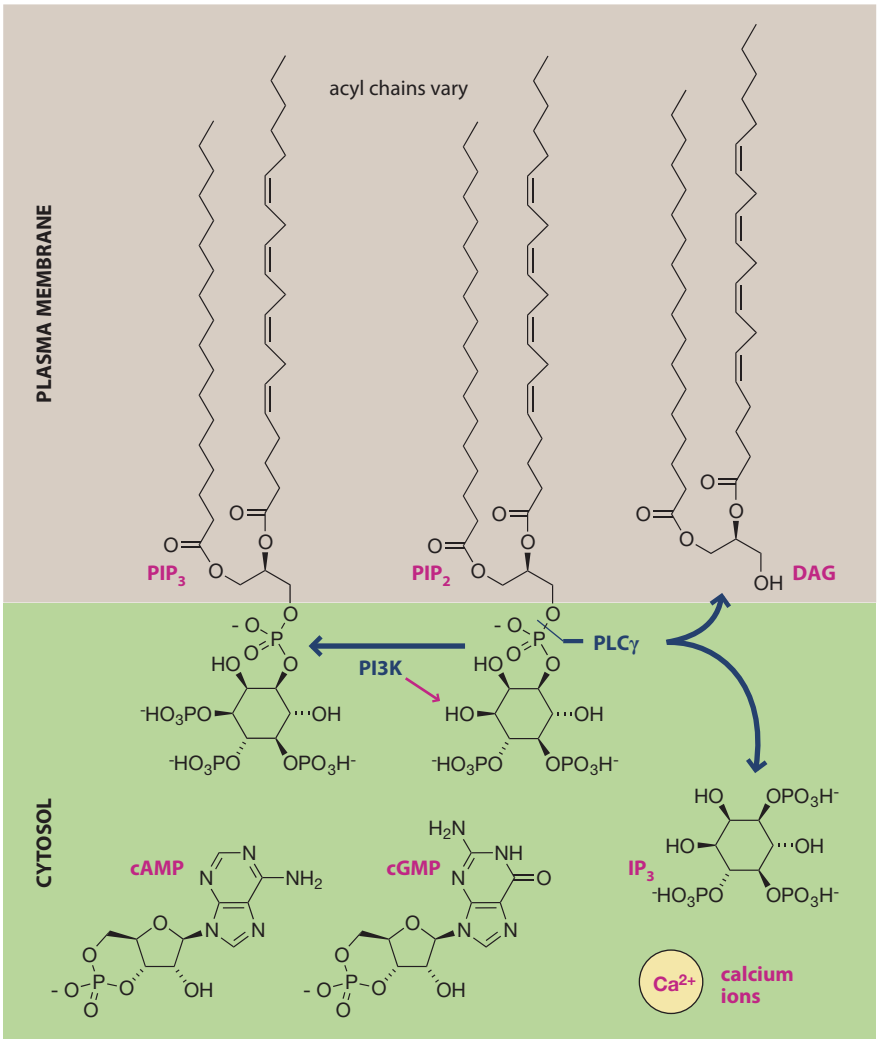



**Figure 9.61 Small molecules, big effects.** Abundant small, nonprotein molecules are important in signal transduction. Fatty acid derivatives such as $PIP_2$, $PIP_3$, and DAG are localized in the plasma membrane, whereas polar ionic compounds are localized in the cytosol.

(16–20 carbons) and unsaturation (0–4 double bonds), but an arachidonyl group is most common at the 2 position of glycerol, and a stearoyl group is most common at the 1 position. There is some selectivity among the various enzymes that process phosphatidylinositols; for example, some prefer arachidonyl groups at the S-2 position of glycerol; however, there seems to be a wide tolerance for different fatty acyl chains. Given the heterogeneity of lipids, we cannot adequately represent all phosphatidylinositols with a single structure, any more than we can represent a voter with a single human face. Each time a phosphatidylinositol is drawn it is up to the reader to envision a collection of molecules with variations in the fatty acid chains.

Diacylglycerol remains localized within the plasma membrane, where it interacts with membrane-bound proteins such as protein kinase C. Similarly, phosphatidylinositol 1,4,5-triphosphate ($PIP_3$) also remains localized within the membrane, but it dangles the inositol subunit—like bait on a hook. The phosphoinositol head group of $PIP_3$ is irresistible to proteins with pleckstrin homology domains, such as the kinase PDK1 (**Figure 9.62**).

Small molecules that diffuse through the cytosol include cyclic adenosine monophosphate (cAMP), inositol 1,4,5-triphosphate ($IP_3$), and metal cations, particularly $Ca^{2+}$. Several of these small molecules are associated with common effects among the various types of human cells. cAMP usually promotes cell growth via the relatively promiscuous kinase cAMP-dependent protein kinase (PKA) and a transcription factor called the cAMP response element binding protein. $IP_3$ opens calcium storage vessels, resulting in an immediate burst in $Ca^{2+}$ ion concentration. An increase in intracellular $Ca^{2+}$ levels induces the contraction of muscle cells (such as skeletal, cardiac, vascular, bronchial, and ciliary) and the release of messengers from vesicles (such as neurons

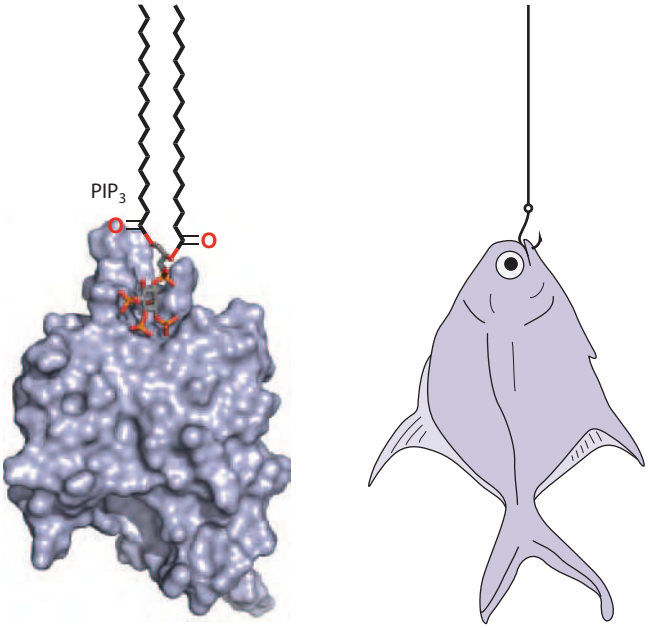



**Figure 9.62 Fishing for proteins.** The pleckstrin homology domain of PDK1 binds to the phosphoinositide head group of $PIP_3$ (PDB 1W1D). As a result, the entire kinase is localized at the cell membrane.

and white blood cells). Cyclic guanosine monophosphate (cGMP) inhibits the contraction of smooth muscle cells. Most studies of signal transduction pathways that lead to muscle contraction have been carried out on vascular smooth muscle (high blood pressure, inflammation, and asthma) and cardiac muscle (cardiovascular disease).

## Receptor tyrosine kinases turn on calcium signaling pathways via phospholipase C

The phosphorylated form of Tyr760 on the FGF receptor binds specifically to the SH2 domain of the γ isoform of phospholipase C (PLCγ). Phospholipase C leads to increases in the concentrations of $Ca^{2+}$ ions and diacylglycerol—both of these molecules are involved in the activation of protein kinase C (**Figure 9.63**). Phospholipase C

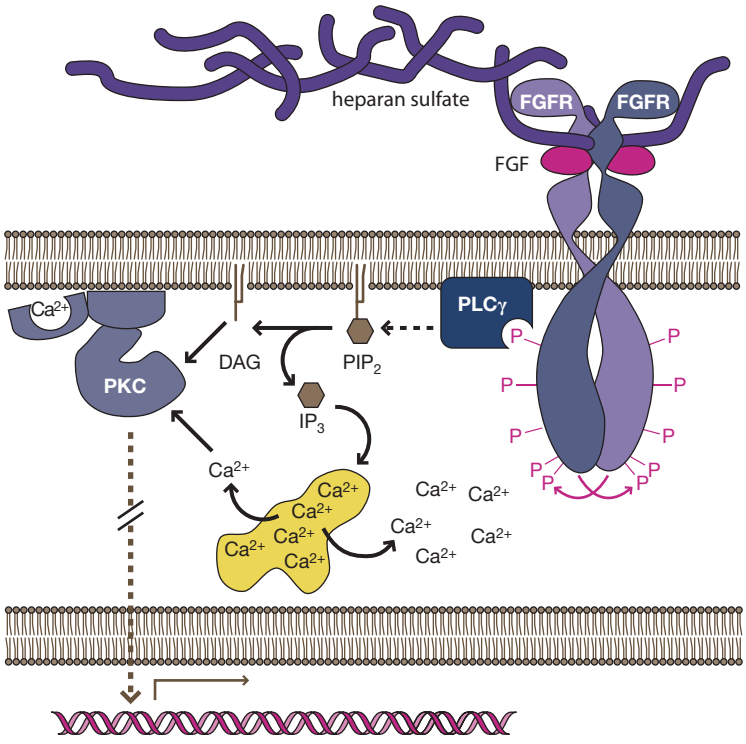



**Figure 9.63 The PLCγ subpathway.** PLCγ catalyzes the hydrolysis of $PIP_2$ to give DAG and $IP_3$. DAG directly and selectively activates protein kinase C, whereas $IP_3$ indirectly activates PKC by increasing cellular $Ca^{2+}$ levels.

cleaves the diacylglycerol moiety of $PIP_2$, generating the membrane-bound diacylglycerol and cytosolic $IP_3$. $IP_3$ opens $Ca^{2+}$ channels in calcium storage vesicles, resulting in a burst of cytosolic $Ca^{2+}$ ions. Recall from Chapter 6 that $Ca^{2+}$ ions bind to a domain of protein kinase C, which otherwise blocks a membrane-localization domain. Once $Ca^{2+}$ unblocks the membrane localization domain, PKC binds to the plasma membrane, where the concentration of diacylglycerol is high. The binding of diacylglycerol to PKC leads to a fully activated form of the kinase localized at the cell membrane.

## Problem 9.12

The active site of phospholipase Cγ contains three key catalytic residues: two histidines and one arginine. Suggest a plausible arrow-pushing mechanism for the enzyme-catalyzed hydrolysis of $PIP_2$ to $IP_3$ and DAG that includes a role for these three key residues and that involves neighboring-group participation by analogy with ribonuclease A.

## Receptor tyrosine kinases broadcast both proliferative and anti-apoptotic signals via Akt

The phosphotyrosines that result from activation of the growth factor receptors are recognized by proteins with SH2 domains. Grb2 (pronounced "grab two") is a key mediator of growth factor receptor signals. In urothelial cells, the SH2 domain of Grb2 binds to phosphotyrosine 724 of the FGF receptor, further recruiting the proteins Gab1 and PI3K to the cell membrane (**Figure 9.64**). Once PI3K is localized at the cell membrane it can catalyze the transfer of a phosphate group from ATP to $PIP_2$ to generate $PIP_3$. As the concentrations of $PIP_3$ increase, the $PIP_3$ molecules recruit two protein kinases to the membrane, PDK1 and Akt, each of which possesses a pleckstrin homology domain (**Figure 9.65**). Binding to $PIP_3$ in the plasma membrane fixes both the location and orientation of proteins, allowing PDK1 to phosphorylate Thr308 and Ser473 in Akt, leading to a fully activated form of Akt that targets many proteins.

Most of the proteins that are phosphorylated by Akt are inhibited, but because those proteins inhibit mitosis and differentiation, Akt ultimately *activates* pathways that permit cell proliferation. The result of this boolean logic is reminiscent of the

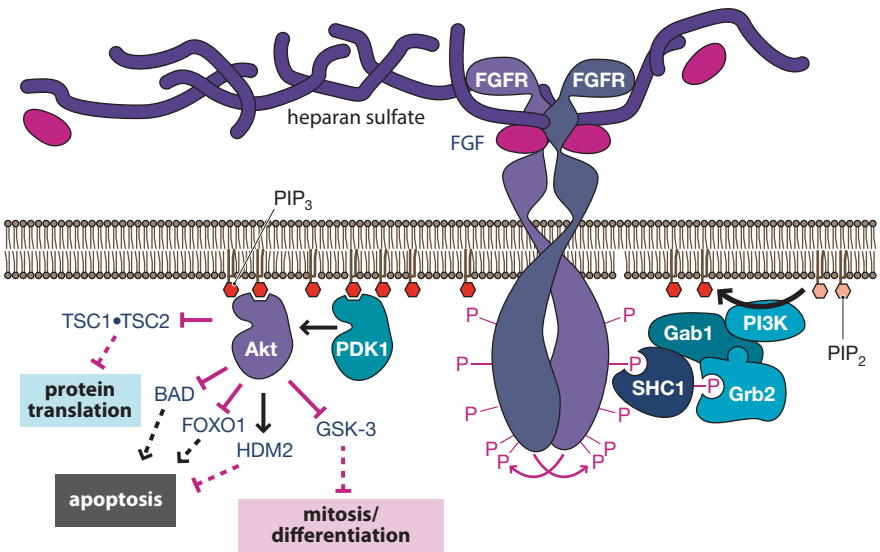



**Figure 9.64 The PI3K/Akt subpathway.** The kinase Akt binds to a lipid derivative PI3K. Once localized to the plasma membrane Akt exerts three effects: (1) enhancement of protein translation, (2) inhibition of apoptosis, and (3) promotion of proliferation and differentiation. The dashed arrows suggest that we have omitted many of the intervening steps.

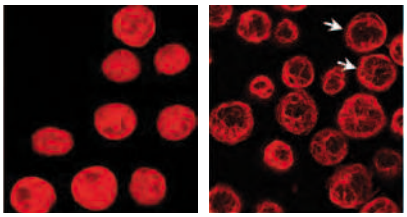
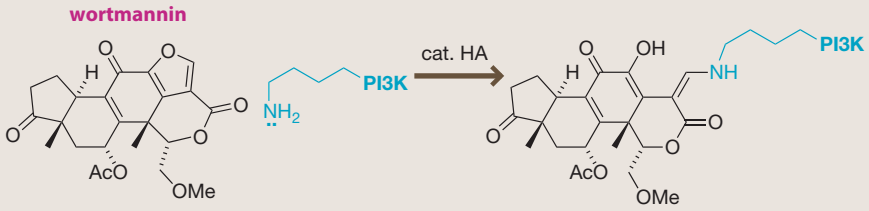


**Figure 9.65 Akt at the membrane.** The localization of Akt in a fibroblast cell line can be determined by staining fixed cells with a red fluorescent antibody. In the absence of mitogenic signals (top) Akt is distributed throughout the cytoplasm. In the presence of mitogenic signaling (bottom), Akt is localized mainly at the plasma membrane. (From G. Guo et al., *Nature* 29:3845–3853, 2010. With permission from Macmillan Publishers Ltd.)

ancient proverb "the enemy of my enemy is my friend." Akt phosphorylates the proteins TSC1 and TSC2, preventing them from inhibiting protein translation. By releasing the brakes on protein translation, the cell is ready for rapid growth. Akt suppresses the suicidal tendencies of the cell by deactivating BAD and FOXO1 and by activating HDM2. Left unchecked, these three proteins would otherwise couple any attempts at cell division with apoptosis. Finally, Akt deactivates glycogen synthase kinase 3 (GSK-3) through phosphorylation. When GSK-3 is active, it suppresses cell division. By releasing the brakes, Akt allows cell division and differentiation. In the next section, we discuss how GSK-3 leads to the degradation of β-catenin, ultimately affecting transcription.

## Problem 9.13

Wortmannin is a highly specific covalent inhibitor of PI3K, forming a covalent bond with Lys802 in the ATP-binding pocket. Suggest a plausible arrow-pushing mechanism for an acid-catalyzed reaction of wortmannin with PI3K.




## The differences between various receptor tyrosine kinase pathways are less important than the similarities

We began our discussion of receptor tyrosine kinases by listing the many types of growth factor signaling pathways (EGF, VEGF, PDGF, NGF, and so on) and contrasting the different structures of the receptor–ligand complexes. Our decision to focus on the FGF receptor was inconsequential (for readers of this book). We would have discussed the same common downstream pathways—STAT, MAP kinase, and PLCγ and PI3K—regardless of which receptor tyrosine kinase was chosen. Given the complexities of those divergent pathways, it is more profitable to focus on the similarities rather than the differences between the various receptor tyrosine kinase pathways. The overall message is that receptor tyrosine kinases activate the transcription of genes leading to cell growth and cell division; equally importantly, receptor tyrosine kinases also inhibit pathways that lead to cell death.

## Chemical methods for isolation and identification of kinase substrates

Often, one isolates a kinase protein or gene for a kinase for which the substrate is not yet known. One way to identify target peptide sequences for kinases is to apply the kinase reaction to peptide libraries and then sequence only the phosphopeptide products. However, it is nontrivial to selectively remove anionic phosphopeptides from random peptide mixtures, because peptides abundant in Asp and Glu will be polyanionic, just like phosphorylated peptides. One highly successful approach to this problem involves the use of $Fe^{3+}$ resins for the affinity isolation of phosphopeptides from peptide libraries. When a mixture of phosphorylated (less than 1% of the mixture) and nonphosphorylated peptides is mixed with ferric iminodiacetate resin beads, the phosphate groups bind selectively to the ferric ions (**Figure 9.66**). More than 99.9% of

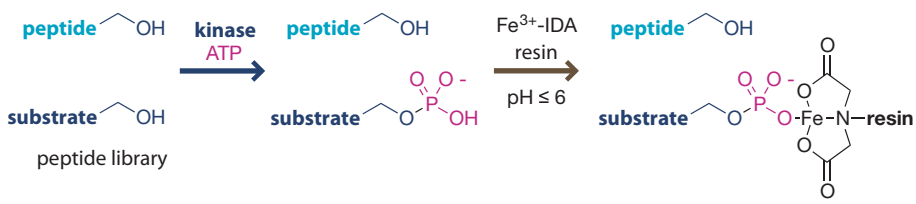



**Figure 9.66 Fishing for phosphoproteins, part 1.** The Cantley soluble oriented peptide library method allows one to isolate phosphorylated peptides from large peptide libraries on the basis of the affinity of phosphate anions for $Fe^{3+}$. IDA, imino diacetic acid.

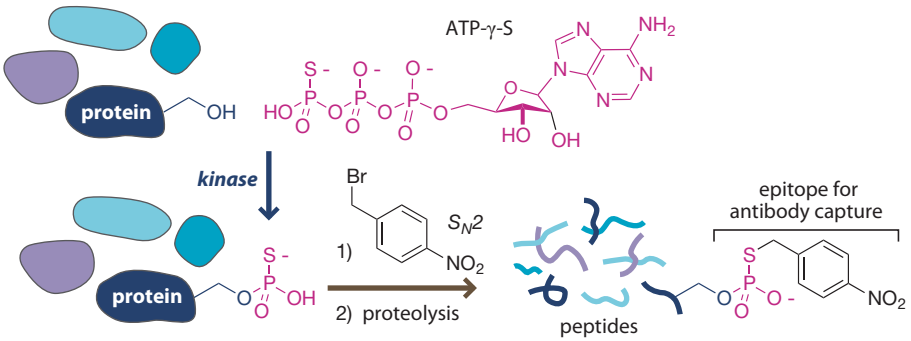



**Figure 9.67 Fishing for phosphoproteins, part 2.** The Shokat strategy for the isolation of kinase products from complex mixtures of proteins involves an ATP analog, ATP-γ-S.

the nonphosphorylated peptides can be washed off by elution with buffers at or below pH 6.0. With a concentrated high-pH buffer (500 mM $NH_4HCO_3$, pH 8.0), more than 90% of the phosphopeptides can be eluted. The entire phosphopeptide mixture is then sequenced to determine which amino acids are preferred at each of the positions of the preferred substrates.

An alternative approach for identifying substrates for kinases is to phosphorylate protein mixtures by using a mixture of kinase and ATP-γ-S. The sulfur atom in a phosphorothioate prefers the anionic form, even at neutral pH, and like the cysteine thiol it is highly nucleophilic. Both cysteine and phosphorothioserines are alkylated, but antibodies against the anionic nitrobenzylphosphorothio group readily distinguish between an alkylated cysteine and an alkylated phosphorothioserine. This method can even be applied to mixtures of full-length proteins from cell lysates. The protein can be digested and the sequences of the peptides determined by mass spectrometry (**Figure 9.67**). Once you know the sequence of the peptide, you can search through databases to identify the full-length protein.

## 9.6 G PROTEIN-COUPLED RECEPTORS

### Seven-transmembrane domain G protein-coupled receptors can respond to a wide range of ligands with high dynamic range

Seven-transmembrane domain G protein-coupled receptors (7TM GPCRs, or GPCRs for short) are prosaically named. Recall from Chapter 5 that they consist of a membrane-bound receptor, with seven membrane-spanning α helices, coupled to a GTP hydrolase (called a G protein) that serves as a timing device. The more apt term "heptahelical" is slowly entering the vernacular as a replacement for the clumsy term "seven-transmembrane domain," but we will continue to use the 7TM description to reinforce a connection to the large body of GPCR literature (**Figure 9.68**). There are more than 900 different genes for GPCRs in the human genome, about 5% of the total number of genes. Almost half of the genes for 7TM GPCRs code for olfactory receptors that recognize diverse odorants. It is common to group G proteins into superfamilies based on genetic sequence. In order of population, the human GPCR families are: Rhodopsin (R), Adhesion (A), Frizzled/Taste2 (F), Glutamate (G), and Secretin (S) (**Table 9.3**).

**Table 9.3 Families of human 7TM GPCRs.**

| Family | Number of receptors |
|---|---|
| Rhodopsin | 284 non-olfactory |
| Rhodopsin | 388 olfactory |
| Adhesion | 33 |
| Taste2 | 25 |
| Glutamate | 22 (includes Taste1) |
| Secretin | 15 |
| Frizzled | 11 |

**Figure 9.68 Structure of a GPCR.** The $\beta_2$-adrenergic receptor with the synthetic antagonist carazolol (van der Waals spheres) bound in place of epinephrine. The transmembrane helices are rendered in color. The structure is lacking the lower cytoplasmic loop between the yellow and orange helices. (PDB 2RH1)

It is not surprising that the majority of pharmaceutical drugs that were developed in the previous millennium were ultimately found to target GPCRs. Small molecules have the best oral availability, and 7TM receptors are the most common receptor for small-molecule signals in humans. Furthermore it is relatively easy to identify small-molecule ligands (for example histamine, adrenaline, and thyrotropin releasing hormone) and to design medicinally useful analogs, even if the structure of the protein target is unknown. Major advances were made in genomics and structural biology in the 1990s that emboldened pharmaceutical chemists to target signal transduction pathways inside cells. Thus, the focus of pharmaceutical chemists has broadened from membrane-bound receptors such as GPCRs to include many intracellular targets as well.

## High-affinity ligand–receptor interactions lead to slow response times and low dynamic range

An ideal sensor system should have high sensitivity, fast response time, and wide dynamic range. Designing highly sensitive receptors using reversible ligand–receptor interactions leads to slow response times and low dynamic range. To understand the origin of these long response times, let us return to the high-affinity streptavidin–biotin interaction that we discussed in Chapter 6. Recall that the half-life for the avidin–biotin complex is over half a year long. The half-life for the closely related streptavidin-biotin is several days. Thus, streptavidin could be a highly sensitive receptor for biotin and bind to a minuscule amount of biotin. However, the streptavidin receptor cannot respond to *changes* in concentration on a timescale that is useful for a cell.

To understand the origin of the slow off-rates of high-affinity receptors, we need to return to a kinetic picture of equilibrium binding that was introduced in Chapter 6. Recall that smaller dissociation constants correlate with tighter binding. The dissociation constant, like any equilibrium constant, can be expressed as the ratio of the rate constant for the off-rate and the rate constant for the on-rate (**Figure 9.69**). There is usually little variation in the on-rate constants for small-molecule ligands and receptors (this simplifying assumption only works well for small-molecule ligands). However, off-rate constants vary considerably. The half-life for dissociation is determined by the equation $t_{1/2} = (\ln 2)/k_{off}$ (**Figure 9.70**). At pH 7.4, a streptavidin–biotin interaction with an off-rate of $3 \times 10^{-5}\ s^{-1}$ has a half-life of more than 6 hours.

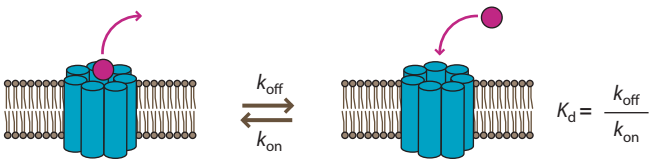



**Figure 9.69 Receptor-on, receptor-off.** The off-rate ($k_{off}$) is an important determinant of binding affinity.

The limited dynamic range or lack of responsiveness to different ligand conditions by high-affinity receptors such as streptavidin is more difficult to appreciate. Recall that biological responses correlate with the percentage of bound receptor. A larger response results from a higher percentage of receptors bound to the ligand. Thus, reversible ligand–receptor interactions are highly sensitive to changes in ligand concentration only when the ligand concentration is close to the $K_d$ for the receptor–ligand interaction. At this concentration, 50% of the receptors are bound to ligand, and 50% of the receptors remain empty; this situation provides empty receptors to react to an influx of new ligand, and also bound receptors ready to dissociate in response to a decrease in the concentration of the ligand. For the streptavidin–biotin interaction, only concentrations of biotin in the subpicomolar range lead to differences in the percentage of biotin bound to streptavidin.

When the ligand concentration is close to the $K_d$ for the receptor–ligand interaction, 10-fold changes in ligand concentration lead to marked changes in the percentage of bound receptor; however, outside that narrow window, 10-fold changes in ligand concentrations lead to very small changes in the percentage of bound receptors (**Figure 9.71**). For example, if the $K_d$ for a receptor–ligand interaction is 1 μM, the

$$t_{1/2} = \frac{\ln 2}{k_{off}}$$

| $k_{off}$ | $t_{1/2}$ |
|---|---|
| $1\ s^{-1}$ | 1 s |
| $10^{-3}\ s^{-1}$ | 12 min |
| $10^{-6}\ s^{-1}$ | 8 days |
| $10^{-9}\ s^{-1}$ | 22 years |

**Figure 9.70 Binding and half-lives.** Dissociation of a ligand from the receptor–ligand complex is a first-order or unimolecular reaction. The top equation relates the rate constant to the half-life for a reaction (lower table).

percentage of bound receptor is close to 100% regardless of whether the concentration of ligand is 100 μM, 1000 μM, or 10,000 μM; similarly, the percentage of bound receptor is close to 0% regardless of whether the concentration of ligand is 0.01 μM, 0.001 μM, or 0.0001 μM.

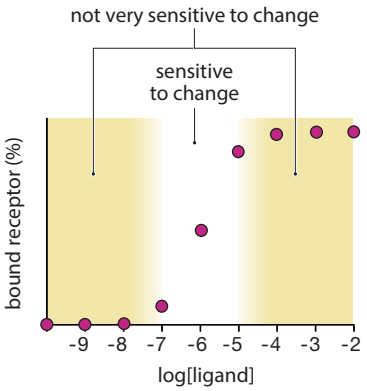



**Figure 9.71 Limited window of sensitivity.** As seen in this dose–response curve, reversible receptor–ligand interactions provide high dynamic sensitivity to changes in ligand concentration only when the concentration of the ligand is close to the $K_d$ for the receptor–ligand interaction (the unshaded portion of the graph).

## G proteins allow low-affinity receptors to have high sensitivity

G proteins are common elements of signal transduction pathways that must modulate the strength of the incoming signal. G proteins slowly catalyze the hydrolysis of GTP to GDP (**Figure 9.72**). The GTP-bound state turns on downstream signaling; in the GDP-bound state, signaling is shut off. G proteins are not enzymes in a traditional sense, because the catalytic rate is low and product dissociation is slow. However, each of these phases of the catalytic cycle is accelerated by a specific class of GTP-binding-protein regulators. **Guanine nucleotide exchange factors** (GEFs) catalyze the exchange of GDP for GTP; GEFs therefore increase signal output. **GTPase-activating proteins** (GAPs) accelerate hydrolysis of the bound GTP; GAPs therefore decrease signal output. Thus, the ratio of GEF to GAP determines the strength of the signal (**Figure 9.73**). In addition, regulation through binding to other proteins and lipids, along with phosphorylation and subcellular localization, controls the activity of GEFs and GAPs. Such regulation can effectively tie the G protein signaling to other cell signaling pathways.

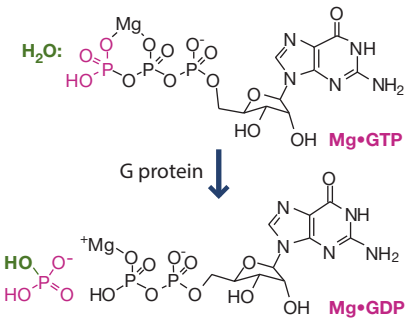



**Figure 9.72 Shutting off a G protein.** G proteins are deactivated when they hydrolyze Mg•GTP to GDP.

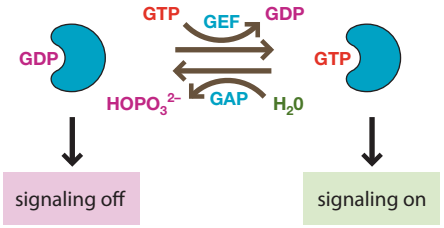



**Figure 9.73 The volume knob.** The ratio of GEF to GAP, both of which are catalysts, determines the strength of G protein signaling. (Adapted from D.P. Siderovski and F.S. Willard, *Int. J. Biol. Sci*. 1:51–66, 2005.)

Given their control over cell growth pathways, G proteins are a common site of oncogenic mutations. For example, mutations to the G protein Ras (Section 9.5) can prevent GTP hydrolysis. The resultant cells have growth signals permanently turned on. A third of all human cancers have been found to have such mutations in Ras. Furthermore, the small-molecule analogs of genetic mutations can alter the course of signal transduction pathways. For example, the small molecule brefeldin A binds to the G protein Arf1 and glues the protein to a GEF, effectively preventing the dissociation of GDP from Arf1. The activity prevents the binding of Arf1 to coat protein found in the membrane of the Golgi complex (**Figure 9.74**). Thus, brefeldin A blocks the transport of vesicles between the endoplasmic reticulum and the Golgi complex; this ability makes brefeldin A a widely used tool for the study of vesicle trafficking in the cell.

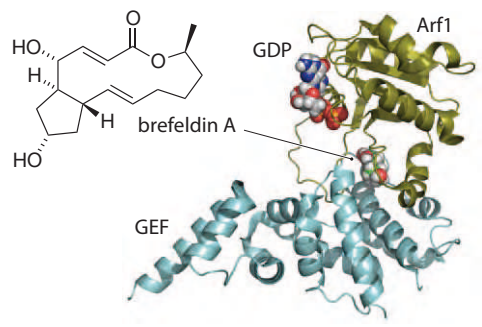



**Figure 9.74 Blocking anterograde transport of vesicles between the ER and the Golgi.** The G protein Arf1 has an essential role in Golgi vesicle trafficking. The polyketide natural product brefeldin A binds to and stabilizes the complex between a GEF and Arf1, inhibiting the release of GDP. (PDB 1RE0)

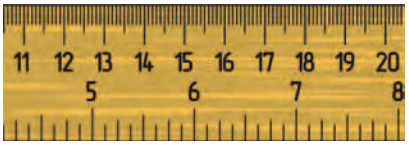



**Figure 9.75 Low dynamic range.** A ruler is exceptional for measuring objects on the scale of millimeters to centimeters, but worthless for measuring objects in the range of nanometers or kilometers.

## Seven-transmembrane domain G protein-coupled receptors can respond to a wide range of ligands with high dynamic range

G protein-coupled receptors respond with high sensitivity, fast response times, and high dynamic range. They can bind ligands with modest affinity—briefly—and then release them. During the short time that 7TM receptors bind to their cognate ligands, they act as guanine nucleotide exchange factors (GEFs), and release active G proteins. The power of a G protein-coupled receptor is that the lifetime of the signal is not limited by the amount of time that the receptor is bound! Instead, the lifetime of the signal is determined by the ratio of GEFs to GAPs, which the cell can control. This powerful concept bears repeating: cells can modulate the sensitivity of G protein signaling pathways. A detector that can accurately distinguish responses with both high and low sensitivity is said to have high (or wide) dynamic range (**Figure 9.75**). For example, your eyes rely on 7TM G protein-coupled receptors to detect light. Having photon receptors with a wide dynamic range allows you to distinguish objects in a dimly lit room as easily as you distinguish them under sunlight that is a billion times brighter.

### Problem 9.14

G proteins hydrolyze a bound Mg•GTP, using water as a nucleophile. Draw an arrow-pushing mechanism for the reaction. Mixtures of $AlF_3$ and GDP can lead to persistent activation of G proteins *in vitro*. Suggest a mechanism for this activation of G proteins.

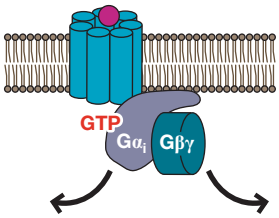



**Figure 9.76 Fork in the road.** Dissociation of Gα and Gβγ subunits allows 7TM GPCRs to activate two different pathways.

## Heterotrimeric G proteins are designed to generate divergent signals

The G proteins that couple to 7TM receptors are composed of three different protein chains: an α subunit and a complex between β and γ subunits. Such G proteins are referred to as heterotrimeric G proteins, and are distinguished from smaller G proteins such as Ras. Activation of a 7TM GPCR induces dissociation of the α subunit from the βγ subunit, and each of these turns on a separate pathway (**Figure 9.76**). The α subunit often modulates pathways that generate small molecules (cAMP or DAG and $IP_3$), which ultimately affect transcription. The βγ subunit modulates rapid cellular responses (such as the contraction of muscle cells or the firing of neurons), usually based on **ion channels**: for example GIRK1 potassium channels (cardiac myocytes) and CaCn calcium channels (Purkinje neurons).

To understand the divergence of signals from heterotrimeric G proteins, it is instructive to examine how the vagus nerve decreases the contractile force of heart muscle cells. Vagal neurons release the neurotransmitter acetylcholine, which binds to a 7TM GPCR, the M2 acetylcholine receptor, on cardiac myocytes (**Figure 9.77**). Those acetylcholine receptors are coupled to a heterotrimeric G protein, leading to an exchange of GDP for GTP within the α subunit. The α subunit, $G\alpha_i$•GTP, dissociates from the βγ subunit, Gβγ. Gβγ activates GIRK1 potassium channels in the cell membrane, and the leakage of $K^+$ ions from the cell decreases the force of contractions. The $K^+$ gradient is essential for myocyte function; in the United States, death by lethal injection is performed by infusion of potassium chloride, which stops the heart. The $G\alpha_i$•GTP subunit inhibits adenylyl cyclase, slowing the rate at which it cyclizes ATP into cAMP (see Figure 9.77). The levels of cAMP in the cell ultimately affect the transcription of genes. The conversion of $G\alpha_i$•GTP to $G\alpha_i$•GDP decreases the affinity for adenylyl cyclase, freeing the $G\alpha_i$•GDP to bind to an unligated acetylcholine receptor.

## Some elements of signal transduction pathways can integrate inputs

The principal feature of a cardiac myocyte that distinguishes it from other cells is ceaseless rhythmic contraction that moves blood around the body. The force and frequency of myocyte contraction are dependent on a wide range of chemical signals: the neurotransmitter acetylcholine, the circulating hormones epinephrine (also known as adrenaline) and norepinephrine (also known as noradrenaline), and circulating

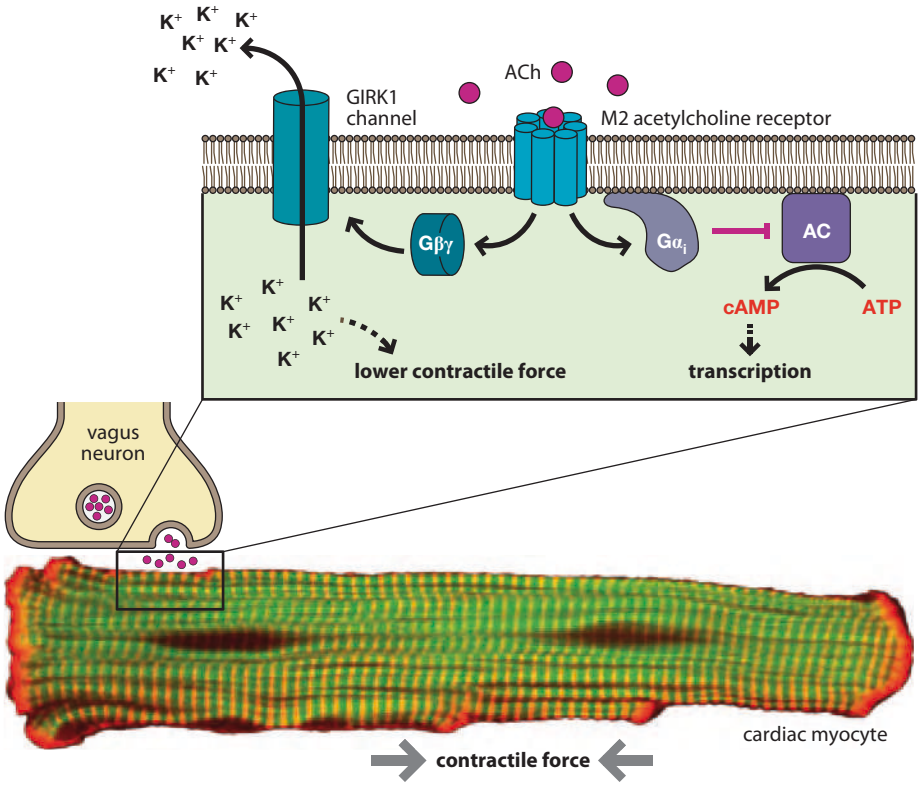



**Figure 9.77 Dual effects.** The M2 acetylcholine receptor exerts two effects on the cell via dissociation of the heterotrimeric G protein. It decreases the contractile force and affects transcription. AC, adenylyl cyclase. (Cardiac myocyte, courtesy of Nicholas Geisse, Harvard University. With permission from Kevin Kit Parker, Harvard University.)

peptide hormones such as endothelin I and angiotensin II. An overworked heart cell has relatively few ways in which to ease the burden. Cell proliferation is not an option because cardiac myocytes are terminally differentiated, without the capacity for proliferation. Instead of proliferating to share the workload, individual cardiac myocytes grow larger, leading to a visibly enlarged heart (**Figure 9.78**). The genes for myocyte hypertrophy are controlled by a transcription factor called the cyclic AMP response element binding protein (CREB). Thus, any signaling pathways that affect levels of cyclic AMP can affect transcription by CREB. The diterpene natural product forskolin activates adenylyl cyclase, mimicking agonists of 7TM GPCRs coupled to $G\alpha_s$ (**Figure 9.79**).

**Figure 9.78 Enlarged heart.** Alcoholism and high blood pressure overwork the heart, causing it to enlarge. (Courtesy of Photo Researchers.)

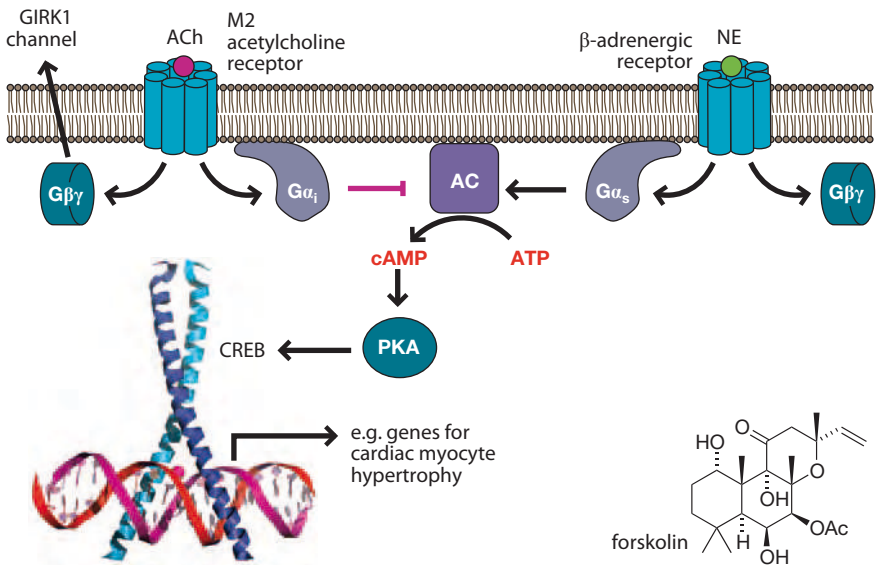



**Figure 9.79 Two paths converge.** Adenylyl cyclase (AC) integrates input from two different 7TM GPCRs. Inset: the small molecule forskolin activates adenylyl cyclase. NE, norepinephrine; AC, adenyl cyclase.

In cardiac myocytes, adenylyl cyclase integrates responses from two different types of 7TM GPCRs: one for circulating norepinephrine and the other for acetylcholine (released from the vagus nerve). The G protein $G\alpha_i$ subunit, released from the M2 acetylcholine receptor, inhibits adenylyl cyclase, whereas the $G\alpha_s$ subunit, released from the β-adrenergic receptor, activates adenylyl cyclase. In the human genome, there are 23 different genes for α subunits, 7 different genes for β subunits, and 12 different genes for γ subunits; each GPCR in a cell usually binds to a distinct combination of heterotrimeric G protein subunits. The Gβγ subunit released by the β-adrenergic receptor in cardiac myocytes does not affect the GIRK1 potassium channel.

Activation of adenylyl cyclase causes levels of cAMP to increase, but only up to a certain point. Cardiac myocytes express a phosphodiesterase PDE4 that begins to hydrolyze cAMP efficiently as the concentration approaches micromolar concentrations (the $K_m$ for PDE4D is 1.5 μM). Ultimately, cAMP generated by adenylyl cyclase activates cAMP-dependent protein kinase (PKA), which phosphorylates the transcription factor CREB at Ser133. CREB is also an integrator of signals, because Ser133 can also be phosphorylated by kinases from other pathways. CREB affects genes other than those for hypertrophy. Studies have shown that long-term activation of M2 acetylcholine receptors in atrial myocytes leads to the downregulation of M2 receptors, presumably through repression of the gene.

## Contraction of endothelial smooth muscle is controlled by $G\alpha_q$

On inhalation of noxious compounds or dangerous pathogens, the body responds by decreasing the volume of inspired air. Airway smooth muscle cells that surround the bronchiolar passages decrease respiratory volume by contracting (**Figure 9.80**). The smooth muscle cells respond to several different signaling molecules—the neurotransmitter acetylcholine, the cysteinyl leukotriene LTD4, the peptide endothelin, and the inflammatory signal histamine—each through a specific 7TM GPCR. These GPCRs are coupled to heterotrimeric G proteins containing $G\alpha_q$ subunits. Whereas $G\alpha_i$ and $G\alpha_s$ subunits affect adenylyl cyclase, $G\alpha_q$ activates the membrane-bound enzyme phospholipase Cβ, and no Gα subunits inhibit PLCβ. Recall from Chapter 8 that PLCβ cleaves the lipid phosphatidylinositol 4,5-bisphosphate to generate inositol triphosphate, which opens ion channels in calcium vesicles, and diacylglycerol, which remains associated with the membrane, where it activates protein kinase C.

$G\alpha_q$ affects the transcription of genes in several ways. Protein kinase C phosphorylates a wide range of signaling proteins that ultimately affect transcription, often associated with cell proliferation. $Ca^{2+}$ ions exert both rapid and slow effects on muscle cells. The $Ca^{2+}$-induced contraction of muscle cells is a rapid effect that does not

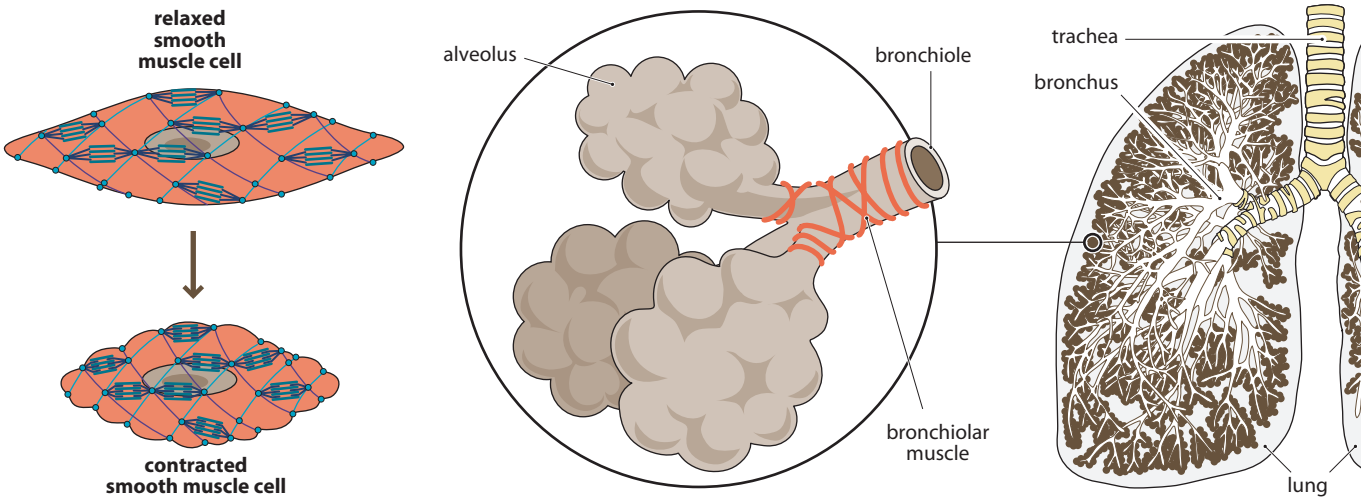



**Figure 9.80 Response to noxious stimuli.** Contraction of smooth muscle cells in the bronchiolar airway reduces the diameter of bronchioles and restricts airflow.

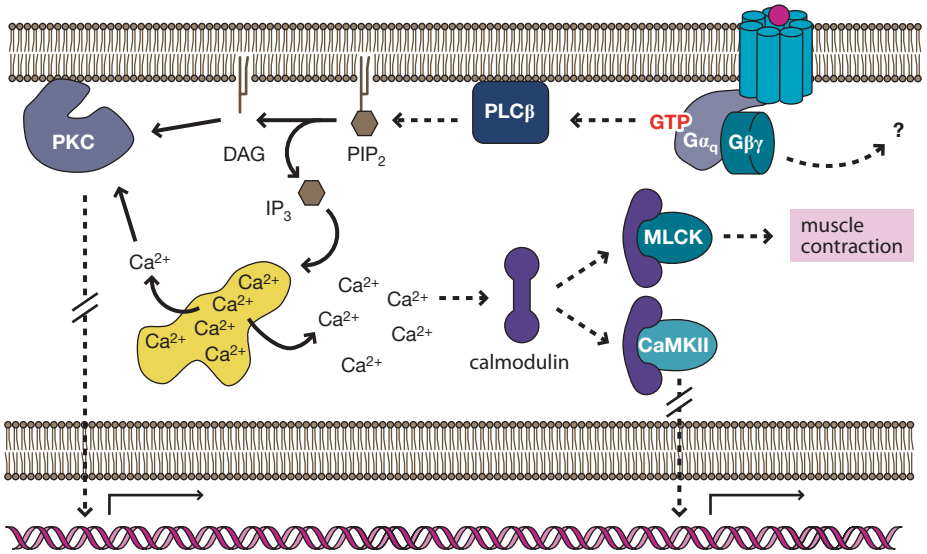



**Figure 9.81 Signaling contraction.** Histamine induces the contraction of smooth muscle cells in the airway by increasing the levels of $Ca^{2+}$ ions in the cell. MLCK, myosin light chain kinase.

involve transcription. $Ca^{2+}$ ions also affect transcription via protein kinases. Recall from our discussion of protein kinase C in Chapter 6 that $Ca^{2+}$ ions promote the association of protein kinase C with diacylglycerol groups in the plasma membrane. Once localized at the plasma membrane, protein kinase C phosphorylates proteins that ultimately lead to cell growth, including various subdomains of G proteins, and other regulatory proteins and receptors. Furthermore, $Ca^{2+}$ levels are monitored within the cell by a protein called calmodulin, which has binding sites for four $Ca^{2+}$ ions. In the presence of $Ca^{2+}$, calmodulin undergoes a marked conformational change that allows the protein to activate $Ca^{2+}$/calmodulin-dependent protein kinase II (CaMKII). CaMKII phosphorylates cytoskeletal proteins and other proteins that ultimately affect the transcription of genes such as endothelial nitric oxide synthase (**Figure 9.81**).

## Some bacterial toxins reprogram Gα subunits, with deadly results

Although humans live in harmony with many strains of bacteria, some bacteria are highly pathogenic and produce sophisticated protein toxins that are responsible for the most overt disease symptoms. Often these bacterial toxins are composed of two components: an A subunit that catalyzes a modification of an intracellular protein, and a B subunit that acts as a delivery vehicle for getting across the host membrane. The bacterium *Vibrio cholerae* and some strains of *Escherichia coli* produce a similar protein toxin (**Figure 9.82**). The B subunit of cholera toxin is a hexameric complex that binds multivalently to GM1 gangliosides on the surface of human cells. The A subunit of the cholera toxin is a glycosyltransferase that uses the high concentrations of $NAD^+$ in the cell as a substrate for modifying the G protein $G\alpha_s$ at Arg201; the by-product is the pyridine derivative nicotinamide. The modified Gα subunit cannot hydrolyze GTP; this leads to constitutive activation of adenylyl cyclase (**Figure 9.83**). In the epithelial cells of the gastrointestinal tract, constitutive $G\alpha_s$ activation leads to the opening of chloride channels. The massive export of $Cl^-$ ions is accompanied by a loss of water through uncontrollable diarrhea. The latter incident ensures that *V. cholerae* re-enters the environment and often the drinking water, where it can find more victims to infect.

Pertussis toxin, produced by the bacterium responsible for whooping cough, ADP-ribosylates a specific cysteine residue, four residues from the C terminus in $G\alpha_i$, $G\alpha_o$, and transducin, the G protein coupled to rhodopsin. In the ciliary cells of the respiratory tract, ADP-ribosylation of this cysteine residue of G proteins prevents the exchange of GDP, leaving the G protein in an inactive form. Without the ability to inhibit **adenylate cyclase**, the levels of cAMP and the activity of cAMP-dependent protein kinase are maintained at uncontrollably high levels. Ultimately, the effect on airway epithelial cells leads to a characteristic coughing pattern and ultimately to a whooping sound when the victim finally inhales.

**Figure 9.82 Death's dispensary.** In 1854, Dr. John Snow founded the field of epidemiology by tracking incidents during a cholera outbreak to a single pump providing water in the Soho neighborhood of London. Removing the handle from the pump ended the epidemic. (George J. Pinwell, Death's Dispensary, 1866.)

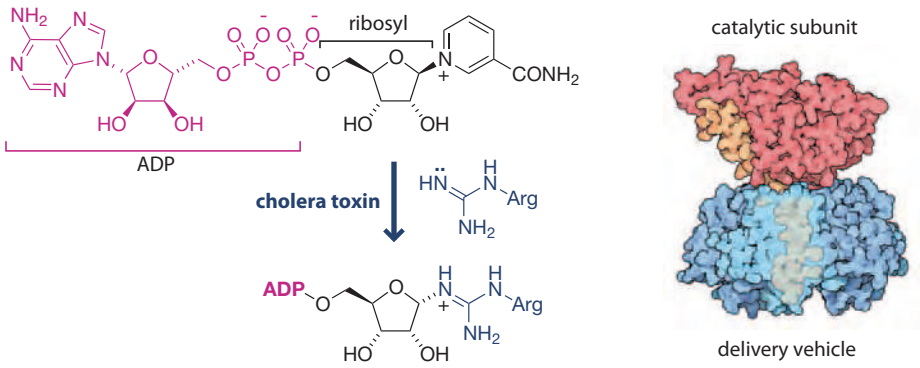



**Figure 9.83 G protein assassin.** Left: cholera toxin catalyzes the transfer of ADP-ribose from $NAD^+$ to a specific arginine side chain of G proteins. Right: the structure of cholera toxin with the B subunit rendered in blue and the catalytic A subunit rendered in red. (Right, courtesy of David S. Goodsell and RCSB Protein Data Bank.)

G proteins are not the sole substrates for bacterial ribosylating toxins. Both diphtheria toxin and *Pseudomonas* toxin A ADP-ribosylate a key piece of the protein translational machinery, elongation factor II. *Botulinum* C2 toxin ADP-ribosylates Arg177 of G-actin, leading to depolymerization of actin filaments. Because pertussis toxin and cholera toxin have orthogonal substrate selectivity, the toxins served for many years as diagnostic reagents for characterizing various G protein signaling pathways.

## Problem 9.15

Forskolin induces the formation of new synapses between rat hippocampal neurons growing in cell culture. The effect is inhibited by the active ingredient in marijuana ($\Delta^9$-tetrahydrocannabinol) which acts on another type of GPCR called the CB1 receptor. Which type of Gα subunit is coupled to the CB1 receptor: $G\alpha_i$, $G\alpha_q$, or $G\alpha_s$? How could you use an ADP-ribosylation toxin to confirm your answer?

## Adenylyl cyclase and phospholipase Cβ are the most common mediators of 7TM GPCRs

7TM GPCRs have a significant role in the function of all human cells, regardless of whether the cell is an vascular smooth muscle cell, a cardiac muscle cell, a neuron, or a white blood cell. If you understand how 7TM GPCRs modulate the activity of adenylyl cyclase (via $G\alpha_s$ or $G\alpha_i$) or phospholipase Cβ (via $G\alpha_q$), you understand how most cells in the human body respond to small-molecule ligands. In a few cell types, Gα subunits modulate the phosphodiesterase PDE6 or a small GTPase called Rho, but we will not allow ourselves to be distracted by these fascinating exceptions. For many 7TM GPCRs, no one has yet determined what kind of Gα subtype transduces the receptor signal. Much less is known about the Gβγ subunits. It is tempting to assume that Gβγ subunits serve as buffers that sequester Gα subunits, but the ability of Gβγ to activate GIRK1 potassium ion channels suggests that they have much more interesting roles.

The remainder of our discussion of 7TM-GPCR signaling will focus on the 7TM receptors and their ligands. Because the ligands are often small molecules, they offer the best opportunity to exploit our knowledge of chemistry at the level of atoms and bonds. Before proceeding, consider that all 7TM GPCRs have the capacity for fast response times, wide dynamic range, and high sensitivity. In all cases, signals from GPCRs are usually being integrated with signals from other GPCRs and even other types of receptors.

## Many pharmaceuticals act on 7TM GPCRs that respond to ligands derived from amino acids

Historically, 7TM GPCRs have been the most popular target for the development of pharmaceuticals. Even if nothing is known about the structure of the receptor, the endogenous ligand is usually a sensible lead for analog design. Drugs that target GPCRs do not have to enter cells; however, the pill-a-day paradigm for the development of

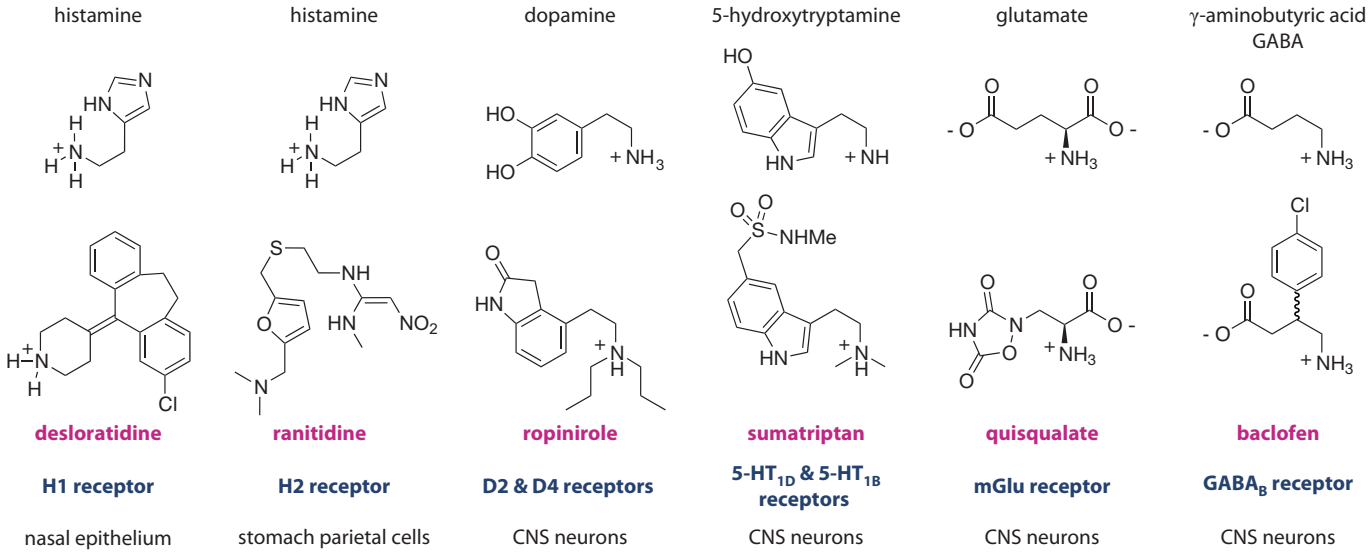



**Figure 9.84 7TM GPCR ligands and their drug analogs.** A variety of signaling molecules that act on 7TM GPCRs are derived from amino acids (upper). From the structures you cannot tell which drugs activate the GPCRs and which drugs inhibit them (lower).

blockbuster drugs requires one to develop a pharmaceutical that can enter the bloodstream by first crossing the gastrointestinal lumen. Such drugs require a low molecular mass and a balance between hydrophobic character and polarity. Neurotransmitters derived from amino acids are particularly amenable to analog synthesis. A large number of pharmaceuticals target neurotransmitter GPCRs.

Another significant challenge to targeting GPCRs is that one usually wants to target just a single subtype. For example, both stomach cells and cells in the nasal epithelium respond to histamine via GPCRs; however, the stomach cells express the H2 histamine receptor subtype, whereas the cells in the nasal epithelium express the H1 histamine receptor subtype. These two different GPCRs have different protein sequences and different ligand-binding sites. Thus, the H1 receptor antagonist desloratidine decreases the allergic response in the nasal epithelium (and other tissues) but not in the stomach. In contrast, the anti-ulcer drug ranitidine acts on H2 receptors in parietal cells in the stomach, preventing the release of acid. GPCRs also have an important role in neuronal signaling by neurotransmitters, including dopamine, 5-hydroxytryptamine, glutamate, and γ-aminobutyric acid (**Figure 9.84**).

The neurotransmitters dopamine, 5-hydroxytryptamine (5-HT; also known as serotonin), glutamate, and glycine also act on 7TM GPCRs within the central nervous system. Most of these receptors are located within synapses. Synapses usually transmit neuronal signals in one direction. One neuron releases neurotransmitters into the synaptic cleft and the other neuron has 7TM GPCRs for that neurotransmitter. For synapses in the brain, the signal is usually terminated by specialized proteins that pump the neurotransmitter back into the neuron that released it. Neuroscientists have developed some drugs that act on 7TM GPCRs, but the blockbuster drugs for depression and schizophrenia inhibit reuptake pumps rather than activating the receptors. If the reuptake pump is blocked, when a neuron releases a neurotransmitter the concentration of neurotransmitter within the synaptic cleft stays high for a longer period of time.

Structurally, dopamine is a phenethylamine, and many phenethylamine derivatives interfere with signaling at dopaminergic synapses (see Figure 9.84). Methamphetamine, sold legally as Adderall® but commonly known as "speed," is the archetype of these drugs, which include legal drugs such as Ritalin™ and drugs of abuse such as ecstasy. LSD (lysergic acid diethylamide) is the most notorious drug that interferes with 5-HT signaling, but many drugs for schizophrenia and migraines also target 5-HT receptors. A food additive, the amino acid glutamate (monosodium glutamate, or MSG) tastes like meat, and imparts a pleasant savory flavor, which coats the tongue with umami taste. Some individuals get headaches after ingesting large amounts of MSG, but because the compound is a common flavoring agent in foods, particularly Asian cuisine, restaurants sometimes advertise with signs that say "no MSG."

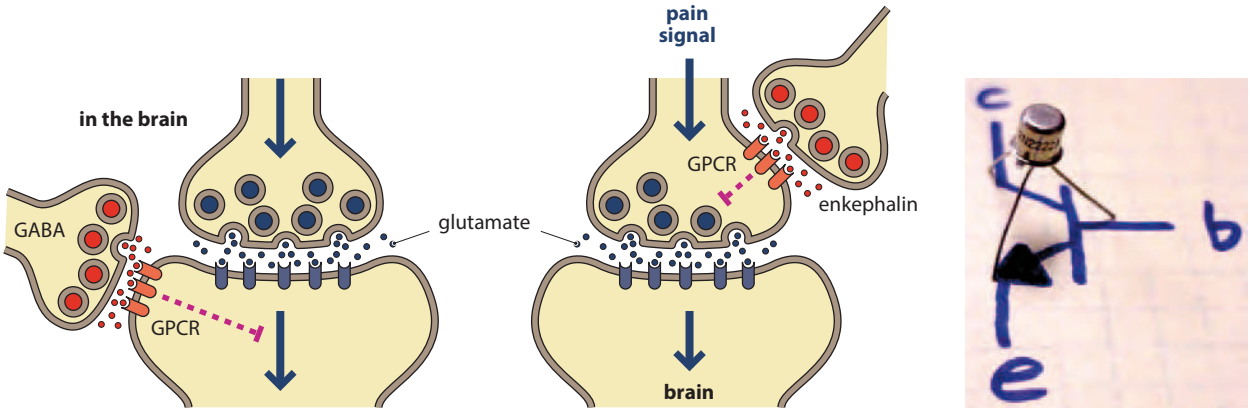



**Figure 9.85 A biological transistor.** Some inhibitory synapses involve GPCRs that modulate the strength of signals. GABA reduces the response of the postsynaptic neuron that receives the signal. The neuropeptide enkephalin reduces the response of the presynaptic neuron that sends the signal. These types of modulated neuronal connections resemble a transistor. (Right image, from Coltecnica, Ltda, 2012. http://www.coltecnica.com.)

Excitatory neurotransmitters (such as dopamine, 5-HT, or glutamate) are released at synapses, inducing an action potential in the next neuron. γ-Aminobutyric acid (GABA) is the only inhibitory amino acid neurotransmitter. When GABA is released at an inhibitory synapse, it acts on a 7TM GPCR called the $GABA_B$ receptor, inhibiting the effect of excitatory amino acids on that same neuron. By modulating neuronal signaling, GABA controls the flow of signals much like a transistor does (**Figure 9.85**). The GABA derivative baclofen also inhibits synaptic transmission and is used to diminish uncontrolled jerky movements (spasticity) (see Figure 9.84).

## Opioids act on 7TM GPCRs that bind to neuropeptides

Morphine is the most enduring example of a small molecule that is used to activate signal transduction pathways. The Sumerians described the collection of opium, which contains morphine, in cuneiform script on tablets dating back to 5000 BC. The immediacy and potency of the effect made it easy to assign a direct cause-and-effect relationship between the latex of the poppy and analgesia (**Figure 9.86**). Morphine acts on μ-opioid receptors, GPCRs that normally respond to neuropeptides called enkephalins. The analgesic properties of morphine are due to μ-opioid receptors found in sensory neurons in the spinal cord that lead to pain centers in the brain. Like GABA, opioids inhibit synaptic transmission by other neurons, but whereas GABA acts on the neuron that receives the signal, opioids act on neurons that send the signal. Opioids do not act through pathways involving transcriptional control. Enkephalins

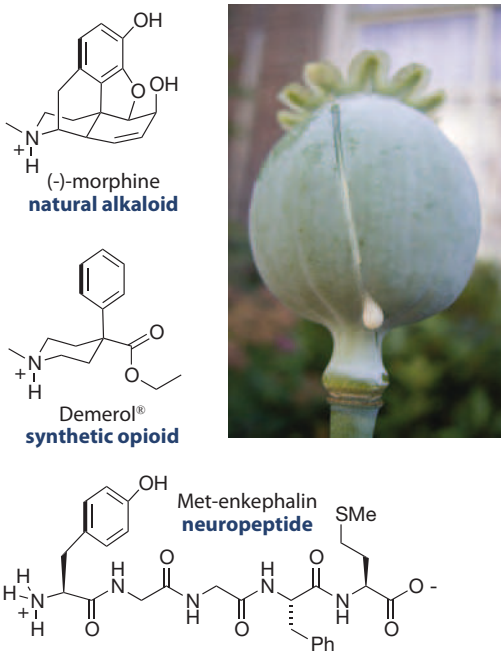
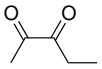
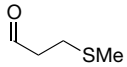



**Figure 9.86 Ligands for the μ-opioid receptor.** Opioids act upon the μ-opioid receptor. Morphine is present in the latex of the opium poppy, *Papaver somniferum*. Morphine and synthetic opioids such as Demerol mimic the effects of enkephalin. Met-enkephalin has a methionine at the C terminus; Leu-enkephalin has a leucine at the C terminus. (Image courtesy of Wikimedia.)

have a tyrosine residue at the N terminus of the peptide, and the aryl ring of nonpeptidic opioids mimics this tyrosine residue. A wide range of other peptides also act via 7TM GPCRs, for example secretin, vasoactive intestinal peptide, neuropeptide Y, and substance P.

## Smell and taste involve 7TM GPCRs

All of the human senses involve signal transduction through 7TM GPCRs. The aroma of cooked foods is usually associated with a large number of compounds. For example, the enticing aroma of freshly roasted peanuts can be attributed to at least 27 volatile compounds (**Figure 9.87**). The most significant contributor is methanethiol, not because it is most abundant but because we have odorant receptors that are extremely sensitive to thiols. Each of the 5 million olfactory neurons in the nasal epithelium encodes just one of the 900 different odorant receptors, all of which are 7TM GPCRs. If you inhale a pure compound, such as camphor, it binds tightly to some receptors and weakly to others. The olfactory bulb processes this pattern of signals and your brain interprets it as a single odor (**Figure 9.88**). This kind of multiplexed receptor mapping allows you to distinguish more than 10,000 different odors with only 900 odorant receptors.

| | conc. ($\mu g\ kg^{-1}$) | odor threshold ($\mu g\ kg^{-1}$) | odor activity value |
|---|---|---|---|
| —SH | 113 | 0.06 | 1889 |
|  | 83 | 0.3 | 286 |
|  | 40 | 0.2 | 200 |
| | 637 | 5.4 | 118 |
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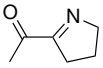
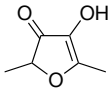
|  | 8.9 | 0.1 | 89 |
|  | 1953 | 25 | 78 |

**Figure 9.87 Peanuts!** The relative intensities of odors in fresh roasted peanuts are due to both the concentration of the odorants and our sensitivity to those odorants.

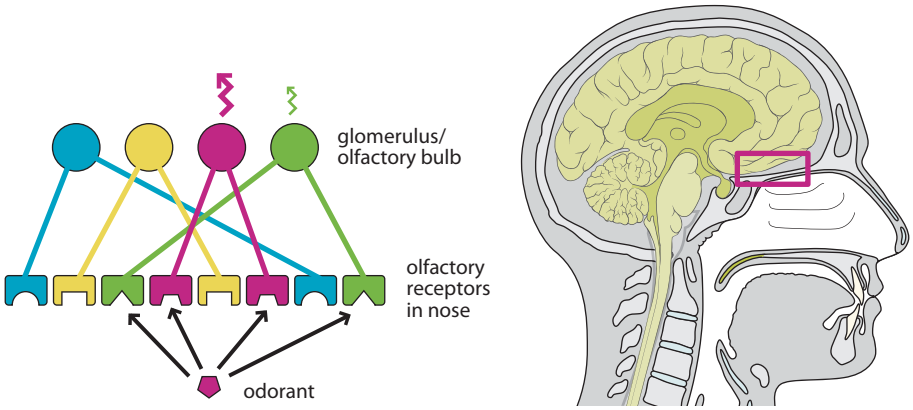



**Figure 9.88 Odorant receptor mapping.** You distinguish individual odors through spatial mapping of olfactory neurons in the nasal epithelium. Your brain decodes the pattern of neuronal signals as an odor. The olfactory region of the brain is highlighted. (Adapted from K. Touhara, *Microsc. Res. Tech.* 58:135–141, 2002. With permission from John Wiley & Sons.)

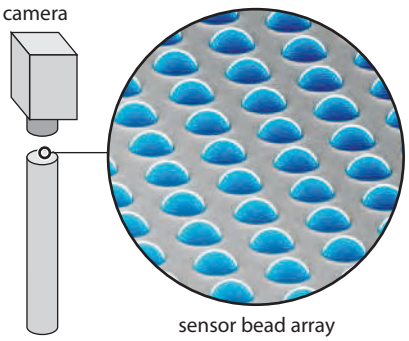
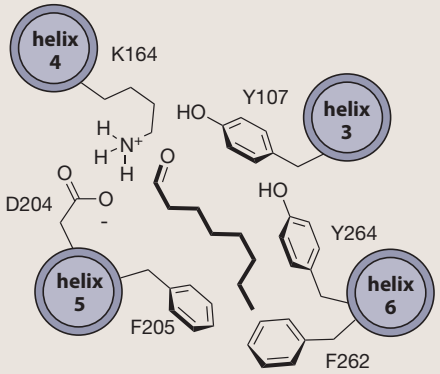



**Figure 9.89 Robo-smell.** An artificial nose can be made by spatially mapping the response of random sensor bead arrays on the ends of optical fiber bundles to various analytes, followed by computer decoding. (From M. Wadman, *Nature* 444:256, 2006. With permission from Macmillan Publishers Ltd.)

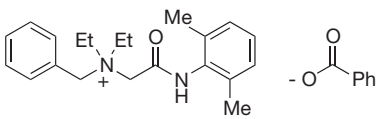
Multiplexed spatial mapping has been used to create an artificial nose capable of detecting a wide range of analytes. By randomly adsorbing a library of different beads, each coated with a different fluorescence-based sensor, on the ends of an optical fiber bundle, it is possible to individually monitor each of the beads in response to a pure analyte or a complex mixture of analytes. The beads can be derivatized with small molecules, peptides, oligonucleotides—anything. A computer can "memorize" the image of the pure analyte and then, using pattern recognition algorithms, look for the pattern produced by the analyte even when the array is exposed to a complex mixture. This technique works with proteins, DNA, and odor molecules (**Figure 9.89**).

## Problem 9.16

There is no crystal structure for an odorant receptor, but modeling studies suggest that the I7 odorant receptor tightly binds aliphatic aldehydes through hydrogen bonding with the ammonium ion on K164. Suggest an alternative mode of binding that would explain the high affinity of the ligand.




At one time, the ability to taste different flavors was thought to be associated with different regions of the tongue. However, the receptors for taste are not spatially segregated. The ability to distinguish tastes is based on a mixture of 7TM GPCRs and ligand-gated ion channels; however, the taste receptors in the human tongue do not have the level of diversity of the odorant receptors. In contrast with the large number of odorant receptors, humans have a relatively small repertoire of taste receptors, fewer than 50 7TM GPCRs for savory, sweet, and bitter tastes, and a few ion channels for salty and sour tastes. The sweet-tasting proteins thaumatin and monellin mentioned in Chapter 7 bind to the human taste receptor hT1R2 (human taste receptor, type 1, member 2). There are more than 25 members of the human taste receptor hT2R family that respond to bitter flavors. The ammonium ion denatonium binds to the hT2R44 receptor and is the bitterest compound known to man (**Figure 9.90**). Denatonium benzoate is added to alcohol to denature it, and is used to make household chemicals unpalatable to young children. The bitter properties of denatonium were discovered in 1958 by accident, but chemists are now building tasty molecules on purpose. The same techniques of cell biology, medicinal chemistry, and pharmacology that were once reserved for medicines are now being applied to the development of safe, selective, super-potent flavoring molecules.




denatonium benzoate
Bitrex®

**Figure 9.90 Super bitter.** Denatonium benzoate is the most bitter known compound, binding potently to the hT2R44 receptor. When added to household chemicals, it deters curious toddlers from ingesting them.

## How do you bind to a photon?

The human eye can sense as few as five photons. This remarkable sensitivity for photons relies on a 7TM GPCR and is based on the photochemical isomerization of an olefin from *cis* to *trans*. Specifically, the 7TM rhodopsin responds to the isomerization of 11-*cis*-retinal to 11-*trans*-retinal (**Figure 9.91**). The aldehyde retinal is covalently attached as an iminium ion to Lys296 on rhodopsin. The photon-catalyzed change in double-bond configuration changes the shape of the retinal ligand drastically; as far as the receptor is concerned, the binding site thus undergoes a shift from an unbound state to a bound state.

Most studies on the molecular basis of vision have been conducted with bacteriorhodopsin from the bacterium *Halobacterium halobium*. Bacteriorhodopsin differs from mammalian rhodopsin in two important ways. In bacteriorhodopsin, retinal is

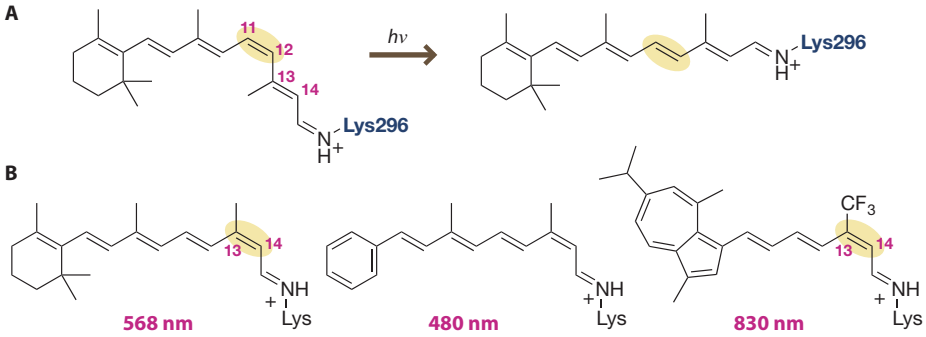



**Figure 9.91 Retinal and its analogs.** (A) In mammalian rhodopsin, the photoisomerization of 11-*cis*-retinal iminium ion into the 11-*trans* isomer drives human vision and leads to the same effect as binding a new ligand. (B) Chemical analogs of retinal in bacteriorhodopsin allow bacteria to "see" at different wavelengths.

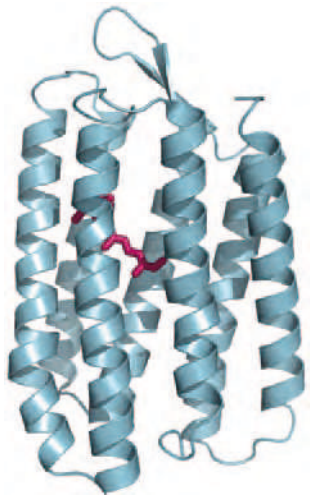
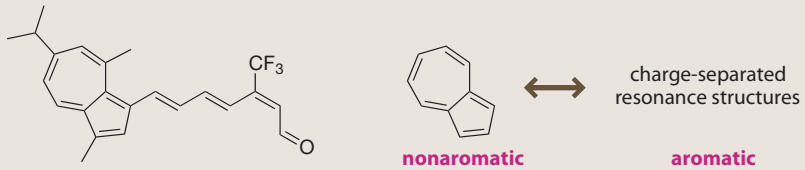


**Figure 9.92 Bacteriorhodopsin.** The 7TM protein bacteriorhodopsin (blue) opens a proton channel in response to light, causing the *cis–trans* isomerization of retinal (red). (PDB 1C3W)

bound as the 13-*cis* isomer instead of the 11-*cis* isomer of mammalian rhodopsin. Furthermore, bacteriorhodopsin is a light-driven proton pump, not a G protein-coupled receptor. For many years, the crystal structure of bacteriorhodopsin was the only precise model for a 7TM receptor that was available to chemical biologists (**Figure 9.92**). Chemical analogs of retinal have been synthesized and incorporated into bacteriorhodopsin, leading to receptors with altered sensitivity to light of different wavelengths (see Figure 9.91).

## Problem 9.17

The azulene ring system of the retinoid analog below interacts favorably with cation-stabilizing residues in bacteriorhodopsin. The neutral resonance structure for azulene is nonaromatic as a result of cross-conjugation. Draw six charge-separated aromatic resonance structures for azulene, and determine which ring bears the positive charge.




## The decision between immortality and destiny involves the protein Wnt and the β-catenin pathway

In the classic children's story *The 500 Hats of Bartholomew Cubbins*, a young boy finds that each time he removes his feathered hat a new one appears in its place. After the 450th hat, each of the following hats becomes more ornate than the previous one. The process is exhausted with the 500th and most splendorous hat, leaving him bareheaded (**Figure 9.93**). Human stem cells follow a similar trend. They have unlimited capacity for replication, but once they have received the right set of signals they begin to differentiate. Differentiation pushes the cell toward a useful function at a significant cost—loss of immortality. The inevitable fate of all fully differentiated cells (for example cells found in the leg muscle, heart, kidney, and brain) is death, because such cells have a limited capacity for mitotic cell division. Thus, each mitotic event involves a weighty decision to produce identical daughter cells or differentiated daughter cells. The Wnt pathway has a key role in that choice. The choice of immortality is made by cancer cells; the choice to fulfill a mortal destiny is made by cells in developing embryos.

**Figure 9.93 Regenerative hats.** Developmental biologists hope to control proliferation and differentiation, much like Bartholomew Cubbins in the classic work by Dr. Seuss. (Courtesy of Dr. Seuss Enterprises, L.P. With permission from Random House, Inc.)

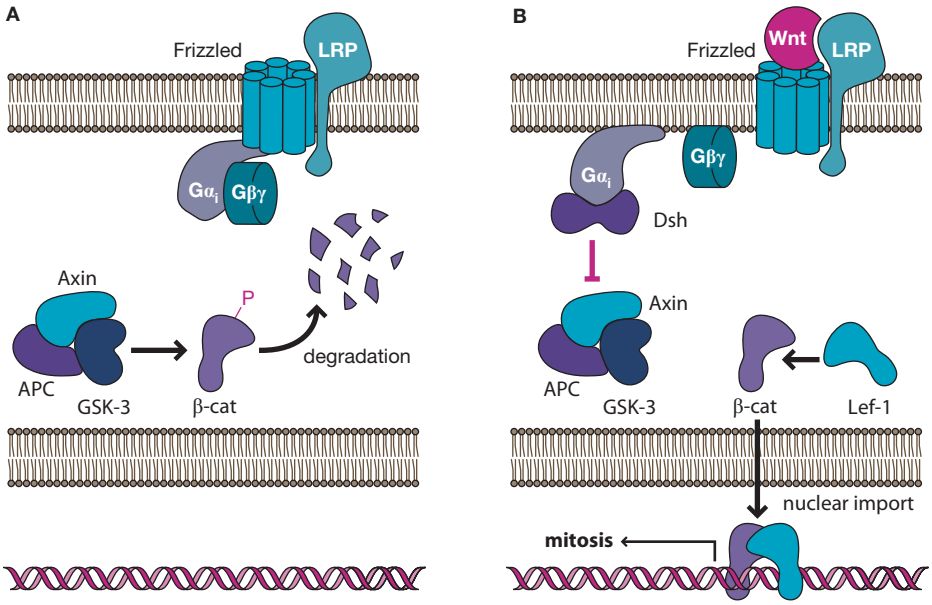



**Figure 9.94 Control through degradation.** In the presence of Wnt, the transcription factor β-catenin (β-cat) is continuously degraded (A); however, in the presence of Wnt, Dishevelled (Dsh) prevents the degradation of β-catenin, allowing it to complex with Lef-1, turning on mitotic genes (B).

To understand the effect of Wnt signaling on cells, it is essential to understand the basal metabolism of the transcription factor β-catenin. In the absence of Wnt, β-catenin is continuously marked for proteolytic degradation by phosphorylation. The phosphorylation of β-catenin is performed by a trimeric complex composed of axin, APC, and glycogen synthase kinase 3 (**Figure 9.94A**). In the presence of Wnt, the trimeric kinase complex is inhibited, and concentrations of β-catenin increase until it affects transcription by forming a complex with Lef-1 (Figure 9.94B). In colon cancers, the β-catenin–Lef-1 complex turns on the transcription of genes such as those encoding c-Myc and cyclin D1, both related to mitosis. How does Wnt inhibit the trimeric kinase complex? The signaling protein Wnt binds to a 7TM GPCR called Frizzled (instead of Wnt receptor) which causes the protein Dishevelled to inhibit the trimeric kinase complex. An additional protein, LRP, is required for activation, and seems to serve a role as a co-receptor. Thus, Wnt signaling prevents the degradation of β-catenin, allowing it to affect transcription. The intermediacy of a heterotrimeric G protein in the activation of Dishevelled was long suspected but only recently demonstrated; the mechanism by which the G protein affects Dishevelled has not been fully elucidated.

## A seven-transmembrane receptor that controls development does not bind to an extracellular ligand

The cyclops was a frightening figure in Greek and Roman mythology, with just one eye positioned in the middle of its head. The inspiration for the cyclops may come from cyclopia, which is a specific form of a rare birth defect called holoprosencephaly. The first clues to the molecular origin of holoprosencephaly came from Idaho sheep ranchers who noted sheep born with cyclopia (**Figure 9.95**). The congenital deformities were later linked to the consumption of *Veratrum californicum*, also known as the California corn lily or California false hellebore. The magical properties of medicinal hellebores, easily confused with false hellebores, date back to ancient times. *Veratrum californicum* produces a group of steroidal alkaloids such as cyclopamine, with potent teratogenic (causing birth defects) activity. Cyclopamine does not kill cells indiscriminately; it interferes with a specific GPCR that controls cell differentiation during fetal development. Steroidal ligands have an important role in this cell differentiation pathway.

**Figure 9.95 Cyclops.** This lamb was born to a ewe that had fed on California false hellebore, which contains the alkaloid cyclopamine. (From R.F. Keeler, *Lipids* 13:708–715, 1978. With permission from Springer.)

In *Drosophila*, mutations to a gene named *Hedgehog* result in larvae with a profusion of spiky tentacles. At a molecular level, *Veratrum* alkaloids such as cyclopamine exert teratogenic effects similar to the *Hedgehog* mutations in fruit flies. The Hedgehog

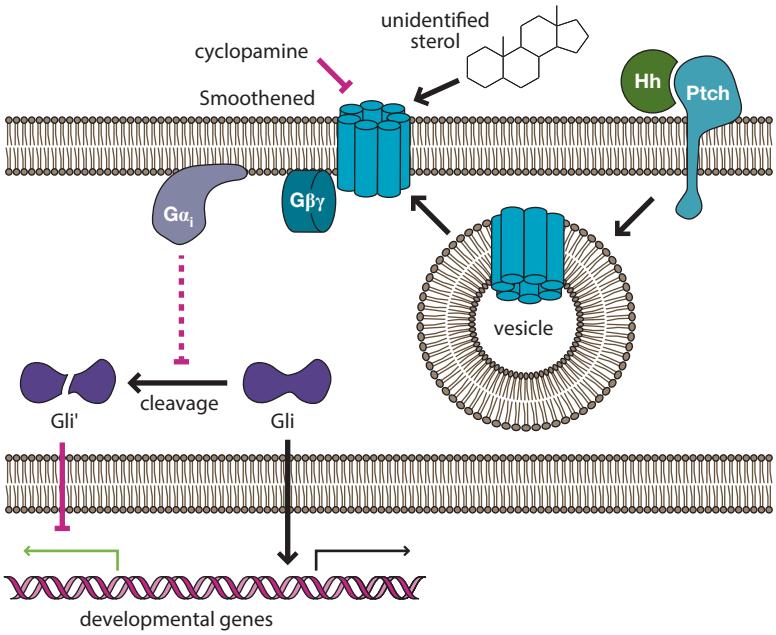



**Figure 9.96 Cyclopamine repression of genes required for normal development.** Gli can affect two different sets of genes required for normal development. Full-length Gli activates one set of genes. Truncated Gli′ represses another set of genes. When Smoothened is activated, one set of developmental genes is activated and another is not repressed. Hh, Hedgehog; Ptch, Patched.

protein is modified post-translationally with a cholesterol molecule, and then packaged into lipoprotein complexes for transport around the developing organism. When Hedgehog binds to the receptor Patched, it allows the GPCR Smoothened to move from endocytic vesicles to the cell membrane (**Figure 9.96**). Once in the cell membrane, Smoothened acts as a receptor for some kind of steroidal alcohol (sterol) that has not yet been identified; cyclopamine binds to Smoothened, which ultimately prevents the cleavage of a transcription factor called Gli. At the time of writing of this book, many details of this pathway are incomplete. Full-length Gli activates the transcription of genes for differentiation. In the absence of Hedgehog, Gli is cleaved and the truncated form of Gli acts as a repressor of gene transcription.

There are vague similarities between the β-catenin pathway and the Hedgehog signaling pathway. Activation of the 7TM receptors Frizzled and Smoothened requires proteins other than heterotrimeric G proteins. Both pathways involve the constitutive proteolysis of a transcription factor in the absence of ligand. Both pathways affect the differentiation of cells.

## 9.7 ION CHANNEL RECEPTORS

### Ion channel receptors provide an ultra-fast response to stimuli

Human cells maintain an imbalance of ion concentrations between the inside and the outside of the cell. At a moment's notice, ion channels are poised to open in rapid succession, allowing a flood of ions to permeate the cytoplasm. This type of ionic flooding is the fastest mechanism for communicating an event to all parts of the cell. Neuronal signaling is the most significant physiological event that involves ion channel receptors. As you read the words on this page, hundreds of millions of ion channels are opening and closing. If you choose to memorize the ion concentrations in **Table 9.4** to the right, you will probably have to look at the data several times … ruminate on it and savor it. Repetitive neuronal signaling will eventually change the neurons in those pathways at a transcriptional level.

Neurons come in many shapes and sizes, but a common morphological theme is that they have a cell body that receives chemical signals, and long extensions called axons that transmit the cellular signal toward a synaptic junction (**Figure 9.97**). The axons in the nerves in your arms and legs can be up to 1 meter in length. However, in communicating the signal from one end of the cell to the other, no molecules travel

**Table 9.4 Approximate ion concentrations inside and outside human cells.**

| Ion | Concentration (mM) | |
|---|---|---|
| | Inside | Outside |
| $Na^+$ | 10 | 145 |
| $K^+$ | 140 | 5 |
| $Cl^-$ | 10 | 110 |
| $HCO_3^-$ | 10 | 25 |
| $H_2PO_4^-$ | – | 1 |
| $Mg^{2+}$ | 0.5 | 2 |
| $Ca^{2+}$ | 0 | 2 |

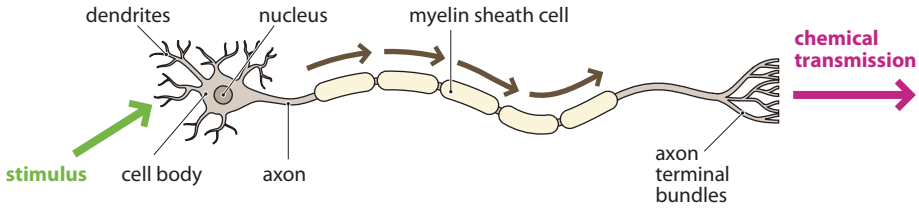



**Figure 9.97 Anatomy of a neuron.** Dendrites at the cell body receive signals that are propagated along the axons.

the length of the cell. Diffusion across such distances would be impossibly slow for a simple action such as catching a baseball.

Ion channels in the cell membrane can be activated by ligands or transmembrane ion potentials, and are called ligand-gated ion channels and voltage-gated ion channels respectively. There are three important classes of ligand-gated ion channels based on structure: Cys-loop receptors, ionotropic glutamate receptors, and ATP-gated channels. Voltage-gated ion channels are not activated by extracellular ligands, but grouping ligand-gated and voltage-gated ion channels into one section will help us to organize the information better.

## A human cell is a bag of potassium in a salty ocean

Human serum, the liquid that surrounds the cells in whole blood, is a reasonable approximation to liquid that surrounds the other cells in the human body (see Table 9.4). Serum has many of the same ions as seawater, with about one-third of the salinity (sodium chloride concentration). In contrast, the cytoplasm of human cells has very little sodium, very little chloride, and a relatively high concentration of potassium. Experiments meant to mimic the extracellular environment are performed in sodium chloride buffers such as phosphate-buffered saline (PBS). Experiments meant to mimic intracellular conditions should be performed in low-saline, potassium-rich buffers.

To maintain this ionic imbalance between the inside of the cell and the outside of the cell, energy-driven pumps pull $K^+$ ions in, and drive $Na^+$ and $Cl^-$ ions out (**Figure 9.98**). Hydrolysis of ATP to ADP provides the energy for $Na^+/K^+$ pumps. Recall that contraction of muscle cells involves a burst of cytosolic $Ca^{2+}$ ions. Pumps also actively pump $Ca^{2+}$ ions out of the cell and/or into cellular compartments such as the endoplasmic reticulum (or the sarcoplasmic reticulum of muscle cells). In these compartments, the concentration of $Ca^{2+}$ ions (about 0.1 mM) is 1000 times the concentration in the cytosol. Various stimuli can open calcium channels in calcium vesicles and in the plasma membrane, leading to a rapid influx of $Ca^{2+}$ that triggers the release of neurotransmitters, for example.

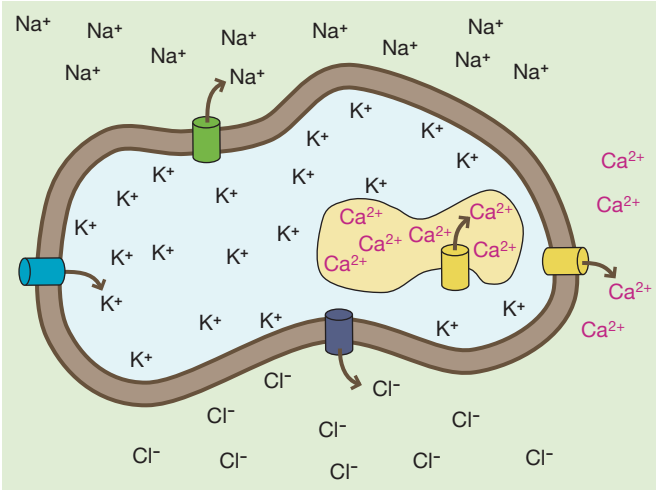



**Figure 9.98 Ions in, ions out.** Pumps maintain the concentration of anions and cations inside the cell. The imbalance of ions creates a tension that can be rapidly relieved through the opening of membrane ion channels (not shown).

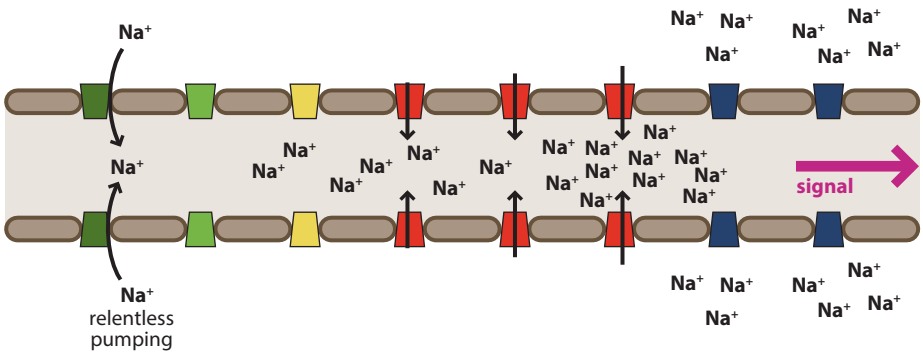



**Figure 9.99 Firing the neuron.** Sodium ion channels open in response to intracellular $Na^+$ ions, but only for a brief period. Energy-driven pumps (left) along the length of the axon then expel the $Na^+$ ions from the axon.

We measure the differences in ion concentration using the same units used by electricians to measure an imbalance in electrons—volts—but try not to confuse the chemical potential created by an imbalance of cations with the electrician's concept of an electromagnetic potential.

## Voltage-gated ion channels are activated by transmembrane differences in ion concentrations

How does a signal travel along an axon? Energy-driven export pumps relentlessly pump $Na^+$ ions out of the axon. As a result of this concentration difference, $Na^+$ ions yearn to enter the axon from outside the cell. At intervals along the axon, voltage-gated sodium channels are poised to open and allow a rush of $Na^+$ ions into the axon. Ironically, the opening of these sodium channels is initiated by a high concentration of intracellular $Na^+$. Thus, after one channel allows $Na^+$ ions to enter, neighboring sodium channels follow suit. The opening of these voltage-gated sodium channels is temporary, and they rapidly close. Once the voltage-gated sodium channels close, the intracellular concentration of $Na^+$ begins to decrease, as the energy-driven $Na^+$ pumps continue their relentless expulsion of $Na^+$ ions out of the cell. The region of high intracellular $Na^+$ concentration travels along the axon at impressive rates, up to 1 m $s^{-1}$, even though each individual $Na^+$ ion travels only a microscopic distance into the cell and then back out again (**Figure 9.99**).

At the end of the axon, voltage-gated calcium channels allow a wave of $Ca^{2+}$ ions to enter, causing the release of neurotransmitters into a slender gap between neurons called the synapse. Grayanotoxin, found in rhododendrons, is an inhibitor of voltage-gated sodium channels, keeping them in an open state. Significant amounts of grayanotoxin can accumulate in the honey of bees that feed on rhododendron nectar (**Figure 9.100**). Eating honey contaminated with grayanotoxin leads to a variety of symptoms, including altered mental states. Xenophon, an Athenian author and military commander, first described the phenomenon of "mad honey poisoning" in 401 BC.

### Problem 9.18

Fura-2 is non-fluorescent in the absence of $Ca^{2+}$ because a key nitrogen lone pair quenches the fluorescence through resonance. However, fura-2 tightly grips $Ca^{2+}$ ions as an octadentate ligand, and in so doing becomes highly fluorescent. Suggest a plausible structure for the fura-2•$Ca^{2+}$ complex.

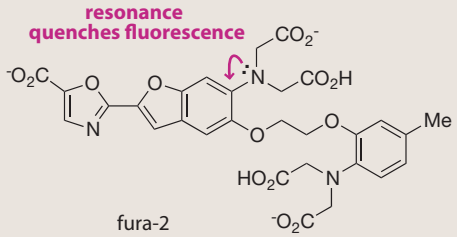


**Figure 9.100 Dangerous honey.** Honey produced by bees feeding on rhododendrons can become contaminated with grayanotoxin 1, an inhibitor of voltage-gated sodium channels in humans. (Courtesy of W.P. Coombs, Jr., Western New England University.)

Voltage-gated ion channels control processes other than neuronal firing. In muscle cells, voltage-gated calcium channels on the membranes of storage vesicles contribute to the burst of $Ca^{2+}$ initiated by inositol triphosphate. These voltage-gated ion channels are usually classified on the basis of the types of non-native small molecules

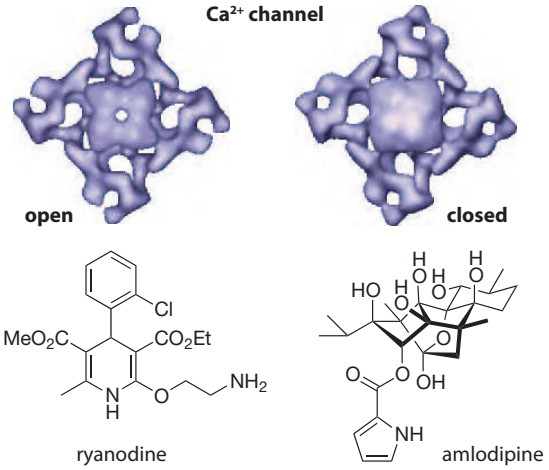



**Figure 9.101 Calcium channel states and their ligands.** Ryanodine binds to the open state of calcium channels. Amlodipine binds to the closed state of calcium channels. Ryanodine and amlodipine don't bind to the same type of calcium channel. (Top, from E.V. Orlova et al., *Nat. Struct. Biol.* 3:547–552, 1996. With permission from Macmillan Publishers Ltd.)

that can be used by pharmacologists to open them. Ryanodine receptors, are ion channels which allow $Ca^{2+}$ to escape from intracellular storage vesicles, and are named for their response to the plant alkaloid ryanodine (**Figure 9.101**). Under normal circumstances, the ryanodine receptors open in response to $Ca^{2+}$. Dihydropyridine receptors are so named because they respond to synthetic dihydropyridine drugs like the blockbuster amlodipine. Channels with similar functions account for the monogamous relationship between egg and sperm. Once a spermatozoon penetrates the exterior of the ovum, $IP_3$ initiates a wave of $Ca^{2+}$ ions that, with blinding speed, makes the egg impenetrable to all other suitors.

Voltage-gated potassium channels also have a role in cell signaling. For example, immunological T cells have voltage-gated potassium channels. Forty out of the 78 human genes for potassium channels are voltage-gated. A variety of protein toxins, from scorpions, sea anemones, snakes, and cone snails, block voltage-gated potassium channels, often by plugging the $K^+$ exit site by inserting a cationic lysine side chain into the narrow orifice of the channel (**Figure 9.102**). One such toxin, charybdotoxin from the death stalker scorpion, blocks $Ca^{2+}$-activated potassium channels in aortic smooth muscle. $Ca^{2+}$-activated potassium channels are, in a functional sense, a hybrid between voltage-gated ion channels and ligand-gated ion channels because the interactions with $Ca^{2+}$ are highly specific (see Figure 9.102). Another toxin that targets the $Ca^{2+}$-activated potassium channel is lolitrem B, a small molecule from the fungus *Epichloë*. Lolitrem B is a potent inhibitor ($IC_{50}$ = 4 nM) of the $Ca^{2+}$-activated potassium channel that is present on the presynaptic membrane of Purkinje neurons.

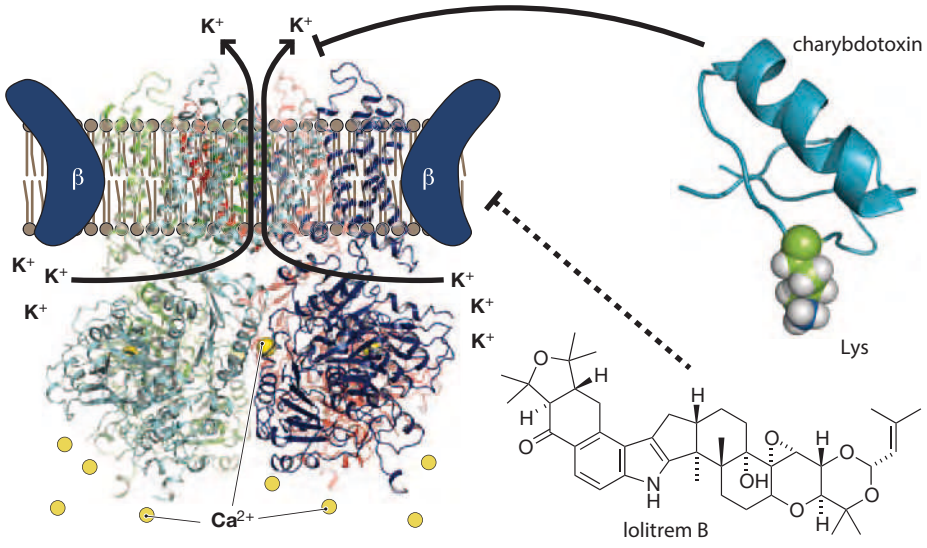



**Figure 9.102 Control over the calcium-activated potassium channel.** The binding of $Ca^{2+}$ ions to the calcium-activated potassium channel allows $K^+$ ions to leak out of cells. The effect of $Ca^{2+}$ is modulated by β subunits. The channels are inhibited by both protein toxins, such as charybdotoxin (PDB 2A9H), and small molecules, such as lolitrem B. Lolitrem B acts on the modulatory β subunits. (Adapted from P. Yuan et al., *Science* 329:182–186, 2010.)

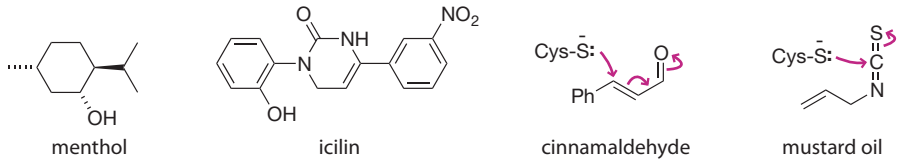


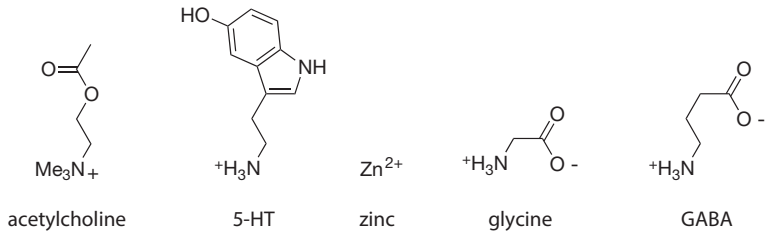

**Figure 9.103 Compounds that activate the TRPA1 cation channel.** Cinnamaldehyde and mustard oil are electrophiles, which covalently modify TRP channels for long-lasting effects.

Hapless sheep that feed on ryegrass infected with *Epichloë* develop a neurological condition known as ryegrass staggers, characterized by tremors and a lack of coordination. The action of lolitrem B is dependent on a regulatory protein that modulates the activity of $Ca^{2+}$-activated potassium channels.

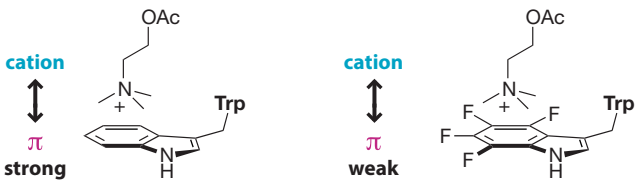
**Transient receptor potential (TRP) cation channels** are stimulated by cold, and directly activate neurons in pain pathways. Menthol, for example, creates a cooling sensation by activating TRP channels. The synthetic compound icilin is 200-fold more potent in producing these effects. Surprisingly, electrophiles such as enones, isothiocyanates, α-iodoacetamide, and thiosulfonates act at the TRP channel TRPA1. These compounds covalently modify cysteines on the channel, leading to persistent activation (**Figure 9.103**).

## Pentameric Cys-loop receptors are gated by neurotransmitters

Cys-loop receptors are ligand-gated ion channels that have a common structure based on five protein subunits. These veritable skyscrapers extend far above and below the cell membrane (**Figure 9.104**). The excitatory neurotransmitter 5-HT acts on both 7TM GPCRs and on a ligand-gated cation channel called the 5-HT3 receptor (**Figure 9.105**). GABA modulates synaptic transmission by acting on the 7TM GPCR called the $GABA_B$ receptor. However, in other neurons it acts on a ligand-gated chloride channel called the $GABA_A$ receptor.




The binding site for the nicotinic acetylcholine receptor has no anionic side chains. Instead, acetylcholine binds to an active site composed of three tyrosine side chains and two tryptophan side chains. One, and only one, of these aromatic groups is important for the ability of acetylcholine to open the channel. The importance of a cation–π interaction between the quaternary ammonium ion and tryptophan was demonstrated through the creation and assay of a series of fluorotryptophan mutants by loading tryptophan analogs onto suppressor tRNAs (Chapter 4). By this method a series of fluorinated tryptophans were synthesized and attached to mimics of tRNA that recognize the rare amber stop codon, UAG. The amber stop codon TAG was easily introduced into the muscle-type nicotinic acetylcholine receptor gene through oligonucleotide-directed mutagenesis. Fluorine substituents on the key tryptophan residue were found to strongly decrease the ability of acetylcholine to open the channel (**Figure 9.106**). Such electron-withdrawing functionalities can decrease the strength



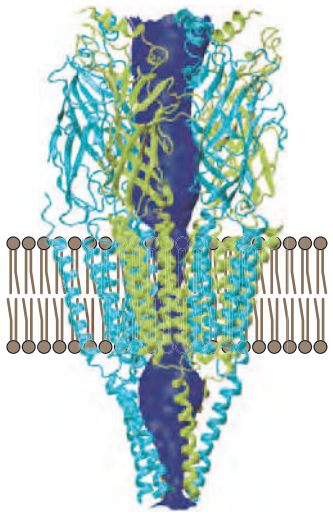



**Figure 9.104 Pentameric nicotinic acetylcholine receptor.** The five subunits of the nicotinic acetylcholine receptor are rendered in cyan and green. The cation channel is rendered in blue. (PDB 2BG9) (Adapted from S.M. Sine and A.G. Engel, *Nature* 440:448–455, 2006. With permission from Macmillan Publishers Ltd.)

**Figure 9.105 Examples of ligands that act at Cys-loop receptor ion channels.**

**Figure 9.106 Exploring the biological implications of a cation–π interaction with synthetic chemistry and mutagenesis.** A cation–π interaction is important in the binding of acetylcholine, as shown by much weaker binding after replacement in the nicotinic acetylcholine receptor with a fluorinated tryptophan. The fluorinated tryptophan substitution, however, has only a weak effect on nicotine binding.

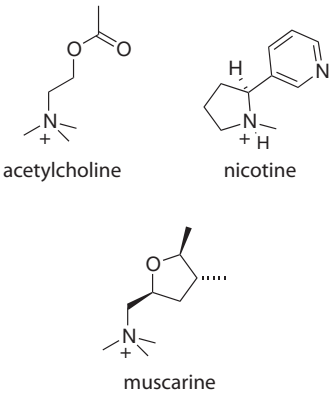



**Figure 9.107 Some of the common ligands for acetylcholine receptors.**

of the cation–π interaction between the aromatic ring of the tryptophan indole group and the quaternary amine of acetylcholine. Surprisingly, the effect of the introduced fluorines on the agonist ability of nicotine was small, suggesting that nicotine might interact with the active site very differently from acetylcholine.

## The nicotinic acetylcholine receptor is a popular target for toxins

Before the ready availability of sequencing methods, pharmacologists used non-natural ligand susceptibility to distinguish between various subtypes of receptors for neurotransmitters (analogous to the naming of the ryanodine-sensitive voltage-gated ion channels described above). Acetylcholine receptors can be distinguished by their sensitivity to nicotine or muscarine (**Figure 9.107**). Nicotinic acetylcholine receptors are found between neurons in the autonomic ganglia and at the neuromuscular junction; such receptors control primitive responses such as "fight or flight" and "rest and digest." The nicotinic acetylcholine receptor has two binding sites for the neurotransmitter acetylcholine and a wide range of natural toxins target these binding sites. For example, curare, the poison used by Amazonian hunters to coat the tips of hunting darts, targets the acetylcholine-binding site. A variety of animal venoms contain proteins that jam the nicotinic acetylcholine receptor into its open conformation (**Figure 9.108**): erabutoxin (sea snake), α-bungarotoxin (krait, *Bungarus caeruleus*), α-conotoxin (*Conus* snails), and α-cobratoxin (cobra, *Naja naja kaouthia)*.

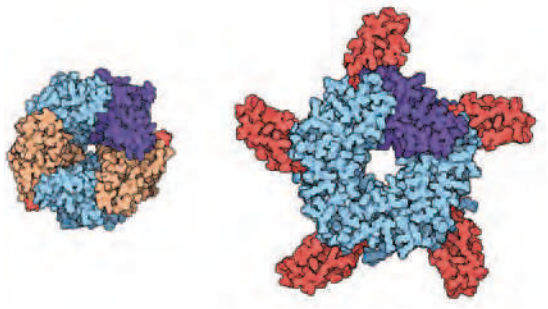


**Figure 9.108 Open wide.** These structures of the nicotinic acetylcholine receptor depict the closed form (left) with acetylcholine-binding sites in red, and the open form (right) bound to five molecules of α-cobratoxin. (Courtesy of David S. Goodsell and the RCSB Protein Database.)

Many of the natural product ligands that target the acetylcholine receptor have a common pyrrolidine ring (**Figure 9.109**). The blue-green algae *Anabaena flos-aquae* produces a toxic substance that irreversibly binds to the nicotinic acetylcholine receptor. The substance was initially referred to as "very fast death factor" but was later named anatoxin A. Surprisingly, other toxins that act at nicotinic receptors have potentially beneficial applications. For example, the compound epibatidine, isolated from the Ecuadoran frog *Epipedobates tricolor*, is a potent analgesic, 200 times more powerful than morphine in alleviating pain. Sadly, when *E. tricolor* is raised in the laboratory, the frogs no longer produce epibatidine. The production of epibatidine is a consequence of the frog's diet or a symbiotic relationship with a microorganism unique to its native habitat.

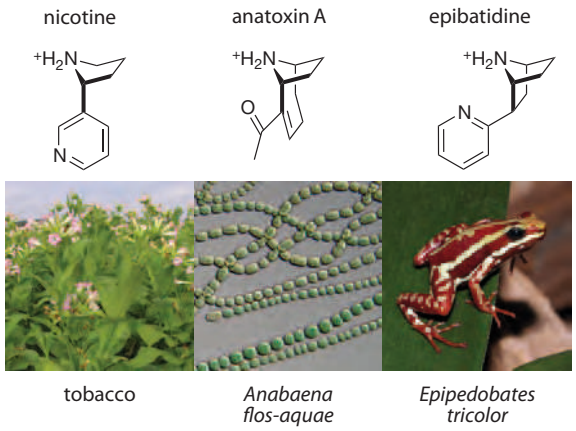



**Figure 9.109 Nicotine and structural variants.** Natural products that can target the nicotinic acetylcholine receptor share common structural features. The source of each compound is shown below its structure. (Bottom left, courtesy of Joachim Mullerchen, Wikimedia; middle, courtesy of the Cultural Collection of Autotrophic Organisms; right, courtesy of H. Krisp, Wikimedia.)

### Tetrameric glutamate receptors are defined by their specificity for glutamate analogs

You may recall that glutamic acid is a ligand for some 7TM GPCRs, imparting umami flavor to food. More commonly, glutamic acid is a ligand for ion channel receptors. Twenty different subtypes of glutamate ion channel receptors are known, and each has four protein subunits. The major classes of glutamate receptors are categorized on the basis of their susceptibility to the glutamate analogs *N*-methyl-D-aspartate (NMDA), α-amino-3-hydroxy-5-methyl-4-isoxazole propionic acid (AMPA), and kainate (**Figure 9.110**). The glutamate receptors include AMPA, kainate, and NMDA receptors. Such receptors all respond to the excitatory amino acids glutamic acid and aspartic acid. The popular flavoring agent MSG (monosodium glutamate) is actually a pure, crystalline form of the neurotransmitter glutamate.

Most of the computing power of the brain is controlled by excitatory amino acid receptor pathways. Activation of ligand-gated ion channels does not always lead to changes in transcription. Often, such activation merely induces a fast response like neuronal depolarization. However, many processes that result in transcriptional activation, such as neuronal changes associated with memory, are initiated by ligand-gated ion channels. Repeated stimulation of NMDA receptors (by glutamate) in hippocampal neurons leads to $Ca^{2+}$ influx. In these neurons, $Ca^{2+}$ ions lead to the activation of protein kinase C (PKC) and CaMKII. These kinases affect neuronal function in two ways. First, PKC phosphorylates the AMPA receptors, making them more sensitive to glutamate. Second, the kinases PKC and CaMKII eventually activate the ERK pathway in these neurons, leading to changes in gene expression.

Kainic acid is named after the red algae *Digenea*, referred to as kaininso in Japan. *Digenea* has been used for centuries as an antihelminthic, or treatment for parasitic worms. Recently, a structurally related natural product, domoic acid, was discovered as the active agent in amnesic shellfish poisoning (ASP) (see Figure 9.110). Domoic acid is produced not by shellfish but by the diatom *Pseudonitzschia*, which can be accumulated by shellfish. Domoic acid activates AMPA and kainate receptors, leading to persistent $Ca^{2+}$ influx. In serious cases, domoic acid poisoning results in the permanent loss of short-term memory.

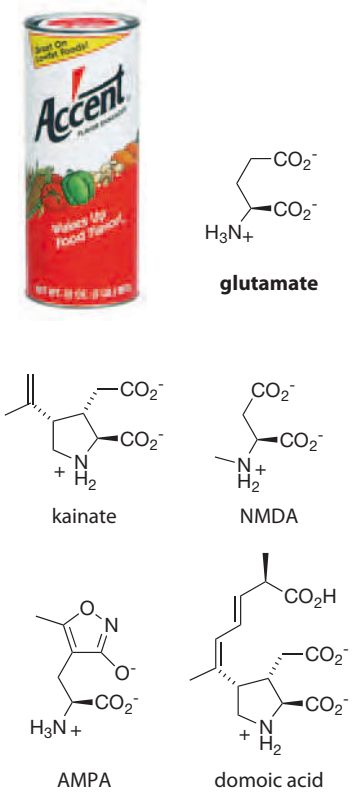
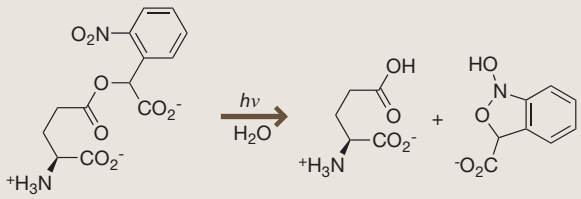



**Figure 9.110 Glutamate and structural variants.** Glutamate is a popular flavoring agent. A variety of nonendogenous ligands are selective for different subtypes of glutamate ion channel receptors. (Top, courtesy of B&G Foods, Inc.)

**Problem 9.19**

The following "caged" glutamate derivative can be induced to release glutamate on activation with light. Suggest a plausible arrow-pushing mechanism for the reaction.




## 9.8 TRIMERIC DEATH RECEPTORS

### Tumor necrosis factor binding to TNF receptors triggers diverse, cell-dependent responses

Recall from Chapter 3 that rapidly dividing cells are poised to self-destruct if they fail to pass through various mitotic checkpoints. Such apoptotic pathways are mediated by the tetrameric transcription factor p53. However, even non-mitotic cells can be induced to self-destruct on receipt of the correct signals. Such signals are mainly mediated by a group of homotrimeric receptors that bind to homotrimeric ligands, either expressed on cell surfaces or in soluble forms. This three-fold symmetry of

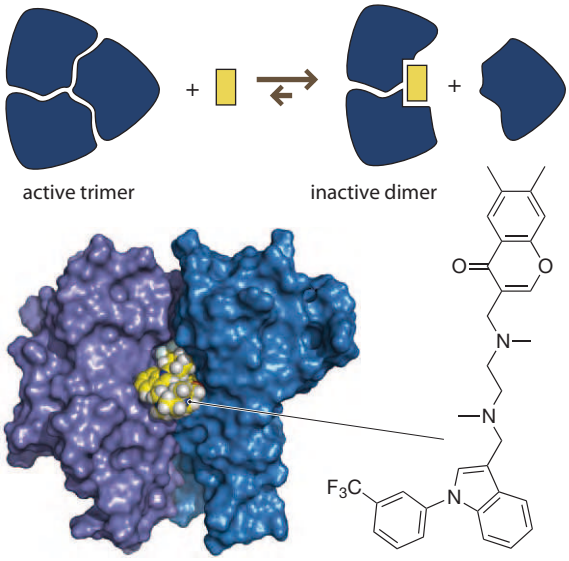



**Figure 9.111 Blocking a protein–protein interaction.** A combination of screening and design led to a small molecule that can bind to a dimer of TNFα, preventing the formation of the symmetrical TNFα trimer. (PDB 2AZ5)

**trimeric death receptors** is a distinctive feature of these ligands and receptors. The FasL-promoted protease cascade discussed in Chapter 6 is an example of such a pathway. A second important pathway leading to apoptosis is mediated by the ligand tumor necrosis factor (TNF). Of the two TNF isoforms, TNFα and TNFβ, TNFα has more potent effects.

Short bursts of TNFα induce acute-phase inflammatory responses in many cells, but chronic administration leads to nonspecific cytotoxicity. The **therapeutic index** (the window between beneficial and harmful doses) is so narrow that TNFα has not found use against human tumors. However, TNFα is highly cytotoxic toward many tumor cell lines grown in the laboratory; the cytokine does not operate via the mitotic checkpoints and p53-dependent pathways that we discussed in Chapter 3. From a clinical viewpoint, the most significant role of TNFα is the undesired proinflammatory effects seen in autoimmune diseases such as arthritis and Crohn's disease. Drugs that interfere with TNFα can prevent an overzealous immune system from attacking chondrocytes in the joints (causing arthritis) or tissue in the gastrointestinal tract (causing Crohn's disease). Small molecules have been identified that bind to and destabilize the TNFα trimer (**Figure 9.111**). However, the best drugs targeting TNFα on the market are proteins that sequester TNFα or block the receptor.

A wide range of cells produce TNFα, but the protein is mainly associated with cells of the immune system. For example, when macrophages recognize bacteria, the macrophages release large amounts of soluble TNFα. TNF is initially present as a homotrimer on the surface of cells. Then the protease TACE cleaves the membrane-bound form of TNFα to produce a soluble form. TNFα exists as the biologically active trimer only when its concentrations reach or exceed the nanomolar range.

There are two types of trimeric receptors for TNFα. Most cells express both TNFR-1 and TNFR-2 on their cell surfaces. TNFR-2 can only be activated by the membrane-bound form of TNF. TNFR-1 binds to FADD via an adaptor protein. Recall from the discussion of caspase cascades in Chapter 6 that FADD leads to DNA fragmentation via a mitochondrion-dependent caspase cascade.

TNFα induces apoptosis via TNFR-1, but it has the opposite effect on immune cells via TNFR-2. TNFR-2 does not associate with proteins that display death domains. When activated by TNFα, TNFR-2 binds to the adaptor protein TRAF2, which activates the kinase NF-κB-inducing kinase (NIK). NIK phosphorylates another kinase, IκB kinase (IKK), which in turn phosphorylates IκB. IκB normally sequesters a transcription factor—from the NF-κB family—in the cytoplasm, but once phosphorylated, IκB is targeted for destruction by ubiquitination, releasing NF-κB. Free NF-κB is cleaved by a protease, allowing it to dimerize and translocate to the nucleus, upregulating proinflammatory pathways (**Figure 9.112** and **Figure 9.113**). In T cells, activation of NF-κB induces differentiation and proliferation—the exact opposite of the apoptotic effects mediated by the TNFR-1 receptor.

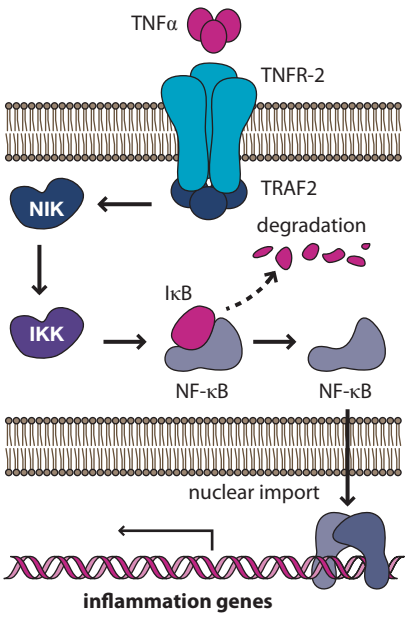
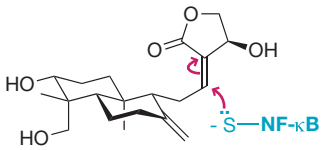



**Figure 9.112 IκB sequestering the transcription factor NF-κB.** When TNFα binds to TNFR-2 in T cells it leads to the degradation of IκB, releasing the transcription factor.

**Figure 9.113 NF-κB (p50) binding to DNA as a dimer.** In this structure of the C62A mutant (PDB 1SVC), the mutated residue is rendered as yellow spheres. The gaps in the DNA double helix arise from the fact that some of the DNA bases were not base-paired in the crystallized molecule.




andrographolide

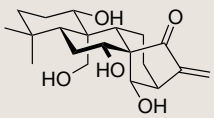
**Figure 9.114 Covalent modification of NF-κB.** The α,β-unsaturated lactone of andrographolide forms a covalent bond with NF-κB.

The natural product andrographolide was isolated from an *in vitro* screen of plant extracts for inhibitors of NF-κB transcriptional activity. Andrographolide was shown to form a covalent bond with Cys62 of the p50 variant of NF-κB (**Figure 9.114**). Electrophiles such as andrographolide can react with the many cysteines found inside the cell, but the reporter assay reveals only those that affect transcription of the gene under control of NF-κB (in this case, a luciferase reporter gene.)

**Problem 9.20**

The natural product kamebakaurin also targets the same reactive cysteine on NF-κB as andrographolide. Draw a structure of the cysteine adduct of kamebakaurin.

kamebakaurin




## 9.9 PATHWAYS CONTROLLED BY SMALL DIFFUSIBLE GAS MOLECULES

### Oxygen levels are monitored through HIF-1α

Human cells need to monitor levels of oxygen. Under aerobic conditions, human cells degrade glucose oxidatively to generate carbon dioxide, like a slow flame except that the energy is harnessed through the production of ATP. Under anaerobic conditions, a different set of enzymes extract energy from glucose, ultimately producing lactic acid. The switch between genes for aerobic respiration and anaerobic respiration is controlled by a transcription factor called hypoxia inducible factor-1α (HIF-1α). Under aerobic conditions, HIF-1α has a half-life of about 5 minutes because Pro402 and Pro564 on HIF-1α are constitutively oxidized to hydroxyprolines by a prolyl hydroxylase enzyme. The hydroxylated prolines flag HIF-1α for enzymatic ubiquitination and proteolysis. Thus, the hydroxylase enzyme serves as a kind of oxygen receptor, because oxygen binds as a substrate. When oxygen levels are low, the oxidation of HIF-1α is prevented and concentrations of unmodified HIF-1α build up, eventually becoming sufficient to form a transcriptional complex involving many proteins, including a key protein CBP/p300.

The genes controlled by HIF-1α depend on the cell type. In muscle cells, HIF-1α controls genes for aerobic respiration. In kidney cells, HIF-1α promotes expression of the cytokine erythropoietin, which leads to the production of oxygen-carrying red blood cells. HIF-1α inhibits genes for cell proliferation; cell division does not make sense in the absence of nutrients. This effect is particularly important for solid tumors, which are chronically deficient in oxygen. A screen for small molecules that could inhibit the interaction of CBP/p300 with HIF-1α revealed that the natural product

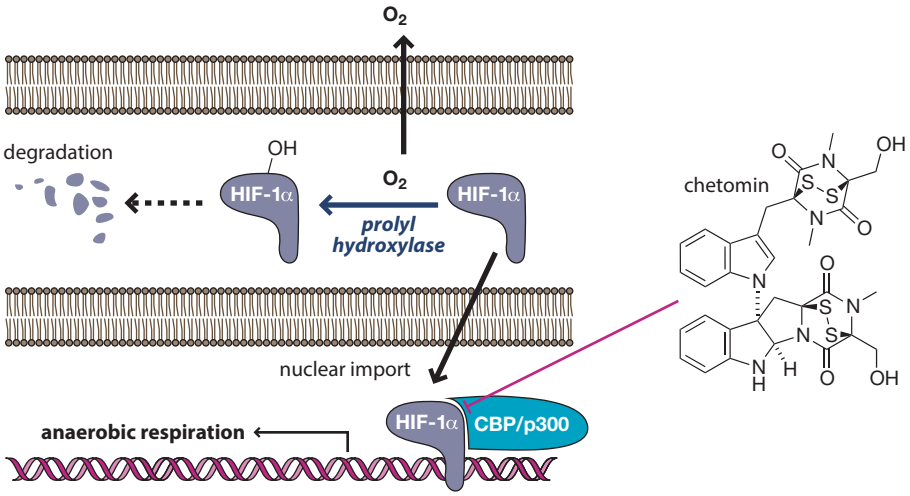



**Figure 9.115 Oxygen sensor.** The transcription factor HIF-1α is a substrate for oxidation with $O_2$, catalyzed by a prolyl hydroxylase. Right: chetomin is a potent inhibitor of the HIF-1α interaction with CBP/p300.

chetomin is a potent inhibitor of the interaction (**Figure 9.115**). Chetomin tricks cells into thinking that there is no oxygen, and causes the cells to stop proliferating.

## A nitric oxide receptor induces the production of cGMP

Vascular smooth muscle cells control blood pressure by contracting or relaxing. Nonmuscle cells of the vascular endothelium can exert highly localized control over vascular smooth muscle by generating nitric oxide (NO), which diffuses freely out of endothelial cells and into nearby muscle cells; however, NO is highly reactive and undergoes rapid and promiscuous reactions with many molecules, including thiols and molecular oxygen. Thus, NO can only exert local effects, unlike hormones that circulate throughout the body. Many diatomic gas molecules (for example CO, $O_2$, and NO) form tight complexes with iron porphyrins. Vascular smooth muscle cells contain a form of **guanylate cyclase** that is regulated by an iron–porphyrin complex. When nitric oxide binds to the iron atom of the porphyrin (**Figure 9.116**), the enzyme undergoes an activating conformational change that allows it to convert GTP into cyclic guanosine monophosphate (cGMP), in direct analogy to the reaction catalyzed by adenylyl cyclase. cGMP binds to cGMP-dependent protein kinase I, which phosphorylates calcium channels in the cell membrane and in calcium storage vesicles. Phosphorylation inhibits these channels from opening even if other pathways are generating $IP_3$. The cGMP-dependent protein kinase also phosphorylates proteins that, in turn, affect phosphorylation of myosin, directly inhibiting muscle contraction. By inhibiting the contraction of vascular smooth muscles, NO leads to vasodilation. cGMP levels are diminished by the action of cGMP phosphodiesterase 5 (PDE5), which hydrolyzes cGMP to guanosine monophosphate (**Figure 9.117** and **Figure 9.118**).

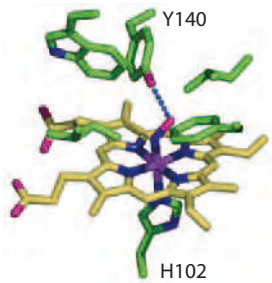



**Figure 9.116 Dr. NO.** NO binds to the iron porphyrin cofactor of guanylate cyclase. (Courtesy of C.S. Raman and Pierre Nioche, *Science* 306:1550–1553, 2004.)

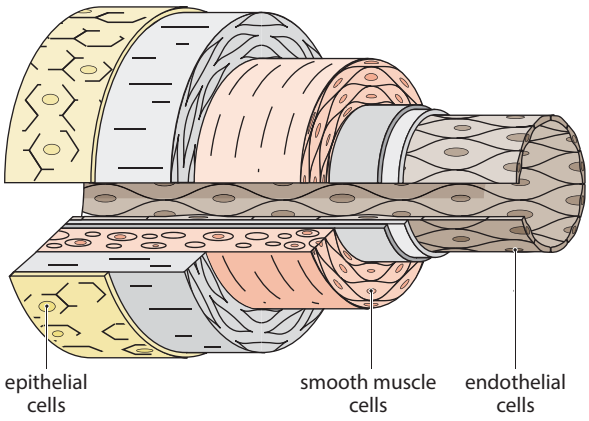



**Figure 9.117 Cutaway view of an arterial blood vessel showing layers of cells.** (Adapted from S. Fox, Human Physiology, 12th ed. McGraw-Hill, 2010.)

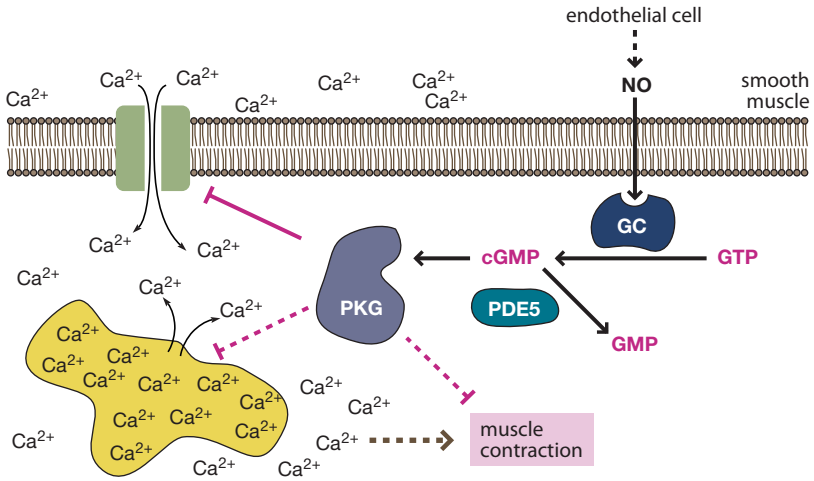



**Figure 9.118 Control by NO.** In blood vessels, endothelial cells inhibit the contraction of smooth muscle cells by releasing NO, which activates guanylate cyclase (GC) an enzyme that generates cGMP. cGMP-activated protein kinase (PKG) phosphorylates ion channels, preventing them from opening.

The vascular smooth muscle cells that supply blood to the penis are also controlled by cGMP levels. Medications for erectile dysfunction (such as Viagra®, Cialis®, and Levitra®) promote vasodilation by inhibiting PDE5; the resultant high levels of cGMP ultimately depress contraction of the blood vessels (**Figure 9.119**). Nitric oxide also acts as a diffusible signal for neurons and macrophages. Since 1879, long before its mechanism was understood, nitroglycerin has been used as a medication to increase blood flow to the heart. As shown more recently, aldehyde dehydrogenase generates nitric oxide from nitroglycerin, and the NO causes vasodilation of coronary arteries.

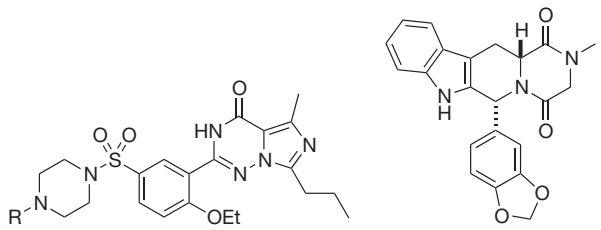



**Figure 9.119 The compound in the blue pill.** Inhibitors of PDE5 lead to relaxation of the vascular smooth muscle and are used to treat erectile dysfunction.

## 9.10 SUMMARY

The cells of the human body are engaged in an endless conversation mediated by secreted signals and displayed signals. Some signals initiate rapid responses, such as contraction or secretion; other signals initiate slower responses involving the synthesis of new proteins according to the central dogma: transcription of DNA into RNA, and translation of RNA into proteins. Those transcriptional processes can redefine the constitution and capabilities of a human cell. In this chapter we have taught you how extracellular signals are transduced into transcriptional changes, dropping you into a vast sea of acronyms. To help you chart your course we described signal transduction pathways mediated by seven types of receptors: nuclear receptors, cytokine receptors, receptor tyrosine kinases, trimeric death receptors, G protein-coupled receptors, ion channel receptors, and diffusible gas receptors. These seven flotillas of signaling proteins will help you to confidently navigate the primary research literature.

We have come full circle. We started this book by describing how the transcription of genes leads to all of the downstream biooligomers in the human cell. Filling in the details took up most of the book. With this chapter we have concluded our journey by describing how small molecules and macromolecules, ultimately produced by biooligomers, can control the transcription of other biooligomers in the cell. Moreover, we covered many examples where synthetic small molecules could be used to control

signal transduction processes, many of which involve transcription. Biooligomers are an architectural paradigm for the efficient generation of molecular diversity—an essential ingredient in the recipe for evolution. You are now empowered to understand and control biology at the highest level of relevant detail—the level of atoms and bonds. At the level of atoms and bonds you can engineer the medicines of the future or probe the secrets of cells that make us healthy, unhealthy, and human.

## LEARNING OUTCOMES

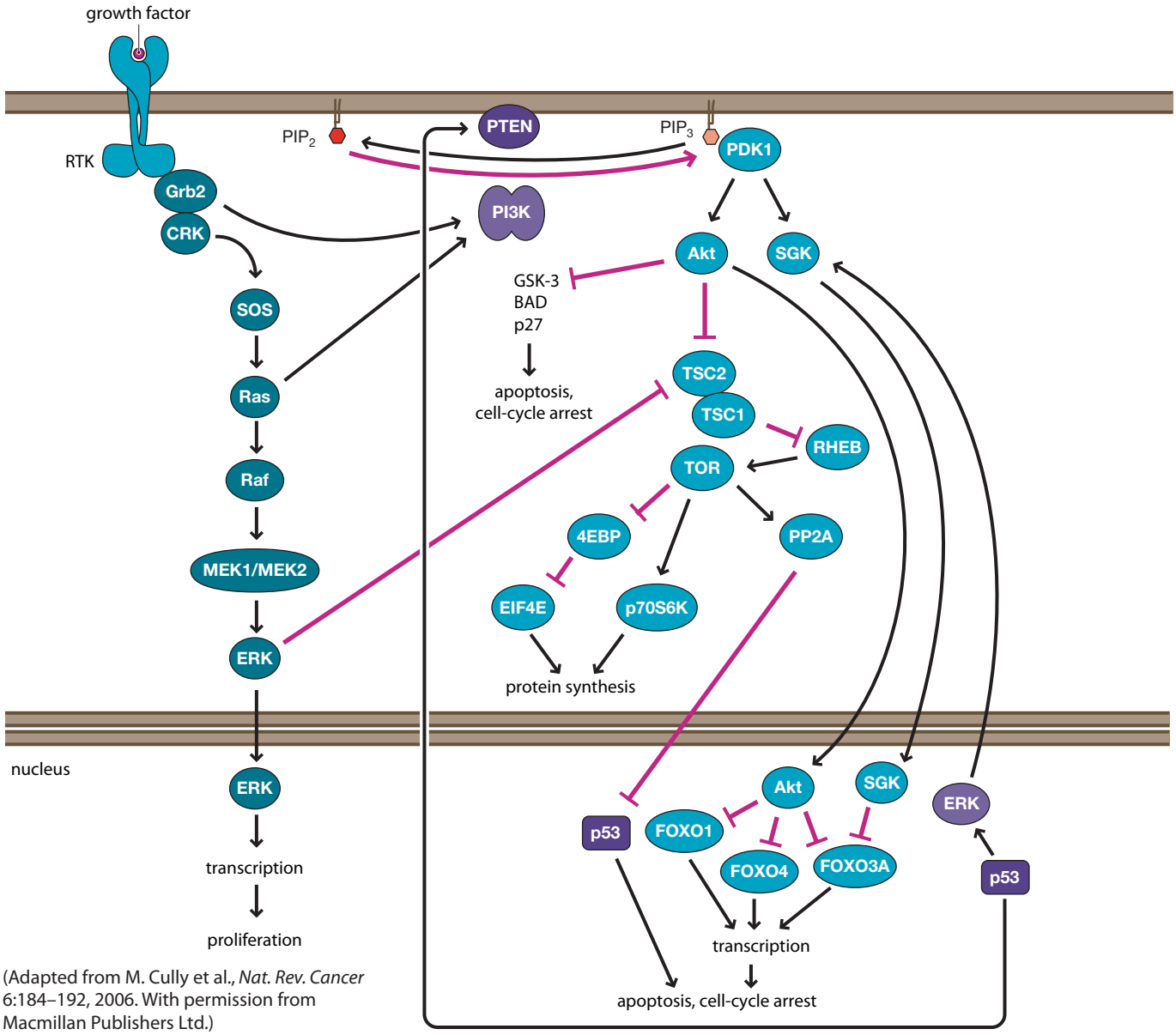
- Interpret the boolean logic of signal transduction diagrams as it applies to control of gene expression.
- Predict the effects of inhibitors and activators on signal transduction pathways.
- Understand the connection between $Ca^{2+}$ ions and two rapid nontranscriptional processes: muscle contraction and exocytosis.
- Distinguish the seven major signal transduction pathways that control transcription in human cells.
- Recognize the high-affinity ligands for nuclear receptors.
- Know the three types of ligands for human two-component pathways: interleukins, interferons, and TGF-$\beta$.
- Recognize growth factors as the ligands for receptor tyrosine kinases.
- Distinguish between the four subpathways downstream of receptor tyrosine kinases: STAT, MAP kinases, PLC$\gamma$, and PI3K/Akt.
- Distinguish the death pathways (TNF$\alpha$ and FADD) from receptor tyrosine kinase pathways.
- Recognize the common classes of ligands that act at GPCRs.
- Distinguish between adenylyl cyclase and PLC$\beta$ pathways controlled by 7TM GPCRs.
- Recognize the imbalance between $K^+$, $Na^+$, and $Ca^{2+}$ ions in human cells.
- Understand how proteolytic degradation is modulated through signaling.
- Recognize the two diffusible gases that are important in human cell signaling: NO and $O_2$.

# PROBLEMS

**9.21** Draw the Lewis dot structure for nitric oxide.

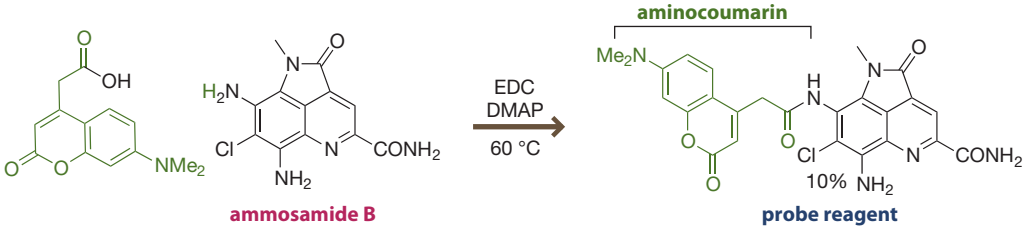
**9.22** Using cell signaling diagrams throughout the chapter, rank each of the seven-signal transduction pathways according to the maximum number of steps required to affect transcription within the nucleus.

***9.23** Consider the action of the following small molecules on the following signal transduction pathway (see figure on the next page).
tyrphostin, a specific inhibitor of tyrosine kinases
AZD8330, a specific inhibitor of MEK1 and MEK2
wortmannin, a specific inhibitor of PI3K
rapamycin, a specific inhibitor of TOR
FR180204, a specific inhibitor of ERK
A-443654, a specific inhibitor of PKB
**A** Which compound would most effectively prevent uncontrolled proliferation?
**B** Which compound best inhibits apoptotic signals without inhibiting proliferative signals?
**C** Which compound would most selectively inhibit protein synthesis?

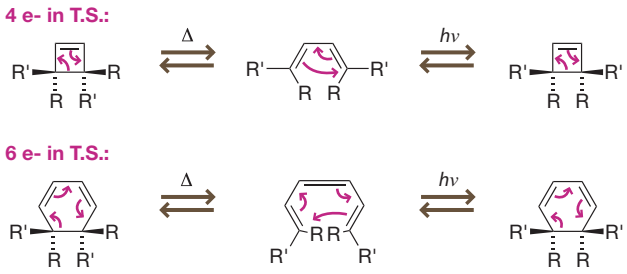



(Adapted from M. Cully et al., *Nat. Rev. Cancer* 6:184–192, 2006. With permission from Macmillan Publishers Ltd.)

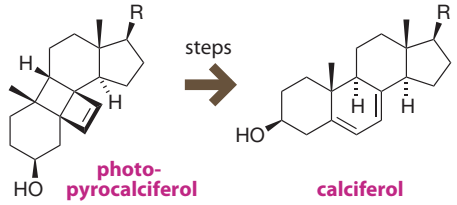
**9.24** The red natural product ammosamide B is produced by *Streptomyces* strain CNR-698, isolated from ocean sediments in the Bahamas. It is cytotoxic to various cultured cancer cells with an $IC_{50}$ value as low as 20 nM. To identify a molecular target, an aminocoumarin fluorophore was added by using a carbodiimide coupling in low, but useful, yield. The carbodiimide EDC (1-ethyl-3-[dimethylaminopropyl] carbodiimide•HCl) is a water-soluble carbodiimide that works just like DCC and DIC. In cells treated with the probe reagent, fluorescence was shown to localize within lysosomal vesicles. The inability to wash away or compete away the fluorescence with free ammosamide B suggested a covalent or tight binding interaction between ammosamide and the molecular target. Suggest a plausible arrow-pushing mechanism for the carbodiimide coupling reaction.



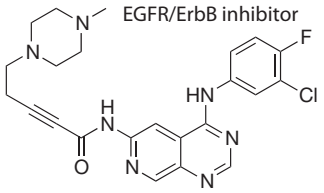
***9.25** Electrocyclic ring-opening and ring-closing reactions are stereospecific. The pattern of stereoselection depends on the number of electrons in the transition state (T.S.) and on whether the reaction is promoted by heat (Δ; over 100 °C) or light.



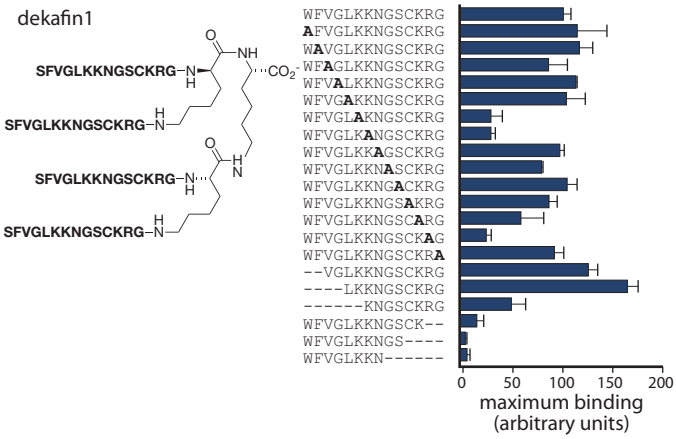

Given the patterns of stereospecificity above, suggest conditions for a series of electrocyclic reactions that would efficiently convert photopyrocalciferol into calciferol.




**9.26** The heterocycle below is a potent, selective, irreversible inhibitor of the ErbB family of EGF receptor tyrosine kinases. The inhibitor is believed to form a covalent adduct with a cysteine residue (Cys773 in ErbB1, Cys784 in ErbB2, and Cys778 in ErbB4) near the entrance to the ATP-binding site of the tyrosine kinase domain. Suggest a plausible structure for the adduct between the inhibitor and the cysteine residue.



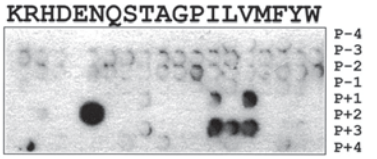

***9.27** A synthetic tetramer, derived from a short segment of FGF, was shown to bind to the FGF receptor *in vitro* and activate the FGF receptor pathway in neuronal cells. An alanine scan was used to assess the contribution of each of the residues to the interaction with the receptor.




What do the alanine scanning results suggest about the residues with which the tetramer interacts with the FGF receptor?

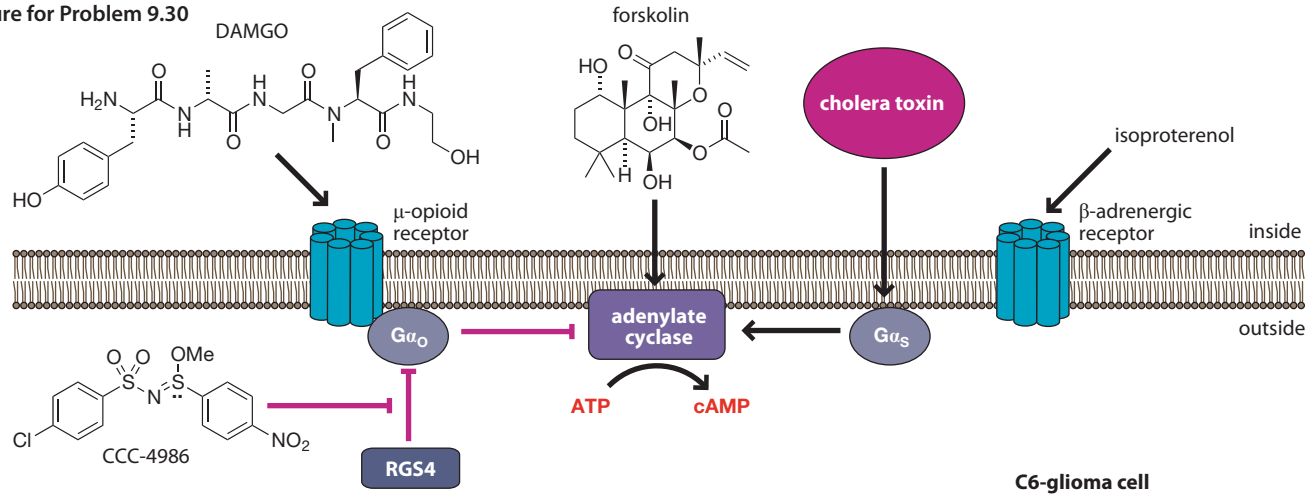
**9.28** The natural product wortmannin inhibits PI3K. What effect does wortmannin exert on protein translation, apoptosis, and mitosis in urothelial cells?

***9.29** An array of phosphotyrosine nonapeptides was spotted onto a paper membrane, and then subjected to binding with Grb2, which has an SH2 domain. Each spot was composed of a mixture of peptides, with two defined positions. Thus, strong binding of Grb2 to a spot composed of peptides with the sequence XXWXpYXXXX suggests that tryptophan is strongly preferred two residues before the phosphotyrosine (the P – 2 position). Using the data below, what consensus peptide sequence would be best for binding to the SH2 domain of Grb2? Use X to represent positions with no preference. (From M. Rodriguez et al., *J. Biol. Chem.* 279:8802–8807, 2004. With permission from The American Society for Biochemistry and Molecular Biology.)




**9.30** A screen to identify molecules that affect G protein signaling identified CCC-4986, which was shown to inhibit RGS4, a GTPase-activating protein. As summarized in the following signal transduction diagram, CCC-4986 was tested in a C6-glioma cell to determine the effect on cAMP production.

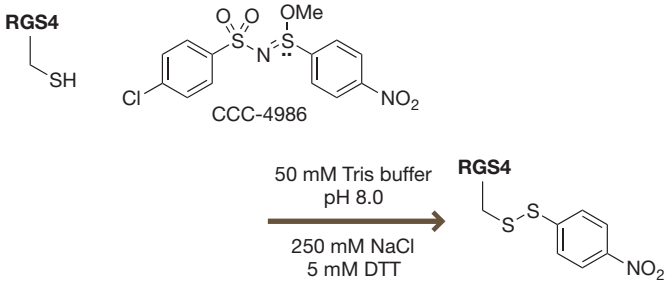
**Figure for Problem 9.30**



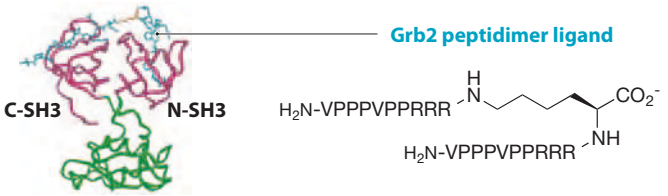
**A** For each of the following compounds, predict whether it will **increase** or **decrease** cAMP production:

i. isoproterenol (also known as isoprenaline)
ii. DAMGO
iii. forskolin
iv. cholera toxin
v. CCC-4986

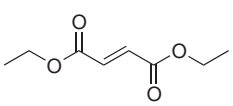
**B** Subsequent studies showed that CCC-4986 covalently modifies RGS4 by reacting with the thiolate side chain of Cys132. On the basis of mass spectrometry, the adduct was proposed to be a disulfide. If you feel up to it, suggest a plausible arrow-pushing mechanism for the reaction.




***9.31** Grb2 has two SH3 domains, one at the N terminus and one at the C terminus. Molecular modeling suggests that a simple dimer of a polyproline sequence VPPPVPPRRR would be able to bind simultaneously to both of the SH3 domains of Grb2. How would you synthesize the peptidimer ligand by solid-phase synthesis? Clearly indicate which protecting groups you would use in the synthesis.

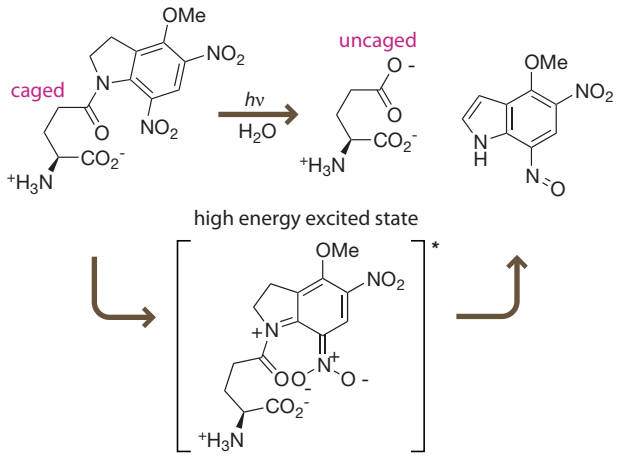



**9.32** The transcription factor Pap1 in fission yeast is normally exported from the nucleus because it has a nuclear export sequence—a 19 amino acid sequence near the N terminus (515–533). In response to hydrogen peroxide, export of Pap1 is inhibited, and it remains in the nucleus where it initiates transcription of a variety of genes. When two cysteine residues in the nuclear export sequence are mutated (C523A and C532D), nuclear export is abolished and the transcription factor accumulates in the nucleus, where it initiates transcription. Surprisingly, the non-oxidative reagent diethyl maleate exerts an effect similar to that of hydrogen peroxide. Suggest a plausible chemical mechanism accounting for the similar effect of hydrogen peroxide and diethyl maleate.

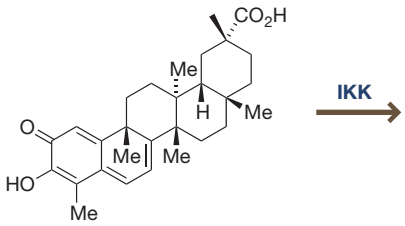


diethyl maleate

***9.33** The following "caged" glutamate derivative can be induced to release glutamate on activation by ultraviolet radiation. Suggest a plausible arrow-pushing mechanism for the reaction that goes through the intermediate shown.

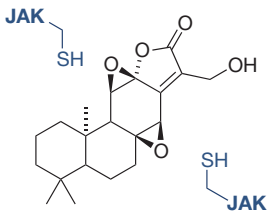



**9.34** Celastrol, isolated from the thunder god vine (*Tripterygium wilfordii*), forms an aromatic adduct by reacting with Cys179 in IKK. Suggest a plausible structure for the covalent adduct.

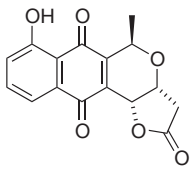



celastrol

***9.35** 17-Hydroxy-jolkinolide, isolated from *Euphorbia fischeriana* Steud, forms covalent crosslinks between cysteines in Janus kinases. Suggest a plausible structure for the covalent adduct between 17-hydroxy-jolkinolide and the cysteine residues of two Janus kinases.



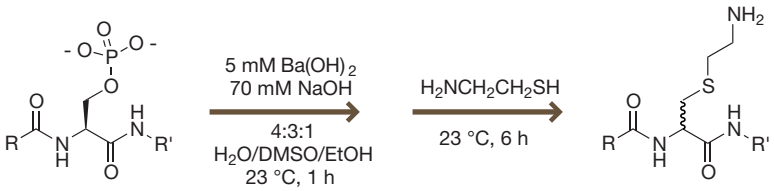

**9.36** The pyranonaphthoquinone antibiotic kalafungin inhibits the kinase Akt2 by reacting with a thiol in the active site. Which atom of kalafungin would you expect to be most reactive toward nucleophilic protein side chains (like cysteine thiolates) under bioreductive conditions?




kalafungin

$IC_{50}$ 70 nM in MDA468 cells

***9.37** A method for the detection of phosphoserine residues in proteins involves the two-step, one-pot reaction sequence below. Suggest a plausible arrow-pushing mechanism for each step. Which




of the ribosomally incorporated amino acids does the final product resemble?

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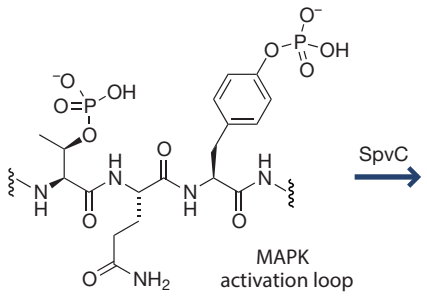
**9.38** The *Salmonella* toxin SpvC catalytically removes one of the phosphates from the activation loop of MAP kinases (for example FMTpEYpVA) to produce an inactive MAP kinase. The modified kinase cannot be phosphorylated by any MAP kinase kinase. On the basis of mass spectrometry (which detects the neutral forms of phosphate esters), the SpvC-catalyzed dephosphorylation leads to a loss of 98 Da, instead of the 80 Da difference expected for hydrolysis. Suggest a plausible structure for the modification induced by SpvC.

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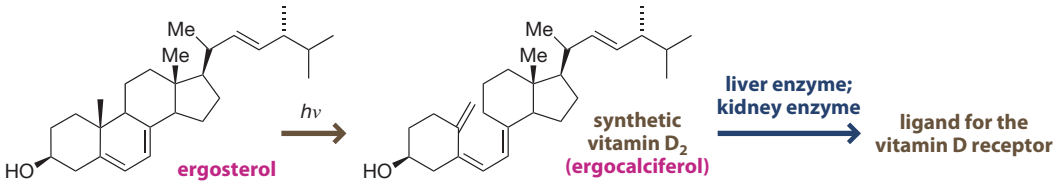
***9.39** The fungal steroid ergosterol, which is plentiful in brewer's yeast, can be converted into an analog of vitamin $D_3$ called ergocalciferol (also known as vitamin $D_2$). Ergocalciferol is widely used as a dietary vitamin D supplement. The human body transforms ergocalciferol into a ligand for the vitamin D receptor by using the same enzymes that metabolize vitamin $D_3$. Draw the structure of the tris-hydroxy form of vitamin $D_2$ that would bind to the vitamin D receptor.



# Glossary

**α-amylase** an enzyme that catalyzes the hydrolysis of amylose into glucose.

**β barrel** a cylindrical protein structural motif composed of antiparallel β strands.

**β hairpin** a protein structural motif composed of four amino acids and a hydrogen bond between the *i* and *i* + 3 residue. This type of turn structure leads to a 180° change in the direction of the peptide backbone.

**activator** a transcription factor responsible for initiating and promoting transcription of a gene or genes.

**acyl carrier protein domain (ACP)** the domain of a polyketide synthase that attaches to the nascent polyketide and carries the intermediate compound to different domains of the polyketide synthase.

**adenylate cyclase** an enzyme responsible for catalysis of the reaction converting ATP into cAMP.

**alanine scanning** systematic substitution of specific residues in a protein with the amino acid alanine. The approach is used to quantify contributions made by side chain atoms past the β-carbon.

**annealing** hybridization of strands of DNA by incubating the two at a high temperature (> 95 °C) for a short time and then slowly cooling the solution.

**anomer** a diastereomer of a monosaccharide with either an α or β configuration at the acetal or ketal carbon.

**anomeric center (anomeric carbon)** the acetal or ketal carbon in monosaccharides or glycoconjugates.

**anomeric effect** a stereoelectronic preference of anomeric alkoxy substituents in pyranose rings to adopt axial orientations over equatorial orientations.

**antibody** proteins of the immune system charged with binding to nonhuman molecules from pathogenic organisms.

**antisense strand** the DNA strand that provides the template for the sense strand and for RNA polymerase during transcription.

**apoprotein** proteins lacking a key cofactor.

**apoptosis** an orchestrated process leading to the death of the cell.

**apoptosome** a complex formed by cytochrome *c* with Apaf-1 and dATP, which activates caspase-9 by cleaving procaspase-9 during apoptosis.

**aptamer** a DNA or RNA sequence with affinity for specific target molecules.

**Argonaute** a ribonuclease that binds to siRNA and destroys any RNA transcripts complementary to the siRNA sequence.

**association constant ($K_a$)** the equilibrium constant for the association of two molecules, such as a ligand and receptor, to give a complex.

**autocrine signaling** hormonal communication between cells of the same type.

**autophosphorylation** in a dimer of protein kinases, reciprocal phosphorylation of side chains on the partner kinase.

**bacteriophage** viruses that infect only bacteria.

**bait protein** *see* yeast two-hybrid.

**base** the heterocyclic substituent that uniquely distinguishes each of the four DNA (or RNA) subunits. Each DNA or RNA molecule is composed of many nucleotide subunits and each subunit has a distinctive nitrogen heterocycle substituent. The term base is often used to refer to the entire nucleotide subunit. In the context of chemistry, the term base is used to mean a proton acceptor.

**base pairs** two DNA or RNA bases held together by hydrogen bonds, such as A•T or G•C.

**biooligomers** polymers produced by a cell that are constructed through the connection of subunits to form chain-like molecules, including proteins, DNA, RNA, glycans, lipids, and terpenes.

**bioreductive activation** a mechanism for converting an unreactive prodrug into an active compound through a reduction reaction that takes place inside the cell.

**C terminus** the end of the protein backbone with the carboxy functional group.

**carbohydrates** "hydrates of carbon" or compounds with the empirical formula $C_n(H_2O)_n$. This term is reserved for molecules composed exclusively of one or more monosaccharides, and excludes glycoproteins and other glycans.

**central dogma of molecular biology** an organizing principle to describe the encoding of sequence information in biooligomers. In the classical definition, it states that the DNA provides a blueprint for the molecules synthesized by the cell; the DNA is transcribed into RNA, which is then translated into proteins.

**checkpoint** during eukaryotic cell division, a conditional test that determines whether the cell is allowed to continue the process of cell division.

**chemical biology** the use of tools from chemistry to understand and manipulate biology at the level of atoms and bonds.

**chemical genetics** using small molecules as probes of protein function in the cell. For example, protein activity can be shut off through a specific inhibitor of the target protein.

**chitin** a β(1–4)-linked polymer of *N*-acetyl-D-glucosamine, which provides the exoskeletons for arthropods.

**cloning** making identical copies of a cell or gene. In the context of molecular biology, this process involves placing the gene into a plasmid for expression of the encoded protein or the production of multiple copies of the plasmid that contains the gene.

**coenzyme A** an enzyme co-factor attached through a thioester to provide a handle for enzyme binding to two- and three-carbon building blocks (acetyl and propionyl, respectively).

**combinatorial chemistry** a method for the synthesis of collections of molecules each containing the same number of subunits, typically accomplished using a modular approach to join subunits together.

**complementation assay** an approach for testing an interaction based on the restoration of a wild-type phenotype. For example, a split GFP assay will result in functional, fluorescent GFP if the two pieces of the protein can reassemble as a result of a mediating interaction.

**covalent (bond)** a type of bond in which the interaction energy is dominated by the interaction of filled orbitals with unfilled orbitals, the third term in Equation 2.2, for example $H_3C—H$ or $H_3C—CH_3$.

**cross conjugated (cross conjugation)** two or more π bonds that cannot participate in resonance due to an intervening C=C bond.

**cyclodextrins** cyclic oligomers of glucose, $[\text{-Glc}\alpha(1,4)\text{-}]_n$.

**cystine** a dimeric, oxidized form of cysteine linked by a sulfur–sulfur disulfide bond.

**cytokines** proteins used in extracellular cell–cell signaling. Inside the cell, their signals are transduced by Janus kinases and STAT transcription factors.

**deacetylases** enzymes responsible for removing an acetyl functionality from a protein or oligosaccharide.

**dehydratase domain (DH)** the domain of a polyketide synthase that catalyzes a dehydration reaction on the intermediate polyketide, which results in the formation of an enone intermediate polyketide.

**developmental biology** the study of how cells differentiate to grow from a fertilized egg to a complete organism.

**dicer** a nuclease charged with hydrolyzing pre-miRNAs to yield siRNAs.

**dispersion force** the attractive interactions between molecules or functional groups that is embodied by the third term of Equation 2.3.

**dissociation constant ($K_d$)** the equilibrium constant for the dissociation of a molecular complex to give the individual components, such as a ligand and receptor.

**DNA ligase** an enzyme that can repair nicked DNA by forming a phosphate ester bond from the 3′ hydroxyl of one strand and the 5′ phosphate of another.

**DNA primase** an enzyme charged with synthesizing short strands of RNA to provide a primer for DNA polymerization of the lagging strand during replication.

**DNA shuffling** a method for engineering new genes based on the hybridization of single-stranded fragments of homologous genes from different organisms or other sources during PCR-based amplification.

**domains** small regions of proteins (typically with fewer than 100 amino acid residues) that are capable of folding independently when removed from the larger protein structure.

**dose–response curve** a plot of drug concentration versus biological response, used to determine $EC_{50}$ values.

**DTT** dithiothreitol. A synthetic reducing agent used to cleave disulfide bonds in peptides and proteins.

**Edman degradation** a chemical method for determining the amino acid sequence of a protein, based upon chemical cleavage and identification of the N-terminal amino acid.

**efflux pump** a membrane-bound protein that harnesses the hydrolysis of ATP to push small molecules out of the cell. This class of proteins provides a mechanism for antibacterial drug resistance.

**enoyl reductase domain (ER)** the domain of a polyketide synthase that catalyzes a conjugate reduction reaction on an enone polyketide intermediate.

**enzyme** a protein capable of catalyzing a chemical transformation.

**epigenetics** the study of heritable factors that are not encoded by the organism's DNA sequence. For example, DNA methylation or modification of the histones can influence otherwise identical sequences of DNA.

**error-prone PCR** a technique for generating a collection of DNA sequences based upon the addition of DMSO, $Mn^{2+}$ or other factors, which can disrupt the fidelity of DNA polymerase during PCR.

**exons** the parts of a eukaryotic RNA transcript that remain after the removal of introns. The remaining part of the transcript is translated into a protein.

**exonuclease** an enzyme or subdomain of a larger enzyme responsible for the hydrolysis of DNA or RNA strands by cleaving one base at a time from the end of the strand.

**fatty acid synthase** the type I polyketide synthase protein complex responsible for the synthesis of fatty acids. It iteratively introduces each two-carbon subunit through the processive action of each catalytic domain.

**fatty acid** an unbranched alkanoic acid with an even number of carbons produced by a polyketide synthase. They vary in length and level of unsaturation.

**flow cytometry** a method for analyzing a population of cells. The method involves passing a hydrodynamically focused stream of the cells through a fluorimeter. The fluorescence of each cell, introduced by fluorescently tagged antibodies, fluorogenic agents, or fluorescent proteins, can be quantified. In FACS, the flow cytometry is used to separate out a population of cells using a fluorescence-based criterion.

**fluorophore** a fluorescent chromophore that absorbs photons and emits photons with lower energy and higher wavelength.

**forward genetics** the process of introducing random mutations into an organism to identify interesting phenotypes. DNA sequencing can then identify the associated genotype, thus implicating a specific gene required for biological activity. In forward chemical genetics, collections of small molecules are used in place of genetic mutations. Specific phenotypes from treatment with the small molecule can be identified, and the target of the small molecule can reveal a contribution to a cellular pathway.

**frontier molecular orbitals** the most reactive orbitals in a chemical species, usually corresponding to the highest occupied molecular orbital and the lowest occupied molecular orbital.

**G proteins** a class of proteins that bind and hydrolyze GTP. Many such proteins interact with 7-transmembrane domain receptors to transduce signaling information from the extracellular milieu to the cytoplasm.

**G protein-coupled receptors (GPCRs)** a common class of 7-transmembrane domain receptors that transduce extracellular hormone signals into intracellular actions like contraction, secretion, or mitotic proliferation.

**gene** a DNA sequence encoding a protein and a promoter.

**genetic code** the correlation between DNA or RNA triplets (codons) and the resultant amino acid in the protein.

**genome** the genetic material that is necessary and sufficient to encode an organism.

**genotype** a gene or combination of genes.

**glycans** molecules that contain carbohydrate functional groups, such as glycolipids, glycoproteins, and polysaccharides.

**glycome** the collection of glycans in a cell, tissue, or organism.

**glycoproteins** proteins post-translationally modified through the attachment of oligosaccharides.

**glycosphingolipids** oligosaccharides attached to the lipid ceramide.

**glycosylation** the enzyme-catalyzed transfer of carbohydrates to the surface of proteins, lipids and oligosaccharides.

**glycosylhydrolases** a class of enzymes that break glycosidic bonds through the addition of water.

**glycosyltransferases** the class of enzymes charged with forming glycosidic bonds during the synthesis of glycans and carbohydrates. The glycosyl donor substrates for this enzyme have nucleotide phosphates as the leaving group at the anomeric position.

**GSH** a symbolic abbreviation for the tripeptide glutathione that emphasizes the thiol functional group.

**GSSG** a symbolic abbreviation for the oxidized dimer of glutathione that emphasizes the disulfide functional group.

**GTPase-activating proteins (GAPs)** a regulatory protein that accelerates the hydrolysis of GTP to GDP by G proteins.

**guanine nucleotide exchange factors (GEFs)** a regulatory protein that catalyzes the ejection of the spent guanine nucleotide GDP from a G protein. The binding of a new GTP substrate regenerates an activated G protein–GTP complex.

**guanylate cyclase** an enzyme that converts GTP into cyclic guanosine monophosphate (cGMP).

**heat shock** raising the temperature of an organism to provoke transcriptional changes in response.

**histone acetyltransferase (HAT)** an enzyme responsible for transferring an acetyl group to specific lysine side chains of histones.

**histone deacetylase (HDAC)** an enzyme that hydrolyzes the acetyl amide functionalities from the lysine side chains of modified histones.

**hotspot of binding energy** found on the surface of a protein, a cluster of amino acid side chains that provide the bulk of binding energy to mediate the protein's noncovalent binding.

**hybridization** forming double-stranded DNA by incubating two complementary single-stranded DNA sequences.

**hydrolase** an enzyme that catalyzes the hydrolysis of bonds, such as amides and glycosides, through the addition of water.

**hydrophobic effect** the tendency of water molecules to maximize interactions between themselves while minimizing interactions with solutes.

**inhibitory constant ($K_i$)** the dissociation constant for an inhibitor–enzyme complex. For typical inhibitors that bind at the same site as the substrate, this equilibrium constant is independent of the inhibitor concentration. In contrast, $IC_{50}$ is dependent on substrate concentration. When the substrate concentration is much lower than $K_m$, $IC_{50}$ is equal to $K_m$.

**inteins** internal segments of proteins that are removed autocatalytically from a protein through a reaction involving an *N* to *S* acyl shift at a cysteine residue. The remaining proteins are referred to as exteins.

**intercalation** a sandwich-like interaction between DNA bases and small molecules involving an aromatic compound sliding between the π-stacked DNA bases.

**interleukins** a class of cytokines that are important in the differentiation and development of blood cells in the hematopoietic lineage.

**introns** the parts of a eukaryotic RNA transcript that are removed from the transcript and are not translated into proteins.

**ion channels** a class of proteins that form pores in cell membranes. After opening in response to a regulatory signal, these proteins allow the passage of ions through the pore. This class of proteins includes both ion-specific and nonspecific channels.

**ionic (bond)** a type of bond in which the interaction energy is dominated by the Coulombic terms of Equation 2.2 or 2.3. For example K–Br, Cs–OH.

**isomerase** an enzyme that catalyzes a reaction leading to the isomerization of a starting material without changing the molecular formula of the substrate.

**ketoreductase domain (KR)** the domain of a polyketide synthase that catalyzes the reduction of a β-ketone to an alcohol.

**ketosynthase domain (KS)** the domain of a polyketide synthase that catalyzes the Claisen condensation of malonyl and acetyl thioesters, which results in the formation of a new carbon-carbon bond.

**kinase** a class of transferase enzymes responsible for transferring a phosphoryl group to a hydroxyl group, using ATP as the phosphate donor.

**lability** refers to chemical reactivity, and is the opposite of chemical stability. In chemical biology, the term implies susceptibility to reactions that lead to loss of biological function.

**ligase** an enzyme that catalyzes the joining (ligation) of two independent molecules, typically in an energy-dependent process. For example, subtiligase is a variant of the enzyme subtilisin, and joins together two peptide fragments.

**lipid** a cellular molecule that can dissolve in nonpolar organic solvent and not in water.

**lipid raft** microdomains within the cell membrane that are particularly rich in glycosphingolipids and cholesterol. Diverse transmembrane receptors and membrane-associated proteins cluster with this microdomain, which thus provides a focal site for organizing cell signaling.

**lipidome** the collection of all lipids found in the cell.

**lyase** an enzyme that cleaves bonds through mechanisms other than hydrolysis or redox reactions.

**monosaccharides** monomeric sugars used as the fuel for metabolism and the building blocks for glycans and other structures in the cell (for example RNA and DNA).

**mutation** a change to a DNA sequence. Silent mutations result in no change to the protein sequence encoded by the DNA.

**N terminus** the end of the protein backbone with the amino functional group.

**native chemical ligation** a method for joining together unprotected peptide fragments.

**neighboring group participation** adjacent functionalities or atoms contributing to the reactivity of a functional group.

**nick** in double-stranded DNA, a site where adjacent nucleotides are not covalently bonded.

**nonenzymic glycation** post-translational reactions between carbohydrates and proteins. This reaction does not require catalysis by enzymes, and can lead to protein crosslinking and other deleterious products. For example, the open-chain (aldehyde) form of glucose condenses with the amino functionality of lysine side chains.

**nuclear localization sequence** a short peptide fused to a protein, which directs the protein into the nucleus, through binding to karyopherin β.

**nuclear receptors** the class of receptors that bind directly to hydrophobic small molecules, including steroids and thyroid hormones. The resultant receptor–ligand complex can enter the nucleus and act as a transcription factor through binding to DNA.

**nucleoside** a ribose or deoxyribose sugar covalently linked through a glycosidic bond to a purine or pyrimidine base.

**nucleotide** a nucleoside with one or more phosphate groups attached to a ribose or deoxyribose moiety through a phosphate ester bond.

**oligosaccharides** linear or branched chains of monosaccharides covalently linked through glycosidic bonds. These glycan chains are distinct from polysaccharides, which are composed of one or more monosaccharides connected as repeating units.

**oncoprotein** a protein associated with cancer as a result of mutations in its encoding gene.

**operator** a region of a DNA responsible for binding to transcription factors.

**operon** a cluster of genes meant to be transcribed together to provide a complete set of transcripts to encode proteins that then work together.

**origin of replication (ORI or ori)** a DNA sequence within a plasmid or chromosome instructing the cell to make copies of the plasmid or chromosome.

**oxidoreductase** an enzyme that catalyzes oxidation/reduction processes.

**oxocarbenium ion** a structure with a carbocation adjacent to an oxygen atom.

**oxonium ion** a species containing a positively charged oxygen atom with three bonds.

**paracrine signaling** hormonal communication between cells of a different type.

**PEG** polyethylene glycol, a polymer with the chemical structure $[CH_2CH_2O]_n$.

**peptide** an oligomer of α-amino acids, distinguished from proteins by their short length and lack of complex folded structure.

**peptide nucleic acids (PNAs)** synthetic analogs of DNA with the bases linked together by amide bonds in place of the phosphodiester and deoxyribose backbone of DNA.

**phenotype** the observable characteristics resulting from a genotype.

**phospholipase** an enzyme that catalyzes the hydrolysis of the acyl group from the glycerol core of triacylglycerols.

**phosphoramidite** a key reagent for oligonucleotide synthesis possessing a trivalent phosphorus atom with a highly reactive P–N bond.

**plasma membrane** the lipid barrier between the extracellular milieu and the cytoplasm of the cell.

**plasmid** a relatively short, circular piece of DNA with an origin of replication and one or more genes.

**polar covalent (bond)** a type of bond between atoms with substantially different electronegativities and in which the interaction energy is dependent on both the first and third terms of equation 2.2, for example HO–Na or $H_3C$–F.

**polyketide** a class of natural products biosynthesized from two- and three-carbon building blocks through a series of Claisen condensations.

**polysaccharides** polymers of monosaccharides covalently linked through glycosidic bonds (for example starch, cellulose, and amylose).

**post-translational modifications** covalent alterations to proteins that occur after the protein has been synthesized by the ribosome. These protein modifications include trimming, splicing, phosphorylation, glycosylation, oxidation, addition of membrane anchors, fusion with other proteins, alkylation, and acetylation.

**prebiotic chemistry** a subfield of chemistry that attempts to identify the most probable chemical reactions that led from inanimate molecules to living organisms.

**prey protein** *see* yeast two-hybrid.

**primary metabolite** molecules considered essential for normal cellular function, such as ATP, glyceraldehyde 3-phosphate, or cholesterol.

**probe** an oligonucleotide designed to hybridize to a specific sequence of DNA or RNA.

**promoter** a DNA sequence found near the beginning of an open reading frame that can recruit a transcription factor to initiate the transcription of a gene into RNA.

**proteasome** a large complex of proteins found in the cell that hydrolyzes proteins marked for destruction, usually by ubiquitinylation.

**protein** an oligomer of α-amino acids translated by the ribosome. Structurally, they are distinguished from peptides by the presence of polypeptides with at least two elements of interacting secondary structure.

**proteolysis** hydrolysis of amide bonds to degrade a protein to its constituent peptides and/or amino acids.

**proteome** the collection of all proteins found in a cell, tissue, organ, or organism.

**purine** the nitrogen-containing bicyclic aromatic ring system found in adenosine or guanosine.

**pyrimidine** the nitrogen containing aromatic ring found in cytidine, thymidine, or uridine.

**pyrosequencing** a technique for sequencing DNA based on the sensitive detection of pyrophosphate, which is produced by the successful incorporation of a dNTP into the nascent DNA strand during synthesis by DNA polymerase.

**quaternary structure** a level of protein structure defined by the three-dimensional arrangement of multiple folded protein chains.

**Ramachandran plot** a plot of the $\phi$ versus $\psi$ angles for a polypeptide, protein, or group of proteins. Certain regions of the plot are associated with secondary structures such as α helices and β sheets.

**receptor tyrosine kinases** a class of cell-surface receptors that usually bind to growth factor proteins. These proteins undergo dimerization upon ligand binding, and then autophosphorylate the dimerizing partners.

**replication** the process by which a copy of a DNA molecule is made.

**repressor** a transcription factor responsible for shutting down RNA transcription.

**retrovirus** an RNA-based virus.

**reverse genetics** examining a specific protein by making mutations to its encoding gene, and then examining the resultant function. In reverse chemical genetics, a small molecule with a known cellular target is used as a probe to understand contributions to a cellular phenotype by the target.

**reverse transcriptase** an enzyme that converts RNA to DNA.

**riboswitches** structured regions of RNA that can exert control over transcription, splicing or translation.

**ribozymes** RNA molecules capable of catalyzing biological reactions.

**RNA interference** RNA molecules that elicit a destructive response toward homologous RNA transcripts.

**secondary metabolite** molecules considered nonessential for the immediate survival of a cell. Such compounds are often used for the chemical defense of the cell.

**secondary structure** elements of biooligomer structure with well-defined patterns of hydrogen bonds: in proteins, helices and sheets formed through hydrogen bonds with the peptide backbone; in DNA, stem loops and other structures formed through base pairing.

**selectant** a biooligomer of defined sequence (DNA, RNA, or protein) that is selected from a library of molecules on the basis of a unique chemical characteristic such as binding, reactivity, or catalysis.

**sense strand** the DNA strand that resembles the resultant RNA transcript. This strand of DNA hybridizes with the antisense strand.

**sequencing** determining the order of subunits in a biooligomer such as DNA or a protein.

**seven-transmembrane domain receptors (7TM receptors)** *see* G protein-coupled receptors (GPCRs).

**signal transduction** the conversion of one type of signal in a chemical or physical form into another type of chemical signal (for example a photon or small molecule converted into a protein conformational change or the conversion of extracellular information into the transcription of genes).

**single nucleotide polymorphism (SNP)** pronounced "snip." In humans, SNPs are the single nucleotide differences that exist between otherwise identical homologous genes, like pairs of homologous genes within a chromosome or homologous genes in genetically related individuals.

**solid-phase peptide synthesis (SPPS)** a method for the chemical synthesis of peptides, based upon the attachment of the C terminus of the peptide to an insoluble polymer followed by iterative amino acid coupling reactions.

**spliceosome** a multiprotein–RNA complex that is responsible for the removal of introns in RNA transcripts.

**splicing** removing an internal segment from a biopolymer.

**stem cell** a type of cell capable of differentiating into most or all different tissue types.

**stereoelectronic effect** a difference in conformation or reactivity that is attributable to the donation of filled orbitals into unfilled orbitals.

**sticky end** a segment of single-stranded DNA that extends from one of the strands of the duplex.

**substrate** the starting material for an enzyme-catalyzed transformation.

**substrate-assisted catalysis** catalysis by an enzyme that includes functional groups from the reaction starting material or substrate.

**sulfotransferases** a class of enzymes that adds sulfate ($SO^{3-}$) groups to hydroxyl groups or amines. Modifies glycans through the addition of a sulfate functionality.

**supercoil (supercoiling)** in DNA, the coils that form as the double helix of DNA wraps upon itself.

**synthase** an enzyme that catalyzes the synthesis of a biomolecule, such as terpenes by terpene synthase.

***t*-Boc protecting group** the tert-butoxycarbonyl protecting group used to mask amino functionalities.

**TCEP** tris-carboxyethylphosphine. A synthetic reducing agent used to cleave disulfide bonds in peptides and proteins. The reagent has three carboxylate groups that confer solubility in water.

**terminator** a DNA sequence that appears at the end of a gene to end transcription.

**terpene** a class of natural products biosynthesized from isoprene building blocks through the addition of carbocations to olefins.

**tertiary structure** elements of biooligomer structure resulting from the three-dimensional interactions of secondary structure with each other. In RNA, this level of structure results from the interactions of stems and loops with each other.

**therapeutic index** a comparison of the dosage of a drug that has a beneficial effect to the dosage that induces toxic effects.

**thioesterase (TE) domain** the domain of a polyketide synthase that catalyzes the hydrolysis of the thioester bond responsible for attaching the intermediate polyketide to the acyl carrier protein.

**topoisomerase** an enzyme that changes the topology of DNA but not the bond connectivity. Topology is a characteristic that distinguishes the vast number of unique knots that can be tied with a single piece of string.

**transacylase domain (MAT)** the domain of a polyketide synthase that catalyzes the transfer of malonyl and acetyl groups from coenzyme A to the acyl carrier protein.

**transannular** meaning "across the ring;" refers to reactions or interactions between atoms on opposite sides of a ring. Such interactions are particularly prominent in medium-sized rings composed of eight to eleven atoms.

**transcribe** to make a complementary strand of RNA based on a single-stranded DNA template by the enzyme RNA polymerase.

**transcription factor** a protein binding to a DNA sequence necessary to initiate or suppress RNA transcription by recruiting or blocking binding by RNA polymerase.

**transcriptome** the collection of all RNA transcripts from a cell, tissue, organ, or organism.

**transferase** an enzyme that catalyzes the transfer of a functional group from one compound to another.

**transient receptor potential (TRP) cation channels** channels for cationic ions that respond to heat, cold, or compounds that bind noncovalently or covalently.

**transition-state analog** a stable small molecule that mimics the transition state for an enzymatic reaction.

**translate** to synthesize a protein of defined sequence based on the sequence in a messenger RNA.

**trimeric death receptors** found at the cell surface, a class of homotrimeric receptors that transduce signals leading to apoptotic pathways.

**type I polyketide synthase** composed of a single, long polypeptide, a multidomain complex responsible for the synthesis of polyketides.

**type II polyketide synthase** composed of multiple, individual proteins, a multidomain complex responsible for the synthesis of polyketides.

**type III polyketide synthase** a protein responsible for the synthesis of polyketides, which lacks the multidomain organization of the type I and II polyketide synthases.

**van der Waals interaction** non-Coulombic interactions between nonbonded molecules.

**yeast two-hybrid** a method for investigating protein interactions inside yeast cells, based upon splitting the GAL4 transcription factor into DNA-binding and activation domains for fusion to the bait and prey proteins, respectively.

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## 5.4 THE CHEMISTRY OF DISULFIDE CROSSLINKS

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## CHAPTER 6 PROBLEMS


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## 8.1 THE CLAISEN REACTION IN POLYKETIDE BIOSYNTHESIS

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## 8.3 THE BIOLOGICAL ROLE OF HUMAN POLYKETIDES

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## 9.2 AN OVERVIEW OF SIGNAL TRANSDUCTION PATHWAYS IN HUMAN CELLS

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## 9.4 CELL-SURFACE RECEPTORS THAT INTERACT DIRECTLY WITH TRANSCRIPTION FACTORS

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## 9.5 RECEPTOR TYROSINE KINASES

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